

lyche-numerical-linear-algebra: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00001	7	1.000	$\boldsymbol{A}$	$\boldsymbol{A}$	• <i>System of linear equations.</i> Given a (square) matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$. Find a
lyche-numerical-linear-algebra_F00002	7	1.000	$\boldsymbol{b}$	$\boldsymbol{b}$	• <i>System of linear equations.</i> Given a (square) matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$. Find a
lyche-numerical-linear-algebra_F00003	7	0.993	$\boldsymbol{x}$	$\boldsymbol{x}$	vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{b}$.
lyche-numerical-linear-algebra_F00004	7	0.993	$\boldsymbol{Ax} = \boldsymbol{b}$	$\boldsymbol{Ax} = \boldsymbol{b}$	vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{b}$.
lyche-numerical-linear-algebra_F00005	7	1.000	$\boldsymbol{b} - \boldsymbol{Ax}$	$\boldsymbol{b} - \boldsymbol{Ax}$	such that the sum of squares of the components of $\boldsymbol{b} - \boldsymbol{Ax}$ is as small as possible.
lyche-numerical-linear-algebra_F00006	7	1.000	λ	λ	• <i>Eigenvalues and eigenvectors.</i> Given a (square) matrix $\boldsymbol{A}$. Find a number λ
lyche-numerical-linear-algebra_F00007	7	0.997	$\boldsymbol{Ax} = \lambda \boldsymbol{x}$	$\boldsymbol{Ax} = \lambda \boldsymbol{x}$	and/or a nonzero vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \lambda \boldsymbol{x}$.
lyche-numerical-linear-algebra_F00008	10	1.000	$\mathbb{R}^n$	$\mathbb{R}^n$	1.2.3 The Vector Spaces $\mathbb{R}^n$ and $\mathbb{C}^n$
lyche-numerical-linear-algebra_F00009	10	1.000	$\mathbb{C}^n$	$\mathbb{C}^n$	1.2.3 The Vector Spaces $\mathbb{R}^n$ and $\mathbb{C}^n$
lyche-numerical-linear-algebra_F00010	13	1.000	$\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^*$	$\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^*$	7.1.1 The Matrices $\boldsymbol{A}^* \boldsymbol{A}, \boldsymbol{A} \boldsymbol{A}^*$
lyche-numerical-linear-algebra_F00011	13	1.000	p	p	8.2.3 The Operator p -Norms
lyche-numerical-linear-algebra_F00012	15	1.000	ω	ω	12.5 The Optimal SOR Parameter ω
lyche-numerical-linear-algebra_F00013	16	1.000	λ_m	λ_m	14.5.2 Approximating λ_m
lyche-numerical-linear-algebra_F00014	17	1.000	T	T	Fig. 1.1 The triangle T defined by the three points P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00015	17	1.000	P_1, P_2	P_1, P_2	Fig. 1.1 The triangle T defined by the three points P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00016	17	1.000	P_3	P_3	Fig. 1.1 The triangle T defined by the three points P_1, P_2 and P_3
lyche-numerical-linear-algebra_F00017	17	1.000	$f(x) = \arctan(10x) + \pi/2$	$f(x) = \arctan(10x) + \pi/2$	$f(x) = \arctan(10x) + \pi/2$ on $[-1, 1]$. See text
lyche-numerical-linear-algebra_F00018	17	1.000	$[-1, 1]$	$[-1, 1]$	$f(x) = \arctan(10x) + \pi/2$ on $[-1, 1]$. See text
lyche-numerical-linear-algebra_F00019	17	1.000	$n=14$	$n = 14$	$f(x) = \arctan(10x) + \pi/2$ at $n = 14$ uniform points on
lyche-numerical-linear-algebra_F00020	17	1.000	$f(x) = x^4$	$f(x) = x^4$	Fig. 2.3 A cubic spline with one knot interpolating $f(x) = x^4$ on
lyche-numerical-linear-algebra_F00021	17	0.994	5×5	5×5	Fig. 3.2 Lower triangular 5×5 band matrices: $d = 1$
lyche-numerical-linear-algebra_F00022	17	0.994	$d=1$	$d = 1$	Fig. 3.2 Lower triangular 5×5 band matrices: $d=1$
lyche-numerical-linear-algebra_F00023	17	1.000	$d=2$	$d = 2$	(left) and $d = 2$ right
lyche-numerical-linear-algebra_F00024	17	0.997	$\boldsymbol{v}_1$	$\boldsymbol{v}_1$	Fig. 5.1 The construction of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in Gram-Schmidt. The

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra_F00025	17	■ 0.997	$\backslash\boldsymbol{v}_{_2}$	$\boldsymbol{v}_2$	Fig. 5.1 The construction of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in Gram-Schmidt. The
lyche-numerical-linear-algebra_F00026	17	■ 1.000	c	c	constant c is given by $c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$
lyche-numerical-linear-algebra_F00027	17	■ 1.000	$c:=\left\langle\boldsymbol{s}_2,\boldsymbol{v}_1\right\rangle /\left\langle\boldsymbol{v}_1,\boldsymbol{v}_1\right\rangle$	$c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$	constant c is given by $c := \langle \boldsymbol{s}_2, \boldsymbol{v}_1 \rangle / \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle$
lyche-numerical-linear-algebra_F00028	17	■ 1.000	$\boldsymbol{s}+\boldsymbol{t}$	$\boldsymbol{s} + \boldsymbol{t}$	Fig. 5.2 The orthogonal projections of $\boldsymbol{s} + \boldsymbol{t}$ into $\mathcal{S}$ and $\mathcal{T}$
lyche-numerical-linear-algebra_F00029	17	■ 1.000	$\mathcal{S}$	$\mathcal{S}$	Fig. 5.2 The orthogonal projections of $\boldsymbol{s} + \boldsymbol{t}$ into $\mathcal{S}$ and $\mathcal{T}$
lyche-numerical-linear-algebra_F00030	17	■ 1.000	$\mathcal{T}$	$\mathcal{T}$	Fig. 5.2 The orthogonal projections of $\boldsymbol{s} + \boldsymbol{t}$ into $\mathcal{S}$ and $\mathcal{T}$
lyche-numerical-linear-algebra_F00031	17	■ 0.683	$y_1^2 / 9+y_2^2=1$	$y_1^2/9+y_2^2=1$	Fig. 7.1 The ellipse $y_1^2/9+y_2^2=1$ (left) and the rotated ellipse $\mathcal{AS}$
lyche-numerical-linear-algebra_F00032	17	■ 0.683	$\boldsymbol{A} \ \boldsymbol{\mathcal{S}}$	$\mathcal{AS}$	Fig. 7.1 The ellipse $y_1^2/9+y_2^2=1$ (left) and the rotated ellipse $\mathcal{AS}$
lyche-numerical-linear-algebra_F00033	18	■ 0.995	$n=100$	$n=100$	$(n=100)$
lyche-numerical-linear-algebra_F00034	18	■ 1.000	$\alpha \rightarrow\left 1-\alpha \lambda_1\right $	$\alpha \rightarrow\left 1-\alpha \lambda_1\right $	Fig. 12.1 The functions $\alpha \rightarrow\left 1-\alpha \lambda_1\right $ and $\alpha \rightarrow\left 1-\alpha \lambda_n\right $
lyche-numerical-linear-algebra_F00035	18	■ 1.000	$\alpha \rightarrow\left 1-\alpha \lambda_n\right $	$\alpha \rightarrow\left 1-\alpha \lambda_n\right $	Fig. 12.1 The functions $\alpha \rightarrow\left 1-\alpha \lambda_1\right $ and $\alpha \rightarrow\left 1-\alpha \lambda_n\right $
lyche-numerical-linear-algebra_F00036	18	■ 0.737	$12.2 \ \rho\left(\boldsymbol{G}_{\omega}\right)$	$12.2 \rho\left(\boldsymbol{G}_{\omega}\right)$	Fig. 12.2 $\rho\left(\boldsymbol{G}_{\omega}\right)$ with $\omega \in[0,2]$ for $n=100$, (lower curve) and
lyche-numerical-linear-algebra_F00037	18	■ 0.737	$\omega \in[0,2]$	$\omega \in[0,2]$	Fig. 12.2 $\rho\left(\boldsymbol{G}_{\omega}\right)$ with $\omega \in[0,2]$ for $n=100$, (lower curve) and
lyche-numerical-linear-algebra_F00038	18	■ 1.000	$n=2500$	$n=2500$	$n=2500$ (upper curve)
lyche-numerical-linear-algebra_F00039	18	■ 0.987	$Q(x, y)$	$Q(x, y)$	Fig. 13.1 Level curves for $Q(x, y)$ given by (13.4). Also shown
lyche-numerical-linear-algebra_F00040	18	■ 1.000	Q	Q	gradient iteration (right) to find the minimum of Q (cf
lyche-numerical-linear-algebra_F00041	18	■ 1.000	$\boldsymbol{x}-\boldsymbol{x}_{_0}$	$\boldsymbol{x}-\boldsymbol{x}_0$	Fig. 13.2 The orthogonal projection of $\boldsymbol{x}-\boldsymbol{x}_0$ into $\mathbb{W}_k$
lyche-numerical-linear-algebra_F00042	18	■ 1.000	$\mathbb{W}_k$	$\mathbb{W}_k$	Fig. 13.2 The orthogonal projection of $\boldsymbol{x}-\boldsymbol{x}_0$ into $\mathbb{W}_k$
lyche-numerical-linear-algebra_F00043	18	■ 0.999	$k=3 \ . \ f \equiv Q-Q^{*}$	$k=3 . f \equiv Q-Q^{*}$	$k=3 . f \equiv Q-Q^{*}$ has a double zero at μ_1 and one zero
lyche-numerical-linear-algebra_F00044	18	■ 0.999	μ_1	μ_1	$k=3 . f \equiv Q-Q^{*}$ has a double zero at μ_1 and one zero
lyche-numerical-linear-algebra_F00045	18	■ 1.000	μ_2	μ_2	between μ_2 and μ_3
lyche-numerical-linear-algebra_F00046	18	■ 1.000	μ_3	μ_3	between μ_2 and μ_3
lyche-numerical-linear-algebra_F00047	18	■ 0.991	R_i	R_i	Fig. 14.1 The Gershgorin disk R_i
lyche-numerical-linear-algebra_F00048	19	■ 1.000	k_n	k_n	Table 12.1 The number of iterations k_n to solve the discrete Poisson
lyche-numerical-linear-algebra_F00049	19	■ 1.000	n	n	problem with n unknowns using the methods of Jacobi,
lyche-numerical-linear-algebra_F00050	19	■ 1.000	10^{-8}	10^{-8}	Gauss-Seidel, and SOR (see text) with a tolerance 10^{-8}
lyche-numerical-linear-algebra_F00051	19	■ 1.000	$\boldsymbol{G}_J, \boldsymbol{G}_1, \boldsymbol{G}_{\omega^{*}}$	$\boldsymbol{G}_J, \boldsymbol{G}_1, \boldsymbol{G}_{\omega^{*}}$	Table 12.2 Spectral radial for $\boldsymbol{G}_J, \boldsymbol{G}_1, \boldsymbol{G}_{\omega^{*}}$ and the smallest integer
lyche-numerical-linear-algebra_F00052	19	■ 1.000	$\rho\left(\boldsymbol{G}\right)^{k_n} \leq 10^{-8}$	$\rho\left(\boldsymbol{G}\right)^{k_n} \leq 10^{-8}$	k_n such that $\rho\left(\boldsymbol{G}\right)^{k_n} < 10^{-8}$

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lyche-numerical-linear-algebra_F00053	19	■ 1.000	K	K	Table 13.1 The number of iterations K for the averaging problem on a
lyche-numerical-linear-algebra_F00054	19	■ 1.000	$\sqrt{n} \times \sqrt{n}$	$\sqrt{n} \times \sqrt{n}$	$\sqrt{n} \times \sqrt{n}$ grid for various n
lyche-numerical-linear-algebra_F00055	19	■ 0.926	K_{pre}	K_{pre}	Table 13.3 The number of iterations K (no preconditioning) and K_{pre}
lyche-numerical-linear-algebra_F00056	21	■ 1.000	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	complex numbers are denoted by $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, respectively.
lyche-numerical-linear-algebra_F00057	21	■ 1.000	$v := e$	$v := e$	2. We use the “colon equal” symbol $v := e$ to indicate that the symbol v is defined
lyche-numerical-linear-algebra_F00058	21	■ 1.000	v	v	2. We use the “colon equal” symbol $v := e$ to indicate that the symbol v is defined
lyche-numerical-linear-algebra_F00059	21	■ 1.000	e	e	by the expression e .
lyche-numerical-linear-algebra_F00060	21	■ 1.000	$\mathbf{x} \in \mathbb{R}^n$	$\mathbf{x} \in \mathbb{R}^n$	column vectors. Thus $\mathbf{x} \in \mathbb{R}^n$ means
lyche-numerical-linear-algebra_F00061	21	■ 1.000	$x_i \in \mathbb{R}$	$x_i \in \mathbb{R}$	where $x_i \in \mathbb{R}$ for $i = 1, \dots, n$. Row vectors are normally identified using the
lyche-numerical-linear-algebra_F00062	21	■ 1.000	$i=1, \dots, n$	$i = 1, \dots, n$	where $x_i \in \mathbb{R}$ for $i = 1, \dots, n$. Row vectors are normally identified using the
lyche-numerical-linear-algebra_F00063	21	■ 1.000	$\mathbf{x}^T$	$\mathbf{x}^T$	transpose operation. Thus if $\mathbf{x} \in \mathbb{R}^n$ then $\mathbf{x}$ is a column vector and $\mathbf{x}^T$ is a row
lyche-numerical-linear-algebra_F00064	22	■ 1.000	$\mathbb{R}^{m \times n}$	$\mathbb{R}^{m \times n}$	5. $\mathbb{R}^{m \times n}$ is the set of matrices A with real elements. The integers m and n are the
lyche-numerical-linear-algebra_F00065	22	■ 1.000	m	m	5. $\mathbb{R}^{m \times n}$ is the set of matrices A with real elements. The integers m and n are the
lyche-numerical-linear-algebra_F00066	22	■ 0.998	i	i	The element in the i th row and j th column of A will be denoted by $a_{i,j}, a_{ij}$,
lyche-numerical-linear-algebra_F00067	22	■ 0.998	j	j	The element in the i th row and j th column of A will be denoted by $a_{i,j}, a_{ij}$,
lyche-numerical-linear-algebra_F00068	22	■ 0.998	$a_{i,j}, a_{ij}$	$a_{i,j}, a_{ij}$	The element in the i th row and j th column of A will be denoted by $a_{i,j}, a_{ij}$,
lyche-numerical-linear-algebra_F00069	22	■ 1.000	$A(i,j)$	$A(i,j)$	$A(i,j)$ or $(A)_{i,j}$. We use the notations
lyche-numerical-linear-algebra_F00070	22	■ 1.000	$(A)_{i,j}$	$(A)_{i,j}$	$A(i,j)$ or $(A)_{i,j}$. We use the notations
lyche-numerical-linear-algebra_F00071	22	■ 1.000	$\mathbf{a}_{:j}$	$\mathbf{a}_{:j}$	for the columns $\mathbf{a}_{:j}$ and rows $\mathbf{a}_{i:}^T$ of A . We often drop the colon and write $\mathbf{a}_j$
lyche-numerical-linear-algebra_F00072	22	■ 1.000	$\mathbf{a}_{i:}^T$	$\mathbf{a}_{i:}^T$	for the columns $\mathbf{a}_{:j}$ and rows $\mathbf{a}_{i:}^T$ of A . We often drop the colon and write $\mathbf{a}_j$
lyche-numerical-linear-algebra_F00073	22	■ 1.000	$\mathbf{a}_j$	$\mathbf{a}_j$	for the columns $\mathbf{a}_{:j}$ and rows $\mathbf{a}_{i:}^T$ of A . We often drop the colon and write $\mathbf{a}_j$
lyche-numerical-linear-algebra_F00074	22	■ 1.000	$\mathbf{a}_i^T$	$\mathbf{a}_i^T$	and $\mathbf{a}_i^T$ with the risk of some confusion. If $m = 1$ then A is a row vector, if
lyche-numerical-linear-algebra_F00075	22	■ 1.000	$m=1$	$m = 1$	and $\mathbf{a}_i^T$ with the risk of some confusion. If $m = 1$ then A is a row vector, if
lyche-numerical-linear-algebra_F00076	22	■ 1.000	$n=1$	$n = 1$	$n = 1$ then A is a column vector, while if $m = n$ then A is a square matrix.
lyche-numerical-linear-algebra_F00077	22	■ 1.000	$m=n$	$m = n$	$n = 1$ then A is a column vector, while if $m = n$ then A is a square matrix.
lyche-numerical-linear-algebra_F00078	22	■ 0.999	$A, B, C, \dots$	$A, B, C, \dots$	In this text we will denote matrices by boldface capital letters $A, B, C, \dots$ and
lyche-numerical-linear-algebra_F00079	22	■ 1.000	$\mathbf{x}, \mathbf{y}, \mathbf{z}, \dots$	$\mathbf{x}, \mathbf{y}, \mathbf{z}, \dots$	vectors most often by boldface lower case letters $\mathbf{x}, \mathbf{y}, \mathbf{z}, \dots$.

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lyche-numerical-linear-algebra-F00080	22	■ 1.000	$x=a+i\ b$	$x = a + ib$	6. A complex number is a number written in the form $x = a + ib$, where a, b
lyche-numerical-linear-algebra-F00081	22	■ 1.000	$a,\ b$	a, b	6. A complex number is a number written in the form $x = a + ib$, where a, b
lyche-numerical-linear-algebra-F00082	22	■ 1.000	$i^2=-1$	$i^2 = -1$	are real numbers and i , the imaginary unit , satisfies $i^2 = -1$. The set of all
lyche-numerical-linear-algebra-F00083	22	■ 1.000	$\mathbb{C}$	$\mathbb{C}$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra-F00084	22	■ 1.000	$a=\operatorname{Re} x$	$a = \operatorname{Re} x$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra-F00085	22	■ 1.000	$b=\operatorname{Im} x$	$b = \operatorname{Im} x$	such numbers is denoted by $\mathbb{C}$. The numbers $a = \operatorname{Re} x$ and $b = \operatorname{Im} x$ are the
lyche-numerical-linear-algebra-F00086	22	■ 1.000	x	x	real and imaginary part of x . The number $\bar{x} := a - ib$ is called the complex
lyche-numerical-linear-algebra-F00087	22	■ 1.000	$\bar{x}:=a-i\ b$	$\bar{x} := a - ib$	real and imaginary part of x . The number $\bar{x} := a - ib$ is called the complex
lyche-numerical-linear-algebra-F00088	22	■ 1.000	$ x :=\sqrt{\bar{x}x}\ x=\sqrt{a^2+b^2}$	$ x := \sqrt{\bar{x}x} = \sqrt{a^2 + b^2}$	conjugate of $x = a + ib$, and $ x := \sqrt{\bar{x}x} = \sqrt{a^2 + b^2}$ the absolute value or
lyche-numerical-linear-algebra-F00089	23	■ 1.000	$e^{x+y}=e^x\ e^y$	$e^{x+y} = e^x e^y$	We have $e^{x+y} = e^x e^y$ for all $x, y \in \mathbb{C}$. The polar form of a complex number is
lyche-numerical-linear-algebra-F00090	23	■ 1.000	$x,\ y\in\mathbb{C}$	$x, y \in \mathbb{C}$	We have $e^{x+y} = e^x e^y$ for all $x, y \in \mathbb{C}$. The polar form of a complex number is
lyche-numerical-linear-algebra-F00091	23	■ 1.000	$A\in\mathbb{C}^{m\times n}$	$A \in \mathbb{C}^{m \times n}$	7. For matrices and vectors with complex elements we use the notation $A \in \mathbb{C}^{m \times n}$
lyche-numerical-linear-algebra-F00092	23	■ 1.000	$x\in\mathbb{C}^n$	$x \in \mathbb{C}^n$	and $x \in \mathbb{C}^n$. We define complex row vectors using either the transpose x^T or
lyche-numerical-linear-algebra-F00093	23	■ 1.000	$x^*:=\overline{x^T}=\left[\bar{x}_1,\ldots,\bar{x}_n\right]$	$x^* := \bar{x}^T = [\bar{x}_1, \dots, \bar{x}_n]$	the conjugate transpose operation $x^* := \bar{x}^T = [\bar{x}_1, \dots, \bar{x}_n]$. If $x \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra-F00094	23	■ 0.809	$x^*=x^T$	$x^* = x^T$	$x^* = x^T$.
lyche-numerical-linear-algebra-F00095	23	■ 1.000	$x,y\in\mathbb{C}^n$	$x, y \in \mathbb{C}^n$	8. For $x, y \in \mathbb{C}^n$ and $a \in \mathbb{C}$ the operations of vector addition and scalar
lyche-numerical-linear-algebra-F00096	23	■ 1.000	$a\in\mathbb{C}$	$a \in \mathbb{C}$	8. For $x, y \in \mathbb{C}^n$ and $a \in \mathbb{C}$ the operations of vector addition and scalar
lyche-numerical-linear-algebra-F00097	23	■ 1.000	$C:=A+B$	$C := A + B$	• matrix addition $C := A + B$ if A, B, C are matrices of the same size, i.e.,
lyche-numerical-linear-algebra-F00098	23	■ 1.000	A,B,C	A, B, C	• matrix addition $C := A + B$ if A, B, C are matrices of the same size, i.e.,
lyche-numerical-linear-algebra-F00099	23	■ 1.000	$c_{ij}=a_{ij}+b_{ij}$	$c_{ij} := a_{ij} + b_{ij}$	with the same number of rows and columns, and $c_{ij} := a_{ij} + b_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00100	23	■ 1.000	$i,\ j$	i, j	with the same number of rows and columns, and $c_{ij} := a_{ij} + b_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00101	23	■ 1.000	$C:=\alpha\ A$	$C := \alpha A$	• multiplication by a scalar $C := \alpha A$, where $c_{ij} := \alpha a_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00102	23	■ 1.000	$c_{ij}:=\alpha\ a_{ij}$	$c_{ij} := \alpha a_{ij}$	• multiplication by a scalar $C := \alpha A$, where $c_{ij} := \alpha a_{ij}$ for all i, j .
lyche-numerical-linear-algebra-F00103	23	■ 1.000	$C:=AB,\ C=A\cdot B$	$C := AB, C = A \cdot B$	• matrix multiplication $C := AB, C = A \cdot B$ or $C = A * B$, where $A \in$
lyche-numerical-linear-algebra-F00104	23	■ 1.000	$C=A*B$	$C = A * B$	• matrix multiplication $C := AB, C = A \cdot B$ or $C = A * B$, where $A \in$
lyche-numerical-linear-algebra-F00105	23	■ 1.000	$A\in$	$A \in$	• matrix multiplication $C := AB, C = A \cdot B$ or $C = A * B$, where $A \in$
lyche-numerical-linear-algebra-F00106	23	■ 1.000	$\mathbb{C}^{m\times p},\ B\in\mathbb{C}^{p\times n},\ C\in\mathbb{C}^{m\times n}$	$\mathbb{C}^{m \times p}, B \in \mathbb{C}^{p \times n}, C \in \mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times p}, B \in \mathbb{C}^{p \times n}, C \in \mathbb{C}^{m \times n}$, and $c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$ for $i = 1, \dots, m$,

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lyche-numerical-linear-algebra-F00107	23	■ 1.000	$c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$	$c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$	$\mathbb{C}^{m \times p}$, $\mathbf{B} \in \mathbb{C}^{p \times n}$, $\mathbf{C} \in \mathbb{C}^{m \times n}$, and $c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$ for $i = 1, \dots, m$,
lyche-numerical-linear-algebra-F00108	23	■ 1.000	$i=1, \ldots, m$	$i = 1, \dots, m$	$\mathbb{C}^{m \times p}$, $\mathbf{B} \in \mathbb{C}^{p \times n}$, $\mathbf{C} \in \mathbb{C}^{m \times n}$, and $c_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$ for $i = 1, \dots, m$,
lyche-numerical-linear-algebra-F00109	23	■ 1.000	$j=1, \ldots, n$	$j = 1, \dots, n$	$j = 1, \dots, n$.
lyche-numerical-linear-algebra-F00110	23	■ 1.000	$\boldsymbol{C} := \boldsymbol{A} \times \boldsymbol{B}, \boldsymbol{D} := \boldsymbol{A} / \boldsymbol{B}$	$\mathbf{C} := \mathbf{A} \times \mathbf{B}, \mathbf{D} := \mathbf{A} / \mathbf{B}$	element-by-element matrix operations $\mathbf{C} := \mathbf{A} \times \mathbf{B}, \mathbf{D} := \mathbf{A} / \mathbf{B}$, and $\mathbf{E} :=$
lyche-numerical-linear-algebra-F00111	23	■ 1.000	$\boldsymbol{E} :=$	$\mathbf{E} :=$	element-by-element matrix operations $\mathbf{C} := \mathbf{A} \times \mathbf{B}, \mathbf{D} := \mathbf{A} / \mathbf{B}$, and $\mathbf{E} :=$
lyche-numerical-linear-algebra-F00112	23	■ 1.000	$\boldsymbol{A} \wedge r$	$\mathbf{A} \wedge r$	$\mathbf{A} \wedge r$ where all matrices are of the same size and $c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$
lyche-numerical-linear-algebra-F00113	23	■ 1.000	$c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$	$c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$	$\mathbf{A} \wedge r$ where all matrices are of the same size and $c_{ij} := a_{ij} b_{ij}, d_{ij} := a_{ij} / b_{ij}$
lyche-numerical-linear-algebra-F00114	23	■ 1.000	$e_{ij} := a_{ij}^r$	$e_{ij} := a_{ij}^r$	and $e_{ij} := a_{ij}^r$ for all i, j and suitable r . For the division $\mathbf{A} / \mathbf{B}$ we assume that
lyche-numerical-linear-algebra-F00115	23	■ 1.000	r	r	and $e_{ij} := a_{ij}^r$ for all i, j and suitable r . For the division $\mathbf{A} / \mathbf{B}$ we assume that
lyche-numerical-linear-algebra-F00116	23	■ 1.000	$\boldsymbol{A} / \boldsymbol{B}$	$\mathbf{A} / \mathbf{B}$	and $e_{ij} := a_{ij}^r$ for all i, j and suitable r . For the division $\mathbf{A} / \mathbf{B}$ we assume that
lyche-numerical-linear-algebra-F00117	23	■ 1.000	$\boldsymbol{B}$	$\mathbf{B}$	all elements of $\mathbf{B}$ are nonzero. The element-by-element product $\mathbf{C} = \mathbf{A} \times \mathbf{B}$
lyche-numerical-linear-algebra-F00118	23	■ 1.000	$\boldsymbol{C} = \boldsymbol{A} \times \boldsymbol{B}$	$\mathbf{C} = \mathbf{A} \times \mathbf{B}$	all elements of $\mathbf{B}$ are nonzero. The element-by-element product $\mathbf{C} = \mathbf{A} \times \mathbf{B}$
lyche-numerical-linear-algebra-F00119	23	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{m \times n}$	$\mathbf{A} \in \mathbb{R}^{m \times n}$	10. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ or $\mathbf{A} \in \mathbb{C}^{m \times n}$. The transpose $\mathbf{A}^T$ and conjugate transpose
lyche-numerical-linear-algebra-F00120	23	■ 1.000	$\boldsymbol{A}^T$	$\mathbf{A}^T$	10. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ or $\mathbf{A} \in \mathbb{C}^{m \times n}$. The transpose $\mathbf{A}^T$ and conjugate transpose
lyche-numerical-linear-algebra-F00121	23	■ 1.000	$\boldsymbol{A}^*$	$\mathbf{A}^*$	$\mathbf{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00122	23	■ 1.000	$n \times m$	$n \times m$	$\mathbf{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00123	23	■ 1.000	$a_{ij}^T := a_{ji}$	$a_{ij}^T := a_{ji}$	$\mathbf{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00124	23	■ 1.000	$a_{ij}^* := \bar{a}_{ji}$	$a_{ij}^* := \bar{a}_{ji}$	$\mathbf{A}^*$ are $n \times m$ matrices with elements $a_{ij}^T := a_{ji}$ and $a_{ij}^* := \bar{a}_{ji}$, respectively.
lyche-numerical-linear-algebra-F00125	23	■ 1.000	n, p	n, p	If $\mathbf{B}$ is an n, p matrix then $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ and $(\mathbf{AB})^* = \mathbf{B}^* \mathbf{A}^*$. A matrix
lyche-numerical-linear-algebra-F00126	23	■ 1.000	$(\boldsymbol{A} \boldsymbol{B})^T = \boldsymbol{B}^T \boldsymbol{A}^T$	$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$	If $\mathbf{B}$ is an n, p matrix then $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ and $(\mathbf{AB})^* = \mathbf{B}^* \mathbf{A}^*$. A matrix
lyche-numerical-linear-algebra-F00127	23	■ 1.000	$(\boldsymbol{A} \boldsymbol{B})^* = \boldsymbol{B}^* \boldsymbol{A}^*$	$(\mathbf{AB})^* = \mathbf{B}^* \mathbf{A}^*$	If $\mathbf{B}$ is an n, p matrix then $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$ and $(\mathbf{AB})^* = \mathbf{B}^* \mathbf{A}^*$. A matrix
lyche-numerical-linear-algebra-F00128	23	■ 1.000	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$	$\mathbf{A} \in \mathbb{C}^{n \times n}$	$\mathbf{A} \in \mathbb{C}^{n \times n}$ is symmetric if $\mathbf{A}^T = \mathbf{A}$ and Hermitian if $\mathbf{A}^* = \mathbf{A}$.
lyche-numerical-linear-algebra-F00129	23	■ 1.000	$\boldsymbol{A}^T = \boldsymbol{A}$	$\mathbf{A}^T = \mathbf{A}$	$\mathbf{A} \in \mathbb{C}^{n \times n}$ is symmetric if $\mathbf{A}^T = \mathbf{A}$ and Hermitian if $\mathbf{A}^* = \mathbf{A}$.
lyche-numerical-linear-algebra-F00130	23	■ 1.000	$\boldsymbol{A}^* = \boldsymbol{A}$	$\mathbf{A}^* = \mathbf{A}$	$\mathbf{A} \in \mathbb{C}^{n \times n}$ is symmetric if $\mathbf{A}^T = \mathbf{A}$ and Hermitian if $\mathbf{A}^* = \mathbf{A}$.
lyche-numerical-linear-algebra-F00131	23	■ 1.000	$\boldsymbol{I}_n = \boldsymbol{I} := \left[\delta_{ij} \right]_{i,j=1}^n$	$\mathbf{I}_n = \mathbf{I} := [\delta_{ij}]_{i,j=1}^n$	while $\mathbf{I}_n = \mathbf{I} := [\delta_{ij}]_{i,j=1}^n$, where
lyche-numerical-linear-algebra-F00132	23	■ 1.000	$\boldsymbol{I}$	$\mathbf{I}$	rows of $\mathbf{I}$ are the unit vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$.

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lyche-numerical-linear-algebra_F00133	23	■ 1.000	$\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n$	$\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n$	rows of $\boldsymbol{I}$ are the unit vectors $\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n$.
lyche-numerical-linear-algebra_F00134	24	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$ or $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. Then $\boldsymbol{A}$ is
lyche-numerical-linear-algebra_F00135	24	■ 1.000	$a_{ij} = 0$	$a_{ij} = 0$	• diagonal if $a_{ij} = 0$ for $i \neq j$.
lyche-numerical-linear-algebra_F00136	24	■ 1.000	$i \neq j$	$i \neq j$	• diagonal if $a_{ij} = 0$ for $i \neq j$.
lyche-numerical-linear-algebra_F00137	24	■ 1.000	$i > j$	$i > j$	• upper triangular or right triangular if $a_{ij} = 0$ for $i > j$.
lyche-numerical-linear-algebra_F00138	24	■ 1.000	$i < j$	$i < j$	• lower triangular or left triangular if $a_{ij} = 0$ for $i < j$.
lyche-numerical-linear-algebra_F00139	24	■ 1.000	$i > j + 1$	$i > j + 1$	• upper Hessenberg if $a_{ij} = 0$ for $i > j + 1$.
lyche-numerical-linear-algebra_F00140	24	■ 1.000	$i < j + 1$	$i < j + 1$	• lower Hessenberg if $a_{ij} = 0$ for $i < j + 1$.
lyche-numerical-linear-algebra_F00141	24	■ 1.000	$ i - j > 1$	$ i - j > 1$	• tridiagonal if $a_{ij} = 0$ for $ i - j > 1$.
lyche-numerical-linear-algebra_F00142	24	■ 0.999	d	d	• d-banded if $a_{ij} = 0$ for $ i - j > d$.
lyche-numerical-linear-algebra_F00143	24	■ 0.999	$ i - j > d$	$ i - j > d$	• d-banded if $a_{ij} = 0$ for $ i - j > d$.
lyche-numerical-linear-algebra_F00144	24	■ 1.000	$n \times n$	$n \times n$	13. We use the following notations for diagonal- and tridiagonal $n \times n$ matrices
lyche-numerical-linear-algebra_F00145	24	■ 1.000	$b_{ii} := d_i$	$b_{ii} := d_i$	Here $b_{ii} := d_i$ for $i = 1, \ldots, n$, $b_{i+1,i} := a_i$, $b_{i,i+1} := c_i$ for $i = 1, \ldots, n - 1$,
lyche-numerical-linear-algebra_F00146	24	■ 1.000	$i=1, \ldots, n, b_{i+1,i}:=a_i, b_{i,i+1}:=c_i$	$i = 1, \ldots, n, b_{i+1,i} := a_i, b_{i,i+1} := c_i$	Here $b_{ii} := d_i$ for $i = 1, \ldots, n, b_{i+1,i} := a_i, b_{i,i+1} := c_i$ for $i = 1, \ldots, n - 1$,
lyche-numerical-linear-algebra_F00147	24	■ 1.000	$i=1, \ldots, n-1$	$i = 1, \ldots, n - 1$	Here $b_{ii} := d_i$ for $i = 1, \ldots, n, b_{i+1,i} := a_i, b_{i,i+1} := c_i$ for $i = 1, \ldots, n - 1$,
lyche-numerical-linear-algebra_F00148	24	■ 1.000	$b_{ij} := 0$	$b_{ij} := 0$	and $b_{ij} := 0$ otherwise.
lyche-numerical-linear-algebra_F00149	24	■ 1.000	$1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j_1 < j_2 < \cdots < j_c \leq n$	$1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j_1 < j_2 < \cdots < j_c \leq n$	14. Suppose $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $1 \leq i_1 < i_2 < \cdots < i_r \leq m, 1 \leq j_1 < j_2 < \cdots < j_c \leq n$.
lyche-numerical-linear-algebra_F00150	24	■ 1.000	$j_c \leq n$	$j_c \leq n$	$j_c \leq n$. The matrix $\boldsymbol{A}(i, j) \in \mathbb{C}^{r \times c}$ is the submatrix of $\boldsymbol{A}$ consisting of rows
lyche-numerical-linear-algebra_F00151	24	■ 1.000	$\boldsymbol{A}(i, j) \in \mathbb{C}^{r \times c}$	$\boldsymbol{A}(i, j) \in \mathbb{C}^{r \times c}$	$j_c \leq n$. The matrix $\boldsymbol{A}(i, j) \in \mathbb{C}^{r \times c}$ is the submatrix of $\boldsymbol{A}$ consisting of rows
lyche-numerical-linear-algebra_F00152	24	■ 1.000	$i := [i_1, \ldots, i_r]$	$i := [i_1, \ldots, i_r]$	$i := [i_1, \ldots, i_r]$ and columns $j := [j_1, \ldots, j_c]$
lyche-numerical-linear-algebra_F00153	24	■ 1.000	$j := [j_1, \ldots, j_c]$	$j := [j_1, \ldots, j_c]$	$i := [i_1, \ldots, i_r]$ and columns $j := [j_1, \ldots, j_c]$
lyche-numerical-linear-algebra_F00154	25	■ 1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	Many mathematical systems have analogous properties to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$.
lyche-numerical-linear-algebra_F00155	25	■ 1.000	$\mathbb{R}^3$	$\mathbb{R}^3$	Many mathematical systems have analogous properties to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$.
lyche-numerical-linear-algebra_F00156	25	■ 1.000	$\mathcal{V}$	$\mathcal{V}$	Definition 1.1 (Real Vector Space) A real vector space is a nonempty set $\mathcal{V}$,
lyche-numerical-linear-algebra_F00157	25	■ 1.000	$+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$	$+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$	whose objects are called vectors , together with two operations $+: \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V}$
lyche-numerical-linear-algebra_F00158	25	■ 0.992	$\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$	$\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$	and $\cdot: \mathbb{R} \times \mathcal{V} \longrightarrow \mathcal{V}$, called addition and scalar multiplication , satisfying the
lyche-numerical-linear-algebra_F00159	25	■ 1.000	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$	following axioms for all vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.

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lyche-numerical-linear-algebra-F00160	25	■ 1.000	c, d	c, d	following axioms for all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.
lyche-numerical-linear-algebra-F00161	25	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	following axioms for all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathcal{V}$ and scalars c, d in $\mathbb{R}$.
lyche-numerical-linear-algebra-F00162	25	■ 1.000	$\mathbf{u} + \mathbf{v}$	$\mathbf{u} + \mathbf{v}$	(V1) The sum $\mathbf{u} + \mathbf{v}$ is in $\mathcal{V}$,
lyche-numerical-linear-algebra-F00163	25	■ 0.960	$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$	$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$	(V2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$,
lyche-numerical-linear-algebra-F00164	25	■ 0.994	$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$	$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$	(V3) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$,
lyche-numerical-linear-algebra-F00165	25	■ 0.999	$\mathbf{0}$	$\mathbf{0}$	(V4) There is a zero vector $\mathbf{0}$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$,
lyche-numerical-linear-algebra-F00166	25	■ 0.999	$\mathbf{u} + \mathbf{0} = \mathbf{u}$	$\mathbf{u} + \mathbf{0} = \mathbf{u}$	(V4) There is a zero vector $\mathbf{0}$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$,
lyche-numerical-linear-algebra-F00167	25	■ 1.000	$\mathbf{u}$	$\mathbf{u}$	(V5) For each $\mathbf{u}$ in $\mathcal{V}$ there is a vector $-\mathbf{u}$ in $\mathcal{V}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$,
lyche-numerical-linear-algebra-F00168	25	■ 1.000	$-\mathbf{u}$	$-\mathbf{u}$	(V5) For each $\mathbf{u}$ in $\mathcal{V}$ there is a vector $-\mathbf{u}$ in $\mathcal{V}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$,
lyche-numerical-linear-algebra-F00169	25	■ 1.000	$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$	$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$	(V5) For each $\mathbf{u}$ in $\mathcal{V}$ there is a vector $-\mathbf{u}$ in $\mathcal{V}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$,
lyche-numerical-linear-algebra-F00170	25	■ 1.000	$c \cdot \mathbf{u}$	$c \cdot \mathbf{u}$	(S1) The scalar multiple $c \cdot \mathbf{u}$ is in $\mathcal{V}$,
lyche-numerical-linear-algebra-F00171	25	■ 1.000	$c \cdot (\mathbf{u} + \mathbf{v}) = c \cdot \mathbf{u} + c \cdot \mathbf{v}$	$c \cdot (\mathbf{u} + \mathbf{v}) = c \cdot \mathbf{u} + c \cdot \mathbf{v}$	(S2) $c \cdot (\mathbf{u} + \mathbf{v}) = c \cdot \mathbf{u} + c \cdot \mathbf{v}$,
lyche-numerical-linear-algebra-F00172	25	■ 1.000	$(c + d) \cdot \mathbf{u} = c \cdot \mathbf{u} + d \cdot \mathbf{u}$	$(c + d) \cdot \mathbf{u} = c \cdot \mathbf{u} + d \cdot \mathbf{u}$	(S3) $(c + d) \cdot \mathbf{u} = c \cdot \mathbf{u} + d \cdot \mathbf{u}$,
lyche-numerical-linear-algebra-F00173	25	■ 1.000	$c \cdot (d \cdot \mathbf{u}) = (cd) \cdot \mathbf{u}$	$c \cdot (d \cdot \mathbf{u}) = (cd) \cdot \mathbf{u}$	(S4) $c \cdot (d \cdot \mathbf{u}) = (cd) \cdot \mathbf{u}$,
lyche-numerical-linear-algebra-F00174	25	■ 1.000	$1 \cdot \mathbf{u} = \mathbf{u}$	$1 \cdot \mathbf{u} = \mathbf{u}$	(S5) $1 \cdot \mathbf{u} = \mathbf{u}$.
lyche-numerical-linear-algebra-F00175	25	■ 0.589	$c \mathbf{v}$	$c\mathbf{v}$	The scalar multiplication symbol $\cdot$ is often omitted, writing $c\mathbf{v}$ instead of $c \cdot \mathbf{v}$. We
lyche-numerical-linear-algebra-F00176	25	■ 0.589	$c \cdot \mathbf{v}$	$c \cdot \mathbf{v}$	The scalar multiplication symbol $\cdot$ is often omitted, writing $c\mathbf{v}$ instead of $c \cdot \mathbf{v}$. We
lyche-numerical-linear-algebra-F00177	25	■ 1.000	$\mathbf{u} - \mathbf{v} := \mathbf{u} + (-\mathbf{v})$	$\mathbf{u} - \mathbf{v} := \mathbf{u} + (-\mathbf{v})$	define $\mathbf{u} - \mathbf{v} := \mathbf{u} + (-\mathbf{v})$. We call $\mathcal{V}$ a complex vector space if the scalars consist
lyche-numerical-linear-algebra-F00178	25	■ 0.974	$\mathbf{u} \in \mathcal{V}$	$\mathbf{u} \in \mathcal{V}$	2. For each $\mathbf{u} \in \mathcal{V}$ the negative $-\mathbf{u}$ of $\mathbf{u}$ is unique.
lyche-numerical-linear-algebra-F00179	25	■ 1.000	$0\mathbf{u} = \mathbf{0}, c\mathbf{0} = \mathbf{0}$	$0\mathbf{u} = \mathbf{0}, c\mathbf{0} = \mathbf{0}$	3. $0\mathbf{u} = \mathbf{0}, c\mathbf{0} = \mathbf{0}$, and $-\mathbf{u} = (-1)\mathbf{u}$.
lyche-numerical-linear-algebra-F00180	25	■ 1.000	$-\mathbf{u} = (-1)\mathbf{u}$	$-\mathbf{u} = (-1)\mathbf{u}$	3. $0\mathbf{u} = \mathbf{0}, c\mathbf{0} = \mathbf{0}$, and $-\mathbf{u} = (-1)\mathbf{u}$.
lyche-numerical-linear-algebra-F00181	25	■ 1.000	$n \in \mathbb{N}$	$n \in \mathbb{N}$	1. The spaces $\mathbb{R}^n$ and $\mathbb{C}^n$, where $n \in \mathbb{N}$, are real and complex vector spaces,
lyche-numerical-linear-algebra-F00182	25	■ 1.000	$\mathcal{D}$	$\mathcal{D}$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\mathbf{f}, \mathbf{g} : \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra-F00183	25	■ 1.000	$d \in \mathbb{N}$	$d \in \mathbb{N}$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\mathbf{f}, \mathbf{g} : \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra-F00184	25	■ 1.000	$\mathbf{f}, \mathbf{g} : \mathcal{D} \rightarrow \mathbb{R}^d$	$\mathbf{f}, \mathbf{g} : \mathcal{D} \rightarrow \mathbb{R}^d$	2. Let $\mathcal{D}$ be a subset of $\mathbb{R}$ and $d \in \mathbb{N}$. The set $\mathcal{V}$ of all functions $\mathbf{f}, \mathbf{g} : \mathcal{D} \rightarrow \mathbb{R}^d$ is a
lyche-numerical-linear-algebra-F00185	25	■ 1.000	$\mathbf{f}, \mathbf{g}$	$\mathbf{f}, \mathbf{g}$	Two functions $\mathbf{f}, \mathbf{g}$ in $\mathcal{V}$ are equal if $\mathbf{f}(t) = \mathbf{g}(t)$ for all $t \in \mathcal{D}$. The zero
lyche-numerical-linear-algebra-F00186	25	■ 1.000	$\mathbf{f}(t) = \mathbf{g}(t)$	$\mathbf{f}(t) = \mathbf{g}(t)$	Two functions $\mathbf{f}, \mathbf{g}$ in $\mathcal{V}$ are equal if $\mathbf{f}(t) = \mathbf{g}(t)$ for all $t \in \mathcal{D}$. The zero
lyche-numerical-linear-algebra-F00187	25	■ 1.000	$t \in \mathcal{D}$	$t \in \mathcal{D}$	Two functions $\mathbf{f}, \mathbf{g}$ in $\mathcal{V}$ are equal if $\mathbf{f}(t) = \mathbf{g}(t)$ for all $t \in \mathcal{D}$. The zero

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lyche-numerical-linear-algebra-F00188	25	■ 0.998	$\backslash\boldsymbol{\mathrm{f}}(t)=\mathbf{f}\{0\}$	$\boldsymbol{f}(t) = \mathbf{0}$	element is the zero function given by $\boldsymbol{f}(t) = \mathbf{0}$ for all $t \in \mathcal{D}$ and the negative
lyche-numerical-linear-algebra-F00189	25	■ 1.000	$\backslash\boldsymbol{\mathrm{f}}\}$	$\boldsymbol{f}$	of $\boldsymbol{f}$ is given by $-\boldsymbol{f} = (-1)\boldsymbol{f}$. In the following we will use boldface letters for
lyche-numerical-linear-algebra-F00190	25	■ 1.000	$-\backslash\boldsymbol{\mathrm{f}}\} = (-1) \ \backslash\boldsymbol{\mathrm{f}}\}$	$-\boldsymbol{f} = (-1)\boldsymbol{f}$	of $\boldsymbol{f}$ is given by $-\boldsymbol{f} = (-1)\boldsymbol{f}$. In the following we will use boldface letters for
lyche-numerical-linear-algebra-F00191	25	■ 1.000	$d>1$	$d > 1$	functions only if $d > 1$.
lyche-numerical-linear-algebra-F00192	25	■ 0.997	$n \ \backslash\geq \ 0$	$n \geq 0$	3. For $n \geq 0$ the space Π_n of polynomials of degree at most n consists of all
lyche-numerical-linear-algebra-F00193	25	■ 0.997	$\backslash\Pi_{\{n\}}$	Π_n	3. For $n \geq 0$ the space Π_n of polynomials of degree at most n consists of all
lyche-numerical-linear-algebra-F00194	25	■ 1.000	$p: \ \backslash\mathbb{R}\} \ \backslash\rightarrow \ \backslash\mathbb{R}\}, \ p: \ \backslash\mathbb{R}\} \ \backslash\rightarrow \ \backslash\mathbb{C}\}$	$p: \mathbb{R} \rightarrow \mathbb{R}, p: \mathbb{R} \rightarrow \mathbb{C}$	polynomials $p: \mathbb{R} \rightarrow \mathbb{R}$, $p: \mathbb{R} \rightarrow \mathbb{C}$, or $p: \mathbb{C} \rightarrow \mathbb{C}$ of the form
lyche-numerical-linear-algebra-F00195	25	■ 1.000	$p: \ \backslash\mathbb{C}\} \ \backslash\rightarrow \ \backslash\mathbb{C}\}$	$p: \mathbb{C} \rightarrow \mathbb{C}$	polynomials $p: \mathbb{R} \rightarrow \mathbb{R}$, $p: \mathbb{R} \rightarrow \mathbb{C}$, or $p: \mathbb{C} \rightarrow \mathbb{C}$ of the form
lyche-numerical-linear-algebra-F00196	26	■ 1.000	$a_{\{0\}}, \ \backslash\ldots, \ a_{\{n\}}$	$a_0, \dots, a_n$	where the coefficients $a_0, \dots, a_n$ are real or complex numbers. p is called the
lyche-numerical-linear-algebra-F00197	26	■ 1.000	$0 \ \backslash\leq \ k \ \backslash\leq \ n$	$0 \leq k \leq n$	integer $0 \leq k \leq n$ such that $p(t) = a_0 + \dots + a_k t^k$ with $a_k \neq 0$. The degree of
lyche-numerical-linear-algebra-F00198	26	■ 1.000	$p(t)=a_{\{0\}}+\backslash\cdots+a_{\{k\}} \ t^{\{k\}}$	$p(t) = a_0 + \dots + a_k t^k$	integer $0 \leq k \leq n$ such that $p(t) = a_0 + \dots + a_k t^k$ with $a_k \neq 0$. The degree of
lyche-numerical-linear-algebra-F00199	26	■ 1.000	$a_{\{k\}} \ \backslash\neq \ 0$	$a_k \neq 0$	integer $0 \leq k \leq n$ such that $p(t) = a_0 + \dots + a_k t^k$ with $a_k \neq 0$. The degree of
lyche-numerical-linear-algebra-F00200	26	■ 1.000	$n \ \backslash\geq \ 1$	$n \geq 1$	Definition 1.2 (Linear Combination) For $n \geq 1$ let $\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ be a set of
lyche-numerical-linear-algebra-F00201	26	■ 1.000	$\backslash\mathcal{X}\}:=\backslash\left\{\backslash\boldsymbol{\mathrm{x}}_{\{1\}}, \ \backslash\ldots, \ \backslash\boldsymbol{\mathrm{x}}_{\{n\}}\right\}$	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$	Definition 1.2 (Linear Combination) For $n \geq 1$ let $\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ be a set of
lyche-numerical-linear-algebra-F00202	26	■ 1.000	$c_{\{1\}}, \ \backslash\ldots, \ c_{\{n\}}$	$c_1, \dots, c_n$	vectors in a vector space $\mathcal{V}$ and let $c_1, \dots, c_n$ be scalars.
lyche-numerical-linear-algebra-F00203	26	■ 1.000	$c_{\{1\}} \ \backslash\boldsymbol{\mathrm{x}}_{\{1\}}+\backslash\cdots+c_{\{n\}} \ \backslash\boldsymbol{\mathrm{x}}_{\{n\}}$	$c_1 \boldsymbol{x}_1 + \dots + c_n \boldsymbol{x}_n$	1. The sum $c_1 \boldsymbol{x}_1 + \dots + c_n \boldsymbol{x}_n$ is called a linear combination of $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$.
lyche-numerical-linear-algebra-F00204	26	■ 1.000	$\backslash\boldsymbol{\mathrm{x}}_{\{1\}}, \ \backslash\ldots, \ \backslash\boldsymbol{\mathrm{x}}_{\{n\}}$	$\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$	1. The sum $c_1 \boldsymbol{x}_1 + \dots + c_n \boldsymbol{x}_n$ is called a linear combination of $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$.
lyche-numerical-linear-algebra-F00205	26	■ 1.000	$c_{\{j\}} \ \backslash\boldsymbol{\mathrm{x}}_{\{j\}} \ \backslash\neq \ \backslash\mathbf{0}\}$	$c_j \boldsymbol{x}_j \neq \mathbf{0}$	2. The linear combination is nontrivial if $c_j \boldsymbol{x}_j \neq \mathbf{0}$ for at least one j .
lyche-numerical-linear-algebra-F00206	26	■ 0.999	$\backslash\mathcal{X}\}$	$\mathcal{X}$	3. The set of all linear combinations of elements in $\mathcal{X}$ is denoted $\text{span}(\mathcal{X})$.
lyche-numerical-linear-algebra-F00207	26	■ 0.999	$\backslash\operatorname{span}\}(\backslash\mathcal{X}\})$	$\text{span}(\mathcal{X})$	3. The set of all linear combinations of elements in $\mathcal{X}$ is denoted $\text{span}(\mathcal{X})$.
lyche-numerical-linear-algebra-F00208	26	■ 1.000	$\backslash\left\{\backslash\boldsymbol{\mathrm{x}}_{\{1\}}, \ \backslash\ldots, \ \backslash\boldsymbol{\mathrm{x}}_{\{n\}}\right\} \ \backslash\right\}$	$\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$	exists $n \in \mathbb{N}$ and $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ in $\mathcal{V}$ such that $\mathcal{V} = \text{span}(\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\})$.
lyche-numerical-linear-algebra-F00209	26	■ 1.000	$\backslash\mathcal{V}\}=\backslash\operatorname{span}\}\backslash\left\{\backslash\left\{\backslash\boldsymbol{\mathrm{x}}_{\{1\}}, \ \backslash\ldots, \ \backslash\boldsymbol{\mathrm{x}}_{\{n\}}\right\} \ \backslash\right\} \ \backslash\right\}$	$\mathcal{V} = \text{span}(\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\})$	exists $n \in \mathbb{N}$ and $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ in $\mathcal{V}$ such that $\mathcal{V} = \text{span}(\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\})$.
lyche-numerical-linear-algebra-F00210	26	■ 1.000	$\backslash\boldsymbol{\mathrm{x}}\}=\backslash\left[\boldsymbol{x}_{\{1\}}, \ \backslash\ldots, \ \boldsymbol{x}_{\{m\}}\right]^{\{T\}}$	$\boldsymbol{x} = [x_1, \dots, x_m]^T$	1. Any $\boldsymbol{x} = [x_1, \dots, x_m]^T$ in $\mathbb{C}^m$ can be written as a linear combination of the unit
lyche-numerical-linear-algebra-F00211	26	■ 1.000	$\backslash\mathbb{C}\}^{\{m\}}$	$\mathbb{C}^m$	1. Any $\boldsymbol{x} = [x_1, \dots, x_m]^T$ in $\mathbb{C}^m$ can be written as a linear combination of the unit
lyche-numerical-linear-algebra-F00212	26	■ 0.991	$\backslash\boldsymbol{\mathrm{x}}\}=\boldsymbol{x}_{\{1\}} \ \backslash\boldsymbol{\mathrm{e}}_{\{1\}}+\boldsymbol{x}_{\{2\}} \ \backslash\boldsymbol{\mathrm{e}}_{\{2\}}+\backslash\cdots+\boldsymbol{x}_{\{m\}} \ \backslash\boldsymbol{\mathrm{e}}_{\{m\}}$	$\boldsymbol{x} = x_1 \boldsymbol{e}_1 + x_2 \boldsymbol{e}_2 + \dots + x_m \boldsymbol{e}_m$	vectors as $\boldsymbol{x} = x_1 \boldsymbol{e}_1 + x_2 \boldsymbol{e}_2 + \dots + x_m \boldsymbol{e}_m$. Thus, $\mathbb{C}^m = \text{span}(\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_m\})$ and
lyche-numerical-linear-algebra-F00213	26	■ 0.991	$\backslash\mathbb{C}\}^{\{m\}}=\backslash\operatorname{span}\}\backslash\left\{\backslash\left\{\backslash\boldsymbol{\mathrm{e}}_{\{1\}}, \ \backslash\ldots, \ \backslash\boldsymbol{\mathrm{e}}_{\{m\}}\right\} \ \backslash\right\} \ \backslash\right\}$	$\mathbb{C}^m = \text{span}(\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_m\})$	vectors as $\boldsymbol{x} = x_1 \boldsymbol{e}_1 + x_2 \boldsymbol{e}_2 + \dots + x_m \boldsymbol{e}_m$. Thus, $\mathbb{C}^m = \text{span}(\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_m\})$ and

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lyche-numerical-linear-algebra-F00214	26	■ 1.000	$\mathbb{R}^m$	$\mathbb{R}^m$	$\mathbb{C}^m$ is finite dimensional. Similarly $\mathbb{R}^m$ is finite dimensional.
lyche-numerical-linear-algebra-F00215	26	■ 1.000	$\Pi = \cup_n \Pi_n$	$\Pi = \cup_n \Pi_n$	2. Let $\Pi = \cup_n \Pi_n$ be the space of all polynomials. Π is a vector space that
lyche-numerical-linear-algebra-F00216	26	■ 1.000	Π	Π	2. Let $\Pi = \cup_n \Pi_n$ be the space of all polynomials. Π is a vector space that
lyche-numerical-linear-algebra-F00217	26	■ 1.000	$\Pi =$	$\Pi =$	is not finite dimensional. For suppose Π is finite dimensional. Then $\Pi =$
lyche-numerical-linear-algebra-F00218	26	■ 0.986	$\operatorname{span}\left(\left\{p_1, \dots, p_m\right\}\right)$	$\operatorname{span}\left(\left\{p_1, \dots, p_m\right\}\right)$	$\operatorname{span}\left(\left\{p_1, \dots, p_m\right\}\right)$ for some polynomials $p_1, \dots, p_m$. Let d be an integer such
lyche-numerical-linear-algebra-F00219	26	■ 0.986	$p_1, \dots, p_m$	$p_1, \dots, p_m$	$\operatorname{span}\left(\left\{p_1, \dots, p_m\right\}\right)$ for some polynomials $p_1, \dots, p_m$. Let d be an integer such
lyche-numerical-linear-algebra-F00220	26	■ 1.000	p_j	p_j	that the degree of p_j is less than d for $j = 1, \dots, m$. A polynomial of degree d
lyche-numerical-linear-algebra-F00221	26	■ 1.000	$j=1, \dots, m$	$j = 1, \dots, m$	that the degree of p_j is less than d for $j = 1, \dots, m$. A polynomial of degree d
lyche-numerical-linear-algebra-F00222	26	■ 0.958	$\mathcal{X} = \left\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\right\}$	$\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$	Definition 1.3 (Linear Independence) A set $\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ of nonzero
lyche-numerical-linear-algebra-F00223	26	■ 1.000	$\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$	$\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$	1. If $\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$ then the scalars $c_1, \dots, c_n$ in the representation $\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \dots +$
lyche-numerical-linear-algebra-F00224	26	■ 1.000	$\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \dots +$	$\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \dots +$	1. If $\boldsymbol{x} \in \operatorname{span}(\mathcal{X})$ then the scalars $c_1, \dots, c_n$ in the representation $\boldsymbol{x} = c_1 \boldsymbol{x}_1 + \dots +$
lyche-numerical-linear-algebra-F00225	26	■ 0.995	$c_n \boldsymbol{x}_n$	$c_n \boldsymbol{x}_n$	$c_n \boldsymbol{x}_n$ are unique.
lyche-numerical-linear-algebra-F00226	26	■ 0.995	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$	Lemma 1.1 (Linear Independence and Span) Suppose $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ span a vector
lyche-numerical-linear-algebra-F00227	26	■ 1.000	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$	space $\mathcal{V}$ and that $\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$ are linearly independent vectors in $\mathcal{V}$. Then $k \leq n$.
lyche-numerical-linear-algebra-F00228	26	■ 1.000	$k \leq n$	$k \leq n$	space $\mathcal{V}$ and that $\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$ are linearly independent vectors in $\mathcal{V}$. Then $k \leq n$.
lyche-numerical-linear-algebra-F00229	27	■ 1.000	$k > n$	$k > n$	Proof Suppose $k > n$. Write $\boldsymbol{w}_1$ as a linear combination of elements from
lyche-numerical-linear-algebra-F00230	27	■ 1.000	$\boldsymbol{w}_1$	$\boldsymbol{w}_1$	Proof Suppose $k > n$. Write $\boldsymbol{w}_1$ as a linear combination of elements from
lyche-numerical-linear-algebra-F00231	27	■ 1.000	$\mathcal{X}_0 := \left\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\right\}$	$\mathcal{X}_0 := \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	the set $\mathcal{X}_0 := \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$, say $\boldsymbol{w}_1 = c_1 \boldsymbol{v}_1 + \dots + c_n \boldsymbol{v}_n$. Since $\boldsymbol{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra-F00232	27	■ 1.000	$\boldsymbol{w}_1 = c_1 \boldsymbol{v}_1 + \dots + c_n \boldsymbol{v}_n$	$\boldsymbol{w}_1 = c_1 \boldsymbol{v}_1 + \dots + c_n \boldsymbol{v}_n$	the set $\mathcal{X}_0 := \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$, say $\boldsymbol{w}_1 = c_1 \boldsymbol{v}_1 + \dots + c_n \boldsymbol{v}_n$. Since $\boldsymbol{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra-F00233	27	■ 1.000	$\boldsymbol{w}_1 \neq \mathbf{0}$	$\boldsymbol{w}_1 \neq \mathbf{0}$	the set $\mathcal{X}_0 := \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$, say $\boldsymbol{w}_1 = c_1 \boldsymbol{v}_1 + \dots + c_n \boldsymbol{v}_n$. Since $\boldsymbol{w}_1 \neq \mathbf{0}$
lyche-numerical-linear-algebra-F00234	27	■ 1.000	c_{i_1}	c_{i_1}	not all the c 's are equal to zero. Pick a nonzero c , say c_{i_1} . Then $\boldsymbol{v}_{i_1}$ can be
lyche-numerical-linear-algebra-F00235	27	■ 1.000	$\boldsymbol{v}_{i_1}$	$\boldsymbol{v}_{i_1}$	not all the c 's are equal to zero. Pick a nonzero c , say c_{i_1} . Then $\boldsymbol{v}_{i_1}$ can be
lyche-numerical-linear-algebra-F00236	27	■ 1.000	$\boldsymbol{v}$	$\boldsymbol{v}$	expressed as a linear combination of $\boldsymbol{w}_1$ and the remaining $\boldsymbol{v}$'s. So the set $\mathcal{X}_1 :=$
lyche-numerical-linear-algebra-F00237	27	■ 1.000	$\mathcal{X}_1 :=$	$\mathcal{X}_1 :=$	expressed as a linear combination of $\boldsymbol{w}_1$ and the remaining $\boldsymbol{v}$'s. So the set $\mathcal{X}_1 :=$
lyche-numerical-linear-algebra-F00238	27	■ 1.000	$\left\{\boldsymbol{w}_1, \boldsymbol{v}_1, \dots, \boldsymbol{v}_{i_1-1}, \boldsymbol{v}_{i_1+1}, \dots, \boldsymbol{v}_n\right\}$	$\{\boldsymbol{w}_1, \boldsymbol{v}_1, \dots, \boldsymbol{v}_{i_1-1}, \boldsymbol{v}_{i_1+1}, \dots, \boldsymbol{v}_n\}$	$\{\boldsymbol{w}_1, \boldsymbol{v}_1, \dots, \boldsymbol{v}_{i_1-1}, \boldsymbol{v}_{i_1+1}, \dots, \boldsymbol{v}_n\}$ must also be a spanning set for $\mathcal{V}$. We repeat this
lyche-numerical-linear-algebra-F00239	27	■ 1.000	$\boldsymbol{w}_2$	$\boldsymbol{w}_2$	for $\boldsymbol{w}_2$ and $\mathcal{X}_1$. In the linear combination $\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$, we must
lyche-numerical-linear-algebra-F00240	27	■ 1.000	$\mathcal{X}_1$	$\mathcal{X}_1$	for $\boldsymbol{w}_2$ and $\mathcal{X}_1$. In the linear combination $\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$, we must

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lyche-numerical-linear-algebra_F00241	27	■ 1.000	$\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$	$\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$	for $\boldsymbol{w}_2$ and $\mathcal{X}_1$. In the linear combination $\boldsymbol{w}_2 = d_{i_1} \boldsymbol{w}_1 + \sum_{j \neq i_1} d_j \boldsymbol{v}_j$, we must
lyche-numerical-linear-algebra_F00242	27	■ 0.900	$d_{i_2} \neq 0$	$d_{i_2} \neq 0$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_F00243	27	■ 0.900	i_2	i_2	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_F00244	27	■ 0.900	$i_2 \neq i_1$	$i_2 \neq i_1$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_F00245	27	■ 0.900	$\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$	$\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$	have $d_{i_2} \neq 0$ for some i_2 with $i_2 \neq i_1$. For otherwise $\boldsymbol{w}_2 = d_1 \boldsymbol{w}_1$ contradicting
lyche-numerical-linear-algebra_F00246	27	■ 1.000	$\boldsymbol{w}$	$\boldsymbol{w}$	the linear independence of the $\boldsymbol{w}$'s. So the set $\mathcal{X}_2$ consisting of the $\boldsymbol{v}$'s with $\boldsymbol{v}_{i_1}$
lyche-numerical-linear-algebra_F00247	27	■ 1.000	$\mathcal{X}_2$	$\mathcal{X}_2$	the linear independence of the $\boldsymbol{w}$'s. So the set $\mathcal{X}_2$ consisting of the $\boldsymbol{v}$'s with $\boldsymbol{v}_{i_1}$
lyche-numerical-linear-algebra_F00248	27	■ 0.998	$\boldsymbol{v}_{i_2}$	$\boldsymbol{v}_{i_2}$	replaced by $\boldsymbol{w}_1$ and $\boldsymbol{v}_{i_2}$ replaced by $\boldsymbol{w}_2$ is again a spanning set for $\mathcal{V}$. Repeating this
lyche-numerical-linear-algebra_F00249	27	■ 1.000	$n - 2$	$n - 2$	process $n - 2$ more times we obtain a spanning set $\mathcal{X}_n$ where $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ have been
lyche-numerical-linear-algebra_F00250	27	■ 1.000	$\mathcal{X}_n$	$\mathcal{X}_n$	process $n - 2$ more times we obtain a spanning set $\mathcal{X}_n$ where $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ have been
lyche-numerical-linear-algebra_F00251	27	■ 1.000	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$	replaced by $\boldsymbol{w}_1, \dots, \boldsymbol{w}_k$. Since $k > n$ we can then write $\boldsymbol{w}_k$ as a linear combination
lyche-numerical-linear-algebra_F00252	27	■ 1.000	$\boldsymbol{w}_k$	$\boldsymbol{w}_k$	replaced by $\boldsymbol{w}_1, \dots, \boldsymbol{w}_n$. Since $k > n$ we can then write $\boldsymbol{w}_k$ as a linear combination
lyche-numerical-linear-algebra_F00253	27	—	$\square$	$\square$	—
lyche-numerical-linear-algebra_F00254	27	■ 1.000	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	Definition 1.4 (Basis) A finite set of vectors $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ in a vector space $\mathcal{V}$ is a
lyche-numerical-linear-algebra_F00255	27	■ 1.000	$\operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\} = \mathcal{V}$	$\operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\} = \mathcal{V}$	1. $\operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\} = \mathcal{V}$.
lyche-numerical-linear-algebra_F00256	27	■ 0.999	$\{\boldsymbol{v}_{i_1}, \dots, \boldsymbol{v}_{i_k}\}$	$\{\boldsymbol{v}_{i_1}, \dots, \boldsymbol{v}_{i_k}\}$	that $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ is a spanning set for $\mathcal{V}$. Then we can find a subset $\{\boldsymbol{v}_{i_1}, \dots, \boldsymbol{v}_{i_k}\}$
lyche-numerical-linear-algebra_F00257	27	■ 1.000	$\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	$\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	Proof Let $\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ be a finite dimensional vector space. By Theo-
lyche-numerical-linear-algebra_F00258	27	■ 1.000	$\operatorname{dim} \mathcal{V}$	$\operatorname{dim} \mathcal{V}$	space and denoted $\operatorname{dim} \mathcal{V}$.
lyche-numerical-linear-algebra_F00259	27	■ 1.000	$\mathcal{X} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	$\mathcal{X} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$	Proof Suppose $\mathcal{X} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ and $\mathcal{Y} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_k\}$ are two bases for $\mathcal{V}$. By
lyche-numerical-linear-algebra_F00260	27	■ 1.000	$\mathcal{Y} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_k\}$	$\mathcal{Y} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_k\}$	Proof Suppose $\mathcal{X} = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$ and $\mathcal{Y} = \{\boldsymbol{w}_1, \dots, \boldsymbol{w}_k\}$ are two bases for $\mathcal{V}$. By
lyche-numerical-linear-algebra_F00261	27	■ 1.000	$\mathcal{Y}$	$\mathcal{Y}$	Lemma 1.1 we have $k \leq n$. Using the same Lemma with $\mathcal{X}$ and $\mathcal{Y}$ switched we
lyche-numerical-linear-algebra_F00262	27	■ 1.000	$n \leq k$	$n \leq k$	obtain $n \leq k$. We conclude that $n = k$.
lyche-numerical-linear-algebra_F00263	27	■ 1.000	$n = k$	$n = k$	obtain $n \leq k$. We conclude that $n = k$.
lyche-numerical-linear-algebra_F00264	27	■ 1.000	$\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_n\}$	$\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_n\}$	The set of unit vectors $\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_n\}$ form a basis for both $\mathbb{R}^n$ and $\mathbb{C}^n$.
lyche-numerical-linear-algebra_F00265	27	■ 0.996	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$	vectors $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$ in a finite dimensional vector space $\mathcal{V}$ can be enlarged to a
lyche-numerical-linear-algebra_F00266	28	■ 1.000	$\boldsymbol{v}_{k+1}$	$\boldsymbol{v}_{k+1}$	Proof If $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$ does not span $\mathcal{V}$ we can enlarge the set by one vector $\boldsymbol{v}_{k+1}$
lyche-numerical-linear-algebra_F00267	28	■ 1.000	$\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$	$\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$	(V1) The sum $\boldsymbol{u} + \boldsymbol{v}$ is in $\mathcal{S}$ for any $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{S}$.
lyche-numerical-linear-algebra_F00268	28	■ 1.000	$c \boldsymbol{u}$	$c \boldsymbol{u}$	(S1) The scalar multiple $c \boldsymbol{u}$ is in $\mathcal{S}$ for any scalar c and any $\boldsymbol{u} \in \mathcal{S}$.

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lyche-numerical-linear-algebra-F00269	28	■ 1.000	$\boldsymbol{u} \in \mathcal{S}$	$\boldsymbol{u} \in \mathcal{S}$	(S1) The scalar multiple $c\boldsymbol{u}$ is in $\mathcal{S}$ for any scalar c and any $\boldsymbol{u} \in \mathcal{S}$.
lyche-numerical-linear-algebra-F00270	28	■ 1.000	$V_1 - V_5$	$V_1 - V_5$	axioms $V_1 - V_5$ and $S_1 - S_5$ are satisfied for $\mathcal{S}$. In particular, $\mathcal{S}$ must contain the
lyche-numerical-linear-algebra-F00271	28	■ 1.000	$S_1 - S_5$	$S_1 - S_5$	axioms $V_1 - V_5$ and $S_1 - S_5$ are satisfied for $\mathcal{S}$. In particular, $\mathcal{S}$ must contain the
lyche-numerical-linear-algebra-F00272	28	■ 1.000	$\{\mathbf{0}\}$	$\{\mathbf{0}\}$	1. $\{\mathbf{0}\}$, where $\mathbf{0}$ is the zero vector is a subspace, the trivial subspace . The dimension
lyche-numerical-linear-algebra-F00273	28	■ 1.000	$\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\} \subseteq \mathcal{V}$	$\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\} \subseteq \mathcal{V}$	3. $\text{span}(\mathcal{X})$ is a subspace of $\mathcal{V}$ for any $\mathcal{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\} \subseteq \mathcal{V}$. Indeed, it is easy to
lyche-numerical-linear-algebra-F00274	28	■ 1.000	$\mathcal{S} \oplus \mathcal{T}$	$\mathcal{S} \oplus \mathcal{T}$	denoted $\mathcal{S} \oplus \mathcal{T}$ if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$.
lyche-numerical-linear-algebra-F00275	28	■ 1.000	$\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$	$\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$	denoted $\mathcal{S} \oplus \mathcal{T}$ if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$.
lyche-numerical-linear-algebra-F00276	29	■ 1.000	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\}$	Proof Let $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra-F00277	28	■ 1.000	$\mathcal{S} \cap \mathcal{T}$	$\mathcal{S} \cap \mathcal{T}$	Proof Let $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra-F00278	28	■ 1.000	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\} = \emptyset$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\} = \emptyset$	Proof Let $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\}$ be a basis for $\mathcal{S} \cap \mathcal{T}$, where $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p\} = \emptyset$, the empty
lyche-numerical-linear-algebra-F00279	29	■ 1.000	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$ for $\mathcal{S}$ and a basis $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$ for $\mathcal{T}$. Every
lyche-numerical-linear-algebra-F00280	29	■ 1.000	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q\}$ for $\mathcal{S}$ and a basis $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$ for $\mathcal{T}$. Every
lyche-numerical-linear-algebra-F00281	29	■ 1.000	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$	$\boldsymbol{x} \in \mathcal{S} + \mathcal{T}$ can be written as a linear combination of
lyche-numerical-linear-algebra-F00282	29	■ 1.000	$\mathcal{S} + \mathcal{T}$	$\mathcal{S} + \mathcal{T}$	so these vectors span $\mathcal{S} + \mathcal{T}$. We show that they are linearly independent and hence
lyche-numerical-linear-algebra-F00283	29	■ 1.000	$\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$	$\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$	a basis. Suppose $\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$, where $\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$, and
lyche-numerical-linear-algebra-F00284	29	■ 1.000	$\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$	a basis. Suppose $\boldsymbol{u} + \boldsymbol{s} + \boldsymbol{t} = \mathbf{0}$, where $\boldsymbol{u} := \sum_{j=1}^p \alpha_j \boldsymbol{u}_j$, $\boldsymbol{s} := \sum_{j=1}^q \rho_j \boldsymbol{s}_j$, and
lyche-numerical-linear-algebra-F00285	29	■ 1.000	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra-F00286	29	■ 1.000	$\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$	$\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra-F00287	29	■ 1.000	$\boldsymbol{s} \in$	$\boldsymbol{s} \in$	$\boldsymbol{t} := \sum_{j=1}^r \sigma_j \boldsymbol{t}_j$. Now $\boldsymbol{s} = -(\boldsymbol{u} + \boldsymbol{t})$ belongs to both $\mathcal{S}$ and to $\mathcal{T}$ and hence $\boldsymbol{s} \in$
lyche-numerical-linear-algebra-F00288	29	■ 1.000	$\boldsymbol{s}$	$\boldsymbol{s}$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F00289	29	■ 1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F00290	29	■ 1.000	$\boldsymbol{s} :=$	$\boldsymbol{s} :=$	$\mathcal{S} \cap \mathcal{T}$. Therefore $\boldsymbol{s}$ can be written as a linear combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_p$ say $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F00291	29	■ 1.000	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$. But then $\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$ and since
lyche-numerical-linear-algebra-F00292	29	■ 1.000	$\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$	$\sum_{j=1}^p \beta_j \boldsymbol{u}_j$. But then $\mathbf{0} = \sum_{j=1}^p \beta_j \boldsymbol{u}_j - \sum_{j=1}^q \rho_j \boldsymbol{s}_j$ and since
lyche-numerical-linear-algebra-F00293	29	■ 1.000	$\beta_1 = \dots = \beta_p = \rho_1 = \dots = \rho_q = 0$	$\beta_1 = \dots = \beta_p = \rho_1 = \dots = \rho_q = 0$	is linearly independent we must have $\beta_1 = \dots = \beta_p = \rho_1 = \dots = \rho_q = 0$
lyche-numerical-linear-algebra-F00294	29	■ 1.000	$\boldsymbol{s} = \mathbf{0}$	$\boldsymbol{s} = \mathbf{0}$	and hence $\boldsymbol{s} = \mathbf{0}$. We then have $\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$ and by linear independence of

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lyche-numerical-linear-algebra_F00295	29	■ 1.000	$\boldsymbol{u}+\boldsymbol{t}=\mathbf{0}$	$\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$	and hence $\boldsymbol{s} = \mathbf{0}$. We then have $\boldsymbol{u} + \boldsymbol{t} = \mathbf{0}$ and by linear independence of
lyche-numerical-linear-algebra_F00296	29	■ 1.000	$\alpha_1=\cdots=\alpha_p=\sigma_1=\cdots=\sigma_r=0$	$\alpha_1 = \cdots = \alpha_p = \sigma_1 = \cdots = \sigma_r = 0$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$ we obtain $\alpha_1 = \cdots = \alpha_p = \sigma_1 = \cdots = \sigma_r = 0$. We
lyche-numerical-linear-algebra_F00297	29	■ 1.000	$\left\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_q, \boldsymbol{t}_1, \ldots, \boldsymbol{t}_r\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$	have shown that the vectors $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_p, \boldsymbol{s}_1, \dots, \boldsymbol{s}_q, \boldsymbol{t}_1, \dots, \boldsymbol{t}_r\}$ constitute a basis
lyche-numerical-linear-algebra_F00298	29	■ 1.000	$\operatorname{dim}\{\mathbf{0}\}=0$	$\dim\{\mathbf{0}\} = 0$	and (1.7) follows. Equation (1.7) implies (1.8) since $\dim\{\mathbf{0}\} = 0$.
lyche-numerical-linear-algebra_F00299	29	■ 1.000	$\left\{\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n\right\}$	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$	n -dimensional vector space $\mathcal{V}$ and let $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ be a basis for $\mathcal{S}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$
lyche-numerical-linear-algebra_F00300	29	■ 1.000	$\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\right\}$	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$	n -dimensional vector space $\mathcal{V}$ and let $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ be a basis for $\mathcal{S}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$
lyche-numerical-linear-algebra_F00301	29	■ 1.000	$\boldsymbol{s}_j$	$\boldsymbol{s}_j$	a basis for $\mathcal{V}$. Then each $\boldsymbol{s}_j$ can be expressed as a linear combination of $\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$,
lyche-numerical-linear-algebra_F00302	29	■ 1.000	$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$	a basis for $\mathcal{V}$. Then each $\boldsymbol{s}_j$ can be expressed as a linear combination of $\boldsymbol{v}_1, \dots, \boldsymbol{v}_m$.
lyche-numerical-linear-algebra_F00303	30	■ 1.000	$\boldsymbol{x} \in \mathcal{S}$	$\boldsymbol{x} \in \mathcal{S}$	If $\boldsymbol{x} \in \mathcal{S}$ then $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some coefficients $\boldsymbol{b} :=$
lyche-numerical-linear-algebra_F00304	30	■ 1.000	$\boldsymbol{x}=\sum_{j=1}^n c_j \boldsymbol{s}_j=\sum_{i=1}^m b_i \boldsymbol{v}_i$	$\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$	If $\boldsymbol{x} \in \mathcal{S}$ then $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some coefficients $\boldsymbol{b} :=$
lyche-numerical-linear-algebra_F00305	30	■ 1.000	$\boldsymbol{b}:=$	$\boldsymbol{b} :=$	If $\boldsymbol{x} \in \mathcal{S}$ then $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some coefficients $\boldsymbol{b} :=$
lyche-numerical-linear-algebra_F00306	30	■ 1.000	$\left[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_m\right]^T, \boldsymbol{c}:=\left[c_1, \ldots, c_n\right]^T$	$[\boldsymbol{b}_1, \dots, \boldsymbol{b}_m]^T, \boldsymbol{c} := [c_1, \dots, c_n]^T$	$[\boldsymbol{b}_1, \dots, \boldsymbol{b}_m]^T, \boldsymbol{c} := [c_1, \dots, c_n]^T$. Moreover $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$, where $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$ is
lyche-numerical-linear-algebra_F00307	30	■ 1.000	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{c}$	$\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$	$[\boldsymbol{b}_1, \dots, \boldsymbol{b}_m]^T, \boldsymbol{c} := [c_1, \dots, c_n]^T$. Moreover $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$, where $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$ is
lyche-numerical-linear-algebra_F00308	30	■ 1.000	$\boldsymbol{A}=\left[a_{i j}\right] \in \mathbb{C}^{m \times n}$	$\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$	$[\boldsymbol{b}_1, \dots, \boldsymbol{b}_m]^T, \boldsymbol{c} := [c_1, \dots, c_n]^T$. Moreover $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$, where $\boldsymbol{A} = [a_{ij}] \in \mathbb{C}^{m \times n}$ is
lyche-numerical-linear-algebra_F00309	30	■ 0.999	$a_{i j}$	a_{ij}	Proof Equation (1.9) holds for some a_{ij} since $\boldsymbol{s}_j \in \mathcal{V}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ spans $\mathcal{V}$.
lyche-numerical-linear-algebra_F00310	30	■ 0.999	$\boldsymbol{s}_j \in \mathcal{V}$	$\boldsymbol{s}_j \in \mathcal{V}$	Proof Equation (1.9) holds for some a_{ij} since $\boldsymbol{s}_j \in \mathcal{V}$ and $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ spans $\mathcal{V}$.
lyche-numerical-linear-algebra_F00311	30	■ 0.980	$\left(c_j\right)$	(c_j)	be written $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some scalars (c_j) and (b_i) . But then
lyche-numerical-linear-algebra_F00312	30	■ 0.980	$\left(b_i\right)$	(b_i)	be written $\boldsymbol{x} = \sum_{j=1}^n c_j \boldsymbol{s}_j = \sum_{i=1}^m b_i \boldsymbol{v}_i$ for some scalars (c_j) and (b_i) . But then
lyche-numerical-linear-algebra_F00313	30	■ 1.000	$b_i=\sum_{j=1}^n a_{i j} c_j$	$b_i = \sum_{j=1}^n a_{ij} c_j$	Since $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ is linearly independent it follows that $b_i = \sum_{j=1}^n a_{ij} c_j$ for $i =$
lyche-numerical-linear-algebra_F00314	30	■ 1.000	$i=$	$i =$	Since $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_m\}$ is linearly independent it follows that $b_i = \sum_{j=1}^n a_{ij} c_j$ for $i =$
lyche-numerical-linear-algebra_F00315	30	■ 1.000	$1, \ldots, m$	$1, \dots, m$	$1, \dots, m$ or $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{c}$. Finally, to show that $\boldsymbol{A}$ has linearly independent columns
lyche-numerical-linear-algebra_F00316	30	■ 1.000	$\boldsymbol{b}:=\boldsymbol{A} \boldsymbol{c}=\mathbf{0}$	$\boldsymbol{b} := \boldsymbol{A}\boldsymbol{c} = \mathbf{0}$	suppose $\boldsymbol{b} := \boldsymbol{A}\boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \dots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_F00317	30	■ 1.000	$\boldsymbol{c}=\left[c_1, \ldots, c_n\right]^T$	$\boldsymbol{c} = [c_1, \dots, c_n]^T$	suppose $\boldsymbol{b} := \boldsymbol{A}\boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \dots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_F00318	30	■ 1.000	$\boldsymbol{x}:=\sum_{j=1}^n c_j \boldsymbol{s}_j$	$\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$	suppose $\boldsymbol{b} := \boldsymbol{A}\boldsymbol{c} = \mathbf{0}$ for some $\boldsymbol{c} = [c_1, \dots, c_n]^T$. Define $\boldsymbol{x} := \sum_{j=1}^n c_j \boldsymbol{s}_j$. Then
lyche-numerical-linear-algebra_F00319	30	■ 0.995	$\boldsymbol{x}=\sum_{i=1}^m b_i \boldsymbol{v}_i$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_n\}$ is linearly

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lyche-numerical-linear-algebra_F00320	30	0.995	$\boldsymbol{b}=\mathbf{0}$	$\boldsymbol{b} = \mathbf{0}$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{s_1, \dots, s_n\}$ is linearly
lyche-numerical-linear-algebra_F00321	30	0.995	$\boldsymbol{x}=\mathbf{0}$	$\boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x} = \sum_{i=1}^m b_i \boldsymbol{v}_i$ and since $\boldsymbol{b} = \mathbf{0}$ we have $\boldsymbol{x} = \mathbf{0}$. But since $\{s_1, \dots, s_n\}$ is linearly
lyche-numerical-linear-algebra_F00322	30	1.000	$\boldsymbol{c}=\mathbf{0}$	$\boldsymbol{c} = \mathbf{0}$	independent it follows that $\boldsymbol{c} = \mathbf{0}$.
lyche-numerical-linear-algebra_F00323	30	1.000	$\mathcal{V}=\mathbb{R}^m$	$\mathcal{V} = \mathbb{R}^m$	When $\mathcal{V} = \mathbb{R}^m$ or $\mathbb{C}^m$ we can think of n vectors in $\mathcal{V}$, say $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n$, as a set
lyche-numerical-linear-algebra_F00324	30	1.000	$m \times n$	$m \times n$	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ or as the columns of an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$. A
lyche-numerical-linear-algebra_F00325	30	1.000	$\boldsymbol{X}=\left[\boldsymbol{x}_{1}, \ldots, \boldsymbol{x}_{n}\right]$	$\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$	$\mathcal{X} := \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ or as the columns of an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$. A
lyche-numerical-linear-algebra_F00326	30	0.998	$\boldsymbol{X} \boldsymbol{c}$	$\boldsymbol{X} \boldsymbol{c}$	linear combination can then be written as a matrix times vector $\boldsymbol{X} \boldsymbol{c}$, where $\boldsymbol{c} =$
lyche-numerical-linear-algebra_F00327	30	0.998	$\boldsymbol{c} =$	$\boldsymbol{c} =$	linear combination can then be written as a matrix times vector $\boldsymbol{X} \boldsymbol{c}$, where $\boldsymbol{c} =$
lyche-numerical-linear-algebra_F00328	30	1.000	$\left[c_1, \ldots, c_n\right]^T$	$[c_1, \dots, c_n]^T$	$[c_1, \dots, c_n]^T$ is the vector of scalars. Thus
lyche-numerical-linear-algebra_F00329	30	1.000	$\boldsymbol{x}_j \in \mathcal{V}, j=1, \ldots, n$	$\boldsymbol{x}_j \in \mathcal{V}, j = 1, \dots, n$	ated with an $m \times n$ matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n]$, where $\boldsymbol{x}_j \in \mathcal{V}, j = 1, \dots, n$ are the
lyche-numerical-linear-algebra_F00330	30	1.000	$\mathcal{R}(\boldsymbol{X})$	$\mathcal{R}(\boldsymbol{X})$	1. The subspace $\mathcal{R}(\boldsymbol{X})$ is called the column space of $\boldsymbol{X}$. It is the smallest subspace
lyche-numerical-linear-algebra_F00331	30	1.000	$\boldsymbol{X}$	$\boldsymbol{X}$	1. The subspace $\mathcal{R}(\boldsymbol{X})$ is called the column space of $\boldsymbol{X}$. It is the smallest subspace
lyche-numerical-linear-algebra_F00332	30	1.000	$\mathcal{R}\left(\boldsymbol{X}^T\right)$	$\mathcal{R}\left(\boldsymbol{X}^T\right)$	2. $\mathcal{R}\left(\boldsymbol{X}^T\right)$ is called the row space of $\boldsymbol{X}$. It is generated by the rows of $\boldsymbol{X}$ written as
lyche-numerical-linear-algebra_F00333	30	1.000	$\mathcal{N}(\boldsymbol{X}):=\left\{\boldsymbol{y} \in \mathbb{R}^n: \boldsymbol{X} \boldsymbol{y}=\mathbf{0}\right\}$	$\mathcal{N}(\boldsymbol{X}) := \{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X} \boldsymbol{y} = \mathbf{0}\}$	3. The subspace $\mathcal{N}(\boldsymbol{X}) := \{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{X} \boldsymbol{y} = \mathbf{0}\}$ is called the null space or kernel
lyche-numerical-linear-algebra_F00334	30	1.000	$\mathcal{N}(\boldsymbol{X})$	$\mathcal{N}(\boldsymbol{X})$	space of $\boldsymbol{X}$. The dimension of $\mathcal{N}(\boldsymbol{X})$ is called the nullity of $\boldsymbol{X}$ and denoted
lyche-numerical-linear-algebra_F00335	30	1.000	$\text{null}(\boldsymbol{X})$	$\text{null}(\boldsymbol{X})$	$\text{null}(\boldsymbol{X})$.
lyche-numerical-linear-algebra_F00336	31	0.922	$\boldsymbol{X} \in$	$\boldsymbol{X} \in$	Theorem 1.5 (Counting Dimensions of Fundamental Subspaces) Suppose $\boldsymbol{X} \in$
lyche-numerical-linear-algebra_F00337	31	1.000	$\mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times n}$	$\mathbb{C}^{m \times n}$. Then
lyche-numerical-linear-algebra_F00338	31	1.000	$\text{rank}(\boldsymbol{X})=\text{rank}\left(\boldsymbol{X}^*\right)$	$\text{rank}(\boldsymbol{X}) = \text{rank}(\boldsymbol{X}^*)$	1. $\text{rank}(\boldsymbol{X}) = \text{rank}(\boldsymbol{X}^*)$.
lyche-numerical-linear-algebra_F00339	31	1.000	$\text{rank}(\boldsymbol{X})+\text{null}(\boldsymbol{X})=n$	$\text{rank}(\boldsymbol{X}) + \text{null}(\boldsymbol{X}) = n$	2. $\text{rank}(\boldsymbol{X}) + \text{null}(\boldsymbol{X}) = n$.
lyche-numerical-linear-algebra_F00340	31	1.000	$\text{rank}(\boldsymbol{X})+\text{null}\left(\boldsymbol{X}^*\right)=m$	$\text{rank}(\boldsymbol{X}) + \text{null}(\boldsymbol{X}^*) = m$	3. $\text{rank}(\boldsymbol{X}) + \text{null}(\boldsymbol{X}^*) = m$.
lyche-numerical-linear-algebra_F00341	31	1.000	x_j	x_j	x_j , and the components b_i of the right hand side are real or complex numbers. The
lyche-numerical-linear-algebra_F00342	31	1.000	b_i	b_i	x_j , and the components b_i of the right hand side are real or complex numbers. The
lyche-numerical-linear-algebra_F00343	32	0.992	$\boldsymbol{a}_j=\left[a_{1 j}, \ldots, a_{m j}\right]^T \in \mathbb{C}^m$	$\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$	where $\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$ for $j = 1, \dots, n$ and $\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$.
lyche-numerical-linear-algebra_F00344	32	0.992	$\boldsymbol{b}=\left[b_1, \ldots, b_m\right]^T \in \mathbb{C}^m$	$\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$	where $\boldsymbol{a}_j = [a_{1j}, \dots, a_{mj}]^T \in \mathbb{C}^m$ for $j = 1, \dots, n$ and $\boldsymbol{b} = [b_1, \dots, b_m]^T \in \mathbb{C}^m$.
lyche-numerical-linear-algebra_F00345	32	1.000	$m < n, m = n$	$m < n, m = n$	square , or overdetermined if $m < n, m = n$, or $m > n$, respectively.
lyche-numerical-linear-algebra_F00346	32	1.000	$m > n$	$m > n$	square , or overdetermined if $m < n, m = n$, or $m > n$, respectively.

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lyche-numerical-linear-algebra-F00347	32	1.000	$m < n$	$m < n$	Lemma 1.3 (Underdetermined System) Suppose $A \in \mathbb{R}^{m \times n}$ with $m < n$. Then
lyche-numerical-linear-algebra-F00348	32	0.570	$\boldsymbol{Ax} = \mathbf{0}$	$\boldsymbol{Ax} = \mathbf{0}$	there is a nonzero $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \mathbf{0}$.
lyche-numerical-linear-algebra-F00349	32	1.000	$\boldsymbol{Ax} = \mathbf{0}$	$\boldsymbol{Ax} = \mathbf{0}$	nonzero $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Ax} = \mathbf{0}$.
lyche-numerical-linear-algebra-F00350	32	1.000	$\mathbb{R}^{n \times n}$	$\mathbb{R}^{n \times n}$	$\mathbb{R}^{n \times n}$ is said to be nonsingular if the only real solution of the homogeneous system
lyche-numerical-linear-algebra-F00351	32	0.658	$\boldsymbol{b} \in \mathbb{R}^n$	$\boldsymbol{b} \in \mathbb{R}^n$	The linear system $\boldsymbol{Ax} = \boldsymbol{b}$ has a unique solution $\boldsymbol{x} \in \mathbb{R}^n$ for any $\boldsymbol{b} \in \mathbb{R}^n$ if and only
lyche-numerical-linear-algebra-F00352	32	1.000	$\boldsymbol{B} = [\boldsymbol{A} \ \boldsymbol{b}] \in \mathbb{R}^{n \times (n+1)}$	$\boldsymbol{B} = [\boldsymbol{A} \ \boldsymbol{b}] \in \mathbb{R}^{n \times (n+1)}$	Proof Suppose A is nonsingular. We define $\boldsymbol{B} = [\boldsymbol{A} \ \boldsymbol{b}] \in \mathbb{R}^{n \times (n+1)}$ by adding a
lyche-numerical-linear-algebra-F00353	32	0.997	$\boldsymbol{z} \in \mathbb{R}^{n+1}$	$\boldsymbol{z} \in \mathbb{R}^{n+1}$	column to A . By Lemma 1.3 there is a nonzero $\boldsymbol{z} \in \mathbb{R}^{n+1}$ such that $\boldsymbol{Bz} = \mathbf{0}$. If we
lyche-numerical-linear-algebra-F00354	32	0.997	$\boldsymbol{Bz} = \mathbf{0}$	$\boldsymbol{Bz} = \mathbf{0}$	column to A . By Lemma 1.3 there is a nonzero $\boldsymbol{z} \in \mathbb{R}^{n+1}$ such that $\boldsymbol{Bz} = \mathbf{0}$. If we
lyche-numerical-linear-algebra-F00355	32	0.896	$\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$	$\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$	write $\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$ where $\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra-F00356	32	0.896	$\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$	$\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$	write $\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$ where $\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra-F00357	32	0.896	$z_{n+1} \in \mathbb{R}$	$z_{n+1} \in \mathbb{R}$	write $\boldsymbol{z} = \begin{bmatrix} \tilde{\boldsymbol{z}} \\ z_{n+1} \end{bmatrix}$ where $\tilde{\boldsymbol{z}} = [z_1, \dots, z_n]^T \in \mathbb{R}^n$ and $z_{n+1} \in \mathbb{R}$, then
lyche-numerical-linear-algebra-F00358	33	0.737	$z_{n+1} = 0$	$z_{n+1} = 0$	We cannot have $z_{n+1} = 0$ for then $A\tilde{\boldsymbol{z}} = \mathbf{0}$ for a nonzero $\tilde{\boldsymbol{z}}$, contradicting the
lyche-numerical-linear-algebra-F00359	33	0.737	$A\tilde{\boldsymbol{z}} = \mathbf{0}$	$A\tilde{\boldsymbol{z}} = \mathbf{0}$	We cannot have $z_{n+1} = 0$ for then $A\tilde{\boldsymbol{z}} = \mathbf{0}$ for a nonzero $\tilde{\boldsymbol{z}}$, contradicting the
lyche-numerical-linear-algebra-F00360	33	0.737	$\tilde{\boldsymbol{z}}$	$\tilde{\boldsymbol{z}}$	We cannot have $z_{n+1} = 0$ for then $A\tilde{\boldsymbol{z}} = \mathbf{0}$ for a nonzero $\tilde{\boldsymbol{z}}$, contradicting the
lyche-numerical-linear-algebra-F00361	33	0.997	$\boldsymbol{x} := -\tilde{\boldsymbol{z}}/z_{n+1}$	$\boldsymbol{x} := -\tilde{\boldsymbol{z}}/z_{n+1}$	nonsingularity of A . Define $\boldsymbol{x} := -\tilde{\boldsymbol{z}}/z_{n+1}$. Then
lyche-numerical-linear-algebra-F00362	33	1.000	$\boldsymbol{Ay} = \boldsymbol{b}$	$\boldsymbol{Ay} = \boldsymbol{b}$	Suppose $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ay} = \boldsymbol{b}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then $A(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$ and since A is
lyche-numerical-linear-algebra-F00363	33	1.000	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$	$\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$	Suppose $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ay} = \boldsymbol{b}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then $A(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$ and since A is
lyche-numerical-linear-algebra-F00364	33	1.000	$A(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$	$A(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$	Suppose $\boldsymbol{Ax} = \boldsymbol{b}$ and $\boldsymbol{Ay} = \boldsymbol{b}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then $A(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0}$ and since A is
lyche-numerical-linear-algebra-F00365	33	1.000	$\boldsymbol{x} - \boldsymbol{y} = \mathbf{0}$	$\boldsymbol{x} - \boldsymbol{y} = \mathbf{0}$	nonsingular we conclude that $\boldsymbol{x} - \boldsymbol{y} = \mathbf{0}$ or $\boldsymbol{x} = \boldsymbol{y}$. Thus the solution is unique.
lyche-numerical-linear-algebra-F00366	33	1.000	$\boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{x} = \boldsymbol{y}$	nonsingular we conclude that $\boldsymbol{x} - \boldsymbol{y} = \mathbf{0}$ or $\boldsymbol{x} = \boldsymbol{y}$. Thus the solution is unique.
lyche-numerical-linear-algebra-F00367	33	1.000	$m <$	$m <$	Lemma 1.4 (Complex Underdetermined System) Suppose $A \in \mathbb{C}^{m \times n}$ with $m <$
lyche-numerical-linear-algebra-F00368	33	1.000	$\boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{b} \in \mathbb{C}^n$ if and only if the matrix A is nonsingular.
lyche-numerical-linear-algebra-F00369	33	1.000	$\boldsymbol{B} \in \mathbb{C}^{n \times n}$	$\boldsymbol{B} \in \mathbb{C}^{n \times n}$	Suppose $A \in \mathbb{C}^{n \times n}$ is a square matrix. A matrix $\boldsymbol{B} \in \mathbb{C}^{n \times n}$ is called a right inverse
lyche-numerical-linear-algebra-F00370	33	1.000	$\boldsymbol{AB} = \boldsymbol{I}$	$\boldsymbol{AB} = \boldsymbol{I}$	of A if $\boldsymbol{AB} = \boldsymbol{I}$. A matrix $\boldsymbol{C} \in \mathbb{C}^{n \times n}$ is said to be a left inverse of A if $\boldsymbol{CA} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-F00371	33	1.000	$\boldsymbol{C} \in \mathbb{C}^{n \times n}$	$\boldsymbol{C} \in \mathbb{C}^{n \times n}$	of A if $\boldsymbol{AB} = \boldsymbol{I}$. A matrix $\boldsymbol{C} \in \mathbb{C}^{n \times n}$ is said to be a left inverse of A if $\boldsymbol{CA} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-F00372	33	1.000	$\boldsymbol{CA} = \boldsymbol{I}$	$\boldsymbol{CA} = \boldsymbol{I}$	of A if $\boldsymbol{AB} = \boldsymbol{I}$. A matrix $\boldsymbol{C} \in \mathbb{C}^{n \times n}$ is said to be a left inverse of A if $\boldsymbol{CA} = \boldsymbol{I}$.

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lyche-numerical-linear-algebra_F00373	33	1.000	$\boldsymbol{C}$	$\boldsymbol{C}$	inverse $\boldsymbol{B}$ and a left inverse $\boldsymbol{C}$ then
lyche-numerical-linear-algebra_F00374	33	1.000	$\boldsymbol{A}^{-1}$	$\boldsymbol{A}^{-1}$	and this common inverse is called the inverse of $\boldsymbol{A}$ and denoted by $\boldsymbol{A}^{-1}$. Thus the
lyche-numerical-linear-algebra_F00375	33	1.000	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$ $\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{I}$	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$	inverse satisfies $\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$.
lyche-numerical-linear-algebra_F00376	33	0.998	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$ $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	Theorem 1.8 (Product of Nonsingular Matrices) If $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{AB} =$
lyche-numerical-linear-algebra_F00377	33	0.998	$\boldsymbol{AB} =$	$\boldsymbol{AB} =$	Theorem 1.8 (Product of Nonsingular Matrices) If $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{AB} =$
lyche-numerical-linear-algebra_F00378	33	1.000	$\boldsymbol{BA} = \boldsymbol{I}$	$\boldsymbol{BA} = \boldsymbol{I}$	if either $\boldsymbol{AB} = \boldsymbol{I}$ or $\boldsymbol{BA} = \boldsymbol{I}$ then $\boldsymbol{A}$ is nonsingular and $\boldsymbol{A}^{-1} = \boldsymbol{B}$.
lyche-numerical-linear-algebra_F00379	33	1.000	$\boldsymbol{A}^{-1} = \boldsymbol{B}$	$\boldsymbol{A}^{-1} = \boldsymbol{B}$	if either $\boldsymbol{AB} = \boldsymbol{I}$ or $\boldsymbol{BA} = \boldsymbol{I}$ then $\boldsymbol{A}$ is nonsingular and $\boldsymbol{A}^{-1} = \boldsymbol{B}$.
lyche-numerical-linear-algebra_F00380	33	1.000	$\boldsymbol{Cx} = \mathbf{0}$	$\boldsymbol{Cx} = \mathbf{0}$	Proof Suppose both $\boldsymbol{A}$ and $\boldsymbol{B}$ are nonsingular and let $\boldsymbol{Cx} = \mathbf{0}$. Then $\boldsymbol{ABx} = \mathbf{0}$ and
lyche-numerical-linear-algebra_F00381	33	1.000	$\boldsymbol{ABx} = \mathbf{0}$	$\boldsymbol{ABx} = \mathbf{0}$	Proof Suppose both $\boldsymbol{A}$ and $\boldsymbol{B}$ are nonsingular and let $\boldsymbol{Cx} = \mathbf{0}$. Then $\boldsymbol{ABx} = \mathbf{0}$ and
lyche-numerical-linear-algebra_F00382	33	1.000	$\boldsymbol{Bx} = \mathbf{0}$	$\boldsymbol{Bx} = \mathbf{0}$	since $\boldsymbol{A}$ is nonsingular we see that $\boldsymbol{Bx} = \mathbf{0}$. Since $\boldsymbol{B}$ is nonsingular we have $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra_F00383	34	1.000	$\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\mathbf{0} = \mathbf{0}$ $\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\mathbf{0} = \mathbf{0}$	$\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\mathbf{0} = \mathbf{0}$	vector so that $\boldsymbol{Bx} = \mathbf{0}$. But then $\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\mathbf{0} = \mathbf{0}$ so $\boldsymbol{C}$ is
lyche-numerical-linear-algebra_F00384	34	1.000	$\tilde{\boldsymbol{x}}$	$\tilde{\boldsymbol{x}}$	singular. Finally suppose $\boldsymbol{B}$ is nonsingular, but $\boldsymbol{A}$ is singular. Let $\tilde{\boldsymbol{x}}$ be a nonzero
lyche-numerical-linear-algebra_F00385	34	1.000	$\boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$	$\boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$	vector such that $\boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$. By Theorem 1.7 there is a vector $\boldsymbol{x}$ such that $\boldsymbol{Bx} = \tilde{\boldsymbol{x}}$ and
lyche-numerical-linear-algebra_F00386	34	1.000	$\boldsymbol{Bx} = \tilde{\boldsymbol{x}}$	$\boldsymbol{Bx} = \tilde{\boldsymbol{x}}$	vector such that $\boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$. By Theorem 1.7 there is a vector $\boldsymbol{x}$ such that $\boldsymbol{Bx} = \tilde{\boldsymbol{x}}$ and
lyche-numerical-linear-algebra_F00387	34	1.000	$\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$ $\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$	$\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$	$\boldsymbol{x}$ is nonzero since $\tilde{\boldsymbol{x}}$ is nonzero. But then $\boldsymbol{Cx} = (\boldsymbol{AB})\boldsymbol{x} = \boldsymbol{A}(\boldsymbol{Bx}) = \boldsymbol{A}\tilde{\boldsymbol{x}} = \mathbf{0}$ for a
lyche-numerical-linear-algebra_F00388	34	1.000	$\boldsymbol{Ab}_i = \boldsymbol{e}_i$	$\boldsymbol{Ab}_i = \boldsymbol{e}_i$	systems $\boldsymbol{Ab}_i = \boldsymbol{e}_i$ has a unique solution $\boldsymbol{b}_i$ for $i = 1, \dots, n$. Let $\boldsymbol{B} = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$.
lyche-numerical-linear-algebra_F00389	34	1.000	$\boldsymbol{b}_i$	$\boldsymbol{b}_i$	systems $\boldsymbol{Ab}_i = \boldsymbol{e}_i$ has a unique solution $\boldsymbol{b}_i$ for $i = 1, \dots, n$. Let $\boldsymbol{B} = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$.
lyche-numerical-linear-algebra_F00390	34	1.000	$\boldsymbol{B} = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$	$\boldsymbol{B} = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$	systems $\boldsymbol{Ab}_i = \boldsymbol{e}_i$ has a unique solution $\boldsymbol{b}_i$ for $i = 1, \dots, n$. Let $\boldsymbol{B} = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_n]$.
lyche-numerical-linear-algebra_F00391	34	0.968	$\boldsymbol{AB} = [\boldsymbol{Ab}_1, \dots, \boldsymbol{Ab}_n] = [\boldsymbol{e}_1, \dots, \boldsymbol{e}_n] = \boldsymbol{I}$ $\boldsymbol{AB} = [\boldsymbol{Ab}_1, \dots, \boldsymbol{Ab}_n] = [\boldsymbol{e}_1, \dots, \boldsymbol{e}_n] = \boldsymbol{I}$	$\boldsymbol{AB} = [\boldsymbol{Ab}_1, \dots, \boldsymbol{Ab}_n] = [\boldsymbol{e}_1, \dots, \boldsymbol{e}_n] = \boldsymbol{I}$	Then $\boldsymbol{AB} = [\boldsymbol{Ab}_1, \dots, \boldsymbol{Ab}_n] = [\boldsymbol{e}_1, \dots, \boldsymbol{e}_n] = \boldsymbol{I}$ so that $\boldsymbol{A}$ has a right inverse $\boldsymbol{B}$.
lyche-numerical-linear-algebra_F00392	34	1.000	$1.8 \boldsymbol{B}$	$1.8 \boldsymbol{B}$	By Theorem 1.8 $\boldsymbol{B}$ is nonsingular since $\boldsymbol{I}$ is nonsingular and $\boldsymbol{AB} = \boldsymbol{I}$. Since $\boldsymbol{B}$ is
lyche-numerical-linear-algebra_F00393	34	1.000	$\boldsymbol{BC} = \boldsymbol{I}$	$\boldsymbol{BC} = \boldsymbol{I}$	inverse $\boldsymbol{C}$, i.e. $\boldsymbol{BC} = \boldsymbol{I}$. But then $\boldsymbol{AB} = \boldsymbol{BC} = \boldsymbol{I}$ so $\boldsymbol{B}$ has both a right inverse and
lyche-numerical-linear-algebra_F00394	34	1.000	$\boldsymbol{AB} = \boldsymbol{BC} = \boldsymbol{I}$	$\boldsymbol{AB} = \boldsymbol{BC} = \boldsymbol{I}$	inverse $\boldsymbol{C}$, i.e. $\boldsymbol{BC} = \boldsymbol{I}$. But then $\boldsymbol{AB} = \boldsymbol{BC} = \boldsymbol{I}$ so $\boldsymbol{B}$ has both a right inverse and
lyche-numerical-linear-algebra_F00395	34	1.000	$\boldsymbol{A} = \boldsymbol{C}$	$\boldsymbol{A} = \boldsymbol{C}$	a left inverse which must be equal so $\boldsymbol{A} = \boldsymbol{C}$. Since $\boldsymbol{BC} = \boldsymbol{I}$ we have $\boldsymbol{BA} = \boldsymbol{I}$, so
lyche-numerical-linear-algebra_F00396	34	0.995	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$	Corollary 1.2 (Basic Properties of the Inverse Matrix) Suppose $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$

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lyche-numerical-linear-algebra_F00397	34	■ 1.000	$\left(\boldsymbol{A}^{-1}\right)^{-1}=\boldsymbol{A}$	$(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$	1. $\boldsymbol{A}^{-1}$ is nonsingular and $(\boldsymbol{A}^{-1})^{-1} = \boldsymbol{A}$.
lyche-numerical-linear-algebra_F00398	34	■ 1.000	$\boldsymbol{C}=\boldsymbol{A} \ \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$	2. $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$ is nonsingular and $\boldsymbol{C}^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00399	34	■ 1.000	$\boldsymbol{C}^{-1}=\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$	$\boldsymbol{C}^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$	2. $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$ is nonsingular and $\boldsymbol{C}^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00400	34	■ 1.000	$\left(\boldsymbol{A}^T\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^T=:\boldsymbol{A}^{-T}$	$(\boldsymbol{A}^T)^{-1} = (\boldsymbol{A}^{-1})^T =: \boldsymbol{A}^{-T}$	3. $\boldsymbol{A}^T$ is nonsingular and $(\boldsymbol{A}^T)^{-1} = (\boldsymbol{A}^{-1})^T =: \boldsymbol{A}^{-T}$.
lyche-numerical-linear-algebra_F00401	34	■ 1.000	$\left(\boldsymbol{A}^*\right)^{-1}=\left(\boldsymbol{A}^{-1}\right)^*=:\boldsymbol{A}^{-*}$	$(\boldsymbol{A}^*)^{-1} = (\boldsymbol{A}^{-1})^* =: \boldsymbol{A}^{-*}$	4. $\boldsymbol{A}^*$ is nonsingular and $(\boldsymbol{A}^*)^{-1} = (\boldsymbol{A}^{-1})^* =: \boldsymbol{A}^{-*}$.
lyche-numerical-linear-algebra_F00402	34	■ 1.000	$c \ \boldsymbol{A}$	$c\boldsymbol{A}$	5. $c\boldsymbol{A}$ is nonsingular and $(c\boldsymbol{A})^{-1} = \frac{1}{c}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00403	34	■ 1.000	$(c \ \boldsymbol{A})^{-1}=\frac{1}{c} \ \boldsymbol{A}^{-1}$	$(c\boldsymbol{A})^{-1} = \frac{1}{c}\boldsymbol{A}^{-1}$	5. $c\boldsymbol{A}$ is nonsingular and $(c\boldsymbol{A})^{-1} = \frac{1}{c}\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra_F00404	34	■ 1.000	$\boldsymbol{A}^{-1} \boldsymbol{A}=\boldsymbol{I}$	$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$	1. Since $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$ the matrix $\boldsymbol{A}$ is a right inverse of $\boldsymbol{A}^{-1}$. Thus $\boldsymbol{A}^{-1}$ is nonsingular
lyche-numerical-linear-algebra_F00405	34	■ 1.000	$\left(\boldsymbol{B}^{-1} \boldsymbol{A}^{-1}\right)(\boldsymbol{A} \boldsymbol{B})=\boldsymbol{B}^{-1}\left(\boldsymbol{A}^{-1} \boldsymbol{A}\right) \boldsymbol{B}=\boldsymbol{B}^{-1} \boldsymbol{B}=\boldsymbol{I}$	$(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{A}\boldsymbol{B}) = \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} = \boldsymbol{B}^{-1}\boldsymbol{B} = \boldsymbol{I}$	2. We note that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{A}\boldsymbol{B}) = \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} = \boldsymbol{B}^{-1}\boldsymbol{B} = \boldsymbol{I}$. Thus $\boldsymbol{A}\boldsymbol{B}$ is
lyche-numerical-linear-algebra_F00406	34	■ 1.000	$\boldsymbol{A} \ \boldsymbol{B}$	$\boldsymbol{A}\boldsymbol{B}$	2. We note that $(\boldsymbol{B}^{-1}\boldsymbol{A}^{-1})(\boldsymbol{A}\boldsymbol{B}) = \boldsymbol{B}^{-1}(\boldsymbol{A}^{-1}\boldsymbol{A})\boldsymbol{B} = \boldsymbol{B}^{-1}\boldsymbol{B} = \boldsymbol{I}$. Thus $\boldsymbol{A}\boldsymbol{B}$ is
lyche-numerical-linear-algebra_F00407	34	■ 1.000	$\boldsymbol{I}=\boldsymbol{I}^T=\left(\boldsymbol{A}^{-1} \boldsymbol{A}\right)^T=\boldsymbol{A}^T\left(\boldsymbol{A}^{-1}\right)^T$	$\boldsymbol{I} = \boldsymbol{I}^T = (\boldsymbol{A}^{-1}\boldsymbol{A})^T = \boldsymbol{A}^T(\boldsymbol{A}^{-1})^T$	3. Now $\boldsymbol{I} = \boldsymbol{I}^T = (\boldsymbol{A}^{-1}\boldsymbol{A})^T = \boldsymbol{A}^T(\boldsymbol{A}^{-1})^T$ showing that $(\boldsymbol{A}^{-1})^T$ is a right inverse
lyche-numerical-linear-algebra_F00408	34	■ 1.000	$\left(\boldsymbol{A}^{-1}\right)^T$	$(\boldsymbol{A}^{-1})^T$	3. Now $\boldsymbol{I} = \boldsymbol{I}^T = (\boldsymbol{A}^{-1}\boldsymbol{A})^T = \boldsymbol{A}^T(\boldsymbol{A}^{-1})^T$ showing that $(\boldsymbol{A}^{-1})^T$ is a right inverse
lyche-numerical-linear-algebra_F00409	34	■ 1.000	$\frac{1}{c} \ \boldsymbol{A}^{-1}$	$\frac{1}{c}\boldsymbol{A}^{-1}$	4. The matrix $\frac{1}{c}\boldsymbol{A}^{-1}$ is a one sided inverse of $c\boldsymbol{A}$.
lyche-numerical-linear-algebra_F00410	35	■ 1.000	$n!$	$n!$	This sum ranges of all $n!$ permutations of $\{1, 2, \dots, n\}$. Moreover, $\text{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00411	35	■ 1.000	$\{1, 2, \ldots, n\}$	$\{1, 2, \dots, n\}$	This sum ranges of all $n!$ permutations of $\{1, 2, \dots, n\}$. Moreover, $\text{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00412	35	■ 1.000	$\operatorname{sign}(\sigma)$	$\text{sign}(\sigma)$	This sum ranges of all $n!$ permutations of $\{1, 2, \dots, n\}$. Moreover, $\text{sign}(\sigma)$ equals
lyche-numerical-linear-algebra_F00413	35	■ 1.000	σ	σ	the number of times a bigger integer precedes a smaller one in σ . We also denote
lyche-numerical-linear-algebra_F00414	35	■ 1.000	ϵ	ϵ	The first term on the right corresponds to the identity permutation ϵ given by $\epsilon(i) =$
lyche-numerical-linear-algebra_F00415	35	■ 1.000	$\epsilon(i)=$	$\epsilon(i) =$	The first term on the right corresponds to the identity permutation ϵ given by $\epsilon(i) =$
lyche-numerical-linear-algebra_F00416	35	■ 1.000	$i, \ i=1,2$	$i, i = 1, 2$	$i, i = 1, 2$. The second term comes from the permutation $\sigma = \{2, 1\}$. For $n = 3$
lyche-numerical-linear-algebra_F00417	35	■ 1.000	$\sigma=\{2,1\}$	$\sigma = \{2, 1\}$	$i, i = 1, 2$. The second term comes from the permutation $\sigma = \{2, 1\}$. For $n = 3$
lyche-numerical-linear-algebra_F00418	35	■ 1.000	$n=3$	$n = 3$	$i, i = 1, 2$. The second term comes from the permutation $\sigma = \{2, 1\}$. For $n = 3$
lyche-numerical-linear-algebra_F00419	35	■ 0.719	$\{1, 2, 3\}$	$\{1, 2, 3\}$	there are six permutations of $\{1, 2, 3\}$. Then
lyche-numerical-linear-algebra_F00420	35	■ 0.999	$\operatorname{sign}(\{1, 2, 3\})=\operatorname{sign}(\{2, 3, 1\})=\operatorname{sign}(\{3, 1, 2\})=1$	$\text{sign}(\{1, 2, 3\}) = \text{sign}(\{2, 3, 1\}) = \text{sign}(\{3, 1, 2\}) = 1$	This follows since $\text{sign}(\{1, 2, 3\}) = \text{sign}(\{2, 3, 1\}) = \text{sign}(\{3, 1, 2\}) = 1$, and
lyche-numerical-linear-algebra_F00421	35	■ 0.749	$\operatorname{sign}(\{2, 1, 3\})=\operatorname{sign}(\{3, 2, 1\})=\operatorname{sign}(\{1, 3, 2\})=-1$	$\text{sign}(\{2, 1, 3\}) = \text{sign}(\{3, 2, 1\}) = \text{sign}(\{1, 3, 2\}) = -1$	$\text{sign}(\{2, 1, 3\}) = \text{sign}(\{3, 2, 1\}) = \text{sign}(\{1, 3, 2\}) = -1$.

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lyche-numerical-linear-algebra_F00422	35	■ 1.000	$\operatorname{det}(\boldsymbol{A})=$	$\det(\boldsymbol{A}) =$	to reduce it to a simpler form. For example, if $\boldsymbol{A}$ is triangular then $\det(\boldsymbol{A}) =$
lyche-numerical-linear-algebra_F00423	36	■ 1.000	$a_{11} a_{22} \cdots a_{nn}$	$a_{11}a_{22}\cdots a_{nn}$	$a_{11}a_{22}\cdots a_{nn}$, the product of the diagonal elements. In particular, for the identity
lyche-numerical-linear-algebra_F00424	36	■ 1.000	$\operatorname{det}(\boldsymbol{I})=1$	$\det(\boldsymbol{I}) = 1$	matrix $\det(\boldsymbol{I}) = 1$. The elementary operations using either rows or columns are
lyche-numerical-linear-algebra_F00425	36	■ 1.000	$\operatorname{det}(\boldsymbol{B})=-\operatorname{det}(\boldsymbol{A})$	$\det(\boldsymbol{B}) = -\det(\boldsymbol{A})$	1. Interchanging two rows(columns): $\det(\boldsymbol{B}) = -\det(\boldsymbol{A})$,
lyche-numerical-linear-algebra_F00426	36	■ 1.000	$\alpha, \operatorname{det}(\boldsymbol{B})=\alpha \operatorname{det}(\boldsymbol{A})$	$\alpha, \det(\boldsymbol{B}) = \alpha \det(\boldsymbol{A})$	2. Multiply a row(column) by a scalar: α , $\det(\boldsymbol{B}) = \alpha \det(\boldsymbol{A})$,
lyche-numerical-linear-algebra_F00427	36	■ 1.000	$\boldsymbol{A}_{i, j}$	$\boldsymbol{A}_{i,j}$	Here $\boldsymbol{A}_{i,j}$ denotes the submatrix of $\boldsymbol{A}$ obtained by deleting the i th row and j th
lyche-numerical-linear-algebra_F00428	36	■ 1.000	$1 \leq i, j \leq n$	$1 \leq i, j \leq n$	column of $\boldsymbol{A}$. For $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ and $1 \leq i, j \leq n$ the determinant $\det(\boldsymbol{A}_{ij})$ is called the
lyche-numerical-linear-algebra_F00429	36	■ 1.000	$\operatorname{det}(\boldsymbol{A}_{i j})$	$\det(\boldsymbol{A}_{ij})$	column of $\boldsymbol{A}$. For $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ and $1 \leq i, j \leq n$ the determinant $\det(\boldsymbol{A}_{ij})$ is called the
lyche-numerical-linear-algebra_F00430	36	■ 0.969	$\left(x_1, y_1\right)$	(x_1, y_1)	line through two points (x_1, y_1) and (x_2, y_2) in the plane can be written as the
lyche-numerical-linear-algebra_F00431	36	■ 0.969	$\left(x_2, y_2\right)$	(x_2, y_2)	line through two points (x_1, y_1) and (x_2, y_2) in the plane can be written as the
lyche-numerical-linear-algebra_F00432	36	■ 1.000	$\operatorname{det}(\boldsymbol{A})=0$	$\det(\boldsymbol{A}) = 0$	Rearranging the equation $\det(\boldsymbol{A}) = 0$ we obtain
lyche-numerical-linear-algebra_F00433	37	■ 1.000	$\boldsymbol{A}, \boldsymbol{B}$	$\boldsymbol{A}, \boldsymbol{B}$	$\boldsymbol{A}, \boldsymbol{B}$ are square matrices of order n with real or complex elements, then
lyche-numerical-linear-algebra_F00434	37	■ 0.984	$\operatorname{det}(\boldsymbol{A B})=\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{B})$	$\det(\boldsymbol{AB}) = \det(\boldsymbol{A}) \det(\boldsymbol{B})$	1. $\det(\boldsymbol{AB}) = \det(\boldsymbol{A}) \det(\boldsymbol{B})$.
lyche-numerical-linear-algebra_F00435	37	■ 1.000	$\operatorname{det}\left(\boldsymbol{A}^T\right)=\operatorname{det}(\boldsymbol{A})$	$\det\left(\boldsymbol{A}^T\right) = \det(\boldsymbol{A})$	2. $\det(\boldsymbol{A}^T) = \det(\boldsymbol{A})$, and $\det(\boldsymbol{A}^*) = \overline{\det(\boldsymbol{A})}$, (complex conjugate).
lyche-numerical-linear-algebra_F00436	37	■ 1.000	$\operatorname{det}\left(\boldsymbol{A}^*\right)=\overline{\operatorname{det}(\boldsymbol{A})}$	$\det\left(\boldsymbol{A}^*\right) = \overline{\det(\boldsymbol{A})}$	2. $\det(\boldsymbol{A}^T) = \det(\boldsymbol{A})$, and $\det(\boldsymbol{A}^*) = \overline{\det(\boldsymbol{A})}$, (complex conjugate).
lyche-numerical-linear-algebra_F00437	37	■ 1.000	$\operatorname{det}(a \boldsymbol{A})=a^n \operatorname{det}(\boldsymbol{A})$	$\det(a\boldsymbol{A}) = a^n \det(\boldsymbol{A})$	3. $\det(a\boldsymbol{A}) = a^n \det(\boldsymbol{A})$, for $a \in \mathbb{C}$.
lyche-numerical-linear-algebra_F00438	37	■ 0.943	$\boldsymbol{A}=\left[\begin{array}{ll}\boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E}\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$	5. If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$.
lyche-numerical-linear-algebra_F00439	37	■ 0.943	$\boldsymbol{C}, \boldsymbol{E}$	$\boldsymbol{C}, \boldsymbol{E}$	5. If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$.
lyche-numerical-linear-algebra_F00440	37	■ 0.943	$\operatorname{det}(\boldsymbol{A})=\operatorname{det}(\boldsymbol{C}) \operatorname{det}(\boldsymbol{E})$	$\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$	5. If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{C} & \boldsymbol{D} \\ \mathbf{0} & \boldsymbol{E} \end{bmatrix}$ for some square matrices $\boldsymbol{C}, \boldsymbol{E}$ then $\det(\boldsymbol{A}) = \det(\boldsymbol{C}) \det(\boldsymbol{E})$.
lyche-numerical-linear-algebra_F00441	37	■ 1.000	$\boldsymbol{x}=$	$\boldsymbol{x} =$	6. Cramer's rule Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is nonsingular and $\boldsymbol{b} \in \mathbb{C}^n$. Let $\boldsymbol{x} =$
lyche-numerical-linear-algebra_F00442	37	■ 0.982	$\left[x_1, x_2, \ldots, x_n\right]^T$	$[x_1, x_2, \dots, x_n]^T$	$[x_1, x_2, \dots, x_n]^T$ be the unique solution of $\boldsymbol{Ax} = \boldsymbol{b}$. Then
lyche-numerical-linear-algebra_F00443	37	■ 1.000	$\boldsymbol{A}_j(\boldsymbol{b})$	$\boldsymbol{A}_j(\boldsymbol{b})$	where $\boldsymbol{A}_j(\boldsymbol{b})$ denote the matrix obtained from $\boldsymbol{A}$ by replacing the j th column of
lyche-numerical-linear-algebra_F00444	37	■ 0.999	$\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	$\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	where the matrix $\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$ with elements $\operatorname{adj}(\boldsymbol{A})_{i, j} = (-1)^{i+j} \det(\boldsymbol{A}_{j, i})$
lyche-numerical-linear-algebra_F00445	37	■ 0.999	$\operatorname{adj}(\boldsymbol{A})_{i, j}=(-1)^{i+j} \operatorname{det}\left(\boldsymbol{A}_{j, i}\right)$	$\operatorname{adj}(\boldsymbol{A})_{i, j} = (-1)^{i+j} \det(\boldsymbol{A}_{j, i})$	where the matrix $\operatorname{adj}(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$ with elements $\operatorname{adj}(\boldsymbol{A})_{i, j} = (-1)^{i+j} \det(\boldsymbol{A}_{j, i})$
lyche-numerical-linear-algebra_F00446	37	■ 1.000	$\boldsymbol{A}_{j, i}$	$\boldsymbol{A}_{j, i}$	is called the adjoint of $\boldsymbol{A}$. Moreover, $\boldsymbol{A}_{j, i}$ denotes the submatrix of $\boldsymbol{A}$ obtained

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F00447	37	1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$	$\boldsymbol{A} \in \mathbb{C}^{m \times p}, \boldsymbol{B} \in \mathbb{C}^{p \times n}$	8. Cauchy-Binet formula: Let $\boldsymbol{A} \in \mathbb{C}^{m \times p}$, $\boldsymbol{B} \in \mathbb{C}^{p \times n}$ and $\boldsymbol{C} = \boldsymbol{AB}$. Suppose
lyche-numerical-linear-algebra-F00448	37	1.000	$1 \leq r \leq \min\{m, n, p\}$	$1 \leq r \leq \min\{m, n, p\}$	$r \leq \min\{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \dots, i_r\}$ and $\boldsymbol{j} = \{j_1, \dots, j_r\}$ be integers
lyche-numerical-linear-algebra-F00449	37	1.000	$\boldsymbol{i} = \{i_1, \dots, i_r\}$	$\boldsymbol{i} = \{i_1, \dots, i_r\}$	$1 \leq r \leq \min\{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \dots, i_r\}$ and $\boldsymbol{j} = \{j_1, \dots, j_r\}$ be integers
lyche-numerical-linear-algebra-F00450	37	1.000	$\boldsymbol{j} = \{j_1, \dots, j_r\}$	$\boldsymbol{j} = \{j_1, \dots, j_r\}$	$1 \leq r \leq \min\{m, n, p\}$ and let $\boldsymbol{i} = \{i_1, \dots, i_r\}$ and $\boldsymbol{j} = \{j_1, \dots, j_r\}$ be integers
lyche-numerical-linear-algebra-F00451	37	1.000	$1 \leq i_1 < i_2 < \dots < i_r \leq m$	$1 \leq i_1 < i_2 < \dots < i_r \leq m$	with $1 \leq i_1 < i_2 < \dots < i_r \leq m$ and $1 \leq j_1 < j_2 < \dots < j_r \leq n$. Then
lyche-numerical-linear-algebra-F00452	37	1.000	$1 \leq j_1 < j_2 < \dots < j_r \leq n$	$1 \leq j_1 < j_2 < \dots < j_r \leq n$	with $1 \leq i_1 < i_2 < \dots < i_r \leq m$ and $1 \leq j_1 < j_2 < \dots < j_r \leq n$. Then
lyche-numerical-linear-algebra-F00453	37	0.999	$\boldsymbol{k} = \{k_1, \dots, k_r\}$	$\boldsymbol{k} = \{k_1, \dots, k_r\}$	where we sum over all $\boldsymbol{k} = \{k_1, \dots, k_r\}$ with $1 \leq k_1 < k_2 < \dots < k_r \leq p$.
lyche-numerical-linear-algebra-F00454	37	0.999	$1 \leq k_1 < k_2 < \dots < k_r \leq p$	$1 \leq k_1 < k_2 < \dots < k_r \leq p$	where we sum over all $\boldsymbol{k} = \{k_1, \dots, k_r\}$ with $1 \leq k_1 < k_2 < \dots < k_r \leq p$.
lyche-numerical-linear-algebra-F00455	38	0.937	$\lambda \in \mathbb{C}$	$\lambda \in \mathbb{C}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is a square matrix, $\lambda \in \mathbb{C}$ and $\boldsymbol{x} \in \mathbb{C}^n$. We say that $(\lambda, \boldsymbol{x})$ is an
lyche-numerical-linear-algebra-F00456	38	0.937	$\lambda, \boldsymbol{x}$	$\lambda, \boldsymbol{x}$	Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is a square matrix, $\lambda \in \mathbb{C}$ and $\boldsymbol{x} \in \mathbb{C}^n$. We say that $(\lambda, \boldsymbol{x})$ is an
lyche-numerical-linear-algebra-F00457	38	1.000	$\sigma(\boldsymbol{A})$	$\sigma(\boldsymbol{A})$	of $\boldsymbol{A}$ and is denoted by $\sigma(\boldsymbol{A})$. For example, $\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$.
lyche-numerical-linear-algebra-F00458	38	1.000	$\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$	$\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$	of $\boldsymbol{A}$ and is denoted by $\sigma(\boldsymbol{A})$. For example, $\sigma(\boldsymbol{I}) = \{1, \dots, 1\} = \{1\}$.
lyche-numerical-linear-algebra-F00459	38	1.000	$\lambda \in$	$\lambda \in$	Lemma 1.5 (Characteristic Equation) For any $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ we have $\lambda \in$
lyche-numerical-linear-algebra-F00460	38	0.995	$\sigma(\boldsymbol{A}) \iff \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$	$\sigma(\boldsymbol{A}) \iff \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$	$\sigma(\boldsymbol{A}) \iff \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$.
lyche-numerical-linear-algebra-F00461	38	0.699	$(\lambda, \boldsymbol{x})$	$(\lambda, \boldsymbol{x})$	Proof Suppose $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}$. The equation $\boldsymbol{Ax} = \lambda \boldsymbol{x}$ can be written
lyche-numerical-linear-algebra-F00462	38	1.000	$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \mathbf{0}$. Since $\boldsymbol{x}$ is nonzero the matrix $\boldsymbol{A} - \lambda \boldsymbol{I}$ must be singular with a
lyche-numerical-linear-algebra-F00463	38	1.000	$\boldsymbol{A} - \lambda \boldsymbol{I}$	$\boldsymbol{A} - \lambda \boldsymbol{I}$	$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{x} = \mathbf{0}$. Since $\boldsymbol{x}$ is nonzero the matrix $\boldsymbol{A} - \lambda \boldsymbol{I}$ must be singular with a
lyche-numerical-linear-algebra-F00464	38	1.000	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$	zero determinant. Conversely, if $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$ then $\boldsymbol{A} - \lambda \boldsymbol{I}$ is singular and
lyche-numerical-linear-algebra-F00465	38	1.000	$\det(\boldsymbol{A} - \lambda \boldsymbol{I})$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I})$	The expression $\det(\boldsymbol{A} - \lambda \boldsymbol{I})$ is a polynomial of exact degree n in λ . For $n = 3$
lyche-numerical-linear-algebra-F00466	38	1.000	$r(\lambda)$	$r(\lambda)$	where each term in $r(\lambda)$ has at most $n - 2$ factors containing λ . It follows that r is
lyche-numerical-linear-algebra-F00467	38	0.998	$n - 2, \det(\boldsymbol{A} - \lambda \boldsymbol{I})$	$n - 2, \det(\boldsymbol{A} - \lambda \boldsymbol{I})$	a polynomial of degree at most $n - 2$, $\det(\boldsymbol{A} - \lambda \boldsymbol{I})$ is a polynomial of exact degree
lyche-numerical-linear-algebra-F00468	38	1.000	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (-1)^n \det(\lambda \boldsymbol{I} - \boldsymbol{A})$	$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (-1)^n \det(\lambda \boldsymbol{I} - \boldsymbol{A})$	We observe that $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = (-1)^n \det(\lambda \boldsymbol{I} - \boldsymbol{A})$ so $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$ if and
lyche-numerical-linear-algebra-F00469	38	1.000	$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$	$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$	only if $\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = 0$.
lyche-numerical-linear-algebra-F00470	38	0.958	$\pi_{\boldsymbol{A}} : \mathbb{C} \rightarrow$	$\pi_{\boldsymbol{A}} : \mathbb{C} \rightarrow$	Definition 1.9 (Characteristic Polynomial of a Matrix) The function $\pi_{\boldsymbol{A}} : \mathbb{C} \rightarrow$
lyche-numerical-linear-algebra-F00471	38	1.000	$\pi_{\boldsymbol{A}}(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I})$	$\pi_{\boldsymbol{A}}(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I})$	$\mathbb{C}$ given by $\pi_{\boldsymbol{A}}(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I})$ is called the characteristic polynomial of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F00472	39	0.994	$\lambda_1, \dots, \lambda_n$	$\lambda_1, \dots, \lambda_n$	ties, precisely n eigenvalues $\lambda_1, \dots, \lambda_n$ some of which might be complex even if $\boldsymbol{A}$
lyche-numerical-linear-algebra-F00473	39	1.000	$\boldsymbol{A}\bar{\boldsymbol{x}} = \bar{\lambda}\boldsymbol{x}$	$\boldsymbol{A}\bar{\boldsymbol{x}} = \bar{\lambda}\boldsymbol{x}$	$\boldsymbol{A}$ real gives $\boldsymbol{A}\bar{\boldsymbol{x}} = \bar{\lambda}\boldsymbol{x}$.

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lyche-numerical-linear-algebra_F00474	39	■ 0.945	$\pi_{\mathbf{A}}$	$\pi_{\mathbf{A}}$	Proof We compare two different expansions of $\pi_{\mathbf{A}}$. On the one hand from (1.16)
lyche-numerical-linear-algebra_F00475	39	■ 1.000	$c_{n-1} = (-1)^{n-1} \operatorname{trace}(\mathbf{A})$	$c_{n-1} = (-1)^{n-1} \operatorname{trace}(\mathbf{A})$	where $c_{n-1} = (-1)^{n-1} \operatorname{trace}(\mathbf{A})$ and $c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$. On the other hand
lyche-numerical-linear-algebra_F00476	39	■ 1.000	$c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$	$c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$	where $c_{n-1} = (-1)^{n-1} \operatorname{trace}(\mathbf{A})$ and $c_0 = \pi_{\mathbf{A}}(0) = \det(\mathbf{A})$. On the other hand
lyche-numerical-linear-algebra_F00477	39	■ 1.000	$d_{n-1} = (-1)^{n-1} (\lambda_1 + \cdots + \lambda_n)$	$d_{n-1} = (-1)^{n-1} (\lambda_1 + \cdots + \lambda_n)$	where $d_{n-1} = (-1)^{n-1} (\lambda_1 + \cdots + \lambda_n)$ and $d_0 = \lambda_1 \cdots \lambda_n$. Since $c_j = d_j$ for all j
lyche-numerical-linear-algebra_F00478	39	■ 1.000	$d_0 = \lambda_1 \cdots \lambda_n$	$d_0 = \lambda_1 \cdots \lambda_n$	where $d_{n-1} = (-1)^{n-1} (\lambda_1 + \cdots + \lambda_n)$ and $d_0 = \lambda_1 \cdots \lambda_n$. Since $c_j = d_j$ for all j
lyche-numerical-linear-algebra_F00479	39	■ 1.000	$c_j = d_j$	$c_j = d_j$	where $d_{n-1} = (-1)^{n-1} (\lambda_1 + \cdots + \lambda_n)$ and $d_0 = \lambda_1 \cdots \lambda_n$. Since $c_j = d_j$ for all j
lyche-numerical-linear-algebra_F00480	39	■ 1.000	$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	Thus, if $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ then $\operatorname{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$ so that $\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$.
lyche-numerical-linear-algebra_F00481	39	■ 1.000	$\operatorname{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$	$\operatorname{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$	Thus, if $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ then $\operatorname{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$ so that $\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$.
lyche-numerical-linear-algebra_F00482	39	■ 1.000	$\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$	$\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$	Thus, if $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ then $\operatorname{trace}(\mathbf{A}) = 4$, $\det(\mathbf{A}) = 3$ so that $\pi_{\mathbf{A}}(\lambda) = \lambda^2 - 4\lambda + 3$.
lyche-numerical-linear-algebra_F00483	39	■ 1.000	$\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$	$\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$	Since $\mathbf{A}$ is singular $\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$, some $\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$, some $\mathbf{x} \neq$
lyche-numerical-linear-algebra_F00484	39	■ 1.000	$\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$	$\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$	Since $\mathbf{A}$ is singular $\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$, some $\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$, some $\mathbf{x} \neq$
lyche-numerical-linear-algebra_F00485	39	■ 1.000	$\mathbf{x} \neq$	$\mathbf{x} \neq$	Since $\mathbf{A}$ is singular $\Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}$, some $\mathbf{x} \neq \mathbf{0} \Longleftrightarrow \mathbf{A}\mathbf{x} = \mathbf{0}\mathbf{x}$, some $\mathbf{x} \neq$
lyche-numerical-linear-algebra_F00486	39	■ 0.981	$0 \Longleftrightarrow$	$0 \Longleftrightarrow$	$0 \Longleftrightarrow$ zero is an eigenvalue of $\mathbf{A}$, we obtain
lyche-numerical-linear-algebra_F00487	40	■ 0.848	$\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{D} \\ 0 & \mathbf{C} \end{bmatrix}$	$\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{D} \\ 0 & \mathbf{C} \end{bmatrix}$	Theorem 1.12 (Eigenvalues of a Block Triangular Matrix) If $\mathbf{A} = \begin{bmatrix} \mathbf{B} & \mathbf{D} \\ 0 & \mathbf{C} \end{bmatrix}$ is
lyche-numerical-linear-algebra_F00488	40	■ 0.998	$\pi_{\mathbf{A}} = \pi_{\mathbf{B}} \cdot \pi_{\mathbf{C}}$	$\pi_{\mathbf{A}} = \pi_{\mathbf{B}} \cdot \pi_{\mathbf{C}}$	block triangular then $\pi_{\mathbf{A}} = \pi_{\mathbf{B}} \cdot \pi_{\mathbf{C}}$.
lyche-numerical-linear-algebra_F00489	40	■ 1.000	$\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$	$\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$	a) With $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$, how many arithmetic operations are required to form the
lyche-numerical-linear-algebra_F00490	40	■ 1.000	$2n \times 2n$	$2n \times 2n$	b) Consider the $2n \times 2n$ block matrix
lyche-numerical-linear-algebra_F00491	40	■ 1.000	$\mathbf{A}, \dots, \mathbf{Z}$	$\mathbf{A}, \dots, \mathbf{Z}$	where all matrices $\mathbf{A}, \dots, \mathbf{Z}$ are in $\mathbb{R}^{n \times n}$. How many operations does it take to
lyche-numerical-linear-algebra_F00492	40	■ 1.000	$\mathbf{W}, \mathbf{X}, \mathbf{Y}$	$\mathbf{W}, \mathbf{X}, \mathbf{Y}$	compute $\mathbf{W}, \mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$ by the obvious algorithm?
lyche-numerical-linear-algebra_F00493	40	■ 1.000	$\mathbf{Z}$	$\mathbf{Z}$	compute $\mathbf{W}, \mathbf{X}, \mathbf{Y}$ and $\mathbf{Z}$ by the obvious algorithm?
lyche-numerical-linear-algebra_F00494	40	■ 1.000	$m \times m$	$m \times m$	matrices $\mathbf{A}$ and $\mathbf{B}$ of size $m \times m$, with $m = 2^k$ for some $k \geq 0$, calculates the
lyche-numerical-linear-algebra_F00495	40	■ 1.000	$m = 2^k$	$m = 2^k$	matrices $\mathbf{A}$ and $\mathbf{B}$ of size $m \times m$, with $m = 2^k$ for some $k \geq 0$, calculates the
lyche-numerical-linear-algebra_F00496	40	■ 1.000	$k \geq 0$	$k \geq 0$	matrices $\mathbf{A}$ and $\mathbf{B}$ of size $m \times m$, with $m = 2^k$ for some $k \geq 0$, calculates the
lyche-numerical-linear-algebra_F00497	40	■ 1.000	$\mathcal{O}(m^{\log_2(7)})$	$\mathcal{O}(m^{\log_2(7)})$	e) Show that the operation count of the recursive algorithm is $\mathcal{O}(m^{\log_2(7)})$. Note
lyche-numerical-linear-algebra_F00498	40	■ 1.000	$\log_2(7) \approx 2.8 < 3$	$\log_2(7) \approx 2.8 < 3$	that $\log_2(7) \approx 2.8 < 3$, so this is less costly than straightforward matrix
lyche-numerical-linear-algebra_F00499	41	■ 1.000	2×2	2×2	Exercise 1.2 (The Inverse of a General 2×2 Matrix) Show that
lyche-numerical-linear-algebra_F00500	41	■ 1.000	a, b, c, d	a, b, c, d	for any a, b, c, d such that $ad - bc \neq 0$.

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lyche-numerical-linear-atgela_F00501	41	■ 1.000	a d-b c \neq 0	$ad - bc \neq 0$	for any a, b, c, d such that $ad - bc \neq 0$.
lyche-numerical-linear-atgela_F00502	41	■ 1.000	\boldsymbol{B}, \boldsymbol{C} \in	$B, C \in$	Exercise 1.4 (Sherman-Morrison Formula) Suppose $A \in \mathbb{C}^{n \times n}$, and $B, C \in$
lyche-numerical-linear-atgela_F00503	41	■ 1.000	\mathbb{R}^{n \times m}	$\mathbb{R}^{n \times m}$	$\mathbb{R}^{n \times m}$ for some $n, m \in \mathbb{N}$. If $(I + C^T A^{-1} B)^{-1}$ exists then
lyche-numerical-linear-atgela_F00504	41	■ 1.000	n, m \in \mathbb{N}	$n, m \in \mathbb{N}$	$\mathbb{R}^{n \times m}$ for some $n, m \in \mathbb{N}$. If $(I + C^T A^{-1} B)^{-1}$ exists then
lyche-numerical-linear-atgela_F00505	41	■ 1.000	\left(\boldsymbol{I}+\boldsymbol{C}^T\boldsymbol{A}^{-1}\boldsymbol{B}\right)^{-1}	$(I + C^T A^{-1} B)^{-1}$	$\mathbb{R}^{n \times m}$ for some $n, m \in \mathbb{N}$. If $(I + C^T A^{-1} B)^{-1}$ exists then
lyche-numerical-linear-atgela_F00506	41	■ 1.000	\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n	$u, v \in \mathbb{R}^n$	a) Let $u, v \in \mathbb{R}^n$ and suppose $v^T u \neq 1$. Show that $I - uv^T$ has an inverse given
lyche-numerical-linear-atgela_F00507	41	■ 1.000	\boldsymbol{v}^T\boldsymbol{u} \neq 1	$v^T u \neq 1$	a) Let $u, v \in \mathbb{R}^n$ and suppose $v^T u \neq 1$. Show that $I - uv^T$ has an inverse given
lyche-numerical-linear-atgela_F00508	41	■ 1.000	\boldsymbol{I}-\boldsymbol{u}\boldsymbol{v}^T	$I - uv^T$	a) Let $u, v \in \mathbb{R}^n$ and suppose $v^T u \neq 1$. Show that $I - uv^T$ has an inverse given
lyche-numerical-linear-atgela_F00509	41	■ 0.813	\boldsymbol{I}-\tau\boldsymbol{u}\boldsymbol{v}^T	$I - \tau uv^T$	by $I - \tau uv^T$, where $\tau = 1/(v^T u - 1)$.
lyche-numerical-linear-atgela_F00510	41	■ 0.813	\tau=1/\left(\boldsymbol{v}^T\boldsymbol{u}-1\right)	$\tau = 1/(v^T u - 1)$	by $I - \tau uv^T$, where $\tau = 1/(v^T u - 1)$.
lyche-numerical-linear-atgela_F00511	41	■ 1.000	\boldsymbol{C}:=\boldsymbol{A}^{-1}	$C := A^{-1}$	b) Let $A \in \mathbb{R}^{n \times n}$ be nonsingular with inverse $C := A^{-1}$, and let $a \in \mathbb{R}^n$. Let $\bar{A}$ be
lyche-numerical-linear-atgela_F00512	41	■ 1.000	\boldsymbol{a} \in \mathbb{R}^n	$a \in \mathbb{R}^n$	b) Let $A \in \mathbb{R}^{n \times n}$ be nonsingular with inverse $C := A^{-1}$, and let $a \in \mathbb{R}^n$. Let $\bar{A}$ be
lyche-numerical-linear-atgela_F00513	41	■ 1.000	\overline{\boldsymbol{A}}	$\bar{A}$	b) Let $A \in \mathbb{R}^{n \times n}$ be nonsingular with inverse $C := A^{-1}$, and let $a \in \mathbb{R}^n$. Let $\bar{A}$ be
lyche-numerical-linear-atgela_F00514	41	■ 1.000	\boldsymbol{a}^T	a^T	the matrix which differs from A by exchanging the i th row of A with a^T , i.e.,
lyche-numerical-linear-atgela_F00515	41	■ 0.995	\overline{\boldsymbol{A}}=\boldsymbol{A}-\boldsymbol{e}_i\left(\boldsymbol{e}_i^T\boldsymbol{A}-\boldsymbol{a}^T\right)	$\bar{A} = A - e_i(e_i^T A - a^T)$	$\bar{A} = A - e_i(e_i^T A - a^T)$, where e_i is the i th column in the identity matrix I .
lyche-numerical-linear-atgela_F00516	41	■ 0.995	\boldsymbol{e}_i	e_i	$\bar{A} = A - e_i(e_i^T A - a^T)$, where e_i is the i th column in the identity matrix I .
lyche-numerical-linear-atgela_F00517	41	■ 1.000	\overline{\boldsymbol{C}}=\overline{\boldsymbol{A}}^{-1}	$\bar{C} = \bar{A}^{-1}$	then $\bar{A}$ has an inverse $\bar{C} = \bar{A}^{-1}$ given by
lyche-numerical-linear-atgela_F00518	41	■ 1.000	\boldsymbol{a}	a	c) Write an algorithm which to given C and a checks if (1.20) holds and computes
lyche-numerical-linear-atgela_F00519	41	■ 0.999	\overline{\boldsymbol{C}}	$\bar{C}$	$\bar{C}$ provided $\lambda \neq 0$. (hint: Use (1.21) to find formulas for computing each column
lyche-numerical-linear-atgela_F00520	41	■ 0.999	\lambda \neq 0	$\lambda \neq 0$	$\bar{C}$ provided $\lambda \neq 0$. (hint: Use (1.21) to find formulas for computing each column
lyche-numerical-linear-atgela_F00521	41	■ 1.000	\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{E} \in	$A, B, C, E \in$	Exercise 1.6 (Matrix Products (Exam Exercise 2009-1)) Let $A, B, C, E \in$
lyche-numerical-linear-atgela_F00522	41	■ 1.000	\boldsymbol{B}\boldsymbol{C}	BC	a) How many operations are required to compute the matrix product BC ? How
lyche-numerical-linear-atgela_F00523	41	■ 1.000	\boldsymbol{L} \in \mathbb{R}^{n \times n}	$L \in \mathbb{R}^{n \times n}$	b) Show that there exists a lower triangular matrix $L \in \mathbb{R}^{n \times n}$ such that $A = L + L^T$.
lyche-numerical-linear-atgela_F00524	41	■ 1.000	\boldsymbol{A}=\boldsymbol{L}+\boldsymbol{L}^T	$A = L + L^T$	b) Show that there exists a lower triangular matrix $L \in \mathbb{R}^{n \times n}$ such that $A = L + L^T$.
lyche-numerical-linear-atgela_F00525	42	■ 1.000	\boldsymbol{E}^T\boldsymbol{A}\boldsymbol{E}=\boldsymbol{S}+\boldsymbol{S}^T	$E^T A E = S + S^T$	c) We have $E^T A E = S + S^T$ where $S = E^T L E$. How many operations are
lyche-numerical-linear-atgela_F00526	42	■ 1.000	\boldsymbol{S}=\boldsymbol{E}^T\boldsymbol{L}\boldsymbol{E}	$S = E^T L E$	c) We have $E^T A E = S + S^T$ where $S = E^T L E$. How many operations are

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lyche-numerical-linear-algebra_F00527	42	■ 1.000	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E}$	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E}$	required to compute $\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E}$ in this way?
lyche-numerical-linear-algebra_F00528	42	■ 1.000	$\left(x_1, y_1, z_1\right), \left(x_2, y_2, z_2\right), \left(x_3, y_3, z_3\right)$	$\left(x_1, y_1, z_1\right), \left(x_2, y_2, z_2\right), \left(x_3, y_3, z_3\right)$	is the equation for a plane through three points $\left(x_1, y_1, z_1\right), \left(x_2, y_2, z_2\right), \left(x_3, y_3, z_3\right)$
lyche-numerical-linear-algebra_F00529	43	■ 1.000	$P_i = \left(x_i, y_i\right), i = 1, 2, 3$	$P_i = \left(x_i, y_i\right), i = 1, 2, 3$	Exercise 1.10 (Signed Area of a Triangle) Let $P_i = \left(x_i, y_i\right), i = 1, 2, 3$, be three
lyche-numerical-linear-algebra_F00530	43	■ 0.816	T	T	points in the plane defining a triangle T . Show that the area of T is ²
lyche-numerical-linear-algebra_F00531	43	■ 0.999	$\prod_{i>j}\left(x_i-x_j\right)=\prod_{i=2}^n\left(x_i-x_1\right)\left(x_i-x_2\right) \cdots\left(x_i-x_{i-1}\right)$	$\prod_{i>j}\left(x_i-x_j\right)=\prod_{i=2}^n\left(x_i-x_1\right)\left(x_i-x_2\right) \cdots\left(x_i-x_{i-1}\right)$	where $\prod_{i>j}\left(x_i-x_j\right)=\prod_{i=2}^n\left(x_i-x_1\right)\left(x_i-x_2\right) \cdots\left(x_i-x_{i-1}\right)$. This determinant
lyche-numerical-linear-algebra_F00532	43	■ 1.000	$\boldsymbol{\alpha}=\left[\alpha_1, \ldots, \alpha_n\right]^T, \boldsymbol{\beta}=\left[\beta_1, \ldots, \beta_n\right]^T$	$\boldsymbol{\alpha}=\left[\alpha_1, \ldots, \alpha_n\right]^T, \boldsymbol{\beta}=\left[\beta_1, \ldots, \beta_n\right]^T$	Exercise 1.12 (Cauchy Determinant (1842)) Let $\boldsymbol{\alpha}=\left[\alpha_1, \ldots, \alpha_n\right]^T, \boldsymbol{\beta}=\left[\beta_1, \ldots, \beta_n\right]^T$ be in $\mathbb{R}^n$.
lyche-numerical-linear-algebra_F00533	43	■ 1.000	$a_{i, j}=1 /\left(\alpha_i+\beta_j\right), i, j=1, 2, \ldots, n$	$a_{i, j}=1 /\left(\alpha_i+\beta_j\right), i, j=1, 2, \ldots, n$	a) Consider the matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ with elements $a_{i, j}=1 /\left(\alpha_i+\beta_j\right), i, j=1, 2, \ldots, n$. Show that
lyche-numerical-linear-algebra_F00534	43	■ 0.996	$A(T)=A\left(A B P_3 P_1\right)+A\left(P_3 B C P_2\right)-A\left(P_1 A C P_2\right)$	$A(T)=A\left(A B P_3 P_1\right)+A\left(P_3 B C P_2\right)-A\left(P_1 A C P_2\right)$	² Hint: $A(T)=A\left(A B P_3 P_1\right)+A\left(P_3 B C P_2\right)-A\left(P_1 A C P_2\right)$ c.f. Fig. 1.1.
lyche-numerical-linear-algebra_F00536	43	■ 0.997	x_n^k	x_n^k	³ Hint: subtract x_n^k times column k from column $k+1$ for $k=n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00537	43	■ 0.997	k	k	³ Hint: subtract x_n^k times column k from column $k+1$ for $k=n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00538	43	■ 0.997	$k+1$	$k+1$	³ Hint: subtract x_n^k times column k from column $k+1$ for $k=n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00539	43	■ 0.997	$k=n-1, n-2, \ldots, 1$	$k=n-1, n-2, \ldots, 1$	³ Hint: subtract x_n^k times column k from column $k+1$ for $k=n-1, n-2, \ldots, 1$.
lyche-numerical-linear-algebra_F00540	43	■ 0.997	$P=\prod_{i=1}^n \prod_{j=1}^n a_{i j}$	$P=\prod_{i=1}^n \prod_{j=1}^n a_{i j}$	where $P=\prod_{i=1}^n \prod_{j=1}^n a_{i j}$, and for $\boldsymbol{\gamma}=\left[\gamma_1, \ldots, \gamma_n\right]^T$
lyche-numerical-linear-algebra_F00541	44	■ 1.000	$\boldsymbol{\gamma}=\left[\gamma_1, \ldots, \gamma_n\right]^T$	$\boldsymbol{\gamma}=\left[\gamma_1, \ldots, \gamma_n\right]^T$	where $P=\prod_{i=1}^n \prod_{j=1}^n a_{i j}$, and for $\boldsymbol{\gamma}=\left[\gamma_1, \ldots, \gamma_n\right]^T$
lyche-numerical-linear-algebra_F00542	44	■ 1.000	$\prod_{j=1}^n\left(\alpha_i+\beta_j\right)$	$\prod_{j=1}^n\left(\alpha_i+\beta_j\right)$	Hint: Multiply the i th row of $\boldsymbol{A}$ by $\prod_{j=1}^n\left(\alpha_i+\beta_j\right)$ for $i=1, 2, \ldots, n$. Call the
lyche-numerical-linear-algebra_F00543	44	■ 1.000	$i=1, 2, \ldots, n$	$i=1, 2, \ldots, n$	Hint: Multiply the i th row of $\boldsymbol{A}$ by $\prod_{j=1}^n\left(\alpha_i+\beta_j\right)$ for $i=1, 2, \ldots, n$. Call the
lyche-numerical-linear-algebra_F00544	44	■ 0.999	$n-1$	$n-1$	resulting matrix $\boldsymbol{C}$. Each element of $\boldsymbol{C}$ is a product of $n-1$ factors $\alpha_r+\beta_s$. Hence
lyche-numerical-linear-algebra_F00545	44	■ 0.999	$\alpha_r+\beta_s$	$\alpha_r+\beta_s$	resulting matrix $\boldsymbol{C}$. Each element of $\boldsymbol{C}$ is a product of $n-1$ factors $\alpha_r+\beta_s$. Hence
lyche-numerical-linear-algebra_F00546	44	■ 1.000	$\det(\boldsymbol{C})$	$\det(\boldsymbol{C})$	$\det(\boldsymbol{C})$ is a sum of terms where each term contain precisely $n(n-1)$ factors
lyche-numerical-linear-algebra_F00547	44	■ 1.000	$n(n-1)$	$n(n-1)$	$\det(\boldsymbol{C})$ is a sum of terms where each term contain precisely $n(n-1)$ factors
lyche-numerical-linear-algebra_F00548	44	■ 0.997	$\det(\boldsymbol{C})=q(\alpha, \beta)$	$\det(\boldsymbol{C})=q(\alpha, \beta)$	$\alpha_r+\beta_s$. Thus $\det(\boldsymbol{C})=q(\alpha, \beta)$ where q is a polynomial of degree at most
lyche-numerical-linear-algebra_F00549	44	■ 0.997	q	q	$\alpha_r+\beta_s$. Thus $\det(\boldsymbol{C})=q(\alpha, \beta)$ where q is a polynomial of degree at most
lyche-numerical-linear-algebra_F00550	44	■ 1.000	α_i	α_i	$n(n-1)$ in α_i and β_j . Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i=\alpha_j$ for
lyche-numerical-linear-algebra_F00551	44	■ 1.000	β_j	β_j	$n(n-1)$ in α_i and β_j . Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i=\alpha_j$ for
lyche-numerical-linear-algebra_F00552	44	■ 1.000			

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lyche-numerical-linear-algebra-F00553	44	1.000	$\operatorname{det}(\boldsymbol{A})$	$\det(\boldsymbol{A})$	$n(n-1)$ in α_i and β_j . Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i = \alpha_j$ for
lyche-numerical-linear-algebra-F00554	44	1.000	$\alpha_i = \alpha_j$	$\alpha_i = \alpha_j$	$n(n-1)$ in α_i and β_j . Since $\det(\boldsymbol{A})$ and therefore $\det(\boldsymbol{C})$ vanishes if $\alpha_i = \alpha$ for
lyche-numerical-linear-algebra-F00555	44	1.000	$\beta_r = \beta_s$	$\beta_r = \beta_s$	some $i \neq j$ or $\beta_r = \beta_s$ for some $r \neq s$, we have that $q(\alpha, \beta)$ must be divisible
lyche-numerical-linear-algebra-F00556	44	1.000	$r \neq s$	$r \neq s$	some $i \neq j$ or $\beta_r = \beta_s$ for some $r \neq s$, we have that $q(\alpha, \beta)$ must be divisible
lyche-numerical-linear-algebra-F00557	44	1.000	$q(\boldsymbol{\alpha}, \boldsymbol{\beta})$	$q(\alpha, \beta)$	some $i \neq j$ or $\beta_r = \beta_s$ for some $r \neq s$, we have that $q(\alpha, \beta)$ must be divisible
lyche-numerical-linear-algebra-F00558	44	1.000	$g(\boldsymbol{\alpha})$	$g(\alpha)$	by each factor in $g(\alpha)$ and $g(\beta)$. Since $g(\alpha)$ and $g(\beta)$ is a polynomial of degree
lyche-numerical-linear-algebra-F00559	44	1.000	$g(\boldsymbol{\beta})$	$g(\beta)$	by each factor in $g(\alpha)$ and $g(\beta)$. Since $g(\alpha)$ and $g(\beta)$ is a polynomial of degree
lyche-numerical-linear-algebra-F00560	44	1.000	$\boldsymbol{\alpha}$	α	for some constant k independent of α and β . Show that $k = 1$ by choosing
lyche-numerical-linear-algebra-F00561	44	1.000	$\boldsymbol{\beta}$	β	for some constant k independent of α and β . Show that $k = 1$ by choosing
lyche-numerical-linear-algebra-F00562	44	1.000	$k=1$	$k = 1$	for some constant k independent of α and β . Show that $k = 1$ by choosing
lyche-numerical-linear-algebra-F00563	44	1.000	$\beta_i + \alpha_i = 0, i = 1, 2, \dots, n$	$\beta_i + \alpha_i = 0, i = 1, 2, \dots, n$	$\beta_i + \alpha_i = 0, i = 1, 2, \dots, n$.
lyche-numerical-linear-algebra-F00564	44	1.000	$\boldsymbol{A}^{-1} = (b_{j,k})$	$\boldsymbol{A}^{-1} = (b_{j,k})$	the elements of $\boldsymbol{A}^{-1} = (b_{j,k})$. Answer:
lyche-numerical-linear-algebra-F00565	44	1.000	$\boldsymbol{H}_n = (h_{i,j})$	$\boldsymbol{H}_n = (h_{i,j})$	Exercise 1.13 (Inverse of the Hilbert Matrix) Let $\boldsymbol{H}_n = (h_{i,j})$ be the $n \times n$
lyche-numerical-linear-algebra-F00566	44	1.000	$h_{i,j} = 1/(i+j-1)$	$h_{i,j} = 1/(i+j-1)$	matrix with elements $h_{i,j} = 1/(i+j-1)$. Use Exercise 1.12 to show that the elements
lyche-numerical-linear-algebra-F00567	44	1.000	$t_{i,j}^n$	$t_{i,j}^n$	$t_{i,j}^n$ in $\boldsymbol{T}_n = \boldsymbol{H}_n^{-1}$ are given by
lyche-numerical-linear-algebra-F00568	44	1.000	$\boldsymbol{T}_n = \boldsymbol{H}_n^{-1}$	$\boldsymbol{T}_n = \boldsymbol{H}_n^{-1}$	$t_{i,j}^n$ in $\boldsymbol{T}_n = \boldsymbol{H}_n^{-1}$ are given by
lyche-numerical-linear-algebra-F00569	45	1.000	$\boldsymbol{L}$	$\boldsymbol{L}$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A} =$
lyche-numerical-linear-algebra-F00570	45	1.000	$\boldsymbol{U}$	$\boldsymbol{U}$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A} =$
lyche-numerical-linear-algebra-F00571	45	1.000	$\boldsymbol{A} =$	$\boldsymbol{A} =$	triangular matrix $\boldsymbol{L}$ and an upper triangular matrix $\boldsymbol{U}$ resulting in the product $\boldsymbol{A} =$
lyche-numerical-linear-algebra-F00572	45	1.000	$\boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{LU}$	$\boldsymbol{LU}$. We also consider the factorization $\boldsymbol{A} = \boldsymbol{LDU}$, where $\boldsymbol{L}$ is lower triangular,
lyche-numerical-linear-algebra-F00573	45	1.000	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{A} = \boldsymbol{LDU}$	$\boldsymbol{LU}$. We also consider the factorization $\boldsymbol{A} = \boldsymbol{LDU}$, where $\boldsymbol{L}$ is lower triangular,
lyche-numerical-linear-algebra-F00574	45	1.000	$\boldsymbol{D}$	$\boldsymbol{D}$	$\boldsymbol{D}$ is diagonal and $\boldsymbol{U}$ is upper triangular. Moreover, $\boldsymbol{L}$ and $\boldsymbol{U}$ have ones on their
lyche-numerical-linear-algebra-F00575	45	1.000	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$	$\boldsymbol{A} = \boldsymbol{QR}$	on QR factorization. Here $\boldsymbol{A} = \boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q} = \boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra-F00576	45	1.000	$\boldsymbol{Q}$	$\boldsymbol{Q}$	on QR factorization. Here $\boldsymbol{A} = \boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q} = \boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra-F00577	45	1.000	$\boldsymbol{Q}^* \boldsymbol{Q} = \boldsymbol{I}$	$\boldsymbol{Q}^* \boldsymbol{Q} = \boldsymbol{I}$	on QR factorization. Here $\boldsymbol{A} = \boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q} = \boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra-F00578	45	1.000	$\boldsymbol{R}$	$\boldsymbol{R}$	on QR factorization. Here $\boldsymbol{A} = \boldsymbol{QR}$, where $\boldsymbol{Q}$ is unitary, i.e., $\boldsymbol{Q}^* \boldsymbol{Q} = \boldsymbol{I}$, and $\boldsymbol{R}$
lyche-numerical-linear-algebra-F00579	46	0.996	$[a, b], n+1 \geq 2$	$[a, b], n+1 \geq 2$	Given an interval $[a, b], n+1 \geq 2$ equidistant sites in $[a, b]$

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lyche-numerical-linear-algebra_F00580	46	■ 0.996	[a, b]	$[a, b]$	Given an interval $[a, b]$, $n + 1 \geq 2$ equidistant sites in $[a, b]$
lyche-numerical-linear-algebra_F00581	46	■ 1.000	y	y	and y values $\mathbf{y} := [y_1, \dots, y_{n+1}]^T \in \mathbb{R}^{n+1}$. We seek a function $g : [a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_F00582	46	■ 1.000	$\boldsymbol{y} := \left[y_1, \ldots, y_{n+1} \right]^T \in \mathbb{R}^{n+1}$	$\mathbf{y} := [y_1, \dots, y_{n+1}]^T \in \mathbb{R}^{n+1}$	and y values $\mathbf{y} := [y_1, \dots, y_{n+1}]^T \in \mathbb{R}^{n+1}$. We seek a function $g : [a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_F00583	46	■ 1.000	$g : [a, b] \rightarrow \mathbb{R}$	$g : [a, b] \rightarrow \mathbb{R}$	and y values $\mathbf{y} := [y_1, \dots, y_{n+1}]^T \in \mathbb{R}^{n+1}$. We seek a function $g : [a, b] \rightarrow \mathbb{R}$ such
lyche-numerical-linear-algebra_F00584	47	■ 1.000	$a \leq x_1 < x_2 < \dots < x_{n+1} \leq b$	$a \leq x_1 < x_2 < \dots < x_{n+1} \leq b$	$a \leq x_1 < x_2 < \dots < x_{n+1} \leq b$.
lyche-numerical-linear-algebra_F00585	47	■ 1.000	n+1	$n + 1$	Since there are $n + 1$ interpolation conditions in (2.2) a natural choice for a function
lyche-numerical-linear-algebra_F00586	47	■ 1.000	g	g	g is a polynomial of degree n . As shown in most books on numerical methods such
lyche-numerical-linear-algebra_F00587	47	■ 1.000	n=1, g	$n = 1, g$	when $n = 1$, g is the straight line
lyche-numerical-linear-algebra_F00588	47	■ 1.000	f	f	The function f and the polynomial g of degree at most 13 satisfying (2.2) with
lyche-numerical-linear-algebra_F00589	47	■ 1.000	[a, b]=[-1,1]	$[a, b] = [-1, 1]$	$[a, b] = [-1, 1]$ and $y_i = f(x_i), i = 1, \dots, 14$ is shown in Fig. 2.1. The interpolant
lyche-numerical-linear-algebra_F00590	47	■ 1.000	$y_i = f(x_i), i = 1, \dots, 14$	$y_i = f(x_i), i = 1, \dots, 14$	$[a, b] = [-1, 1]$ and $y_i = f(x_i), i = 1, \dots, 14$ is shown in Fig. 2.1. The interpolant
lyche-numerical-linear-algebra_F00591	48	■ 1.000	p_i	p_i	where each p_i is a polynomial of degree ≤ 1 . In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra_F00592	48	■ 1.000	≤ 1	≤ 1	where each p_i is a polynomial of degree ≤ 1 . In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra_F00593	48	■ 1.000	p_1	p_1	where each p_i is a polynomial of degree ≤ 1 . In particular, p_1 is given in (2.3) and
lyche-numerical-linear-algebra_F00594	48	■ 0.656	≤ 2	≤ 2	continuous derivatives of order ≤ 2 (C^2). We consider here the following functions
lyche-numerical-linear-algebra_F00595	48	■ 0.656	C^2	C^2	continuous derivatives of order ≤ 2 (C^2). We consider here the following functions
lyche-numerical-linear-algebra_F00596	49	■ 0.950	D_2	D_2	Definition 2.1 (The D_2-Spline Problem) Given $n \in \mathbb{N}$, an interval $[a, b]$, $y \in$
lyche-numerical-linear-algebra_F00597	49	■ 0.950	$\mathbf{y} \in$	$\mathbf{y} \in$	Definition 2.1 (The D_2-Spline Problem) Given $n \in \mathbb{N}$, an interval $[a, b]$, $y \in$
lyche-numerical-linear-algebra_F00598	49	■ 1.000	$\mathbb{R}^{n+1}$	$\mathbb{R}^{n+1}$	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \dots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1} . The problem
lyche-numerical-linear-algebra_F00599	49	■ 1.000	$x_1, \dots, x_{n+1}$	$x_1, \dots, x_{n+1}$	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \dots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1} . The problem
lyche-numerical-linear-algebra_F00600	49	■ 1.000	μ_1, μ_{n+1}	μ_1, μ_{n+1}	$\mathbb{R}^{n+1}$, knots (sites) $x_1, \dots, x_{n+1}$ given by (2.1) and numbers μ_1, μ_{n+1} . The problem
lyche-numerical-linear-algebra_F00601	49	■ 1.000	$g \in C^2[a, b]$	$g \in C^2[a, b]$	• smoothness: $g \in C^2[a, b]$, i.e., derivatives of order ≤ 2 are continuous on $\mathbb{R}$,
lyche-numerical-linear-algebra_F00602	49	■ 1.000	$g(x_i) = y_i, \quad i = 1, 2, \dots, n+1$	$g(x_i) = y_i, \quad i = 1, 2, \dots, n+1$	• interpolation: $g(x_i) = y_i, \quad i = 1, 2, \dots, n+1$.
lyche-numerical-linear-algebra_F00603	49	■ 1.000	D_2	D_2	• D_2 boundary conditions: $g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$.
lyche-numerical-linear-algebra_F00604	49	■ 1.000	$g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$	$g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$	• D_2 boundary conditions: $g''(a) = \mu_1, \quad g''(b) = \mu_{n+1}$.
lyche-numerical-linear-algebra_F00605	49	■ 1.000	N	N	We call g a D_2 -spline. It is called an N -spline or natural spline if $\mu_1 = \mu_{n+1} = 0$.
lyche-numerical-linear-algebra_F00606	49	■ 1.000	$\mu_1 = \mu_{n+1} = 0$	$\mu_1 = \mu_{n+1} = 0$	We call g a D_2 -spline. It is called an N -spline or natural spline if $\mu_1 = \mu_{n+1} = 0$.

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lyche-numerical-linear-algebra_F00607	49	■ 1.000	n=2	$n = 2$	<i>Example 2.1 (A D_2-Spline)</i> Suppose we choose $n = 2$ and sample data from the
lyche-numerical-linear-algebra_F00608	49	■ 1.000	$f:[0,2] \rightarrow \mathbb{R}$	$f : [0, 2] \rightarrow \mathbb{R}$	function $f : [0, 2] \rightarrow \mathbb{R}$ given by $f(x) = x^4$. Thus we consider the D_2 -spline
lyche-numerical-linear-algebra_F00609	49	■ 1.000	$[a, b]=[0,2], \boldsymbol{y}:=[0,1,16]^T$	$[a, b] = [0, 2], \boldsymbol{y} := [0, 1, 16]^T$	problem with $[a, b] = [0, 2], \boldsymbol{y} := [0, 1, 16]^T$ and $\mu_1 = g''(0) = 0, \mu_3 = g''(2) =$
lyche-numerical-linear-algebra_F00610	49	■ 1.000	$\mu_1=g^{\{\prime \prime\}}(0)=0, \mu_3=g^{\{\prime \prime\}}(2)=$	$\mu_1 = g''(0) = 0, \mu_3 = g''(2) =$	problem with $[a, b] = [0, 2], \boldsymbol{y} := [0, 1, 16]^T$ and $\mu_1 = g''(0) = 0, \mu_3 = g''(2) =$
lyche-numerical-linear-algebra_F00611	49	■ 1.000	$x_{\{1\}}=0, x_{\{2\}}=1$	$x_1 = 0, x_2 = 1$	48. The knots are $x_1 = 0, x_2 = 1$ and $x_3 = 2$. The function g given by
lyche-numerical-linear-algebra_F00612	49	■ 1.000	$x_{\{3\}}=2$	$x_3 = 2$	48. The knots are $x_1 = 0, x_2 = 1$ and $x_3 = 2$. The function g given by
lyche-numerical-linear-algebra_F00613	49	■ 1.000	$p_{\{2\}}$	p_2	is a D_2 -spline solving this problem. Indeed, p_1 and p_2 are cubic polynomials. For
lyche-numerical-linear-algebra_F00614	49	■ 1.000	$p_{\{1\}}(1)=p_{\{2\}}(1)=1, p_{\{1\}}^{\{\prime \prime\}}(1)=p_{\{2\}}^{\{\prime \prime\}}(1)=4, p_{\{1\}}^{\{\prime \prime\prime\}}(1)=p_{\{2\}}^{\{\prime \prime\prime\}}(1)=$	$p_1(1) = p_2(1) = 1, p_1'(1) = p_2'(1) = 4, p_1''(1) = p_2''(1) =$	smoothness we find $p_1(1) = p_2(1) = 1, p_1'(1) = p_2'(1) = 4, p_1''(1) = p_2''(1) =$
lyche-numerical-linear-algebra_F00615	49	■ 1.000	$g \in C^{\{2\}}[0,2]$	$g \in C^2[0, 2]$	9 which implies that $g \in C^2[0, 2]$. Finally we check that the interpolation and
lyche-numerical-linear-algebra_F00616	49	■ 1.000	$g(0)=p_{\{1\}}(0)=0, g(1)=p_{\{2\}}(1)=1, g(2)=$	$g(0) = p_1(0) = 0, g(1) = p_2(1) = 1, g(2) =$	boundary conditions hold. Indeed, $g(0) = p_1(0) = 0, g(1) = p_2(1) = 1, g(2) =$
lyche-numerical-linear-algebra_F00617	49	■ 1.000	$p_{\{2\}}(2)=16, g^{\{\prime \prime\}}(0)=p_{\{1\}}^{\{\prime \prime\}}(0)=0$	$p_2(2) = 16, g''(0) = p_1''(0) = 0$	$p_2(2) = 16, g''(0) = p_1''(0) = 0$ and $g''(2) = p_2''(2) = 48$. Note that $p_1'''(x) = 9 \neq$
lyche-numerical-linear-algebra_F00618	49	■ 1.000	$g^{\{\prime \prime\}}(2)=p_{\{2\}}^{\{\prime \prime\}}(2)=48$	$g''(2) = p_2''(2) = 48$	$p_2(2) = 16, g''(0) = p_1''(0) = 0$ and $g''(2) = p_2''(2) = 48$. Note that $p_1'''(x) = 9 \neq$
lyche-numerical-linear-algebra_F00619	49	■ 1.000	$p_{\{1\}}^{\{\prime \prime\prime\}}(x)=9 \neq$	$p_1'''(x) = 9 \neq$	$p_2(2) = 16, g''(0) = p_1''(0) = 0$ and $g''(2) = p_2''(2) = 48$. Note that $p_1'''(x) = 9 \neq$
lyche-numerical-linear-algebra_F00620	49	■ 1.000	$39=p_{\{2\}}^{\{\prime \prime\prime\}}(x)$	$39 = p_2'''(x)$	$39 = p_2'''(x)$ showing that the third derivative of g is piecewise constant with a jump
lyche-numerical-linear-algebra_F00621	49	■ 1.000	$3(n-1) C^{\{2\}}$	$3(n - 1)C^2$	interpolant. Indeed counting requirements we have $3(n - 1) C^2$ conditions,
lyche-numerical-linear-algebra_F00622	49	■ 1.000	$4 n$	$4n$	$n + 1$ conditions (2.2), and two boundary conditions, adding to $4n$. Since a
lyche-numerical-linear-algebra_F00623	49	■ 1.000	$p_{\{1\}}, \ldots, p_{\{n\}}$	$p_1, \dots, p_n$	coefficients of the n polynomials $p_1, \dots, p_n$ and give hope for uniqueness of the
lyche-numerical-linear-algebra_F00624	50	■ 0.829	$[0,2]$	$[0, 2]$	Fig. 2.3 A cubic spline with one knot interpolating $f(x) = x^4$ on $[0, 2]$
lyche-numerical-linear-algebra_F00625	50	■ 1.000	$(0,2)$	$(0, 2)$	Lemma 2.1 (Representing Each p_i Using $(0, 2)$ Interpolation) Given $a < b$,
lyche-numerical-linear-algebra_F00626	50	■ 1.000	$a<b$	$a < b$	Lemma 2.1 (Representing Each p_i Using $(0, 2)$ Interpolation) Given $a < b$,
lyche-numerical-linear-algebra_F00627	50	■ 1.000	$h=(b-a) / n$	$h = (b - a)/n$	$h = (b-a)/n$ with $n \geq 2, x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \dots, n+1$.
lyche-numerical-linear-algebra_F00628	50	■ 1.000	$n \geq 2, x_{\{i\}}=a+(i-1) h$	$n \geq 2, x_i = a + (i - 1)h$	$h = (b-a)/n$ with $n \geq 2, x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \dots, n+1$.
lyche-numerical-linear-algebra_F00629	50	■ 1.000	$y_{\{i\}}, \mu_{\{i\}}$	y_i, μ_i	$h = (b-a)/n$ with $n \geq 2, x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \dots, n+1$.
lyche-numerical-linear-algebra_F00630	50	■ 1.000	$i=1, \ldots, n+1$	$i = 1, \dots, n + 1$	$h = (b-a)/n$ with $n \geq 2, x_i = a+(i-1)h$, and numbers y_i, μ_i for $i = 1, \dots, n+1$.
lyche-numerical-linear-algebra_F00631	51	■ 1.000	$1 \leq i \leq n$	$1 \leq i \leq n$	Proof Consider p_i in the form (2.7) for some $1 \leq i \leq n$. Evoking (2.6) we find
lyche-numerical-linear-algebra_F00632	51	■ 1.000	$p_{\{i\}}(x_{\{i\}})=c_{\{i, 1\}}=y_{\{i\}}$	$p_i(x_i) = c_{i,1} = y_i$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3} + 6c_{i,4}(x - x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$

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lyche-numerical-linear-algebra_F00633	51	1.000	$p_{i+1}'(x) = 2c_{i,3} + 6c_{i,4}(x - x_i)$	$p_i''(x) = 2c_{i,3} + 6c_{i,4}(x - x_i)$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3} + 6c_{i,4}(x - x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$
lyche-numerical-linear-algebra_F00634	51	1.000	$c_{i,3}$	$c_{i,3}$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3} + 6c_{i,4}(x - x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$
lyche-numerical-linear-algebra_F00635	51	1.000	$p_{i+1}'(x) =$	$p_i''(x_i) =$	$p_i(x_i) = c_{i,1} = y_i$. Since $p_i''(x) = 2c_{i,3} + 6c_{i,4}(x - x_i)$ we obtain $c_{i,3}$ from $p_i''(x_i) =$
lyche-numerical-linear-algebra_F00636	51	1.000	$2c_{i,3} = \mu_i$	$2c_{i,3} = \mu_i$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_F00637	51	1.000	$c_{i,4}$	$c_{i,4}$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_F00638	51	1.000	$p_{i+1}'(x) = \mu_i + 6hc_{i,4}$	$p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$	$2c_{i,3} = \mu_i$ (a moment), and then $c_{i,4}$ from $p_i''(x_{i+1}) = \mu_i + 6hc_{i,4} = \mu_{i+1}$. Finally
lyche-numerical-linear-algebra_F00639	51	1.000	$c_{i,2}$	$c_{i,2}$	we find $c_{i,2}$ by solving $p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1} - \mu_i}{6h}h^3 = y_{i+1}$. For
lyche-numerical-linear-algebra_F00640	51	1.000	$p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1} - \mu_i}{6h}h^3 = y_{i+1}$	$p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1} - \mu_i}{6h}h^3 = y_{i+1}$	we find $c_{i,2}$ by solving $p_i(x_{i+1}) = y_i + c_{i,2}h + \frac{\mu_i}{2}h^2 + \frac{\mu_{i+1} - \mu_i}{6h}h^3 = y_{i+1}$. For
lyche-numerical-linear-algebra_F00641	51	1.000	$j = 0, 1, 2, 3$	$j = 0, 1, 2, 3$	$j = 0, 1, 2, 3$ the shifted powers $(x - x_i)^j$ constitute a basis for cubic polynomials
lyche-numerical-linear-algebra_F00642	51	1.000	$(x - x_i)^j$	$(x - x_i)^j$	$j = 0, 1, 2, 3$ the shifted powers $(x - x_i)^j$ constitute a basis for cubic polynomials
lyche-numerical-linear-algebra_F00643	51	0.999	$\mu_1, \dots$	$\mu_1, \dots$	Theorem 2.1 (Constructing a D_2-Spline) Suppose for some moments $\mu_1, \dots$,
lyche-numerical-linear-algebra_F00644	51	1.000	μ_{n+1}	μ_{n+1}	μ_{n+1} that each p_i is given as in Lemma 2.1 for $i = 1, \dots, n$. If in addition
lyche-numerical-linear-algebra_F00645	51	0.973	$\mu_1, \dots, \mu_{n+1}$	$\mu_1, \dots, \mu_{n+1}$	$\mu_1, \dots, \mu_{n+1}$. Consider the C^2 requirement. Since $p_{i-1}(x_i) = p_i(x_i) = y_i$ and
lyche-numerical-linear-algebra_F00646	51	0.973	$p_{i-1}(x_i) = p_i(x_i) = y_i$	$p_{i-1}(x_i) = p_i(x_i) = y_i$	$\mu_1, \dots, \mu_{n+1}$. Consider the C^2 requirement. Since $p_{i-1}(x_i) = p_i(x_i) = y_i$ and
lyche-numerical-linear-algebra_F00647	51	1.000	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$ for $i = 2, \dots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_F00648	51	1.000	$i = 2, \dots, n$	$i = 2, \dots, n$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$ for $i = 2, \dots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_F00649	51	1.000	$g \in C^2$	$g \in C^2$	$p_{i-1}'(x_i) = p_i'(x_i) = \mu_i$ for $i = 2, \dots, n$ it follows that $g \in C^2$ if and only if
lyche-numerical-linear-algebra_F00650	51	1.000	$p_{i-1}'(x_i) = p_i'(x_i)$	$p_{i-1}'(x_i) = p_i'(x_i)$	$p_{i-1}'(x_i) = p_i'(x_i)$ for $i = 2, \dots, n$. By (2.7)
lyche-numerical-linear-algebra_F00651	51	1.000	$g(x_i) = y_i, i = 1, \dots, n+1, g'(a) = p_1'(x_1) = \mu_1$	$g(x_i) = y_i, i = 1, \dots, n+1, g''(a) = p_1''(x_1) = \mu_1$	By construction $g(x_i) = y_i, i = 1, \dots, n+1, g''(a) = p_1''(x_1) = \mu_1$ and $g''(b) =$
lyche-numerical-linear-algebra_F00652	51	1.000	$g'(b) =$	$g''(b) =$	By construction $g(x_i) = y_i, i = 1, \dots, n+1, g''(a) = p_1''(x_1) = \mu_1$ and $g''(b) =$
lyche-numerical-linear-algebra_F00653	51	1.000	$p_n'(x_{n+1}) = \mu_{n+1}$	$p_n''(x_{n+1}) = \mu_{n+1}$	$p_n''(x_{n+1}) = \mu_{n+1}$. It follows that g solves the D_2 -spline problem.
lyche-numerical-linear-algebra_F00654	51	1.000	$\mu_2, \dots, \mu_n$	$\mu_2, \dots, \mu_n$	In order for the D_2 -spline to exist we need to show that $\mu_2, \dots, \mu_n$ always can be
lyche-numerical-linear-algebra_F00655	51	1.000	$n \geq 3$	$n \geq 3$	determined from (2.9). For $n \geq 3$ and with μ_1 and μ_{n+1} given (2.9) can be written
lyche-numerical-linear-algebra_F00656	52	1.000	$\mathbf{A} = [a_{ij}] \in \mathbb{C}^{n \times n}$	$\mathbf{A} = [a_{ij}] \in \mathbb{C}^{n \times n}$	Definition 2.2 (Strict Diagonal Dominance) The matrix $\mathbf{A} = [a_{ij}] \in \mathbb{C}^{n \times n}$ is
lyche-numerical-linear-algebra_F00657	52	1.000	$ x_k = \max_i x_i $	$ x_k = \max_i x_i $	that $ x_k = \max_i x_i $. Then

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lyche-numerical-linear-algebra-F00658	52	■ 0.999	$\max_{1 \leq i \leq n} \left \left x_i \right \right = \left \left x_k \right \right \leq \frac{ b_k }{\sigma_k} \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right)$		and this implies $\max_{1 \leq i \leq n} x_i = x_k \leq \frac{ b_k }{\sigma_k} \leq \max_{1 \leq i \leq n} \left(\frac{ b_i }{\sigma_i} \right)$. For the
lyche-numerical-linear-algebra-F00659	52	■ 0.999	$\max_{1 \leq i \leq n} \left \left x_i \right \right \leq 0$	$\max_{1 \leq i \leq n} x_i \leq 0$	nonsingularity, if $A\mathbf{x} = \mathbf{0}$, then $\max_{1 \leq i \leq n} x_i \leq 0$ by (2.13), and so $\mathbf{x} = \mathbf{0}$.
lyche-numerical-linear-algebra-F00660	52	■ 1.000	$g_{\{1\}}$	g_1	uniqueness suppose g_1 and g_2 are two D_2 -splines interpolating the same data. Then
lyche-numerical-linear-algebra-F00661	52	■ 1.000	$g_{\{2\}}$	g_2	uniqueness suppose g_1 and g_2 are two D_2 -splines interpolating the same data. Then
lyche-numerical-linear-algebra-F00662	52	■ 1.000	$g:=g_{\{1\}}-g_{\{2\}}$	$g := g_1 - g_2$	$g := g_1 - g_2$ is an N -spline satisfying (2.2) with $\mathbf{y} = \mathbf{0}$. The solution $[\mu_2, \dots, \mu_n]^T$
lyche-numerical-linear-algebra-F00663	52	■ 1.000	$\boldsymbol{y}=\mathbf{0}$	$\mathbf{y} = \mathbf{0}$	$g := g_1 - g_2$ is an N -spline satisfying (2.2) with $\mathbf{y} = \mathbf{0}$. The solution $[\mu_2, \dots, \mu_n]^T$
lyche-numerical-linear-algebra-F00664	52	■ 1.000	$\left[\mu_2, \ldots, \mu_n \right]^T$	$[\mu_2, \dots, \mu_n]^T$	$g := g_1 - g_2$ is an N -spline satisfying (2.2) with $\mathbf{y} = \mathbf{0}$. The solution $[\mu_2, \dots, \mu_n]^T$
lyche-numerical-linear-algebra-F00665	52	■ 1.000	$\mu_1=\mu_{n+1}$	$\mu_1 = \mu_{n+1}$	of (2.11) and also $\mu_1 = \mu_{n+1}$ are zero. It follows from (2.8) that all coefficients $c_{i,j}$
lyche-numerical-linear-algebra-F00666	52	■ 1.000	$c_{\{i, j\}}$	$c_{i,j}$	of (2.11) and also $\mu_1 = \mu_{n+1}$ are zero. It follows from (2.8) that all coefficients $c_{i,j}$
lyche-numerical-linear-algebra-F00667	52	■ 1.000	$g=0$	$g = 0$	are zero. We conclude that $g = 0$ and $g_1 = g_2$.
lyche-numerical-linear-algebra-F00668	52	■ 1.000	$g_{\{1\}}=g_{\{2\}}$	$g_1 = g_2$	are zero. We conclude that $g = 0$ and $g_1 = g_2$.
lyche-numerical-linear-algebra-F00669	52	■ 1.000	B	B	<i>Example 2.2 (Cubic B-Spline)</i> For the N -spline with $[a, b] = [0, 4]$ and $\mathbf{y} =$
lyche-numerical-linear-algebra-F00670	52	■ 1.000	$[a, b]=[0,4]$	$[a, b] = [0, 4]$	<i>Example 2.2 (Cubic B-Spline)</i> For the N -spline with $[a, b] = [0, 4]$ and $\mathbf{y} =$
lyche-numerical-linear-algebra-F00671	52	■ 1.000	$\boldsymbol{y}=$	$\mathbf{y} =$	<i>Example 2.2 (Cubic B-Spline)</i> For the N -spline with $[a, b] = [0, 4]$ and $\mathbf{y} =$
lyche-numerical-linear-algebra-F00672	52	■ 0.997	$\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0 \right]$	$\left[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0 \right]$	$[0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}, 0]$ the linear system (2.9) takes the form
lyche-numerical-linear-algebra-F00673	53	■ 1.000	$\mu_2=\mu_4=1, \mu_3=-2$	$\mu_2 = \mu_4 = 1, \mu_3 = -2$	The solution is $\mu_2 = \mu_4 = 1, \mu_3 = -2$. The knotset is $\{0, 1, 2, 3, 4\}$. Using (2.8)
lyche-numerical-linear-algebra-F00674	53	■ 1.000	$\{0, 1, 2, 3, 4\}$	$\{0, 1, 2, 3, 4\}$	The solution is $\mu_2 = \mu_4 = 1, \mu_3 = -2$. The knotset is $\{0, 1, 2, 3, 4\}$. Using (2.8)
lyche-numerical-linear-algebra-F00675	53	■ 1.000	$(0,4)$	$(0, 4)$	A plot of this spline is shown in Fig. 2.4. On $(0, 4)$ the function g equals the nonzero
lyche-numerical-linear-algebra-F00676	53	■ 1.000	$D^{\{2\}}$	D^2	To find the D^2 -spline g we have to solve the triangular system (2.11). Consider
lyche-numerical-linear-algebra-F00677	53	■ 1.000	$\boldsymbol{A}=\operatorname{tridiag}\left(a_{\{i\}}, d_{\{i\}}, c_{\{i\}}\right) \in$	$\mathbf{A} = \operatorname{tridiag}(a_i, d_i, c_i) \in$	solving a general tridiagonal linear system $A\mathbf{x} = \mathbf{b}$ where $\mathbf{A} = \operatorname{tridiag}(a_i, d_i, c_i) \in$
lyche-numerical-linear-algebra-F00678	54	■ 1.000	$\mathbb{C}^n \times n$	$\mathbb{C}^{n \times n}$	$\mathbb{C}^{n \times n}$. Instead of using Gaussian elimination directly, we can construct two matrices
lyche-numerical-linear-algebra-F00679	54	■ 1.000	$\boldsymbol{A}=\boldsymbol{L} \boldsymbol{U}$	$\mathbf{A} = \mathbf{L} \mathbf{U}$	$\mathbf{L}$ and $\mathbf{U}$ such that $\mathbf{A} = \mathbf{L} \mathbf{U}$. Since $A\mathbf{x} = \mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{b}$ we can find $\mathbf{x}$ by solving
lyche-numerical-linear-algebra-F00680	54	■ 1.000	$\boldsymbol{A} \boldsymbol{x}=\boldsymbol{L} \boldsymbol{U} \boldsymbol{x}=\boldsymbol{b}$	$\mathbf{A} \mathbf{x} = \mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{b}$	$\mathbf{L}$ and $\mathbf{U}$ such that $\mathbf{A} = \mathbf{L} \mathbf{U}$. Since $A\mathbf{x} = \mathbf{L} \mathbf{U} \mathbf{x} = \mathbf{b}$ we can find $\mathbf{x}$ by solving
lyche-numerical-linear-algebra-F00681	54	■ 1.000	$\boldsymbol{L} \boldsymbol{z}=\boldsymbol{b}$	$\mathbf{L} \mathbf{z} = \mathbf{b}$	two systems $\mathbf{L} \mathbf{z} = \mathbf{b}$ and $\mathbf{U} \mathbf{x} = \mathbf{z}$. Moreover $\mathbf{L}$ and $\mathbf{U}$ are both triangular and
lyche-numerical-linear-algebra-F00682	54	■ 1.000	$\boldsymbol{U} \boldsymbol{x}=\boldsymbol{z}$	$\mathbf{U} \mathbf{x} = \mathbf{z}$	two systems $\mathbf{L} \mathbf{z} = \mathbf{b}$ and $\mathbf{U} \mathbf{x} = \mathbf{z}$. Moreover $\mathbf{L}$ and $\mathbf{U}$ are both triangular and
lyche-numerical-linear-algebra-F00683	54	■ 1.000	$u_{\{1\}}, u_{\{2\}}, \ldots, u_{\{n-1\}}$	$u_1, u_2, \dots, u_{n-1}$	then $\mathbf{A} = \mathbf{L} \mathbf{U}$. If $u_1, u_2, \dots, u_{n-1}$ are nonzero then (2.16) is well defined. If in

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F00684	54	■ 0.999	$u_{\{n\}} \neq 0$	$u_n \neq 0$	addition $u_n \neq 0$ then we can solve $\mathbf{Lz} = \mathbf{b}$ and $\mathbf{Ux} = \mathbf{z}$ for $\mathbf{z}$ and $\mathbf{x}$. We formulate
lyche-numerical-linear-algebra-F00685	54	■ 0.999	$\boldsymbol{z}$	$\mathbf{z}$	addition $u_n \neq 0$ then we can solve $\mathbf{Lz} = \mathbf{b}$ and $\mathbf{Ux} = \mathbf{z}$ for $\mathbf{z}$ and $\mathbf{x}$. We formulate
lyche-numerical-linear-algebra-F00686	54	■ 1.000	$\boldsymbol{l} \in \mathbb{C}^{n-1}, \boldsymbol{u} \in \mathbb{C}^n$	$\mathbf{l} \in \mathbb{C}^{n-1}, \mathbf{u} \in \mathbb{C}^n$	this as two algorithms. In <code>trifactor</code> , vectors $\mathbf{l} \in \mathbb{C}^{n-1}, \mathbf{u} \in \mathbb{C}^n$ are computed
lyche-numerical-linear-algebra-F00687	54	■ 0.852	$\boldsymbol{a}, \boldsymbol{c} \in \mathbb{C}^{n-1}, \boldsymbol{d} \in \mathbb{C}^n$	$\mathbf{a}, \mathbf{c} \in \mathbb{C}^{n-1}, \mathbf{d} \in \mathbb{C}^n$	from $\mathbf{a}, \mathbf{c} \in \mathbb{C}^{n-1}, \mathbf{d} \in \mathbb{C}^n$. This implements the LU factorization of a tridiagonal
lyche-numerical-linear-algebra-F00688	55	■ 1.000	$\boldsymbol{l} \in \mathbb{C}^{n-1}$	$\mathbf{l} \in \mathbb{C}^{n-1}$	computed from a previous call to <code>trifactor</code> . Here $\mathbf{l} \in \mathbb{C}^{n-1}$ and $\mathbf{u} \in \mathbb{C}^n$ are
lyche-numerical-linear-algebra-F00689	55	■ 1.000	$\boldsymbol{u} \in \mathbb{C}^n$	$\mathbf{u} \in \mathbb{C}^n$	computed from a previous call to <code>trifactor</code> . Here $\mathbf{l} \in \mathbb{C}^{n-1}$ and $\mathbf{u} \in \mathbb{C}^n$ are
lyche-numerical-linear-algebra-F00690	55	■ 1.000	$\boldsymbol{b} \in \mathbb{C}^{n,r}$	$\mathbf{b} \in \mathbb{C}^{n,r}$	output from <code>trifactor</code> and $\mathbf{b} \in \mathbb{C}^{n,r}$ for some $r \in \mathbb{N}$:
lyche-numerical-linear-algebra-F00691	55	■ 1.000	$r \in \mathbb{N}$	$r \in \mathbb{N}$	output from <code>trifactor</code> and $\mathbf{b} \in \mathbb{C}^{n,r}$ for some $r \in \mathbb{N}$:
lyche-numerical-linear-algebra-F00692	55	■ 1.000	$u_{\{k\}}$	u_k	Proof We show that the u_k 's in (2.16) are nonzero for $k = 1, \dots, n$. For this it is
lyche-numerical-linear-algebra-F00693	55	■ 1.000	$k=1, \dots, n$	$k = 1, \dots, n$	Proof We show that the u_k 's in (2.16) are nonzero for $k = 1, \dots, n$. For this it is
lyche-numerical-linear-algebra-F00694	55	■ 1.000	$a_{\{0\}} := c_{\{n\}} := 0$	$a_0 := c_n := 0$	and where $a_0 := c_n := 0$. By assumption $ u_1 = d_1 = \sigma_1 + c_1 $. Suppose $ u_k \geq$
lyche-numerical-linear-algebra-F00695	55	■ 1.000	$ u_1 = d_1 = \sigma_1 + c_1 $	$ u_1 = d_1 = \sigma_1 + c_1 $	and where $a_0 := c_n := 0$. By assumption $ u_1 = d_1 = \sigma_1 + c_1 $. Suppose $ u_k \geq$
lyche-numerical-linear-algebra-F00696	55	■ 1.000	$ u_k \geq$	$ u_k \geq$	and where $a_0 := c_n := 0$. By assumption $ u_1 = d_1 = \sigma_1 + c_1 $. Suppose $ u_k \geq$
lyche-numerical-linear-algebra-F00697	55	■ 0.999	$\sigma_k + c_k $	$\sigma_k + c_k $	$\sigma_k + c_k $ for some $1 \leq k \leq n - 1$. Then $ c_k / u_k < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra-F00698	55	■ 0.999	$1 \leq k \leq n - 1$	$1 \leq k \leq n - 1$	$\sigma_k + c_k $ for some $1 \leq k \leq n - 1$. Then $ c_k / u_k < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra-F00699	55	■ 0.999	$ c_k / u_k < 1$	$ c_k / u_k < 1$	$\sigma_k + c_k $ for some $1 \leq k \leq n - 1$. Then $ c_k / u_k < 1$ and by (2.16) and strict
lyche-numerical-linear-algebra-F00700	56	■ 1.000	$u_{\{1\}} = d_{\{1\}}$	$u_1 = d_1$	given by (2.16). Then (2.17) holds, $u_1 = d_1$ and
lyche-numerical-linear-algebra-F00701	56	■ 1.000	$l_{\{k\}}$	l_k	we cannot have severe growth in the computed elements u_k and l_k .
lyche-numerical-linear-algebra-F00702	56	■ 1.000	$3n - 3$	$3n - 3$	onal matrix of order n using (2.16) is $3n - 3$, while the number of arithmetic oper-
lyche-numerical-linear-algebra-F00703	56	■ 1.000	$r(5n - 4)$	$r(5n - 4)$	ations for Algorithm <code>trisolve</code> is $r(5n - 4)$, where r is the number of right-hand
lyche-numerical-linear-algebra-F00704	56	■ 1.000	$O(n)$	$O(n)$	sides. This means that the complexity to solve a tridiagonal system is $O(n)$, or
lyche-numerical-linear-algebra-F00705	56	■ 1.000	$8n - 7$	$8n - 7$	more precisely $8n - 7$ when $r = 1$, and this number only grows linearly ² with n .
lyche-numerical-linear-algebra-F00706	56	■ 1.000	$r=1$	$r = 1$	more precisely $8n - 7$ when $r=1$, and this number only grows linearly ² with n .
lyche-numerical-linear-algebra-F00707	56	■ 0.939	$[0, 1]$	$[0, 1]$	where f is a given continuous function on $[0, 1]$ and u is an unknown function. This
lyche-numerical-linear-algebra-F00708	56	■ 0.939	u	u	where f is a given continuous function on $[0, 1]$ and u is an unknown function. This
lyche-numerical-linear-algebra-F00709	56	■ 1.000	$u(0) = u(1) =$	$u(0) = u(1) =$	integration constants so that the homogeneous boundary conditions $u(0) = u(1) =$
lyche-numerical-linear-algebra-F00710	56	■ 1.000	$f(x) = 1$	$f(x) = 1$	0 are satisfied. For example, if $f(x) = 1$ then $u(x) = x(x - 1)/2$ is the solution.

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lyche-numerical-linear-algebra_F00711	56	■ 1.000	$u(x)=x(x-1) / 2$	$u(x) = x(x - 1)/2$	0 are satisfied. For example, if $f(x) = 1$ then $u(x) = x(x - 1)/2$ is the solution.
lyche-numerical-linear-algebra_F00712	56	■ 1.000	$O\left(n^3\right)$	$O\left(n^3\right)$	² We show in Sect. 3.3.2 that Gaussian elimination on a full $n \times n$ system is an $O(n^3)$ process.
lyche-numerical-linear-algebra_F00713	57	■ 1.000	$h:=1 /(m+1)$	$h := 1/(m + 1)$	positive integer m , let $h := 1/(m + 1)$ be the discretization parameter, and replace
lyche-numerical-linear-algebra_F00714	57	■ 0.816	$x_{\{j\}}:=j \ h$	$x_j := jh$	the interval $[0, 1]$ by grid points $x_j := jh$ for $j = 0, 1, \dots, m + 1$. We then obtain
lyche-numerical-linear-algebra_F00715	57	■ 0.816	$j=0,1, \ldots, m+1$	$j = 0, 1, \dots, m + 1$	the interval $[0, 1]$ by grid points $x_j := jh$ for $j = 0, 1, \dots, m + 1$. We then obtain
lyche-numerical-linear-algebra_F00716	57	■ 0.958	$v_{\{j\}}$	v_j	approximations v_j to the exact solution $u(x_j)$ for $j = 1, \dots, m$ by replacing the
lyche-numerical-linear-algebra_F00717	57	■ 0.958	$u\left(x_{\{j\}}\right)$	$u\left(x_j\right)$	approximations v_j to the exact solution $u(x_j)$ for $j = 1, \dots, m$ by replacing the
lyche-numerical-linear-algebra_F00718	57	■ 1.000	$h^{\{2\}}$	h^2	Moving the h^2 factor to the right hand side this can be written as an $m \times m$ linear
lyche-numerical-linear-algebra_F00719	57	■ 1.000	$\boldsymbol{T}$	$\boldsymbol{T}$	The matrix $\boldsymbol{T}$ is called the second derivative matrix and will occur frequently in
lyche-numerical-linear-algebra_F00720	57	■ 0.999	$\boldsymbol{T}=\operatorname{tridiag}\left(a_{\{i\}}, d_{\{i\}}, c_{\{i\}}\right) \text { in }$	$\boldsymbol{T} = \operatorname{tridiag}\left(a_i, d_i, c_i\right) \in$	this book. It is our second example of a tridiagonal matrix, $\boldsymbol{T} = \operatorname{tridiag}\left(a_i, d_i, c_i\right) \in$
lyche-numerical-linear-algebra_F00721	57	■ 1.000	$\mathbb{R}^{\{m \times m\}}$	$\mathbb{R}^{m \times m}$	$\mathbb{R}^{m \times m}$, where in this case $a_i = c_i = -1$ and $d_i = 2$ for all i .
lyche-numerical-linear-algebra_F00722	57	■ 1.000	$a_{\{i\}}=c_{\{i\}}=-1$	$a_i = c_i = -1$	$\mathbb{R}^{m \times m}$, where in this case $a_i = c_i = -1$ and $d_i = 2$ for all i .
lyche-numerical-linear-algebra_F00723	57	■ 1.000	$d_{\{i\}}=2$	$d_i = 2$	$\mathbb{R}^{m \times m}$, where in this case $a_i = c_i = -1$ and $d_i = 2$ for all i .
lyche-numerical-linear-algebra_F00724	58	■ 1.000	$\boldsymbol{A}_{\{1\}}$	$\boldsymbol{A}_1$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra_F00725	58	■ 1.000	$\boldsymbol{A}_{\{2\}}$	$\boldsymbol{A}_2$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra_F00726	58	■ 1.000	$\boldsymbol{A}_{\{3\}}$	$\boldsymbol{A}_3$	They are all weakly diagonally dominant, but $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are singular, while $\boldsymbol{A}_3$ is
lyche-numerical-linear-algebra_F00727	58	■ 1.000	$\operatorname{det}\left(\boldsymbol{A}_{\{3\}}\right)=4 \neq 0$	$\det\left(\boldsymbol{A}_3\right) = 4 \neq 0$	has a zero row, and $\det(\boldsymbol{A}_3) = 4 \neq 0$. It follows that for the nonsingularity and
lyche-numerical-linear-algebra_F00728	58	■ 1.000	$\left d_{\{1\}}\right >\left c_{\{1\}}\right $	$ d_1 > c_1 $	$\mathbb{C}^{n \times n}$ is tridiagonal and weakly diagonally dominant. If in addition $ d_1 > c_1 $ and
lyche-numerical-linear-algebra_F00729	58	■ 0.574	$a_{\{i\}} \neq 0$	$a_i \neq 0$	$a_i \neq 0$ for $i = 1, \dots, n - 2$, then $\boldsymbol{A}$ has a unique LU factorization (2.15). If in
lyche-numerical-linear-algebra_F00730	58	■ 0.574	$i=1, \ldots, n-2$	$i = 1, \dots, n - 2$	$a_i \neq 0$ for $i = 1, \dots, n - 2$, then $\boldsymbol{A}$ has a unique LU factorization (2.15). If in
lyche-numerical-linear-algebra_F00731	58	■ 0.574	L U	LU	$a_i \neq 0$ for $i = 1, \dots, n - 2$, then $\boldsymbol{A}$ has a unique LU factorization (2.15). If in
lyche-numerical-linear-algebra_F00732	58	■ 1.000	$d_{\{n\}} \neq 0$	$d_n \neq 0$	addition $d_n \neq 0$, then $\boldsymbol{A}$ is nonsingular.
lyche-numerical-linear-algebra_F00733	58	■ 0.999	$k=1, \ldots, n-1$	$k = 1, \dots, n - 1$	factorization if the u_k 's in (2.16) are nonzero for $k = 1, \dots, n - 1$. For this it is
lyche-numerical-linear-algebra_F00734	58	■ 0.991	$\left u_{\{k\}}\right >\left c_{\{k\}}\right $	$ u_k > c_k $	sufficient to show by induction that $ u_k > c_k $ for $k = 1, \dots, n - 1$. By assumption
lyche-numerical-linear-algebra_F00735	58	■ 0.798	$\left u_{\{1\}}\right =\left d_{\{1\}}\right >\left c_{\{1\}}\right $	$ u_1 = d_1 > c_1 $	$ u_1 = d_1 > c_1 $. Suppose $ u_k > c_k $ for some $1 \leq k \leq n - 2$. Then $ c_k / u_k < 1$
lyche-numerical-linear-algebra_F00736	58	■ 0.798	$1 \leq k \leq n-2$	$1 \leq k \leq n - 2$	$ u_1 = d_1 > c_1 $. Suppose $ u_k > c_k $ for some $1 \leq k \leq n - 2$. Then $ c_k / u_k < 1$
lyche-numerical-linear-algebra_F00737	58	■ 1.000	$k=n-1$	$k = n - 1$	This also holds for $k = n - 1$ if $a_{n-1} \neq 0$. By (2.23) and weak diagonal
lyche-numerical-linear-algebra_F00738	58	■ 1.000	$a_{\{n-1\}} \neq 0$	$a_{n-1} \neq 0$	This also holds for $k = n - 1$ if $a_{n-1} \neq 0$. By (2.23) and weak diagonal

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lyche-numerical-linear-algebra_F00766	59	1.000	$F=\frac{\pi^2}{L^2}R$	$F = \frac{\pi^2 R}{L^2}$	find $F = \frac{\pi^2 R}{L^2}$.
lyche-numerical-linear-algebra_F00767	59	1.000	$m \in \mathbb{N}, h:=1/(m+1)$	$m \in \mathbb{N}, h := 1/(m+1)$	We can approximate this eigenvalue numerically. Choosing $m \in \mathbb{N}, h := 1/(m+1)$
lyche-numerical-linear-algebra_F00768	60	1.000	$v_j \approx u(jh)$	$v_j \approx u(jh)$	where $v_j \approx u(jh)$ for $j = 0, \dots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_F00769	60	1.000	$j=0, \dots, m+1$	$j = 0, \dots, m+1$	where $v_j \approx u(jh)$ for $j = 0, \dots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_F00770	60	1.000	$\lambda:=h^2 K$	$\lambda := h^2 K$	where $v_j \approx u(jh)$ for $j = 0, \dots, m+1$. If we define $\lambda := h^2 K$ then we obtain the
lyche-numerical-linear-algebra_F00771	60	1.000	$\lambda=4 \sin^2(\pi h/2)$	$\lambda = 4 \sin^2(\pi h/2)$	that the smallest eigenvalue of (2.26) is given by $\lambda = 4 \sin^2(\pi h/2)$. Since $\lambda =$
lyche-numerical-linear-algebra_F00772	60	1.000	$\lambda=$	$\lambda =$	that the smallest eigenvalue of (2.26) is given by $\lambda = 4 \sin^2(\pi h/2)$. Since $\lambda =$
lyche-numerical-linear-algebra_F00773	60	1.000	$h^2 K=\frac{h^2 F L^2}{R}$	$h^2 K = \frac{h^2 F L^2}{R}$	$h^2 K = \frac{h^2 F L^2}{R}$ we can solve for F to obtain
lyche-numerical-linear-algebra_F00774	60	1.000	h	h	For small h this is a good approximation to the value $\frac{\pi^2 R}{L^2}$ we computed above.
lyche-numerical-linear-algebra_F00775	60	1.000	$\frac{\pi^2}{L^2}R$	$\frac{\pi^2 R}{L^2}$	For small h this is a good approximation to the value $\frac{\pi^2 R}{L^2}$ we computed above.
lyche-numerical-linear-algebra_F00776	60	0.997	$\mathbf{T}=\text{tridiag}(-1,2,-1)$	$\mathbf{T} = \text{tridiag}(-1, 2, -1)$	The second derivative matrix $\mathbf{T} = \text{tridiag}(-1, 2, -1)$ is a special case of the
lyche-numerical-linear-algebra_F00777	60	0.990	$a, d \in \mathbb{R}$	$a, d \in \mathbb{R}$	where $a, d \in \mathbb{R}$. We call this the 1D test matrix . It is symmetric and strictly
lyche-numerical-linear-algebra_F00778	60	0.990	$\mathbf{1D}$	1D	where $a, d \in \mathbb{R}$. We call this the 1D test matrix . It is symmetric and strictly
lyche-numerical-linear-algebra_F00779	60	1.000	$ d >2 a $	$ d > 2 a $	diagonally dominant if $ d > 2 a $.
lyche-numerical-linear-algebra_F00780	61	1.000	$m=3$	$m = 3$	For $m=3$,
lyche-numerical-linear-algebra_F00781	61	1.000	$\mathbf{T}_1=\left(t_{kj}\right)_{k,j=1}^m$	$\mathbf{T}_1 = (t_{kj})_{k,j=1}^m =$	Lemma 2.2 (Eigenpairs of 1D Test Matrix) Suppose $\mathbf{T}_1 = (t_{kj})_{k,j=1}^m =$
lyche-numerical-linear-algebra_F00782	61	0.999	$\text{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$	$\text{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$	$\text{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ with $m \geq 2, a, d \in \mathbb{R}$, and let $h = 1/(m+1)$.
lyche-numerical-linear-algebra_F00783	61	0.999	$m \geq 2, a, d \in \mathbb{R}$	$m \geq 2, a, d \in \mathbb{R}$	$\text{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ with $m \geq 2, a, d \in \mathbb{R}$ and let $h = 1/(m+1)$.
lyche-numerical-linear-algebra_F00784	61	0.999	$h=1/(m+1)$	$h = 1/(m+1)$	$\text{tridiag}(a, d, a) \in \mathbb{R}^{m \times m}$ with $m \geq 2, a, d \in \mathbb{R}$, and let $h = 1/(m+1)$.
lyche-numerical-linear-algebra_F00785	61	1.000	$\mathbf{T}_1 \mathbf{s}_j = \lambda_j \mathbf{s}_j$	$\mathbf{T}_1 \mathbf{s}_j = \lambda_j \mathbf{s}_j$	1. We have $\mathbf{T}_1 \mathbf{s}_j = \lambda_j \mathbf{s}_j$ for $j = 1, \dots, m$, where
lyche-numerical-linear-algebra_F00786	61	1.000	$1 < k < m$	$1 < k < m$	Proof We find for $1 < k < m$
lyche-numerical-linear-algebra_F00787	61	1.000	$k=1, m$	$k = 1, m$	This also holds for $k = 1, m$, and part 1 follows. Since $j\pi h = j\pi/(m+1) \in (0, \pi)$
lyche-numerical-linear-algebra_F00788	61	1.000	$j \pi h=j \pi /(m+1) \in(0, \pi)$	$j \pi h = j \pi /(m+1) \in (0, \pi)$	This also holds for $k = 1, m$, and part 1 follows. Since $j\pi h = j\pi/(m+1) \in (0, \pi)$
lyche-numerical-linear-algebra_F00789	61	0.988	$(0, \pi)$	$(0, \pi)$	for $j = 1, \dots, m$ and the cosine function is strictly monotone decreasing on $(0, \pi)$
lyche-numerical-linear-algebra_F00790	61	1.000	$\mathbf{T}_1$	$\mathbf{T}_1$	the eigenvalues are distinct, and since $\mathbf{T}_1$ is symmetric it follows from Lemma 2.3
lyche-numerical-linear-algebra_F00791	62	1.000	$i=\sqrt{-1}$	$i = \sqrt{-1}$	complex exponentials. With $i = \sqrt{-1}$ we find

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lyche-numerical-linear-algebra-F00792	62	1.000	$\boldsymbol{A}^* := \overline{\boldsymbol{A}}^T$	$\boldsymbol{A}^* := \overline{\boldsymbol{A}}^T$	Recall that the conjugate transpose of a matrix is defined by $\boldsymbol{A}^* := \overline{\boldsymbol{A}}^T$, where
lyche-numerical-linear-algebra-F00793	62	1.000	$\boldsymbol{x} \neq 0$	$\boldsymbol{x} \neq 0$	Proof Suppose $\boldsymbol{A}^* = \boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq 0$. We multiply both sides of
lyche-numerical-linear-algebra-F00794	62	0.998	$\boldsymbol{x}^*$	$\boldsymbol{x}^*$	$\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ from the left by $\boldsymbol{x}^*$ and divide by $\boldsymbol{x}^*\boldsymbol{x}$ to obtain $\lambda = \frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}$. Taking
lyche-numerical-linear-algebra-F00795	62	0.998	$\boldsymbol{x}^* \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{x}$	$\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ from the left by $\boldsymbol{x}^*$ and divide by $\boldsymbol{x}^*\boldsymbol{x}$ to obtain $\lambda = \frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}$. Taking
lyche-numerical-linear-algebra-F00796	62	0.998	$\lambda = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	$\lambda = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	$\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ from the left by $\boldsymbol{x}^*$ and divide by $\boldsymbol{x}^*\boldsymbol{x}$ to obtain $\lambda = \frac{\boldsymbol{x}^*\boldsymbol{A}\boldsymbol{x}}{\boldsymbol{x}^*\boldsymbol{x}}$. Taking
lyche-numerical-linear-algebra-F00797	62	0.996	$\bar{\lambda} = \lambda^* = \frac{(\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^*}{(\boldsymbol{x}^* \boldsymbol{x})^*} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \lambda$	$\bar{\lambda} = \lambda^* = \frac{(\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^*}{(\boldsymbol{x}^* \boldsymbol{x})^*} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \lambda$	complex conjugates we find $\bar{\lambda} = \lambda^* = \frac{(\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^*}{(\boldsymbol{x}^* \boldsymbol{x})^*} = \frac{\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \lambda$, and λ is real.
lyche-numerical-linear-algebra-F00798	62	0.725	$(\mu, \boldsymbol{y})$	$(\mu, \boldsymbol{y})$	Suppose that $(\lambda, \boldsymbol{x})$ and $(\mu, \boldsymbol{y})$ are two eigenpairs for $\boldsymbol{A}$ with $\mu \neq \lambda$. Multiplying
lyche-numerical-linear-algebra-F00799	62	0.725	$\mu \neq \lambda$	$\mu \neq \lambda$	Suppose that $(\lambda, \boldsymbol{x})$ and $(\mu, \boldsymbol{y})$ are two eigenpairs for $\boldsymbol{A}$ with $\mu \neq \lambda$. Multiplying
lyche-numerical-linear-algebra-F00800	62	1.000	$\boldsymbol{y}^*$	$\boldsymbol{y}^*$	$\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ by $\boldsymbol{y}^*$ gives
lyche-numerical-linear-algebra-F00801	62	1.000	μ	μ	using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^* \boldsymbol{x} = 0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra-F00802	62	1.000	$\lambda \neq \mu$	$\lambda \neq \mu$	using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^* \boldsymbol{x} = 0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra-F00803	62	1.000	$\boldsymbol{y}^* \boldsymbol{x} = 0$	$\boldsymbol{y}^* \boldsymbol{x} = 0$	using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^* \boldsymbol{x} = 0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra-F00804	62	1.000	$\boldsymbol{y}$	$\boldsymbol{y}$	using that μ is real. Since $\lambda \neq \mu$ it follows that $\boldsymbol{y}^* \boldsymbol{x} = 0$, which means that $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra-F00805	63	0.997	$\left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	(i) $\left[\begin{array}{cc} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right],$
lyche-numerical-linear-algebra-F00806	63	0.949	$\left[\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \boldsymbol{a}_{:3} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$[\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \boldsymbol{a}_{:3}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	(ii) $[\boldsymbol{a}_{:1}, \boldsymbol{a}_{:2}, \boldsymbol{a}_{:3}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right],$
lyche-numerical-linear-algebra-F00807	63	0.125	$\left[\begin{array}{c} \boldsymbol{a}_1^T \\ \boldsymbol{a}_2^T \\ \boldsymbol{a}_3^T \end{array} \right] = \left[\begin{array}{c} 123 \\ 45 \\ 789 \end{array} \right]$	$\left[\begin{array}{c} \boldsymbol{a}_1^T \\ \boldsymbol{a}_2^T \\ \boldsymbol{a}_3^T \end{array} \right] = \left[\begin{array}{c} 123 \\ 45 \\ 789 \end{array} \right]$	(iii) $\left[\begin{array}{c} \boldsymbol{a}_1^T \\ \boldsymbol{a}_2^T \\ \boldsymbol{a}_3^T \end{array} \right] = \left[\begin{array}{c} 123 \\ 45 \\ 789 \end{array} \right],$
lyche-numerical-linear-algebra-F00808	63	0.673	$\left[\boldsymbol{A}_{11}, \boldsymbol{A}_{12} \right] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$[\boldsymbol{A}_{11}, \boldsymbol{A}_{12}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	(iv) $[\boldsymbol{A}_{11}, \boldsymbol{A}_{12}] = \left[\begin{array}{c cc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right].$

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lyche-numerical-linear-algebra-F00809	63	0.902	(i i)	(ii)	while in (ii) and (iii) A has been partitioned into columns and rows, respectively.
lyche-numerical-linear-algebra-F00810	63	0.902	(i i i) $\boldsymbol{A}$	(iii) A	while in (ii) and (iii) A has been partitioned into columns and rows, respectively.
lyche-numerical-linear-algebra-F00811	63	1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times p}$	$A \in \mathbb{C}^{m \times p}$	In the following we assume that $A \in \mathbb{C}^{m \times p}$ and $B \in \mathbb{C}^{p \times n}$. Here are some rules
lyche-numerical-linear-algebra-F00812	63	1.000	$\boldsymbol{B} \in \mathbb{C}^{p \times n}$	$B \in \mathbb{C}^{p \times n}$	In the following we assume that $A \in \mathbb{C}^{m \times p}$ and $B \in \mathbb{C}^{p \times n}$. Here are some rules
lyche-numerical-linear-algebra-F00813	63	1.000	$\boldsymbol{B} = \left[\boldsymbol{b}_{:1}, \ldots, \boldsymbol{b}_{:n} \right]$	$B = [b_{:1}, \dots, b_{:n}]$	1. If $B = [b_{:1}, \dots, b_{:n}]$ is partitioned into columns then the partition of the product
lyche-numerical-linear-algebra-F00814	63	1.000	$\boldsymbol{A} \boldsymbol{e}_j$	Ae_j	and we see that column j of A can be written Ae_j for $j = 1, \dots, p$.
lyche-numerical-linear-algebra-F00815	63	1.000	$j=1, \ldots, p$	$j = 1, \dots, p$	and we see that column j of A can be written Ae_j for $j = 1, \dots, p$.
lyche-numerical-linear-algebra-F00816	63	1.000	$\boldsymbol{A} = \boldsymbol{I}$	$A = I$	and taking $A = I$ it follows that row i of B can be written $e_i^T B$ for $i = 1, \dots, m$.
lyche-numerical-linear-algebra-F00817	63	1.000	$\boldsymbol{e}_i^T \boldsymbol{B}$	$e_i^T B$	and taking $A = I$ it follows that row i of B can be written $e_i^T B$ for $i = 1, \dots, m$.
lyche-numerical-linear-algebra-F00818	63	1.000	$\boldsymbol{A} \boldsymbol{x}$	Ax	3. It is often useful to write the matrix-vector product Ax as a linear combination
lyche-numerical-linear-algebra-F00819	64	1.000	$\boldsymbol{B} = \left[\boldsymbol{B}_1, \boldsymbol{B}_2 \right]$	$B = [B_1, B_2]$	4. If $B = [B_1, B_2]$, where $B_1 \in \mathbb{C}^{p \times r}$ and $B_2 \in \mathbb{C}^{p \times (n-r)}$ then
lyche-numerical-linear-algebra-F00820	64	1.000	$\boldsymbol{B}_1 \in \mathbb{C}^{p \times r}$	$B_1 \in \mathbb{C}^{p \times r}$	4. If $B = [B_1, B_2]$, where $B_1 \in \mathbb{C}^{p \times r}$ and $B_2 \in \mathbb{C}^{p \times (n-r)}$ then
lyche-numerical-linear-algebra-F00821	64	1.000	$\boldsymbol{B}_2 \in \mathbb{C}^{p \times (n-r)}$	$B_2 \in \mathbb{C}^{p \times (n-r)}$	4. If $B = [B_1, B_2]$, where $B_1 \in \mathbb{C}^{p \times r}$ and $B_2 \in \mathbb{C}^{p \times (n-r)}$ then
lyche-numerical-linear-algebra-F00822	64	0.993	$\boldsymbol{A} = \left[\begin{array}{l} \boldsymbol{A}_1 \\ \boldsymbol{A}_2 \end{array} \right]$	$A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$	5. If $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{k \times p}$ and $A_2 \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra-F00823	64	0.993	$\boldsymbol{A}_1 \in \mathbb{C}^{k \times p}$	$A_1 \in \mathbb{C}^{k \times p}$	5. If $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{k \times p}$ and $A_2 \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra-F00824	64	0.993	$\boldsymbol{A}_2 \in \mathbb{C}^{(m-k) \times p}$	$A_2 \in \mathbb{C}^{(m-k) \times p}$	5. If $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{k \times p}$ and $A_2 \in \mathbb{C}^{(m-k) \times p}$ then
lyche-numerical-linear-algebra-F00825	64	0.985	$\boldsymbol{A} = \left[\boldsymbol{A}_1, \boldsymbol{A}_2 \right]$	$A = [A_1, A_2]$	6. If $A = [A_1, A_2]$ and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{m \times s}$, $A_2 \in \mathbb{C}^{m \times (p-s)}$, $B_1 \in$
lyche-numerical-linear-algebra-F00826	64	0.985	$\boldsymbol{B} = \left[\begin{array}{l} \boldsymbol{B}_1 \\ \boldsymbol{B}_2 \end{array} \right]$	$B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$	6. If $A = [A_1, A_2]$ and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{m \times s}$, $A_2 \in \mathbb{C}^{m \times (p-s)}$, $B_1 \in$
lyche-numerical-linear-algebra-F00827	64	0.985	$\boldsymbol{A}_1 \in \mathbb{C}^{m \times s}, \boldsymbol{A}_2 \in \mathbb{C}^{m \times (p-s)}, \boldsymbol{B}_1 \in$	$A_1 \in \mathbb{C}^{m \times s}, A_2 \in \mathbb{C}^{m \times (p-s)}, B_1 \in$	6. If $A = [A_1, A_2]$ and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, where $A_1 \in \mathbb{C}^{m \times s}, A_2 \in \mathbb{C}^{m \times (p-s)}, B_1 \in$
lyche-numerical-linear-algebra-F00828	64	1.000	$\mathbb{C}^{s \times n}$	$\mathbb{C}^{s \times n}$	$\mathbb{C}^{s \times n}$ and $B_2 \in \mathbb{C}^{(p-s) \times n}$ then
lyche-numerical-linear-algebra-F00829	64	1.000	$\boldsymbol{B}_2 \in \mathbb{C}^{(p-s) \times n}$	$B_2 \in \mathbb{C}^{(p-s) \times n}$	$\mathbb{C}^{s \times n}$ and $B_2 \in \mathbb{C}^{(p-s) \times n}$ then
lyche-numerical-linear-algebra-F00830	64	1.000	$\boldsymbol{A} = \left[\begin{array}{ll} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{22} \end{array} \right]$	$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$	7. If $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ then

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lyche-numerical-linear-algebra_F00831	64	1.000	$\boldsymbol{B}=\left[\begin{array}{ll}\boldsymbol{B}_{11}&\boldsymbol{B}_{12}\\\boldsymbol{B}_{21}&\boldsymbol{B}_{22}\end{array}\right]$	$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$	7. If $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ and $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$ then
lyche-numerical-linear-algebra_F00832	64	1.000	$\boldsymbol{A}_{11}$	A_{11}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}
lyche-numerical-linear-algebra_F00833	64	1.000	$\boldsymbol{A}_{21}$	A_{21}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}
lyche-numerical-linear-algebra_F00834	64	1.000	$\boldsymbol{B}_{11}$	B_{11}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}
lyche-numerical-linear-algebra_F00835	64	1.000	$\boldsymbol{B}_{12}$	B_{12}	number of columns in A_{11} and A_{21} equals the number of rows in B_{11} and B_{12}
lyche-numerical-linear-algebra_F00836	65	1.000	$\boldsymbol{A}_{i\ k}\ \boldsymbol{B}_{k\ j}$	$A_{ik}B_{kj}$	8. Consider finally the general case. If all the matrix products $A_{ik}B_{kj}$ in
lyche-numerical-linear-algebra_F00837	65	1.000	$\boldsymbol{A},\ \boldsymbol{A}_{11}$	A, A_{11}	where A, A_{11} and A_{22} are square matrices. Then A is nonsingular if and only if
lyche-numerical-linear-algebra_F00838	65	1.000	$\boldsymbol{A}_{22}$	A_{22}	where A, A_{11} and A_{22} are square matrices. Then A is nonsingular if and only if
lyche-numerical-linear-algebra_F00839	65	1.000	$\boldsymbol{B}:=\boldsymbol{A}^{-1}$	$B := A^{-1}$	Proof Suppose A is nonsingular. We partition $B := A^{-1}$ conformally with A and
lyche-numerical-linear-algebra_F00840	66	1.000	$\boldsymbol{B}_{21} =$	$B_{21} =$	The first equation implies that A_{11} is nonsingular, this in turn implies that $B_{21} =$
lyche-numerical-linear-algebra_F00841	66	0.577	$\mathbf{0}\ \boldsymbol{A}_{11}^{-1} = \mathbf{0}$	$0A_{11}^{-1} = 0$	$0A_{11}^{-1} = 0$ in the second equation, and then the third equation simplifies to
lyche-numerical-linear-algebra_F00842	66	1.000	$\boldsymbol{B}_{22}\ \boldsymbol{A}_{22} = \boldsymbol{I}$	$B_{22}A_{22} = I$	$B_{22}A_{22} = I$. We conclude that also A_{22} is nonsingular. From the fourth equation
lyche-numerical-linear-algebra_F00843	66	1.000	a_{ii}	a_{ii}	$A = [a_{ij}] \in \mathbb{C}^{n \times n}$ is nonsingular if and only if the diagonal elements a_{ii} ,
lyche-numerical-linear-algebra_F00844	66	1.000	$a_{ii}^{-1}, i=1, \ldots, n$	$a_{ii}^{-1}, i=1, \dots, n$	diagonal elements $a_{ii}^{-1}, i=1, \dots, n$.
lyche-numerical-linear-algebra_F00845	66	0.785	a_{11}	a_{11}	$[a_{11}]$ is nonsingular if and only if $a_{11} \neq 0$ and in that case $A^{-1} = [a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra_F00846	66	0.785	$a_{11} \neq \mathbf{0}$	$a_{11} \neq 0$	$[a_{11}]$ is nonsingular if and only if $a_{11} \neq 0$ and in that case $A^{-1} = [a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra_F00847	66	0.785	$\boldsymbol{A}^{-1} = \left[a_{11}^{-1}\right]$	$A^{-1} = [a_{11}^{-1}]$	$[a_{11}]$ is nonsingular if and only if $a_{11} \neq 0$ and in that case $A^{-1} = [a_{11}^{-1}]$. Suppose
lyche-numerical-linear-algebra_F00848	66	1.000	$\boldsymbol{A} \in \mathbb{C}^{(k+1) \times (k+1)}$	$A \in \mathbb{C}^{(k+1) \times (k+1)}$	the result holds for $n = k$ and let $A \in \mathbb{C}^{(k+1) \times (k+1)}$ be upper triangular. We partition
lyche-numerical-linear-algebra_F00849	66	1.000	$\boldsymbol{A}_k \in \mathbb{C}^{k \times k}$	$A_k \in \mathbb{C}^{k \times k}$	and note that $A_k \in \mathbb{C}^{k \times k}$ is upper triangular. By Lemma 2.4 A is nonsingular if and
lyche-numerical-linear-algebra_F00850	66	1.000	2.4 $\boldsymbol{A}$	2.4 A	and note that $A_k \in \mathbb{C}^{k \times k}$ is upper triangular. By Lemma 2.4 A is nonsingular if and
lyche-numerical-linear-algebra_F00851	66	1.000	$\boldsymbol{A}_k$	A_k	only if A_k and $(a_{k+1,k+1})$ are nonsingular and in that case
lyche-numerical-linear-algebra_F00852	66	1.000	$\left(a_{k+1,k+1}\right)$	$(a_{k+1,k+1})$	only if A_k and $a_{k+1,k+1}$ are nonsingular and in that case
lyche-numerical-linear-algebra_F00853	66	1.000	$\boldsymbol{c} \in \mathbb{C}^n$	$c \in \mathbb{C}^n$	for some $c \in \mathbb{C}^n$. By the induction hypothesis A_k is nonsingular if and only if the
lyche-numerical-linear-algebra_F00854	66	1.000	$a_{11}, \ldots, a_{kk}$	$a_{11}, \dots, a_{kk}$	diagonal elements $a_{11}, \dots, a_{kk}$ of A_k are nonzero and in that case A_k^{-1} is upper
lyche-numerical-linear-algebra_F00855	66	1.000	$\boldsymbol{A}_k^{-1}$	A_k^{-1}	diagonal elements $a_{11}, \dots, a_{kk}$ of A_k are nonzero and in that case A_k^{-1} is upper

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lyche-numerical-linear-algebra_F00882	68	■ 1.000	$\boldsymbol{y} \in \mathbb{R}^{n+1}$	$\boldsymbol{y} \in \mathbb{R}^{n+1}$	an interval $[a, b]$, a vector $\boldsymbol{y} \in \mathbb{R}^{n+1}$ and μ_1, μ_{n+1} . The vector $\boldsymbol{x} = [x_1, \dots, x_n]$ and
lyche-numerical-linear-algebra_F00883	68	■ 1.000	$\boldsymbol{x} = \left[x_1, \dots, x_n \right]$	$\boldsymbol{x} = [x_1, \dots, x_n]$	an interval $[a, b]$, a vector $\boldsymbol{y} \in \mathbb{R}^{n+1}$ and μ_1, μ_{n+1} . The vector $\boldsymbol{x} = [x_1, \dots, x_n]$ and
lyche-numerical-linear-algebra_F00884	69	■ 1.000	$\boldsymbol{C} \in \mathbb{R}^{n \times 4}$	$\boldsymbol{C} \in \mathbb{R}^{n \times 4}$	the coefficient matrix $\boldsymbol{C} \in \mathbb{R}^{n \times 4}$ in (2.7) are returned in the following algorithm. It
lyche-numerical-linear-algebra_F00885	69	■ 1.000	$g(r_j)$	$g(r_j)$	form (2.4) we need to compute values $g(r_j)$ at a number of sites $\boldsymbol{r} = [r_1, \dots, r_m] \in$
lyche-numerical-linear-algebra_F00886	69	■ 1.000	$\boldsymbol{r} = \left[r_1, \dots, r_m \right] \in$	$\boldsymbol{r} = [r_1, \dots, r_m] \in$	form (2.4) we need to compute values $g(r_j)$ at a number of sites $\boldsymbol{r} = [r_1, \dots, r_m] \in$
lyche-numerical-linear-algebra_F00887	69	■ 1.000	i_j	i_j	find an integer i_j so that $g(r_j) = p_{i_j}(r_j)$.
lyche-numerical-linear-algebra_F00888	69	■ 1.000	$g(r_j) = p_{i_j}(r_j)$	$g(r_j) = p_{i_j}(r_j)$	find an integer i_j so that $g(r_j) = p_{i_j}(r_j)$.
lyche-numerical-linear-algebra_F00889	69	■ 1.000	$k \in \mathbb{N}, \boldsymbol{t} = \left[t_1, \dots, t_k \right]$	$k \in \mathbb{N}, \boldsymbol{t} = [t_1, \dots, t_k]$	Given $k \in \mathbb{N}, \boldsymbol{t} = [t_1, \dots, t_k]$ and a real number x . We consider the problem
lyche-numerical-linear-algebra_F00890	69	■ 1.000	$i=1$	$i = 1$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00891	69	■ 1.000	$x < t_2, i=k$	$x < t_2, i = k$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00892	69	■ 1.000	$x \geq t_k$	$x \geq t_k$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00893	69	■ 1.000	$t_i \leq$	$t_i \leq$	of computing an integer i so that $i = 1$ if $x < t_2$, $i = k$ if $x \geq t_k$, and $t_i \leq$
lyche-numerical-linear-algebra_F00894	69	■ 1.000	$x < t_{i+1}$	$x < t_{i+1}$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m -vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F00895	69	■ 1.000	$\boldsymbol{x} \in \mathbb{R}^m$	$\boldsymbol{x} \in \mathbb{R}^m$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m -vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F00896	69	■ 1.000	$\boldsymbol{i}$	$\boldsymbol{i}$	$x < t_{i+1}$ otherwise. If $\boldsymbol{x} \in \mathbb{R}^m$ is a vector then an m -vector $\boldsymbol{i}$ should be computed,
lyche-numerical-linear-algebra_F00897	69	■ 1.000	$\boldsymbol{i} = \left[i_1, \dots, i_m \right]$	$\boldsymbol{i} = [i_1, \dots, i_m]$	The following MATLAB function determines $\boldsymbol{i} = [i_1, \dots, i_m]$. It uses the built in
lyche-numerical-linear-algebra_F00898	70	■ 1.000	$\boldsymbol{x}, \boldsymbol{C}$	$\boldsymbol{x}, \boldsymbol{C}$	Given output $\boldsymbol{x}, \boldsymbol{C}$ of splineint, defining a cubic spline g , and a vector $\boldsymbol{X}$,
lyche-numerical-linear-algebra_F00899	70	■ 1.000	$\boldsymbol{G} = g(\boldsymbol{X})$	$\boldsymbol{G} = g(\boldsymbol{X})$	splineval computes the vector $\boldsymbol{G} = g(\boldsymbol{X})$.
lyche-numerical-linear-algebra_F00900	71	■ 1.000	$f \in C^2[x-h, x+h]$	$f \in C^2[x-h, x+h]$	a) Show using Taylor expansion that if $f \in C^2[x-h, x+h]$ then for some η_2
lyche-numerical-linear-algebra_F00901	71	■ 1.000	η_2	η_2	a) Show using Taylor expansion that if $f \in C^2[x-h, x+h]$ then for some η_2
lyche-numerical-linear-algebra_F00902	71	■ 1.000	$f \in C^4[x-h, x+h]$	$f \in C^4[x-h, x+h]$	b) Show that if $f \in C^4[x-h, x+h]$ then for some η_4
lyche-numerical-linear-algebra_F00903	71	■ 1.000	η_4	η_4	b) Show that if $f \in C^4[x-h, x+h]$ then for some η_4
lyche-numerical-linear-algebra_F00904	71	■ 1.000	$\delta^2 f(x)$	$\delta^2 f(x)$	$\delta^2 f(x)$ is known as the central difference approximation to the second derivative
lyche-numerical-linear-algebra_F00905	71	■ 1.000	f, q	f, q	We assume that the given functions f, q and r are continuous on $[a, b]$ and that
lyche-numerical-linear-algebra_F00906	71	■ 1.000	$q(x) \geq 0$	$q(x) \geq 0$	$q(x) \geq 0$ for $x \in [a, b]$. It can then be shown that (2.36) has a unique solution u .
lyche-numerical-linear-algebra_F00907	71	■ 1.000	$x \in [a, b]$	$x \in [a, b]$	$q(x) \geq 0$ for $x \in [a, b]$. It can then be shown that (2.36) has a unique solution u .
lyche-numerical-linear-algebra_F00908	71	■ 1.000	$m \in \mathbb{N}, h = (b-a)/(m+1), x_j = a + jh$	$m \in \mathbb{N}, h = (b-a)/(m+1), x_j = a + jh$	To solve (2.36) numerically we choose $m \in \mathbb{N}, h = (b-a)/(m+1), x_j = a + jh$

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lyche-numerical-linear-algebra_F00909	71	■ 0.993	$v_{\{0\}}=g_{\{0\}}$	$v_0 = g_0$	with $v_0 = g_0$ and $v_{m+1} = g_1$.
lyche-numerical-linear-algebra_F00910	71	■ 0.993	$v_{\{m+1\}}=g_{\{1\}}$	$v_{m+1} = g_1$	with $v_0 = g_0$ and $v_{m+1} = g_1$.
lyche-numerical-linear-algebra_F00911	71	■ 1.000	$\boldsymbol{A} \boldsymbol{v} = \boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{v} = \boldsymbol{b}$	a) Show that (2.37) leads to a tridiagonal linear system $\boldsymbol{A} \boldsymbol{v} = \boldsymbol{b}$, where $\boldsymbol{A} =$
lyche-numerical-linear-algebra_F00912	71	■ 0.523	$\operatorname{tridiag}(\boldsymbol{a}_{\{j\}}, \boldsymbol{d}_{\{j\}}, \boldsymbol{c}_{\{j\}}) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}(a_j, d_j, c_j) \in \mathbb{R}^{m \times m}$	$\operatorname{tridiag}(a_j, d_j, c_j) \in \mathbb{R}^{m \times m}$ has elements
lyche-numerical-linear-algebra_F00913	72	■ 1.000	$\frac{h}{2} r(x) < 1$	$\frac{h}{2} r(x) < 1$	h is so small that $\frac{h}{2} r(x) < 1$ for all $x \in [a, b]$.
lyche-numerical-linear-algebra_F00914	72	■ 1.000	$v_{\{1\}}, \ldots, v_{\{m\}}$	$v_1, \dots, v_m$	(c) Propose a method to find $v_1, \dots, v_m$.
lyche-numerical-linear-algebra_F00915	72	■ 1.000	$r=0, f=q=1$	$r = 0, f = q = 1$	a) Consider the problem (2.36) with $r = 0, f = q = 1$ and boundary conditions
lyche-numerical-linear-algebra_F00916	72	■ 1.000	$u(0)=1, u(1)=0$	$u(0) = 1, u(1) = 0$	$u(0) = 1, u(1) = 0$. The exact solution is $u(x) = 1 - \sinh x / \sinh 1$. Write a
lyche-numerical-linear-algebra_F00917	72	■ 1.000	$u(x)=1-\sinh x / \sinh 1$	$u(x) = 1 - \sinh x / \sinh 1$	$u(0) = 1, u(1) = 0$. The exact solution is $u(x) = 1 - \sinh x / \sinh 1$. Write a
lyche-numerical-linear-algebra_F00918	72	■ 1.000	$h=0.1, 0.05, 0.025, 0.0125$	$h = 0.1, 0.05, 0.025, 0.0125$	computer program to solve (2.37) for $h = 0.1, 0.05, 0.025, 0.0125$, and compute
lyche-numerical-linear-algebra_F00919	72	■ 1.000	$\max_{1 \leq j \leq m} u(x_j) - v_j $	$\max_{1 \leq j \leq m} u(x_j) - v_j $	the “error” $\max_{1 \leq j \leq m} u(x_j) - v_j $ for each h .
lyche-numerical-linear-algebra_F00920	72	■ 1.000	$v_{\{j\}}, j=$	$v_j, j =$	b) Make a combined plot of the solution u and the computed points $v_j, j =$
lyche-numerical-linear-algebra_F00921	72	■ 1.000	$0, \ldots, m+1$	$0, \dots, m+1$	$0, \dots, m+1$ for $h = 0.1$.
lyche-numerical-linear-algebra_F00922	72	■ 1.000	$h=0.1$	$h = 0.1$	$0, \dots, m+1$ for $h = 0.1$.
lyche-numerical-linear-algebra_F00923	72	■ 1.000	$h^{\{p\}}$	h^p	c) One can show that the error is proportional to h^p for some integer p . Estimate p
lyche-numerical-linear-algebra_F00924	72	■ 1.000	$a_{\{i, i+1\}} a_{\{i+1, i\}} > 0$	$a_{i,i+1} a_{i+1,i} > 0$	tridiagonal and suppose $a_{i,i+1} a_{i+1,i} > 0$ for $i = 1, \dots, n-1$. Show that there
lyche-numerical-linear-algebra_F00925	72	■ 1.000	$\boldsymbol{D} = \operatorname{diag}(\boldsymbol{d}_{\{1\}}, \ldots, \boldsymbol{d}_{\{n\}})$	$\boldsymbol{D} = \operatorname{diag}(d_1, \dots, d_n)$	exists a diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, \dots, d_n)$ with $d_i > 0$ for all i such that
lyche-numerical-linear-algebra_F00926	72	■ 1.000	$\boldsymbol{d}_{\{i\}} > 0$	$d_i > 0$	exists a diagonal matrix $\boldsymbol{D} = \operatorname{diag}(d_1, \dots, d_n)$ with $d_i > 0$ for all i such that
lyche-numerical-linear-algebra_F00927	72	■ 1.000	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$	$\boldsymbol{B} := \boldsymbol{D} \boldsymbol{A} \boldsymbol{D}^{-1}$ is symmetric.
lyche-numerical-linear-algebra_F00928	73	■ 1.000	$\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$	Exercise 2.14 (LU Factorization of 2. Derivative Matrix) Show that $\boldsymbol{T} = \boldsymbol{L} \boldsymbol{U}$,
lyche-numerical-linear-algebra_F00929	73	■ 0.998	$\boldsymbol{S} \in \mathbb{R}^{m \times m}$	$\boldsymbol{S} \in \mathbb{R}^{m \times m}$	Exercise 2.15 (Inverse of the 2. Derivative Matrix) Let $\boldsymbol{S} \in \mathbb{R}^{m \times m}$ have elements
lyche-numerical-linear-algebra_F00930	73	■ 1.000	$s_{\{i j\}}$	s_{ij}	s_{ij} given by
lyche-numerical-linear-algebra_F00931	73	■ 1.000	$\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$	$\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$	Show that $\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$ and conclude that $\boldsymbol{T}^{-1} = \boldsymbol{S}$.
lyche-numerical-linear-algebra_F00932	73	■ 1.000	$\boldsymbol{T}^{-1} = \boldsymbol{S}$	$\boldsymbol{T}^{-1} = \boldsymbol{S}$	Show that $\boldsymbol{S} \boldsymbol{T} = \boldsymbol{I}$ and conclude that $\boldsymbol{T}^{-1} = \boldsymbol{S}$.
lyche-numerical-linear-algebra_F00933	73	■ 1.000	$a_{\{i j\}} = \boldsymbol{e}_{\{i\}}^T \boldsymbol{A} \boldsymbol{e}_{\{j\}}$	$a_{ij} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j$	that $a_{ij} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_j$ for all i, j .
lyche-numerical-linear-algebra_F00934	73	■ 1.000	$\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_i \boldsymbol{e}_j^T$	$\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_i \boldsymbol{e}_j^T$	show that $\boldsymbol{A} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} \boldsymbol{e}_i \boldsymbol{e}_j^T$.
lyche-numerical-linear-algebra_F00935	73	■ 0.942	$\boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{A}^T \boldsymbol{A}$	Exercise 2.18 (The Product $\boldsymbol{A}^T \boldsymbol{A}$) Let $\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$. Explain why this product is

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lyche-numerical-linear-algebra_F00936	73	■ 0.942	$\boldsymbol{B}=\boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$	Exercise 2.18 (The Product $\boldsymbol{A}^T \boldsymbol{A}$) Let $\boldsymbol{B} = \boldsymbol{A}^T \boldsymbol{A}$. Explain why this product is
lyche-numerical-linear-algebra_F00937	73	■ 1.000	$b_{ij}=\boldsymbol{a}_{:i}^T \boldsymbol{a}_{:j}$	$b_{ij} = \boldsymbol{a}_{:i}^T \boldsymbol{a}_{:j}$	defined for any matrix $\boldsymbol{A}$. Show that $b_{ij} = \boldsymbol{a}_{:i}^T \boldsymbol{a}_{:j}$ for all i, j .
lyche-numerical-linear-algebra_F00938	73	■ 1.000	$\boldsymbol{B} \in \mathbb{R}^{p \times n}$	$\boldsymbol{B} \in \mathbb{R}^{p \times n}$	Exercise 2.19 (Outer Product Expansion) For $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{B} \in \mathbb{R}^{p \times n}$ show
lyche-numerical-linear-algebra_F00939	73	■ 1.000	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$, and $\boldsymbol{X} \in \mathbb{R}^{n \times p}$. Show that
lyche-numerical-linear-algebra_F00940	73	■ 1.000	$\boldsymbol{X} \in \mathbb{R}^{n \times p}$	$\boldsymbol{X} \in \mathbb{R}^{n \times p}$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{B} \in \mathbb{R}^{m \times p}$, and $\boldsymbol{X} \in \mathbb{R}^{n \times p}$. Show that
lyche-numerical-linear-algebra_F00941	74	■ 0.989	$\boldsymbol{B} =$	$\boldsymbol{B} =$	Exercise 2.21 (Block Multiplication Example) Suppose $\boldsymbol{A} = [\boldsymbol{A}_1, \boldsymbol{A}_2]$ and $\boldsymbol{B} =$
lyche-numerical-linear-algebra_F00942	74	■ 1.000	$\left[\begin{array}{c} \boldsymbol{B}_1 \\ \mathbf{0} \end{array}\right]$	$\left[\begin{array}{c} \boldsymbol{B}_1 \\ \mathbf{0} \end{array}\right]$	$\left[\begin{array}{c} \boldsymbol{B}_1 \\ \mathbf{0} \end{array}\right]$. When is $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{A}_1\boldsymbol{B}_1$?
lyche-numerical-linear-algebra_F00943	74	■ 1.000	$\boldsymbol{A} \boldsymbol{B} = \boldsymbol{A}_1 \boldsymbol{B}_1$	$\boldsymbol{A}\boldsymbol{B} = \boldsymbol{A}_1\boldsymbol{B}_1$	$\left[\begin{array}{c} \boldsymbol{B}_1 \\ \mathbf{0} \end{array}\right]$. When is $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{A}_1\boldsymbol{B}_1$?
lyche-numerical-linear-algebra_F00944	74	■ 0.999	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in$	$\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in$	Exercise 2.22 (Another Block Multiplication Example) Suppose $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} \in$
lyche-numerical-linear-algebra_F00945	74	■ 1.000	$\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1 \in \mathbb{R}^{(n-1) \times (n-1)}$	$\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1 \in \mathbb{R}^{(n-1) \times (n-1)}$	where $\boldsymbol{A}_1, \boldsymbol{B}_1, \boldsymbol{C}_1 \in \mathbb{R}^{(n-1) \times (n-1)}$. Show that
lyche-numerical-linear-algebra_F00946	74	■ 0.921	$\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$?	$\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$?	$\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$?
lyche-numerical-linear-algebra_F00947	74	■ 1.000	$\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$	2.7.6 Is the matrix $\left[\begin{array}{cc} 1 & 1+i \\ 1+i & 2 \end{array}\right]$ Hermitian? Symmetric?
lyche-numerical-linear-algebra_F00948	75	■ 1.000	$\boldsymbol{P}$	$\boldsymbol{P}$	factorization, if row interchanges are introduced. Here $\boldsymbol{P}$ is a permutation matrix, $\boldsymbol{L}$
lyche-numerical-linear-algebra_F00949	75	■ 0.951	3×3	3×3	linear equations in n unknowns. ¹ We first recall how it works on a 3×3 system.
lyche-numerical-linear-algebra_F00950	76	■ 1.000	$a_{11}^{(1)} \neq 0$	$a_{11}^{(1)} \neq 0$	To solve this system by Gaussian elimination suppose $a_{11}^{(1)} \neq 0$. We subtract $l_{21}^{(1)} :=$
lyche-numerical-linear-algebra_F00951	76	■ 1.000	$l_{21}^{(1)} :=$	$l_{21}^{(1)} :=$	To solve this system by Gaussian elimination suppose $a_{11}^{(1)} \neq 0$. We subtract $l_{21}^{(1)} :=$
lyche-numerical-linear-algebra_F00952	76	■ 0.997	$a_{21}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)} / a_{11}^{(1)}$ times equation I from equation II and $l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$ times equation I
lyche-numerical-linear-algebra_F00953	76	■ 0.997	$l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$	$l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$	$a_{21}^{(1)} / a_{11}^{(1)}$ times equation I from equation II and $l_{31}^{(1)} := a_{31}^{(1)} / a_{11}^{(1)}$ times equation I
lyche-numerical-linear-algebra_F00954	76	■ 1.000	$b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$	$b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00955	76	■ 1.000	$i = 2, 3$	$i = 2, 3$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00956	76	■ 1.000	$a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$	$a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00957	76	■ 1.000	$i, j = 2, 3$	$i, j = 2, 3$	where $b_i^{(2)} = b_i^{(1)} - l_{i1}^{(1)} b_1^{(1)}$ for $i = 2, 3$ and $a_{ij}^{(2)} = a_{ij}^{(1)} - l_{i1}^{(1)} a_{1j}^{(1)}$ for $i, j = 2, 3$.
lyche-numerical-linear-algebra_F00958	76	■ 1.000	$a_{11}^{(1)} = 0$	$a_{11}^{(1)} = 0$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$
lyche-numerical-linear-algebra_F00959	76	■ 1.000	$a_{21}^{(1)} \neq 0$	$a_{21}^{(1)} \neq 0$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$

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lyche-numerical-linear-algebra_F00960	76	■ 1.000	$a_{11}^{(1)} =$	$a_{11}^{(1)} =$	If $a_{11}^{(1)} = 0$ and $a_{21}^{(1)} \neq 0$ we first interchange equation I and equation II. If $a_{11}^{(1)} =$
lyche-numerical-linear-algebra_F00961	76	■ 0.997	$a_{21}^{(1)} = 0$	$a_{21}^{(1)} = 0$	$a_{21}^{(1)} = 0$ we interchange equation I and III. Since the system is nonsingular the first
lyche-numerical-linear-algebra_F00962	76	■ 0.705	$a_{22}^{(2)} \neq 0$	$a_{22}^{(2)} \neq 0$	If $a_{22}^{(2)} \neq 0$ we subtract $l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$ times equation II' from equation III' to
lyche-numerical-linear-algebra_F00963	76	■ 0.705	$l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$	$l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$	If $a_{22}^{(2)} \neq 0$ we subtract $l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$ times equation II' from equation III' to
lyche-numerical-linear-algebra_F00964	76	■ 0.705	$\{ \}^{\prime}$	$'$	If $a_{22}^{(2)} \neq 0$ we subtract $l_{32}^{(2)} := a_{32}^{(2)} / a_{22}^{(2)}$ times equation II' from equation III' to
lyche-numerical-linear-algebra_F00965	76	■ 1.000	$a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$	$a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00966	76	■ 1.000	$b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$	$b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00967	76	■ 1.000	$a_{22}^{(2)} = 0$	$a_{22}^{(2)} = 0$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00968	76	■ 1.000	$a_{32}^{(2)} \neq 0$	$a_{32}^{(2)} \neq 0$	where $a_{33}^{(3)} = a_{33}^{(2)} - l_{32}^{(2)} a_{23}^{(2)}$ and $b_3^{(3)} = b_3^{(2)} - l_{32}^{(2)} b_2^{(2)}$. If $a_{22}^{(2)} = 0$ then $a_{32}^{(2)} \neq 0$
lyche-numerical-linear-algebra_F00969	76	■ 1.000	$a_{kk}^{(k)} \neq 0, k=1,2$	$a_{kk}^{(k)} \neq 0, k=1,2$	Indeed, if $a_{kk}^{(k)} \neq 0, k=1,2$ then
lyche-numerical-linear-algebra_F00970	77	■ 1.000	$\boldsymbol{A}^{(1)}$	$\boldsymbol{A}^{(1)}$	$\boldsymbol{A}^{(1)}$ (cf. the proof of Theorem 3.2).
lyche-numerical-linear-algebra_F00971	77	■ 1.000	$\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$	$\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ and generate a sequence of equivalent systems $\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$ for $k =$
lyche-numerical-linear-algebra_F00972	77	■ 1.000	$k =$	$k =$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ and generate a sequence of equivalent systems $\boldsymbol{A}^{(k)} \boldsymbol{x} = \boldsymbol{b}^{(k)}$ for $k =$
lyche-numerical-linear-algebra_F00973	77	■ 1.000	$1, \ldots, n$	$1, \ldots, n$	$1, \ldots, n$, where $\boldsymbol{A}^{(1)} = \boldsymbol{A}$, $\boldsymbol{b}^{(1)} = \boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00974	77	■ 1.000	$\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$	$\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$	$1, \ldots, n$, where $\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00975	77	■ 1.000	$\boldsymbol{A}^{(k)}$	$\boldsymbol{A}^{(k)}$	$1, \ldots, n$, where $\boldsymbol{A}^{(1)} = \boldsymbol{A}, \boldsymbol{b}^{(1)} = \boldsymbol{b}$, and $\boldsymbol{A}^{(k)}$ has zeros under the diagonal in its
lyche-numerical-linear-algebra_F00976	77	■ 1.000	$k-1$	$k-1$	first $k-1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00977	77	■ 1.000	$\boldsymbol{A}^{(n)}$	$\boldsymbol{A}^{(n)}$	first $k-1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00978	77	■ 1.000	$\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$	$\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$	first $k-1$ columns. Thus $\boldsymbol{A}^{(n)}$ is upper triangular and the system $\boldsymbol{A}^{(n)} \boldsymbol{x} = \boldsymbol{b}^{(n)}$ is
lyche-numerical-linear-algebra_F00979	77	■ 1.000	$\boldsymbol{A}^{(k+1)}$	$\boldsymbol{A}^{(k+1)}$	The process transforming $\boldsymbol{A}^{(k)}$ into $\boldsymbol{A}^{(k+1)}$ for $k=1, \ldots, n-1$ can be described
lyche-numerical-linear-algebra_F00980	78	■ 1.000	$j=k$	$j=k$	For $j=k$ it follows from (3.2) that $a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kk}^{(k)} = 0$ for $i =$
lyche-numerical-linear-algebra_F00981	78	■ 1.000	$a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kk}^{(k)} = 0$	$a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kk}^{(k)} = 0$	For $j=k$ it follows from (3.2) that $a_{ik}^{(k+1)} = a_{ik}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kk}^{(k)} = 0$ for $i =$
lyche-numerical-linear-algebra_F00982	78	■ 1.000	$k+1, \ldots, n$	$k+1, \ldots, n$	$k+1, \ldots, n$. Thus $\boldsymbol{A}^{(k+1)}$ will have zeros under the diagonal in its first k columns
lyche-numerical-linear-algebra_F00983	78	■ 1.000	$l_{ik}^{(k)}$	$l_{ik}^{(k)}$	and the elimination is carried one step further. The numbers $l_{ik}^{(k)}$ in (3.2) are called

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lyche-numerical-linear-algebra_F00984	78	■ 1.000	$a_{k\ k}^{\{(k)\}}$	$a_{kk}^{(k)}$	$a_{kk}^{(k)}$ are nonzero for $k = 1, \dots, n - 1$. This depends on certain submatrices of A
lyche-numerical-linear-algebra_F00985	78	■ 1.000	$\boldsymbol{A}_{\{[k]\}} \in \mathbb{C}^{k \times k}$	$A_{[k]} \in \mathbb{C}^{k \times k}$	Definition 3.1 (Principal Submatrix) For $k = 1, \dots, n$ the matrices $A_{[k]} \in \mathbb{C}^{k \times k}$
lyche-numerical-linear-algebra_F00986	78	■ 0.997	$\boldsymbol{B} \in \mathbb{C}^{k \times k}$	$B \in \mathbb{C}^{k \times k}$	matrix $B \in \mathbb{C}^{k \times k}$ is called a principal submatrix of A if $B = A(r, r)$, where
lyche-numerical-linear-algebra_F00987	78	■ 0.997	$\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r})$	$B = A(r, r)$	matrix $B \in \mathbb{C}^{k \times k}$ is called a principal submatrix of A if $B = A(r, r)$, where
lyche-numerical-linear-algebra_F00988	78	■ 1.000	$\boldsymbol{r} = \left[r_1, \dots, r_k \right]$	$r = [r_1, \dots, r_k]$	$r = [r_1, \dots, r_k]$ for some $1 \leq r_1 < \dots < r_k \leq n$. Thus,
lyche-numerical-linear-algebra_F00989	78	■ 1.000	$1 \leq r_1 < \dots < r_k \leq n$	$1 \leq r_1 < \dots < r_k \leq n$	$r = [r_1, \dots, r_k]$ for some $1 \leq r_1 < \dots < r_k \leq n$. Thus,
lyche-numerical-linear-algebra_F00990	78	■ 1.000	$r_{\{j\}} = j$	$r_j = j$	A principal submatrix is leading if $r_j = j$ for $j = 1, \dots, k$. Also a principal
lyche-numerical-linear-algebra_F00991	78	■ 1.000	$j = 1, \dots, k$	$j = 1, \dots, k$	A principal submatrix is leading if $r_j = j$ for $j = 1, \dots, k$. Also a principal
lyche-numerical-linear-algebra_F00992	78	■ 1.000	$\boldsymbol{A} = \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$	$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	<i>Example 3.2 (Principal Submatrices)</i> The principal submatrices of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$
lyche-numerical-linear-algebra_F00993	78	■ 1.000	$a_{k\ k}^{\{(k)\}} \neq 0$	$a_{k,k}^{(k)} \neq 0$	Theorem 3.1 We have $a_{k,k}^{(k)} \neq 0$ for $k = 1, \dots, n - 1$ if and only if the leading
lyche-numerical-linear-algebra_F00994	78	■ 1.000	$\boldsymbol{A}_{\{[k]\}}$	$A_{[k]}$	principal submatrices $A_{[k]}$ of A are nonsingular for $k = 1, \dots, n - 1$. Moreover
lyche-numerical-linear-algebra_F00995	78	■ 1.000	$\boldsymbol{B}_k = \boldsymbol{A}_{\{k-1\}}^{\{(k)\}}$	$B_k = A_{k-1}^{(k)}$	Proof Let $B_k = A_{k-1}^{(k)}$ be the upper left $k - 1$ corner of $A^{(k)}$ given by (3.1). Observe
lyche-numerical-linear-algebra_F00996	79	■ 1.000	$\boldsymbol{B}_k$	B_k	that the elements of B_k are computed from A by using only elements from $A_{[k-1]}$.
lyche-numerical-linear-algebra_F00997	79	■ 1.000	$\boldsymbol{A}_{\{[k-1]\}}$	$A_{[k-1]}$	that the elements of B_k are computed from A by using only elements from $A_{[k-1]}$.
lyche-numerical-linear-algebra_F00998	79	■ 1.000	$\boldsymbol{B}_{k+1}$	B_{k+1}	of diagonal elements of B_{k+1} and (3.3) follows. But then $a_{11}^{(1)} \dots a_{kk}^{(k)} \neq 0$ for $k =$
lyche-numerical-linear-algebra_F00999	79	■ 1.000	$a_{11}^{\{(1)\}} \cdots a_{kk}^{\{(k)\}} \neq 0$	$a_{11}^{(1)} \dots a_{kk}^{(k)} \neq 0$	of diagonal elements of B_{k+1} and (3.3) follows. But then $a_{11}^{(1)} \dots a_{kk}^{(k)} \neq 0$ for $k =$
lyche-numerical-linear-algebra_F01000	79	■ 1.000	$1, \dots, n-1$	$1, \dots, n - 1$	$1, \dots, n - 1$ if and only if $\det(A_{[k]}) \neq 0$ for $k = 1, \dots, n - 1$, or equivalently $A_{[k]}$
lyche-numerical-linear-algebra_F01001	79	■ 1.000	$\operatorname{det}(\boldsymbol{A}_{\{[k]\}}) \neq 0$	$\det(A_{[k]}) \neq 0$	$1, \dots, n - 1$ if and only if $\det(A_{[k]}) \neq 0$ for $k = 1, \dots, n - 1$, or equivalently $A_{[k]}$
lyche-numerical-linear-algebra_F01002	79	■ 1.000	$l_{ij}^{\{(j)\}}$	$l_{ij}^{(j)}$	where the $l_{ij}^{(j)}$ and $a_{ij}^{(i)}$ are given by (3.2).
lyche-numerical-linear-algebra_F01003	79	■ 1.000	$a_{ij}^{\{(i)\}}$	$a_{ij}^{(i)}$	where the $l_{ij}^{(j)}$ and $a_{ij}^{(i)}$ are given by (3.2).
lyche-numerical-linear-algebra_F01004	79	■ 1.000	$i \leq j$	$i \leq j$	Thus for $i \leq j$ we find
lyche-numerical-linear-algebra_F01005	79	■ 1.000	$a_{nn}^{\{(n)\}} = 0$	$a_{nn}^{(n)} = 0$	matrix U is then singular, and we must have $a_{nn}^{(n)} = 0$ when A is singular.
lyche-numerical-linear-algebra_F01006	80	■ 1.000	$\boldsymbol{L} \boldsymbol{U} \boldsymbol{x} = \boldsymbol{b}$	$LUx = b$	Since $LUx = b$ we have $Ly = b$, where $y := Ux$. We first solve $Ly = b$, for y
lyche-numerical-linear-algebra_F01007	80	■ 1.000	$\boldsymbol{L} \boldsymbol{y} = \boldsymbol{b}$	$Ly = b$	Since $LUx = b$ we have $Ly = b$, where $y := Ux$. We first solve $Ly = b$, for y
lyche-numerical-linear-algebra_F01008	80	■ 1.000	$\boldsymbol{y} := \boldsymbol{U} \boldsymbol{x}$	$y := Ux$	Since $LUx = b$ we have $Ly = b$, where $y := Ux$. We first solve $Ly = b$, for y

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lyche-numerical-linear-algebra-F01009	80	1.000	$\backslash\boldsymbol{\mathrm{U}} \ \backslash\boldsymbol{\mathrm{x}}=\backslash\boldsymbol{\mathrm{y}}$	$Ux = y$	and then $Ux = y$ for x .
lyche-numerical-linear-algebra-F01010	80	0.999	$2.5 \ \backslash\boldsymbol{\mathrm{A}}$	$2.5A$	A nonsingular triangular linear system $Ax = b$ is easy to solve. By Lemma 2.5 A
lyche-numerical-linear-algebra-F01011	80	1.000	$x_{\{1\}}=b_{\{1\}} \ / \ a_{\{11\}}$	$x_1 = b_1/a_{11}$	From the first equation we find $x_1 = b_1/a_{11}$. Solving the second equation for x_2 we
lyche-numerical-linear-algebra-F01012	80	1.000	$x_{\{2\}}$	x_2	From the first equation we find $x_1 = b_1/a_{11}$. Solving the second equation for x_2 we
lyche-numerical-linear-algebra-F01013	80	0.999	$x_{\{2\}}=\backslash\left(b_{\{2\}}-a_{\{21\}} \ x_{\{1\}}\backslash\right) \ / \ a_{\{22\}}$	$x_2 = (b_2 - a_{21}x_1)/a_{22}$	obtain $x_2 = (b_2 - a_{21}x_1)/a_{22}$. Finally the third equation gives $x_3 = (b_3 - a_{31}x_1 -$
lyche-numerical-linear-algebra-F01014	80	0.999	$x_{\{3\}}=\backslash\left(b_{\{3\}}-a_{\{31\}} \ x_{\{1\}}-\backslash\right).$	$x_3 = (b_3 - a_{31}x_1 -$	obtain $x_2 = (b_2 - a_{21}x_1)/a_{22}$. Finally the third equation gives $x_3 = (b_3 - a_{31}x_1 -$
lyche-numerical-linear-algebra-F01015	80	1.000	$\backslash\left.a_{\{32\}} \ x_{\{2\}}\backslash\right) \ / \ a_{\{33\}}$	$a_{32}x_2)/a_{33}$	$a_{32}x_2)/a_{33}$. This process is known as forward substitution. In general
lyche-numerical-linear-algebra-F01016	80	1.000	$a_{\{k, \ j\}}=0$	$a_{k,j} = 0$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k + 1, \dots, k$ for $k = 1, 2, \dots, n$, and where
lyche-numerical-linear-algebra-F01017	80	1.000	$j \ \backslash\mathrm{notin}\backslash\left\{\backslash l_{\{k\}}, \ \backslash l_{\{k\}}+1, \ \backslash\mathrm{ldots}, \ k\backslash\right).$	$j \notin \{l_k, l_k + 1, \dots, k$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k + 1, \dots, k$ for $k = 1, 2, \dots, n$, and where
lyche-numerical-linear-algebra-F01018	80	1.000	$k=1,2, \ \backslash\mathrm{ldots}, \ n$	$k = 1, 2, \dots, n$	so that $a_{k,j} = 0$ for $j \notin \{l_k, l_k + 1, \dots, k$ for $k = 1, 2, \dots, n$, and where
lyche-numerical-linear-algebra-F01019	80	1.000	$\backslash l_{\{k\}}:=\backslash\mathrm{max} \ (1, \ k-d)$	$l_k := \max(1, k - d)$	$l_k := \max(1, k - d)$, see Fig. 3.2. For a lower triangular d -band matrix the
lyche-numerical-linear-algebra-F01020	80	1.000	$d=n$	$d = n$	Note that (3.8) reduces to (3.7) if $d = n$. Letting $A(k, l_k : (k - 1)) * x(l_k : (k - 1))$
lyche-numerical-linear-algebra-F01021	80	1.000	$A\backslash\left(k, \ \backslash l_{\{k\}}:(k-1)\backslash\right) \ * \ x\backslash\left(\backslash l_{\{k\}}:(k-1)\backslash\right)$	$A(k, l_k : (k - 1)) * x(l_k : (k - 1))$	Note that (3.8) reduces to (3.7) if $d = n$. Letting $A(k, l_k : (k - 1)) * x(l_k : (k - 1))$
lyche-numerical-linear-algebra-F01022	80	0.995	$\backslash\mathrm{sum}_{\{j=\backslash l_{\{k\}}\}^{\{k-1\}}} \ a_{\{k \ j\}} \ x_{\{j\}}$	$\sum_{j=l_k}^{k-1} a_{kj}x_j$	denote the sum $\sum_{j=l_k}^{k-1} a_{kj}x_j$ we arrive at the following algorithm, where the initial
lyche-numerical-linear-algebra-F01023	81	1.000	$x_{\{n\}}$	x_n	substitution or 'bottom-up'. We first find x_n from the last equation and then move
lyche-numerical-linear-algebra-F01024	81	1.000	$\backslash\boldsymbol{\mathrm{A}} \ \backslash\mathrm{in} \ \backslash\mathrm{mathbb{C}}^{\{n \ \mathrm{times} \ n\}}, \ \backslash\boldsymbol{\mathrm{b}} \ \backslash\mathrm{in} \ \backslash\mathrm{mathbb{C}}^{\{n\}}$	$A \in \mathbb{C}^{n \times n}, b \in \mathbb{C}^n$	banded matrix $A \in \mathbb{C}^{n \times n}, b \in \mathbb{C}^n$ and d , as input, and returns an $x \in \mathbb{C}^n$ so
lyche-numerical-linear-algebra-F01025	81	1.000	$n-k+1$	$n - k + 1$	This system is of order $n - k + 1$. The unknowns are $x_k, \dots, x_n$.
lyche-numerical-linear-algebra-F01026	81	1.000	$x_{\{k\}}, \ \backslash\mathrm{ldots}, \ x_{\{n\}}$	$x_k, \dots, x_n$	This system is of order $n - k + 1$. The unknowns are $x_k, \dots, x_n$.
lyche-numerical-linear-algebra-F01027	82	1.000	$x_{\{k\}}=b_{\{k\}} \ / \ a_{\{k, \ k\}}$	$x_k = b_k/a_{k,k}$	We see that $x_k = b_k/a_{k,k}$ and eliminating x_k from the remaining equations we
lyche-numerical-linear-algebra-F01028	82	1.000	$x_{\{k\}}$	x_k	We see that $x_k = b_k/a_{k,k}$ and eliminating x_k from the remaining equations we
lyche-numerical-linear-algebra-F01029	82	1.000	$n-k$	$n - k$	obtain a system of order $n - k$ with unknowns $x_{k+1}, \dots, x_n$
lyche-numerical-linear-algebra-F01030	82	1.000	$x_{\{k+1\}}, \ \backslash\mathrm{ldots}, \ x_{\{n\}}$	$x_{k+1}, \dots, x_n$	obtain a system of order $n - k$ with unknowns $x_{k+1}, \dots, x_n$
lyche-numerical-linear-algebra-F01031	82	1.000	$k=1,2, \ \backslash\mathrm{ldots} \ n$	$k = 1, 2, \dots, n$	Thus at the k th step, $k = 1, 2, \dots, n$ we set $x_k = b_k/A(k, k)$ and update b as follows:
lyche-numerical-linear-algebra-F01032	82	1.000	$x_{\{k\}}=b_{\{k\}} \ / \ A(k, \ k)$	$x_k = b_k/A(k, k)$	Thus at the k th step, $k = 1, 2, \dots, n$ we set $x_k = b_k/A(k, k)$ and update b as follows:
lyche-numerical-linear-algebra-F01033	82	1.000	b	b	Thus at the k th step, $k = 1, 2, \dots, n$ we set $x_k = b_k/A(k, k)$ and update b as follows:
lyche-numerical-linear-algebra-F01034	82	1.000	$M, \ D, \ A, \ S$	M, D, A, S	operations. Let M, D, A, S be the number of (complex) multiplications, divisions,
lyche-numerical-linear-algebra-F01035	82	1.000	$a_{\{i \ j\}}^{\{k+1\}}=a_{\{i \ j\}}^{\{(k)\}}-\backslash l_{\{i \ k\}}^{\{(k)\}} \ a_{\{k \ j\}}^{\{(k)\}}$	$a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$	calculation of $a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$ which is carried out $(n - k)^2$ times. Moreover,

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lyche-numerical-linear-algebra-F01036	82	1.000	$(n-k)^2$	$(n-k)^2$	calculation of $a_{ij}^{k+1} = a_{ij}^{(k)} - l_{ik}^{(k)} a_{kj}^{(k)}$ which is carried out $(n-k)^2$ times. Moreover,
lyche-numerical-linear-algebra-F01037	83	1.000	$M+$	$M+$	each calculation involves one subtraction and one multiplication. Thus we find $M+$
lyche-numerical-linear-algebra-F01038	83	1.000	$S=2 \sum_{k=1}^{n-1} (n-k)^2 = 2 \sum_{m=1}^{n-1} m^2 = \frac{2}{3} n(n-1) \left(n - \frac{1}{2} \right)$	$S = 2 \sum_{k=1}^{n-1} (n-k)^2 = 2 \sum_{m=1}^{n-1} m^2 = \frac{2}{3} n(n-1) \left(n - \frac{1}{2} \right)$	$S = 2 \sum_{k=1}^{n-1} (n-k)^2 = 2 \sum_{m=1}^{n-1} m^2 = \frac{2}{3} n(n-1) \left(n - \frac{1}{2} \right)$ For each k there are $n-k$
lyche-numerical-linear-algebra-F01039	83	1.000	$\sum_{k=1}^{n-1} (n-k) = \frac{1}{2} n(n-1)$	$\sum_{k=1}^{n-1} (n-k) = \frac{1}{2} n(n-1)$	divisions giving a sum of $\sum_{k=1}^{n-1} (n-k) = \frac{1}{2} n(n-1)$. Since there are no additions
lyche-numerical-linear-algebra-F01040	83	1.000	N_{LU}	N_{LU}	We are only interested in N_{LU} when n is large and for such n the term $\frac{2}{3}n^3$
lyche-numerical-linear-algebra-F01041	83	1.000	$\frac{2}{3} n^3$	$\frac{2}{3} n^3$	We are only interested in N_{LU} when n is large and for such n the term $\frac{2}{3}n^3$
lyche-numerical-linear-algebra-F01042	83	1.000	G_n	G_n	the number G_n as follows:
lyche-numerical-linear-algebra-F01043	83	0.997	$3.2 \left(G_n := \frac{2}{3} n^3 \right)$	$3.2 \left(G_n := \frac{2}{3} n^3 \right)$	Definition 3.2 ($G_n := \frac{2}{3} n^3$) We define $G_n := \frac{2}{3} n^3$.
lyche-numerical-linear-algebra-F01044	83	0.997	$G_n := \frac{2}{3} n^3$	$G_n := \frac{2}{3} n^3$	Definition 3.2 ($G_n := \frac{2}{3} n^3$) We define $G_n := \frac{2}{3} n^3$.
lyche-numerical-linear-algebra-F01045	83	1.000	$2 n^3 / 3$	$2n^3/3$	There is a quick way to arrive at the leading term $2n^3/3$. We only consider the
lyche-numerical-linear-algebra-F01046	83	1.000	$M+S$	$M+S$	loop contributing to $M+S$. Then we replace sums by integrals letting the summation
lyche-numerical-linear-algebra-F01047	83	1.000	N_S	N_S	Consider next N_S , the number of forward plus backward substitutions. By (3.7)
lyche-numerical-linear-algebra-F01048	83	1.000	$n=10^6$	$n = 10^6$	system requires $O(n^2)$ arithmetic operations. Thus, if $n = 10^6$ and one arith-
lyche-numerical-linear-algebra-F01049	83	1.000	$c=10^{-14}$	$c = 10^{-14}$	metic operation requires $c = 10^{-14}$ seconds of computing time then $cn^3 =$
lyche-numerical-linear-algebra-F01050	83	1.000	$c n^3 =$	$cn^3 =$	metic operation requires $c = 10^{-14}$ seconds of computing time then $cn^3 =$
lyche-numerical-linear-algebra-F01051	83	1.000	10^4	10^4	10^4 seconds ≈ 3 hours and $cn^2 = 0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra-F01052	83	1.000	≈ 3	≈ 3	10^4 seconds ≈ 3 hours and $cn^2 = 0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra-F01053	83	1.000	$c n^2=0.01$	$cn^2 = 0.01$	10^4 seconds ≈ 3 hours and $cn^2 = 0.01$ second, giving dramatic differences in
lyche-numerical-linear-algebra-F01054	84	1.000	$\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{l} 1 \\ 1 \end{array} \right]$	$\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right] = \left[\begin{array}{l} 1 \\ 1 \end{array} \right]$	nonsingular system. A simple example is $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. We show here that any
lyche-numerical-linear-algebra-F01055	84	0.980	(k, k)	(k, k)	as pivoting . The element which is moved to the diagonal position (k, k) is called
lyche-numerical-linear-algebra-F01056	84	1.000	$r_k \geq k$	$r_k \geq k$	1. Choose $r_k \geq k$ so that $a_{r_k,k}^{(k)} \neq 0$.
lyche-numerical-linear-algebra-F01057	84	1.000	$a_{r_k,k}^{(k)} \neq 0$	$a_{r_k,k}^{(k)} \neq 0$	1. Choose $r_k \geq k$ so that $a_{r_k,k}^{(k)} \neq 0$.
lyche-numerical-linear-algebra-F01058	84	1.000	r_k	r_k	2. Interchange rows r_k and k of $A^{(k)}$.
lyche-numerical-linear-algebra-F01059	84	1.000	$a_{ij}^{(k+1)}$	$a_{ij}^{(k+1)}$	3. Eliminate by computing $l_{ik}^{(k)}$ and $a_{ij}^{(k+1)}$ using (3.2).

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lyche-numerical-linear-algebra-F01060	84	1.000	$\boldsymbol{A}^{(1)}=\boldsymbol{A}$	$\boldsymbol{A}^{(1)} = \boldsymbol{A}$	$\boldsymbol{A}^{(1)} = \boldsymbol{A}$ this holds for $k = 1$. By Lemma 2.4 the lower right diagonal block in
lyche-numerical-linear-algebra-F01061	84	1.000	$\boldsymbol{e}_{i_1}, \dots, \boldsymbol{e}_{i_n}$	$\boldsymbol{e}_{i_1}, \dots, \boldsymbol{e}_{i_n}$	where $\boldsymbol{e}_{i_1}, \dots, \boldsymbol{e}_{i_n}$ is a permutation of the unit vectors $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n \in \mathbb{R}^n$.
lyche-numerical-linear-algebra-F01062	84	1.000	$\boldsymbol{e}_{i_1}, \dots, \boldsymbol{e}_{i_n} \in \mathbb{R}^n$	$\boldsymbol{e}_1, \dots, \boldsymbol{e}_n \in \mathbb{R}^n$	where $\boldsymbol{e}_{i_1}, \dots, \boldsymbol{e}_{i_n}$ is a permutation of the unit vectors $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n \in \mathbb{R}^n$.
lyche-numerical-linear-algebra-F01063	84	1.000	$\boldsymbol{p}=[i_1, \dots, i_n]^T$	$\boldsymbol{p} = [i_1, \dots, i_n]^T$	Every permutation $\boldsymbol{p} = [i_1, \dots, i_n]^T$ of the integers $1, 2, \dots, n$ gives rise to a
lyche-numerical-linear-algebra-F01064	85	0.997	$\boldsymbol{A} \boldsymbol{P} = \left(\boldsymbol{A} \boldsymbol{e}_{i_1}, \dots, \boldsymbol{A} \boldsymbol{e}_{i_n} \right) = \boldsymbol{A}(:, \boldsymbol{p})$	$\boldsymbol{A} \boldsymbol{P} = (\boldsymbol{A} \boldsymbol{e}_{i_1}, \dots, \boldsymbol{A} \boldsymbol{e}_{i_n}) = \boldsymbol{A}(:, \boldsymbol{p})$	Indeed, $\boldsymbol{A} \boldsymbol{P} = (\boldsymbol{A} \boldsymbol{e}_{i_1}, \dots, \boldsymbol{A} \boldsymbol{e}_{i_n}) = \boldsymbol{A}(:, \boldsymbol{p})$ and $\boldsymbol{P}^T \boldsymbol{A} = (\boldsymbol{A}^T \boldsymbol{P})^T = (\boldsymbol{A}^T ($
lyche-numerical-linear-algebra-F01065	85	0.997	$\boldsymbol{P}^T \boldsymbol{A} = (\boldsymbol{A}^T \boldsymbol{P})^T = (\boldsymbol{A}^T ($	$\boldsymbol{P}^T \boldsymbol{A} = (\boldsymbol{A}^T \boldsymbol{P})^T = (\boldsymbol{A}^T ($	Indeed, $\boldsymbol{A} \boldsymbol{P} = (\boldsymbol{A} \boldsymbol{e}_{i_1}, \dots, \boldsymbol{A} \boldsymbol{e}_{i_n}) = \boldsymbol{A}(:, \boldsymbol{p})$ and $\boldsymbol{P}^T \boldsymbol{A} = (\boldsymbol{A}^T \boldsymbol{P})^T = (\boldsymbol{A}^T ($
lyche-numerical-linear-algebra-F01066	85	0.773	$\boldsymbol{P}^T \boldsymbol{A} = (\boldsymbol{A}^T \boldsymbol{P})^T = (\boldsymbol{A}^T ($	$\boldsymbol{P}^T \boldsymbol{A} = (\boldsymbol{A}^T \boldsymbol{P})^T = (\boldsymbol{A}^T ($	$\boldsymbol{P}^T \boldsymbol{A} = (\boldsymbol{A}^T \boldsymbol{P})^T = (\boldsymbol{A}^T ($
lyche-numerical-linear-algebra-F01067	85	1.000	$\boldsymbol{P}^T \boldsymbol{P} = \boldsymbol{I}$	$\boldsymbol{P}^T \boldsymbol{P} = \boldsymbol{I}$	Since $\boldsymbol{P}^T \boldsymbol{P} = \boldsymbol{I}$ the inverse of $\boldsymbol{P}$ is equal to its transpose, $\boldsymbol{P}^{-1} = \boldsymbol{P}^T$ and
lyche-numerical-linear-algebra-F01068	85	1.000	$\boldsymbol{P}^{-1} = \boldsymbol{P}^T$	$\boldsymbol{P}^{-1} = \boldsymbol{P}^T$	Since $\boldsymbol{P}^T \boldsymbol{P} = \boldsymbol{I}$ the inverse of $\boldsymbol{P}$ is equal to its transpose, $\boldsymbol{P}^{-1} = \boldsymbol{P}^T$ and
lyche-numerical-linear-algebra-F01069	85	1.000	$\boldsymbol{P} \boldsymbol{P}^T = \boldsymbol{I}$	$\boldsymbol{P} \boldsymbol{P}^T = \boldsymbol{I}$	$\boldsymbol{P} \boldsymbol{P}^T = \boldsymbol{I}$ as well. We will use a particularly simple permutation matrix.
lyche-numerical-linear-algebra-F01070	85	0.997	$\boldsymbol{I}_{jk}$	$\boldsymbol{I}_{jk}$	Definition 3.4 We define a (j, k) -Interchange matrix $\boldsymbol{I}_{jk}$ by interchanging col-
lyche-numerical-linear-algebra-F01071	85	0.997	$\boldsymbol{I}_{jk}$	$\boldsymbol{I}_{jk}$	Definition 3.4 We define a (j, k) -Interchange matrix $\boldsymbol{I}_{jk}$ by interchanging col-
lyche-numerical-linear-algebra-F01072	85	0.999	$\boldsymbol{I}_{jk} = \boldsymbol{I}_{kj}$	$\boldsymbol{I}_{jk} = \boldsymbol{I}_{kj}$	Since $\boldsymbol{I}_{jk} = \boldsymbol{I}_{kj}$, and we obtain the identity by applying $\boldsymbol{I}_{jk}$ twice, we see that
lyche-numerical-linear-algebra-F01073	85	1.000	$\boldsymbol{I}_{jk}^2 = \boldsymbol{I}$	$\boldsymbol{I}_{jk}^2 = \boldsymbol{I}$	$\boldsymbol{I}_{jk}^2 = \boldsymbol{I}$ and an interchange matrix is symmetric and equal to its own inverse. Pre-
lyche-numerical-linear-algebra-F01074	85	1.000	$\boldsymbol{p}_k$	$\boldsymbol{p}_k$	We can keep track of the row interchanges using pivot vectors $\boldsymbol{p}_k$. We define
lyche-numerical-linear-algebra-F01075	85	1.000	$\boldsymbol{p}_{k+1}$	$\boldsymbol{p}_{k+1}$	We obtain $\boldsymbol{p}_{k+1}$ from $\boldsymbol{p}_k$ by interchanging the entries r_k and k in $\boldsymbol{p}_k$. In particular,
lyche-numerical-linear-algebra-F01076	85	0.686	$\boldsymbol{P}_k := \boldsymbol{I}_{r_k, k}$	$\boldsymbol{P}_k := \boldsymbol{I}_{r_k, k}$	interchange matrices $\boldsymbol{P}_k := \boldsymbol{I}_{r_k, k}$. Since $\boldsymbol{P}_k \boldsymbol{I}(\boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{P}_k \boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{p}_{k+1}, :)$
lyche-numerical-linear-algebra-F01077	85	0.686	$\boldsymbol{P}_k \boldsymbol{I}(\boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{P}_k \boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{p}_{k+1}, :)$	$\boldsymbol{P}_k \boldsymbol{I}(\boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{P}_k \boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{p}_{k+1}, :)$	interchange matrices $\boldsymbol{P}_k := \boldsymbol{I}_{r_k, k}$. Since $\boldsymbol{P}_k \boldsymbol{I}(\boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{P}_k \boldsymbol{p}_k, :) = \boldsymbol{I}(\boldsymbol{p}_{k+1}, :)$
lyche-numerical-linear-algebra-F01078	85	1.000	$a_{ij}^{(1)} = a_{ij}$	$a_{ij}^{(1)} = a_{ij}$	pivoting starting with $a_{ij}^{(1)} = a_{ij}$ can be described as follows:
lyche-numerical-linear-algebra-F01079	86	0.994	$\boldsymbol{A} = \boldsymbol{P} \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{A} = \boldsymbol{P} \boldsymbol{L} \boldsymbol{U}$	$\mathbb{C}^{n \times n}$ leads to the factorization $\boldsymbol{A} = \boldsymbol{P} \boldsymbol{L} \boldsymbol{U}$, where $\boldsymbol{P}$ is a permutation matrix, $\boldsymbol{L}$
lyche-numerical-linear-algebra-F01080	86	0.976	$\boldsymbol{P} = \boldsymbol{I}(:, \boldsymbol{p})$	$\boldsymbol{P} = \boldsymbol{I}(:, \boldsymbol{p})$	explicitly, $\boldsymbol{P} = \boldsymbol{I}(:, \boldsymbol{p})$, where $\boldsymbol{p} = \boldsymbol{I}_{r_{n-1}, n-1} \cdots \boldsymbol{I}_{r_1, 1} [1, \dots, n]^T$, and
lyche-numerical-linear-algebra-F01081	86	0.976	$\boldsymbol{p} = \boldsymbol{I}_{r_{n-1}, n-1} \cdots \boldsymbol{I}_{r_1, 1} [1, \dots, n]^T$	$\boldsymbol{p} = \boldsymbol{I}_{r_{n-1}, n-1} \cdots \boldsymbol{I}_{r_1, 1} [1, \dots, n]^T$	explicitly, $\boldsymbol{P} = \boldsymbol{I}(:, \boldsymbol{p})$, where $\boldsymbol{p} = \boldsymbol{I}_{r_{n-1}, n-1} \cdots \boldsymbol{I}_{r_1, 1} [1, \dots, n]^T$, and
lyche-numerical-linear-algebra-F01082	86	1.000	$\boldsymbol{P}^T \boldsymbol{A} = \boldsymbol{L} \boldsymbol{U}$	$\boldsymbol{P}^T \boldsymbol{A} = \boldsymbol{L} \boldsymbol{U}$	The PLU factorization can also be written $\boldsymbol{P}^T \boldsymbol{A} = \boldsymbol{L} \boldsymbol{U}$. This shows that for a
lyche-numerical-linear-algebra-F01083	87	0.998	$\mathrm{fl}(x_1)$	$\mathrm{fl}(x_1)$	solutions $\mathrm{fl}(x_1)$ and $\mathrm{fl}(x_2)$ computed in this way is

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lyche-numerical-linear-algebra-F01084	87	■ 0.998	$\mathrm{fl}\left(x_2\right)$	$\mathrm{fl}\left(x_2\right)$	solutions $\mathrm{fl}(x_1)$ and $\mathrm{fl}(x_2)$ computed in this way is
lyche-numerical-linear-algebra-F01085	87	■ 1.000	x_1	x_1	The computed value 0 of x_1 is completely wrong. Suppose instead we apply
lyche-numerical-linear-algebra-F01086	88	■ 1.000	$\mathrm{fl}\left(x_1\right)=1$	$\mathrm{fl}\left(x_1\right)=1$	In this case rounding each calculation to three digits produces $\mathrm{fl}(x_1)=1$ and
lyche-numerical-linear-algebra-F01087	88	■ 1.000	$\mathrm{fl}\left(x_2\right)=2$	$\mathrm{fl}\left(x_2\right)=2$	$\mathrm{fl}(x_2)=2$ which is quite satisfactory since it is the exact solution rounded to three
lyche-numerical-linear-algebra-F01088	88	■ 1.000	s_k	s_k	coefficients of wildly different sizes. Note that the scaling factors s_k are computed
lyche-numerical-linear-algebra-F01089	88	■ 1.000	r_k, s_k	r_k, s_k	with r_k, s_k the smallest such indices in case of a tie, is known as complete pivoting .
lyche-numerical-linear-algebra-F01090	88	■ 0.967	$\mathbf{L U}$	LU	that $A = LU$ is an LU factorization of $A \in \mathbb{C}^{n \times n}$ if $L \in \mathbb{C}^{n \times n}$ is lower triangular
lyche-numerical-linear-algebra-F01091	88	■ 0.967	$\boldsymbol{L} \in \mathbb{C}^{n \times n}$	$L \in \mathbb{C}^{n \times n}$	that $A = LU$ is an LU factorization of $A \in \mathbb{C}^{n \times n}$ if $L \in \mathbb{C}^{n \times n}$ is lower triangular
lyche-numerical-linear-algebra-F01092	88	■ 1.000	$\boldsymbol{U} \in \mathbb{C}^{n \times n}$	$U \in \mathbb{C}^{n \times n}$	and $U \in \mathbb{C}^{n \times n}$ is upper triangular, i.e.,
lyche-numerical-linear-algebra-F01093	88	■ 0.963	n^2	n^2	To find an LU factorization there is one equation for each of the n^2 elements in A ,
lyche-numerical-linear-algebra-F01094	88	■ 1.000	n^2+n	n^2+n	and L and U contain a total of n^2+n unknown elements. There are several ways to
lyche-numerical-linear-algebra-F01095	89	■ 0.701	$\quad l_{ii}=1$	$l_{ii}=1$	L1U: $l_{ii}=1$ all i ,
lyche-numerical-linear-algebra-F01096	89	■ 0.992	$\quad u_{ii}=1$	$u_{ii}=1$	LU1: $u_{ii}=1$ all i ,
lyche-numerical-linear-algebra-F01097	89	■ 0.998	$\quad \boldsymbol{A}=\boldsymbol{L} \boldsymbol{D}$ $\boldsymbol{U}, l_{ii}=u_{ii}=1$	$A = LDU, l_{ii} = u_{ii} = 1$	LDU: $A = LDU, l_{ii} = u_{ii} = 1$ all i , $D = \mathrm{diag}(d_{11}, \dots, d_{nn})$.
lyche-numerical-linear-algebra-F01098	89	■ 0.998	$i, \boldsymbol{D}=\mathrm{diag}\left(d_{11}, \dots, \right.$ $\left. \dots, d_{nn}\right)$	$i, D = \mathrm{diag}\left(d_{11}, \dots, d_{nn}\right)$	LDU: $A = LDU, l_{ii} = u_{ii} = 1$ all $i, D = \mathrm{diag}(d_{11}, \dots, d_{nn})$.
lyche-numerical-linear-algebra-F01099	89	■ 0.889	$a, b, c, d \in \mathbb{C}$	$a, b, c, d \in \mathbb{C}$	<i>Example 3.5 (L1U of 2×2 Matrix)</i> Let $a, b, c, d \in \mathbb{C}$. An L1U factorization of
lyche-numerical-linear-algebra-F01100	89	■ 0.855	$\boldsymbol{A}=\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]$	$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ must satisfy the equations
lyche-numerical-linear-algebra-F01101	89	■ 0.984	l_1	l_1	for the unknowns l_1 in L and u_1, u_2, u_3 in U . The equations are
lyche-numerical-linear-algebra-F01102	89	■ 0.984	u_1, u_2, u_3	u_1, u_2, u_3	for the unknowns l_1 in L and u_1, u_2, u_3 in U . The equations are
lyche-numerical-linear-algebra-F01103	89	■ 0.999	$a l_1=c$	$al_1=c$	equation $al_1=c$. There are essentially three cases
lyche-numerical-linear-algebra-F01104	89	■ 1.000	$a \neq 0$	$a \neq 0$	1. $a \neq 0$: The matrix has a unique L1U factorization.
lyche-numerical-linear-algebra-F01105	89	■ 1.000	$a=c=0$	$a=c=0$	2. $a=c=0$: The L1U factorization exists, but it is not unique. Any value for l_1
lyche-numerical-linear-algebra-F01106	89	■ 1.000	$a=0, c \neq 0$	$a=0, c \neq 0$	3. $a=0, c \neq 0$: No L1U factorization exists.
lyche-numerical-linear-algebra-F01107	89	■ 1.000	$\boldsymbol{A}_4$	A_4	has no L1U factorization, A_3 has an L1U factorization but it is not unique, and A_4
lyche-numerical-linear-algebra-F01108	90	■ 1.000	$\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}, \boldsymbol{U}_{[k]}$	$A_{[k]}, L_{[k]}, U_{[k]}$	<i>L1U factorization of $A \in \mathbb{C}^{n \times n}$. For $k=1, \dots, n$ let $A_{[k]}, L_{[k]}, U_{[k]}$ be the leading</i>
lyche-numerical-linear-algebra-F01109	90	■ 0.986	$\boldsymbol{A}, \boldsymbol{L}, \boldsymbol{U}$	A, L, U	<i>principal submatrices of A, L, U, respectively. Then $A_{[k]} = L_{[k]}U_{[k]}$ is an L1U</i>
lyche-numerical-linear-algebra-F01110	90	■ 0.986	$\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]}$ $\boldsymbol{U}_{[k]}$	$A_{[k]} = L_{[k]}U_{[k]}$	<i>principal submatrices of A, L, U, respectively. Then $A_{[k]} = L_{[k]}U_{[k]}$ is an L1U</i>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01111	90	1.000	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{T}_k \in \mathbb{C}^{n-k, n-k}$	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{T}_k \in \mathbb{C}^{n-k, n-k}$	where $\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{T}_k \in \mathbb{C}^{n-k, n-k}$. Comparing blocks we find $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]}$. Since
lyche-numerical-linear-algebra-F01112	90	1.000	$\boldsymbol{L}_{[k]}$	$\boldsymbol{L}_{[k]}$	$\boldsymbol{L}_{[k]}$ is unit lower triangular and $\boldsymbol{U}_{[k]}$ is upper triangular this is an L1U factorization
lyche-numerical-linear-algebra-F01113	90	1.000	$\boldsymbol{U}_{[k]}$	$\boldsymbol{U}_{[k]}$	$\boldsymbol{L}_{[k]}$ is unit lower triangular and $\boldsymbol{U}_{[k]}$ is upper triangular this is an L1U factorization
lyche-numerical-linear-algebra-F01114	90	0.597	1×1	1×1	is clearly true for $n = 1$, since the unique L1U factorization of a 1×1 matrix is
lyche-numerical-linear-algebra-F01115	90	1.000	$\left[a_{11}\right] = [1] \left[a_{11}\right]$	$[a_{11}] = [1] [a_{11}]$	$[a_{11}] = [1][a_{11}]$. Suppose that $\boldsymbol{A}_{[n-1]}$ has a unique L1U factorization $\boldsymbol{A}_{[n-1]} =$
lyche-numerical-linear-algebra-F01116	90	1.000	$\boldsymbol{A}_{[n-1]}$	$\boldsymbol{A}_{[n-1]}$	$[a_{11}] = [1][a_{11}]$. Suppose that $\boldsymbol{A}_{[n-1]}$ has a unique L1U factorization $\boldsymbol{A}_{[n-1]} =$
lyche-numerical-linear-algebra-F01117	90	1.000	$\boldsymbol{A}_{[n-1]} =$	$\boldsymbol{A}_{[n-1]} =$	$[a_{11}] = [1][a_{11}]$. Suppose that $\boldsymbol{A}_{[n-1]}$ has a unique L1U factorization $\boldsymbol{A}_{[n-1]} =$
lyche-numerical-linear-algebra-F01118	90	0.992	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By block multiplication
lyche-numerical-linear-algebra-F01119	90	0.992	$\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$	$\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$	$\boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By block multiplication
lyche-numerical-linear-algebra-F01120	90	1.000	$\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	$\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra-F01121	90	1.000	$\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$	$\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra-F01122	90	1.000	$u_{nn} \in \mathbb{C}$	$u_{nn} \in \mathbb{C}$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{U}_{n-1}$ and $\boldsymbol{l}_n, \boldsymbol{u}_n \in \mathbb{C}^{n-1}$ and $u_{nn} \in \mathbb{C}$ are determined
lyche-numerical-linear-algebra-F01123	90	1.000	$\boldsymbol{L}_{n-1}$	$\boldsymbol{L}_{n-1}$	Since $\boldsymbol{A}_{[n-1]}$ is nonsingular it follows that $\boldsymbol{L}_{n-1}$ and $\boldsymbol{U}_{n-1}$ are nonsingular and
lyche-numerical-linear-algebra-F01124	90	1.000	$\boldsymbol{U}_{n-1}$	$\boldsymbol{U}_{n-1}$	Since $\boldsymbol{A}_{[n-1]}$ is nonsingular it follows that $\boldsymbol{L}_{n-1}$ and $\boldsymbol{U}_{n-1}$ are nonsingular and
lyche-numerical-linear-algebra-F01125	90	1.000	$\boldsymbol{l}_n, \boldsymbol{u}_n$	$\boldsymbol{l}_n, \boldsymbol{u}_n$	therefore $\boldsymbol{l}_n, \boldsymbol{u}_n$, and u_{nn} are uniquely given. Thus (3.17) gives a unique L1U
lyche-numerical-linear-algebra-F01126	90	1.000	u_{nn}	u_{nn}	therefore $\boldsymbol{l}_n, \boldsymbol{u}_n$, and u_{nn} are uniquely given. Thus (3.17) gives a unique L1U
lyche-numerical-linear-algebra-F01127	91	1.000	$k \leq n-1$	$k \leq n-1$	$\boldsymbol{A}_{[k]}$ is singular for some $k \leq n-1$. We will show that this leads to a contradiction.
lyche-numerical-linear-algebra-F01128	91	1.000	$\boldsymbol{A}_{[j]}$	$\boldsymbol{A}_{[j]}$	Let k be the smallest integer so that $\boldsymbol{A}_{[k]}$ is singular. Since $\boldsymbol{A}_{[j]}$ is nonsingular for
lyche-numerical-linear-algebra-F01129	91	0.916	$j \leq k-1$	$j \leq k-1$	$j \leq k-1$ it follows from what we have already shown that $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{U}_{[k]}$ is the
lyche-numerical-linear-algebra-F01130	91	1.000	$\boldsymbol{U}_{[k]}^T \boldsymbol{M}_k^T = \boldsymbol{C}_k^T$	$\boldsymbol{U}_{[k]}^T \boldsymbol{M}_k^T = \boldsymbol{C}_k^T$	and $\boldsymbol{L}_{[k]}$ is nonsingular. By (3.16) we have $\boldsymbol{U}_{[k]}^T \boldsymbol{M}_k^T = \boldsymbol{C}_k^T$. This can be written as
lyche-numerical-linear-algebra-F01131	91	1.000	$\boldsymbol{M}_k^T$	$\boldsymbol{M}_k^T$	$n-k$ linear systems for the columns of $\boldsymbol{M}_k^T$. By assumption $\boldsymbol{M}_k^T$ exists, but since
lyche-numerical-linear-algebra-F01132	91	1.000	$\boldsymbol{U}_{[k]}^T$	$\boldsymbol{U}_{[k]}^T$	$\boldsymbol{U}_{[k]}^T$ is singular $\boldsymbol{M}_k$ is not unique, a contradiction.
lyche-numerical-linear-algebra-F01133	91	1.000	$\boldsymbol{M}_k$	$\boldsymbol{M}_k$	$\boldsymbol{U}_{[k]}^T$ is singular $\boldsymbol{M}_k$ is not unique, a contradiction.
lyche-numerical-linear-algebra-F01134	91	1.000	$k=n$	$k=n$	By combining the last two equations in (3.18) we obtain with $k=n$
lyche-numerical-linear-algebra-F01135	91	1.000	$k-d$	$k-d$	is d -banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$
lyche-numerical-linear-algebra-F01136	91	1.000	$\boldsymbol{r}_k$	$\boldsymbol{r}_k$	is d -banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$
lyche-numerical-linear-algebra-F01137	91	1.000	$\boldsymbol{c}_k$	$\boldsymbol{c}_k$	is d -banded then the first $k-d$ components in $\boldsymbol{r}_k$ and $\boldsymbol{c}_k$ are zero so both $\boldsymbol{L}$ and $\boldsymbol{U}$

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lyche-numerical-linear-algebra_F01138	91	1.000	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$	$\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$	to solve the lower triangular system $\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$ for the k th row $\boldsymbol{l}_k^T$ in $\boldsymbol{L}$ and the
lyche-numerical-linear-algebra_F01139	91	1.000	$\boldsymbol{l}_k^T$	$\boldsymbol{l}_k^T$	to solve the lower triangular system $\boldsymbol{U}_{k-1}^T \boldsymbol{l}_k = \boldsymbol{r}_k$ for the k th row $\boldsymbol{l}_k^T$ in $\boldsymbol{L}$ and the
lyche-numerical-linear-algebra_F01140	91	0.699	$\left[\begin{array}{c} \boldsymbol{u}_k \\ u_k \end{array} \right]$	$\left[\begin{array}{c} \boldsymbol{u}_k \\ u_k \end{array} \right]$	k th column $\begin{bmatrix} \boldsymbol{u}_k \\ u_{kk} \end{bmatrix}$ in $\boldsymbol{U}$ for $k = 2, \dots, n$. This leads to the following algorithm to
lyche-numerical-linear-algebra_F01141	91	0.699	$k=2, \ldots, n$	$k = 2, \dots, n$	k th column $\begin{bmatrix} \boldsymbol{u}_k \\ u_{kk} \end{bmatrix}$ in $\boldsymbol{U}$ for $k = 2, \dots, n$. This leads to the following algorithm to
lyche-numerical-linear-algebra_F01142	91	1.000	$d \geq 1$	$d \geq 1$	compute the L1U factorization of a d -banded matrix $\boldsymbol{A}$ with $d \geq 1$. The algorithm
lyche-numerical-linear-algebra_F01143	91	1.000	$O(d^2 n)$	$O(d^2 n)$	the number of arithmetic operation for this algorithm is $O(d^2 n)$.
lyche-numerical-linear-algebra_F01144	91	0.996	$k < n$	$k < n$	factorization even if $\boldsymbol{A}_{[k]}$ is singular for some $k < n$. By Theorem 4.1 such an LU
lyche-numerical-linear-algebra_F01145	91	1.000	$\boldsymbol{A} = \boldsymbol{I} \boldsymbol{A}$	$\boldsymbol{A} = \boldsymbol{I} \boldsymbol{A}$	$\boldsymbol{A}$ is $\boldsymbol{A} = \boldsymbol{I} \boldsymbol{A}$ so it always exists even if $\boldsymbol{A}$ has zeros somewhere on the diagonal.
lyche-numerical-linear-algebra_F01146	91	1.000	a_{kk}	a_{kk}	By Lemma 2.5, if some a_{kk} is zero then $\boldsymbol{A}_{[k]}$ is singular and the L1U factorization
lyche-numerical-linear-algebra_F01147	92	1.000	$\boldsymbol{A}_{ii}$	$\boldsymbol{A}_{ii}$	where each diagonal block $\boldsymbol{A}_{ii}$ is square. We call the factorization
lyche-numerical-linear-algebra_F01148	92	0.434	$\boldsymbol{L}$	$\boldsymbol{L}$	a block L1U factorization of $\boldsymbol{A}$. Here the i th diagonal blocks $\boldsymbol{I}$ and $\boldsymbol{U}_{ii}$ in $\boldsymbol{L}$ and
lyche-numerical-linear-algebra_F01149	92	0.434	$\boldsymbol{U}_{ii}$	$\boldsymbol{U}_{ii}$	a block L1U factorization of $\boldsymbol{A}$. Here the i th diagonal blocks $\boldsymbol{I}$ and $\boldsymbol{U}_{ii}$ in $\boldsymbol{L}$ and
lyche-numerical-linear-algebra_F01150	92	0.999	$k=1, \ldots, m-1$	$k = 1, \dots, m-1$	are nonsingular for $k = 1, \dots, m-1$.
lyche-numerical-linear-algebra_F01151	92	1.000	$\boldsymbol{A}_{\{k\}}$	$\boldsymbol{A}_{\{k\}}$	Proof Suppose $\boldsymbol{A}_{\{k\}}$ is nonsingular for $k = 1, \dots, m-1$. Following the proof
lyche-numerical-linear-algebra_F01152	92	0.993	$\boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{A}_{\{m-1\}}$	in Theorem 3.4 suppose $\boldsymbol{A}_{\{m-1\}}$ has a unique block LU factorization $\boldsymbol{A}_{\{m-1\}} =$
lyche-numerical-linear-algebra_F01153	92	0.993	$\boldsymbol{A}_{\{m-1\}} =$	$\boldsymbol{A}_{\{m-1\}} =$	in Theorem 3.4 suppose $\boldsymbol{A}_{\{m-1\}}$ has a unique block LU factorization $\boldsymbol{A}_{\{m-1\}} =$
lyche-numerical-linear-algebra_F01154	92	0.999	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$, and that $\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$ are nonsingular. Then $\boldsymbol{L}_{\{m-1\}}$ and
lyche-numerical-linear-algebra_F01155	92	0.999	$\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$, and that $\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$ are nonsingular. Then $\boldsymbol{L}_{\{m-1\}}$ and
lyche-numerical-linear-algebra_F01156	92	0.999	$\boldsymbol{L}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}}$	$\boldsymbol{L}_{\{m-1\}} \boldsymbol{U}_{\{m-1\}}$, and that $\boldsymbol{A}_{\{1\}}, \dots, \boldsymbol{A}_{\{m-1\}}$ are nonsingular. Then $\boldsymbol{L}_{\{m-1\}}$ and
lyche-numerical-linear-algebra_F01157	92	0.523	$\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{U}_{\{m-1\}}$	$\boldsymbol{U}_{\{m-1\}}$ are nonsingular and
lyche-numerical-linear-algebra_F01158	93	0.934	$\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}$	$\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}$	that $\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}$ is the unique block LU factorization of $\boldsymbol{A}_{\{k\}}$ for $k = 1, \dots, m$.
lyche-numerical-linear-algebra_F01159	93	0.934	$k=1, \ldots, m$	$k = 1, \dots, m$	that $\boldsymbol{A}_{\{k\}} = \boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}}$ is the unique block LU factorization of $\boldsymbol{A}_{\{k\}}$ for $k = 1, \dots, m$.
lyche-numerical-linear-algebra_F01160	93	1.000	$\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii} \tilde{\boldsymbol{U}}_{ii}$	$\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii} \tilde{\boldsymbol{U}}_{ii}$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii} \tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L} \text{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra_F01161	93	1.000	$\boldsymbol{A} = \hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$	$\boldsymbol{A} = \hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii} \tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L} \text{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra_F01162	93	1.000	$\hat{\boldsymbol{L}} := \boldsymbol{L} \text{diag}(\tilde{\boldsymbol{L}}_{ii})$	$\hat{\boldsymbol{L}} := \boldsymbol{L} \text{diag}(\tilde{\boldsymbol{L}}_{ii})$	factorization $\boldsymbol{U}_{ii} = \tilde{\boldsymbol{L}}_{ii} \tilde{\boldsymbol{U}}_{ii}$. Then $\boldsymbol{A} = \hat{\boldsymbol{L}} \hat{\boldsymbol{U}}$, where $\hat{\boldsymbol{L}} := \boldsymbol{L} \text{diag}(\tilde{\boldsymbol{L}}_{ii})$ and
lyche-numerical-linear-algebra_F01163	93	0.987	$\hat{\boldsymbol{U}} := \text{diag}(\tilde{\boldsymbol{L}}_{ii}^{-1}) \boldsymbol{U}$	$\hat{\boldsymbol{U}} := \text{diag}(\tilde{\boldsymbol{L}}_{ii}^{-1}) \boldsymbol{U}$	$\hat{\boldsymbol{U}} := \text{diag}(\tilde{\boldsymbol{L}}_{ii}^{-1}) \boldsymbol{U}$, and this is an ordinary LU factorization of $\boldsymbol{A}$.

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lyche-numerical-linear-algebra-F01164	93	■ 0.997	$\left[\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\right]$	$[b_1, \dots, b_n]$	$[b_1, \dots, b_n]$ is also lower triangular. The k th column b_k of B is the solution of the
lyche-numerical-linear-algebra-F01165	93	■ 0.997	$\boldsymbol{b}_k$	b_k	$[b_1, \dots, b_n]$ is also lower triangular. The k th column b_k of B is the solution of the
lyche-numerical-linear-algebra-F01166	93	■ 1.000	$\boldsymbol{A} \boldsymbol{b}_k = \boldsymbol{e}_k$	$A b_k = e_k$	linear systems $A b_k = e_k$. Show that $b_k(k) = 1/a(k, k)$ for $k = 1, \dots, n$, and
lyche-numerical-linear-algebra-F01167	93	■ 1.000	$b_k(k) = 1/a(k, k)$	$b_k(k) = 1/a(k, k)$	linear systems $A b_k = e_k$. Show that $b_k(k) = 1/a(k, k)$ for $k = 1, \dots, n$, and
lyche-numerical-linear-algebra-F01168	94	■ 1.000	$k = n, n-1, \dots, 1$	$k = n, n-1, \dots, 1$	for $k = n, n-1, \dots, 1$.
lyche-numerical-linear-algebra-F01169	94	■ 1.000	$\frac{1}{3} n (2n^2 + 1) \approx G_n$	$\frac{1}{3} n (2n^2 + 1) \approx G_n$	cation AB can be done in $\frac{1}{3} n (2n^2 + 1) \approx G_n$ arithmetic operations when $A \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01170	94	■ 1.000	$B \in \mathbb{R}^{n \times n}$	$B \in \mathbb{R}^{n \times n}$	is lower triangular and $B \in \mathbb{R}^{n \times n}$ is upper triangular. What about BA ?
lyche-numerical-linear-algebra-F01171	94	■ 1.000	BA	BA	is lower triangular and $B \in \mathbb{R}^{n \times n}$ is upper triangular. What about BA ?
lyche-numerical-linear-algebra-F01172	94	■ 1.000	P, L, U	P, L, U	Exercise 3.5 (Using PLU for A^*) Suppose we know the PLU factors P, L, U in a
lyche-numerical-linear-algebra-F01173	94	■ 1.000	$A^* x = b$	$A^* x = b$	$A^* x = b$ economically.
lyche-numerical-linear-algebra-F01174	94	■ 1.000	$2G_n$	$2G_n$	$2G_n$ arithmetic operations to compute $A^{-1} = U^{-1} L^{-1} P^T$. Here we have not
lyche-numerical-linear-algebra-F01175	94	■ 1.000	$A^{-1} = U^{-1} L^{-1} P^T$	$A^{-1} = U^{-1} L^{-1} P^T$	$2G_n$ arithmetic operations to compute $A^{-1} = U^{-1} L^{-1} P^T$. Here we have not
lyche-numerical-linear-algebra-F01176	94	■ 1.000	P^T	P^T	counted the final multiplication with P^T which amounts to n row interchanges.
lyche-numerical-linear-algebra-F01177	95	■ 0.993	$k = 1, 2, \dots, n-1$	$k = 1, 2, \dots, n-1$	1. for $k = 1, 2, \dots, n-1$
lyche-numerical-linear-algebra-F01178	95	■ 1.000	$a_{k,j}$	$a_{k,j}$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k+1, \dots, n$
lyche-numerical-linear-algebra-F01179	95	■ 1.000	$a_{k+1,j}$	$a_{k+1,j}$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k+1, \dots, n$
lyche-numerical-linear-algebra-F01180	95	■ 1.000	$j = k, k+1, \dots, n$	$j = k, k+1, \dots, n$	(a) exchange $a_{k,j}$ and $a_{k+1,j}$ for $j = k, k+1, \dots, n$
lyche-numerical-linear-algebra-F01181	95	■ 1.000	$i = k+1, k+2, \dots, n$	$i = k+1, k+2, \dots, n$	(b) for $i = k+1, k+2, \dots, n$
lyche-numerical-linear-algebra-F01182	95	■ 1.000	$a_{i,k} = m_{i,k} = a_{i,k} / a_{k,k}$	$a_{i,k} = m_{i,k} = a_{i,k} / a_{k,k}$	i. $a_{i,k} = m_{i,k} = a_{i,k} / a_{k,k}$
lyche-numerical-linear-algebra-F01183	95	■ 1.000	$a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$	$a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$	ii. $a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$ for $j = k+1, k+2, \dots, n$
lyche-numerical-linear-algebra-F01184	95	■ 1.000	$j = k+1, k+2, \dots, n$	$j = k+1, k+2, \dots, n$	ii. $a_{i,j} = a_{i,j} - m_{i,k} a_{k,j}$ for $j = k+1, k+2, \dots, n$
lyche-numerical-linear-algebra-F01185	95	■ 1.000	b_k	b_k	(a) exchange b_k and b_{k+1}
lyche-numerical-linear-algebra-F01186	95	■ 1.000	b_{k+1}	b_{k+1}	(a) exchange b_k and b_{k+1}
lyche-numerical-linear-algebra-F01187	95	■ 1.000	$b_i = b_i - a_{i,k} b_k$	$b_i = b_i - a_{i,k} b_k$	(b) $b_i = b_i - a_{i,k} b_k$ for $i = k+1, k+2, \dots, n$
lyche-numerical-linear-algebra-F01188	95	■ 1.000	$x_n = b_n / a_{n,n}$	$x_n = b_n / a_{n,n}$	2. $x_n = b_n / a_{n,n}$
lyche-numerical-linear-algebra-F01189	95	■ 0.981	$= 0$	$= 0$	(a) $sum = 0$
lyche-numerical-linear-algebra-F01190	95	■ 0.958	$+a_{k,j} x_j$	$+a_{k,j} x_j$	(b) $sum = sum + a_{k,j} x_j$ for $j = k+1, k+2, \dots, n$

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lyche-numerical-linear-algebra-F01191	95	1.000	$x_{\{k\}}=\left(b_{\{k\}}-\right.$	$x_k = (b_k -$	(c) $x_k = (b_k - \text{sum})/a_{k,k}$
lyche-numerical-linear-algebra-F01192	95	1.000	$\left. \right) / a_{\{k, k\}}$	$)/a_{k,k}$	(c) $x_k = (b_k - \text{sum})/a_{k,k}$
lyche-numerical-linear-algebra-F01193	95	1.000	$\boldsymbol{H} \in \mathbb{R}^{n \times n}$	$\boldsymbol{H} \in \mathbb{R}^{n \times n}$	We say that $\boldsymbol{H} \in \mathbb{R}^{n \times n}$ is unreduced upper Hessenberg if it is upper Hessenberg
lyche-numerical-linear-algebra-F01194	95	1.000	$h_{\{i, i-1\}} \neq 0$	$h_{i,i-1} \neq 0$	and the subdiagonal elements $h_{i,i-1} \neq 0$ for $i = 2, \dots, n$.
lyche-numerical-linear-algebra-F01195	95	0.983	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$	solving the linear system $\boldsymbol{H} \boldsymbol{x} = \boldsymbol{b}$ using suitable specializations of Algorithms 1
lyche-numerical-linear-algebra-F01196	95	1.000	$\boldsymbol{U} \in \mathbb{R}^{n \times n}$	$\boldsymbol{U} \in \mathbb{R}^{n \times n}$	c) Let $\boldsymbol{U} \in \mathbb{R}^{n \times n}$ be upper triangular and nonsingular. We define
lyche-numerical-linear-algebra-F01197	95	1.000	$\boldsymbol{v} \in \mathbb{R}^n$	$\boldsymbol{v} \in \mathbb{R}^n$	where $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{e}_1$ is the first unit vector in $\mathbb{R}^n$. We also let
lyche-numerical-linear-algebra-F01198	95	1.000	$\boldsymbol{e}_1$	$\boldsymbol{e}_1$	where $\boldsymbol{v} \in \mathbb{R}^n$ and $\boldsymbol{e}_1$ is the first unit vector in $\mathbb{R}^n$. We also let
lyche-numerical-linear-algebra-F01199	95	0.999	$\boldsymbol{I}_{\{i, j\}}$	$\boldsymbol{I}_{i,j}$	where the $\boldsymbol{I}_{i,j}$ are obtained from the identity matrix by interchanging rows i and
lyche-numerical-linear-algebra-F01200	95	0.822	$\boldsymbol{E} := \boldsymbol{C} \boldsymbol{P}$	$\boldsymbol{E} := \boldsymbol{C} \boldsymbol{P}$	j . Explain why the matrix $\boldsymbol{E} := \boldsymbol{C} \boldsymbol{P}$ is unreduced upper Hessenberg.
lyche-numerical-linear-algebra-F01201	96	1.000	$\boldsymbol{W} \in \mathbb{R}^n$	$\boldsymbol{W} \in \mathbb{R}^n$	$\boldsymbol{A} = \boldsymbol{L} \boldsymbol{U}$. To a given $\boldsymbol{W} \in \mathbb{R}^n$ we define a rank one modification of $\boldsymbol{A}$ by
lyche-numerical-linear-algebra-F01202	96	1.000	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^T$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^T$	Show that $\boldsymbol{B}$ has the factorization $\boldsymbol{B} = \boldsymbol{L} \boldsymbol{H} \boldsymbol{P}^T$, where $\boldsymbol{L}$ is unit lower triangular,
lyche-numerical-linear-algebra-F01203	96	1.000	$\boldsymbol{H}$	$\boldsymbol{H}$	$\boldsymbol{P}$ is given by (3.29) and $\boldsymbol{H}$ is unreduced upper Hessenberg.
lyche-numerical-linear-algebra-F01204	96	1.000	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{b}$	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{b}$, where $\boldsymbol{B}$ is given by (3.30). We assume that the matrices $\boldsymbol{L}$ and $\boldsymbol{U}$ in
lyche-numerical-linear-algebra-F01205	96	0.999	$1 \leq$	$1 \leq$	Exercise 3.9 (# Operations for Banded Triangular Systems) Show that for $1 \leq$
lyche-numerical-linear-algebra-F01206	96	1.000	$d \leq n$	$d \leq n$	$d \leq n$ Algorithm 3.4, with $A(k, k) = 1$ for $k = 1, \dots, n$ in Algorithm 3.1, requires
lyche-numerical-linear-algebra-F01207	96	1.000	$A(k, k)=1$	$A(k, k) = 1$	$d \leq n$ Algorithm 3.4, with $A(k, k) = 1$ for $k = 1, \dots, n$ in Algorithm 3.1, requires
lyche-numerical-linear-algebra-F01208	96	1.000	$N_{\{L U\}}(n, d) := \left(2 d^2+d\right) n-\left(d^2+d\right)(8 d+1) / 6=O\left(d^2 n\right)$	$N_{LU}(n, d) := (2d^2 + d)n - (d^2 + d)(8d + 1)/6 = O(d^2n)$	exactly $N_{LU}(n, d) := (2d^2 + d)n - (d^2 + d)(8d + 1)/6 = O(d^2n)$ operations. ² In
lyche-numerical-linear-algebra-F01209	96	1.000	$d=n-1$	$d = n - 1$	particular, for a full matrix $d = n - 1$ and we find $N_{LU}(n, n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$
lyche-numerical-linear-algebra-F01210	96	1.000	$N_{\{L U\}}(n, n)=\frac{2}{3} n^3-\frac{1}{2} n^2-\frac{1}{6} n \approx$	$N_{LU}(n, n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$	particular, for a full matrix $d = n - 1$ and we find $N_{LU}(n, n) = \frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n \approx$
lyche-numerical-linear-algebra-F01211	96	1.000	$N_{\{L U\}}(n, 1)=3 n-3=O(n)$	$N_{LU}(n, 1) = 3n - 3 = O(n)$	tridiagonal matrix $N_{LU}(n, 1) = 3n - 3 = O(n)$.
lyche-numerical-linear-algebra-F01212	96	0.985	$\boldsymbol{A}=\left[\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right] 1 \& 1 \backslash \backslash 0 \& 1 \end{array}\right] \right]$	$\boldsymbol{A} = \left[\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \end{array} \right] \right]$	Exercise 3.12 (Row Interchange) Show that $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has a unique L1U
lyche-numerical-linear-algebra-F01213	96	0.999	$2 \leq k \leq d$	$2 \leq k \leq d$	² Hint: Consider the cases $2 \leq k \leq d$ and $d + 1 \leq k \leq n$ separately.
lyche-numerical-linear-algebra-F01214	96	0.999	$d+1 \leq k \leq n$	$d + 1 \leq k \leq n$	² Hint: Consider the cases $2 \leq k \leq d$ and $d + 1 \leq k \leq n$ separately.
lyche-numerical-linear-algebra-F01215	97	1.000	$u_{\{k k\}}$	u_{kk}	u_{kk} in the L1U factorization are
lyche-numerical-linear-algebra-F01216	97	1.000	$\boldsymbol{L}_{\{2,2\}}, \boldsymbol{U}_{\{2,2\}} \in \mathbb{R}^{(n-1) \times (n-1)}$	$\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.
lyche-numerical-linear-algebra-F01217	97	1.000	$\boldsymbol{A}_{\{2,2\}}:=\boldsymbol{L}_{\{2,2\}} \boldsymbol{U}_{\{2,2\}}$	$\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2} \boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.

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lyche-numerical-linear-algebra-F01218	97	■ 1.000	$\backslash\boldsymbol{B}_{2,2}:=\backslash\boldsymbol{A}_{2,2}^{-1}$	$\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$	where $\boldsymbol{L}_{2,2}, \boldsymbol{U}_{2,2} \in \mathbb{R}^{(n-1) \times (n-1)}$. Define $\boldsymbol{A}_{2,2} := \boldsymbol{L}_{2,2}\boldsymbol{U}_{2,2}$ and $\boldsymbol{B}_{2,2} := \boldsymbol{A}_{2,2}^{-1}$.
lyche-numerical-linear-algebra-F01219	97	■ 1.000	$\boldsymbol{l}_{\{i, j\}, i>j}$	$\boldsymbol{l}_{i,j}, i > j$	b) Suppose that the elements $\boldsymbol{l}_{i,j}, i > j$ in $\boldsymbol{L}$ and $\boldsymbol{u}_{i,j}, j \geq i$ in $\boldsymbol{U}$ are stored in
lyche-numerical-linear-algebra-F01220	97	■ 1.000	$\boldsymbol{u}_{\{i, j\}, j \geq i}$	$\boldsymbol{u}_{i,j}, j \geq i$	b) Suppose that the elements $\boldsymbol{l}_{i,j}, i > j$ in $\boldsymbol{L}$ and $\boldsymbol{u}_{i,j}, j \geq i$ in $\boldsymbol{U}$ are stored in
lyche-numerical-linear-algebra-F01221	97	■ 1.000	$\boldsymbol{a}_{\{i, j\}}$	$\boldsymbol{a}_{i,j}$	$\boldsymbol{A}$ with elements $\boldsymbol{a}_{i,j}$. Write an algorithm that overwrites the elements in $\boldsymbol{A}$ with
lyche-numerical-linear-algebra-F01222	97	■ 1.000	$\backslash\boldsymbol{s} \in \backslash\mathbb{R}^n$	$\boldsymbol{s} \in \mathbb{R}^n$	ones in $\boldsymbol{A}^{-1}$. Only one extra vector $\boldsymbol{s} \in \mathbb{R}^n$ should be used.
lyche-numerical-linear-algebra-F01223	97	■ 1.000	$\backslash\boldsymbol{T} \ \backslash\boldsymbol{H} \ \backslash\boldsymbol{x}=\backslash\boldsymbol{b}$	$\boldsymbol{THx} = \boldsymbol{b}$	Exercise 3.18 (Solving $\boldsymbol{THx} = \boldsymbol{b}$ (Exam Exercise 1981-3)) In this exercise we
lyche-numerical-linear-algebra-F01224	97	■ 1.000	$\backslash\boldsymbol{T}, \ \backslash\boldsymbol{H}, \ \backslash\boldsymbol{S} \in \backslash\mathbb{R}^n \ \backslash\mathbb{R}^n \ \backslash\mathbb{R}^n$	$\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$	consider nonsingular matrices $\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$ with $\boldsymbol{T} = (t_{ij})$ upper triangular,
lyche-numerical-linear-algebra-F01225	97	■ 1.000	$\backslash\boldsymbol{T}=\backslash\left(\boldsymbol{t}_{i \ j}\right)$	$\boldsymbol{T} = (t_{ij})$	consider nonsingular matrices $\boldsymbol{T}, \boldsymbol{H}, \boldsymbol{S} \in \mathbb{R}^{n \times n}$ with $\boldsymbol{T} = (t_{ij})$ upper triangular,
lyche-numerical-linear-algebra-F01226	97	■ 1.000	$\backslash\boldsymbol{H}=\backslash\left(\boldsymbol{h}_{i \ j}\right)$	$\boldsymbol{H} = (h_{ij})$	$\boldsymbol{H} = (h_{ij})$ upper Hessenberg and $\boldsymbol{S} := \boldsymbol{TH}$. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra-F01227	97	■ 1.000	$\backslash\boldsymbol{S}:=\backslash\boldsymbol{T} \ \backslash\boldsymbol{H}$	$\boldsymbol{S} := \boldsymbol{TH}$	$\boldsymbol{H} = (h_{ij})$ upper Hessenberg and $\boldsymbol{S} := \boldsymbol{TH}$. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra-F01228	97	■ 1.000	$\backslash\boldsymbol{H}=\backslash\boldsymbol{L} \ \backslash\boldsymbol{U}$	$\boldsymbol{H} = \boldsymbol{LU}$	LU1 factorization $\boldsymbol{H} = \boldsymbol{LU}$ with $\ \boldsymbol{L}\ _{\infty}\ \boldsymbol{U}\ _{\infty} \leq K\ \boldsymbol{H}\ _{\infty}$ for a constant K not
lyche-numerical-linear-algebra-F01229	97	■ 1.000	$\backslash\backslash\boldsymbol{L}\backslash\backslash_{\infty}\backslash\backslash\boldsymbol{U}\backslash\backslash_{\infty}\backslash\leq K\backslash\backslash\boldsymbol{H}\backslash\backslash_{\infty}$	$\ \boldsymbol{L}\ _{\infty}\ \boldsymbol{U}\ _{\infty} \leq K\ \boldsymbol{H}\ _{\infty}$	LU1 factorization $\boldsymbol{H} = \boldsymbol{LU}$ with $\ \boldsymbol{L}\ _{\infty}\ \boldsymbol{U}\ _{\infty} \leq K\ \boldsymbol{H}\ _{\infty}$ for a constant K not
lyche-numerical-linear-algebra-F01230	97	■ 1.000	$\backslash\boldsymbol{S}$	$\boldsymbol{S}$	a) Give an algorithm which computes $\boldsymbol{S}$ from $\boldsymbol{T}$ and $\boldsymbol{H}$ without using the lower parts
lyche-numerical-linear-algebra-F01231	97	■ 0.999	$\backslash\left(\boldsymbol{t}_{i \ j}, i>j\right)$	$(t_{ij}, i > j)$	$(t_{ij}, i > j)$ of $\boldsymbol{T}$ and $(h_{ij}, i > j + 1)$ of $\boldsymbol{H}$. In what order should the elements
lyche-numerical-linear-algebra-F01232	97	■ 0.999	$\backslash\left(\boldsymbol{h}_{i \ j}, i>j+1\right)$	$(h_{ij}, i > j + 1)$	$(t_{ij}, i > j)$ of $\boldsymbol{T}$ and $(h_{ij}, i > j + 1)$ of $\boldsymbol{H}$. In what order should the elements
lyche-numerical-linear-algebra-F01233	98	■ 0.987	$\backslash\boldsymbol{T}, \ \backslash\boldsymbol{H}$	$\boldsymbol{T}, \boldsymbol{H}$	d) Given $\boldsymbol{b} \in \mathbb{R}^n$ and $\boldsymbol{T}, \boldsymbol{H}$ as before. Suppose $\boldsymbol{S}$ and the LU1-factorization are not
lyche-numerical-linear-algebra-F01234	98	■ 1.000	$\backslash\boldsymbol{S} \ \backslash\boldsymbol{x}=\backslash\boldsymbol{b}$	$\boldsymbol{Sx} = \boldsymbol{b}$	computed. We want to find $\boldsymbol{x} \in \mathbb{R}^n$ such that $\boldsymbol{Sx} = \boldsymbol{b}$. We have the 2 following
lyche-numerical-linear-algebra-F01235	98	■ 0.779	$\backslash\boldsymbol{S}=\backslash\boldsymbol{T} \ \backslash\boldsymbol{H}$	$\boldsymbol{S} = \boldsymbol{TH}$	1. $\boldsymbol{S} = \boldsymbol{TH}$
lyche-numerical-linear-algebra-F01236	98	■ 0.991	$\backslash\boldsymbol{T} \ \backslash\boldsymbol{z}=\backslash\boldsymbol{b}$	$\boldsymbol{Tz} = \boldsymbol{b}$	1. Solve $\boldsymbol{Tz} = \boldsymbol{b}$
lyche-numerical-linear-algebra-F01237	98	■ 0.894	$\backslash\boldsymbol{H} \ \backslash\boldsymbol{x}=\backslash\boldsymbol{z}$	$\boldsymbol{Hx} = \boldsymbol{z}$	2. Solve $\boldsymbol{Hx} = \boldsymbol{z}$
lyche-numerical-linear-algebra-F01238	98	■ 1.000	$\backslash\boldsymbol{a}_{\{1\}}, \ \backslash\boldsymbol{a}_{\{2\}}, \ \backslash\text{dots}, \ \backslash\boldsymbol{a}_{\{n\}}$	$\boldsymbol{a}_1, \boldsymbol{a}_2, \dots, \boldsymbol{a}_n$	$\mathbb{R}^{n \times n}$ be nonsingular with columns $\boldsymbol{a}_1, \boldsymbol{a}_2, \dots, \boldsymbol{a}_n$. We assume that $\boldsymbol{A}$ has a unique
lyche-numerical-linear-algebra-F01239	98	■ 1.000	$\boldsymbol{p} \leq \boldsymbol{n}$	$p \leq n$	For a positive integer $p \leq n$ and $\boldsymbol{b} \in \mathbb{R}^n$ we define
lyche-numerical-linear-algebra-F01240	98	■ 1.000	$\backslash\boldsymbol{H}:=\backslash\boldsymbol{L}^{-1} \ \backslash\boldsymbol{B}$	$\boldsymbol{H} := \boldsymbol{L}^{-1}\boldsymbol{B}$	a) Show that $\boldsymbol{H} := \boldsymbol{L}^{-1}\boldsymbol{B}$ is upper Hessenberg. We assume that $\boldsymbol{H}$ has a unique
lyche-numerical-linear-algebra-F01241	98	■ 1.000	$\backslash\boldsymbol{H}=\backslash\boldsymbol{L}_{\boldsymbol{H}} \ \backslash\boldsymbol{U}_{\boldsymbol{H}}$	$\boldsymbol{H} = \boldsymbol{L}_H\boldsymbol{U}_H$	L1U factorization. $\boldsymbol{H} = \boldsymbol{L}_H\boldsymbol{U}_H$.
lyche-numerical-linear-algebra-F01242	98	■ 1.000	$\backslash\boldsymbol{H}:=\backslash\boldsymbol{L}_{\boldsymbol{H}} \ \backslash\boldsymbol{U}_{\boldsymbol{H}}$	$\boldsymbol{H} := \boldsymbol{L}_H\boldsymbol{U}_H$	c) Suppose we have found the L1U factorization $\boldsymbol{H} := \boldsymbol{L}_H\boldsymbol{U}_H$ of $\boldsymbol{H}$. Explain how
lyche-numerical-linear-algebra-F01243	98	■ 1.000	$\backslash\boldsymbol{L}_{\boldsymbol{H}}$	$\boldsymbol{L}_H$	we can find the L1U factorization of $\boldsymbol{B}$ from $\boldsymbol{L}_H$ and $\boldsymbol{U}_H$.
lyche-numerical-linear-algebra-F01244	98	■ 1.000	$\backslash\boldsymbol{U}_{\boldsymbol{H}}$	$\boldsymbol{U}_H$	we can find the L1U factorization of $\boldsymbol{B}$ from $\boldsymbol{L}_H$ and $\boldsymbol{U}_H$.

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lyche-numerical-linear-algebra_F01245	98	■ 0.999	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{L}$	$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{L}$	$\mathbb{R}^{n \times n}$ has a UIL factorization if $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{L}$ for an upper triangular matrix $\boldsymbol{U} \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra_F01246	98	■ 1.000	$\boldsymbol{P} \in \mathbb{R}^{n \times n}$	$\boldsymbol{P} \in \mathbb{R}^{n \times n}$	b) Let the columns of $\boldsymbol{P} \in \mathbb{R}^{n \times n}$ be the unit vectors in reverse order, i.e.,
lyche-numerical-linear-algebra_F01247	98	■ 1.000	$\boldsymbol{P}^T=\boldsymbol{P}$	$\boldsymbol{P}^T = \boldsymbol{P}$	Show that $\boldsymbol{P}^T = \boldsymbol{P}$ and $\boldsymbol{P}^2 = \boldsymbol{I}$. What is the connection between the elements
lyche-numerical-linear-algebra_F01248	99	■ 1.000	$\boldsymbol{P}^2=\boldsymbol{I}$	$\boldsymbol{P}^2 = \boldsymbol{I}$	Show that $\boldsymbol{P}^T = \boldsymbol{P}$ and $\boldsymbol{P}^2 = \boldsymbol{I}$. What is the connection between the elements
lyche-numerical-linear-algebra_F01249	98	■ 1.000	$\boldsymbol{P} \boldsymbol{A}$	$\boldsymbol{P}\boldsymbol{A}$	in $\boldsymbol{A}$ and $\boldsymbol{P}\boldsymbol{A}$?
lyche-numerical-linear-algebra_F01250	99	■ 1.000	$\boldsymbol{B}:=\boldsymbol{P} \boldsymbol{A}$	$\boldsymbol{B} := \boldsymbol{P}\boldsymbol{A}$	c) Let $\boldsymbol{B} := \boldsymbol{P}\boldsymbol{A}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.
lyche-numerical-linear-algebra_F01251	99	■ 1.000	r, s	r, s	c) Let $\boldsymbol{B} := \boldsymbol{P}\boldsymbol{A}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.
lyche-numerical-linear-algebra_F01252	99	■ 1.000	i, j, n	i, j, n	c) Let $\boldsymbol{B} := \boldsymbol{P}\boldsymbol{A}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.
lyche-numerical-linear-algebra_F01253	99	■ 1.000	$b_{i,j}=a_{r, s}$	$b_{i,j} = a_{r,s}$	c) Let $\boldsymbol{B} := \boldsymbol{P}\boldsymbol{A}$. Find integers r, s , depending on i, j, n , such that $b_{i,j} = a_{r,s}$.
lyche-numerical-linear-algebra_F01254	99	■ 1.000	$b_{i,j}$	$b_{i,j}$	The elements $b_{i,j}$ in $\boldsymbol{B}$ should be stored in position i, j in $\boldsymbol{A}$. You should not use
lyche-numerical-linear-algebra_F01255	99	■ 1.000	$w \in \mathbb{R}$	$w \in \mathbb{R}$	other matrices than $\boldsymbol{A}$ and a scalar $w \in \mathbb{R}$.
lyche-numerical-linear-algebra_F01256	99	■ 1.000	$\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}=\boldsymbol{M} \boldsymbol{R}$	$\boldsymbol{P}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{M}\boldsymbol{R}$	e) Let $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{M}\boldsymbol{R}$ be an L1U factorization of $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$, i.e., $\boldsymbol{M}$ is lower triangular
lyche-numerical-linear-algebra_F01257	99	■ 1.000	$\boldsymbol{P} \boldsymbol{A} \boldsymbol{P}$	$\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$	e) Let $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{M}\boldsymbol{R}$ be an L1U factorization of $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$, i.e., $\boldsymbol{M}$ is lower triangular
lyche-numerical-linear-algebra_F01258	99	■ 1.000	$\boldsymbol{M}$	$\boldsymbol{M}$	e) Let $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P} = \boldsymbol{M}\boldsymbol{R}$ be an L1U factorization of $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}$, i.e., $\boldsymbol{M}$ is lower triangular
lyche-numerical-linear-algebra_F01259	99	■ 1.000	$\boldsymbol{M}, \boldsymbol{R}$	$\boldsymbol{M}, \boldsymbol{R}$	$\boldsymbol{L}$ in a UIL factorization of $\boldsymbol{A}$ in terms of $\boldsymbol{M}, \boldsymbol{R}$ and $\boldsymbol{P}$.
lyche-numerical-linear-algebra_F01260	99	■ 0.504	$\hat{\boldsymbol{L}}$	$\hat{\boldsymbol{L}}$	Exercise 3.21 (Making Block LU into LU) Show that $\hat{\boldsymbol{L}}$ is unit lower triangular
lyche-numerical-linear-algebra_F01261	99	■ 1.000	$\hat{\boldsymbol{U}}$	$\hat{\boldsymbol{U}}$	and $\hat{\boldsymbol{U}}$ is upper triangular.
lyche-numerical-linear-algebra_F01262	100	■ 0.997	$a_{ji}=\bar{a}_{ij}$	$a_{ji} = \bar{a}_{ij}$	matrices. Recall that a matrix $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is Hermitian if $\boldsymbol{A}^* = \boldsymbol{A}$, i.e., $a_{ji} = \bar{a}_{ij}$
lyche-numerical-linear-algebra_F01263	100	■ 1.000	$a_{ii}=\bar{a}_{ii}$	$a_{ii} = \bar{a}_{ii}$	for all i, j . A real Hermitian matrix is symmetric. Since $a_{ii} = \bar{a}_{ii}$ the diagonal
lyche-numerical-linear-algebra_F01264	100	■ 1.000	$\boldsymbol{U}=\boldsymbol{L}^*$	$\boldsymbol{U} = \boldsymbol{L}^*$	1. the LDL* factorization which is an LDU factorization with $\boldsymbol{U} = \boldsymbol{L}^*$ and $\boldsymbol{D}$ a
lyche-numerical-linear-algebra_F01265	100	■ 1.000	$l_{ii}>0$	$l_{ii} > 0$	2. the LL* factorization which is an LU factorization with $\boldsymbol{U} = \boldsymbol{L}^*$ and $l_{ii} > 0$
lyche-numerical-linear-algebra_F01266	100	■ 0.999	$\boldsymbol{A}^*=\left(\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*\right)^*=\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^*=\boldsymbol{A}$	$\boldsymbol{A}^* = (\boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^*)^* = \boldsymbol{L}\boldsymbol{D}^*\boldsymbol{L}^* = \boldsymbol{A}$	that $\boldsymbol{A}^* = (\boldsymbol{L}\boldsymbol{D}\boldsymbol{L}^*)^* = \boldsymbol{L}\boldsymbol{D}^*\boldsymbol{L}^* = \boldsymbol{A}$. The LL* factorization is called a Cholesky
lyche-numerical-linear-algebra_F01267	100	■ 0.966	$b \in \mathbb{C}$	$b \in \mathbb{C}$	<i>Example 4.1 (LDL* of 2×2 Hermitian Matrix)</i> Let $a, d \in \mathbb{R}$ and $b \in \mathbb{C}$. An LDL*
lyche-numerical-linear-algebra_F01268	101	■ 1.000	d_1, d_2	d_1, d_2	for the unknowns l_1 in $\boldsymbol{L}$ and d_1, d_2 in $\boldsymbol{D}$. They are determined from
lyche-numerical-linear-algebra_F01269	101	■ 1.000	d_1	d_1	1. $a \neq 0$: The matrix has a unique LDL* factorization. Note that d_1 and d_2 are real.
lyche-numerical-linear-algebra_F01270	101	■ 1.000	d_2	d_2	1. $a \neq 0$: The matrix has a unique LDL* factorization. Note that d_1 and d_2 are real.
lyche-numerical-linear-algebra_F01271	101	■ 1.000	$a=b=0$	$a = b = 0$	2. $a = b = 0$: The LDL* factorization exists, but it is not unique. Any value for l_1

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lyche-numerical-linear-algebra-F01272	101	■ 1.000	$a=0, \ b \neq 0$	$a = 0, b \neq 0$	3. $a = 0, b \neq 0$: No LDL^* factorization exists.
lyche-numerical-linear-algebra-F01273	101	■ 0.830	$\boldsymbol{A}=\boldsymbol{L} \ \boldsymbol{D} \ \boldsymbol{L}^{\ast}$	$\boldsymbol{A} = \boldsymbol{LDL}^*$	Lemma 4.1 (LDL* of Leading Principal Sub Matrices) Suppose $\boldsymbol{A} = \boldsymbol{LDL}^*$
lyche-numerical-linear-algebra-F01274	101	■ 0.954	$\boldsymbol{L} \ \boldsymbol{D} \ \boldsymbol{L}^{\ast}$	$\boldsymbol{LDL}^*$	is an LDL^* factorization of $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. For $k = 1, \dots, n$ let $\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$ and $\boldsymbol{D}_{[k]}$
lyche-numerical-linear-algebra-F01275	101	■ 0.954	$\boldsymbol{A}_{[k]}, \ \boldsymbol{L}_{[k]}$	$\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$	is an LDL^* factorization of $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. For $k = 1, \dots, n$ let $\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$ and $\boldsymbol{D}_{[k]}$
lyche-numerical-linear-algebra-F01276	101	■ 0.954	$\boldsymbol{D}_{[k]}$	$\boldsymbol{D}_{[k]}$	is an LDL^* factorization of $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. For $k = 1, \dots, n$ let $\boldsymbol{A}_{[k]}, \boldsymbol{L}_{[k]}$ and $\boldsymbol{D}_{[k]}$
lyche-numerical-linear-algebra-F01277	101	■ 1.000	$\boldsymbol{A}, \ \boldsymbol{L}$	$\boldsymbol{A}, \boldsymbol{L}$	be the leading principal submatrices of $\boldsymbol{A}, \boldsymbol{L}$ and $\boldsymbol{D}$, respectively. Then $\boldsymbol{A}_{[k]} =$
lyche-numerical-linear-algebra-F01278	101	■ 1.000	$\boldsymbol{A}_{[k]}=$	$\boldsymbol{A}_{[k]} =$	be the leading principal submatrices of $\boldsymbol{A}, \boldsymbol{L}$ and $\boldsymbol{D}$, respectively. Then $\boldsymbol{A}_{[k]} =$
lyche-numerical-linear-algebra-F01279	101	■ 1.000	$\boldsymbol{L}_{[k]} \ \boldsymbol{D}_{[k]} \ \boldsymbol{L}_{[k]}^{\ast}$	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$	$\boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$ is an LDL^* factorization of $\boldsymbol{A}_{[k]}$ for $k = 1, \dots, n$.
lyche-numerical-linear-algebra-F01280	101	■ 1.000	$\boldsymbol{A}=\boldsymbol{L} \ \boldsymbol{D} \ \boldsymbol{L}^{\ast}$	$\boldsymbol{A} = \boldsymbol{LDL}^*$	Proof For $k = 1, \dots, n - 1$ we partition $\boldsymbol{A} = \boldsymbol{LDL}^*$ as follows:
lyche-numerical-linear-algebra-F01281	101	■ 1.000	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$	$\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$	where $\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$. Block multiplication gives $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$.
lyche-numerical-linear-algebra-F01282	101	■ 1.000	$\boldsymbol{A}_{[k]}=\boldsymbol{L}_{[k]} \ \boldsymbol{D}_{[k]} \ \boldsymbol{L}_{[k]}^{\ast}$	$\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$	where $\boldsymbol{F}_k, \boldsymbol{N}_k, \boldsymbol{E}_k \in \mathbb{C}^{n-k, n-k}$. Block multiplication gives $\boldsymbol{A}_{[k]} = \boldsymbol{L}_{[k]} \boldsymbol{D}_{[k]} \boldsymbol{L}_{[k]}^*$.
lyche-numerical-linear-algebra-F01283	101	■ 1.000	$\boldsymbol{A}=\boldsymbol{A}^{\ast}$	$\boldsymbol{A} = \boldsymbol{A}^*$	factorization if and only if $\boldsymbol{A} = \boldsymbol{A}^*$ and $\boldsymbol{A}_{[k]}$ is nonsingular for $k = 1, \dots, n - 1$.
lyche-numerical-linear-algebra-F01284	101	■ 1.000	$\boldsymbol{A}_{[k]}^{\ast}=\boldsymbol{A}_{[k]}$	$\boldsymbol{A}_{[k]}^* = \boldsymbol{A}_{[k]}$	Note that $\boldsymbol{A}_{[k]}^* = \boldsymbol{A}_{[k]}$ for $k = 1, \dots, n$. We use induction on n to show that $\boldsymbol{A}$
lyche-numerical-linear-algebra-F01285	101	■ 0.983	$\left[a_{11}\right]=\left[1\right]\left[a_{11}\right]\left[1\right]$	$[a_{11}] = [1] [a_{11}] [1]$	unique LDL^* factorization of a 1-by-1 matrix is $[a_{11}] = [1][a_{11}][1]$ and a_{11} is real
lyche-numerical-linear-algebra-F01286	101	■ 0.997	$\boldsymbol{L}_{n-1} \ \boldsymbol{D}_{n-1} \ \boldsymbol{L}_{n-1}^{\ast}$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By definition $\boldsymbol{D}_{n-1}$ is
lyche-numerical-linear-algebra-F01287	101	■ 0.997	$\boldsymbol{D}_{n-1}$	$\boldsymbol{D}_{n-1}$	$\boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$, and that $\boldsymbol{A}_{[1]}, \dots, \boldsymbol{A}_{[n-1]}$ are nonsingular. By definition $\boldsymbol{D}_{n-1}$ is
lyche-numerical-linear-algebra-F01288	102	■ 1.000	$\boldsymbol{A}_{[n-1]}=\boldsymbol{L}_{n-1} \ \boldsymbol{D}_{n-1} \ \boldsymbol{L}_{n-1}^{\ast}$	$\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$	if and only if $\boldsymbol{A}_{[n-1]} = \boldsymbol{L}_{n-1} \boldsymbol{D}_{n-1} \boldsymbol{L}_{n-1}^*$, and
lyche-numerical-linear-algebra-F01289	102	■ 1.000	d_{nn}	d_{nn}	nonsingular. Also d_{nn} is real since a_{nn} and $\boldsymbol{D}_{n-1}$ are real.
lyche-numerical-linear-algebra-F01290	102	■ 1.000	a_{nn}	a_{nn}	nonsingular. Also d_{nn} is real since a_{nn} and $\boldsymbol{D}_{n-1}$ are real.
lyche-numerical-linear-algebra-F01291	102	■ 1.000	$\frac{1}{2} \ G_n$	$\frac{1}{2} G_n$	$\frac{1}{2} G_n$, half the number of operations needed for the LU factorization. Indeed, in the
lyche-numerical-linear-algebra-F01292	102	■ 1.000	$\boldsymbol{m}$	$\boldsymbol{m}$	and $\boldsymbol{m}$, while only one such system is needed to find $\boldsymbol{m}$ in the Hermitian case (4.3).
lyche-numerical-linear-algebra-F01293	102	■ 1.000	$f: \mathbb{C}^n \rightarrow \mathbb{R}$	$f: \mathbb{C}^n \rightarrow \mathbb{R}$	Given $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. The function $f: \mathbb{C}^n \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra-F01294	103	■ 0.998	$\overline{f(\boldsymbol{x})}=\overline{\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}}=\left(\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}\right)^{\ast}=\boldsymbol{x}^{\ast} \boldsymbol{A}^{\ast} \boldsymbol{x}=f(\boldsymbol{x})$	$\overline{f(\boldsymbol{x})} = \overline{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}} = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x} = f(\boldsymbol{x})$	$\overline{f(\boldsymbol{x})} = \boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = (\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{x} = f(\boldsymbol{x})$.
lyche-numerical-linear-algebra-F01295	103	■ 1.000	$\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x}>0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} > 0$	(i) positive definite if $\boldsymbol{A}^* = \boldsymbol{A}$ and $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{C}^n$;
lyche-numerical-linear-algebra-F01296	103	■ 1.000	$\boldsymbol{x}^{\ast} \boldsymbol{A} \boldsymbol{x} \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0$	(ii) positive semidefinite if $\boldsymbol{A}^* = \boldsymbol{A}$ and $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} \geq 0$ for all $\boldsymbol{x} \in \mathbb{C}^n$;
lyche-numerical-linear-algebra-F01297	103	■ 1.000	$-\boldsymbol{A}$	$-\boldsymbol{A}$	(iii) negative (semi)definite if $-\boldsymbol{A}$ is positive (semi)definite.

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lyche-numerical-linear-algebra-F01298	103	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \implies \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \implies \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0 \implies \boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra-F01299	103	0.999	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$	3. A positive definite matrix $\boldsymbol{A}$ is nonsingular. For if $\boldsymbol{A} \boldsymbol{x} = \mathbf{0}$ then $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = 0$ and
lyche-numerical-linear-algebra-F01300	103	1.000	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$	and $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra-F01301	103	1.000	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra-F01302	103	1.000	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$	$\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$ for all nonzero $\boldsymbol{x} \in \mathbb{R}^n$ then $\boldsymbol{z}^* \boldsymbol{A} \boldsymbol{z} > 0$ for
lyche-numerical-linear-algebra-F01303	103	0.999	$\boldsymbol{z} \in \mathbb{C}^n$	$\boldsymbol{z} \in \mathbb{C}^n$	all nonzero $\boldsymbol{z} \in \mathbb{C}^n$. For if $\boldsymbol{z} = \boldsymbol{x} + i \boldsymbol{y} \neq \mathbf{0}$ with $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra-F01304	103	0.999	$\boldsymbol{z} = \boldsymbol{x} + i \boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{z} = \boldsymbol{x} + i \boldsymbol{y} \neq \mathbf{0}$	all nonzero $\boldsymbol{z} \in \mathbb{C}^n$. For if $\boldsymbol{z} = \boldsymbol{x} + i \boldsymbol{y} \neq \mathbf{0}$ with $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ then
lyche-numerical-linear-algebra-F01305	103	0.991	$\boldsymbol{x}, \boldsymbol{y}$	$\boldsymbol{x}, \boldsymbol{y}$	and this is positive since at least one of the real vectors $\boldsymbol{x}, \boldsymbol{y}$ is nonzero.
lyche-numerical-linear-algebra-F01306	103	1.000	∇f	∇f	important in nonlinear optimization. Consider (cf. (16.1)) the gradient ∇f and
lyche-numerical-linear-algebra-F01307	103	0.658	$H f$	$H f$	hessian $H f$ of a function $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra-F01308	103	0.658	$f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$	$f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$	hessian $H f$ of a function $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra-F01309	103	1.000	Ω	Ω	We assume that f has continuous first and second order partial derivatives on Ω .
lyche-numerical-linear-algebra-F01310	103	1.000	$\nabla f(\boldsymbol{x}) = \mathbf{0}$	$\nabla f(\boldsymbol{x}) = \mathbf{0}$	that if $\nabla f(\boldsymbol{x}) = \mathbf{0}$ and $H f(\boldsymbol{x})$ is positive definite then $\boldsymbol{x}$ is a local minimum for f .
lyche-numerical-linear-algebra-F01311	103	1.000	$H f(\boldsymbol{x})$	$H f(\boldsymbol{x})$	that if $\nabla f(\boldsymbol{x}) = \mathbf{0}$ and $H f(\boldsymbol{x})$ is positive definite then $\boldsymbol{x}$ is a local minimum for f .
lyche-numerical-linear-algebra-F01312	103	0.933	$\boldsymbol{A}^* \boldsymbol{A}$	$\boldsymbol{A}^* \boldsymbol{A}$	Lemma 4.2 (The Matrix $\boldsymbol{A}^* \boldsymbol{A}$) The matrix $\boldsymbol{A}^* \boldsymbol{A}$ is positive semidefinite for any
lyche-numerical-linear-algebra-F01313	103	1.000	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$	$m, n \in \mathbb{N}$ and $\boldsymbol{A} \in \mathbb{C}^{m \times n}$. It is positive definite if and only if $\boldsymbol{A}$ has linearly
lyche-numerical-linear-algebra-F01314	104	0.986	$\boldsymbol{z} := \boldsymbol{A} \boldsymbol{x}$	$\boldsymbol{z} := \boldsymbol{A} \boldsymbol{x}$	Proof Clearly $\boldsymbol{A}^* \boldsymbol{A}$ is Hermitian. Let $\boldsymbol{x} \in \mathbb{C}^n$ and set $\boldsymbol{z} := \boldsymbol{A} \boldsymbol{x}$. By the definition
lyche-numerical-linear-algebra-F01315	104	1.000	$\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{z} = \ \boldsymbol{z}\ _2^2 = \ \boldsymbol{A} \boldsymbol{x}\ _2^2 \geq 0$	$\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{z} = \ \boldsymbol{z}\ _2^2 = \ \boldsymbol{A} \boldsymbol{x}\ _2^2 \geq 0$	(1.11) of the Euclidean norm we have $\boldsymbol{x}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x} = \boldsymbol{z}^* \boldsymbol{z} = \ \boldsymbol{z}\ _2^2 = \ \boldsymbol{A} \boldsymbol{x}\ _2^2 \geq 0$
lyche-numerical-linear-algebra-F01316	104	0.515	$\boldsymbol{T} =$	$\boldsymbol{T} =$	Lemma 4.3 ($\boldsymbol{T}$ Is Positive Definite) The second derivative matrix $\boldsymbol{T} =$
lyche-numerical-linear-algebra-F01317	104	0.983	$\text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n \times n}$	$\text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n \times n}$	$\text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n \times n}$ is positive definite.
lyche-numerical-linear-algebra-F01318	104	0.994	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$	Thus $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra-F01319	104	0.994	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$	$\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$	Thus $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra-F01320	104	0.994	$x_1 = x_n = 0$	$x_1 = x_n = 0$	Thus $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra-F01321	104	0.994	$x_i = x_{i+1}$	$x_i = x_{i+1}$	Thus $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} \geq 0$ and if $\boldsymbol{x}^T \boldsymbol{T} \boldsymbol{x} = 0$ then $x_1 = x_n = 0$ and $x_i = x_{i+1}$ for $i =$
lyche-numerical-linear-algebra-F01322	104	0.999	$\boldsymbol{x} = \mathbf{0}$	$\boldsymbol{x} = \mathbf{0}$	1, $\dots$, $n - 1$ which implies that $\boldsymbol{x} = \mathbf{0}$. Hence $\boldsymbol{T}$ is positive definite.
lyche-numerical-linear-algebra-F01323	104	1.000	$\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) \in \mathbb{C}^{k \times k}$	$\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) \in \mathbb{C}^{k \times k}$	Recall that a principal submatrix $\boldsymbol{B} = \boldsymbol{A}(\boldsymbol{r}, \boldsymbol{r}) \in \mathbb{C}^{k \times k}$ of a matrix $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has

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lyche-numerical-linear-algebra-F01324	104	■ 1.000	$\mathbf{b}_{\{i, j\}} = \mathbf{a}_{\{r_i\}, r_j\}}$	$b_{i,j} = a_{r_i, r_j}$	elements $b_{i,j} = a_{r_i, r_j}$ for $i, j = 1, \dots, k$, where $1 \leq r_1 < \dots < r_k \leq n$. It is a
lyche-numerical-linear-algebra-F01325	104	■ 1.000	$i, j=1, \ldots, k$	$i, j = 1, \dots, k$	elements $b_{i,j} = a_{r_i, r_j}$ for $i, j = 1, \dots, k$, where $1 \leq r_1 < \dots < r_k \leq n$. It is a
lyche-numerical-linear-algebra-F01326	104	■ 1.000	$\mathbf{r} = [1, 2, \ldots, k]^T$	$\mathbf{r} = [1, 2, \dots, k]^T$	leading principal submatrix , denoted $A_{[k]}$ if $\mathbf{r} = [1, 2, \dots, k]^T$. We have
lyche-numerical-linear-algebra-F01327	104	■ 1.000	$\mathbf{B} := \mathbf{A}(\mathbf{r}, \mathbf{r})$	$\mathbf{B} := \mathbf{A}(\mathbf{r}, \mathbf{r})$	<i>Proof</i> Let X and $\mathbf{B} := \mathbf{A}(\mathbf{r}, \mathbf{r})$ be given by (4.5). If $\mathbf{A}$ is positive semidefinite then
lyche-numerical-linear-algebra-F01328	104	■ 1.000	$\mathbf{y}^* \mathbf{B} \mathbf{y} = 0$	$\mathbf{y}^* \mathbf{B} \mathbf{y} = 0$	Suppose $\mathbf{A}$ is positive definite and $\mathbf{y}^* \mathbf{B} \mathbf{y} = 0$. By (4.6) we have $\mathbf{x} = \mathbf{0}$ and since
lyche-numerical-linear-algebra-F01329	105	■ 1.000	$\mathbf{A}^{-*} := \left(\mathbf{A}^{-1} \right)^* = \left(\mathbf{A}^* \right)^{-1}$	$\mathbf{A}^{-*} := \left(\mathbf{A}^{-1} \right)^* = \left(\mathbf{A}^* \right)^{-1}$	<i>Proof</i> Recall that $\mathbf{A}^{-*} := (\mathbf{A}^{-1})^* = (\mathbf{A}^*)^{-1}$.
lyche-numerical-linear-algebra-F01330	105	■ 1.000	$1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$	$1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$	We show that $1 \Rightarrow 2 \Rightarrow 3 \Rightarrow 1$.
lyche-numerical-linear-algebra-F01331	105	■ 0.999	$1 \Rightarrow$	$1 \Rightarrow$	$1 \Rightarrow 2$: Suppose $\mathbf{A}$ is positive definite. By Lemma 4.4 the leading principal
lyche-numerical-linear-algebra-F01332	105	■ 1.000	$\mathbf{x}_i := \mathbf{L}^{-*} \mathbf{e}_i$	$\mathbf{x}_i := \mathbf{L}^{-*} \mathbf{e}_i$	is positive we note that $\mathbf{x}_i := \mathbf{L}^{-*} \mathbf{e}_i$ is nonzero since $\mathbf{L}^{-*}$ is nonsingular. But
lyche-numerical-linear-algebra-F01333	105	■ 1.000	$\mathbf{L}^{-*}$	$\mathbf{L}^{-*}$	is positive we note that $\mathbf{x}_i := \mathbf{L}^{-*} \mathbf{e}_i$ is nonzero since $\mathbf{L}^{-*}$ is nonsingular. But
lyche-numerical-linear-algebra-F01334	105	■ 1.000	$d_{ii} = \mathbf{e}_i^* \mathbf{D} \mathbf{e}_i = \mathbf{e}_i^* \mathbf{L}^{-1} \mathbf{A} \mathbf{L}^{-*} \mathbf{e}_i = \mathbf{x}_i^* \mathbf{A} \mathbf{x}_i > 0$	$d_{ii} = \mathbf{e}_i^* \mathbf{D} \mathbf{e}_i = \mathbf{e}_i^* \mathbf{L}^{-1} \mathbf{A} \mathbf{L}^{-*} \mathbf{e}_i = \mathbf{x}_i^* \mathbf{A} \mathbf{x}_i > 0$	then $d_{ii} = \mathbf{e}_i^* \mathbf{D} \mathbf{e}_i = \mathbf{e}_i^* \mathbf{L}^{-1} \mathbf{A} \mathbf{L}^{-*} \mathbf{e}_i = \mathbf{x}_i^* \mathbf{A} \mathbf{x}_i > 0$ since $\mathbf{A}$ is positive definite.
lyche-numerical-linear-algebra-F01335	105	■ 0.913	$2 \Rightarrow$	$2 \Rightarrow$	$2 \Rightarrow 3$: Suppose $\mathbf{A}$ has an LDL* factorization $\mathbf{A} = \mathbf{L} \mathbf{D} \mathbf{L}^*$ with positive
lyche-numerical-linear-algebra-F01336	105	■ 0.913	$\mathbf{L} \mathbf{D} \mathbf{L}^*$	$\mathbf{L} \mathbf{D} \mathbf{L}^*$	$2 \Rightarrow 3$: Suppose $\mathbf{A}$ has an LDL* factorization $\mathbf{A} = \mathbf{L} \mathbf{D} \mathbf{L}^*$ with positive
lyche-numerical-linear-algebra-F01337	105	■ 1.000	d_{ii}	d_{ii}	diagonal elements d_{ii} in $\mathbf{D}$. Then $\mathbf{A} = \mathbf{S} \mathbf{S}^*$, where $\mathbf{S} := \mathbf{L} \mathbf{D}^{1/2}$ and $\mathbf{D}^{1/2} :=$
lyche-numerical-linear-algebra-F01338	105	■ 1.000	$\mathbf{A} = \mathbf{S} \mathbf{S}^*$	$\mathbf{A} = \mathbf{S} \mathbf{S}^*$	diagonal elements d_{ii} in $\mathbf{D}$. Then $\mathbf{A} = \mathbf{S} \mathbf{S}^*$, where $\mathbf{S} := \mathbf{L} \mathbf{D}^{1/2}$ and $\mathbf{D}^{1/2} :=$
lyche-numerical-linear-algebra-F01339	105	■ 1.000	$\mathbf{S} := \mathbf{L} \mathbf{D}^{1/2}$	$\mathbf{S} := \mathbf{L} \mathbf{D}^{1/2}$	diagonal elements d_{ii} in $\mathbf{D}$. Then $\mathbf{A} = \mathbf{S} \mathbf{S}^*$, where $\mathbf{S} := \mathbf{L} \mathbf{D}^{1/2}$ and $\mathbf{D}^{1/2} :=$
lyche-numerical-linear-algebra-F01340	105	■ 1.000	$\mathbf{D}^{1/2} :=$	$\mathbf{D}^{1/2} :=$	diagonal elements d_{ii} in $\mathbf{D}$. Then $\mathbf{A} = \mathbf{S} \mathbf{S}^*$, where $\mathbf{S} := \mathbf{L} \mathbf{D}^{1/2}$ and $\mathbf{D}^{1/2} :=$
lyche-numerical-linear-algebra-F01341	105	■ 1.000	$\operatorname{diag}(\sqrt{d_{11}}, \ldots, \sqrt{d_{nn}})$	$\operatorname{diag}(\sqrt{d_{11}}, \ldots, \sqrt{d_{nn}})$	$\operatorname{diag}(\sqrt{d_{11}}, \ldots, \sqrt{d_{nn}})$, and this is a Cholesky factorization of $\mathbf{A}$.
lyche-numerical-linear-algebra-F01342	105	■ 1.000	$3 \Rightarrow$	$3 \Rightarrow$	$3 \Rightarrow 1$: Suppose $\mathbf{A}$ has a Cholesky factorization $\mathbf{A} = \mathbf{L} \mathbf{L}^*$. Clearly $\mathbf{A}^* = \mathbf{A}$.
lyche-numerical-linear-algebra-F01343	105	■ 1.000	$\mathbf{A} = \mathbf{L} \mathbf{L}^*$	$\mathbf{A} = \mathbf{L} \mathbf{L}^*$	$3 \Rightarrow 1$: Suppose $\mathbf{A}$ has a Cholesky factorization $\mathbf{A} = \mathbf{L} \mathbf{L}^*$. Clearly $\mathbf{A}^* = \mathbf{A}$.
lyche-numerical-linear-algebra-F01344	105	■ 1.000	$\mathbf{L} \mathbf{L}^* = \mathbf{S} \mathbf{S}^*$	$\mathbf{L} \mathbf{L}^* = \mathbf{S} \mathbf{S}^*$	For uniqueness suppose $\mathbf{L} \mathbf{L}^* = \mathbf{S} \mathbf{S}^*$ are two Cholesky factorizations of the
lyche-numerical-linear-algebra-F01345	105	■ 1.000	$\mathbf{S}^{-1} \mathbf{L} = \mathbf{S}^* \mathbf{L}^{-*}$	$\mathbf{S}^{-1} \mathbf{L} = \mathbf{S}^* \mathbf{L}^{-*}$	Then $\mathbf{S}^{-1} \mathbf{L} = \mathbf{S}^* \mathbf{L}^{-*}$, where by Lemma 2.5 $\mathbf{S}^{-1} \mathbf{L}$ is lower triangular and $\mathbf{S}^* \mathbf{L}^{-*}$
lyche-numerical-linear-algebra-F01346	105	■ 1.000	$2.5 \mathbf{S}^{-1} \mathbf{L}$	$2.5 \mathbf{S}^{-1} \mathbf{L}$	Then $\mathbf{S}^{-1} \mathbf{L} = \mathbf{S}^* \mathbf{L}^{-*}$, where by Lemma 2.5 $\mathbf{S}^{-1} \mathbf{L}$ is lower triangular and $\mathbf{S}^* \mathbf{L}^{-*}$
lyche-numerical-linear-algebra-F01347	105	■ 1.000	$\mathbf{S}^* \mathbf{L}^{-*}$	$\mathbf{S}^* \mathbf{L}^{-*}$	Then $\mathbf{S}^{-1} \mathbf{L} = \mathbf{S}^* \mathbf{L}^{-*}$, where by Lemma 2.5 $\mathbf{S}^{-1} \mathbf{L}$ is lower triangular and $\mathbf{S}^* \mathbf{L}^{-*}$
lyche-numerical-linear-algebra-F01348	105	■ 1.000	ℓ_{ii} / s_{ii}	ℓ_{ii} / s_{ii}	is upper triangular, with diagonal elements ℓ_{ii} / s_{ii} and s_{ii} / ℓ_{ii} , respectively. But then

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lyche-numerical-linear-algebra- F01349	105	1.000	$s_{\{i\ i\}} / \ell_{\{i\ i\}}$	s_{ii}/ℓ_{ii}	is upper triangular, with diagonal elements ℓ_{ii}/s_{ii} and s_{ii}/ℓ_{ii} , respectively. But then
lyche-numerical-linear-algebra- F01350	105	1.000	$\ell_{\{i\ i\}}^2 = s_{\{i\ i\}}^2$	$\ell_{ii}^2 = s_{ii}^2$	both matrices must be equal to the same diagonal matrix and $\ell_{ii}^2 = s_{ii}^2$. By positivity
lyche-numerical-linear-algebra- F01351	105	1.000	$\ell_{\{i\ i\}} = s_{\{i\ i\}}$	$\ell_{ii} = s_{ii}$	$\ell_{ii} = s_{ii}$ and we conclude that $S^{-1}L = I = S^*L^{-*}$ which means that $L = S$.
lyche-numerical-linear-algebra- F01352	105	1.000	$\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$ $\boldsymbol{S}^{-1} \boldsymbol{L} = \boldsymbol{I} = \boldsymbol{S}^* \boldsymbol{L}^{-*}$	$S^{-1}L = I = S^*L^{-*}$	$\ell_{ii} = s_{ii}$ and we conclude that $S^{-1}L = I = S^*L^{-*}$ which means that $L = S$.
lyche-numerical-linear-algebra- F01353	105	1.000	$\boldsymbol{L} = \boldsymbol{S}$	$L = S$	$\ell_{ii} = s_{ii}$ and we conclude that $S^{-1}L = I = S^*L^{-*}$ which means that $L = S$.
lyche-numerical-linear-algebra- F01354	105	0.993	$\boldsymbol{A} = \boldsymbol{R}^* \boldsymbol{R}$	$A = R^*R$	A Cholesky factorization can also be written in the equivalent form $A = R^*R$,
lyche-numerical-linear-algebra- F01355	105	1.000	$\boldsymbol{R} = \boldsymbol{L}^*$	$R = L^*$	where $R = L^*$ is upper triangular with positive diagonal elements.
lyche-numerical-linear-algebra- F01356	105	0.993	4.4(2 $\times$ 2)	4.4(2 $\times$ 2)	<i>Example 4.4</i> (2 $\times$ 2) The matrix $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has an LDL* and a Cholesky-
lyche-numerical-linear-algebra- F01357	105	0.993	$\boldsymbol{A} = \left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array} \right]$	$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$	<i>Example 4.4</i> (2 $\times$ 2) The matrix $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has an LDL* and a Cholesky-
lyche-numerical-linear-algebra- F01358	106	1.000	$\frac{1}{2} G_n = n^3 / 3$	$\frac{1}{2}G_n = n^3/3$	count $\frac{1}{2}G_n = n^3/3$ for a full matrix.
lyche-numerical-linear-algebra- F01359	106	1.000	$a_{\{i\ i\}} > 0, \left(a_{\{i\ i\}} \geq 0 \right)$	$a_{ii} > 0, (a_{ii} \geq 0)$	1. $a_{ii} > 0, (a_{ii} \geq 0)$,
lyche-numerical-linear-algebra- F01360	106	1.000	$\left \operatorname{Re} \left(a_{\{i\ j\}} \right) \right < \left(a_{\{i\ i\}} + a_{\{j\ j\}} \right) / 2, \left(\left \operatorname{Re} \left(a_{\{i\ j\}} \right) \right \leq \left(a_{\{i\ i\}} + a_{\{j\ j\}} \right) / 2 \right)$	$ \operatorname{Re}(a_{ij}) < (a_{ii} + a_{jj})/2, (\operatorname{Re}(a_{ij}) \leq (a_{ii} + a_{jj})/2)$	2. $ \operatorname{Re}(a_{ij}) < (a_{ii} + a_{jj})/2, (\operatorname{Re}(a_{ij}) \leq (a_{ii} + a_{jj})/2)$,
lyche-numerical-linear-algebra- F01361	106	1.000	$\left a_{\{i\ j\}} \right < \sqrt{a_{\{i\ i\}} a_{\{j\ j\}}}, \left(\left a_{\{i\ j\}} \right \leq \sqrt{a_{\{i\ i\}} a_{\{j\ j\}}} \right)$	$ a_{ij} < \sqrt{a_{ii}a_{jj}}, (a_{ij} \leq \sqrt{a_{ii}a_{jj}})$	3. $ a_{ij} < \sqrt{a_{ii}a_{jj}}, (a_{ij} \leq \sqrt{a_{ii}a_{jj}})$,
lyche-numerical-linear-algebra- F01362	106	1.000	$a_{\{i\ i\}} = 0$	$a_{ii} = 0$	4. If A is positive semidefinite and $a_{ii} = 0$ for some i then $a_{ij} = a_{ji} = 0$ for
lyche-numerical-linear-algebra- F01363	106	1.000	$a_{\{i\ j\}} = a_{\{j\ i\}} = 0$	$a_{ij} = a_{ji} = 0$	4. If A is positive semidefinite and $a_{ii} = 0$ for some i then $a_{ij} = a_{ji} = 0$ for
lyche-numerical-linear-algebra- F01364	106	1.000	$a_{\{i\ i\}} = \boldsymbol{e}_{\{i\}}^T \boldsymbol{A} \boldsymbol{e}_{\{i\}} > (\geq) 0$	$a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$	Proof Clearly $a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra- F01365	106	1.000	$\alpha, \beta \in \mathbb{C}$	$\alpha, \beta \in \mathbb{C}$	Proof Clearly $a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra- F01366	106	1.000	$\alpha \boldsymbol{e}_{\{i\}} + \beta \boldsymbol{e}_{\{j\}} \neq$	$\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$	Proof Clearly $a_{ii} = \boldsymbol{e}_i^T \boldsymbol{A} \boldsymbol{e}_i > (\geq) 0$ and Part 1 follows. If $\alpha, \beta \in \mathbb{C}$ and $\alpha \boldsymbol{e}_i + \beta \boldsymbol{e}_j \neq$
lyche-numerical-linear-algebra- F01367	106	0.999	$\alpha = 1, \beta = \pm 1$	$\alpha = 1, \beta = \pm 1$	Taking $\alpha = 1, \beta = \pm 1$ we obtain $a_{ii} + a_{jj} \pm 2 \operatorname{Re} a_{ij} > 0$ and this implies Part 2.
lyche-numerical-linear-algebra- F01368	106	0.999	$a_{\{i\ i\}} + a_{\{j\ j\}} \pm 2 \operatorname{Re} a_{\{i\ j\}} > 0$	$a_{ii} + a_{jj} \pm 2 \operatorname{Re} a_{ij} > 0$	Taking $\alpha = 1, \beta = \pm 1$ we obtain $a_{ii} + a_{jj} \pm 2 \operatorname{Re} a_{ij} > 0$ and this implies Part 2.
lyche-numerical-linear-algebra- F01369	106	0.834	$\alpha = -a_{\{i\ j\}}, \beta = a_{\{i\ i\}}$	$\alpha = -a_{ij}, \beta = a_{ii}$	We first show 3. when A is positive definite. Taking $\alpha = -a_{ij}, \beta = a_{ii}$ in (4.7) we
lyche-numerical-linear-algebra- F01370	106	1.000	$a_{\{i\ i\}} > 0$	$a_{ii} > 0$	Since $a_{ii} > 0$ Part 3 follows in the positive definite case.
lyche-numerical-linear-algebra- F01371	106	1.000	$\varepsilon > 0$	$\varepsilon > 0$	Suppose now A is positive semidefinite. For $\varepsilon > 0$ we define $\boldsymbol{B} := \boldsymbol{A} + \varepsilon \boldsymbol{I}$. The
lyche-numerical-linear-algebra- F01372	106	1.000	$\boldsymbol{B} := \boldsymbol{A} + \varepsilon \boldsymbol{I}$	$\boldsymbol{B} := \boldsymbol{A} + \varepsilon \boldsymbol{I}$	Suppose now A is positive semidefinite. For $\varepsilon > 0$ we define $\boldsymbol{B} := \boldsymbol{A} + \varepsilon \boldsymbol{I}$. The
lyche-numerical-linear-algebra- F01373	106	1.000	$\boldsymbol{x}^* \boldsymbol{B} \boldsymbol{x} \geq \varepsilon \ \boldsymbol{x}\ _2^2 > 0$	$\boldsymbol{x}^* \boldsymbol{B} \boldsymbol{x} \geq \varepsilon \ \boldsymbol{x}\ _2^2 > 0$	matrix $\boldsymbol{B}$ is positive definite since it is Hermitian and $\boldsymbol{x}^* \boldsymbol{B} \boldsymbol{x} \geq \varepsilon \ \boldsymbol{x}\ _2^2 > 0$ for any

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lyche-numerical-linear-algebra-F01374	107	1.000	$\left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$	$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	The matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ enjoys all the necessary conditions in Theorem 4.3. But to
lyche-numerical-linear-algebra-F01375	107	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$	if $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ and $\boldsymbol{x}$ is nonzero, then $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$. This implies that $\lambda > 0 (\geq 0)$
lyche-numerical-linear-algebra-F01376	107	1.000	$\lambda > 0 (\geq 0)$	$\lambda > 0 (\geq 0)$	if $\boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}$ and $\boldsymbol{x}$ is nonzero, then $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{x}^* \boldsymbol{x}$. This implies that $\lambda > 0 (\geq 0)$
lyche-numerical-linear-algebra-F01377	107	1.000	$\boldsymbol{x}^* \boldsymbol{x} = \ \boldsymbol{x}\ _2^2 > 0$	$\boldsymbol{x}^* \boldsymbol{x} = \ \boldsymbol{x}\ _2^2 > 0$	since $\boldsymbol{A}$ is positive (semi)definite and $\boldsymbol{x}^* \boldsymbol{x} = \ \boldsymbol{x}\ _2^2 > 0$. Conversely, suppose
lyche-numerical-linear-algebra-F01378	107	1.000	$\boldsymbol{U}^* \boldsymbol{U} =$	$\boldsymbol{U}^* \boldsymbol{U} =$	Theorem 6.9 (the spectral theorem) there is a matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{U}^* \boldsymbol{U} =$
lyche-numerical-linear-algebra-F01379	107	1.000	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra-F01380	107	1.000	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra-F01381	107	1.000	$\boldsymbol{z} :=$	$\boldsymbol{z} :=$	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$ such that $\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\boldsymbol{x} \in \mathbb{C}^n$ and define $\boldsymbol{z} :=$
lyche-numerical-linear-algebra-F01382	107	1.000	$\boldsymbol{U}^* \boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$	$\boldsymbol{U}^* \boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$	$\boldsymbol{U}^* \boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$. Then $\boldsymbol{x} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{x} = \boldsymbol{U} \boldsymbol{z}$ and by the spectral theorem
lyche-numerical-linear-algebra-F01383	107	1.000	$\boldsymbol{x} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{x} = \boldsymbol{U} \boldsymbol{z}$	$\boldsymbol{x} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{x} = \boldsymbol{U} \boldsymbol{z}$	$\boldsymbol{U}^* \boldsymbol{x} = [z_1, \dots, z_n]^T \in \mathbb{C}^n$. Then $\boldsymbol{x} = \boldsymbol{U} \boldsymbol{U}^* \boldsymbol{x} = \boldsymbol{U} \boldsymbol{z}$ and by the spectral theorem
lyche-numerical-linear-algebra-F01384	107	0.982	$\boldsymbol{U}^*$	$\boldsymbol{U}^*$	It follows that $\boldsymbol{A}$ is positive semidefinite. Since $\boldsymbol{U}^*$ is nonsingular we see that $\boldsymbol{z} =$
lyche-numerical-linear-algebra-F01385	107	0.982	$\boldsymbol{z} =$	$\boldsymbol{z} =$	It follows that $\boldsymbol{A}$ is positive semidefinite. Since $\boldsymbol{U}^*$ is nonsingular we see that $\boldsymbol{z} =$
lyche-numerical-linear-algebra-F01386	107	1.000	$\boldsymbol{U}^* \boldsymbol{x}$	$\boldsymbol{U}^* \boldsymbol{x}$	$\boldsymbol{U}^* \boldsymbol{x}$ is nonzero if $\boldsymbol{x}$ is nonzero, and therefore $\boldsymbol{A}$ is positive definite.
lyche-numerical-linear-algebra-F01387	108	1.000	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$	4. $\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$ for a nonsingular $\boldsymbol{B} \in \mathbb{C}^{n \times n}$.
lyche-numerical-linear-algebra-F01388	108	0.793	$\boldsymbol{B} = \boldsymbol{L}$	$\boldsymbol{B} = \boldsymbol{L}$	and by Theorem 4.2 a unique Cholesky factorization $\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$ with $\boldsymbol{B} = \boldsymbol{L}$.
lyche-numerical-linear-algebra-F01389	108	0.973	$4 \Rightarrow$	$4 \Rightarrow$	$4 \Rightarrow 1$: This follows from Lemma 4.2.
lyche-numerical-linear-algebra-F01390	108	1.000	$\boldsymbol{A} := \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.
lyche-numerical-linear-algebra-F01391	108	0.991	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + 2x_2^2 + (x_1 + x_2)^2 > 0$	$\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + 2x_2^2 + (x_1 + x_2)^2 > 0$	1. We have $\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} = 2x_1^2 + 2x_2^2 + (x_1 + x_2)^2 > 0$ for all nonzero $\boldsymbol{x}$ showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra-F01392	108	1.000	$\lambda_1 = 2$	$\lambda_1 = 2$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 4$. They are positive showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra-F01393	108	1.000	$\lambda_2 = 4$	$\lambda_2 = 4$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 4$. They are positive showing that $\boldsymbol{A}$
lyche-numerical-linear-algebra-F01394	108	0.998	$\det(\boldsymbol{A}_{[1]}) = 3$	$\det(\boldsymbol{A}_{[1]}) = 3$	3. We find $\det(\boldsymbol{A}_{[1]}) = 3$ and $\det(\boldsymbol{A}_{[2]}) = 8$ showing again that $\boldsymbol{A}$ is positive definite
lyche-numerical-linear-algebra-F01395	108	0.998	$\det(\boldsymbol{A}_{[2]}) = 8$	$\det(\boldsymbol{A}_{[2]}) = 8$	3. We find $\det(\boldsymbol{A}_{[1]}) = 3$ and $\det(\boldsymbol{A}_{[2]}) = 8$ showing again that $\boldsymbol{A}$ is positive definite
lyche-numerical-linear-algebra-F01396	109	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \ \boldsymbol{L}^* \boldsymbol{x}\ _2^2 \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \ \boldsymbol{L}^* \boldsymbol{x}\ _2^2 \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \ \boldsymbol{L}^* \boldsymbol{x}\ _2^2 \geq 0$ and $\boldsymbol{A}$ is positive semidefinite. For the converse we use
lyche-numerical-linear-algebra-F01397	109	1.000	$\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$	$\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$	There are two cases. Suppose first $\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$. We claim that $\boldsymbol{C} := \boldsymbol{B} -$
lyche-numerical-linear-algebra-F01398	109	1.000	$\boldsymbol{C} := \boldsymbol{B} -$	$\boldsymbol{C} := \boldsymbol{B} -$	There are two cases. Suppose first $\alpha = \boldsymbol{e}_1^* \boldsymbol{A} \boldsymbol{e}_1 > 0$. We claim that $\boldsymbol{C} := \boldsymbol{B} -$
lyche-numerical-linear-algebra-F01399	109	1.000	$\boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\boldsymbol{v} \boldsymbol{v}^* / \alpha$ is positive semidefinite. $\boldsymbol{C}$ is Hermitian since $\boldsymbol{B}$ is. To show that $\boldsymbol{C}$ is positive

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lyche-numerical-linear-algebra- F01400	109	1.000	$\boldsymbol{y} \in \mathbb{C}^{n-1}$	$\boldsymbol{y} \in \mathbb{C}^{n-1}$	semidefinite we consider any $\boldsymbol{y} \in \mathbb{C}^{n-1}$ and define $\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$. Then
lyche-numerical-linear-algebra- F01401	109	1.000	$\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$	$\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$	semidefinite we consider any $\boldsymbol{y} \in \mathbb{C}^{n-1}$ and define $\boldsymbol{x}^* := [-\boldsymbol{y}^* \boldsymbol{v} / \alpha, \boldsymbol{y}^*] \in \mathbb{C}^n$. Then
lyche-numerical-linear-algebra- F01402	109	1.000	$\boldsymbol{C} \in \mathbb{C}^{(n-1) \times (n-1)}$	$\boldsymbol{C} \in \mathbb{C}^{(n-1) \times (n-1)}$	So $\boldsymbol{C} \in \mathbb{C}^{(n-1) \times (n-1)}$ is positive semidefinite and by the induction hypothesis it has
lyche-numerical-linear-algebra- F01403	109	1.000	$\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	$\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	a semi-Cholesky factorization $\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$. The matrix
lyche-numerical-linear-algebra- F01404	109	1.000	$\alpha = 0$	$\alpha = 0$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\boldsymbol{v} = \mathbf{0}$. Moreover, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra- F01405	109	1.000	$\boldsymbol{v} = \mathbf{0}$	$\boldsymbol{v} = \mathbf{0}$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\boldsymbol{v} = \mathbf{0}$. Moreover, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra- F01406	109	1.000	$\boldsymbol{B} \in$	$\boldsymbol{B} \in$	If $\alpha = 0$ then part 4 of Theorem 4.3 implies that $\boldsymbol{v} = \mathbf{0}$. Moreover, $\boldsymbol{B} \in$
lyche-numerical-linear-algebra- F01407	109	1.000	$\mathbb{C}^{(n-1) \times (n-1)}$	$\mathbb{C}^{(n-1) \times (n-1)}$	$\mathbb{C}^{(n-1) \times (n-1)}$ in (4.8) is positive semidefinite and therefore has a semi-Cholesky
lyche-numerical-linear-algebra- F01408	110	0.554	$\boldsymbol{B} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	$\boldsymbol{B} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	factorization $\boldsymbol{B} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$. But then $\boldsymbol{L} \boldsymbol{L}^*$, where $\boldsymbol{L} = \begin{bmatrix} \mathbf{0} & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra- F01409	110	0.554	$\boldsymbol{L}^*$	$\boldsymbol{L}^*$	factorization $\boldsymbol{B} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$. But then $\boldsymbol{L} \boldsymbol{L}^*$, where $\boldsymbol{L} = \begin{bmatrix} \mathbf{0} & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra- F01410	110	0.554	$\boldsymbol{L} = \begin{bmatrix} \mathbf{0} & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$	$\boldsymbol{L} = \begin{bmatrix} \mathbf{0} & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$	factorization $\boldsymbol{B} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$. But then $\boldsymbol{L} \boldsymbol{L}^*$, where $\boldsymbol{L} = \begin{bmatrix} \mathbf{0} & \mathbf{0}^* \\ \mathbf{0} & \boldsymbol{L}_1 \end{bmatrix}$ is a semi-Cholesky
lyche-numerical-linear-algebra- F01411	110	1.000	$\boldsymbol{v}^* = [\boldsymbol{u}^*, \mathbf{0}^*]$	$\boldsymbol{v}^* = [\boldsymbol{u}^*, \mathbf{0}^*]$	Proof Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is d -banded. Then $\boldsymbol{v}^* = [\boldsymbol{u}^*, \mathbf{0}^*]$ in (4.8), where $\boldsymbol{u} \in \mathbb{C}^d$,
lyche-numerical-linear-algebra- F01412	110	1.000	$\boldsymbol{u} \in \mathbb{C}^d$	$\boldsymbol{u} \in \mathbb{C}^d$	Proof Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ is d -banded. Then $\boldsymbol{v}^* = [\boldsymbol{u}^*, \mathbf{0}^*]$ in (4.8), where $\boldsymbol{u} \in \mathbb{C}^d$,
lyche-numerical-linear-algebra- F01413	110	1.000	$\boldsymbol{C} := \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\boldsymbol{C} := \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$	and therefore $\boldsymbol{C} := \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$ differs from $\boldsymbol{B}$ only in the upper left $d \times d$ corner. It
lyche-numerical-linear-algebra- F01414	110	1.000	$d \times d$	$d \times d$	and therefore $\boldsymbol{C} := \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$ differs from $\boldsymbol{B}$ only in the upper left $d \times d$ corner. It
lyche-numerical-linear-algebra- F01415	110	1.000	$n, \boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	$n, \boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$	follows that $\boldsymbol{C}$ has the same bandwidth as $\boldsymbol{B}$ and $\boldsymbol{A}$. By induction on n , $\boldsymbol{C} = \boldsymbol{L}_1 \boldsymbol{L}_1^*$,
lyche-numerical-linear-algebra- F01416	110	0.999	$\boldsymbol{L}_1^*$	$\boldsymbol{L}_1^*$	where $\boldsymbol{L}_1^*$ has the same bandwidth as $\boldsymbol{C}$. But then $\boldsymbol{L}$ in (4.10) has the same bandwidth
lyche-numerical-linear-algebra- F01417	110	1.000	$[\beta, \boldsymbol{v}^* / \beta]^*$	$[\beta, \boldsymbol{v}^* / \beta]^*$	of $\boldsymbol{L}$ is $[\beta, \boldsymbol{v}^* / \beta]^*$ if $\alpha > 0$. If $\alpha = 0$ then by 4 in Theorem 4.3 the first column of
lyche-numerical-linear-algebra- F01418	110	1.000	$\alpha > 0$	$\alpha > 0$	of $\boldsymbol{L}$ is $[\beta, \boldsymbol{v}^* / \beta]^*$ if $\alpha > 0$. If $\alpha = 0$ then by 4 in Theorem 4.3 the first column of
lyche-numerical-linear-algebra- F01419	110	0.947	$\boldsymbol{C} = \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\boldsymbol{C} = \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$	$\boldsymbol{C} = \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$ in the lower part of $\boldsymbol{A}(2 : n, 2 : n)$.
lyche-numerical-linear-algebra- F01420	110	0.947	$\boldsymbol{A}(2 : n, 2 : n)$	$\boldsymbol{A}(2 : n, 2 : n)$	$\boldsymbol{C} = \boldsymbol{B} - \boldsymbol{v} \boldsymbol{v}^* / \alpha$ in the lower part of $\boldsymbol{A}(2 : n, 2 : n)$.
lyche-numerical-linear-algebra- F01421	110	1.000	$\min(i + d, n)$	$\min(i + d, n)$	replace all occurrences of n by $\min(i + d, n)$. Continuing the reduction we arrive
lyche-numerical-linear-algebra- F01422	111	1.000	$\ell_{kk} = 0$	$\ell_{kk} = 0$	Column k of $\boldsymbol{L}$ is zero for those k where $\ell_{kk} = 0$. We reduce round-off noise by
lyche-numerical-linear-algebra- F01423	111	0.910	$1 \Longleftarrow 2$	$1 \Longleftrightarrow 2$	$1 \Longleftrightarrow 2$: This follows from Lemma 4.5.
lyche-numerical-linear-algebra- F01424	111	0.858	$1 \Longleftarrow$	$1 \Longleftrightarrow$	$1 \Longleftrightarrow 4$: This follows from Theorem 4.5

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lyche-numerical-linear-algebra- F01425	111	0.933	$1 \Longleftarrow 3$	$1 \iff 3$	$1 \iff 3$: We refer to page 567 of [15], where it is shown that $4 \implies 3$ (and
lyche-numerical-linear-algebra- F01426	111	0.933	$4 \longrightarrow 3$	$4 \implies 3$	$1 \iff 3$: We refer to page 567 of [15], where it is shown that $4 \implies 3$ (and
lyche-numerical-linear-algebra- F01427	111	0.835	$1 \longrightarrow 3$	$1 \implies 3$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra- F01428	111	0.835	$1 \Longleftrightarrow 4$	$1 \iff 4$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra- F01429	111	0.835	$3 \longrightarrow 1$	$3 \implies 1$	therefore $1 \implies 3$ since $1 \iff 4$) and $3 \implies 1$.
lyche-numerical-linear-algebra- F01430	112	0.997	$\boldsymbol{A} := \left[\begin{array}{ll} 1 & 1 \\ 1 & 1 \end{array} \right]$	$\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	matrix $\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F01431	112	0.996	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = x_1^2 + x_2^2 + x_1 x_2 + x_2 x_1 = (x_1 + x_2)^2 \geq 0$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = x_1^2 + x_2^2 + x_1 x_2 + x_2 x_1 = (x_1 + x_2)^2 \geq 0$	1. We have $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = x_1^2 + x_2^2 + x_1 x_2 + x_2 x_1 = (x_1 + x_2)^2 \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^2$
lyche-numerical-linear-algebra- F01432	112	0.996	$\boldsymbol{x} \in \mathbb{R}^2$	$\boldsymbol{x} \in \mathbb{R}^2$	1. We have $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = x_1^2 + x_2^2 + x_1 x_2 + x_2 x_1 = (x_1 + x_2)^2 \geq 0$ for all $\boldsymbol{x} \in \mathbb{R}^2$
lyche-numerical-linear-algebra- F01433	112	1.000	$\lambda_2 = 0$	$\lambda_2 = 0$	2. The eigenvalues of $\boldsymbol{A}$ are $\lambda_1 = 2$ and $\lambda_2 = 0$ and they are nonnegative showing
lyche-numerical-linear-algebra- F01434	112	0.998	$\operatorname{det}(\left[\begin{array}{c} a_{11} \end{array} \right]) =$	$\det([a_{11}]) =$	3. There are three principal sub matrices, and they have determinants $\det([a_{11}]) =$
lyche-numerical-linear-algebra- F01435	112	0.971	$\operatorname{det}(\left[\begin{array}{c} a_{22} \end{array} \right]) = 1$	$\det([a_{22}]) = 1$	1, $\det([a_{22}]) = 1$ and $\det(\boldsymbol{A}) = 0$ and showing again that $\boldsymbol{A}$ is positive
lyche-numerical-linear-algebra- F01436	112	0.575	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$	$\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$	4. Finally $\boldsymbol{A}$ is positive semidefinite since $\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$, where $\boldsymbol{B} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$.
lyche-numerical-linear-algebra- F01437	112	0.575	$\boldsymbol{B} = \left[\begin{array}{ll} 1 & 0 \\ 1 & 0 \end{array} \right]$	$\boldsymbol{B} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	4. Finally $\boldsymbol{A}$ is positive semidefinite since $\boldsymbol{A} = \boldsymbol{B} \boldsymbol{B}^*$, where $\boldsymbol{B} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$.
lyche-numerical-linear-algebra- F01438	112	0.546	$\boldsymbol{A} := \left[\begin{array}{rr} 0 & 0 \\ 0 & -1 \end{array} \right]$	$\boldsymbol{A} := \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$	is not enough consider the matrix $\boldsymbol{A} := \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$. The leading principal minors are
lyche-numerical-linear-algebra- F01439	112	1.000	$\operatorname{det}(\boldsymbol{A}) > 0$	$\det(\boldsymbol{A}) > 0$	4. $\det(\boldsymbol{A}) > 0$,
lyche-numerical-linear-algebra- F01440	112	1.000	$a_{ii} a_{jj} > a_{ij} a_{ji}$	$a_{ii} a_{jj} > a_{ij} a_{ji}$	5. $a_{ii} a_{jj} > a_{ij} a_{ji}$, for $i \neq j$.
lyche-numerical-linear-algebra- F01441	113	0.752	$\lambda = \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} > 0$	$\lambda = \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} > 0$	$\lambda = \frac{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^T \boldsymbol{x}} > 0$.
lyche-numerical-linear-algebra- F01442	113	0.981	$\left[\begin{array}{cc} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{array} \right]$	$\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$	5. The principal submatrix $\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$ has a positive determinant.
lyche-numerical-linear-algebra- F01443	113	1.000	$a \in \mathbb{R}$	$a \in \mathbb{R}$	is positive definite for any $a \in \mathbb{R}$. Indeed, for any nonzero $\boldsymbol{x} \in \mathbb{R}^2$
lyche-numerical-linear-algebra- F01444	113	1.000	$\boldsymbol{A}[a]$	$\boldsymbol{A}[a]$	The eigenvalues of $\boldsymbol{A}[a]$ are positive for $a \in [1 - \frac{\sqrt{5}}{2}, 1 + \frac{\sqrt{5}}{2}]$ and complex for
lyche-numerical-linear-algebra- F01445	113	1.000	$a \in \left[1 - \frac{\sqrt{5}}{2}, 1 + \frac{\sqrt{5}}{2} \right]$	$a \in \left[1 - \frac{\sqrt{5}}{2}, 1 + \frac{\sqrt{5}}{2} \right]$	The eigenvalues of $\boldsymbol{A}[a]$ are positive for $a \in [1 - \frac{\sqrt{5}}{2}, 1 + \frac{\sqrt{5}}{2}]$ and complex for
lyche-numerical-linear-algebra- F01446	113	1.000	a	a	other values of a .
lyche-numerical-linear-algebra- F01447	114	1.000	$\boldsymbol{A} := \left[\begin{array}{cc} 1 & 0 \\ -2 & 1 \end{array} \right]$	$\boldsymbol{A} := \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$	the matrix $\boldsymbol{A} := \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F01448	114	1.000	$\boldsymbol{E} \in \mathbb{R}^{n \times n}$	$\boldsymbol{E} \in \mathbb{R}^{n \times n}$	a) Let $\boldsymbol{E} \in \mathbb{R}^{n \times n}$ be of the form $\boldsymbol{E} = \boldsymbol{I} + \boldsymbol{u} \boldsymbol{u}^T$, where $\boldsymbol{u} \in \mathbb{R}^n$. Show that $\boldsymbol{E}$ is

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lyche-numerical-linear-algebra-F01449	114	1.000	$\boldsymbol{E}=\boldsymbol{I}+\boldsymbol{u}\boldsymbol{u}^T$	$\boldsymbol{E} = \boldsymbol{I} + \boldsymbol{u}\boldsymbol{u}^T$	a) Let $\boldsymbol{E} \in \mathbb{R}^{n \times n}$ be of the form $\boldsymbol{E} = \boldsymbol{I} + \boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u} \in \mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra-F01450	114	1.000	$\boldsymbol{u} \in \mathbb{R}^n$	$\boldsymbol{u} \in \mathbb{R}^n$	a) Let $\boldsymbol{E} \in \mathbb{R}^{n \times n}$ be of the form $\boldsymbol{E} = \boldsymbol{I} + \boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u} \in \mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra-F01451	114	1.000	$\boldsymbol{E}$	$\boldsymbol{E}$	a) Let $\boldsymbol{E} \in \mathbb{R}^{n \times n}$ be of the form $\boldsymbol{E} = \boldsymbol{I} + \boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{u} \in \mathbb{R}^n$. Show that $\boldsymbol{E}$ is
lyche-numerical-linear-algebra-F01452	114	0.834	$\boldsymbol{E}^{-1}$	$\boldsymbol{E}^{-1}$	symmetric and positive definite, and find an expression for $\boldsymbol{E}^{-1}$. ¹
lyche-numerical-linear-algebra-F01453	114	1.000	$\boldsymbol{A}=\boldsymbol{B}+\boldsymbol{u}\boldsymbol{u}^T$	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{u}\boldsymbol{u}^T$	b) Let $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ be of the form $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{u}\boldsymbol{u}^T$, where $\boldsymbol{B} \in \mathbb{R}^{n \times n}$ is symmetric and
lyche-numerical-linear-algebra-F01454	114	1.000	$\boldsymbol{A}=\boldsymbol{L}\boldsymbol{L}^T$	$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^T$	symmetric positive definite matrix with a known Cholesky factorization $\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^T$.
lyche-numerical-linear-algebra-F01455	114	0.998	$\boldsymbol{A}_+$	$\boldsymbol{A}_+$	Furthermore, let $\boldsymbol{A}_+$ be a corresponding $(n+1) \times (n+1)$ matrix of the form
lyche-numerical-linear-algebra-F01456	114	0.998	$(n+1) \times (n+1)$	$(n+1) \times (n+1)$	Furthermore, let $\boldsymbol{A}_+$ be a corresponding $(n+1) \times (n+1)$ matrix of the form
lyche-numerical-linear-algebra-F01457	114	1.000	α	α	where $\boldsymbol{a}$ is a vector in $\mathbb{R}^n$, and α is a real number. We assume that the matrix $\boldsymbol{A}_+$ is
lyche-numerical-linear-algebra-F01458	114	1.000	$\boldsymbol{A}_+ = \boldsymbol{L}_+ \boldsymbol{L}_+^T$	$\boldsymbol{A}_+ = \boldsymbol{L}_+ \boldsymbol{L}_+^T$	a) Show that if $\boldsymbol{A}_+ = \boldsymbol{L}_+ \boldsymbol{L}_+^T$ is the Cholesky factorization of $\boldsymbol{A}_+$, then $\boldsymbol{L}_+$ is of the
lyche-numerical-linear-algebra-F01459	114	1.000	$\boldsymbol{L}_+$	$\boldsymbol{L}_+$	a) Show that if $\boldsymbol{A}_+ = \boldsymbol{L}_+ \boldsymbol{L}_+^T$ is the Cholesky factorization of $\boldsymbol{A}_+$, then $\boldsymbol{L}_+$ is of the
lyche-numerical-linear-algebra-F01460	114	1.000	$\alpha > \ \boldsymbol{L}^{-1} \boldsymbol{a}\ _2^2$	$\alpha > \ \boldsymbol{L}^{-1} \boldsymbol{a}\ _2^2$	b) Explain why $\alpha > \ \boldsymbol{L}^{-1} \boldsymbol{a}\ _2^2$.
lyche-numerical-linear-algebra-F01461	114	1.000	$\boldsymbol{E}^{-1}$	$\boldsymbol{E}^{-1}$	¹ Hint: The matrix $\boldsymbol{E}^{-1}$ is of the form $\boldsymbol{E}^{-1} = \boldsymbol{I} + \boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$ for some $\boldsymbol{a} \in \mathbb{R}$.
lyche-numerical-linear-algebra-F01462	114	1.000	$\boldsymbol{E}^{-1}=\boldsymbol{I}+\boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$	$\boldsymbol{E}^{-1} = \boldsymbol{I} + \boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$	¹ Hint: The matrix $\boldsymbol{E}^{-1}$ is of the form $\boldsymbol{E}^{-1} = \boldsymbol{I} + \boldsymbol{a}\boldsymbol{u}\boldsymbol{u}^T$ for some $\boldsymbol{a} \in \mathbb{R}$.
lyche-numerical-linear-algebra-F01463	115	0.493	$\begin{bmatrix} 10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5 \end{bmatrix}$	$\begin{bmatrix} 10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5 \end{bmatrix}$	4.6.3 Is the matrix $\begin{bmatrix} 10 & 4 & 3 \\ 4 & 0 & 2 \\ 3 & 2 & 5 \end{bmatrix}$ positive definite?
lyche-numerical-linear-algebra-F01464	117	1.000	$\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$	$\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$	a function $\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$ satisfying for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$ the following
lyche-numerical-linear-algebra-F01465	117	1.000	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$	a function $\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$ satisfying for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$ the following
lyche-numerical-linear-algebra-F01466	117	1.000	$\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$	$\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$	a function $\mathcal{V} \times \mathcal{V} \rightarrow \mathbb{C}$ satisfying for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and all $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{C}$ the following
lyche-numerical-linear-algebra-F01467	117	1.000	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$	1. $\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$ with equality if and only if $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra-F01468	117	1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$	2. $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra-F01469	117	1.000	$\langle \boldsymbol{a}\boldsymbol{x} + \boldsymbol{b}\boldsymbol{y}, \boldsymbol{z} \rangle = \boldsymbol{a}\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \boldsymbol{b}\langle \boldsymbol{y}, \boldsymbol{z} \rangle$	$\langle \boldsymbol{a}\boldsymbol{x} + \boldsymbol{b}\boldsymbol{y}, \boldsymbol{z} \rangle = \boldsymbol{a}\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \boldsymbol{b}\langle \boldsymbol{y}, \boldsymbol{z} \rangle$	3. $\langle \boldsymbol{a}\boldsymbol{x} + \boldsymbol{b}\boldsymbol{y}, \boldsymbol{z} \rangle = \boldsymbol{a}\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \boldsymbol{b}\langle \boldsymbol{y}, \boldsymbol{z} \rangle$.
lyche-numerical-linear-algebra-F01470	117	0.996	$(\mathcal{V}, \langle \cdot, \cdot \rangle)$	$(\mathcal{V}, \langle \cdot, \cdot \rangle)$	The pair $(\mathcal{V}, \langle \cdot, \cdot \rangle)$ is called an inner product space .
lyche-numerical-linear-algebra-F01471	117	0.975	$\ \boldsymbol{x}\ = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$	$\ \boldsymbol{x}\ = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$	$\ \boldsymbol{x}\ = \sqrt{\boldsymbol{x}^* \boldsymbol{x}}$.
lyche-numerical-linear-algebra-F01472	118	1.000	$0\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$	$0\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$	Proof If $\boldsymbol{y} = \mathbf{0}$ then $0\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$ and $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent. Moreover the
lyche-numerical-linear-algebra-F01473	118	1.000	$\langle \boldsymbol{x}, \mathbf{0} \rangle = 0$	$\langle \boldsymbol{x}, \mathbf{0} \rangle = 0$	inequality holds with equality since $\langle \boldsymbol{x}, \mathbf{0} \rangle = 0$ and $\ \mathbf{0}\ = 0$.
lyche-numerical-linear-algebra-F01474	118	1.000	$\ \mathbf{0}\ = 0$	$\ \mathbf{0}\ = 0$	inequality holds with equality since $\ \mathbf{0}\ = 0$ and $\langle \mathbf{0}, \mathbf{0} \rangle = 0$.

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lyche-numerical-linear-algebra- F01475	118	1.000	$\boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{y} \neq \mathbf{0}$	So assume $\boldsymbol{y} \neq \mathbf{0}$. Define
lyche-numerical-linear-algebra- F01476	118	0.862	$\langle \boldsymbol{z}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - a \langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$	$\langle \boldsymbol{z}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - a \langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$	By linearity $\langle \boldsymbol{z}, \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle - a \langle \boldsymbol{y}, \boldsymbol{y} \rangle = 0$ so that by 2. and (5.1)
lyche-numerical-linear-algebra- F01477	118	1.000	$\ \boldsymbol{y}\ ^2$	$\ \boldsymbol{y}\ ^2$	Multiplying by $\ \boldsymbol{y}\ ^2$ gives (5.3). We have equality if and only if $\boldsymbol{z} = \mathbf{0}$, which means
lyche-numerical-linear-algebra- F01478	118	1.000	$\boldsymbol{z} = \mathbf{0}$	$\boldsymbol{z} = \mathbf{0}$	Multiplying by $\ \boldsymbol{y}\ ^2$ gives (5.3). We have equality if and only if $\boldsymbol{z} = \mathbf{0}$, which means
lyche-numerical-linear-algebra- F01479	118	1.000	$\ \boldsymbol{x}\ \geq 0$	$\ \boldsymbol{x}\ \geq 0$	1. $\ \boldsymbol{x}\ \geq 0$ with equality if and only if $\boldsymbol{x} = \mathbf{0}$.
lyche-numerical-linear-algebra- F01480	118	1.000	$ a \ \boldsymbol{x}\ = \ a\boldsymbol{x}\ $	$\ a\boldsymbol{x}\ = a \ \boldsymbol{x}\ $	2. $\ a\boldsymbol{x}\ = a \ \boldsymbol{x}\ $.
lyche-numerical-linear-algebra- F01481	118	1.000	$\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $	$\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $	3. $\ \boldsymbol{x} + \boldsymbol{y}\ \leq \ \boldsymbol{x}\ + \ \boldsymbol{y}\ $,
lyche-numerical-linear-algebra- F01482	118	1.000	$\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	$\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$	where $\ \boldsymbol{x}\ := \sqrt{\langle \boldsymbol{x}, \boldsymbol{x} \rangle}$
lyche-numerical-linear-algebra- F01483	118	1.000	$\ \cdot\ : \mathbb{C}^n \rightarrow \mathbb{R}$	$\ \cdot\ : \mathbb{C}^n \rightarrow \mathbb{R}$	In general a function $\ \cdot\ : \mathbb{C}^n \rightarrow \mathbb{R}$ that satisfies these three properties is called a
lyche-numerical-linear-algebra- F01484	119	1.000	$\ \boldsymbol{x} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y} \rangle$	$\ \boldsymbol{x} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y} \rangle$	second one follows from (5.1). Expanding $\ \boldsymbol{x} + a\boldsymbol{y}\ ^2 = \langle \boldsymbol{x} + a\boldsymbol{y}, \boldsymbol{x} + a\boldsymbol{y} \rangle$ using
lyche-numerical-linear-algebra- F01485	119	1.000	$a = 1$	$a = 1$	Now (5.5) with $a = 1$ and the Cauchy-Schwarz inequality implies
lyche-numerical-linear-algebra- F01486	119	0.999	$-1 \leq \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$	$-1 \leq \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$	In the real case the Cauchy-Schwarz inequality implies that $-1 \leq \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\ \boldsymbol{x}\ \ \boldsymbol{y}\ } \leq 1$
lyche-numerical-linear-algebra- F01487	119	1.000	θ	θ	for nonzero $\boldsymbol{x}$ and $\boldsymbol{y}$, so there is a unique angle θ in $[0, \pi]$ such that
lyche-numerical-linear-algebra- F01488	119	1.000	$[0, \pi]$	$[0, \pi]$	for nonzero $\boldsymbol{x}$ and $\boldsymbol{y}$, so there is a unique angle θ in $[0, \pi]$ such that
lyche-numerical-linear-algebra- F01489	119	1.000	$\boldsymbol{x} \perp \boldsymbol{y}$	$\boldsymbol{x} \perp \boldsymbol{y}$	uct space are orthogonal or perpendicular , denoted as $\boldsymbol{x} \perp \boldsymbol{y}$, if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$. The
lyche-numerical-linear-algebra- F01490	119	1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$	uct space are orthogonal or perpendicular , denoted as $\boldsymbol{x} \perp \boldsymbol{y}$, if $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0$. The
lyche-numerical-linear-algebra- F01491	119	1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$	$\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$	vectors are orthonormal if in addition $\ \boldsymbol{x}\ = \ \boldsymbol{y}\ = 1$.
lyche-numerical-linear-algebra- F01492	119	1.000	$\theta = \pi/2$	$\theta = \pi/2$	or $\mathbb{C}^n$ it follows that $\boldsymbol{x} \perp \boldsymbol{y}$ if and only if $\theta = \pi/2$.
lyche-numerical-linear-algebra- F01493	119	0.997	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$	orthogonal basis for $\mathcal{S}$ if it is a basis for $\mathcal{S}$ and $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$ for $i \neq j$. It is an
lyche-numerical-linear-algebra- F01494	119	0.734	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij}$	$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij}$	orthonormal basis for $\mathcal{S}$ if it is a basis for $\mathcal{S}$ and $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij}$ for all i, j .
lyche-numerical-linear-algebra- F01495	120	0.999	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$	$\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$	Theorem 5.4 (Gram-Schmidt) Let $\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_k\}$ be a basis for a real or complex
lyche-numerical-linear-algebra- F01496	120	0.833	$(\mathcal{S}, \langle \cdot, \cdot \rangle)$	$(\mathcal{S}, \langle \cdot, \cdot \rangle)$	inner product space $(\mathcal{S}, \langle \cdot, \cdot \rangle)$. Define
lyche-numerical-linear-algebra- F01497	120	0.999	$S_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$	$S_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$	k . Define subspaces $S_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$ for $j = 1, \dots, k$. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$
lyche-numerical-linear-algebra- F01498	120	0.999	$\boldsymbol{v}_1 = \boldsymbol{s}_1$	$\boldsymbol{v}_1 = \boldsymbol{s}_1$	k . Define subspaces $S_j := \text{span}\{\boldsymbol{s}_1, \dots, \boldsymbol{s}_j\}$ for $j = 1, \dots, k$. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$
lyche-numerical-linear-algebra- F01499	120	1.000	S_1	S_1	is an orthogonal basis for S_1 . Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra- F01500	120	1.000	$j \geq 2$	$j \geq 2$	is an orthogonal basis for S_1 . Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$ is an

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lyche-numerical-linear-algebra-F01501	120	1.000	$\backslash\boldsymbol{v}_{_1}, \backslash\ldots, \backslash\boldsymbol{v}_{_j-1}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$	is an orthogonal basis for S_1 . Suppose for some $j \geq 2$ that $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$ is an
lyche-numerical-linear-algebra-F01502	120	1.000	$S_{_j-1}$	S_{j-1}	orthogonal basis for S_{j-1} and let $\boldsymbol{v}_j$ be given by (5.8) as a linear combination of
lyche-numerical-linear-algebra-F01503	120	1.000	$\backslash\boldsymbol{v}_{_j}$	$\boldsymbol{v}_j$	orthogonal basis for S_{j-1} and let $\boldsymbol{v}_j$ be given by (5.8) as a linear combination of
lyche-numerical-linear-algebra-F01504	120	1.000	$\backslash\boldsymbol{v}_{_i}$	$\boldsymbol{v}_i$	$\boldsymbol{s}_j$ and $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$. Now each of these $\boldsymbol{v}_i$ is a linear combination of $\boldsymbol{s}_1, \dots, \boldsymbol{s}_i$,
lyche-numerical-linear-algebra-F01505	120	1.000	$\backslash\boldsymbol{s}_{_1}, \backslash\ldots, \backslash\boldsymbol{s}_{_i}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_i$	$\boldsymbol{s}_j$ and $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{j-1}$. Now each of these $\boldsymbol{v}_i$ is a linear combination of $\boldsymbol{s}_1, \dots, \boldsymbol{s}_i$,
lyche-numerical-linear-algebra-F01506	120	1.000	$\backslash\boldsymbol{v}_{_j}=\sum_{i=1}^j a_{_i} \backslash\boldsymbol{s}_{_i}$	$\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra-F01507	120	1.000	$a_{_0}, \backslash\ldots, a_{_j}$	$a_0, \dots, a_j$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra-F01508	120	1.000	$a_{_j}=1$	$a_j = 1$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra-F01509	120	1.000	$\backslash\boldsymbol{s}_{_1}, \backslash\ldots, \backslash\boldsymbol{s}_{_j}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$	and we obtain $\boldsymbol{v}_j = \sum_{i=1}^j a_i \boldsymbol{s}_i$ for some $a_0, \dots, a_j$ with $a_j = 1$. Since $\boldsymbol{s}_1, \dots, \boldsymbol{s}_j$
lyche-numerical-linear-algebra-F01510	120	1.000	$a_{_j} \neq 0$	$a_j \neq 0$	are linearly independent and $a_j \neq 0$ we deduce that $\boldsymbol{v}_j \neq 0$. By the induction
lyche-numerical-linear-algebra-F01511	120	1.000	$\backslash\boldsymbol{v}_{_j} \neq 0$	$\boldsymbol{v}_j \neq 0$	are linearly independent and $a_j \neq 0$ we deduce that $\boldsymbol{v}_j \neq 0$. By the induction
lyche-numerical-linear-algebra-F01512	120	1.000	$l=1, \backslash\ldots, j-1$	$l = 1, \dots, j-1$	for $l = 1, \dots, j-1$. Thus $\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$ is an orthogonal basis for S_j .
lyche-numerical-linear-algebra-F01513	120	1.000	$\backslash\boldsymbol{v}_{_1}, \backslash\ldots, \backslash\boldsymbol{v}_{_j}$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$	for $l = 1, \dots, j-1$. Thus $\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$ is an orthogonal basis for S_j .
lyche-numerical-linear-algebra-F01514	120	1.000	$S_{_j}$	S_j	for $l = 1, \dots, j-1$. Thus $\boldsymbol{v}_1, \dots, \boldsymbol{v}_j$ is an orthogonal basis for S_j .
lyche-numerical-linear-algebra-F01515	120	1.000	$\backslash\left\{\backslash\boldsymbol{u}_{_1}, \backslash\ldots, \backslash\boldsymbol{u}_{_k}\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\}$	If $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$ is an orthogonal basis for $\mathcal{S}$ then clearly $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_k\}$ is an
lyche-numerical-linear-algebra-F01516	121	1.000	$\backslash\mathcal{S} \subset \backslash\mathcal{T}$	$\mathcal{S} \subset \mathcal{T}$	Theorem 5.5 (Orthogonal Extension of Basis) Suppose $\mathcal{S} \subset \mathcal{T}$ are finite dimen-
lyche-numerical-linear-algebra-F01517	121	1.000	$\operatorname{dim} \backslash\mathcal{S}:=k<\operatorname{dim} \backslash\mathcal{T}=n$	$\dim \mathcal{S} := k < \dim \mathcal{T} = n$	Proof Suppose $\dim \mathcal{S} := k < \dim \mathcal{T} = n$. Using Theorem 1.3 we first extend
lyche-numerical-linear-algebra-F01518	121	1.000	$\backslash\boldsymbol{s}_{_1}, \backslash\ldots, \backslash\boldsymbol{s}_{_k}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$	an orthogonal basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$ for $\mathcal{S}$ to a basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \dots, \boldsymbol{s}_n$ for $\mathcal{T}$, and
lyche-numerical-linear-algebra-F01519	121	1.000	$\backslash\boldsymbol{s}_{_1}, \backslash\ldots, \backslash\boldsymbol{s}_{_k}, \backslash\boldsymbol{s}_{_k+1}, \backslash\ldots, \backslash\boldsymbol{s}_{_n}$	$\boldsymbol{s}_1, \dots, \boldsymbol{s}_k, \boldsymbol{s}_{k+1}, \dots, \boldsymbol{s}_n$	an orthogonal basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_k$ for $\mathcal{S}$ to a basis $\boldsymbol{s}_1, \dots, \boldsymbol{s}_r, \boldsymbol{s}_{r+1}, \dots, \boldsymbol{s}_n$ for $\mathcal{T}$, and
lyche-numerical-linear-algebra-F01520	121	0.955	$\backslash\boldsymbol{v}_{_i}=\backslash\boldsymbol{s}_{_i}$	$\boldsymbol{v}_i = \boldsymbol{s}_i$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ for $\mathcal{T}$. This is an extension of the basis for $\mathcal{S}$ since $\boldsymbol{v}_i = \boldsymbol{s}_i$ for $i =$
lyche-numerical-linear-algebra-F01521	121	1.000	$1, \backslash\ldots, k$	$1, \dots, k$	$1, \dots, k$. We show this by induction. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$. Suppose for some $2 \leq r < k$
lyche-numerical-linear-algebra-F01522	121	1.000	$2 \leq r < k$	$2 \leq r < k$	$1, \dots, k$. We show this by induction. Clearly $\boldsymbol{v}_1 = \boldsymbol{s}_1$. Suppose for some $2 \leq r < k$
lyche-numerical-linear-algebra-F01523	121	1.000	$\backslash\boldsymbol{v}_{_j}=\backslash\boldsymbol{s}_{_j}$	$\boldsymbol{v}_j = \boldsymbol{s}_j$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra-F01524	121	1.000	$j=1, \backslash\ldots, r-1$	$j = 1, \dots, r-1$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra-F01525	121	1.000	$j=r$	$j = r$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra-F01526	121	1.000	$\backslash\left\langle\backslash\boldsymbol{s}_{_r}, \backslash\boldsymbol{v}_{_i}\right\rangle$	$\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$	that $\boldsymbol{v}_j = \boldsymbol{s}_j$ for $j = 1, \dots, r-1$. Consider (5.8) for $j = r$. Since $\langle \boldsymbol{s}_r, \boldsymbol{v}_i \rangle =$
lyche-numerical-linear-algebra-F01527	121	0.999	$\backslash\left\langle\backslash\boldsymbol{s}_{_r}, \backslash\boldsymbol{s}_{_i}\right\rangle=0$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$	$\langle \boldsymbol{s}_r, \boldsymbol{s}_i \rangle = 0$ for $i < r$ we obtain $\boldsymbol{v}_r = \boldsymbol{s}_r$.

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lyche-numerical-linear-algebra-F01528	121	■ 0.999	$i < r$	$i < r$	$\langle s_r, s_i \rangle = 0$ for $i < r$ we obtain $v_r = s_r$.
lyche-numerical-linear-algebra-F01529	121	■ 0.999	$\boldsymbol{v}_r = \boldsymbol{s}_r$	$v_r = s_r$	$\langle s_r, s_i \rangle = 0$ for $i < r$ we obtain $v_r = s_r$.
lyche-numerical-linear-algebra-F01530	121	■ 0.999	$\mathcal{S} = \operatorname{span}(\boldsymbol{s}_1, \dots, \boldsymbol{s}_k)$	$\mathcal{S} = \operatorname{span}(s_1, \dots, s_k)$	Letting $\mathcal{S} = \operatorname{span}(s_1, \dots, s_k)$ and $\mathcal{T}$ be $\mathbb{R}^n$ or $\mathbb{C}^n$ we obtain
lyche-numerical-linear-algebra-F01531	121	■ 1.000	$1 \leq k < n$	$1 \leq k < n$	Corollary 5.1 (Extending Orthogonal Vectors to a Basis) For $1 \leq k < n$ a
lyche-numerical-linear-algebra-F01532	121	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle$	$\langle x, y \rangle$	an inner product $\langle x, y \rangle$. We define
lyche-numerical-linear-algebra-F01533	121	■ 1.000	$\mathcal{S} + \mathcal{T} := \{\boldsymbol{s} + \boldsymbol{t} : \boldsymbol{s} \in \mathcal{S}, \boldsymbol{t} \in \mathcal{T}\}$	$\mathcal{S} + \mathcal{T} := \{s + t : s \in \mathcal{S}$	• Sum: $\mathcal{S} + \mathcal{T} := \{s + t : s \in \mathcal{S}$ and $t \in \mathcal{T}\}$,
lyche-numerical-linear-algebra-F01534	121	■ 1.000	$\boldsymbol{t} \in \mathcal{T}\}$	$t \in \mathcal{T}\}$	• Sum: $\mathcal{S} + \mathcal{T} := \{s + t : s \in \mathcal{S}$ and $t \in \mathcal{T}\}$,
lyche-numerical-linear-algebra-F01535	121	■ 0.967	$\boldsymbol{\operatorname{sum}} \mathcal{S} \oplus \mathcal{T}$	$\operatorname{sum} \mathcal{S} \oplus \mathcal{T}$	• direct sum $\mathcal{S} \oplus \mathcal{T}$: a sum where $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$,
lyche-numerical-linear-algebra-F01536	121	■ 0.998	$\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	$\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle s, t \rangle = 0$ for all $s \in \mathcal{S}$ and $t \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01537	121	■ 0.998	$\langle \boldsymbol{s}, \boldsymbol{t} \rangle = 0$	$\langle s, t \rangle = 0$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle s, t \rangle = 0$ for all $s \in \mathcal{S}$ and $t \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01538	121	■ 0.998	$\boldsymbol{s} \in \mathcal{S}$	$s \in \mathcal{S}$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle s, t \rangle = 0$ for all $s \in \mathcal{S}$ and $t \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01539	121	■ 0.998	$\boldsymbol{t} \in \mathcal{T}$	$t \in \mathcal{T}$	• orthogonal sum $\mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$: a sum where $\langle s, t \rangle = 0$ for all $s \in \mathcal{S}$ and $t \in \mathcal{T}$.
lyche-numerical-linear-algebra-F01540	121	■ 1.000	$\boldsymbol{v} \in \mathcal{S} \oplus \mathcal{T}$	$v \in \mathcal{S} \oplus \mathcal{T}$	• Every $v \in \mathcal{S} \oplus \mathcal{T}$ can be decomposed uniquely in the form $v = s + t$, where
lyche-numerical-linear-algebra-F01541	121	■ 1.000	$\boldsymbol{v} = \boldsymbol{s} + \boldsymbol{t}$	$v = s + t$	• Every $v \in \mathcal{S} \oplus \mathcal{T}$ can be decomposed uniquely in the form $v = s + t$, where
lyche-numerical-linear-algebra-F01542	121	■ 1.000	$\boldsymbol{v} = \boldsymbol{s}_1 + \boldsymbol{t}_1 = \boldsymbol{s}_2 + \boldsymbol{t}_2$	$v = s_1 + t_1 = s_2 + t_2$	$s \in \mathcal{S}$ and $t \in \mathcal{T}$. For if $v = s_1 + t_1 = s_2 + t_2$ for $s_1, s_2 \in \mathcal{S}$ and $t_1, t_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra-F01543	121	■ 1.000	$\boldsymbol{s}_1, \boldsymbol{s}_2 \in \mathcal{S}$	$s_1, s_2 \in \mathcal{S}$	$s \in \mathcal{S}$ and $t \in \mathcal{T}$. For if $v = s_1 + t_1 = s_2 + t_2$ for $s_1, s_2 \in \mathcal{S}$ and $t_1, t_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra-F01544	121	■ 1.000	$\boldsymbol{t}_1, \boldsymbol{t}_2 \in \mathcal{T}$	$t_1, t_2 \in \mathcal{T}$	$s \in \mathcal{S}$ and $t \in \mathcal{T}$. For if $v = s_1 + t_1 = s_2 + t_2$ for $s_1, s_2 \in \mathcal{S}$ and $t_1, t_2 \in \mathcal{T}$,
lyche-numerical-linear-algebra-F01545	121	■ 0.969	$\mathbf{0} = \boldsymbol{s}_1 - \boldsymbol{s}_2 + \boldsymbol{t}_1 - \boldsymbol{t}_2$	$\mathbf{0} = s_1 - s_2 + t_1 - t_2$	then $\mathbf{0} = s_1 - s_2 + t_1 - t_2$ or $s_1 - s_2 = t_2 - t_1$. It follows that $s_1 - s_2$ and $t_2 - t_1$
lyche-numerical-linear-algebra-F01546	121	■ 0.969	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1$	$s_1 - s_2 = t_2 - t_1$	then $\mathbf{0} = s_1 - s_2 + t_1 - t_2$ or $s_1 - s_2 = t_2 - t_1$. It follows that $s_1 - s_2$ and $t_2 - t_1$
lyche-numerical-linear-algebra-F01547	121	■ 0.969	$\boldsymbol{s}_1 - \boldsymbol{s}_2$	$s_1 - s_2$	then $\mathbf{0} = s_1 - s_2 + t_1 - t_2$ or $s_1 - s_2 = t_2 - t_1$. It follows that $s_1 - s_2$ and $t_2 - t_1$
lyche-numerical-linear-algebra-F01548	121	■ 0.969	$\boldsymbol{t}_2 - \boldsymbol{t}_1$	$t_2 - t_1$	then $\mathbf{0} = s_1 - s_2 + t_1 - t_2$ or $s_1 - s_2 = t_2 - t_1$. It follows that $s_1 - s_2$ and $t_2 - t_1$
lyche-numerical-linear-algebra-F01549	121	■ 1.000	$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \boldsymbol{t}_2 - \boldsymbol{t}_1 = \mathbf{0}$	$s_1 - s_2 = t_2 - t_1 = \mathbf{0}$	belong to both $\mathcal{S}$ and $\mathcal{T}$ and hence to $\mathcal{S} \cap \mathcal{T}$. But then $s_1 - s_2 = t_2 - t_1 = \mathbf{0}$ so
lyche-numerical-linear-algebra-F01550	121	■ 1.000	$\boldsymbol{s}_1 = \boldsymbol{s}_2$	$s_1 = s_2$	$s_1 = s_2$ and $t_2 = t_1$.
lyche-numerical-linear-algebra-F01551	121	■ 1.000	$\boldsymbol{t}_2 = \boldsymbol{t}_1$	$t_2 = t_1$	$s_1 = s_2$ and $t_2 = t_1$.
lyche-numerical-linear-algebra-F01552	121	■ 1.000	$\boldsymbol{v} \in \mathcal{S} \cap \mathcal{T}$	$v \in \mathcal{S} \cap \mathcal{T}$	• An orthogonal sum is a direct sum. For if $v \in \mathcal{S} \cap \mathcal{T}$ then v is orthogonal to itself,
lyche-numerical-linear-algebra-F01553	121	■ 1.000	$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0$	$\langle v, v \rangle = 0$	$\langle v, v \rangle = 0$, which implies that $v = 0$. We often write $\mathcal{T} := \mathcal{S}^\perp$.
lyche-numerical-linear-algebra-F01554	121	■ 1.000	$\boldsymbol{v} = 0$	$v = 0$	$\langle v, v \rangle = 0$, which implies that $v = 0$. We often write $\mathcal{T} := \mathcal{S}^\perp$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01555	121	1.000	$\mathcal{T}:=\mathcal{S}^{\perp}$	$\mathcal{T} := \mathcal{S}^{\perp}$	$\langle v, v \rangle = 0$, which implies that $v = 0$. We often write $\mathcal{T} := \mathcal{S}^{\perp}$.
lyche-numerical-linear-algebra-F01556	121	1.000	$\boldsymbol{v}=\boldsymbol{s}_{\{0\}}+\boldsymbol{t}_{\{0\}} \in \mathcal{S} \oplus \mathcal{T}$	$v = s_0 + t_0 \in \mathcal{S} \oplus \mathcal{T}$	• Suppose $v = s_0 + t_0 \in \mathcal{S} \oplus \mathcal{T}$, where $s_0 \in \mathcal{S}$ and $t_0 \in \mathcal{T}$. The vector s_0 is
lyche-numerical-linear-algebra-F01557	121	1.000	$\boldsymbol{s}_{\{0\}} \in \mathcal{S}$	$s_0 \in \mathcal{S}$	• Suppose $v = s_0 + t_0 \in \mathcal{S} \oplus \mathcal{T}$, where $s_0 \in \mathcal{S}$ and $t_0 \in \mathcal{T}$. The vector s_0 is
lyche-numerical-linear-algebra-F01558	121	1.000	$\boldsymbol{t}_{\{0\}} \in \mathcal{T}$	$t_0 \in \mathcal{T}$	• Suppose $v = s_0 + t_0 \in \mathcal{S} \oplus \mathcal{T}$, where $s_0 \in \mathcal{S}$ and $t_0 \in \mathcal{T}$. The vector s_0 is
lyche-numerical-linear-algebra-F01559	121	1.000	$\boldsymbol{s}_{\{0\}}$	s_0	• Suppose $v = s_0 + t_0 \in \mathcal{S} \oplus \mathcal{T}$, where $s_0 \in \mathcal{S}$ and $t_0 \in \mathcal{T}$. The vector s_0 is
lyche-numerical-linear-algebra-F01560	121	1.000	$\boldsymbol{t}_{\{0\}}$	t_0	called the oblique projection of v into $\mathcal{S}$ along $\mathcal{T}$. Similarly, The vector t_0 is
lyche-numerical-linear-algebra-F01561	122	1.000	$\mathcal{T}=\mathcal{S}^{\perp}$	$\mathcal{T} = \mathcal{S}^{\perp}$	called the orthogonal projection of v in $\mathcal{T} = \mathcal{S}^{\perp}$. The orthogonal projections
lyche-numerical-linear-algebra-F01562	122	1.000	$\langle \cdot, \cdot \rangle$	$\langle \cdot, \cdot \rangle$	<i>dimensional real or complex vector space $\mathcal{V}$ with an inner product $\langle \cdot, \cdot \rangle$. The</i>
lyche-numerical-linear-algebra-F01563	122	0.984	$\boldsymbol{v} \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	$v \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$	<i>orthogonal projections s_0 of $v \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$ into $\mathcal{S}$ and t_0 of $v \in \mathcal{S} \stackrel{\perp}{\oplus} \mathcal{T}$ into $\mathcal{T}$ satisfy</i>
lyche-numerical-linear-algebra-F01564	122	1.000	$\boldsymbol{v}=\boldsymbol{s}_{\{0\}}+\boldsymbol{t}_{\{0\}}$	$v = s_0 + t_0$	$v = s_0 + t_0$, and
lyche-numerical-linear-algebra-F01565	122	1.000	$\left\langle \boldsymbol{s}_{\{0\}}, \boldsymbol{s} \right\rangle, \left\langle \boldsymbol{s}, \boldsymbol{t}_{\{0\}} \right\rangle, \left\langle \boldsymbol{v}-\boldsymbol{t}_{\{0\}}, \boldsymbol{s} \right\rangle, \left\langle \boldsymbol{s}, \boldsymbol{v}-\boldsymbol{t}_{\{0\}} \right\rangle, \left\langle \boldsymbol{v}-\boldsymbol{t}_{\{0\}}, \boldsymbol{v}-\boldsymbol{t}_{\{0\}} \right\rangle$	$\langle s_0, s \rangle = \langle v - t_0, s \rangle = \langle v, s \rangle$	Proof We have $\langle s_0, s \rangle = \langle v - t_0, s \rangle = \langle v, s \rangle$, since $\langle t_0, s \rangle = 0$ for all $s \in \mathcal{S}$ and
lyche-numerical-linear-algebra-F01566	122	1.000	$\left\langle \boldsymbol{t}_{\{0\}}, \boldsymbol{s} \right\rangle = 0$	$\langle t_0, s \rangle = 0$	Proof We have $\langle s_0, s \rangle = \langle v - t_0, s \rangle = \langle v, s \rangle$, since $\langle t_0, s \rangle = 0$ for all $s \in \mathcal{S}$ and
lyche-numerical-linear-algebra-F01567	123	1.000	$\ \boldsymbol{v}\ :=\sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}$	$\ v\ := \sqrt{\langle v, v \rangle}$	$\ v\ := \sqrt{\langle v, v \rangle}$. If $s_0 \in \mathcal{S}$ is the orthogonal projection of $v \in \mathcal{V}$ then
lyche-numerical-linear-algebra-F01568	123	1.000	$\boldsymbol{v} \in \mathcal{V}$	$v \in \mathcal{V}$	$\ v\ := \sqrt{\langle v, v \rangle}$. If $s_0 \in \mathcal{S}$ is the orthogonal projection of $v \in \mathcal{V}$ then
lyche-numerical-linear-algebra-F01569	123	0.975	$s_{\{0\}} \neq s \in \mathcal{S}$	$s_0 \neq s \in \mathcal{S}$	Proof Let $s_0 \neq s \in \mathcal{S}$ and $0 \neq u := s_0 - s \in \mathcal{S}$. It follows from (5.9) that
lyche-numerical-linear-algebra-F01570	123	0.975	$0 \neq \boldsymbol{u}:=s_{\{0\}}-s \in \mathcal{S}$	$0 \neq u := s_0 - s \in \mathcal{S}$	Proof Let $s_0 \neq s \in \mathcal{S}$ and $0 \neq u := s_0 - s \in \mathcal{S}$. It follows from (5.9) that
lyche-numerical-linear-algebra-F01571	123	1.000	$\left\langle \boldsymbol{v}-\boldsymbol{s}_0, \boldsymbol{u} \right\rangle = 0$	$\langle v - s_0, u \rangle = 0$	$\langle v - s_0, u \rangle = 0$. By (5.7) (Pythagoras) we obtain
lyche-numerical-linear-algebra-F01572	123	1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle:=\boldsymbol{y}^* \boldsymbol{x}=\sum_{j=1}^n x_j \overline{y_j}$	$\langle x, y \rangle := y^* x = \sum_{j=1}^n x_j \overline{y_j}$	$\mathbb{C}^n$ given by $\langle x, y \rangle := y^* x = \sum_{j=1}^n x_j \overline{y_j}$. For symmetric and Hermitian matrices
lyche-numerical-linear-algebra-F01573	123	1.000	$\boldsymbol{A}^T=\boldsymbol{A} \Longleftrightarrow \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle=\langle \boldsymbol{x}, \overline{\boldsymbol{A} \boldsymbol{y}} \rangle$	$A^T = A \iff \langle Ax, y \rangle = \langle x, \overline{Ay} \rangle$	1. $A^T = A \iff \langle Ax, y \rangle = \langle x, Ay \rangle$ for all $x, y \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-F01574	123	1.000	$\boldsymbol{A}^*=\boldsymbol{A} \Longleftrightarrow \langle \boldsymbol{A} \boldsymbol{x}, \boldsymbol{y} \rangle=\langle \boldsymbol{x}, \boldsymbol{A} \boldsymbol{y} \rangle$	$A^* = A \iff \langle Ax, y \rangle = \langle x, Ay \rangle$	2. $A^* = A \iff \langle Ax, y \rangle = \langle x, Ay \rangle$ for all $x, y \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-F01575	123	1.000	$\boldsymbol{x}=\boldsymbol{e}_{\{j\}}$	$x = e_j$	For the converse we take $x = e_j$ and $y = e_i$ for some i, j and obtain
lyche-numerical-linear-algebra-F01576	123	1.000	$\boldsymbol{y}=\boldsymbol{e}_{\{i\}}$	$y = e_i$	For the converse we take $x = e_j$ and $y = e_i$ for some i, j and obtain
lyche-numerical-linear-algebra-F01577	123	0.815	$\boldsymbol{A}=\boldsymbol{A}^T$	$A = A^T$	Thus, $A = A^T$ since they have the same i, j element for all i, j . The proof of 2. is
lyche-numerical-linear-algebra-F01578	123	1.000	$\boldsymbol{U}^* \boldsymbol{U}=\boldsymbol{I}$	$U^* U = I$	A square matrix $U \in \mathbb{C}^{n \times n}$ is unitary if $U^* U = I$. If U is real then $U^T U =$

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lyche-numerical-linear-algebra- F01579	123	1.000	$\boldsymbol{U}^T \boldsymbol{U} =$	$\boldsymbol{U}^T \boldsymbol{U} =$	A square matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ is unitary if $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$. If $\boldsymbol{U}$ is real then $\boldsymbol{U}^T \boldsymbol{U} =$
lyche-numerical-linear-algebra- F01580	123	1.000	$\boldsymbol{U}^{-1} = \boldsymbol{U}^*$	$\boldsymbol{U}^{-1} = \boldsymbol{U}^*$	If $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$ the matrix $\boldsymbol{U}$ is nonsingular, $\boldsymbol{U}^{-1} = \boldsymbol{U}^*$ and therefore $\boldsymbol{U} \boldsymbol{U}^* =$
lyche-numerical-linear-algebra- F01581	123	1.000	$\boldsymbol{U} \boldsymbol{U}^* =$	$\boldsymbol{U} \boldsymbol{U}^* =$	If $\boldsymbol{U}^* \boldsymbol{U} = \boldsymbol{I}$ the matrix $\boldsymbol{U}$ is nonsingular, $\boldsymbol{U}^{-1} = \boldsymbol{U}^*$ and therefore $\boldsymbol{U} \boldsymbol{U}^* =$
lyche-numerical-linear-algebra- F01582	123	1.000	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$	$\boldsymbol{U} \boldsymbol{U}^* = \boldsymbol{I}$ as well. Moreover, both the columns and rows of a unitary matrix of
lyche-numerical-linear-algebra- F01583	123	1.000	$\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$	$\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1 \boldsymbol{U}_2)^* (\boldsymbol{U}_1 \boldsymbol{U}_2) =$
lyche-numerical-linear-algebra- F01584	123	1.000	$\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1 \boldsymbol{U}_2)^* (\boldsymbol{U}_1 \boldsymbol{U}_2) =$
lyche-numerical-linear-algebra- F01585	123	1.000	$(\boldsymbol{U}_1 \boldsymbol{U}_2)^* (\boldsymbol{U}_1 \boldsymbol{U}_2) =$	$(\boldsymbol{U}_1 \boldsymbol{U}_2)^* (\boldsymbol{U}_1 \boldsymbol{U}_2) =$	matrices is unitary. Indeed, if $\boldsymbol{U}_1^* \boldsymbol{U}_1 = \boldsymbol{I}$ and $\boldsymbol{U}_2^* \boldsymbol{U}_2 = \boldsymbol{I}$ then $(\boldsymbol{U}_1 \boldsymbol{U}_2)^* (\boldsymbol{U}_1 \boldsymbol{U}_2) =$
lyche-numerical-linear-algebra- F01586	123	0.984	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2 = \boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2 = \boldsymbol{I}$	$\boldsymbol{U}_2^* \boldsymbol{U}_1^* \boldsymbol{U}_1 \boldsymbol{U}_2 = \boldsymbol{I}$.
lyche-numerical-linear-algebra- F01587	124	0.997	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$. In particular, if $\boldsymbol{U}$ is unitary then $\ \boldsymbol{U} \boldsymbol{x}\ _2 =$
lyche-numerical-linear-algebra- F01588	124	0.997	$\ \boldsymbol{U} \boldsymbol{x}\ _2 =$	$\ \boldsymbol{U} \boldsymbol{x}\ _2 =$	$\langle \boldsymbol{U} \boldsymbol{x}, \boldsymbol{U} \boldsymbol{y} \rangle = \langle \boldsymbol{x}, \boldsymbol{y} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$. In particular, if $\boldsymbol{U}$ is unitary then $\ \boldsymbol{U} \boldsymbol{x}\ _2 =$
lyche-numerical-linear-algebra- F01589	124	1.000	$\ \boldsymbol{x}\ _2$	$\ \boldsymbol{x}\ _2$	$\ \boldsymbol{x}\ _2$ for all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra- F01590	124	1.000	$i, j = 1, \dots, n$	$i, j = 1, \dots, n$	$i, j = 1, \dots, n$
lyche-numerical-linear-algebra- F01591	124	1.000	$(\boldsymbol{U}^* \boldsymbol{U})_{i,j} = \delta_{i,j}$	$(\boldsymbol{U}^* \boldsymbol{U})_{i,j} = \delta_{i,j}$	so that $(\boldsymbol{U}^* \boldsymbol{U})_{i,j} = \delta_{i,j}$ for all i, j . The last part of the theorem follows immediately
lyche-numerical-linear-algebra- F01592	124	1.000	$\boldsymbol{y} = \boldsymbol{x}$	$\boldsymbol{y} = \boldsymbol{x}$	by taking $\boldsymbol{y} = \boldsymbol{x}$:
lyche-numerical-linear-algebra- F01593	124	1.000	$\boldsymbol{H} \in \mathbb{C}^{n \times n}$	$\boldsymbol{H} \in \mathbb{C}^{n \times n}$	Definition 5.4 (Householder Transformation) A matrix $\boldsymbol{H} \in \mathbb{C}^{n \times n}$ of the form
lyche-numerical-linear-algebra- F01594	124	1.000	$\boldsymbol{H}^* = (\boldsymbol{I} -$	$\boldsymbol{H}^* = (\boldsymbol{I} -$	A Householder transformation is Hermitian and unitary. Indeed, $\boldsymbol{H}^* = (\boldsymbol{I} -$
lyche-numerical-linear-algebra- F01595	124	1.000	$\boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{H}$	$\boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{H}$	$\boldsymbol{u} \boldsymbol{u}^*) = \boldsymbol{H}$ and
lyche-numerical-linear-algebra- F01596	124	1.000	$\boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^*$	$\boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^*$	used $\boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}^* \boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra- F01597	124	1.000	$\boldsymbol{u}^* \boldsymbol{u} = 1$	$\boldsymbol{u}^* \boldsymbol{u} = 1$	used $\boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}^* \boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra- F01598	124	1.000	$\boldsymbol{v} \in \mathbb{C}^n$	$\boldsymbol{v} \in \mathbb{C}^n$	used $\boldsymbol{I} - 2 \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u}^* \boldsymbol{u} = 1$. For any nonzero $\boldsymbol{v} \in \mathbb{C}^n$ the matrix
lyche-numerical-linear-algebra- F01599	125	1.000	$\boldsymbol{H} = \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*$	$\boldsymbol{H} = \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*$	is a Householder transformation. Indeed, $\boldsymbol{H} = \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u} := \sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$ has
lyche-numerical-linear-algebra- F01600	125	1.000	$\boldsymbol{u} := \sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$	$\boldsymbol{u} := \sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$	is a Householder transformation. Indeed, $\boldsymbol{H} = \boldsymbol{I} - \boldsymbol{u} \boldsymbol{u}^*$, where $\boldsymbol{u} := \sqrt{2} \frac{\boldsymbol{v}}{\ \boldsymbol{v}\ _2}$ has
lyche-numerical-linear-algebra- F01601	125	1.000	$\sqrt{2}$	$\sqrt{2}$	length $\sqrt{2}$.
lyche-numerical-linear-algebra- F01602	125	1.000	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2, \boldsymbol{y}^* \boldsymbol{x}$	$\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2, \boldsymbol{y}^* \boldsymbol{x}$	Lemma 5.2 Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ are two vectors such that $\ \boldsymbol{x}\ _2 = \ \boldsymbol{y}\ _2$, $\boldsymbol{y}^* \boldsymbol{x}$ is
lyche-numerical-linear-algebra- F01603	125	0.993	$\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$	$\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$	real and $\boldsymbol{v} := \boldsymbol{x} - \boldsymbol{y} \neq \mathbf{0}$. Then $(\boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}) \boldsymbol{x} = \boldsymbol{y}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F01604	125	■ 0.993	$\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{y}$	$\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{y}$	real and $\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y}\neq\boldsymbol{0}$. Then $\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{y}$.
lyche-numerical-linear-algebra- F01605	125	■ 1.000	$\boldsymbol{x}^*\boldsymbol{x}=\boldsymbol{y}^*\boldsymbol{y}$	$\boldsymbol{x}^*\boldsymbol{x}=\boldsymbol{y}^*\boldsymbol{y}$	Proof Since $\boldsymbol{x}^*\boldsymbol{x}=\boldsymbol{y}^*\boldsymbol{y}$ and $\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=\boldsymbol{y}^*\boldsymbol{x}$ we have
lyche-numerical-linear-algebra- F01606	125	■ 1.000	$\operatorname{Re}\left(\boldsymbol{y}^*\boldsymbol{x}\right)=\boldsymbol{y}^*\boldsymbol{x}$	$\operatorname{Re}\left(\boldsymbol{y}^*\boldsymbol{x}\right)=\boldsymbol{y}^*\boldsymbol{x}$	Proof Since $\boldsymbol{x}^*\boldsymbol{x}=\boldsymbol{y}^*\boldsymbol{y}$ and $\operatorname{Re}(\boldsymbol{y}^*\boldsymbol{x})=\boldsymbol{y}^*\boldsymbol{x}$ we have
lyche-numerical-linear-algebra- F01607	125	■ 0.998	$\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{x}-\frac{2}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v}\boldsymbol{x}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$	$\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{x}-\frac{2\boldsymbol{v}^*\boldsymbol{x}}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$	But the $\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^*}{\boldsymbol{v}^*\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{x}-\frac{2\boldsymbol{v}^*\boldsymbol{x}}{\boldsymbol{v}^*\boldsymbol{v}}\boldsymbol{v}=\boldsymbol{x}-\boldsymbol{v}=\boldsymbol{y}$.
lyche-numerical-linear-algebra- F01608	125	■ 1.000	$\boldsymbol{H}\boldsymbol{x}$	$\boldsymbol{H}\boldsymbol{x}$	If $\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n$ it follows that $\boldsymbol{H}\boldsymbol{x}$ is the reflected image of $\boldsymbol{x}$. The “mirror” $\mathcal{M}:=$
lyche-numerical-linear-algebra- F01609	125	■ 1.000	$\mathcal{M}:=$	$\mathcal{M}:=$	If $\boldsymbol{x},\boldsymbol{y}\in\mathbb{R}^n$ it follows that $\boldsymbol{H}\boldsymbol{x}$ is the reflected image of $\boldsymbol{x}$. The “mirror” $\mathcal{M}:=$
lyche-numerical-linear-algebra- F01610	125	■ 1.000	$\left\{\boldsymbol{w}\in\mathbb{R}^n:\boldsymbol{w}^*\boldsymbol{v}=0\right\}$	$\{\boldsymbol{w}\in\mathbb{R}^n:\boldsymbol{w}^*\boldsymbol{v}=0\}$	$\{\boldsymbol{w}\in\mathbb{R}^n:\boldsymbol{w}^*\boldsymbol{v}=0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra- F01611	125	■ 1.000	$(\boldsymbol{x}+\boldsymbol{y})/2$	$(\boldsymbol{x}+\boldsymbol{y})/2$	$\{\boldsymbol{w}\in\mathbb{R}^n:\boldsymbol{w}^*\boldsymbol{v}=0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra- F01612	125	■ 1.000	$\boldsymbol{x}-\boldsymbol{y}$	$\boldsymbol{x}-\boldsymbol{y}$	$\{\boldsymbol{w}\in\mathbb{R}^n:\boldsymbol{w}^*\boldsymbol{v}=0\}$ contains the vector $(\boldsymbol{x}+\boldsymbol{y})/2$ and has normal $\boldsymbol{x}-\boldsymbol{y}$. This is
lyche-numerical-linear-algebra- F01613	126	■ 1.000	$\boldsymbol{x}:=[1,0,1]^T$	$\boldsymbol{x}:=[1,0,1]^T$	Example 5.1 (Reflector) Suppose $\boldsymbol{x}:=[1,0,1]^T$ and $\boldsymbol{y}:=[-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra- F01614	126	■ 1.000	$\boldsymbol{y}:=[-1,0,1]^T$	$\boldsymbol{y}:=[-1,0,1]^T$	Example 5.1 (Reflector) Suppose $\boldsymbol{x}:=[1,0,1]^T$ and $\boldsymbol{y}:=[-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra- F01615	126	■ 1.000	$\boldsymbol{v} =$	$\boldsymbol{v} =$	Example 5.1 (Reflector) Suppose $\boldsymbol{x}:=[1,0,1]^T$ and $\boldsymbol{y}:=[-1,0,1]^T$. Then $\boldsymbol{v} =$
lyche-numerical-linear-algebra- F01616	126	■ 1.000	$\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^T$	$\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^T$	$\boldsymbol{x}-\boldsymbol{y}=[2,0,0]^T$ and
lyche-numerical-linear-algebra- F01617	126	■ 0.999	$\boldsymbol{y}\boldsymbol{z}$	$\boldsymbol{y}\boldsymbol{z}$	is the $\boldsymbol{y}\boldsymbol{z}$ plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$
lyche-numerical-linear-algebra- F01618	126	■ 0.999	$\boldsymbol{H}\boldsymbol{x}=\boldsymbol{x}+[-1,0,1]^T=\boldsymbol{y}$	$\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$	is the $\boldsymbol{y}\boldsymbol{z}$ plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$
lyche-numerical-linear-algebra- F01619	126	■ 0.999	$\boldsymbol{P}\boldsymbol{x}=\boldsymbol{x}+[0,0,1]^T=$	$\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$	is the $\boldsymbol{y}\boldsymbol{z}$ plane (cf. Fig. 5.3), $\boldsymbol{H}\boldsymbol{x}=[-1,0,1]^T=\boldsymbol{y}$, and $\boldsymbol{P}\boldsymbol{x}=[0,0,1]^T=$
lyche-numerical-linear-algebra- F01620	126	■ 1.000	$(\boldsymbol{x}+\boldsymbol{y})/2\in\mathcal{M}$	$(\boldsymbol{x}+\boldsymbol{y})/2\in\mathcal{M}$	$(\boldsymbol{x}+\boldsymbol{y})/2\in\mathcal{M}$.
lyche-numerical-linear-algebra- F01621	126	■ 1.000	$\boldsymbol{H}\boldsymbol{x}=\boldsymbol{0}$	$\boldsymbol{H}\boldsymbol{x}=\boldsymbol{0}$	Proof If $\boldsymbol{x}=\boldsymbol{0}$ then $\boldsymbol{H}\boldsymbol{x}=\boldsymbol{0}$ and $a=0$. Any $\boldsymbol{u}$ with $\ \boldsymbol{u}\ _2=\sqrt{2}$ will work, and
lyche-numerical-linear-algebra- F01622	126	■ 1.000	$a=0$	$a=0$	Proof If $\boldsymbol{x}=\boldsymbol{0}$ then $\boldsymbol{H}\boldsymbol{x}=\boldsymbol{0}$ and $a=0$. Any $\boldsymbol{u}$ with $\ \boldsymbol{u}\ _2=\sqrt{2}$ will work, and
lyche-numerical-linear-algebra- F01623	126	■ 1.000	$\ \boldsymbol{u}\ _2=\sqrt{2}$	$\ \boldsymbol{u}\ _2=\sqrt{2}$	Proof If $\boldsymbol{x}=\boldsymbol{0}$ then $\boldsymbol{H}\boldsymbol{x}=\boldsymbol{0}$ and $a=0$. Any $\boldsymbol{u}$ with $\ \boldsymbol{u}\ _2=\sqrt{2}$ will work, and
lyche-numerical-linear-algebra- F01624	126	■ 1.000	$\boldsymbol{u}:=\sqrt{2}\boldsymbol{e}_1$	$\boldsymbol{u}:=\sqrt{2}\boldsymbol{e}_1$	we choose $\boldsymbol{u}:=\sqrt{2}\boldsymbol{e}_1$ in this case. For $\boldsymbol{x}\neq\boldsymbol{0}$ we define
lyche-numerical-linear-algebra- F01625	126	■ 1.000	$\boldsymbol{x}\neq\boldsymbol{0}$	$\boldsymbol{x}\neq\boldsymbol{0}$	we choose $\boldsymbol{u}:=\sqrt{2}\boldsymbol{e}_1$ in this case. For $\boldsymbol{x}\neq\boldsymbol{0}$ we define
lyche-numerical-linear-algebra- F01626	126	■ 0.971	$ \rho =1$	$ \rho =1$	Since $ \rho =1$ we have $\rho\ \boldsymbol{x}\ _2\boldsymbol{z}= \rho ^2\boldsymbol{x}=\boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2=1$ and $z_1=$
lyche-numerical-linear-algebra- F01627	126	■ 0.971	$\rho\ \boldsymbol{x}\ _2\boldsymbol{z}= \rho ^2\boldsymbol{x}=\boldsymbol{x}$	$\rho\ \boldsymbol{x}\ _2\boldsymbol{z}= \rho ^2\boldsymbol{x}=\boldsymbol{x}$	Since $ \rho =1$ we have $\rho\ \boldsymbol{x}\ _2\boldsymbol{z}= \rho ^2\boldsymbol{x}=\boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2=1$ and $z_1=$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F01628	126	■ 0.971	$\ \boldsymbol{z}\ _2=1$	$\ \boldsymbol{z}\ _2 = 1$	Since $ \rho = 1$ we have $\rho\ \boldsymbol{x}\ _2\boldsymbol{z} = \rho ^2\boldsymbol{x} = \boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2 = 1$ and $z_1 =$
lyche-numerical-linear-algebra- F01629	126	■ 0.971	$z_{\{1\}}=$	$z_1 =$	Since $ \rho = 1$ we have $\rho\ \boldsymbol{x}\ _2\boldsymbol{z} = \rho ^2\boldsymbol{x} = \boldsymbol{x}$. Moreover, $\ \boldsymbol{z}\ _2 = 1$ and $z_1 =$
lyche-numerical-linear-algebra- F01630	126	■ 0.969	$\left x_{\{1\}}\right /\ \boldsymbol{x}\ _2$	$ x_1 /\ \boldsymbol{x}\ _2$	$ x_1 /\ \boldsymbol{x}\ _2$ is real so that $\boldsymbol{u}^*\boldsymbol{u} = \frac{(z+\boldsymbol{e}_1)^*(z+\boldsymbol{e}_1)}{1+z_1} = \frac{2+2z_1}{1+z_1} = 2$. Finally,
lyche-numerical-linear-algebra- F01631	126	■ 0.969	$\boldsymbol{u}^*\boldsymbol{u}=\frac{\left(z+\boldsymbol{e}_1\right)^*\left(z+\boldsymbol{e}_1\right)}{1+z_1}=\frac{2+2 z_{\{1\}}}{1+z_{\{1\}}}=2$	$\boldsymbol{u}^*\boldsymbol{u} = \frac{(z+\boldsymbol{e}_1)^*(z+\boldsymbol{e}_1)}{1+z_1} = \frac{2+2z_1}{1+z_1} = 2$	$ x_1 /\ \boldsymbol{x}\ _2$ is real so that $\boldsymbol{u}^*\boldsymbol{u} = \frac{(z+\boldsymbol{e}_1)^*(z+\boldsymbol{e}_1)}{1+z_1} = \frac{2+2z_1}{1+z_1} = 2$. Finally,
lyche-numerical-linear-algebra- F01632	127	■ 1.000	$\boldsymbol{u}^*\boldsymbol{u}=2$	$\boldsymbol{u}^*\boldsymbol{u} = 2$	adapted from [17]. To any given $\boldsymbol{x} \in \mathbb{C}^n$, a number a and a vector $\boldsymbol{u}$ with $\boldsymbol{u}^*\boldsymbol{u} = 2$
lyche-numerical-linear-algebra- F01633	127	■ 0.986	$\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*\right) \boldsymbol{x}=a \boldsymbol{e}_{\{1\}}$	$(\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)\boldsymbol{x} = a\boldsymbol{e}_1$	is computed so that $(\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)\boldsymbol{x} = a\boldsymbol{e}$:
lyche-numerical-linear-algebra- F01634	127	■ 0.999	$z_{\{1\}}=\left x_{\{1\}}\right /\ \boldsymbol{x}\ _2 \geq 0$	$z_1 = x_1 /\ \boldsymbol{x}\ _2 \geq 0$	• In Theorem 5.8 the first component of $\boldsymbol{z}$ is $z_1 = x_1 /\ \boldsymbol{x}\ _2 \geq 0$. Since $\ \boldsymbol{z}\ _2 =$
lyche-numerical-linear-algebra- F01635	127	■ 0.999	$\ \boldsymbol{z}\ _2=$	$\ \boldsymbol{z}\ _2 =$	• In Theorem 5.8 the first component of $\boldsymbol{z}$ is $z_1 = x_1 /\ \boldsymbol{x}\ _2 \geq 0$. Since $\ \boldsymbol{z}\ _2 =$
lyche-numerical-linear-algebra- F01636	127	■ 1.000	$1 \leq 1+z_{\{1\}} \leq 2$	$1 \leq 1 + z_1 \leq 2$	1 we have $1 \leq 1 + z_1 \leq 2$. It follows that we avoid cancelation error when
lyche-numerical-linear-algebra- F01637	127	■ 1.000	$1+z_{\{1\}}$	$1 + z_1$	computing $1 + z_1$ and $\boldsymbol{u}$ and a are computed in a numerically stable way.
lyche-numerical-linear-algebra- F01638	127	■ 0.887	$\boldsymbol{H} \boldsymbol{x}=\left(\boldsymbol{I}-\boldsymbol{u} \boldsymbol{u}^*\right) \boldsymbol{x}=\boldsymbol{x}-\left(\boldsymbol{u}^* \boldsymbol{x}\right) \boldsymbol{u}$	$\boldsymbol{H}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)\boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u}$	Indeed, $\boldsymbol{H}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)\boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$
lyche-numerical-linear-algebra- F01639	127	■ 0.887	$\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$	$\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$	Indeed, $\boldsymbol{H}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)\boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$
lyche-numerical-linear-algebra- F01640	127	■ 0.887	$2 m$	$2m$	Indeed, $\boldsymbol{H}\boldsymbol{x} = (\boldsymbol{I} - \boldsymbol{u}\boldsymbol{u}^*)\boldsymbol{x} = \boldsymbol{x} - (\boldsymbol{u}^*\boldsymbol{x})\boldsymbol{u}$. If $\boldsymbol{u}, \boldsymbol{x} \in \mathbb{R}^m$ this requires $2m$
lyche-numerical-linear-algebra- F01641	127	■ 0.994	$\boldsymbol{u}^T \boldsymbol{x}, m$	$\boldsymbol{u}^T \boldsymbol{x}, m$	operations to find $\boldsymbol{u}^T \boldsymbol{x}$, m operations for $(\boldsymbol{u}^T \boldsymbol{x})\boldsymbol{u}$ and m operations for the final
lyche-numerical-linear-algebra- F01642	127	■ 0.994	$\left(\boldsymbol{u}^T \boldsymbol{x}\right) \boldsymbol{u}$	$(\boldsymbol{u}^T \boldsymbol{x})\boldsymbol{u}$	operations to find $\boldsymbol{u}^T \boldsymbol{x}$, m operations for $(\boldsymbol{u}^T \boldsymbol{x})\boldsymbol{u}$ and m operations for the final
lyche-numerical-linear-algebra- F01643	127	■ 1.000	$4 m$	$4m$	subtraction of the two vectors, a total of $4m$ arithmetic operations. If $\boldsymbol{A} \in \mathbb{R}^{m \times n}$
lyche-numerical-linear-algebra- F01644	127	■ 1.000	$4 m n$	$4mn$	then $4mn$ operations are required for $\boldsymbol{H} \boldsymbol{A}=\boldsymbol{A}-\left(\boldsymbol{u}^T \boldsymbol{A}\right) \boldsymbol{u}$, i.e., $4m$ operations
lyche-numerical-linear-algebra- F01645	127	■ 1.000	$\boldsymbol{H} \boldsymbol{A}=\boldsymbol{A}-\left(\boldsymbol{u}^T \boldsymbol{A}\right) \boldsymbol{u}$	$\boldsymbol{H}\boldsymbol{A} = \boldsymbol{A} - (\boldsymbol{u}^T \boldsymbol{A})\boldsymbol{u}$	then $4mn$ operations are required for $\boldsymbol{H} \boldsymbol{A} = \boldsymbol{A} - (\boldsymbol{u}^T \boldsymbol{A})\boldsymbol{u}$, i.e., $4m$ operations
lyche-numerical-linear-algebra- F01646	127	■ 1.000	$\boldsymbol{x}^T:=\left[\boldsymbol{y}, \boldsymbol{z}\right]^T$	$\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$	a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \leq k <$
lyche-numerical-linear-algebra- F01647	127	■ 1.000	$\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$	$\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$	a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \leq k <$
lyche-numerical-linear-algebra- F01648	127	■ 1.000	$1 \leq k <$	$1 \leq k <$	a vector. Suppose $\boldsymbol{x}^T := [\boldsymbol{y}, \boldsymbol{z}]^T$, where $\boldsymbol{y} \in \mathbb{C}^k, \boldsymbol{z} \in \mathbb{C}^{n-k}$ for some $1 \leq k <$
lyche-numerical-linear-algebra- F01649	127	■ 0.915	$[\hat{\boldsymbol{u}}, a]:=\operatorname{housegen}(\boldsymbol{z})$	$[\hat{\boldsymbol{u}}, a] := \text{housegen}(\boldsymbol{z})$	n . The command $[\hat{\boldsymbol{u}}, a] := \text{housegen}(\boldsymbol{z})$ defines a Householder transformation
lyche-numerical-linear-algebra- F01650	127	■ 0.384	$\hat{\boldsymbol{H}}=\boldsymbol{I}-\hat{\boldsymbol{u}} \hat{\boldsymbol{u}}^*$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}}\boldsymbol{z} = a\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$ we see that
lyche-numerical-linear-algebra- F01651	127	■ 0.384	$\hat{\boldsymbol{H}} \boldsymbol{z}=a \boldsymbol{e}_1 \in \mathbb{C}^{n-k}$	$\hat{\boldsymbol{H}}\boldsymbol{z} = a\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}}\boldsymbol{z} = a\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$ we see that
lyche-numerical-linear-algebra- F01652	127	■ 0.384	$\boldsymbol{u}:=\left[\begin{array}{c} \mathbf{0} \\ \hat{\boldsymbol{u}} \end{array}\right] \in \mathbb{C}^n$	$\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$	$\hat{\boldsymbol{H}} = \boldsymbol{I} - \hat{\boldsymbol{u}}\hat{\boldsymbol{u}}^*$ so that $\hat{\boldsymbol{H}}\boldsymbol{z} = a\boldsymbol{e}_1 \in \mathbb{C}^{n-k}$. With $\boldsymbol{u} := \begin{bmatrix} 0 \\ \hat{\boldsymbol{u}} \end{bmatrix} \in \mathbb{C}^n$ we see that

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lyche-numerical-linear-algebra F01653	127	1.000	$\boldsymbol{u}^* \boldsymbol{u} = \hat{\boldsymbol{u}}^* \hat{\boldsymbol{u}} = 2$	$\boldsymbol{u}^* \boldsymbol{u} = \hat{\boldsymbol{u}}^* \hat{\boldsymbol{u}} = 2$	$\hat{\boldsymbol{u}}^* \hat{\boldsymbol{u}} = \hat{\boldsymbol{u}}^* \hat{\boldsymbol{u}} = 2$, and
lyche-numerical-linear-algebra F01654	128	1.000	$\boldsymbol{R} \in \mathbb{C}^{m \times n}$	$\boldsymbol{R} \in \mathbb{C}^{m \times n}$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m \times n}$ is upper trapezoidal , if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra F01655	128	1.000	$r_{\{i, j\}} = 0$	$r_{i,j} = 0$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m \times n}$ is upper trapezoidal , if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra F01656	128	1.000	$j < i$	$j < i$	We say that a matrix $\boldsymbol{R} \in \mathbb{C}^{m \times n}$ is upper trapezoidal , if $r_{i,j} = 0$ for $j < i$ and
lyche-numerical-linear-algebra F01657	128	1.000	$i = 2, 3 \ldots, m$	$i = 2, 3 \ldots, m$	$\equiv 2, 3 \ldots, m$ Upper trapezoidal matrices corresponding to $m < n$, $m = n$, and
lyche-numerical-linear-algebra F01658	128	1.000	$m \leq n$	$m \leq n$	We treat the cases $m > n$ and $m \leq n$ separately and consider first $m > n$. We
lyche-numerical-linear-algebra F01659	128	0.996	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_n$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_n$	describe how to find a sequence $\boldsymbol{H}_1, \dots, \boldsymbol{H}_n$ of Householder transformations such
lyche-numerical-linear-algebra F01660	128	1.000	$\boldsymbol{R}_1$	$\boldsymbol{R}_1$	and where $\boldsymbol{R}_1$ is square and upper triangular. We define
lyche-numerical-linear-algebra F01661	129	1.000	$\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$	$\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$	Let $\hat{\boldsymbol{H}}_k := \boldsymbol{I} - \hat{\boldsymbol{u}}_k \hat{\boldsymbol{u}}_k^*$ be a Householder transformation that maps the first column
lyche-numerical-linear-algebra F01662	129	1.000	$\left[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)} \right]^T$	$\left[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)} \right]^T$	$[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra F01663	129	1.000	$\boldsymbol{D}_k$	$\boldsymbol{D}_k$	$[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra F01664	129	1.000	$\boldsymbol{e}_1, \hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$	$\boldsymbol{e}_1, \hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$	$[a_{k,k}^{(k)}, \dots, a_{m,k}^{(k)}]^T$ of $\boldsymbol{D}_k$ to a multiple of $\boldsymbol{e}_1$, $\hat{\boldsymbol{H}}_k(\boldsymbol{D}_k \boldsymbol{e}_1) = a_k \boldsymbol{e}_1$. Using Algorithm 5.1
lyche-numerical-linear-algebra F01665	129	0.865	$\left[\hat{\boldsymbol{u}}_k, a_k \right] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$	$[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$	we have $[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$. Then $\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$ is a Householder
lyche-numerical-linear-algebra F01666	129	0.865	$\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$	$\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$	we have $[\hat{\boldsymbol{u}}_k, a_k] = \text{housegen}(\boldsymbol{D}_k \boldsymbol{e}_1)$. Then $\boldsymbol{H}_k := \begin{bmatrix} \boldsymbol{I}_{k-1} & \mathbf{0} \\ \mathbf{0} & \hat{\boldsymbol{H}}_k \end{bmatrix}$ is a Householder
lyche-numerical-linear-algebra F01667	129	1.000	$\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$	$\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra F01668	129	1.000	$\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$	$\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra F01669	129	1.000	$\boldsymbol{A}_{k+1}$	$\boldsymbol{A}_{k+1}$	where $\boldsymbol{B}_{k+1} \in \mathbb{C}^{k \times k}$ is upper triangular and $\boldsymbol{D}_{k+1} \in \mathbb{C}^{(m-k) \times (n-k)}$. Thus $\boldsymbol{A}_{k+1}$ is
lyche-numerical-linear-algebra F01670	129	1.000	$\boldsymbol{R} := \boldsymbol{A}_{n+1} = \begin{bmatrix} \boldsymbol{R}_1 \\ \mathbf{0} \end{bmatrix}$	$\boldsymbol{R} := \boldsymbol{A}_{n+1} = \begin{bmatrix} \boldsymbol{R}_1 \\ \mathbf{0} \end{bmatrix}$	further. At the end $\boldsymbol{R} := \boldsymbol{A}_{n+1} = \begin{bmatrix} \boldsymbol{R}_1 \\ \mathbf{0} \end{bmatrix}$, where $\boldsymbol{R}_1$ is upper triangular.
lyche-numerical-linear-algebra F01671	129	1.000	$m > 1$	$m > 1$	already in upper trapezoidal form. Suppose $m > 1$. In this case $m - 1$ Householder
lyche-numerical-linear-algebra F01672	129	1.000	$m - 1$	$m - 1$	already in upper trapezoidal form. Suppose $m > 1$. In this case $m - 1$ Householder
lyche-numerical-linear-algebra F01673	129	0.999	$\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$	transformations will suffice and $\boldsymbol{H}_{m-1} \cdots \boldsymbol{H}_1 \boldsymbol{A}$ is upper trapezoidal.
lyche-numerical-linear-algebra F01674	129	1.000	$\hat{\boldsymbol{u}}_k = [u_{kk}, \dots, u_{mk}]^T$	$\hat{\boldsymbol{u}}_k = [u_{kk}, \dots, u_{mk}]^T$	In an algorithm we can store most of the vector $\hat{\boldsymbol{u}}_k = [u_{kk}, \dots, u_{mk}]^T$ and the
lyche-numerical-linear-algebra F01675	129	1.000	$u_{k,k}$	$u_{k,k}$	matrix and $\boldsymbol{A}_k$ in $\boldsymbol{A}$. However, the elements $u_{k,k}$ and $a_k = r_{k,k}$ have to compete for
lyche-numerical-linear-algebra F01676	129	1.000	$a_k = r_{k,k}$	$a_k = r_{k,k}$	matrix and $\boldsymbol{A}_k$ in $\boldsymbol{A}$. However, the elements $u_{k,k}$ and $a_k = r_{k,k}$ have to compete for

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lyche-numerical-linear-algebra-F01677	129	1.000	m=4	$m = 4$	the diagonal in A . For $m = 4$ and $n = 3$ the two possibilities look as follows:
lyche-numerical-linear-algebra-F01678	129	1.000	$\boldsymbol{B} \in \mathbb{C}^{m \times r}$	$B \in \mathbb{C}^{m \times r}$	$B \in \mathbb{C}^{m \times r}$ as input, and uses <code>housegen</code> to compute Householder transformations
lyche-numerical-linear-algebra-F01679	129	1.000	$\boldsymbol{H}_1, \ldots, \boldsymbol{H}_s$	$H_1, \dots, H_s$	$H_1, \dots, H_s$ so that $R = H_s \cdots H_1 A$ is upper trapezoidal, and $C = H_s \cdots H_1 B$.
lyche-numerical-linear-algebra-F01680	129	1.000	$\boldsymbol{R} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{A}$	$R = H_s \cdots H_1 A$	$H_1, \dots, H_s$ so that $R = H_s \cdots H_1 A$ is upper trapezoidal, and $C = H_s \cdots H_1 B$.
lyche-numerical-linear-algebra-F01681	129	1.000	$\boldsymbol{C} = \boldsymbol{H}_s \cdots \boldsymbol{H}_1 \boldsymbol{B}$	$C = H_s \cdots H_1 B$	$H_1, \dots, H_s$ so that $R = H_s \cdots H_1 A$ is upper trapezoidal, and $C = H_s \cdots H_1 B$.
lyche-numerical-linear-algebra-F01682	129	1.000	$r_{k, k}$	$r_{k,k}$	matrix with m rows and 0 columns. $r_{k,k}$ is stored in A , and $u_{k,k}$ is stored in a
lyche-numerical-linear-algebra-F01683	129	0.961	$\operatorname{side}(\boldsymbol{s}) \boldsymbol{B}$	side(s) B	and least squares problems with right hand side(s) B , and to compute the product of
lyche-numerical-linear-algebra-F01684	129	0.999	$\boldsymbol{B} = \boldsymbol{I}$	$B = I$	the Householder transformations by choosing $B = I$.
lyche-numerical-linear-algebra-F01685	130	0.998	$v = \hat{u}_k$	$v = \hat{u}_k$	Here $v = \hat{u}_k$ and the update is computed as $\hat{H}_k C = (I - vv^*)C = C - v(v^*C)$.
lyche-numerical-linear-algebra-F01686	130	0.998	$\hat{H}_k C = (I - vv^*)C = C - v(v^*C)$	$\hat{H}_k C = (I - vv^*)C = C - v(v^*C)$	Here $v = \hat{u}_k$ and the update is computed as $\hat{H}_k C = (I - vv^*)C = C - v(v^*C)$.
lyche-numerical-linear-algebra-F01687	130	1.000	$n+1, \ldots, m$	$n+1, \dots, m$	zeros in rows $n+1, \dots, m$.
lyche-numerical-linear-algebra-F01688	130	1.000	$C - v(v^*C)$	$C - v(v^*C)$	The bulk of the work in Algorithm 5.2 is the computation of $C - v(v^*C)$ for each
lyche-numerical-linear-algebra-F01689	130	1.000	$C \in \mathbb{C}^{(m-k+1) \times (n+r-k)}$	$C \in \mathbb{C}^{(m-k+1) \times (n+r-k)}$	k . Since in Algorithm 5.2, $C \in \mathbb{C}^{(m-k+1) \times (n+r-k)}$ and $m \geq n$ the cost of computing
lyche-numerical-linear-algebra-F01690	130	1.000	$m \geq n$	$m \geq n$	k . Since in Algorithm 5.2, $C \in \mathbb{C}^{(m-k+1) \times (n+r-k)}$ and $m \geq n$ the cost of computing
lyche-numerical-linear-algebra-F01691	130	1.000	$C - v(v^T C)$	$C - v(v^T C)$	the update $C - v(v^T C)$ in the real case is $4(m-k+1)(n+r-k)$ arithmetic
lyche-numerical-linear-algebra-F01692	130	1.000	$4(m-k+1)(n+r-k)$	$4(m-k+1)(n+r-k)$	the update $C - v(v^T C)$ in the real case is $4(m-k+1)(n+r-k)$ arithmetic
lyche-numerical-linear-algebra-F01693	130	1.000	$r=0$	$r = 0$	For $m = n$ and $r = 0$ this gives $4n^3/3 = 2G_n$ for the number of arithmetic
lyche-numerical-linear-algebra-F01694	130	1.000	$4n^3/3 = 2G_n$	$4n^3/3 = 2G_n$	For $m = n$ and $r = 0$ this gives $4n^3/3 = 2G_n$ for the number of arithmetic
lyche-numerical-linear-algebra-F01695	130	1.000	$Rx = c$	$Rx = c$	obtain an upper triangular system $Rx = c$ that is upper triangular and nonsingular if
lyche-numerical-linear-algebra-F01696	131	1.000	$4n^3/3 =$	$4n^3/3 =$	reason is that the number of arithmetic operations in (5.16) when $m = n$ is $4n^3/3 =$
lyche-numerical-linear-algebra-F01697	131	0.998	$R = H_{n-1} \cdots H_1 A$	$R = H_{n-1} \cdots H_1 A$	Algorithm 5.2 gives $R = H_{n-1} \cdots H_1 A$ implying the factorization $A = QR$,
lyche-numerical-linear-algebra-F01698	131	1.000	$Q = H_1 \cdots H_{n-1}$	$Q = H_1 \cdots H_{n-1}$	where $Q = H_1 \cdots H_{n-1}$ is unitary and R is upper triangular. This is known as a
lyche-numerical-linear-algebra-F01699	131	0.982	$\mathbf{Q} R$	QR	$A = QR$ is a QR decomposition of A if $Q \in \mathbb{C}^{m,m}$ is square and unitary and
lyche-numerical-linear-algebra-F01700	131	0.982	$Q \in \mathbb{C}^{m,m}$	$Q \in \mathbb{C}^{m,m}$	$A = QR$ is a QR decomposition of A if $Q \in \mathbb{C}^{m,m}$ is square and unitary and
lyche-numerical-linear-algebra-F01701	131	1.000	$R_1 \in \mathbb{C}^{n \times n}$	$R_1 \in \mathbb{C}^{n \times n}$	where $R_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n,n}$ is the zero matrix with $m - n$
lyche-numerical-linear-algebra-F01702	131	1.000	$\mathbf{0}_{m-n,n}$	$\mathbf{0}_{m-n,n}$	where $R_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n,n}$ is the zero matrix with $m - n$

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lyche-numerical-linear-algebra-F01703	131	1.000	$m - n$	$m - n$	where $R_1 \in \mathbb{C}^{n \times n}$ is upper triangular and $\mathbf{0}_{m-n,n}$ is the zero matrix with $m - n$
lyche-numerical-linear-algebra-F01704	131	0.961	$\boldsymbol{A} = \boldsymbol{Q}_1 \boldsymbol{R}_1$	$A = Q_1 R_1$	rows and n columns. For $m \geq n$ we call $A = Q_1 R_1$ a QR factorization of A if
lyche-numerical-linear-algebra-F01705	131	1.000	$\boldsymbol{Q}_1 \in \mathbb{C}^{m \times n}$	$Q_1 \in \mathbb{C}^{m \times n}$	$Q_1 \in \mathbb{C}^{m \times n}$ has orthonormal columns and $R_1 \in \mathbb{C}^{n \times n}$ is upper triangular.
lyche-numerical-linear-algebra-F01706	131	1.000	$\boldsymbol{Q} \boldsymbol{R}$	QR	QR by simply using the first n columns of Q and the first n rows of R . Indeed, if
lyche-numerical-linear-algebra-F01707	131	1.000	$\left[\boldsymbol{Q}_1, \boldsymbol{Q}_2 \right]$	$[Q_1, Q_2]$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ \mathbf{0} \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01708	131	1.000	$\boldsymbol{R} = \left[\begin{array}{c} \boldsymbol{R}_1 \\ \mathbf{0} \end{array} \right]$	$R = \begin{bmatrix} R_1 \\ \mathbf{0} \end{bmatrix}$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ \mathbf{0} \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01709	131	1.000	$\boldsymbol{Q}_1 \in \mathbb{R}^{m \times n}$	$Q_1 \in \mathbb{R}^{m \times n}$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ \mathbf{0} \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01710	131	1.000	$\boldsymbol{R}_1 \in \mathbb{R}^{n \times n}$	$R_1 \in \mathbb{R}^{n \times n}$	we partition Q as $[Q_1, Q_2]$ and $R = \begin{bmatrix} R_1 \\ \mathbf{0} \end{bmatrix}$, where $Q_1 \in \mathbb{R}^{m \times n}$ and $R_1 \in \mathbb{R}^{n \times n}$
lyche-numerical-linear-algebra-F01711	131	1.000	$\left\{ \boldsymbol{q}_1, \dots, \boldsymbol{q}_n \right\}$	$\{q_1, \dots, q_n\}$	columns $\{q_1, \dots, q_n\}$ of Q_1 into an orthonormal basis $\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$
lyche-numerical-linear-algebra-F01712	131	1.000	$\boldsymbol{Q}_1$	Q_1	columns $\{q_1, \dots, q_n\}$ of Q_1 into an orthonormal basis $\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$
lyche-numerical-linear-algebra-F01713	131	1.000	$\left\{ \boldsymbol{q}_1, \dots, \boldsymbol{q}_n, \boldsymbol{q}_{n+1}, \dots, \boldsymbol{q}_m \right\}$	$\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$	columns $\{q_1, \dots, q_n\}$ of Q_1 into an orthonormal basis $\{q_1, \dots, q_n, q_{n+1}, \dots, q_m\}$
lyche-numerical-linear-algebra-F01714	131	1.000	$\boldsymbol{Q} = \left[\boldsymbol{q}_1, \dots, \boldsymbol{q}_m \right]$	$Q = [q_1, \dots, q_m]$	$A = QR$, where $Q = [q_1, \dots, q_m]$ and $R = \begin{bmatrix} R_1 \\ \mathbf{0} \end{bmatrix}$.
lyche-numerical-linear-algebra-F01715	132	0.743	$\boldsymbol{Q}^T \boldsymbol{Q} = \boldsymbol{I}$	$Q^T Q = I$	Since $Q^T Q = I$ and R is upper trapezoidal, this is a QR decomposition of A . A
lyche-numerical-linear-algebra-F01716	132	1.000	$\boldsymbol{Q} \boldsymbol{R}$	QR	$m, n \in \mathbb{N}$ has a QR decomposition.
lyche-numerical-linear-algebra-F01717	132	1.000	$\boldsymbol{A} = [\boldsymbol{1}] \boldsymbol{A}$	$A = [1]A$	Proof If $m = 1$ then A is already in upper trapezoidal form and $A = [1]A$ is a
lyche-numerical-linear-algebra-F01718	132	1.000	$s := \min(m-1, n)$	$s := \min(m-1, n)$	QR decomposition of A . Suppose $m > 1$ and set $s := \min(m-1, n)$. Note that
lyche-numerical-linear-algebra-F01719	132	0.999	$\boldsymbol{R} = \boldsymbol{C} \boldsymbol{A}$	$R = CA$	for any vector x . With $B = I$ in Algorithm 5.2 we obtain $R = CA$ and $C =$
lyche-numerical-linear-algebra-F01720	132	0.999	$\boldsymbol{C} =$	$C =$	for any vector x . With $B = I$ in Algorithm 5.2 we obtain $R = CA$ and $C =$
lyche-numerical-linear-algebra-F01721	132	0.615	$\boldsymbol{H}_s \cdots \boldsymbol{H}_2 \boldsymbol{H}_1$	$H_s \cdots H_2 H_1$	$H_s \cdots H_2 H_1$. Thus $A = QR$ is a QR decomposition of A since $Q := C^* =$
lyche-numerical-linear-algebra-F01722	132	0.615	$\boldsymbol{Q} := \boldsymbol{C}^* =$	$Q := C^* =$	$H_s \cdots H_2 H_1$. Thus $A = QR$ is a QR decomposition of A since $Q := C^* =$
lyche-numerical-linear-algebra-F01723	132	1.000	$\boldsymbol{H}_1 \cdots \boldsymbol{H}_s$	$H_1 \cdots H_s$	$H_1 \cdots H_s$ is a product of unitary matrices and therefore unitary.
lyche-numerical-linear-algebra-F01724	132	1.000	$\boldsymbol{A}^* \boldsymbol{A} =$	$A^* A =$	Proof Let $A = Q_1 R_1$ be a QR factorization of $A \in \mathbb{C}^{m \times n}$. Now $A^* A =$
lyche-numerical-linear-algebra-F01725	132	1.000	$\boldsymbol{R}_1^* \boldsymbol{Q}_1^* \boldsymbol{Q}_1 \boldsymbol{R}_1 = \boldsymbol{R}_1^* \boldsymbol{R}_1$	$R_1^* Q_1^* Q_1 R_1 = R_1^* R_1$	$R_1^* Q_1^* Q_1 R_1 = R_1^* R_1$. By Lemma 4.2 the matrix $A^* A$ is positive definite, the
lyche-numerical-linear-algebra-F01726	132	1.000	$\boldsymbol{R}_1^* \boldsymbol{R}_1$	$R_1^* R_1$	matrix R_1 is nonsingular, and if its diagonal elements are positive then $R_1^* R_1$ is the
lyche-numerical-linear-algebra-F01727	132	1.000	$\boldsymbol{Q}_1 = \boldsymbol{A} \boldsymbol{R}_1^{-1}$	$Q_1 = AR_1^{-1}$	that R_1 is unique and since necessarily $Q_1 = AR_1^{-1}$, it must also be unique.
lyche-numerical-linear-algebra-F01728	132	1.000	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A} = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$	$B := A^T A = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$	uniqueness proof of Theorem 5.10. We find $B := A^T A = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$. The Cholesky

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lyche-numerical-linear-algebra- F01729	132	1.000	$\boldsymbol{B}=\boldsymbol{R}^T \boldsymbol{R}$	$\boldsymbol{B}=\boldsymbol{R}^T \boldsymbol{R}$	factorization of $\boldsymbol{B}=\boldsymbol{R}^T \boldsymbol{R}$ is given by $\boldsymbol{R}=\frac{1}{\sqrt{5}}\begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$. Now $\boldsymbol{R}^{-1}=\frac{1}{3\sqrt{5}}\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$
lyche-numerical-linear-algebra- F01730	132	1.000	$\boldsymbol{R}=\frac{1}{\sqrt{5}}\begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$	$\boldsymbol{R}=\frac{1}{\sqrt{5}}\begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$	factorization of $\boldsymbol{B}=\boldsymbol{R}^T \boldsymbol{R}$ is given by $\boldsymbol{R}=\frac{1}{\sqrt{5}}\begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$. Now $\boldsymbol{R}^{-1}=\frac{1}{3\sqrt{5}}\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$
lyche-numerical-linear-algebra- F01731	132	1.000	$\boldsymbol{R}^{-1}=\frac{1}{3\sqrt{5}}\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$	$\boldsymbol{R}^{-1}=\frac{1}{3\sqrt{5}}\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$	factorization of $\boldsymbol{B}=\boldsymbol{R}^T \boldsymbol{R}$ is given by $\boldsymbol{R}=\frac{1}{\sqrt{5}}\begin{bmatrix} 5 & -4 \\ 0 & 3 \end{bmatrix}$. Now $\boldsymbol{R}^{-1}=\frac{1}{3\sqrt{5}}\begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$
lyche-numerical-linear-algebra- F01732	132	0.889	$\boldsymbol{Q}=\boldsymbol{A} \boldsymbol{R}^{-1}=\frac{1}{\sqrt{5}}\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$	$\boldsymbol{Q}=\boldsymbol{A} \boldsymbol{R}^{-1}=\frac{1}{\sqrt{5}}\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$	so $\boldsymbol{Q}=\boldsymbol{A} \boldsymbol{R}^{-1}=\frac{1}{\sqrt{5}}\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$. Since $\boldsymbol{A}$ is square $\boldsymbol{A}=\boldsymbol{Q} \boldsymbol{R}$ is both the QR
lyche-numerical-linear-algebra- F01733	133	0.992	$\boldsymbol{a}_1, \dots, \boldsymbol{a}_n$	$\boldsymbol{a}_1, \dots, \boldsymbol{a}_n$	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ be the result of applying Gram Schmidt to the columns $\boldsymbol{a}_1, \dots, \boldsymbol{a}_n$ of $\boldsymbol{A}$,
lyche-numerical-linear-algebra- F01734	133	1.000	$\mathcal{R}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A})$	has orthonormal columns, since $\{\boldsymbol{q}_1, \dots, \boldsymbol{q}_n\}$ is an orthonormal basis for $\mathcal{R}(\boldsymbol{A})$ by
lyche-numerical-linear-algebra- F01735	133	1.000	$\boldsymbol{A}=\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}=\begin{bmatrix} \boldsymbol{a}_1 \\ \boldsymbol{a}_2 \end{bmatrix}$	$\boldsymbol{A}=\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}=\begin{bmatrix} \boldsymbol{a}_1 \\ \boldsymbol{a}_2 \end{bmatrix}$	and factorization of the matrix $\boldsymbol{A}=\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}=\begin{bmatrix} \boldsymbol{a}_1 \\ \boldsymbol{a}_2 \end{bmatrix}$ using Gram-Schmidt.
lyche-numerical-linear-algebra- F01736	133	0.999	$\boldsymbol{v}_1=\boldsymbol{a}_1$	$\boldsymbol{v}_1=\boldsymbol{a}_1$	Using (5.17) we find $\boldsymbol{v}_1=\boldsymbol{a}_1$ and $\boldsymbol{v}_2=\boldsymbol{a}_2-\frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1=\frac{3}{5}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Thus $\boldsymbol{Q}=[\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-algebra- F01737	133	0.999	$\boldsymbol{v}_2=\boldsymbol{a}_2-\frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1=\frac{3}{5}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\boldsymbol{v}_2=\boldsymbol{a}_2-\frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1=\frac{3}{5}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	Using (5.17) we find $\boldsymbol{v}_1=\boldsymbol{a}_1$ and $\boldsymbol{v}_2=\boldsymbol{a}_2-\frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1=\frac{3}{5}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Thus $\boldsymbol{Q}=[\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-algebra- F01738	133	0.999	$\boldsymbol{Q}=\begin{bmatrix} \boldsymbol{q}_1 \\ \boldsymbol{q}_2 \end{bmatrix}$	$\boldsymbol{Q}=[\boldsymbol{q}_1, \boldsymbol{q}_2]$	Using (5.17) we find $\boldsymbol{v}_1=\boldsymbol{a}_1$ and $\boldsymbol{v}_2=\boldsymbol{a}_2-\frac{\boldsymbol{a}_2^T \boldsymbol{v}_1}{\boldsymbol{v}_1^T \boldsymbol{v}_1} \boldsymbol{v}_1=\frac{3}{5}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Thus $\boldsymbol{Q}=[\boldsymbol{q}_1, \boldsymbol{q}_2]$,
lyche-numerical-linear-algebra- F01739	133	1.000	$\boldsymbol{q}_1=\frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$	$\boldsymbol{q}_1=\frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$	where $\boldsymbol{q}_1=\frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\boldsymbol{q}_2=\frac{1}{\sqrt{5}}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. By (5.18) we find
lyche-numerical-linear-algebra- F01740	133	1.000	$\boldsymbol{q}_2=\frac{1}{\sqrt{5}}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\boldsymbol{q}_2=\frac{1}{\sqrt{5}}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	where $\boldsymbol{q}_1=\frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\boldsymbol{q}_2=\frac{1}{\sqrt{5}}\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. By (5.18) we find
lyche-numerical-linear-algebra- F01741	134	1.000	$n=4$	$n=4$	by a Wilkinson diagram for $n=4$
lyche-numerical-linear-algebra- F01742	134	1.000	$\boldsymbol{P} \in \mathbb{R}^{2,2}$	$\boldsymbol{P} \in \mathbb{R}^{2,2}$	a Given's rotation) is a matrix $\boldsymbol{P} \in \mathbb{R}^{2,2}$ of the form
lyche-numerical-linear-algebra- F01743	134	1.000	$\theta \in [0, 2\pi)$	$\theta \in [0, 2\pi)$	A plane rotation is an orthogonal matrix and there is a unique angle $\theta \in [0, 2\pi)$
lyche-numerical-linear-algebra- F01744	134	1.000	$c=\cos \theta$	$c=\cos \theta$	such that $c=\cos \theta$ and $s=\sin \theta$. Moreover, the identity matrix is a plane rotation
lyche-numerical-linear-algebra- F01745	134	1.000	$s=\sin \theta$	$s=\sin \theta$	such that $c=\cos \theta$ and $s=\sin \theta$. Moreover, the identity matrix is a plane rotation
lyche-numerical-linear-algebra- F01746	134	1.000	$\theta=0$	$\theta=0$	corresponding to $\theta=0$. A vector $\boldsymbol{x}$ in the plane is rotated an angle θ clockwise by
lyche-numerical-linear-algebra- F01747	134	1.000	$\boldsymbol{P}=\boldsymbol{R}$	$\boldsymbol{P}=\boldsymbol{R}$	$\boldsymbol{P}=\boldsymbol{R}$. See Exercise 5.16 and Fig. 5.4.
lyche-numerical-linear-algebra- F01748	135	1.000	$\boldsymbol{P}=\boldsymbol{I}$	$\boldsymbol{P}=\boldsymbol{I}$	and we have introduced a zero in $\boldsymbol{x}$. We can take $\boldsymbol{P}=\boldsymbol{I}$ when $\boldsymbol{x}=\mathbf{0}$.
lyche-numerical-linear-algebra- F01749	135	1.000	$1 \leq i < j \leq n$	$1 \leq i < j \leq n$	For an n -vector $\boldsymbol{x} \in \mathbb{R}^n$ and $1 \leq i < j \leq n$ we define a rotation in the i, j-plane
lyche-numerical-linear-algebra- F01750	135	0.998	$\boldsymbol{P}_{ij}=\left(p_{kl}\right) \in \mathbb{R}^{n \times n}$	$\boldsymbol{P}_{ij}=\left(p_{kl}\right) \in \mathbb{R}^{n \times n}$	as a matrix $\boldsymbol{P}_{ij}=\left(p_{kl}\right) \in \mathbb{R}^{n \times n}$ by $p_{kl}=\delta_{kl}$ except for positions ii, jj, ij, ji ,

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lyche-numerical-linear-algebra-135 F01751	135	■ 0.998	$p_{kl}=\delta_{kl}$	$p_{kl} = \delta_{kl}$	as a matrix $\mathbf{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n \times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji ,
lyche-numerical-linear-algebra-135 F01752	135	■ 0.998	$i\ i, \ j\ j, \ i\ j, \ j\ i$	ii, jj, ij, ji	as a matrix $\mathbf{P}_{ij} = (p_{kl}) \in \mathbb{R}^{n \times n}$ by $p_{kl} = \delta_{kl}$ except for positions ii, jj, ij, ji ,
lyche-numerical-linear-algebra-135 F01753	135	■ 1.000	$\boldsymbol{B}=\boldsymbol{P}_{ij}\boldsymbol{A}$	$\mathbf{B} = \mathbf{P}_{ij}\mathbf{A}$	column i and j . In particular, if $\mathbf{B} = \mathbf{P}_{ij}\mathbf{A}$ and $\mathbf{C} = \mathbf{A}\mathbf{P}_{ij}$ then $\mathbf{B}(k, :) = \mathbf{A}(k, :)$,
lyche-numerical-linear-algebra-135 F01754	135	■ 1.000	$\boldsymbol{C}=\boldsymbol{A}\boldsymbol{P}_{ij}$	$\mathbf{C} = \mathbf{A}\mathbf{P}_{ij}$	column i and j . In particular, if $\mathbf{B} = \mathbf{P}_{ij}\mathbf{A}$ and $\mathbf{C} = \mathbf{A}\mathbf{P}_{ij}$ then $\mathbf{B}(k, :) = \mathbf{A}(k, :)$,
lyche-numerical-linear-algebra-135 F01755	135	■ 1.000	$\boldsymbol{B}(k, :)=\boldsymbol{A}(k, :)$	$\mathbf{B}(k, :) = \mathbf{A}(k, :)$	column i and j . In particular, if $\mathbf{B} = \mathbf{P}_{ij}\mathbf{A}$ and $\mathbf{C} = \mathbf{A}\mathbf{P}_{ij}$ then $\mathbf{B}(k, :) = \mathbf{A}(k, :)$,
lyche-numerical-linear-algebra-135 F01756	135	■ 1.000	$\boldsymbol{C}(:, k)=\boldsymbol{A}(:, k)$	$\mathbf{C}(:, k) = \mathbf{A}(:, k)$	$\mathbf{C}(:, k) = \mathbf{A}(:, k)$ for all $k \neq i, j$ and
lyche-numerical-linear-algebra-135 F01757	135	■ 1.000	$k \neq i, j$	$k \neq i, j$	$\mathbf{C}(:, k) = \mathbf{A}(:, k)$ for all $k \neq i, j$ and
lyche-numerical-linear-algebra-135 F01758	135	■ 0.999	$2\ n^{\{3\}}$	$2n^3$	number of arithmetic operations is asymptotically $2n^3$, corresponding to the work
lyche-numerical-linear-algebra-135 F01759	135	■ 1.000	$\boldsymbol{P}_{i, \ i+1}$	$\mathbf{P}_{i, i+1}$	using rotations $\mathbf{P}_{i, i+1}$ for $i = 1, \dots, n - 1$. For $n = 4$ the process can be illustrated
lyche-numerical-linear-algebra-136 F01760	136	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{y}^* \boldsymbol{A}^* \boldsymbol{A} \boldsymbol{x}$	$\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{y}^* \mathbf{A}^* \mathbf{A} \mathbf{x}$	dent columns. Show that $\langle \mathbf{x}, \mathbf{y} \rangle := \mathbf{y}^* \mathbf{A}^* \mathbf{A} \mathbf{x}$ defines an inner product on $\mathbb{C}^n$.
lyche-numerical-linear-algebra-136 F01761	136	■ 0.998	$[0, \ \pi / 2]$	$[0, \pi/2]$	plex case there is a unique angle θ in $[0, \pi/2]$ such that
lyche-numerical-linear-algebra-136 F01762	136	■ 1.000	$\boldsymbol{x}=\boldsymbol{e}_{\{1\}}$	$\mathbf{x} = \mathbf{e}_1$	Exercise 5.4 (What Does Algorithm Housegen Do When $\mathbf{x} = \mathbf{e}_1$?) Determine
lyche-numerical-linear-algebra-136 F01763	136	■ 1.000	$\ \boldsymbol{x}\ _2=\ \boldsymbol{y}\ _2$	$\ \mathbf{x}\ _2 = \ \mathbf{y}\ _2$	$\ \mathbf{x}\ _2 = \ \mathbf{y}\ _2$ and $\mathbf{v} := \mathbf{x} - \mathbf{y} \neq \mathbf{0}$ then it follows from Example 5.1 that
lyche-numerical-linear-algebra-136 F01764	136	■ 0.998	$\left(\boldsymbol{I}-2\frac{\boldsymbol{v}\boldsymbol{v}^T}{\boldsymbol{v}^T\boldsymbol{v}}\right)\boldsymbol{x}=\boldsymbol{y}$	$\left(\mathbf{I} - 2\frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}\right)\mathbf{x} = \mathbf{y}$	$(\mathbf{I} - 2\frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}})\mathbf{x} = \mathbf{y}$. Use this to construct a Householder transformation $\mathbf{H}$ such
lyche-numerical-linear-algebra-136 F01765	136	■ 1.000	$\boldsymbol{H}\boldsymbol{x}=\boldsymbol{y}$	$\mathbf{H}\mathbf{x} = \mathbf{y}$	that $\mathbf{H}\mathbf{x} = \mathbf{y}$ in the following cases.
lyche-numerical-linear-algebra-136 F01766	136	■ 0.995	$\boldsymbol{x}=\left[\begin{array}{l}3\\4\end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l}5\\0\end{array}\right]$	$\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$	a) $\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$.
lyche-numerical-linear-algebra-136 F01767	136	■ 1.000	$\boldsymbol{x}=\left[\begin{array}{l}2\\2\\1\end{array}\right], \quad \boldsymbol{y}=\left[\begin{array}{l}0\\3\\0\end{array}\right]$	$\mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$	b) $\mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$.
lyche-numerical-linear-algebra-137 F01768	137	■ 1.000	$\boldsymbol{x}=[\cos \phi, \sin \phi]^T$	$\mathbf{x} = [\cos \phi, \sin \phi]^T$	Find $\mathbf{H}\mathbf{x}$ if $\mathbf{x} = [\cos \phi, \sin \phi]^T$.
lyche-numerical-linear-algebra-137 F01769	137	■ 0.997	$\boldsymbol{v}:=\boldsymbol{x}-\boldsymbol{y} \neq \mathbf{0}$	$\mathbf{v} := \mathbf{x} - \mathbf{y} \neq \mathbf{0}$	a) Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ with $\ \mathbf{x}\ _2 = \ \mathbf{y}\ _2$ and $\mathbf{v} := \mathbf{x} - \mathbf{y} \neq \mathbf{0}$. Show that
lyche-numerical-linear-algebra-137 F01770	137	■ 1.000	$\boldsymbol{B} \in \mathbb{R}^{\{4,4\}}$	$\mathbf{B} \in \mathbb{R}^{4,4}$	b) Let $\mathbf{B} \in \mathbb{R}^{4,4}$ be given by
lyche-numerical-linear-algebra-137 F01771	137	■ 1.000	$0<\epsilon<1$	$0 < \epsilon < 1$	where $0 < \epsilon < 1$. Compute a Householder transformation $\mathbf{H}$ and a matrix $\mathbf{B}_1$
lyche-numerical-linear-algebra-137 F01772	137	■ 1.000	$\boldsymbol{B}_1$	$\mathbf{B}_1$	where $0 < \epsilon < 1$. Compute a Householder transformation $\mathbf{H}$ and a matrix $\mathbf{B}_1$
lyche-numerical-linear-algebra-137 F01773	137	■ 0.857	$\boldsymbol{B}_1:=\boldsymbol{H}\boldsymbol{B}\boldsymbol{H}$	$\mathbf{B}_1 := \mathbf{H}\mathbf{B}\mathbf{H}$	such that the first column of $\mathbf{B}_1 := \mathbf{H}\mathbf{B}\mathbf{H}$ has a zero in the last two positions.
lyche-numerical-linear-algebra-138 F01774	138	■ 1.000	$\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{\{3 \times 3\}}$	$\mathbf{H}_1, \mathbf{H}_2 \in \mathbb{R}^{3 \times 3}$	Find Householder transformations $\mathbf{H}_1, \mathbf{H}_2 \in \mathbb{R}^{3 \times 3}$ such that $\mathbf{H}_2\mathbf{H}_1\mathbf{A}$ is

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lyche-numerical-linear-algebra-138-F01775	138	1.000	$\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$	Find Householder transformations $\boldsymbol{H}_1, \boldsymbol{H}_2 \in \mathbb{R}^{3 \times 3}$ such that $\boldsymbol{H}_2 \boldsymbol{H}_1 \boldsymbol{A}$ is
lyche-numerical-linear-algebra-138-F01776	138	0.999	$\boldsymbol{A} = [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n] \in \mathbb{C}^{n \times n}$	$\boldsymbol{A} = [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n] \in \mathbb{C}^{n \times n}$	tion to prove a classical determinant inequality. For any $\boldsymbol{A} = [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n] \in \mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra-138-F01777	138	1.000	$ \operatorname{det}(\boldsymbol{Q}) = 1$	$ \det(\boldsymbol{Q}) = 1$	a) Show that if $\boldsymbol{Q}$ is unitary then $ \det(\boldsymbol{Q}) = 1$.
lyche-numerical-linear-algebra-138-F01778	138	1.000	$\boldsymbol{R} = [\boldsymbol{r}_1, \dots, \boldsymbol{r}_n]$	$\boldsymbol{R} = [\boldsymbol{r}_1, \dots, \boldsymbol{r}_n]$	b) Let $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{R}$ be a QR factorization of $\boldsymbol{A}$ and let $\boldsymbol{R} = [\boldsymbol{r}_1, \dots, \boldsymbol{r}_n]$. Show that
lyche-numerical-linear-algebra-138-F01779	138	0.986	$\left(\boldsymbol{A}^* \boldsymbol{A}\right)_{jj} = \ \boldsymbol{a}_j\ _2^2 = (\boldsymbol{R}^* \boldsymbol{R})_{jj} = \ \boldsymbol{r}_j\ _2^2$	$(\boldsymbol{A}^* \boldsymbol{A})_{jj} = \ \boldsymbol{a}_j\ _2^2 = (\boldsymbol{R}^* \boldsymbol{R})_{jj} = \ \boldsymbol{r}_j\ _2^2$	$(\boldsymbol{A}^* \boldsymbol{A})_{jj} = \ \boldsymbol{a}_j\ _2^2 = (\boldsymbol{R}^* \boldsymbol{R})_{jj} = \ \boldsymbol{r}_j\ _2^2$.
lyche-numerical-linear-algebra-138-F01780	138	1.000	$ \operatorname{det}(\boldsymbol{A}) = \prod_{j=1}^n \boldsymbol{r}_{jj} \leq \prod_{j=1}^n \ \boldsymbol{a}_j\ _2$	$ \det(\boldsymbol{A}) = \prod_{j=1}^n \boldsymbol{r}_{jj} \leq \prod_{j=1}^n \ \boldsymbol{a}_j\ _2$	c) Show that $ \det(\boldsymbol{A}) = \prod_{i=1}^n \boldsymbol{r}_{ii} \leq \prod_{j=1}^n \ \boldsymbol{a}_j\ _2$.
lyche-numerical-linear-algebra-138-F01781	138	0.949	$\boldsymbol{B} = \boldsymbol{L}^T \boldsymbol{L}$	$\boldsymbol{B} = \boldsymbol{L}^T \boldsymbol{L}$	form $\boldsymbol{B} = \boldsymbol{L}^T \boldsymbol{L}$, where $\boldsymbol{L}$ is lower triangular with positive diagonal elements (you
lyche-numerical-linear-algebra-138-F01782	138	0.995	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^T$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^T$	$\boldsymbol{B} = \boldsymbol{L} \boldsymbol{L}^T$.
lyche-numerical-linear-algebra-138-F01783	138	1.000	$\boldsymbol{l}_{i,j}$	$\boldsymbol{l}_{i,j}$	a) Suppose $\boldsymbol{B} = \boldsymbol{L}^T \boldsymbol{L}$. Write down the equations to determine the elements $\boldsymbol{l}_{i,j}$ of
lyche-numerical-linear-algebra-138-F01784	138	1.000	$i = n, n-1, \dots, 1$	$i = n, n-1, \dots, 1$	$\boldsymbol{L}$, in the order $i = n, n-1, \dots, 1$ and $j = i, 1, 2, \dots, i-1$.
lyche-numerical-linear-algebra-138-F01785	138	1.000	$j = i, 1, 2, \dots, i-1$	$j = i, 1, 2, \dots, i-1$	$\boldsymbol{L}$, in the order $i = n, n-1, \dots, 1$ and $j = i, 1, 2, \dots, i-1$.
lyche-numerical-linear-algebra-138-F01786	138	1.000	$\boldsymbol{L}^T \boldsymbol{L}$	$\boldsymbol{L}^T \boldsymbol{L}$	b) Explain (without making a detailed algorithm) how the $\boldsymbol{L}^T \boldsymbol{L}$ factorization can be
lyche-numerical-linear-algebra-138-F01787	138	1.000	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{c}$	$\boldsymbol{B} \boldsymbol{x} = \boldsymbol{c}$	used to solve the linear system $\boldsymbol{B} \boldsymbol{x} = \boldsymbol{c}$. Compute $\ \boldsymbol{L}\ _F$. Is the algorithm stable?
lyche-numerical-linear-algebra-138-F01788	138	1.000	$\ \boldsymbol{L}\ _F$	$\ \boldsymbol{L}\ _F$	used to solve the linear system $\boldsymbol{B} \boldsymbol{x} = \boldsymbol{c}$. Compute $\ \boldsymbol{L}\ _F$. Is the algorithm stable?
lyche-numerical-linear-algebra-138-F01789	138	1.000	$\boldsymbol{Q} \boldsymbol{L}$	$\boldsymbol{Q} \boldsymbol{L}$	$\boldsymbol{Q} \boldsymbol{L}$, where $\boldsymbol{Q} \in \mathbb{R}^{n \times n}$ is orthogonal and $\boldsymbol{L} \in \mathbb{R}^{n \times n}$ is lower triangular with
lyche-numerical-linear-algebra-138-F01790	138	1.000	$\boldsymbol{Q} \in \mathbb{R}^{n \times n}$	$\boldsymbol{Q} \in \mathbb{R}^{n \times n}$	$\boldsymbol{Q} \boldsymbol{L}$, where $\boldsymbol{Q} \in \mathbb{R}^{n \times n}$ is orthogonal and $\boldsymbol{L} \in \mathbb{R}^{n \times n}$ is lower triangular with
lyche-numerical-linear-algebra-138-F01791	138	1.000	$\boldsymbol{a} := [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n]^T \in \mathbb{R}^n$	$\boldsymbol{a} := [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n]^T \in \mathbb{R}^n$	a) Given vectors $\boldsymbol{a} := [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n]^T \in \mathbb{R}^n$ and $\boldsymbol{e}_n := [0, \dots, 0, 1]^T$. Find $\boldsymbol{v} \in \mathbb{R}^n$
lyche-numerical-linear-algebra-138-F01792	138	1.000	$\boldsymbol{e}_n := [0, \dots, 0, 1]^T$	$\boldsymbol{e}_n := [0, \dots, 0, 1]^T$	a) Given vectors $\boldsymbol{a} := [\boldsymbol{a}_1, \dots, \boldsymbol{a}_n]^T \in \mathbb{R}^n$ and $\boldsymbol{e}_n := [0, \dots, 0, 1]^T$. Find $\boldsymbol{v} \in \mathbb{R}^n$
lyche-numerical-linear-algebra-138-F01793	138	0.724	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	$\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$	such that the Householder transformation $\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$ satisfies $\boldsymbol{H} \boldsymbol{a} = -\boldsymbol{s} \boldsymbol{e}_n$,
lyche-numerical-linear-algebra-138-F01794	138	0.724	$\boldsymbol{H} \boldsymbol{a} = -\boldsymbol{s} \boldsymbol{e}_n$	$\boldsymbol{H} \boldsymbol{a} = -\boldsymbol{s} \boldsymbol{e}_n$	such that the Householder transformation $\boldsymbol{H} := \boldsymbol{I} - 2 \frac{\boldsymbol{v} \boldsymbol{v}^*}{\boldsymbol{v}^* \boldsymbol{v}}$ satisfies $\boldsymbol{H} \boldsymbol{a} = -\boldsymbol{s} \boldsymbol{e}_n$,
lyche-numerical-linear-algebra-138-F01795	138	1.000	$ \boldsymbol{s} = \ \boldsymbol{a}\ _2$	$ \boldsymbol{s} = \ \boldsymbol{a}\ _2$	where $ \boldsymbol{s} = \ \boldsymbol{a}\ _2$. How should we choose the sign of $\boldsymbol{s}$?
lyche-numerical-linear-algebra-138-F01796	138	1.000	$\boldsymbol{s}$	$\boldsymbol{s}$	where $ \boldsymbol{s} = \ \boldsymbol{a}\ _2$. How should we choose the sign of $\boldsymbol{s}$?
lyche-numerical-linear-algebra-138-F01797	138	0.630	$\Longleftrightarrow \boldsymbol{R}$	$\Longleftrightarrow \boldsymbol{R}$	¹ Show that we have equality $\Longleftrightarrow \boldsymbol{R}$ is diagonal $\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$ is diagonal.
lyche-numerical-linear-algebra-138-F01798	138	0.630	$\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$	$\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$	¹ Show that we have equality $\Longleftrightarrow \boldsymbol{R}$ is diagonal $\Longleftrightarrow \boldsymbol{A}^* \boldsymbol{A}$ is diagonal.
lyche-numerical-linear-algebra-139-F01799	139	1.000	$1 \leq r \leq n, \boldsymbol{v}_r \in \mathbb{R}^r, \boldsymbol{v}_r \neq \mathbf{0}$	$1 \leq r \leq n, \boldsymbol{v}_r \in \mathbb{R}^r, \boldsymbol{v}_r \neq \mathbf{0}$	b) Let $1 < r < n, \boldsymbol{v}_r \in \mathbb{R}^r, \boldsymbol{v}_r \neq \mathbf{0}$, and

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lyche-numerical-linear-algebra-F01800	139	1.000	$\boldsymbol{H} := \left[\begin{array}{cc} \boldsymbol{V}_r & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{array} \right]$	$\boldsymbol{H} := \begin{bmatrix} \boldsymbol{V}_r & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{bmatrix}$	Show that $\boldsymbol{H} := \begin{bmatrix} \boldsymbol{V}_r & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}_{n-r} \end{bmatrix}$ is a Householder transformation. Show also that if
lyche-numerical-linear-algebra-F01801	139	1.000	$a_{i,j} = 0$	$a_{i,j} = 0$	$a_{i,j} = 0$ for $i = 1, \dots, r$ and $j = r + 1, \dots, n$ then the last r columns of $\boldsymbol{A}$ and
lyche-numerical-linear-algebra-F01802	139	1.000	$i=1, \dots, r$	$i = 1, \dots, r$	$a_{i,j} = 0$ for $i = 1, \dots, r$ and $j = r + 1, \dots, n$ then the last r columns of $\boldsymbol{A}$ and
lyche-numerical-linear-algebra-F01803	139	1.000	$j=r+1, \dots, n$	$j = r + 1, \dots, n$	$a_{i,j} = 0$ for $i = 1, \dots, r$ and $j = r + 1, \dots, n$ then the last r columns of $\boldsymbol{A}$ and
lyche-numerical-linear-algebra-F01804	139	1.000	$\boldsymbol{H} \boldsymbol{A}$	$\boldsymbol{H} \boldsymbol{A}$	$\boldsymbol{H} \boldsymbol{A}$ are the same.
lyche-numerical-linear-algebra-F01805	139	1.000	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_{n-1}$	$\boldsymbol{H}_1, \dots, \boldsymbol{H}_{n-1}$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$ can find Householder transformations $\boldsymbol{H}_1, \dots, \boldsymbol{H}_{n-1}$ such that
lyche-numerical-linear-algebra-F01806	139	1.000	$\boldsymbol{H}_{n-1}, \dots, \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_{n-1}, \dots, \boldsymbol{H}_1 \boldsymbol{A}$	$\boldsymbol{H}_{n-1}, \dots, \boldsymbol{H}_1 \boldsymbol{A}$ is lower triangular. Give a $\boldsymbol{QL}$ factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F01807	139	1.000	$d \leq n - 1$	$d \leq n - 1$	$\mathbb{R}^{n \times n}$ be a nonsingular symmetric band matrix with bandwidth $d \leq n - 1$, so that
lyche-numerical-linear-algebra-F01808	139	1.000	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A}$	$\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A}$	$a_{ij} = 0$ for all i, j with $ i - j > d$. We define $\boldsymbol{B} := \boldsymbol{A}^T \boldsymbol{A}$ and let $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$ be the
lyche-numerical-linear-algebra-F01809	139	1.000	$\leq 2d$	$\leq 2d$	b) Show that $\boldsymbol{B}$ has bandwidth $\leq 2d$.
lyche-numerical-linear-algebra-F01810	139	0.426	$\boldsymbol{B} =$	$\boldsymbol{B} =$	c) Write a MATLAB function <code>B=ata(A,d)</code> which computes $\boldsymbol{B}$. You shall
lyche-numerical-linear-algebra-F01811	139	0.426	$(\boldsymbol{A}, d)$	$(\boldsymbol{A}, d)$	c) Write a MATLAB function <code>B=ata(A,d)</code> which computes $\boldsymbol{B}$. You shall
lyche-numerical-linear-algebra-F01812	139	1.000	$\mathcal{O}(cn^2)$	$\mathcal{O}(cn^2)$	exploit the symmetry and the function should only use $\mathcal{O}(cn^2)$ flops, where c
lyche-numerical-linear-algebra-F01813	139	1.000	$\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{R}^T \boldsymbol{R}$	$\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{R}^T \boldsymbol{R}$	e) Show that $\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{R}^T \boldsymbol{R}$.
lyche-numerical-linear-algebra-F01814	139	1.000	$2d$	$2d$	f) Explain why $\boldsymbol{R}$ has upper bandwidth $2d$.
lyche-numerical-linear-algebra-F01815	140	0.928	$\boldsymbol{x} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$	$\boldsymbol{x} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$	Exercise 5.16 (Plane Rotation) Show that if $\boldsymbol{x} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$ then $\boldsymbol{P} \boldsymbol{x} =$
lyche-numerical-linear-algebra-F01816	140	0.928	$\boldsymbol{P} \boldsymbol{x} =$	$\boldsymbol{P} \boldsymbol{x} =$	Exercise 5.16 (Plane Rotation) Show that if $\boldsymbol{x} = \begin{bmatrix} r \cos \alpha \\ r \sin \alpha \end{bmatrix}$ then $\boldsymbol{P} \boldsymbol{x} =$
lyche-numerical-linear-algebra-F01817	140	0.711	$\begin{bmatrix} r \cos(\alpha - \theta) \\ r \sin(\alpha - \theta) \end{bmatrix}$	$\begin{bmatrix} r \cos(\alpha - \theta) \\ r \sin(\alpha - \theta) \end{bmatrix}$	$\begin{bmatrix} r \cos(\alpha - \theta) \\ r \sin(\alpha - \theta) \end{bmatrix}$.
lyche-numerical-linear-algebra-F01818	140	0.988	$\boldsymbol{H} \in \mathbb{R}^{4,4}$	$\boldsymbol{H} \in \mathbb{R}^{4,4}$	Exercise 5.17 (Updating the QR Decomposition) Let $\boldsymbol{H} \in \mathbb{R}^{4,4}$ be upper
lyche-numerical-linear-algebra-F01819	140	1.000	$\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3$	$\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3$	Hessenberg. Find Givens rotation matrices $\boldsymbol{G}_1, \boldsymbol{G}_2, \boldsymbol{G}_3$ such that
lyche-numerical-linear-algebra-F01820	140	1.000	$\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$	$\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$	is upper triangular. (Here each $\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$ for suitable i, j, c og s , and for each k
lyche-numerical-linear-algebra-F01821	140	1.000	i, j, c	i, j, c	is upper triangular. (Here each $\boldsymbol{G}_k = \boldsymbol{P}_{i,j}$ for suitable i, j, c og s , and for each k
lyche-numerical-linear-algebra-F01822	140	0.999	$\boldsymbol{P}_{k,k+1}$	$\boldsymbol{P}_{k,k+1}$	solves the linear system $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ using rotations $\boldsymbol{P}_{k,k+1}$ for $k = 1, \dots, n - 1$. It
lyche-numerical-linear-algebra-F01823	140	0.838	$\boldsymbol{G} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{G} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$	rotation of order 2 has the form $\boldsymbol{G} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, where $s^2 + c^2 = 1$.
lyche-numerical-linear-algebra-F01824	140	0.838	$s^2 + c^2 = 1$	$s^2 + c^2 = 1$	rotation of order 2 has the form $\boldsymbol{G} := \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, where $s^2 + c^2 = 1$.

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lyche-numerical-linear-algebra-F01825	141	1.000	$\boldsymbol{G}$	$\boldsymbol{G}$	a) Is $\boldsymbol{G}$ symmetric and unitary?
lyche-numerical-linear-algebra-F01826	141	1.000	$x_1, x_2 \in \mathbb{R}$	$x_1, x_2 \in \mathbb{R}$	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra-F01827	141	1.000	$r := \sqrt{x_1^2 + x_2^2}$	$r := \sqrt{x_1^2 + x_2^2}$	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra-F01828	141	1.000	y_1, y_2	y_1, y_2	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra-F01829	141	1.000	$y_1 = y_2$	$y_1 = y_2$	b) Given $x_1, x_2 \in \mathbb{R}$ and set $r := \sqrt{x_1^2 + x_2^2}$. Find $\boldsymbol{G}$ and y_1, y_2 so that $y_1 = y_2$,
lyche-numerical-linear-algebra-F01830	141	1.000	$\left[\begin{array}{l} y_1 \\ y_2 \end{array} \right] = \boldsymbol{G} \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right]$	$\left[\begin{array}{l} y_1 \\ y_2 \end{array} \right] = \boldsymbol{G} \left[\begin{array}{l} x_1 \\ x_2 \end{array} \right]$	where $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \boldsymbol{G} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.
lyche-numerical-linear-algebra-F01831	141	1.000	$i \ j$	ij	rotation in the ij -plane is an $m \times m$ -matrix, denoted $\boldsymbol{P}_{i,j}$, which differs from the
lyche-numerical-linear-algebra-F01832	141	1.000	$\boldsymbol{P}_{i,j}$	$\boldsymbol{P}_{i,j}$	rotation in the ij -plane is an $m \times m$ -matrix, denoted $\boldsymbol{P}_{i,j}$, which differs from the
lyche-numerical-linear-algebra-F01833	141	0.938	ii, ij, ji, jj	ii, ij, ji, jj	identity matrix only in the entries ii, ij, ji, jj , which equal
lyche-numerical-linear-algebra-F01834	141	1.000	$\theta \in \mathbb{R}$	$\theta \in \mathbb{R}$	a) For $\theta \in \mathbb{R}$, let $\boldsymbol{P}$ be a Givens rotation of the form
lyche-numerical-linear-algebra-F01835	141	1.000	$\theta \in (-\pi/2, \pi/2]$	$\theta \in (-\pi/2, \pi/2]$	and let $\boldsymbol{x}$ be a fixed vector in $\mathbb{R}^2$. Show that there exists a unique $\theta \in (-\pi/2, \pi/2]$
lyche-numerical-linear-algebra-F01836	141	0.910	$\boldsymbol{P} \boldsymbol{x} = \pm \ \boldsymbol{x}\ _2 \boldsymbol{e}_1$	$\boldsymbol{P} \boldsymbol{x} = \pm \ \boldsymbol{x}\ _2 \boldsymbol{e}_1$	so that $\boldsymbol{P} \boldsymbol{x} = \pm \ \boldsymbol{x}\ _2 \boldsymbol{e}_1$, where $\boldsymbol{e}_1 = (1, 0)^T$.
lyche-numerical-linear-algebra-F01837	141	0.910	$\boldsymbol{e}_1 = (1, 0)^T$	$\boldsymbol{e}_1 = (1, 0)^T$	so that $\boldsymbol{P} \boldsymbol{x} = \pm \ \boldsymbol{x}\ _2 \boldsymbol{e}_1$, where $\boldsymbol{e}_1 = (1, 0)^T$.
lyche-numerical-linear-algebra-F01838	141	1.000	$\boldsymbol{w} \in \mathbb{R}^m$	$\boldsymbol{w} \in \mathbb{R}^m$	b) Show that, for any vector $\boldsymbol{w} \in \mathbb{R}^m$, one can find rotations in the 12-plane, 23-
lyche-numerical-linear-algebra-F01839	141	0.481	$(m-1) \ m$	$(m-1)m$	plane, ..., $(m-1)m$ -plane, so that
lyche-numerical-linear-algebra-F01840	141	0.989	$\alpha = \pm \ \boldsymbol{w}\ _2$	$\alpha = \pm \ \boldsymbol{w}\ _2$	where $\alpha = \pm \ \boldsymbol{w}\ _2$.
lyche-numerical-linear-algebra-F01841	142	1.000	$\boldsymbol{A}_-$	$\boldsymbol{A}_-$	d) Let again $\boldsymbol{A}$ be an $m \times n$ -matrix with $m \geq n$, and let $\boldsymbol{A}_-$ be the matrix obtained
lyche-numerical-linear-algebra-F01842	142	0.916	$\boldsymbol{A}^2$	$\boldsymbol{A}^2$	$\boldsymbol{A}_-$, when we already have a QR decomposition $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$ of $\boldsymbol{A}$. ²
lyche-numerical-linear-algebra-F01843	142	0.936	$\boldsymbol{z}$	$\boldsymbol{z}$	Assume also that $\boldsymbol{z}$ is a given column vector of length n .
lyche-numerical-linear-algebra-F01844	142	0.999	$\boldsymbol{A} + \boldsymbol{z} \boldsymbol{z}^*$	$\boldsymbol{A} + \boldsymbol{z} \boldsymbol{z}^*$	a) Explain why $\boldsymbol{A} + \boldsymbol{z} \boldsymbol{z}^*$ has a unique Cholesky factorization.
lyche-numerical-linear-algebra-F01845	142	1.000	$\boldsymbol{R}^* \boldsymbol{R}$	$\boldsymbol{R}^* \boldsymbol{R}$	factorization $\boldsymbol{R}^* \boldsymbol{R}$.
lyche-numerical-linear-algebra-F01846	142	0.987	$P_{i_1, n+1}, P_{i_2, n+1}, \dots, P_{i_n, n+1}$	$P_{i_1, n+1}, P_{i_2, n+1}, \dots, P_{i_n, n+1}$	c) Explain how one can find plane rotations $P_{i_1, n+1}, P_{i_2, n+1}, \dots, P_{i_n, n+1}$ so that
lyche-numerical-linear-algebra-F01847	142	1.000	$\boldsymbol{R}'$	$\boldsymbol{R}'$	with $\boldsymbol{R}'$ upper triangular, and explain how to obtain a QR decomposition of
lyche-numerical-linear-algebra-F01848	142	0.996	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right]$	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right]$	$\left[\begin{array}{c} \boldsymbol{L}^* \\ \boldsymbol{z}^* \end{array} \right]$ from this. In particular you should write down the numbers $i_1, \dots, i_n$.

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lyche-numerical-linear-algebra- F01849	142	0.996	$i_{\{1\}}, \ldots, i_{\{n\}}$	$i_1, \dots, i_n$	$\begin{bmatrix} L^* \\ z^* \end{bmatrix}$ from this. In particular you should write down the numbers $i_1, \dots, i_n$.
lyche-numerical-linear-algebra- F01850	142	1.000	$\boldsymbol{Q}^T \boldsymbol{A}_-$	$\boldsymbol{Q}^T \boldsymbol{A}_-$	² Consider the matrix $\boldsymbol{Q}^T \boldsymbol{A}_-$.
lyche-numerical-linear-algebra- F01851	146	0.999	$\left(\lambda_k, \boldsymbol{x}_{\{k\}}\right), k=$	$(\lambda_k, \boldsymbol{x}_k), k=$	Proof The proof is by contradiction. Suppose $\boldsymbol{A}$ has eigenpairs $(\lambda_k, \boldsymbol{x}_k), k =$
lyche-numerical-linear-algebra- F01852	146	0.999	$\left\{\boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{m\}}\right\}$	$\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_m\}$	that $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_m\}$ is linearly dependent. Thus $\sum_{j=1}^m c_j \boldsymbol{x}_j = \mathbf{0}$, where at least one
lyche-numerical-linear-algebra- F01853	146	0.999	$\sum_{j=1}^m c_{\{j\}} \boldsymbol{x}_{\{j\}} = \mathbf{0}$	$\sum_{j=1}^m c_j \boldsymbol{x}_j = \mathbf{0}$	that $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_m\}$ is linearly dependent. Thus $\sum_{j=1}^m c_j \boldsymbol{x}_j = \mathbf{0}$, where at least one
lyche-numerical-linear-algebra- F01854	146	1.000	$c_{\{j\}}$	c_j	c_j is nonzero. We must have $m \geq 2$ since eigenvectors are nonzero. We find
lyche-numerical-linear-algebra- F01855	146	1.000	$m \geq 2$	$m \geq 2$	c_j is nonzero. We must have $m \geq 2$ since eigenvectors are nonzero. We find
lyche-numerical-linear-algebra- F01856	146	1.000	$\sum_{j=1}^m c_{\{j\}} \lambda_{\{m\}} \boldsymbol{x}_{\{j\}} = \mathbf{0}$	$\sum_{j=1}^m c_j \lambda_m \boldsymbol{x}_j = \mathbf{0}$	From the last relation we subtract $\sum_{j=1}^m c_j \lambda_m \boldsymbol{x}_j = \mathbf{0}$ and find $\sum_{j=1}^{m-1} c_j (\lambda_j -$
lyche-numerical-linear-algebra- F01857	146	1.000	$\sum_{j=1}^{m-1} c_{\{j\}} \left(\lambda_{\{j\}} - \right.$	$\sum_{j=1}^{m-1} c_j (\lambda_j -$	From the last relation we subtract $\sum_{j=1}^m c_j \lambda_m \boldsymbol{x}_j = \mathbf{0}$ and find $\sum_{j=1}^{m-1} c_j (\lambda_j -$
lyche-numerical-linear-algebra- F01858	146	1.000	$\left. \lambda_{\{m\}} \right) \boldsymbol{x}_{\{j\}} = \mathbf{0}$	$\lambda_m) \boldsymbol{x}_j = \mathbf{0}$	$\lambda_m) \boldsymbol{x}_j = \mathbf{0}$. But since $\lambda_j - \lambda_m \neq 0$ for $j = 1, \dots, m-1$ and at least one
lyche-numerical-linear-algebra- F01859	146	1.000	$\lambda_{\{j\}} - \lambda_{\{m\}} \neq 0$	$\lambda_j - \lambda_m \neq 0$	$\lambda_m) \boldsymbol{x}_j = \mathbf{0}$. But since $\lambda_j - \lambda_m \neq 0$ for $j = 1, \dots, m-1$ and at least one
lyche-numerical-linear-algebra- F01860	146	1.000	$j=1, \ldots, m-1$	$j = 1, \dots, m-1$	$\lambda_m) \boldsymbol{x}_j = \mathbf{0}$. But since $\lambda_j - \lambda_m \neq 0$ for $j = 1, \dots, m-1$ and at least one
lyche-numerical-linear-algebra- F01861	146	1.000	$c_{\{j\}} \neq 0$	$c_j \neq 0$	$c_j \neq 0$ for $j < m$ we see that $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra- F01862	146	1.000	$j < m$	$j < m$	$c_j \neq 0$ for $j < m$ we see that $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra- F01863	146	1.000	$\left\{\boldsymbol{x}_{\{1\}}, \ldots, \boldsymbol{x}_{\{m-1\}}\right\}$	$\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_{m-1}\}$	$c_j \neq 0$ for $j < m$ we see that $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_{m-1}\}$ is linearly dependent, contradicting
lyche-numerical-linear-algebra- F01864	146	1.000	$\boldsymbol{I} \boldsymbol{x} = \boldsymbol{x}$	$\boldsymbol{I} \boldsymbol{x} = \boldsymbol{x}$	Since $\boldsymbol{I} \boldsymbol{x} = \boldsymbol{x}$ and $\lambda_1 = \lambda_2 = 1$ any vector $\boldsymbol{x} \in \mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra- F01865	146	1.000	$\lambda_{\{1\}} = \lambda_{\{2\}} = 1$	$\lambda_1 = \lambda_2 = 1$	Since $\boldsymbol{I} \boldsymbol{x} = \boldsymbol{x}$ and $\lambda_1 = \lambda_2 = 1$ any vector $\boldsymbol{x} \in \mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra- F01866	146	1.000	$\boldsymbol{x} \in \mathbb{C}^2$	$\boldsymbol{x} \in \mathbb{C}^2$	Since $\boldsymbol{I} \boldsymbol{x} = \boldsymbol{x}$ and $\lambda_1 = \lambda_2 = 1$ any vector $\boldsymbol{x} \in \mathbb{C}^2$ is an eigenvector for $\boldsymbol{I}$. In
lyche-numerical-linear-algebra- F01867	146	1.000	$\boldsymbol{e}_{\{2\}}$	$\boldsymbol{e}_2$	particular the two unit vectors $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ are eigenvectors and form an orthonormal
lyche-numerical-linear-algebra- F01868	146	1.000	$\mathbb{C}^2$	$\mathbb{C}^2$	basis for $\mathbb{C}^2$. We conclude that the identity matrix is nondefective. The matrix $\boldsymbol{J}$
lyche-numerical-linear-algebra- F01869	146	1.000	$\boldsymbol{J}$	$\boldsymbol{J}$	basis for $\mathbb{C}^2$. We conclude that the identity matrix is nondefective. The matrix $\boldsymbol{J}$
lyche-numerical-linear-algebra- F01870	146	1.000	$\boldsymbol{J} \boldsymbol{x} = \boldsymbol{x}$	$\boldsymbol{J} \boldsymbol{x} = \boldsymbol{x}$	also has the eigenvalue one with multiplicity two, but since $\boldsymbol{J} \boldsymbol{x} = \boldsymbol{x}$ if and only if
lyche-numerical-linear-algebra- F01871	146	1.000	$x_{\{2\}} = 0$	$x_2 = 0$	$x_2 = 0$, any eigenvector must be a multiple of $\boldsymbol{e}_1$. Thus $\boldsymbol{J}$ is defective.
lyche-numerical-linear-algebra- F01872	146	1.000	$\left[\begin{array}{cc} 2 & -1 \\ -1 & 2 \end{array}\right]$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$	<i>Example 6.2 (Eigenvector Expansion Example)</i> Eigenpairs of $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ are
lyche-numerical-linear-algebra- F01873	146	1.000	$(1, [1, 1]^T)$	$(1, [1, 1]^T)$	$(1, [1, 1]^T)$ and $(3, [1, -1]^T)$. Any $\boldsymbol{x} = [x_1, x_2]^T \in \mathbb{C}^2$ has the eigenvector

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lyche-numerical-linear-algebra- F01874	146	1.000	$\left(3, [1, -1]^T\right)$	$(3, [1, -1]^T)$	$(1, [1, 1]^T)$ and $(3, [1, -1]^T)$. Any $\mathbf{x} = [x_1, x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra- F01875	146	1.000	$\boldsymbol{x}=\left[x_{1}, x_{2}\right]^T \in \mathbb{C}^2$	$\mathbf{x} = [x_1, x_2]^T \in \mathbb{C}^2$	$(1, [1, 1]^T)$ and $(3, [1, -1]^T)$. Any $\mathbf{x} = [x_1, x_2]^T \in \mathbb{C}^2$ has the eigenvector
lyche-numerical-linear-algebra- F01876	147	1.000	$\boldsymbol{S} \in \mathbb{C}^{n \times n}$	$\mathbf{S} \in \mathbb{C}^{n \times n}$	similar if there is a nonsingular matrix $\mathbf{S} \in \mathbb{C}^{n \times n}$ such that $\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$. The
lyche-numerical-linear-algebra- F01877	147	1.000	$\boldsymbol{B}=\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	$\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$	similar if there is a nonsingular matrix $\mathbf{S} \in \mathbb{C}^{n \times n}$ such that $\mathbf{B} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$. The
lyche-numerical-linear-algebra- F01878	147	1.000	$\boldsymbol{A} \rightarrow \boldsymbol{B}$	$\mathbf{A} \rightarrow \mathbf{B}$	transformation $\mathbf{A} \rightarrow \mathbf{B}$ is called a similarity transformation . The columns of $\mathbf{S}$
lyche-numerical-linear-algebra- F01879	147	1.000	$\boldsymbol{s}_1, \boldsymbol{s}_2, \ldots, \boldsymbol{s}_n$	$\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n$	are denoted by $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n$.
lyche-numerical-linear-algebra- F01880	147	1.000	$\operatorname{det}(\boldsymbol{A} \boldsymbol{C})=$	$\det(\mathbf{A}\mathbf{C}) =$	teristic polynomial. Indeed, by the product rule for determinants $\det(\mathbf{A}\mathbf{C}) =$
lyche-numerical-linear-algebra- F01881	147	1.000	$\operatorname{det}(\boldsymbol{A}) \operatorname{det}(\boldsymbol{C})$	$\det(\mathbf{A}) \det(\mathbf{C})$	$\det(\mathbf{A}) \det(\mathbf{C})$ so that
lyche-numerical-linear-algebra- F01882	147	0.866	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}$	$\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$	2. $(\lambda, \mathbf{x})$ is an eigenpair for $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ if and only if $(\lambda, \mathbf{S}\mathbf{x})$ is an eigenpair for $\mathbf{A}$. In
lyche-numerical-linear-algebra- F01883	147	0.866	$\lambda, \boldsymbol{S} \boldsymbol{x}$	$\lambda, \mathbf{S}\mathbf{x}$	2. $(\lambda, \mathbf{x})$ is an eigenpair for $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ if and only if $(\lambda, \mathbf{S}\mathbf{x})$ is an eigenpair for $\mathbf{A}$. In
lyche-numerical-linear-algebra- F01884	147	1.000	$\left(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}\right) \boldsymbol{x}=\lambda \boldsymbol{x}$	$(\mathbf{S}^{-1}\mathbf{A}\mathbf{S})\mathbf{x} = \lambda\mathbf{x}$	fact $(\mathbf{S}^{-1}\mathbf{A}\mathbf{S})\mathbf{x} = \lambda\mathbf{x}$ if and only if $\mathbf{A}(\mathbf{S}\mathbf{x}) = \lambda(\mathbf{S}\mathbf{x})$.
lyche-numerical-linear-algebra- F01885	147	1.000	$\boldsymbol{A}(\boldsymbol{S} \boldsymbol{x})=\lambda(\boldsymbol{S} \boldsymbol{x})$	$\mathbf{A}(\mathbf{S}\mathbf{x}) = \lambda(\mathbf{S}\mathbf{x})$	fact $(\mathbf{S}^{-1}\mathbf{A}\mathbf{S})\mathbf{x} = \lambda\mathbf{x}$ if and only if $\mathbf{A}(\mathbf{S}\mathbf{x}) = \lambda(\mathbf{S}\mathbf{x})$.
lyche-numerical-linear-algebra- F01886	147	1.000	$\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S}=\operatorname{diag}\left(\lambda_1, \ldots, \lambda_n\right)$	$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$	3. If $\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ we can partition $\mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{D}$ by
lyche-numerical-linear-algebra- F01887	147	1.000	$\boldsymbol{A} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{D}$	$\mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{D}$	3. If $\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \mathbf{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ we can partition $\mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{D}$ by
lyche-numerical-linear-algebra- F01888	147	1.000	$\left[\boldsymbol{A} \boldsymbol{s}_1, \ldots, \boldsymbol{A} \boldsymbol{s}_n\right]=\left[\lambda_1 \boldsymbol{s}_1, \ldots, \lambda_n \boldsymbol{s}_n\right]$	$[\mathbf{A}\mathbf{s}_1, \dots, \mathbf{A}\mathbf{s}_n] = [\lambda_1\mathbf{s}_1, \dots, \lambda_n\mathbf{s}_n]$	columns to obtain $[\mathbf{A}\mathbf{s}_1, \dots, \mathbf{A}\mathbf{s}_n] = [\lambda_1\mathbf{s}_1, \dots, \lambda_n\mathbf{s}_n]$. Thus the columns of
lyche-numerical-linear-algebra- F01889	147	1.000	$\boldsymbol{A}, \boldsymbol{C} \in \mathbb{C}^{n \times n}$	$\mathbf{A}, \mathbf{C} \in \mathbb{C}^{n \times n}$	4. For any square matrices $\mathbf{A}, \mathbf{C} \in \mathbb{C}^{n \times n}$ the two products $\mathbf{A}\mathbf{C}$ and $\mathbf{C}\mathbf{A}$ have the
lyche-numerical-linear-algebra- F01890	147	1.000	$\boldsymbol{A} \boldsymbol{C}$	$\mathbf{A}\mathbf{C}$	4. For any square matrices $\mathbf{A}, \mathbf{C} \in \mathbb{C}^{n \times n}$ the two products $\mathbf{A}\mathbf{C}$ and $\mathbf{C}\mathbf{A}$ have the
lyche-numerical-linear-algebra- F01891	147	1.000	$\boldsymbol{C} \boldsymbol{A}$	$\mathbf{C}\mathbf{A}$	4. For any square matrices $\mathbf{A}, \mathbf{C} \in \mathbb{C}^{n \times n}$ the two products $\mathbf{A}\mathbf{C}$ and $\mathbf{C}\mathbf{A}$ have the
lyche-numerical-linear-algebra- F01892	147	1.000	$\boldsymbol{C} \in \mathbb{C}^{n \times m}$	$\mathbf{C} \in \mathbb{C}^{n \times m}$	$\mathbb{C}^{m \times n}$ and $\mathbf{C} \in \mathbb{C}^{n \times m}$, with say $m > n$, the bigger matrix has $m - n$ extra zero
lyche-numerical-linear-algebra- F01893	147	1.000	$n+m$	$n + m$	To show this define for any $m, n \in \mathbb{N}$ block triangular matrices of order $n + m$ by
lyche-numerical-linear-algebra- F01894	147	0.997	$\boldsymbol{S}^{-1}=\left[\begin{array}{cc}\boldsymbol{I} & -\boldsymbol{A} \\ \mathbf{0} & \boldsymbol{I}\end{array}\right]$	$\mathbf{S}^{-1} = \begin{bmatrix} \mathbf{I} & -\mathbf{A} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$	The matrix $\mathbf{S}$ is nonsingular with $\mathbf{S}^{-1} = \begin{bmatrix} \mathbf{I} & -\mathbf{A} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$. Moreover, $\mathbf{E}\mathbf{S} = \mathbf{S}\mathbf{F}$ so
lyche-numerical-linear-algebra- F01895	147	0.997	$\boldsymbol{E} \boldsymbol{S}=\boldsymbol{S} \boldsymbol{F}$	$\mathbf{E}\mathbf{S} = \mathbf{S}\mathbf{F}$	The matrix $\mathbf{S}$ is nonsingular with $\mathbf{S}^{-1} = \begin{bmatrix} \mathbf{I} & -\mathbf{A} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$. Moreover, $\mathbf{E}\mathbf{S} = \mathbf{S}\mathbf{F}$ so
lyche-numerical-linear-algebra- F01896	147	1.000	$\boldsymbol{F}$	$\mathbf{F}$	$\mathbf{E}$ and $\mathbf{F}$ are similar and therefore have the same characteristic polynomials.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F01897	147	1.000	$\pi_{\{\mathbf{E}\}(\lambda)} = \lambda^n \pi_{\{\mathbf{C}\}(\lambda)} = \pi_{\{\mathbf{F}\}(\lambda)} = \lambda^m \pi_{\{\mathbf{C}\}(\lambda)}$	$\pi_{\mathbf{E}}(\lambda) = \lambda^n \pi_{\mathbf{AC}}(\lambda) = \pi_{\mathbf{F}}(\lambda) = \lambda^m \pi_{\mathbf{CA}}(\lambda)$	diagonal blocks. But then $\pi_{\mathbf{F}}(\lambda) = \lambda^n \pi_{\mathbf{AC}}(\lambda) = \pi_{\mathbf{F}}(\lambda) = \lambda^m \pi_{\mathbf{CA}}(\lambda)$. This
lyche-numerical-linear-algebra-F01898	148	1.000	$\lambda_1, \dots, \lambda_k$	$\lambda_1, \dots, \lambda_k$	Suppose $\mathbf{A} \in \mathbb{C}^{n \times n}$ has k distinct eigenvalues $\lambda_1, \dots, \lambda_k$ with multiplicities
lyche-numerical-linear-algebra-F01899	148	1.000	$a_1, \dots, a_k$	$a_1, \dots, a_k$	$a_1, \dots, a_k$ so that
lyche-numerical-linear-algebra-F01900	148	0.890	$a_i = a(\lambda_i) = a_{\mathbf{A}}(\lambda_i)$	$a_i = a(\lambda_i) = a_{\mathbf{A}}(\lambda_i)$	The positive integer $a_i = a(\lambda_i) = a_{\mathbf{A}}(\lambda_i)$ is called the multiplicity , or more
lyche-numerical-linear-algebra-F01901	148	1.000	λ_i	λ_i	precisely the algebraic multiplicity of the eigenvalue λ_i . The multiplicity of an
lyche-numerical-linear-algebra-F01902	148	1.000	a_i	a_i	eigenvalue is simple (double, triple) if a_i is equal to one (two, three).
lyche-numerical-linear-algebra-F01903	148	1.000	$\lambda \in \sigma(\mathbf{A})$	$\lambda \in \sigma(\mathbf{A})$	To define a second kind of multiplicity we consider for each $\lambda \in \sigma(\mathbf{A})$ the
lyche-numerical-linear-algebra-F01904	148	1.000	$g = g(\lambda) =$	$g = g(\lambda) =$	Definition 6.2 (Geometric Multiplicity) The geometric multiplicity $g = g(\lambda) =$
lyche-numerical-linear-algebra-F01905	148	1.000	$g_{\{\mathbf{A}\}(\lambda)}$	$g_{\mathbf{A}}(\lambda)$	$g_{\mathbf{A}}(\lambda)$ of an eigenvalue λ of $\mathbf{A}$ is the dimension of the nullspace $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$.
lyche-numerical-linear-algebra-F01906	148	1.000	$\mathcal{N}(\{\mathbf{A}\} - \lambda \mathbf{I})$	$\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$	$g_{\mathbf{A}}(\lambda)$ of an eigenvalue λ of $\mathbf{A}$ is the dimension of the nullspace $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$.
lyche-numerical-linear-algebra-F01907	148	1.000	$\lambda = 1$	$\lambda = 1$	value $\lambda = 1$ with $\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$. Since $\mathbf{I} - \lambda \mathbf{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra-F01908	148	1.000	$\pi_{\{\mathbf{I}\}(\lambda)} = (1 - \lambda)^n$	$\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$	value $\lambda = 1$ with $\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$. Since $\mathbf{I} - \lambda \mathbf{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra-F01909	148	1.000	$\{\mathbf{I}\} - \lambda \mathbf{I}$	$\mathbf{I} - \lambda \mathbf{I}$	value $\lambda = 1$ with $\pi_{\mathbf{I}}(\lambda) = (1 - \lambda)^n$. Since $\mathbf{I} - \lambda \mathbf{I}$ is the zero matrix when $\lambda = 1$,
lyche-numerical-linear-algebra-F01910	148	1.000	$a = g = n$	$a = g = n$	the nullspace of $\mathbf{I} - \lambda \mathbf{I}$ is all of n -space and it follows that $a = g = n$. On the other
lyche-numerical-linear-algebra-F01911	148	0.985	$\mathbf{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\mathbf{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	hand we saw in Example 6.1 that the matrix $\mathbf{J} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has the eigenvalue $\lambda = 1$
lyche-numerical-linear-algebra-F01912	148	1.000	$a = 2$	$a = 2$	with $a = 2$ and any eigenvector is a multiple of $\mathbf{e}_1$. Thus $g = 1$.
lyche-numerical-linear-algebra-F01913	148	1.000	$g = 1$	$g = 1$	with $a = 2$ and any eigenvector is a multiple of $\mathbf{e}_1$. Thus $g = 1$.
lyche-numerical-linear-algebra-F01914	148	0.991	$\dim \mathcal{N}(\{\mathbf{S}^{-1}\mathbf{A}\} - \lambda \mathbf{I}) = k$	$\dim \mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I}) = k$	$\dim \mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I}) = k$, and $\dim \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) = \ell$. We need to show that
lyche-numerical-linear-algebra-F01915	148	0.991	$\dim \mathcal{N}(\{\mathbf{A}\} - \lambda \mathbf{I}) = \ell$	$\dim \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) = \ell$	$\dim \mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I}) = k$, and $\dim \mathcal{N}(\mathbf{A} - \lambda \mathbf{I}) = \ell$. We need to show that
lyche-numerical-linear-algebra-F01916	148	1.000	$k = \ell$	$k = \ell$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{AS} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01917	148	1.000	$\mathbf{v}_1, \dots, \mathbf{v}_k$	$\mathbf{v}_1, \dots, \mathbf{v}_k$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{AS} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01918	148	1.000	$\mathcal{N}(\{\mathbf{S}^{-1}\mathbf{A}\} - \lambda \mathbf{I})$	$\mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I})$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{AS} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01919	148	1.000	$\mathbf{S}^{-1} \mathbf{AS} \mathbf{v}_i = \lambda \mathbf{v}_i$	$\mathbf{S}^{-1} \mathbf{AS} \mathbf{v}_i = \lambda \mathbf{v}_i$	$k = \ell$. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_k$ is a basis for $\mathcal{N}(\mathbf{S}^{-1} \mathbf{AS} - \lambda \mathbf{I})$. Then $\mathbf{S}^{-1} \mathbf{AS} \mathbf{v}_i = \lambda \mathbf{v}_i$
lyche-numerical-linear-algebra-F01920	148	1.000	$\mathbf{AS} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$	$\mathbf{AS} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$	or $\mathbf{AS} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$. But then $\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$, which
lyche-numerical-linear-algebra-F01921	148	1.000	$\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$	$\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$	or $\mathbf{AS} \mathbf{v}_i = \lambda \mathbf{S} \mathbf{v}_i, i = 1, \dots, k$. But then $\{\mathbf{S} \mathbf{v}_1, \dots, \mathbf{S} \mathbf{v}_k\} \subset \mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$, which

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lyche-numerical-linear-algebra-F01922	148	1.000	$k \leq \ell$	$k \leq \ell$	implies that $k \leq \ell$. Similarly, if $\boldsymbol{w}_1, \dots, \boldsymbol{w}_\ell$ is a basis for $\mathcal{N}(A - \lambda I)$ then
lyche-numerical-linear-algebra-F01923	148	1.000	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_\ell$	$\boldsymbol{w}_1, \dots, \boldsymbol{w}_\ell$	implies that $k \leq \ell$. Similarly, if $\boldsymbol{w}_1, \dots, \boldsymbol{w}_\ell$ is a basis for $\mathcal{N}(A - \lambda I)$ then
lyche-numerical-linear-algebra-F01924	148	1.000	$\left\{ \boldsymbol{S}^{-1} \boldsymbol{w}_1, \dots, \boldsymbol{S}^{-1} \boldsymbol{w}_\ell \right\} \subset \mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda I)$	$\{ \boldsymbol{S}^{-1} \boldsymbol{w}_1, \dots, \boldsymbol{S}^{-1} \boldsymbol{w}_\ell \} \subset \mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda I)$	$\{ \boldsymbol{S}^{-1} \boldsymbol{w}_1, \dots, \boldsymbol{S}^{-1} \boldsymbol{w}_\ell \} \subset \mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda I)$, which implies that $k \geq \ell$. We conclude
lyche-numerical-linear-algebra-F01925	148	1.000	$k \geq \ell$	$k \geq \ell$	$\{ \boldsymbol{S}^{-1} \boldsymbol{w}_1, \dots, \boldsymbol{S}^{-1} \boldsymbol{w}_\ell \} \subset \mathcal{N}(\boldsymbol{S}^{-1} \boldsymbol{A} \boldsymbol{S} - \lambda I)$. which implies that $k \geq \ell$. We conclude
lyche-numerical-linear-algebra-F01926	149	1.000	$\boldsymbol{J}_m(\lambda)$	$\boldsymbol{J}_m(\lambda)$	A Jordan block of order m , denoted $\boldsymbol{J}_m(\lambda)$ is an $m \times m$ matrix of the form
lyche-numerical-linear-algebra-F01927	149	0.764	$\boldsymbol{J}_3(\lambda) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$	$\boldsymbol{J}_3(\lambda) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$	A 3×3 Jordan block has the form $\boldsymbol{J}_3(\lambda) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$. Since a Jordan block is upper
lyche-numerical-linear-algebra-F01928	149	1.000	$\boldsymbol{J}_m(\lambda) \boldsymbol{v} = \lambda \boldsymbol{v}$	$\boldsymbol{J}_m(\lambda) \boldsymbol{v} = \lambda \boldsymbol{v}$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda) \boldsymbol{v} = \lambda \boldsymbol{v}$ for some $\boldsymbol{v} = [v_1, \dots, v_m]$ then $\lambda v_{i-1} + v_i = \lambda v_{i-1}$,
lyche-numerical-linear-algebra-F01929	149	1.000	$\boldsymbol{v} = [v_1, \dots, v_m]$	$\boldsymbol{v} = [v_1, \dots, v_m]$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda) \boldsymbol{v} = \lambda \boldsymbol{v}$ for some $\boldsymbol{v} = [v_1, \dots, v_m]$ then $\lambda v_{i-1} + v_i = \lambda v_{i-1}$,
lyche-numerical-linear-algebra-F01930	149	1.000	$\lambda v_{i-1} + v_i = \lambda v_{i-1}$	$\lambda v_{i-1} + v_i = \lambda v_{i-1}$	$\boldsymbol{e}_1$. Indeed, if $\boldsymbol{J}_m(\lambda) \boldsymbol{v} = \lambda \boldsymbol{v}$ for some $\boldsymbol{v} = [v_1, \dots, v_m]$ then $\lambda v_{i-1} + v_i = \lambda v_{i-1}$,
lyche-numerical-linear-algebra-F01931	149	1.000	$i=2, \dots, m$	$i = 2, \dots, m$	$i = 2, \dots, m$ which shows that $v_2 = \dots = v_m = 0$. Thus, the eigenvalue λ of
lyche-numerical-linear-algebra-F01932	149	1.000	$v_2 = \dots = v_m = 0$	$v_2 = \dots = v_m = 0$	$i = 2, \dots, m$ which shows that $v_2 = \dots = v_m = 0$. Thus, the eigenvalue λ of
lyche-numerical-linear-algebra-F01933	149	0.997	$a=m$	$a = m$	$\boldsymbol{J}_m(\lambda)$ has algebraic multiplicity $a = m$ and geometric multiplicity $g = 1$.
lyche-numerical-linear-algebra-F01934	149	1.000	$g_1, \dots, g_k$	$g_1, \dots, g_k$	$a_1, \dots, a_k$ and geometric multiplicities $g_1, \dots, g_k$. There is a nonsingular matrix
lyche-numerical-linear-algebra-F01935	150	1.000	$\boldsymbol{U}_i$	$\boldsymbol{U}_i$	where each $\boldsymbol{U}_i$ is block diagonal having g_i Jordan blocks along the diagonal
lyche-numerical-linear-algebra-F01936	150	1.000	g_i	g_i	where each $\boldsymbol{U}_i$ is block diagonal having g_i Jordan blocks along the diagonal
lyche-numerical-linear-algebra-F01937	150	1.000	$m_{i,1}, \dots, m_{i,g_i}$	$m_{i,1}, \dots, m_{i,g_i}$	Here $m_{i,1}, \dots, m_{i,g_i}$ are positive integers and they are unique if they are ordered so
lyche-numerical-linear-algebra-F01938	150	1.000	$m_{i,1} \geq m_{i,2} \geq \dots \geq m_{i,g_i}$	$m_{i,1} \geq m_{i,2} \geq \dots \geq m_{i,g_i}$	that $m_{i,1} \geq m_{i,2} \geq \dots \geq m_{i,g_i}$. Moreover, $a_i = \sum_{j=1}^{g_i} m_{i,j}$ for all i .
lyche-numerical-linear-algebra-F01939	150	1.000	$a_i = \sum_{j=1}^{g_i} m_{i,j}$	$a_i = \sum_{j=1}^{g_i} m_{i,j}$	that $m_{i,1} \geq m_{i,2} \geq \dots \geq m_{i,g_i}$. Moreover, $a_i = \sum_{j=1}^{g_i} m_{i,j}$ for all i .
lyche-numerical-linear-algebra-F01940	150	1.000	$\boldsymbol{A} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{J}$	$\boldsymbol{A} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{J}$	They satisfy the matrix equation $\boldsymbol{A} \boldsymbol{S} = \boldsymbol{S} \boldsymbol{J}$.
lyche-numerical-linear-algebra-F01941	150	1.000	$\boldsymbol{U}_1 = \text{diag}(\boldsymbol{J}_3(2), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2))$	$\boldsymbol{U}_1 = \text{diag}(\boldsymbol{J}_3(2), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2))$	• $\boldsymbol{U}_1 = \text{diag}(\boldsymbol{J}_3(2), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2))$ and $\boldsymbol{U}_2 = \boldsymbol{J}_2(3)$.
lyche-numerical-linear-algebra-F01942	150	1.000	$\boldsymbol{U}_2 = \boldsymbol{J}_2(3)$	$\boldsymbol{U}_2 = \boldsymbol{J}_2(3)$	• $\boldsymbol{U}_1 = \text{diag}(\boldsymbol{J}_3(2), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2))$ and $\boldsymbol{U}_2 = \boldsymbol{J}_2(3)$.
lyche-numerical-linear-algebra-F01943	150	1.000	$\lambda=2$	$\lambda = 2$	number of Jordan blocks corresponding to $\lambda = 2$.
lyche-numerical-linear-algebra-F01944	150	1.000	$\boldsymbol{S} = [\boldsymbol{s}_1, \dots, \boldsymbol{s}_8]$	$\boldsymbol{S} = [\boldsymbol{s}_1, \dots, \boldsymbol{s}_8]$	The columns of $\boldsymbol{S} = [\boldsymbol{s}_1, \dots, \boldsymbol{s}_8]$ are determined from the columns of $\boldsymbol{J}$ as
lyche-numerical-linear-algebra-F01945	151	1.000	$\boldsymbol{J}_2(3), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2), \boldsymbol{J}_3(2)$	$\boldsymbol{J}_2(3), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2), \boldsymbol{J}_3(2)$	the 4 Jordan blocks $\boldsymbol{J}_2(3), \boldsymbol{J}_2(2), \boldsymbol{J}_1(2), \boldsymbol{J}_3(2)$ are uniquely given.
lyche-numerical-linear-algebra-F01946	151	1.000	$m_{i,j} \geq 1$	$m_{i,j} \geq 1$	corresponding $\boldsymbol{U}_i$. Moreover each $\boldsymbol{U}_i$ contains g_i Jordan blocks of size $m_{i,j} \geq 1$.

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lyche-numerical-linear-algebra-F01947	151	1.000	$g_{\{i\}} \leq a_{\{i\}}$	$g_i \leq a_i$	Thus $g_i \leq a_i$.
lyche-numerical-linear-algebra-F01948	151	1.000	$\boldsymbol{e}_{\{1\}}, \boldsymbol{e}_{\{4\}}, \boldsymbol{e}_{\{6\}}, \boldsymbol{e}_{\{7\}}$	$\boldsymbol{e}_1, \boldsymbol{e}_4, \boldsymbol{e}_6, \boldsymbol{e}_7$	should then be clear. There are only 4 eigenvectors of $\boldsymbol{J}$, namely $\boldsymbol{e}_1, \boldsymbol{e}_4, \boldsymbol{e}_6, \boldsymbol{e}_7$
lyche-numerical-linear-algebra-F01949	151	1.000	$k=2$	$k = 2$	independent. Moreover there are $k = 2$ distinct eigenvalues and $g_1 + g_2 =$
lyche-numerical-linear-algebra-F01950	151	1.000	$g_{\{1\}}+g_{\{2\}}=$	$g_1 + g_2 =$	independent. Moreover there are $k = 2$ distinct eigenvalues and $g_1 + g_2 =$
lyche-numerical-linear-algebra-F01951	151	1.000	$3+1=4$	$3 + 1 = 4$	$3 + 1 = 4$.
lyche-numerical-linear-algebra-F01952	151	1.000	$\sum_{\{i\}} a_{\{i\}}=n$	$\sum_i a_i = n$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-algebra-F01953	151	1.000	$\sum_{\{i\}} g_{\{i\}}=n$	$\sum_i g_i = n$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-algebra-F01954	151	1.000	$a_{\{i\}}=g_{\{i\}}$	$a_i = g_i$	3. Since $g_i \leq a_i$ for all i and $\sum_i a_i = n$ we have $\sum_i g_i = n$ if and only if $a_i = g_i$
lyche-numerical-linear-algebra-F01955	151	1.000	$i=1, \ldots, k$	$i = 1, \dots, k$	for $i = 1, \dots, k$.
lyche-numerical-linear-algebra-F01956	151	1.000	$\boldsymbol{S}=\boldsymbol{U}$	$\boldsymbol{S} = \boldsymbol{U}$	We turn now to unitary similarity transformations $\boldsymbol{S}^{-1}\boldsymbol{A}\boldsymbol{S}$, where $\boldsymbol{S} = \boldsymbol{U}$ is
lyche-numerical-linear-algebra-F01957	151	1.000	$\boldsymbol{S}^{-1}=\boldsymbol{U}^*$	$\boldsymbol{S}^{-1} = \boldsymbol{U}^*$	unitary. Thus $\boldsymbol{S}^{-1} = \boldsymbol{U}^*$ and a unitary similarity transformation takes the form
lyche-numerical-linear-algebra-F01958	151	1.000	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$.
lyche-numerical-linear-algebra-F01959	152	1.000	$\boldsymbol{R}:=\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{R} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	matrix $\boldsymbol{U} \in \mathbb{C}^{n \times n}$ such that $\boldsymbol{R} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$ is upper triangular.
lyche-numerical-linear-algebra-F01960	152	1.000	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$	call $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{R} \boldsymbol{U}^*$ the Schur factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F01961	152	1.000	$k \times k$	$k \times k$	Assume that the theorem is true for all $k \times k$ matrices, and suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$, where
lyche-numerical-linear-algebra-F01962	152	0.993	$n:=k+1$	$n := k + 1$	$n := k + 1$. Let $(\lambda_1, \boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\ \boldsymbol{v}_1\ _2 = 1$. By Theorem 5.5
lyche-numerical-linear-algebra-F01963	152	0.993	$(\lambda_1, \boldsymbol{v}_1)$	$(\lambda_1, \boldsymbol{v}_1)$	$n := k + 1$. Let $(\lambda_1, \boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\ \boldsymbol{v}_1\ _2 = 1$. By Theorem 5.5
lyche-numerical-linear-algebra-F01964	152	0.993	$\ \boldsymbol{v}_1\ _2=1$	$\ \boldsymbol{v}_1\ _2 = 1$	$n := k + 1$. Let $(\lambda_1, \boldsymbol{v}_1)$ be an eigenpair for $\boldsymbol{A}$ with $\ \boldsymbol{v}_1\ _2 = 1$. By Theorem 5.5
lyche-numerical-linear-algebra-F01965	152	1.000	$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$	$\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$	we can extend $\boldsymbol{v}_1$ to an orthonormal basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ for $\mathbb{C}^n$. The matrix $\boldsymbol{V} :=$
lyche-numerical-linear-algebra-F01966	152	1.000	$\boldsymbol{V}:=$	$\boldsymbol{V} :=$	we can extend $\boldsymbol{v}_1$ to an orthonormal basis $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ for $\mathbb{C}^n$. The matrix $\boldsymbol{V} :=$
lyche-numerical-linear-algebra-F01967	152	1.000	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	$[\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$ is unitary, and
lyche-numerical-linear-algebra-F01968	152	1.000	$\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$	$\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$	By the induction hypothesis there is a unitary matrix $\boldsymbol{W}_1 \in \mathbb{C}^{(n-1) \times (n-1)}$ such that
lyche-numerical-linear-algebra-F01969	152	1.000	$\boldsymbol{W}_1^* \boldsymbol{M} \boldsymbol{W}_1$	$\boldsymbol{W}_1^* \boldsymbol{M} \boldsymbol{W}_1$	$\boldsymbol{W}_1^* \boldsymbol{M} \boldsymbol{W}_1$ is upper triangular. Define
lyche-numerical-linear-algebra-F01970	152	1.000	$\boldsymbol{W}$	$\boldsymbol{W}$	Then $\boldsymbol{W}$ and $\boldsymbol{U}$ are unitary and
lyche-numerical-linear-algebra-F01971	152	0.570	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$	eigenvalues there exists an orthogonal matrix $\boldsymbol{U} \in \mathbb{R}^{n \times n}$ such that $\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}$ is upper
lyche-numerical-linear-algebra-F01972	152	1.000	λ_1	λ_1	Proof Consider the proof of Theorem 6.5. Since $\boldsymbol{A}$ and λ_1 are real the eigenvector
lyche-numerical-linear-algebra-F01973	152	1.000	$\boldsymbol{W}^T \boldsymbol{W}=\boldsymbol{I}$	$\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$	$\boldsymbol{v}_1$ is real and the matrix $\boldsymbol{W}$ is real and $\boldsymbol{W}^T \boldsymbol{W} = \boldsymbol{I}$. By the induction hypothesis $\boldsymbol{V}$

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lyche-numerical-linear-algebra- F01974	152	1.000	$\boldsymbol{V}$	V	v_1 is real and the matrix W is real and $W^T W = I$. By the induction hypothesis V
lyche-numerical-linear-algebra- F01975	152	1.000	$\boldsymbol{V}^T \boldsymbol{V} = \boldsymbol{I}$	$V^T V = I$	is real and $V^T V = I$. But then also $U = VW$ is real and $U^T U = I$.
lyche-numerical-linear-algebra- F01976	152	1.000	$\boldsymbol{U} = \boldsymbol{V} \boldsymbol{W}$	$U = VW$	is real and $V^T V = I$. But then also $U = VW$ is real and $U^T U = I$.
lyche-numerical-linear-algebra- F01977	152	1.000	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$	$U^T U = I$	is real and $V^T V = I$. But then also $U = VW$ is real and $U^T U = I$.
lyche-numerical-linear-algebra- F01978	153	1.000	$n-1 \cdot \boldsymbol{M}$	$n-1 \cdot M$	$n \times n$ matrix A , we obtain a matrix M of order $n-1$. M has the same eigenvalues
lyche-numerical-linear-algebra- F01979	153	1.000	$\boldsymbol{T} := \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$	$T := \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$	numerical work. The second derivative matrix $T := \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ has an eigenpair
lyche-numerical-linear-algebra- F01980	153	0.675	$\left(2, \boldsymbol{x}_1\right)$	$(2, x_1)$	$(2, x_1)$, where $x_1 = [-1, 0, 1]^T$. Find the remaining eigenvalues using deflation.
lyche-numerical-linear-algebra- F01981	153	0.675	$\boldsymbol{x}_1 = [-1, 0, 1]^T$	$x_1 = [-1, 0, 1]^T$	$(2, x_1)$, where $x_1 = [-1, 0, 1]^T$. Find the remaining eigenvalues using deflation.
lyche-numerical-linear-algebra- F01982	153	1.000	$\boldsymbol{x}_1$	x_1	For this we extend x_1 to a basis $\{x_1, x_2, x_3\}$ for $\mathbb{R}^3$ by defining $x_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra- F01983	153	1.000	$\left\{\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3\right\}$	$\{x_1, x_2, x_3\}$	For this we extend x_1 to a basis $\{x_1, x_2, x_3\}$ for $\mathbb{R}^3$ by defining $x_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra- F01984	153	1.000	$\boldsymbol{x}_2 = [0, 1, 0]^T$	$x_2 = [0, 1, 0]^T$	For this we extend x_1 to a basis $\{x_1, x_2, x_3\}$ for $\mathbb{R}^3$ by defining $x_2 = [0, 1, 0]^T$,
lyche-numerical-linear-algebra- F01985	153	1.000	$\boldsymbol{x}_3 = [1, 0, 1]^T$	$x_3 = [1, 0, 1]^T$	$x_3 = [1, 0, 1]^T$. This is already an orthogonal basis and normalizing we obtain the
lyche-numerical-linear-algebra- F01986	153	1.000	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^*$	$A^* A = A A^*$	A matrix $A \in \mathbb{C}^{n \times n}$ is normal if $A^* A = A A^*$. In this section we show that a matrix
lyche-numerical-linear-algebra- F01987	153	1.000	$\boldsymbol{A}^* = -\boldsymbol{A}$	$A^* = -A$	2. $A^* = -A$,
lyche-numerical-linear-algebra- F01988	153	1.000	$\boldsymbol{A}^* = \boldsymbol{A}^{-1}$	$A^* = A^{-1}$	3. $A^* = A^{-1}$,
lyche-numerical-linear-algebra- F01989	153	0.999	$\boldsymbol{A} = \operatorname{diag}(d_1, \dots, d_n)$	$A = \operatorname{diag}(d_1, \dots, d_n)$	4. $A = \operatorname{diag}(d_1, \dots, d_n)$.
lyche-numerical-linear-algebra- F01990	154	0.989	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$	$U^* A U = D$	normal if and only if there exists a unitary matrix $U \in \mathbb{C}^{n \times n}$ such that $U^* A U = D$
lyche-numerical-linear-algebra- F01991	154	1.000	$\boldsymbol{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$	$D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$	is diagonal. If $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ and $U = [u_1, \dots, u_n]$ then (λ_j, u_j) , $j =$
lyche-numerical-linear-algebra- F01992	154	1.000	$\boldsymbol{U} = [\boldsymbol{u}_1, \dots, \boldsymbol{u}_n]$	$U = [u_1, \dots, u_n]$	is diagonal. If $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ and $U = [u_1, \dots, u_n]$ then (λ_j, u_j) , $j =$
lyche-numerical-linear-algebra- F01993	154	1.000	$\left(\lambda_j, \boldsymbol{u}_j\right), j =$	$(\lambda_j, u_j), j =$	is diagonal. If $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ and $U = [u_1, \dots, u_n]$ then (λ_j, u_j) , $j =$
lyche-numerical-linear-algebra- F01994	154	0.996	$\boldsymbol{B} = \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$B = U^* A U$	Proof If $B = U^* A U$, with B diagonal, and $U^* U = I$, then $A = U B U^*$ and
lyche-numerical-linear-algebra- F01995	154	0.996	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{B} \boldsymbol{U}^*$	$A = U B U^*$	Proof If $B = U^* A U$, with B diagonal, and $U^* U = I$, then $A = U B U^*$ and
lyche-numerical-linear-algebra- F01996	154	1.000	$\boldsymbol{B} \boldsymbol{B}^* = \boldsymbol{B}^* \boldsymbol{B}$	$B B^* = B^* B$	Now $B B^* = B^* B$ since B is diagonal, and A is normal.
lyche-numerical-linear-algebra- F01997	154	1.000	$\boldsymbol{B} := \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}$	$B := U^* A U$	I such that $B := U^* A U$ is upper triangular. Since A is normal B is normal. Indeed,
lyche-numerical-linear-algebra- F01998	154	0.458	e_{ii}	e_{ii}	must be diagonal. The diagonal elements e_{ii} in $E := B^* B$ and f_{ii} in $F := B B^*$
lyche-numerical-linear-algebra- F01999	154	0.458	$\boldsymbol{E} := \boldsymbol{B}^* \boldsymbol{B}$	$E := B^* B$	must be diagonal. The diagonal elements e_{ii} in $E := B^* B$ and f_{ii} in $F := B B^*$

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lyche-numerical-linear-algebra- F02000	154	0.458	f_{ii}	f_{ii}	must be diagonal. The diagonal elements e_{ii} in $E := B^*B$ and f_{ii} in $F := BB^*$
lyche-numerical-linear-algebra- F02001	154	0.458	$\boldsymbol{F} := \boldsymbol{B} \boldsymbol{B}^*$	$\boldsymbol{F} := \boldsymbol{B} \boldsymbol{B}^*$	must be diagonal. The diagonal elements e_{ii} in $E := B^*B$ and f_{ii} in $F := BB^*$
lyche-numerical-linear-algebra- F02002	154	1.000	$\left b_{11}\right ^2=\left b_{11}\right ^2+\left b_{12}\right ^2+\cdots+\left b_{1 n}\right ^2$	$ b_{11} ^2 = b_{11} ^2 + b_{12} ^2 + \cdots + b_{1n} ^2$	$i = 1$ we have $ b_{11} ^2 = b_{11} ^2 + b_{12} ^2 + \cdots + b_{1n} ^2$, so $b_{1k} = 0$ for $k = 2, 3, \dots, n$.
lyche-numerical-linear-algebra- F02003	154	1.000	$b_{1 k}=0$	$b_{1k} = 0$	$i = 1$ we have $ b_{11} ^2 = b_{11} ^2 + b_{12} ^2 + \cdots + b_{1n} ^2$, so $b_{1k} = 0$ for $k = 2, 3, \dots, n$.
lyche-numerical-linear-algebra- F02004	154	1.000	$k=2,3, \ldots, n$	$k = 2, 3, \dots, n$	$i = 1$ we have $ b_{11} ^2 = b_{11} ^2 + b_{12} ^2 + \cdots + b_{1n} ^2$, so $b_{1k} = 0$ for $k = 2, 3, \dots, n$.
lyche-numerical-linear-algebra- F02005	154	1.000	$i-1$	$i - 1$	Suppose $\boldsymbol{B}$ is diagonal in its first $i - 1$ rows so that $b_{jk} = 0$ for $j = 1, \dots, i - 1$,
lyche-numerical-linear-algebra- F02006	154	1.000	$b_{j k}=0$	$b_{jk} = 0$	Suppose $\boldsymbol{B}$ is diagonal in its first $i - 1$ rows so that $b_{jk} = 0$ for $j = 1, \dots, i - 1$,
lyche-numerical-linear-algebra- F02007	154	1.000	$j=1, \ldots, i-1$	$j = 1, \dots, i - 1$	Suppose $\boldsymbol{B}$ is diagonal in its first $i - 1$ rows so that $b_{jk} = 0$ for $j = 1, \dots, i - 1$,
lyche-numerical-linear-algebra- F02008	154	0.999	$k=j+1, \ldots, n$	$k = j + 1, \dots, n$	$k = j + 1, \dots, n$. Then
lyche-numerical-linear-algebra- F02009	154	1.000	$b_{i k}=0, k=i+1, \ldots, n$	$b_{ik} = 0, k = i + 1, \dots, n$	and it follows that $b_{ik} = 0, k = i + 1, \dots, n$. By induction on the rows we see that
lyche-numerical-linear-algebra- F02010	154	0.877	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U} =$	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U} =$	<i>Example 6.6</i> The orthogonal diagonalization of $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ is $\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U} =$
lyche-numerical-linear-algebra- F02011	154	0.987	$\operatorname{diag}(1,3)$	$\operatorname{diag}(1, 3)$	$\operatorname{diag}(1, 3)$, where $\boldsymbol{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F02012	154	0.987	$\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & 1 \\ 1 & -1 \end{array}\right]$	$\boldsymbol{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$\operatorname{diag}(1, 3)$, where $\boldsymbol{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.
lyche-numerical-linear-algebra- F02013	155	0.835	$R(\boldsymbol{x})=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}=\lambda$	$R(\boldsymbol{x}) = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \lambda$	If $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}$ then $R(\boldsymbol{x}) = \frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \lambda$.
lyche-numerical-linear-algebra- F02014	155	0.761	$\lambda_j, \boldsymbol{u}_j$	$\lambda_j, \boldsymbol{u}_j$	is normal with orthonormal eigenpairs $(\lambda_j, \boldsymbol{u}_j)$, for $j = 1, 2, \dots, n$. Then the
lyche-numerical-linear-algebra- F02015	155	0.761	$j=1,2, \ldots, n$	$j = 1, 2, \dots, n$	is normal with orthonormal eigenpairs $(\lambda_j, \boldsymbol{u}_j)$, for $j = 1, 2, \dots, n$. Then the
lyche-numerical-linear-algebra- F02016	155	1.000	$\boldsymbol{x}^* \boldsymbol{x}=\sum_{i=1}^n \bar{c}_i \overline{\boldsymbol{u}_i} c_j \boldsymbol{u}_j=\sum_{j=1}^n \bar{\boldsymbol{c}}_{\boldsymbol{j}} \overline{\boldsymbol{u}_{\boldsymbol{j}}} \boldsymbol{c}_{\boldsymbol{j}} \boldsymbol{u}_{\boldsymbol{j}}=$	$\boldsymbol{x}^* \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}_i} c_j \boldsymbol{u}_j =$	Proof By orthonormality of the eigenvectors $\boldsymbol{x}^* \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}_i} c_j \boldsymbol{u}_j =$
lyche-numerical-linear-algebra- F02017	155	1.000	$\sum_{j=1}^n\left c_{j}\right ^2$	$\sum_{j=1}^n c_j ^2$	$\sum_{j=1}^n c_j ^2$. Similarly, $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}_i} c_j \lambda_j \boldsymbol{u}_j = \sum_{i=1}^n \lambda_i c_i ^2$, and
lyche-numerical-linear-algebra- F02018	155	1.000	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}=\sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}_i} c_j \lambda_j \boldsymbol{u}_j=\sum_{i=1}^n \lambda_i\left c_i\right ^2$	$\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}_i} c_j \lambda_j \boldsymbol{u}_j = \sum_{i=1}^n \lambda_i c_i ^2$	$\sum_{j=1}^n c_j ^2$. Similarly, $\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} = \sum_{i=1}^n \sum_{j=1}^n \bar{c}_i \overline{\boldsymbol{u}_i} c_j \lambda_j \boldsymbol{u}_j = \sum_{i=1}^n \lambda_i c_i ^2$, and
lyche-numerical-linear-algebra- F02019	155	1.000	$\sum_{i=1}^n\left c_i\right ^2 / \sum_{j=1}^n\left c_j\right ^2=1$	$\sum_{i=1}^n c_i ^2 / \sum_{j=1}^n c_j ^2 = 1$	combination since $\sum_{i=1}^n c_i ^2 / \sum_{j=1}^n c_j ^2 = 1$.
lyche-numerical-linear-algebra- F02020	155	0.998	$\lambda=\mu+i \nu, \bar{\lambda}=\mu-i \nu$	$\lambda = \mu + i\nu, \bar{\lambda} = \mu - i\nu$	eigenvalues of a real matrix occur in conjugate pairs, $\lambda = \mu + i\nu, \bar{\lambda} = \mu - i\nu$,
lyche-numerical-linear-algebra- F02021	155	0.758	μ, ν	μ, ν	where μ, ν are real. The real 2×2 matrix
lyche-numerical-linear-algebra- F02022	155	0.999	$\lambda=\mu+i \nu$	$\lambda = \mu + i\nu$	has eigenvalues $\lambda = \mu + i\nu$ and $\bar{\lambda} = \mu - i\nu$.

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lyche-numerical-linear-algebra-155 F02023	155	■ 0.999	$\bar{\lambda}=\mu-i \nu$	$\bar{\lambda}=\mu-i\nu$	has eigenvalues $\lambda=\mu+i\nu$ and $\bar{\lambda}=\mu-i\nu$.
lyche-numerical-linear-algebra-156 F02024	156	■ 1.000	$\pi_{\boldsymbol{R}}=$	$\pi_{\boldsymbol{R}}=$	Since $\boldsymbol{R}$ is block triangular the characteristic polynomial of $\boldsymbol{R}$ is given by $\pi_{\boldsymbol{R}}=$
lyche-numerical-linear-algebra-156 F02025	156	■ 0.831	$\boldsymbol{\pi}_{\boldsymbol{D}_1}\boldsymbol{\pi}_{\boldsymbol{D}_2}\boldsymbol{\pi}_{\boldsymbol{D}_3}$	$\boldsymbol{\pi}_{\boldsymbol{D}_1}\boldsymbol{\pi}_{\boldsymbol{D}_2}\boldsymbol{\pi}_{\boldsymbol{D}_3}$	$\boldsymbol{\pi}_{\boldsymbol{D}_1}\boldsymbol{\pi}_{\boldsymbol{D}_2}\boldsymbol{\pi}_{\boldsymbol{D}_3}$. We find
lyche-numerical-linear-algebra-156 F02026	156	■ 1.000	$\boldsymbol{D}_1$	$\boldsymbol{D}_1$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1=2+i, \lambda_2=2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra-156 F02027	156	■ 1.000	$\boldsymbol{D}_3$	$\boldsymbol{D}_3$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1=2+i, \lambda_2=2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra-156 F02028	156	■ 1.000	$\lambda_1=2+i, \lambda_2=2-i$	$\lambda_1=2+i, \lambda_2=2-i$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1=2+i, \lambda_2=2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra-156 F02029	156	■ 1.000	$\boldsymbol{D}_2$	$\boldsymbol{D}_2$	and the eigenvalues $\boldsymbol{D}_1$ and $\boldsymbol{D}_3$ are $\lambda_1=2+i, \lambda_2=2-i$, while $\boldsymbol{D}_2$ obviously has
lyche-numerical-linear-algebra-156 F02030	156	■ 1.000	$\left\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\right\}$	$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$	For any such $\boldsymbol{U}$ the columns $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ of $\boldsymbol{U}$ are orthonormal eigenvectors of $\boldsymbol{A}$
lyche-numerical-linear-algebra-156 F02031	156	■ 1.000	$\boldsymbol{A} \boldsymbol{u}_j=\lambda_j \boldsymbol{u}_j$	$\boldsymbol{A} \boldsymbol{u}_j=\lambda_j \boldsymbol{u}_j$	and $\boldsymbol{A} \boldsymbol{u}_j=\lambda_j \boldsymbol{u}_j$ for $j=1, \ldots, n$.
lyche-numerical-linear-algebra-156 F02032	156	■ 0.995	$\lambda_1, \lambda_2, \ldots, \lambda_n$	$\lambda_1, \lambda_2, \ldots, \lambda_n$	ric. Then $\boldsymbol{A}$ has real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Moreover, there is an orthogonal
lyche-numerical-linear-algebra-157 F02033	157	■ 0.999	$R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	$R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$	terms of the Rayleigh quotient $R(\boldsymbol{x})=R_{\boldsymbol{A}}(\boldsymbol{x}):=\frac{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}$. First we show
lyche-numerical-linear-algebra-157 F02034	157	■ 0.924	$\left(\lambda_j, \boldsymbol{u}_j\right), 1 \leq j \leq n$	$(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$	eigenpairs $(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$, ordered so that $\lambda_1 \geq \cdots \geq \lambda_n$. Let $1 \leq k \leq n$. For
lyche-numerical-linear-algebra-157 F02035	157	■ 0.924	$\lambda_1 \geq \cdots \geq \lambda_n$	$\lambda_1 \geq \cdots \geq \lambda_n$	eigenpairs $(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$, ordered so that $\lambda_1 \geq \cdots \geq \lambda_n$. Let $1 \leq k \leq n$. For
lyche-numerical-linear-algebra-157 F02036	157	■ 0.924	$1 \leq k \leq n$	$1 \leq k \leq n$	eigenpairs $(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq n$, ordered so that $\lambda_1 \geq \cdots \geq \lambda_n$. Let $1 \leq k \leq n$. For
lyche-numerical-linear-algebra-157 F02037	157	■ 1.000	$\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\right)$	$\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\right)$	with equality for $\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\right)$ and $\boldsymbol{x}=\boldsymbol{u}_k$.
lyche-numerical-linear-algebra-157 F02038	157	■ 1.000	$\boldsymbol{x}=\boldsymbol{u}_k$	$\boldsymbol{x}=\boldsymbol{u}_k$	with equality for $\mathcal{S}=\tilde{\mathcal{S}}:=\operatorname{span}\left(\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\right)$ and $\boldsymbol{x}=\boldsymbol{u}_k$.
lyche-numerical-linear-algebra-157 F02039	157	■ 1.000	$\mathcal{S}^{\prime}:=$	$\mathcal{S}^{\prime}:=$	Proof Let $\mathcal{S}$ be any subspace of $\mathbb{C}^n$ of dimension $n-k+1$ and define $\mathcal{S}^{\prime}:=$
lyche-numerical-linear-algebra-157 F02040	157	■ 1.000	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S}+\mathcal{S}^{\prime}:=$
lyche-numerical-linear-algebra-157 F02041	157	■ 1.000	$\boldsymbol{y} \in \mathcal{S}$	$\boldsymbol{y} \in \mathcal{S}$	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S}+\mathcal{S}^{\prime}:=$
lyche-numerical-linear-algebra-157 F02042	157	■ 1.000	$R(\boldsymbol{y}) \geq \lambda_k$	$R(\boldsymbol{y}) \geq \lambda_k$	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S}+\mathcal{S}^{\prime}:=$
lyche-numerical-linear-algebra-157 F02043	157	■ 1.000	$\mathcal{S}+\mathcal{S}^{\prime}:=$	$\mathcal{S}+\mathcal{S}^{\prime}:=$	$\operatorname{span}\left(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_k\right)$. It is enough to find $\boldsymbol{y} \in \mathcal{S}$ so that $R(\boldsymbol{y}) \geq \lambda_k$. Now $\mathcal{S}+\mathcal{S}^{\prime}:=$
lyche-numerical-linear-algebra-157 F02044	157	■ 1.000	$\left\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\right\}$	$\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\}$	$\{\boldsymbol{s}+\boldsymbol{s}^{\prime}: \boldsymbol{s} \in \mathcal{S}, \boldsymbol{s}^{\prime} \in \mathcal{S}^{\prime}\}$ is a subspace of $\mathbb{C}^n$ and by (1.7)
lyche-numerical-linear-algebra-157 F02045	157	■ 0.996	$\mathcal{S} \cap \mathcal{S}^{\prime}$	$\mathcal{S} \cap \mathcal{S}^{\prime}$	It follows that $\mathcal{S} \cap \mathcal{S}^{\prime}$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}=\sum_{j=1}^k c_j \boldsymbol{u}_j$ with $\sum_{j=1}^k\left c_j\right ^2=$
lyche-numerical-linear-algebra-157 F02046	157	■ 0.996	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}=\sum_{j=1}^k c_j \boldsymbol{u}_j$	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}=\sum_{j=1}^k c_j \boldsymbol{u}_j$	It follows that $\mathcal{S} \cap \mathcal{S}^{\prime}$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}=\sum_{j=1}^k c_j \boldsymbol{u}_j$ with $\sum_{j=1}^k\left c_j\right ^2=$
lyche-numerical-linear-algebra-157 F02047	157	■ 0.996	$\sum_{j=1}^k\left c_j\right ^2=$	$\sum_{j=1}^k\left c_j\right ^2=$	It follows that $\mathcal{S} \cap \mathcal{S}^{\prime}$ is nonempty. Let $\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}^{\prime}=\sum_{j=1}^k c_j \boldsymbol{u}_j$ with $\sum_{j=1}^k\left c_j\right ^2=$

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lyche-numerical-linear-algebra-F02048	157	0.997	$c_{\{j\}}=0$	$c_j = 0$	1. Defining $c_j = 0$ for $k + 1 \leq j \leq n$, we obtain by Theorem 6.8
lyche-numerical-linear-algebra-F02049	157	0.997	$k+1 \leq j \leq n$	$k + 1 \leq j \leq n$	1. Defining $c_j = 0$ for $k + 1 < j < n$, we obtain by Theorem 6.8
lyche-numerical-linear-algebra-F02050	157	0.877	$\boldsymbol{z} \in \mathcal{S} = \tilde{\mathcal{S}}$	$z \in \mathcal{S} = \tilde{\mathcal{S}}$	and (6.11) follows. To show equality suppose $z \in \mathcal{S} = \tilde{\mathcal{S}}$. Now $z = \sum_{j=k}^n d_j \boldsymbol{u}_j$ for
lyche-numerical-linear-algebra-F02051	157	0.877	$\boldsymbol{z} = \sum_{j=k}^n d_{\{j\}} \boldsymbol{u}_{\{j\}}$	$z = \sum_{j=k}^n d_j \boldsymbol{u}_j$	and (6.11) follows. To show equality suppose $z \in \mathcal{S} = \tilde{\mathcal{S}}$. Now $z = \sum_{j=k}^n d_j \boldsymbol{u}_j$ for
lyche-numerical-linear-algebra-F02052	157	0.999	$d_{\{k\}}, \ldots, d_{\{n\}}$	$d_k, \dots, d_n$	some $d_k, \dots, d_n$ with $\sum_{j=k}^n d_j ^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$
lyche-numerical-linear-algebra-F02053	157	0.999	$\sum_{j=k}^n d_{\{j\}} ^2 = 1$	$\sum_{j=k}^n d_j ^2 = 1$	some $d_k, \dots, d_n$ with $\sum_{j=k}^n d_j ^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$
lyche-numerical-linear-algebra-F02054	157	0.999	6.8 $R(z) = \sum_{j=k}^n \lambda_{\{j\}} d_{\{j\}} ^2 \leq$	$6.8 R(z) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$	some $d_k, \dots, d_n$ with $\sum_{j=k}^n d_j ^2 = 1$ and by Lemma 6.8 $R(z) = \sum_{j=k}^n \lambda_j d_j ^2 \leq$
lyche-numerical-linear-algebra-F02055	157	0.959	$\lambda_{\{k\}}$	λ_k	λ_k . Since $z \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$ and equality in (6.11)
lyche-numerical-linear-algebra-F02056	157	0.959	$\boldsymbol{z} \in \tilde{\mathcal{S}}$	$z \in \tilde{\mathcal{S}}$	λ_k . Since $z \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$ and equality in (6.11)
lyche-numerical-linear-algebra-F02057	157	0.959	$\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$	$\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$	λ_k . Since $z \in \tilde{\mathcal{S}}$ is arbitrary we have $\max_{\substack{x \in \tilde{\mathcal{S}} \\ x \neq 0}} R(x) \leq \lambda_k$ and equality in (6.11)
lyche-numerical-linear-algebra-F02058	157	1.000	$\mathcal{S} = \tilde{\mathcal{S}}$	$\mathcal{S} = \tilde{\mathcal{S}}$	follows for $\mathcal{S} = \tilde{\mathcal{S}}$. Moreover, $R(\boldsymbol{u}_k) = \lambda_k$.
lyche-numerical-linear-algebra-F02059	157	1.000	$R(\boldsymbol{u}_{\{k\}}) = \lambda_{\{k\}}$	$R(\boldsymbol{u}_k) = \lambda_k$	follows for $\mathcal{S} = \tilde{\mathcal{S}}$. Moreover, $R(\boldsymbol{u}_k) = \lambda_k$.
lyche-numerical-linear-algebra-F02060	158	1.000	$\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_{\{1\}}, \ldots, \boldsymbol{u}_{\{k\}})$	$\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$	with equality for $\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$ and $x = \boldsymbol{u}_k$. Here $(\lambda_j, \boldsymbol{u}_j)$, $1 \leq j \leq$
lyche-numerical-linear-algebra-F02061	158	1.000	$(\lambda_{\{j\}}, \boldsymbol{u}_{\{j\}}), 1 \leq j \leq$	$(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq$	with equality for $\mathcal{S} = \tilde{\mathcal{S}} := \text{span}(\boldsymbol{u}_1, \dots, \boldsymbol{u}_k)$ and $x = \boldsymbol{u}_k$. Here $(\lambda_j, \boldsymbol{u}_j), 1 \leq j \leq$
lyche-numerical-linear-algebra-F02062	158	1.000	$\text{span}(\boldsymbol{u}_{\{k\}}, \ldots, \boldsymbol{u}_{\{n\}})$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and show that $R(y) \leq \lambda_k$ for some $y \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-algebra-F02063	158	1.000	$R(\boldsymbol{y}) \leq \lambda_{\{k\}}$	$R(y) \leq \lambda_k$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and show that $R(y) \leq \lambda_k$ for some $y \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-algebra-F02064	158	1.000	$\boldsymbol{y} \in \mathcal{S} \cap \mathcal{S}'$	$y \in \mathcal{S} \cap \mathcal{S}'$	$\text{span}(\boldsymbol{u}_k, \dots, \boldsymbol{u}_n)$ and show that $R(y) \leq \lambda_k$ for some $y \in \mathcal{S} \cap \mathcal{S}'$. It is easy to see
lyche-numerical-linear-algebra-F02065	158	1.000	$\boldsymbol{y} \in \tilde{\mathcal{S}}$	$y \in \tilde{\mathcal{S}}$	that $R(y) \geq \lambda_k$ for any $y \in \tilde{\mathcal{S}}$.
lyche-numerical-linear-algebra-F02066	158	0.995	$\boldsymbol{A}, \boldsymbol{B} \in$	$\boldsymbol{A}, \boldsymbol{B} \in$	Theorem 6.13 (Eigenvalue Perturbation for Hermitian Matrices) Let $\boldsymbol{A}, \boldsymbol{B} \in$
lyche-numerical-linear-algebra-F02067	158	0.998	$\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$	$\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$	$\mathbb{C}^{n \times n}$ be Hermitian with eigenvalues $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$ and $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$.
lyche-numerical-linear-algebra-F02068	158	0.998	$\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$	$\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$	$\mathbb{C}^{n \times n}$ be Hermitian with eigenvalues $\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n$ and $\beta_1 \geq \beta_2 \geq \cdots \geq \beta_n$.
lyche-numerical-linear-algebra-F02069	158	1.000	$\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_n$	$\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_n$	where $\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_n$ are the eigenvalues of $\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-F02070	158	1.000	$\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$	$\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$	where $\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_n$ are the eigenvalues of $\boldsymbol{E} := \boldsymbol{B} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-F02071	158	0.997	$(\alpha_{\{j\}}, \boldsymbol{u}_{\{j\}}), j=1, \ldots, n$	$(\alpha_j, \boldsymbol{u}_j), j = 1, \dots, n$	eigenvalues are real. Let $(\alpha_j, \boldsymbol{u}_j)$, $j = 1, \dots, n$ be orthonormal eigenpairs for $\boldsymbol{A}$

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lyche-numerical-linear-algebra-F02872	158	■ 1.000	$\mathcal{S} := \operatorname{span}\{\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\}$	$\mathcal{S} := \operatorname{span}\{\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\}$	and let $\mathcal{S} := \operatorname{span}\{\boldsymbol{u}_k, \ldots, \boldsymbol{u}_n\}$. By Theorem 6.11 we obtain
lyche-numerical-linear-algebra-F02873	158	■ 1.000	$\boldsymbol{D} := -\boldsymbol{E}$	$\boldsymbol{D} := -\boldsymbol{E}$	and this proves the upper inequality. For the lower one we define $\boldsymbol{D} := -\boldsymbol{E}$ and
lyche-numerical-linear-algebra-F02874	158	■ 1.000	$-\varepsilon_n$	$-\varepsilon_n$	observe that $-\varepsilon_n$ is the largest eigenvalue of $\boldsymbol{D}$. Since $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$ it follows from the
lyche-numerical-linear-algebra-F02875	158	■ 1.000	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$	$\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$	observe that $-\varepsilon_n$ is the largest eigenvalue of $\boldsymbol{D}$. Since $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{D}$ it follows from the
lyche-numerical-linear-algebra-F02876	158	■ 1.000	$\alpha_k \leq \beta_k - \varepsilon_n$	$\alpha_k \leq \beta_k - \varepsilon_n$	result just proved that $\alpha_k \leq \beta_k - \varepsilon_n$, which is the same as the lower inequality.
lyche-numerical-linear-algebra-F02877	159	■ 1.000	$\mu_1, \ldots, \mu_n$	$\mu_1, \ldots, \mu_n$	<i>normal matrices with eigenvalues $\lambda_1, \ldots, \lambda_n$ and $\mu_1, \ldots, \mu_n$, respectively. Then</i>
lyche-numerical-linear-algebra-F02878	159	■ 0.655	$\boldsymbol{y} \in \mathbb{C}^n$	$\boldsymbol{y} \in \mathbb{C}^n$	$\lambda \in \mathbb{C}$ and $\boldsymbol{y} \in \mathbb{C}^n$ is nonzero. We say that $(\lambda, \boldsymbol{y})$ is a left eigenpair for $\boldsymbol{A}$ if
lyche-numerical-linear-algebra-F02879	159	■ 0.655	$\lambda, \boldsymbol{y}$	$\lambda, \boldsymbol{y}$	$\lambda \in \mathbb{C}$ and $\boldsymbol{y} \in \mathbb{C}^n$ is nonzero. We say that $(\lambda, \boldsymbol{y})$ is a left eigenpair for $\boldsymbol{A}$ if
lyche-numerical-linear-algebra-F02880	159	■ 0.979	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$ or equivalently $\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$. We say that $(\lambda, \boldsymbol{y})$ is a right eigenpair for
lyche-numerical-linear-algebra-F02881	159	■ 0.979	$\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$	$\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$	$\boldsymbol{y}^* \boldsymbol{A} = \lambda \boldsymbol{y}^*$ or equivalently $\boldsymbol{A}^* \boldsymbol{y} = \bar{\lambda} \boldsymbol{y}$. We say that $(\lambda, \boldsymbol{y})$ is a right eigenpair for
lyche-numerical-linear-algebra-F02882	159	■ 0.988	$\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$	$\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$	$\boldsymbol{A}$ if $\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$. If $(\lambda, \boldsymbol{y})$ is a left eigenpair then λ is called a left eigenvalue and
lyche-numerical-linear-algebra-F02883	159	■ 0.988	$(\lambda, \boldsymbol{y})$	$(\lambda, \boldsymbol{y})$	$\boldsymbol{A}$ if $\boldsymbol{A} \boldsymbol{y} = \lambda \boldsymbol{y}$. If $(\lambda, \boldsymbol{y})$ is a left eigenpair then λ is called a left eigenvalue and
lyche-numerical-linear-algebra-F02884	159	■ 0.779	$(\lambda, \boldsymbol{y})$	$(\lambda, \boldsymbol{y})$	$\boldsymbol{y}$ a left eigenvector . Similarly if $(\lambda, \boldsymbol{y})$ is a right eigenpair then λ is called a right
lyche-numerical-linear-algebra-F02885	159	■ 0.999	$\bar{\lambda}$	$\bar{\lambda}$	is an eigenvector of $\boldsymbol{A}^*$. If λ is a left eigenvalue of $\boldsymbol{A}$ then $\bar{\lambda}$ is an eigenvalue of $\boldsymbol{A}^*$
lyche-numerical-linear-algebra-F02886	159	■ 1.000	$\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\}$	$\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\}$	<i>eigenvectors $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\}$ with $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$. Conversely, if $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has</i>
lyche-numerical-linear-algebra-F02887	159	■ 1.000	$\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$	$\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$	<i>eigenvectors $\{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n\}$ with $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$. Conversely, if $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has</i>
lyche-numerical-linear-algebra-F02888	160	■ 1.000	$(\lambda_1, \boldsymbol{x}_1), \ldots, (\lambda_n, \boldsymbol{x}_n)$	$(\lambda_1, \boldsymbol{x}_1), \ldots, (\lambda_n, \boldsymbol{x}_n)$	Proof For any right eigenpairs $(\lambda_1, \boldsymbol{x}_1), \ldots, (\lambda_n, \boldsymbol{x}_n)$ and left eigenpairs
lyche-numerical-linear-algebra-F02889	160	■ 0.998	$(\lambda_1, \boldsymbol{y}_1), \ldots, (\lambda_n, \boldsymbol{y}_n)$	$(\lambda_1, \boldsymbol{y}_1), \ldots, (\lambda_n, \boldsymbol{y}_n)$	$(\lambda_1, \boldsymbol{y}_1), \ldots, (\lambda_n, \boldsymbol{y}_n)$ of $\boldsymbol{A}$ we have $\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}$, $\boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$, where
lyche-numerical-linear-algebra-F02890	160	■ 0.998	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$	$(\lambda_1, \boldsymbol{y}_1), \ldots, (\lambda_n, \boldsymbol{y}_n)$ of $\boldsymbol{A}$ we have $\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D}, \boldsymbol{Y}^* \boldsymbol{A} = \boldsymbol{D} \boldsymbol{Y}^*$, where
lyche-numerical-linear-algebra-F02891	160	■ 1.000	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D} \implies \boldsymbol{A} = \boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \implies \boldsymbol{X}^{-1} \boldsymbol{A} =$	$\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D} \implies \boldsymbol{A} = \boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \implies \boldsymbol{X}^{-1} \boldsymbol{A} =$	Suppose $\boldsymbol{X}$ is nonsingular. Then $\boldsymbol{A} \boldsymbol{X} = \boldsymbol{X} \boldsymbol{D} \implies \boldsymbol{A} = \boldsymbol{X} \boldsymbol{D} \boldsymbol{X}^{-1} \implies \boldsymbol{X}^{-1} \boldsymbol{A} =$
lyche-numerical-linear-algebra-F02892	160	■ 1.000	$\boldsymbol{D} \boldsymbol{X}^{-1}$	$\boldsymbol{D} \boldsymbol{X}^{-1}$	$\boldsymbol{D} \boldsymbol{X}^{-1}$ and it follows that $\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$ contains a collection of left eigenvectors such
lyche-numerical-linear-algebra-F02893	160	■ 1.000	$\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$	$\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$	$\boldsymbol{D} \boldsymbol{X}^{-1}$ and it follows that $\boldsymbol{Y}^* := \boldsymbol{X}^{-1}$ contains a collection of left eigenvectors such
lyche-numerical-linear-algebra-F02894	160	■ 1.000	$\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$	$\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$	that $\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$. Thus the columns of $\boldsymbol{Y}$ are linearly independent and $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$.
lyche-numerical-linear-algebra-F02895	160	■ 1.000	$\boldsymbol{Y}$	$\boldsymbol{Y}$	that $\boldsymbol{Y}^* \boldsymbol{X} = \boldsymbol{I}$. Thus the columns of $\boldsymbol{Y}$ are linearly independent and $\boldsymbol{y}_i^* \boldsymbol{x}_j = \delta_{i,j}$.
lyche-numerical-linear-algebra-F02896	160	■ 1.000	$\boldsymbol{A} \boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*} \boldsymbol{D}$	$\boldsymbol{A} \boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*} \boldsymbol{D}$	Similarly, if $\boldsymbol{Y}$ is nonsingular then $\boldsymbol{A} \boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*} \boldsymbol{D}$ and it follows that $\boldsymbol{X} := \boldsymbol{Y}^{-*}$

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lyche-numerical-linear-algebra-F02097	160	1.000	$\boldsymbol{X} := \boldsymbol{Y}^{-*}$	$\boldsymbol{X} := \boldsymbol{Y}^{-*}$	Similarly, if $\boldsymbol{Y}$ is nonsingular then $\boldsymbol{A}\boldsymbol{Y}^{-*} = \boldsymbol{Y}^{-*}\boldsymbol{D}$ and it follows that $\boldsymbol{X} := \boldsymbol{Y}^{-*}$
lyche-numerical-linear-algebra-F02098	160	0.996	$\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$	$\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$	If $\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$ then $\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$ for
lyche-numerical-linear-algebra-F02099	160	0.996	$\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$	$\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$	If $\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$ then $\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$ for
lyche-numerical-linear-algebra-F02100	160	0.996	$c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$	$c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$	If $\boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{x}_j$ then $\boldsymbol{y}_i^* \boldsymbol{v} = \sum_{j=1}^n c_j \boldsymbol{y}_i^* \boldsymbol{x}_j = c_i \boldsymbol{y}_i^* \boldsymbol{x}_i$, so $c_i = \boldsymbol{y}_i^* \boldsymbol{v} / \boldsymbol{y}_i^* \boldsymbol{x}_i$ for
lyche-numerical-linear-algebra-F02101	160	0.995	$\mu, \boldsymbol{y}$	$\mu, \boldsymbol{y}$	Theorem 6.16 (Biorthogonality) Suppose $(\mu, \boldsymbol{y})$ and $(\lambda, \boldsymbol{x})$ are left and right
lyche-numerical-linear-algebra-F02102	160	1.000	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{y}^* \boldsymbol{x} = \mu \boldsymbol{y}^* \boldsymbol{x}$	$\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{y}^* \boldsymbol{x} = \mu \boldsymbol{y}^* \boldsymbol{x}$	Proof Using the eigenpair relation in two ways we obtain $\boldsymbol{y}^* \boldsymbol{A} \boldsymbol{x} = \lambda \boldsymbol{y}^* \boldsymbol{x} = \mu \boldsymbol{y}^* \boldsymbol{x}$
lyche-numerical-linear-algebra-F02103	160	1.000	$\boldsymbol{y}^* \boldsymbol{x} \neq 0$	$\boldsymbol{y}^* \boldsymbol{x} \neq 0$	eigenpairs of $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. If λ has algebraic multiplicity one then $\boldsymbol{y}^* \boldsymbol{x} \neq 0$.
lyche-numerical-linear-algebra-F02104	160	1.000	$\ \boldsymbol{x}\ _2 = 1$	$\ \boldsymbol{x}\ _2 = 1$	Proof Assume that $\ \boldsymbol{x}\ _2 = 1$. We have (cf. (6.8))
lyche-numerical-linear-algebra-F02105	161	1.000	$\boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{x}$	$\boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{x}$	where $\boldsymbol{V}$ is unitary and $\boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{x}$. We show that if $\boldsymbol{y}^* \boldsymbol{x} = 0$ then λ is also an
lyche-numerical-linear-algebra-F02106	161	1.000	$\boldsymbol{u} := \boldsymbol{V}^* \boldsymbol{y}$	$\boldsymbol{u} := \boldsymbol{V}^* \boldsymbol{y}$	eigenvalue of $\boldsymbol{M}$ contradicting the multiplicity assumption of λ . Let $\boldsymbol{u} := \boldsymbol{V}^* \boldsymbol{y}$.
lyche-numerical-linear-algebra-F02107	161	1.000	$(\bar{\lambda}, \boldsymbol{u})$	$(\bar{\lambda}, \boldsymbol{u})$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$. But then $\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra-F02108	161	1.000	$\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$	$\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$. But then $\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra-F02109	161	1.000	$\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$	$\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$	so $(\bar{\lambda}, \boldsymbol{u})$ is an eigenpair of $\boldsymbol{V}^* \boldsymbol{A}^* \boldsymbol{V}$. But then $\boldsymbol{y}^* \boldsymbol{x} = \boldsymbol{u}^* \boldsymbol{V}^* \boldsymbol{V} \boldsymbol{e}_1 = \boldsymbol{u}^* \boldsymbol{e}_1$. Suppose
lyche-numerical-linear-algebra-F02110	161	0.596	$\boldsymbol{u}^* \boldsymbol{e}_1 = 0$	$\boldsymbol{u}^* \boldsymbol{e}_1 = 0$	that $\boldsymbol{u}^* \boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \begin{bmatrix} 0 \\ \boldsymbol{v} \end{bmatrix}$ for some nonzero $\boldsymbol{v} \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra-F02111	161	0.596	$\boldsymbol{u} = \begin{bmatrix} 0 \\ \boldsymbol{v} \end{bmatrix}$	$\boldsymbol{u} = \begin{bmatrix} 0 \\ \boldsymbol{v} \end{bmatrix}$	that $\boldsymbol{u}^* \boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \begin{bmatrix} 0 \\ \boldsymbol{v} \end{bmatrix}$ for some nonzero $\boldsymbol{v} \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra-F02112	161	0.596	$\boldsymbol{v} \in \mathbb{C}^{n-1}$	$\boldsymbol{v} \in \mathbb{C}^{n-1}$	that $\boldsymbol{u}^* \boldsymbol{e}_1 = 0$, i.e., $\boldsymbol{u} = \begin{bmatrix} 0 \\ \boldsymbol{v} \end{bmatrix}$ for some nonzero $\boldsymbol{v} \in \mathbb{C}^{n-1}$. Then
lyche-numerical-linear-algebra-F02113	161	1.000	$\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\boldsymbol{A} := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has one eigenvalue $\lambda = 1$ of algebraic multiplicity two, one right
lyche-numerical-linear-algebra-F02114	161	1.000	$\boldsymbol{y} = \boldsymbol{e}_2$	$\boldsymbol{y} = \boldsymbol{e}_2$	eigenvector $\boldsymbol{x} = \boldsymbol{e}_1$ and one left eigenvector $\boldsymbol{y} = \boldsymbol{e}_2$. Thus $\boldsymbol{x}$ and $\boldsymbol{y}$ are orthogonal.
lyche-numerical-linear-algebra-F02115	161	1.000	$\det(\boldsymbol{B}^T) =$	$\det(\boldsymbol{B}^T) =$	Exercise 6.2 (Characteristic Polynomial of Transpose) We have $\det(\boldsymbol{B}^T) =$
lyche-numerical-linear-algebra-F02116	161	1.000	$\det(\boldsymbol{B})$	$\det(\boldsymbol{B})$	$\det(\boldsymbol{B})$ and $\det(\overline{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$ for any square matrix $\boldsymbol{B}$. Use this to show that
lyche-numerical-linear-algebra-F02117	161	1.000	$\det(\overline{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$	$\det(\overline{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$	$\det(\boldsymbol{B})$ and $\det(\overline{\boldsymbol{B}}) = \overline{\det(\boldsymbol{B})}$ for any square matrix $\boldsymbol{B}$. Use this to show that
lyche-numerical-linear-algebra-F02118	161	0.991	$\pi_{\boldsymbol{A}^T} = \pi_{\boldsymbol{A}}$	$\pi_{\boldsymbol{A}^T} = \pi_{\boldsymbol{A}}$	a) $\pi_{\boldsymbol{A}^T} = \pi_{\boldsymbol{A}}$,
lyche-numerical-linear-algebra-F02119	161	1.000	$\pi_{\boldsymbol{A}^*}(\bar{\lambda}) = \overline{\pi_{\boldsymbol{A}}(\lambda)}$	$\pi_{\boldsymbol{A}^*}(\bar{\lambda}) = \overline{\pi_{\boldsymbol{A}}(\lambda)}$	b) $\pi_{\boldsymbol{A}^*}(\bar{\lambda}) = \overline{\pi_{\boldsymbol{A}}(\lambda)}$.
lyche-numerical-linear-algebra-F02120	161	1.000	$(\lambda^{-1}, \boldsymbol{x})$	$(\lambda^{-1}, \boldsymbol{x})$	a) If $\boldsymbol{A}$ is nonsingular then $(\lambda^{-1}, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^{-1}$.
lyche-numerical-linear-algebra-F02121	161	0.562	$(\lambda^k, \boldsymbol{x})$	$(\lambda^k, \boldsymbol{x})$	b) $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.

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lyche-numerical-linear-algebra-161-F02122	161	0.562	$\boldsymbol{A}^k$	$\boldsymbol{A}^k$	b) $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra-161-F02123	161	0.562	$k \in \mathbb{Z}$	$k \in \mathbb{Z}$	b) $(\lambda^k, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A}^k$ for $k \in \mathbb{Z}$.
lyche-numerical-linear-algebra-162-F02124	162	1.000	$(\lambda_j, \boldsymbol{x}_j), j=1, \dots, n$	$(\lambda_j, \boldsymbol{x}_j), j=1, \dots, n$	is nondefective with eigenpairs $(\lambda_j, \boldsymbol{x}_j), j=1, \dots, n$ then for any $\boldsymbol{x} \in \mathbb{C}^n$ and
lyche-numerical-linear-algebra-162-F02125	162	1.000	$k \in \mathbb{N}$	$k \in \mathbb{N}$	$k \in \mathbb{N}$
lyche-numerical-linear-algebra-162-F02126	162	1.000	$\boldsymbol{A}^2 =$	$\boldsymbol{A}^2 =$	Exercise 6.5 (Eigenvalues of an Idempotent Matrix) Let $\lambda \in \sigma(\boldsymbol{A})$ where $\boldsymbol{A}^2 =$
lyche-numerical-linear-algebra-162-F02127	162	1.000	$\lambda=0$	$\lambda=0$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$. Show that $\lambda=0$ or $\lambda=1$. (A matrix is called idempotent if $\boldsymbol{A}^2 = \boldsymbol{A}$).
lyche-numerical-linear-algebra-162-F02128	162	1.000	$\boldsymbol{A}^2 = \boldsymbol{A}$	$\boldsymbol{A}^2 = \boldsymbol{A}$	$\boldsymbol{A} \in \mathbb{C}^{n \times n}$. Show that $\lambda=0$ or $\lambda=1$. (A matrix is called idempotent if $\boldsymbol{A}^2 = \boldsymbol{A}$).
lyche-numerical-linear-algebra-162-F02129	162	1.000	$\boldsymbol{A}^k = 0$	$\boldsymbol{A}^k = 0$	Exercise 6.6 (Eigenvalues of a Nilpotent Matrix) Let $\lambda \in \sigma(\boldsymbol{A})$ where $\boldsymbol{A}^k = 0$
lyche-numerical-linear-algebra-162-F02130	162	1.000	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{I}$	$\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{I}$	Exercise 6.7 (Eigenvalues of a Unitary Matrix) Let $\lambda \in \sigma(\boldsymbol{A})$, where $\boldsymbol{A}^* \boldsymbol{A} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-162-F02131	162	1.000	$ \lambda =1$	$ \lambda =1$	Show that $ \lambda =1$.
lyche-numerical-linear-algebra-162-F02132	162	1.000	$\epsilon_0 > 0$	$\epsilon_0 > 0$	$\mathbb{C}^{n \times n}$ is singular. Then we can find $\epsilon_0 > 0$ such that $\boldsymbol{A} + \epsilon \boldsymbol{I}$ is nonsingular for all
lyche-numerical-linear-algebra-162-F02133	162	1.000	$\boldsymbol{A} + \epsilon \boldsymbol{I}$	$\boldsymbol{A} + \epsilon \boldsymbol{I}$	$\mathbb{C}^{n \times n}$ is singular. Then we can find $\epsilon_0 > 0$ such that $\boldsymbol{A} + \epsilon \boldsymbol{I}$ is nonsingular for all
lyche-numerical-linear-algebra-162-F02134	162	0.997	$\epsilon \in \mathbb{C}$	$\epsilon \in \mathbb{C}$	$\epsilon \in \mathbb{C}$ with $ \epsilon < \epsilon_0$. Hint: $\det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n$, where λ_i are the eigenvalues of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-162-F02135	162	0.997	$ \epsilon < \epsilon_0$	$ \epsilon < \epsilon_0$	$\epsilon \in \mathbb{C}$ with $ \epsilon < \epsilon_0$. Hint: $\det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n$, where λ_i are the eigenvalues of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-162-F02136	162	0.997	$\det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n$	$\det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n$	$\epsilon \in \mathbb{C}$ with $ \epsilon < \epsilon_0$. Hint: $\det(\boldsymbol{A}) = \lambda_1 \lambda_2 \cdots \lambda_n$, where λ_i are the eigenvalues of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-162-F02137	162	1.000	$q_0, \dots, q_{n-1} \in \mathbb{C}$	$q_0, \dots, q_{n-1} \in \mathbb{C}$	Exercise 6.9 (Companion Matrix) For $q_0, \dots, q_{n-1} \in \mathbb{C}$ let $p(\lambda) = \lambda^n +$
lyche-numerical-linear-algebra-162-F02138	162	1.000	$p(\lambda) = \lambda^n +$	$p(\lambda) = \lambda^n +$	Exercise 6.9 (Companion Matrix) For $q_0, \dots, q_{n-1} \in \mathbb{C}$ let $p(\lambda) = \lambda^n +$
lyche-numerical-linear-algebra-162-F02139	162	1.000	$q_{n-1} \lambda^{n-1} + \cdots + q_0$	$q_{n-1} \lambda^{n-1} + \cdots + q_0$	$q_{n-1} \lambda^{n-1} + \cdots + q_0$ be a polynomial of degree n in λ . We derive two matrices
lyche-numerical-linear-algebra-162-F02140	162	1.000	$(-1)^n p$	$(-1)^n p$	that have $(-1)^n p$ as its characteristic polynomial.
lyche-numerical-linear-algebra-162-F02141	162	0.998	$p = (-1)^n \pi_{\boldsymbol{A}}$	$p = (-1)^n \pi_{\boldsymbol{A}}$	a) Show that $p = (-1)^n \pi_{\boldsymbol{A}}$ where
lyche-numerical-linear-algebra-162-F02142	162	1.000	$p = (-1)^n \pi_{\boldsymbol{B}}$	$p = (-1)^n \pi_{\boldsymbol{B}}$	b) Show that $p = (-1)^n \pi_{\boldsymbol{B}}$ where
lyche-numerical-linear-algebra-163-F02143	163	0.998	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$. Is $\boldsymbol{A}$ defective?
lyche-numerical-linear-algebra-163-F02144	163	1.000	$\boldsymbol{Y} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{Y} \in \mathbb{R}^{2 \times 2}$	where $a \in \mathbb{R}$. There is a nonsingular matrix $\boldsymbol{Y} \in \mathbb{R}^{2 \times 2}$ and a diagonal matrix
lyche-numerical-linear-algebra-163-F02145	163	1.000	$\boldsymbol{D} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{D} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{D} \in \mathbb{R}^{2 \times 2}$ such that $\boldsymbol{A} = \boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}$?
lyche-numerical-linear-algebra-163-F02146	163	1.000	$\boldsymbol{A} = \boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}$	$\boldsymbol{A} = \boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}$	$\boldsymbol{D} \in \mathbb{R}^{2 \times 2}$ such that $\boldsymbol{A} = \boldsymbol{Y} \boldsymbol{D} \boldsymbol{Y}^{-1}$?
lyche-numerical-linear-algebra-163-F02147	163	1.000	$\boldsymbol{A} \in \mathbb{R}^{n, n}$	$\boldsymbol{A} \in \mathbb{R}^{n, n}$	$\boldsymbol{A} \in \mathbb{R}^{n, n}$ be tridiagonal (i.e. $a_{ij} = 0$ when $ i - j > 1$) and suppose also that

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lyche-numerical-linear-algebra-163-F02148	163	1.000	$a_{i+1, i} a_{i, i+1} > 0$	$a_{i+1, i} a_{i, i+1} > 0$	$a_{i+1, i} a_{i, i+1} > 0$ for $i = 1, \dots, n-1$. Show that the eigenvalues of A are real. ²
lyche-numerical-linear-algebra-163-F02149	163	1.000	$\boldsymbol{A} = \begin{bmatrix} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{bmatrix}$	For the Jordan factorization of the matrix $\boldsymbol{A} = \begin{bmatrix} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{bmatrix}$ we have $\boldsymbol{J} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-163-F02150	163	1.000	$\boldsymbol{J} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\boldsymbol{J} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	For the Jordan factorization of the matrix $\boldsymbol{A} = \begin{bmatrix} 3 & 0 & 1 \\ -4 & 1 & -2 \\ -4 & 0 & -1 \end{bmatrix}$ we have $\boldsymbol{J} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-163-F02151	163	1.000	$\left(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I} \right)^r = \begin{bmatrix} 0 & \boldsymbol{I}_{m-r} \\ 0 & 0 \end{bmatrix}$	$(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^r = \begin{bmatrix} 0 & \boldsymbol{I}_{m-r} \\ 0 & 0 \end{bmatrix}$	Exercise 6.14 (A Nilpotent Matrix) Show that $(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^r = \begin{bmatrix} 0 & \boldsymbol{I}_{m-r} \\ 0 & 0 \end{bmatrix}$ for
lyche-numerical-linear-algebra-163-F02152	163	0.998	$1 \leq r \leq m-1$	$1 \leq r \leq m-1$	$1 \leq r \leq m-1$ and conclude that $(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^m = 0$.
lyche-numerical-linear-algebra-163-F02153	163	0.998	$\left(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I} \right)^m = 0$	$(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^m = 0$	$1 \leq r \leq m-1$ and conclude that $(\boldsymbol{J}_m(\lambda) - \lambda \boldsymbol{I})^m = 0$.
lyche-numerical-linear-algebra-163-F02154	163	1.000	$r=0, 1, 2, \dots, m=2, 3, \dots$	$r = 0, 1, 2, \dots, m = 2, 3, \dots$	Then for $r = 0, 1, 2, \dots, m = 2, 3, \dots$, and any $\lambda \in \mathbb{C}$
lyche-numerical-linear-algebra-163-F02155	163	0.998	$\boldsymbol{A}^r = \boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$	$\boldsymbol{A}^r = \boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$	a) $\boldsymbol{A}^r = \boldsymbol{S} \boldsymbol{J}^r \boldsymbol{S}^{-1}$,
lyche-numerical-linear-algebra-163-F02156	163	1.000	$\boldsymbol{J}^r = \text{diag}(\boldsymbol{U}_1^r, \dots, \boldsymbol{U}_k^r)$	$\boldsymbol{J}^r = \text{diag}(\boldsymbol{U}_1^r, \dots, \boldsymbol{U}_k^r)$	b) $\boldsymbol{J}^r = \text{diag}(\boldsymbol{U}_1^r, \dots, \boldsymbol{U}_k^r)$,
lyche-numerical-linear-algebra-163-F02157	163	1.000	$\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$	$\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$	² Hint: show that there is a diagonal matrix $\boldsymbol{D}$ such that $\boldsymbol{D}^{-1} \boldsymbol{A} \boldsymbol{D}$ is symmetric.
lyche-numerical-linear-algebra-164-F02158	164	0.985	$\boldsymbol{U}_i^r = \text{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i)^r, \dots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)^r)$	$\boldsymbol{U}_i^r = \text{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i)^r, \dots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)^r)$	c) $\boldsymbol{U}_i^r = \text{diag}(\boldsymbol{J}_{m_{i,1}}(\lambda_i)^r, \dots, \boldsymbol{J}_{m_{i,g_i}}(\lambda_i)^r)$,
lyche-numerical-linear-algebra-164-F02159	164	0.968	$\boldsymbol{J}_m(\lambda)^r = (\boldsymbol{E}_m + \lambda \boldsymbol{I}_m)^r = \sum_{k=0}^{\min\{r, m-1\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_m^k$	$\boldsymbol{J}_m(\lambda)^r = (\boldsymbol{E}_m + \lambda \boldsymbol{I}_m)^r = \sum_{k=0}^{\min\{r, m-1\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_m^k$	d) $\boldsymbol{J}_m(\lambda)^r = (\boldsymbol{E}_m + \lambda \boldsymbol{I}_m)^r = \sum_{k=0}^{\min\{r, m-1\}} \binom{r}{k} \lambda^{r-k} \boldsymbol{E}_m^k$.
lyche-numerical-linear-algebra-164-F02160	164	1.000	$\boldsymbol{J}^{100}$	$\boldsymbol{J}^{100}$	Exercise 6.16 (Powers of a Jordan Block) Find $\boldsymbol{J}^{100}$ and $\boldsymbol{A}^{100}$ for the matrix in
lyche-numerical-linear-algebra-164-F02161	164	1.000	$\boldsymbol{A}^{100}$	$\boldsymbol{A}^{100}$	Exercise 6.16 (Powers of a Jordan Block) Find $\boldsymbol{J}^{100}$ and $\boldsymbol{A}^{100}$ for the matrix in
lyche-numerical-linear-algebra-164-F02162	164	1.000	$\mu_{\boldsymbol{A}}(\boldsymbol{A})$	$\mu_{\boldsymbol{A}}(\boldsymbol{A})$	is called the minimal polynomial of $\boldsymbol{A}$. We define the matrix polynomial $\mu_{\boldsymbol{A}}(\boldsymbol{A})$
lyche-numerical-linear-algebra-164-F02163	164	1.000	$\lambda_i - \lambda$	$\lambda_i - \lambda$	by replacing the factors $\lambda_i - \lambda$ by $\lambda_i \boldsymbol{I} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-164-F02164	164	1.000	$\lambda_i \boldsymbol{I} - \boldsymbol{A}$	$\lambda_i \boldsymbol{I} - \boldsymbol{A}$	by replacing the factors $\lambda_i - \lambda$ by $\lambda_i \boldsymbol{I} - \boldsymbol{A}$.
lyche-numerical-linear-algebra-164-F02165	164	1.000	$\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^k \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$	$\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^k \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$	a) We have $\pi_{\boldsymbol{A}}(\lambda) = \prod_{i=1}^k \prod_{j=1}^{g_i} (\lambda_i - \lambda)^{m_{i,j}}$. Use this to show that the minimal
lyche-numerical-linear-algebra-164-F02166	164	0.629	$\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$	$\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$	polynomial divides the characteristic polynomial, i.e., $\pi_{\boldsymbol{A}} = \mu_{\boldsymbol{A}} \nu_{\boldsymbol{A}}$ for some
lyche-numerical-linear-algebra-164-F02167	164	0.953	$\nu_{\boldsymbol{A}}$	$\nu_{\boldsymbol{A}}$	polynomial $\nu_{\boldsymbol{A}}$.
lyche-numerical-linear-algebra-164-F02168	164	0.986	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = 0 \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = 0$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = 0 \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = 0$	b) Show that $\mu_{\boldsymbol{A}}(\boldsymbol{A}) = 0 \iff \mu_{\boldsymbol{A}}(\boldsymbol{J}) = 0$

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lyche-numerical-linear-algebra- F02169	164	■ 1.000	$m_{\{i\}}$	m_i	c) (can be difficult) Use Exercises 6.14, 6.15 and the maximality of m_i to show that
lyche-numerical-linear-algebra- F02170	164	■ 1.000	$\mu_{\{\boldsymbol{A}\}}(\boldsymbol{A})=0$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = 0$	$\mu_{\boldsymbol{A}}(\boldsymbol{A}) = 0$. Thus a matrix satisfies its minimal equation. Finally show that the
lyche-numerical-linear-algebra- F02171	164	■ 1.000	$p(\boldsymbol{A})=\mathbf{0}$	$p(\boldsymbol{A}) = \mathbf{0}$	degree of any polynomial p such that $p(\boldsymbol{A}) = \mathbf{0}$ is at least as large as the degree
lyche-numerical-linear-algebra- F02172	164	■ 1.000	$\pi_{\{\boldsymbol{A}\}}(\boldsymbol{A})=\mathbf{0}$	$\pi_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$	its characteristic equation $\pi_{\boldsymbol{A}}(\boldsymbol{A}) = \mathbf{0}$.
lyche-numerical-linear-algebra- F02173	164	■ 1.000	$p(t):=\sum_{j=0}^r b_{\{j\}} t^{\{j\}}$	$p(t) := \sum_{j=0}^r b_j t^j$	is a polynomial given by $p(t) := \sum_{j=0}^r b_j t^j$, where $b_j \in \mathbb{C}$ and $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. We
lyche-numerical-linear-algebra- F02174	164	■ 1.000	$b_{\{j\}} \in \mathbb{C}$	$b_j \in \mathbb{C}$	is a polynomial given by $p(t) := \sum_{j=0}^r b_j t^j$, where $b_j \in \mathbb{C}$ and $\boldsymbol{A} \in \mathbb{C}^{n \times n}$. We
lyche-numerical-linear-algebra- F02175	164	■ 1.000	$p(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	$p(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$	define the matrix $p(\boldsymbol{A}) \in \mathbb{C}^{n \times n}$ by
lyche-numerical-linear-algebra- F02176	164	■ 1.000	$\boldsymbol{A}^{\{0\}}:=\boldsymbol{I}$	$\boldsymbol{A}^0 := \boldsymbol{I}$	where $\boldsymbol{A}^0 := \boldsymbol{I}$. From this it follows that if $p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$ for some
lyche-numerical-linear-algebra- F02177	164	■ 1.000	$p(t):=\left(t-\alpha_{\{1\}}\right) \cdots\left(t-\alpha_{\{r\}}\right)$	$p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$	where $\boldsymbol{A}^0 := \boldsymbol{I}$. From this it follows that if $p(t) := (t - \alpha_1) \cdots (t - \alpha_r)$ for some
lyche-numerical-linear-algebra- F02178	164	■ 1.000	$\alpha_{\{0\}}, \ldots, \alpha_{\{r\}} \in \mathbb{C}$	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$ then $p(\boldsymbol{A}) = (\boldsymbol{A} - \alpha_1) \cdots (\boldsymbol{A} - \alpha_r)$. We accept this without proof.
lyche-numerical-linear-algebra- F02179	164	■ 1.000	$p(\boldsymbol{A})=\left(\boldsymbol{A}-\alpha_{\{1\}}\right) \cdots\left(\boldsymbol{A}-\alpha_{\{r\}}\right)$	$p(\boldsymbol{A}) = (\boldsymbol{A} - \alpha_1) \cdots (\boldsymbol{A} - \alpha_r)$	$\alpha_0, \dots, \alpha_r \in \mathbb{C}$ then $p(\boldsymbol{A}) = (\boldsymbol{A} - \alpha_1) \cdots (\boldsymbol{A} - \alpha_r)$. We accept this without proof.
lyche-numerical-linear-algebra- F02180	164	■ 1.000	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U}=\boldsymbol{T}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{T}$	Let $\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{T}$, where $\boldsymbol{U}$ is unitary and $\boldsymbol{T}$ upper triangular with the eigenvalues
lyche-numerical-linear-algebra- F02181	164	■ 0.997	$\left[\begin{array}{rr} 2 & 1 \\ -1 & 4 \end{array}\right]$	$\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$	a) Find the characteristic polynomial $\pi_{\boldsymbol{A}}$ to $\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Show that $\pi(\boldsymbol{A}) = \mathbf{0}$.
lyche-numerical-linear-algebra- F02182	164	■ 0.997	$\pi_{\{\boldsymbol{A}\}}(\boldsymbol{A})=\mathbf{0}$	$\pi(\boldsymbol{A}) = \mathbf{0}$	a) Find the characteristic polynomial $\pi_{\boldsymbol{A}}$ to $\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$. Show that $\pi(\boldsymbol{A}) = \mathbf{0}$.
lyche-numerical-linear-algebra- F02183	164	■ 1.000	$p(\boldsymbol{A})=$	$p(\boldsymbol{A}) =$	b) Let now $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ be arbitrary. For any polynomial p show that $p(\boldsymbol{A}) =$
lyche-numerical-linear-algebra- F02184	164	■ 0.877	$\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^*$	$\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^*$	$\boldsymbol{U} p(\boldsymbol{T}) \boldsymbol{U}^*$.
lyche-numerical-linear-algebra- F02185	164	■ 1.000	$n, k \in \mathbb{N}$	$n, k \in \mathbb{N}$	c) Let $n, k \in \mathbb{N}$ with $1 \leq k < n$. Let $\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}$ be upper triangular. Moreover,
lyche-numerical-linear-algebra- F02186	164	■ 1.000	$\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}$	$\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}$	c) Let $n, k \in \mathbb{N}$ with $1 \leq k < n$. Let $\boldsymbol{C}, \boldsymbol{D} \in \mathbb{C}^{n \times n}$ be upper triangular. Moreover,
lyche-numerical-linear-algebra- F02187	164	■ 1.000	$c_{\{i, j\}}=0$	$c_{i, j} = 0$	$c_{i, j} = 0$ for $i, j \leq k$ and $d_{k+1, k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C} \boldsymbol{D}$ and show that $e_{i, j} = 0$
lyche-numerical-linear-algebra- F02188	164	■ 1.000	$i, j \leq k$	$i, j \leq k$	$c_{i, j} = 0$ for $i, j \leq k$ and $d_{k+1, k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C} \boldsymbol{D}$ and show that $e_{i, j} = 0$
lyche-numerical-linear-algebra- F02189	164	■ 1.000	$d_{\{k+1, k+1\}}=0$	$d_{k+1, k+1} = 0$	$c_{i, j} = 0$ for $i, j \leq k$ and $d_{k+1, k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C} \boldsymbol{D}$ and show that $e_{i, j} = 0$
lyche-numerical-linear-algebra- F02190	164	■ 1.000	$\boldsymbol{E}:=\boldsymbol{C} \boldsymbol{D}$	$\boldsymbol{E} := \boldsymbol{C} \boldsymbol{D}$	$c_{i, j} = 0$ for $i, j \leq k$ and $d_{k+1, k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C} \boldsymbol{D}$ and show that $e_{i, j} = 0$
lyche-numerical-linear-algebra- F02191	164	■ 1.000	$e_{\{i, j\}}=0$	$e_{i, j} = 0$	$c_{i, j} = 0$ for $i, j \leq k$ and $d_{k+1, k+1} = 0$. Define $\boldsymbol{E} := \boldsymbol{C} \boldsymbol{D}$ and show that $e_{i, j} = 0$
lyche-numerical-linear-algebra- F02192	164	■ 1.000	$i, j \leq k+1$	$i, j \leq k+1$	for $i, j \leq k+1$.
lyche-numerical-linear-algebra- F02193	165	■ 0.773	$p:=\pi_{\{\boldsymbol{A}\}}$	$p := \pi_{\boldsymbol{A}}$	d) Now let $p := \pi_{\boldsymbol{A}}$ be the characteristic polynomial of $\boldsymbol{A}$. Show that $p(\boldsymbol{T}) = \mathbf{0}$. ³
lyche-numerical-linear-algebra- F02194	165	■ 0.773	$p(\boldsymbol{T})=\mathbf{0}$	$p(\boldsymbol{T}) = \mathbf{0}$. ³	d) Now let $p := \pi_{\boldsymbol{A}}$ be the characteristic polynomial of $\boldsymbol{A}$. Show that $p(\boldsymbol{T}) = \mathbf{0}$. ³

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lyche-numerical-linear-algebra-165 F02195	165	1.000	$\boldsymbol{A}=\left[\begin{array}{ll}1 & 2 \\ 3 & 2\end{array}\right]$	$\boldsymbol{A}=\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$	$\boldsymbol{A}=\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ is $\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}=\begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$, where $\boldsymbol{U}=\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-165 F02196	165	1.000	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}=\left[\begin{array}{cc}-1 & -1 \\ 0 & 4\end{array}\right]$	$\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}=\begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$	$\boldsymbol{A}=\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ is $\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}=\begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$, where $\boldsymbol{U}=\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-165 F02197	165	1.000	$\boldsymbol{U}=\frac{1}{\sqrt{2}}\left[\begin{array}{cc}1 & 1 \\ -1 & 1\end{array}\right]$	$\boldsymbol{U}=\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$	$\boldsymbol{A}=\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ is $\boldsymbol{U}^T \boldsymbol{A} \boldsymbol{U}=\begin{bmatrix} -1 & -1 \\ 0 & 4 \end{bmatrix}$, where $\boldsymbol{U}=\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-165 F02198	165	0.998	$\boldsymbol{C}=\boldsymbol{A}+i \boldsymbol{B}$	$\boldsymbol{C}=\boldsymbol{A}+i \boldsymbol{B}$	Exercise 6.20 (Skew-Hermitian Matrix) Suppose $\boldsymbol{C}=\boldsymbol{A}+i \boldsymbol{B}$, where $\boldsymbol{A}, \boldsymbol{B} \in$
lyche-numerical-linear-algebra-165 F02199	165	1.000	$\boldsymbol{A}^T=-\boldsymbol{A}$	$\boldsymbol{A}^T=-\boldsymbol{A}$	$\mathbb{R}^{n \times n}$. Show that $\boldsymbol{C}$ is skew-Hermitian if and only if $\boldsymbol{A}^T=-\boldsymbol{A}$ and $\boldsymbol{B}^T=\boldsymbol{B}$.
lyche-numerical-linear-algebra-165 F02200	165	1.000	$\boldsymbol{B}^T=\boldsymbol{B}$	$\boldsymbol{B}^T=\boldsymbol{B}$	$\mathbb{R}^{n \times n}$. Show that $\boldsymbol{C}$ is skew-Hermitian if and only if $\boldsymbol{A}^T=-\boldsymbol{A}$ and $\boldsymbol{B}^T=\boldsymbol{B}$.
lyche-numerical-linear-algebra-165 F02201	165	1.000	$\left(\lambda_1, \boldsymbol{u}_1\right), \ldots, \left(\lambda_n, \boldsymbol{u}_n\right)$	$\left(\lambda_1, \boldsymbol{u}_1\right), \ldots, \left(\lambda_n, \boldsymbol{u}_n\right)$	that if the eigenpairs $\lambda_1, \boldsymbol{u}_1), \ldots, (\lambda_n, \boldsymbol{u}_n$ of $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ are orthogonal, i.e.,
lyche-numerical-linear-algebra-165 F02202	165	0.999	$\boldsymbol{u}_j^* \boldsymbol{u}_k=0$	$\boldsymbol{u}_j^* \boldsymbol{u}_k=0$	$\boldsymbol{u}_j^* \boldsymbol{u}_k=0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-algebra-165 F02203	165	0.999	$j \neq k$	$j \neq k$	$\boldsymbol{u}_j^* \boldsymbol{u}_k=0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-algebra-165 F02204	165	0.999	$\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^n$	$\boldsymbol{u}_j^* \boldsymbol{u}_k=0$ for $j \neq k$ then the eigenvector expansions of $\boldsymbol{x}$ and $\boldsymbol{A} \boldsymbol{x} \in \mathbb{C}^n$ take
lyche-numerical-linear-algebra-165 F02205	165	0.996	$\lambda_{\min }$	$\lambda_{\min }$	where $\lambda_{\min }$ is the smallest eigenvalue of $\boldsymbol{A}$, and $R(\boldsymbol{x})$ is the Rayleigh quotient
lyche-numerical-linear-algebra-165 F02206	165	0.996	$R(\boldsymbol{x})$	$R(\boldsymbol{x})$	where $\lambda_{\min }$ is the smallest eigenvalue of $\boldsymbol{A}$, and $R(\boldsymbol{x})$ is the Rayleigh quotient
lyche-numerical-linear-algebra-165 F02207	165	0.997	$\boldsymbol{y} \neq \boldsymbol{0}$	$\boldsymbol{y} \neq \boldsymbol{0}$	b) Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ such that $\ \boldsymbol{x}\ _2=1$ and $\boldsymbol{y} \neq \boldsymbol{0}$. Show that
lyche-numerical-linear-algebra-165 F02208	166	1.000	$t>0$	$t>0$	where $t>0$ is small. ⁴
lyche-numerical-linear-algebra-165 F02209	166	1.000	$\boldsymbol{x}^0 \in B_1$	$\boldsymbol{x}^0 \in B_1$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $\boldsymbol{A} \boldsymbol{x}^0-R\left(\boldsymbol{x}^0\right) \boldsymbol{x}^0 \neq \boldsymbol{0}$, i.e., $\left(R\left(\boldsymbol{x}^0\right), \boldsymbol{x}^0\right)$ is not an
lyche-numerical-linear-algebra-165 F02210	166	1.000	$\boldsymbol{A} \boldsymbol{x}^0-R\left(\boldsymbol{x}^0\right) \boldsymbol{x}^0 \neq \boldsymbol{0}$	$\boldsymbol{A} \boldsymbol{x}^0-R\left(\boldsymbol{x}^0\right) \boldsymbol{x}^0 \neq \boldsymbol{0}$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $\boldsymbol{A} \boldsymbol{x}^0-R\left(\boldsymbol{x}^0\right) \boldsymbol{x}^0 \neq \boldsymbol{0}$, i.e., $\left(R\left(\boldsymbol{x}^0\right), \boldsymbol{x}^0\right)$ is not an
lyche-numerical-linear-algebra-165 F02211	166	1.000	$\left(R\left(\boldsymbol{x}^0\right), \boldsymbol{x}^0\right)$	$\left(R\left(\boldsymbol{x}^0\right), \boldsymbol{x}^0\right)$	Assume that $\boldsymbol{x}^0 \in B_1$ satisfies $\boldsymbol{A} \boldsymbol{x}^0-R\left(\boldsymbol{x}^0\right) \boldsymbol{x}^0 \neq \boldsymbol{0}$, i.e., $\left(R\left(\boldsymbol{x}^0\right), \boldsymbol{x}^0\right)$ is not an
lyche-numerical-linear-algebra-165 F02212	166	1.000	$\boldsymbol{x}^1 \in B_1$	$\boldsymbol{x}^1 \in B_1$	eigenpair for $\boldsymbol{A}$. Explain how we can find a vector $\boldsymbol{x}^1 \in B_1$ such that $R\left(\boldsymbol{x}^1\right)<$
lyche-numerical-linear-algebra-165 F02213	166	1.000	$R\left(\boldsymbol{x}^1\right)<$	$R\left(\boldsymbol{x}^1\right)<$	eigenpair for $\boldsymbol{A}$. Explain how we can find a vector $\boldsymbol{x}^1 \in B_1$ such that $R\left(\boldsymbol{x}^1\right)<$
lyche-numerical-linear-algebra-165 F02214	166	0.993	$R\left(\boldsymbol{x}^0\right)$	$R\left(\boldsymbol{x}^0\right)$	$R\left(\boldsymbol{x}^0\right)$.
lyche-numerical-linear-algebra-165 F02215	166	1.000	$\beta_i \geq \alpha_i$	$\beta_i \geq \alpha_i$	Theorem 6.13, if $\boldsymbol{E}$ is symmetric positive semidefinite then $\beta_i \geq \alpha_i$.
lyche-numerical-linear-algebra-165 F02216	166	1.000	$\boldsymbol{A}:=\left[\begin{array}{ll}0 & 0 \\ 0 & 4\end{array}\right]$	$\boldsymbol{A}:=\begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$	ces $\boldsymbol{A}:=\begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$ and $\boldsymbol{B}:=\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$. Why does this not contradict the Hoffman-
lyche-numerical-linear-algebra-165 F02217	166	1.000	$\boldsymbol{B}:=\left[\begin{array}{cc}-1 & -1 \\ 1 & 1\end{array}\right]$	$\boldsymbol{B}:=\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$	ces $\boldsymbol{A}:=\begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$ and $\boldsymbol{B}:=\begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$. Why does this not contradict the Hoffman-
lyche-numerical-linear-algebra-165 F02218	166	1.000	$\boldsymbol{A}:=\left[\begin{array}{ll}3 & 1 \\ 2 & 2\end{array}\right]$	$\boldsymbol{A}:=\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$	the matrix $\boldsymbol{A}:=\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$ and the two expansions in (6.16) for any $\boldsymbol{v} \in \mathbb{R}^2$.
lyche-numerical-linear-algebra-165 F02219	166	1.000	$\boldsymbol{v} \in \mathbb{R}^2$	$\boldsymbol{v} \in \mathbb{R}^2$	the matrix $\boldsymbol{A}:=\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$ and the two expansions in (6.16) for any $\boldsymbol{v} \in \mathbb{R}^2$.
lyche-numerical-linear-algebra-165 F02220	166	1.000	$\boldsymbol{y}, \boldsymbol{x} \in$	$\boldsymbol{y}, \boldsymbol{x} \in$	Exercise 6.27 (Generalized Rayleigh Quotient) For $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ and any $\boldsymbol{y}, \boldsymbol{x} \in$

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lyche-numerical-linear-algebra-F02221	166	1.000	$R(\mathbf{y}, \mathbf{x}) = R_A(\mathbf{y}, \mathbf{x}) := \frac{\mathbf{y}^* \mathbf{A} \mathbf{x}}{\mathbf{y}^* \mathbf{x}}$	$R(\mathbf{y}, \mathbf{x}) = R_A(\mathbf{y}, \mathbf{x}) := \frac{\mathbf{y}^* \mathbf{A} \mathbf{x}}{\mathbf{y}^* \mathbf{x}}$	$\mathbb{C}^n$ with $\mathbf{y}^* \mathbf{x} \neq 0$ the quantity $R(\mathbf{y}, \mathbf{x}) = R_A(\mathbf{y}, \mathbf{x}) := \frac{\mathbf{y}^* \mathbf{A} \mathbf{x}}{\mathbf{y}^* \mathbf{x}}$ is called a generalized
lyche-numerical-linear-algebra-F02222	166	1.000	$R(\mathbf{y}, \mathbf{x}) = \lambda$	$R(\mathbf{y}, \mathbf{x}) = \lambda$	$R(\mathbf{y}, \mathbf{x}) = \lambda$ for any $\mathbf{y}$ with $\mathbf{y}^* \mathbf{x} \neq 0$. Also show that if $(\lambda, \mathbf{y})$ is a left eigenpair for
lyche-numerical-linear-algebra-F02223	166	1.000	$\mathbf{A}, \mathbf{A}^T$	$\mathbf{A}, \mathbf{A}^T$	6.7.1 Does $\mathbf{A}, \mathbf{A}^T$ and $\mathbf{A}^*$ have the same eigenvalues? What about $\mathbf{A}^* \mathbf{A}$ and
lyche-numerical-linear-algebra-F02224	166	1.000	$\mathbf{A} \mathbf{A}^*$	$\mathbf{A} \mathbf{A}^*$	$\mathbf{A} \mathbf{A}^*$?
lyche-numerical-linear-algebra-F02225	166	0.993	$f(t) = R(\mathbf{x} - t \mathbf{y})$	$f(t) = R(\mathbf{x} - t \mathbf{y})$	⁴ Hint: Use Taylor's theorem for the function $f(t) = R(\mathbf{x} - t \mathbf{y})$.
lyche-numerical-linear-algebra-F02226	168	1.000	$\mathbf{U} := [\mathbf{u}_1, \dots, \mathbf{u}_n] \in \mathbb{C}^{n \times n}$	$\mathbf{U} := [\mathbf{u}_1, \dots, \mathbf{u}_n] \in \mathbb{C}^{n \times n}$	eigenpairs $(\lambda_1, \mathbf{u}_1), \dots, (\lambda_n, \mathbf{u}_n)$. Letting $\mathbf{U} := [\mathbf{u}_1, \dots, \mathbf{u}_n] \in \mathbb{C}^{n \times n}$ and $\mathbf{D} :=$
lyche-numerical-linear-algebra-F02227	168	1.000	$\mathbf{D} :=$	$\mathbf{D} :=$	eigenpairs $(\lambda_1, \mathbf{u}_1), \dots, (\lambda_n, \mathbf{u}_n)$. Letting $\mathbf{U} := [\mathbf{u}_1, \dots, \mathbf{u}_n] \in \mathbb{C}^{n \times n}$ and $\mathbf{D} :=$
lyche-numerical-linear-algebra-F02228	168	1.000	$\text{diag}(\lambda_1, \dots, \lambda_n)$	$\text{diag}(\lambda_1, \dots, \lambda_n)$	$\text{diag}(\lambda_1, \dots, \lambda_n)$ we have the spectral decomposition
lyche-numerical-linear-algebra-F02229	168	1.000	$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$	$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$	$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$, where $\mathbf{U}$ and $\mathbf{V}$ are unitary, and $\mathbf{\Sigma}$ is a nonnegative diagonal matrix,
lyche-numerical-linear-algebra-F02230	168	1.000	$\mathbf{\Sigma}$	$\mathbf{\Sigma}$	$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$, where $\mathbf{U}$ and $\mathbf{V}$ are unitary, and $\mathbf{\Sigma}$ is a nonnegative diagonal matrix,
lyche-numerical-linear-algebra-F02231	168	1.000	$\Sigma_{ij} = 0$	$\Sigma_{ij} = 0$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i . The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra-F02232	168	1.000	$\Sigma_{ii} \geq 0$	$\Sigma_{ii} \geq 0$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i . The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra-F02233	168	1.000	$\sigma_i := \Sigma_{ii}$	$\sigma_i := \Sigma_{ii}$	i.e., $\Sigma_{ij} = 0$ for all $i \neq j$ and $\Sigma_{ii} \geq 0$ for all i . The diagonal elements $\sigma_i := \Sigma_{ii}$ are
lyche-numerical-linear-algebra-F02234	168	1.000	$\sigma_i \geq \sigma_{i+1}$	$\sigma_i \geq \sigma_{i+1}$	$\sigma_i \geq \sigma_{i+1}$ for all i .
lyche-numerical-linear-algebra-F02235	169	1.000	$\sigma_1 = 2$	$\sigma_1 = 2$	mal, and $\mathbf{\Sigma}$ is a nonnegative diagonal matrix with singular values $\sigma_1 = 2$ and
lyche-numerical-linear-algebra-F02236	169	1.000	$\sigma_2 = 1$	$\sigma_2 = 1$	$\sigma_2 = 1$.
lyche-numerical-linear-algebra-F02237	169	1.000	$\mathcal{R}(\mathbf{A}), \mathcal{N}(\mathbf{A})$	$\mathcal{R}(\mathbf{A}), \mathcal{N}(\mathbf{A})$	To start we show that bases for the four fundamental subspaces $\mathcal{R}(\mathbf{A}), \mathcal{N}(\mathbf{A})$,
lyche-numerical-linear-algebra-F02238	169	1.000	$\mathcal{R}(\mathbf{A}^*)$	$\mathcal{R}(\mathbf{A}^*)$	$\mathcal{R}(\mathbf{A}^*)$ and $\mathcal{N}(\mathbf{A}^*)$ of a matrix $\mathbf{A}$ can be determined from the eigenpairs of $\mathbf{A}^* \mathbf{A}$
lyche-numerical-linear-algebra-F02239	169	1.000	$\mathcal{N}(\mathbf{A}^*)$	$\mathcal{N}(\mathbf{A}^*)$	$\mathcal{R}(\mathbf{A}^*)$ and $\mathcal{N}(\mathbf{A}^*)$ of a matrix $\mathbf{A}$ can be determined from the eigenpairs of $\mathbf{A}^* \mathbf{A}$
lyche-numerical-linear-algebra-F02240	169	0.978	$\mathbf{A}^* \mathbf{A}, \mathbf{A} \mathbf{A}^*$	$\mathbf{A}^* \mathbf{A}, \mathbf{A} \mathbf{A}^*$	Theorem 7.1 (The Matrices $\mathbf{A}^* \mathbf{A}, \mathbf{A} \mathbf{A}^*$) Suppose $m, n \in \mathbb{N}$ and $\mathbf{A} \in \mathbb{C}^{m \times n}$.
lyche-numerical-linear-algebra-F02241	169	1.000	$\mathbf{A}^* \mathbf{A} \in \mathbb{C}^{n \times n}$	$\mathbf{A}^* \mathbf{A} \in \mathbb{C}^{n \times n}$	1. The matrices $\mathbf{A}^* \mathbf{A} \in \mathbb{C}^{n \times n}$ and $\mathbf{A} \mathbf{A}^* \in \mathbb{C}^{m \times m}$ have the same nonzero eigenval-
lyche-numerical-linear-algebra-F02242	169	1.000	$\mathbf{A} \mathbf{A}^* \in \mathbb{C}^{m \times m}$	$\mathbf{A} \mathbf{A}^* \in \mathbb{C}^{m \times m}$	1. The matrices $\mathbf{A}^* \mathbf{A} \in \mathbb{C}^{n \times n}$ and $\mathbf{A} \mathbf{A}^* \in \mathbb{C}^{m \times m}$ have the same nonzero eigenval-
lyche-numerical-linear-algebra-F02243	169	0.992	$(\lambda_j, \mathbf{v}_j)$	$(\lambda_j, \mathbf{v}_j)$	3. Let $(\lambda_j, \mathbf{v}_j)$ be orthonormal eigenpairs for $\mathbf{A}^* \mathbf{A}$ with
lyche-numerical-linear-algebra-F02244	169	1.000	$\{\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r\}$	$\{\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r\}$	Then $\{\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r\}$ is an orthogonal basis for the column space $\mathcal{R}(\mathbf{A}) :=$
lyche-numerical-linear-algebra-F02245	169	1.000	$\mathcal{R}(\mathbf{A}) :=$	$\mathcal{R}(\mathbf{A}) :=$	Then $\{\mathbf{A} \mathbf{v}_1, \dots, \mathbf{A} \mathbf{v}_r\}$ is an orthogonal basis for the column space $\mathcal{R}(\mathbf{A}) :=$
lyche-numerical-linear-algebra-F02246	169	1.000	$\{\mathbf{A} \mathbf{y} \in \mathbb{C}^m : \mathbf{y} \in \mathbb{C}^n\}$	$\{\mathbf{A} \mathbf{y} \in \mathbb{C}^m : \mathbf{y} \in \mathbb{C}^n\}$	$\{\mathbf{A} \mathbf{y} \in \mathbb{C}^m : \mathbf{y} \in \mathbb{C}^n\}$ and $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_n\}$ is an orthonormal basis for the

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lyche-numerical-linear-algebra- F02247	169	■ 1.000	$\left\{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_n\right\}$	$\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n\}$	$\{\boldsymbol{A}\boldsymbol{y} \in \mathbb{C}^m : \boldsymbol{y} \in \mathbb{C}^n\}$ and $\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n\}$ is an orthonormal basis for the
lyche-numerical-linear-algebra- F02248	169	■ 0.899	$\mathcal{N}(\boldsymbol{A}) := \left\{\boldsymbol{y} \in \mathbb{C}^n : \boldsymbol{y} \in \mathbb{C}^n : \boldsymbol{A}\boldsymbol{y} = \mathbf{0}\right\}$	$\mathcal{N}(\boldsymbol{A}) := \{\boldsymbol{y} \in \mathbb{C}^n : \boldsymbol{A}\boldsymbol{y} = \mathbf{0}\}$	nullspace $\mathcal{N}(\boldsymbol{A}) := \{\boldsymbol{y} \in \mathbb{C}^n : \boldsymbol{A}\boldsymbol{y} = \mathbf{0}\}$.
lyche-numerical-linear-algebra- F02249	169	■ 0.996	$\left(\lambda_j, \boldsymbol{u}_j\right)$	$(\lambda_j, \boldsymbol{u}_j)$	4. Let $(\lambda_j, \boldsymbol{u}_j)$ be orthonormal eigenpairs for $\boldsymbol{A}\boldsymbol{A}^*$. If $\lambda_j > 0$, $j = 1, \dots, r$ and
lyche-numerical-linear-algebra- F02250	169	■ 0.996	$\lambda_j > 0, j=1, \ldots, r$	$\lambda_j > 0, j = 1, \dots, r$	4. Let $(\lambda_j, \boldsymbol{u}_j)$ be orthonormal eigenpairs for $\boldsymbol{A}\boldsymbol{A}^*$. If $\lambda_j > 0$, $j = 1, \dots, r$ and
lyche-numerical-linear-algebra- F02251	169	■ 1.000	$\lambda_j = 0, j=r+1, \ldots, m$	$\lambda_j = 0, j = r + 1, \dots, m$	$\lambda_j = 0$, $j = r + 1, \dots, m$ then $\{\boldsymbol{A}^*\boldsymbol{u}_1, \dots, \boldsymbol{A}^*\boldsymbol{u}_r\}$ is an orthogonal basis for
lyche-numerical-linear-algebra- F02252	169	■ 1.000	$\left\{\boldsymbol{A}^*\boldsymbol{u}_1, \dots, \boldsymbol{A}^*\boldsymbol{u}_r\right\}$	$\{\boldsymbol{A}^*\boldsymbol{u}_1, \dots, \boldsymbol{A}^*\boldsymbol{u}_r\}$	$\lambda_j = 0$, $j = r + 1, \dots, m$ then $\{\boldsymbol{A}^*\boldsymbol{u}_1, \dots, \boldsymbol{A}^*\boldsymbol{u}_r\}$ is an orthogonal basis for
lyche-numerical-linear-algebra- F02253	169	■ 1.000	$\left\{\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m\right\}$	$\{\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m\}$	the column space $\mathcal{R}(\boldsymbol{A}^*)$ and $\{\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m\}$ is an orthonormal basis for the
lyche-numerical-linear-algebra- F02254	169	■ 1.000	$\pi_{\boldsymbol{A}^*\boldsymbol{A}}$	$\pi_{\boldsymbol{A}^*\boldsymbol{A}}$	1. Consider the characteristic polynomials $\pi_{\boldsymbol{A}^*\boldsymbol{A}}$ and $\pi_{\boldsymbol{A}\boldsymbol{A}^*}$. By (6.1) we have
lyche-numerical-linear-algebra- F02255	169	■ 1.000	$\pi_{\boldsymbol{A}\boldsymbol{A}^*}$	$\pi_{\boldsymbol{A}\boldsymbol{A}^*}$	1. Consider the characteristic polynomials $\pi_{\boldsymbol{A}^*\boldsymbol{A}}$ and $\pi_{\boldsymbol{A}\boldsymbol{A}^*}$. By (6.1) we have
lyche-numerical-linear-algebra- F02256	170	■ 1.000	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\boldsymbol{v} \neq \mathbf{0}$, then
lyche-numerical-linear-algebra- F02257	170	■ 1.000	$\boldsymbol{v} \neq \mathbf{0}$	$\boldsymbol{v} \neq \mathbf{0}$	$\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\boldsymbol{v} \neq \mathbf{0}$, then
lyche-numerical-linear-algebra- F02258	170	■ 1.000	$(\boldsymbol{A}\boldsymbol{v}_j)^*\boldsymbol{A}\boldsymbol{v}_k = \boldsymbol{v}_j^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}_k = \lambda_k\boldsymbol{v}_j^*\boldsymbol{v}_k$	$(\boldsymbol{A}\boldsymbol{v}_j)^*\boldsymbol{A}\boldsymbol{v}_k = \boldsymbol{v}_j^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}_k = \lambda_k\boldsymbol{v}_j^*\boldsymbol{v}_k$	3. By orthonormality of $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ we have $(\boldsymbol{A}\boldsymbol{v}_j)^*\boldsymbol{A}\boldsymbol{v}_k = \boldsymbol{v}_j^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v}_k = \lambda_k\boldsymbol{v}_j^*\boldsymbol{v}_k$
lyche-numerical-linear-algebra- F02259	170	■ 0.578	$\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_n$	$\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_n$	$= 0$ for $j \neq k$, showing that $\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_n$ are orthogonal vectors. Moreover,
lyche-numerical-linear-algebra- F02260	170	■ 1.000	$\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r$	$\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r$	(7.3) implies that $\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r$ are nonzero and $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for $j = r +$
lyche-numerical-linear-algebra- F02261	170	■ 1.000	$\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$	$\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$	(7.3) implies that $\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r$ are nonzero and $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for $j = r +$
lyche-numerical-linear-algebra- F02262	170	■ 1.000	$j = r +$	$j = r +$	(7.3) implies that $\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r$ are nonzero and $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for $j = r +$
lyche-numerical-linear-algebra- F02263	170	■ 1.000	$\mathcal{N}(\boldsymbol{A})$	$\mathcal{N}(\boldsymbol{A})$	linearly independent vectors in $\mathcal{R}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A})$, respectively. The proof will
lyche-numerical-linear-algebra- F02264	170	■ 1.000	$\mathcal{R}(\boldsymbol{A}) \subset \text{span}(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r)$	$\mathcal{R}(\boldsymbol{A}) \subset \text{span}(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r)$	be complete once it is shown that $\mathcal{R}(\boldsymbol{A}) \subset \text{span}(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r)$ and $\mathcal{N}(\boldsymbol{A}) \subset$
lyche-numerical-linear-algebra- F02265	170	■ 1.000	$\mathcal{N}(\boldsymbol{A}) \subset$	$\mathcal{N}(\boldsymbol{A}) \subset$	be complete once it is shown that $\mathcal{R}(\boldsymbol{A}) \subset \text{span}(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r)$ and $\mathcal{N}(\boldsymbol{A}) \subset$
lyche-numerical-linear-algebra- F02266	170	■ 0.999	$\text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$	$\text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$	$\text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$. Suppose $\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$. Then $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$ for some $\boldsymbol{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra- F02267	170	■ 0.999	$\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$	$\text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$. Suppose $\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$. Then $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$ for some $\boldsymbol{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra- F02268	170	■ 0.999	$\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$	$\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$	$\text{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$. Suppose $\boldsymbol{x} \in \mathcal{R}(\boldsymbol{A})$. Then $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y}$ for some $\boldsymbol{y} \in \mathbb{C}^n$,
lyche-numerical-linear-algebra- F02269	170	■ 1.000	$\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j$	$\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j$	Let $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j$ be an eigenvector expansion of $\boldsymbol{y}$. Since $\boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$ for
lyche-numerical-linear-algebra- F02270	170	■ 1.000	$\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{A}\boldsymbol{v}_j = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j \in$	$\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{A}\boldsymbol{v}_j = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j \in$	$j = r + 1, \dots, n$ we obtain $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{A}\boldsymbol{v}_j = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j \in$

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lyche-numerical-linear-algebra-170-F02271	170	1.000	$\operatorname{span}\left(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r\right)$	$\operatorname{span}\left(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r\right)$	$\operatorname{span}(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r)$. Finally, if $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{A})$, then we have
lyche-numerical-linear-algebra-170-F02272	170	1.000	$\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{A})$	$\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{A})$	$\operatorname{span}(\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r)$. Finally, if $\boldsymbol{y} = \sum_{j=1}^n c_j \boldsymbol{v}_j \in \mathcal{N}(\boldsymbol{A})$, then we have
lyche-numerical-linear-algebra-170-F02273	170	1.000	$\boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$	$\boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$	$\boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$, and $c_1 = \dots = c_r = 0$ since $\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r$ are
lyche-numerical-linear-algebra-170-F02274	170	1.000	$c_1 = \dots = c_r = 0$	$c_1 = \dots = c_r = 0$	$\boldsymbol{A}\boldsymbol{y} = \sum_{j=1}^r c_j \boldsymbol{A}\boldsymbol{v}_j = \mathbf{0}$, and $c_1 = \dots = c_r = 0$ since $\boldsymbol{A}\boldsymbol{v}_1, \dots, \boldsymbol{A}\boldsymbol{v}_r$ are
lyche-numerical-linear-algebra-170-F02275	170	1.000	$\boldsymbol{y} = \sum_{j=r+1}^n c_j \boldsymbol{v}_j \in \operatorname{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$	$\boldsymbol{y} = \sum_{j=r+1}^n c_j \boldsymbol{v}_j \in \operatorname{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$	linearly independent. But then $\boldsymbol{y} = \sum_{j=r+1}^n c_j \boldsymbol{v}_j \in \operatorname{span}(\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_n)$.
lyche-numerical-linear-algebra-170-F02276	170	1.000	$\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$	$\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$	4. Since $\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra-170-F02277	170	1.000	$\boldsymbol{B} := \boldsymbol{A}^*$	$\boldsymbol{B} := \boldsymbol{A}^*$	4. Since $\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra-170-F02278	170	1.000	$\boldsymbol{A} = \boldsymbol{B}$	$\boldsymbol{A} = \boldsymbol{B}$	4. Since $\boldsymbol{A}\boldsymbol{A}^* = \boldsymbol{B}^*\boldsymbol{B}$ with $\boldsymbol{B} := \boldsymbol{A}^*$ this follows from part 3 with $\boldsymbol{A} = \boldsymbol{B}$.
lyche-numerical-linear-algebra-170-F02279	170	1.000	$2\boldsymbol{A}^*\boldsymbol{A}$	$2\boldsymbol{A}^*\boldsymbol{A}$	5. By part 1 and 2 $\boldsymbol{A}^*\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^*$ have the same number r of positive eigenvalues
lyche-numerical-linear-algebra-170-F02280	170	1.000	$4r$	$4r$	and by part 3 and $4r$ is the rank of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-170-F02281	170	1.000	$m, n, r \in \mathbb{N}$	$m, n, r \in \mathbb{N}$	Theorem 7.2 (Existence of SVD) Suppose for $m, n, r \in \mathbb{N}$ that $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has
lyche-numerical-linear-algebra-170-F02282	170	1.000	$\lambda_1 \geq \dots \geq$	$\lambda_1 \geq \dots \geq$	rank r , and that $(\lambda_j, \boldsymbol{v}_j)$ are orthonormal eigenpairs for $\boldsymbol{A}^*\boldsymbol{A}$ with $\lambda_1 \geq \dots \geq$
lyche-numerical-linear-algebra-170-F02283	170	1.000	$\lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$	$\lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$	$\lambda_r > 0 = \lambda_{r+1} = \dots = \lambda_n$. Define
lyche-numerical-linear-algebra-170-F02284	170	1.000	$\boldsymbol{V} := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	$\boldsymbol{V} := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$	1. $\boldsymbol{V} := [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n] \in \mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra-170-F02285	170	1.000	$\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$	$\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$	2. $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$
lyche-numerical-linear-algebra-170-F02286	170	1.000	$\sigma_j := \sqrt{\lambda_j}$	$\sigma_j := \sqrt{\lambda_j}$	2. $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$
lyche-numerical-linear-algebra-170-F02287	170	1.000	$j =$	$j =$	2. $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ is a diagonal matrix with diagonal elements $\sigma_j := \sqrt{\lambda_j}$ for $j =$
lyche-numerical-linear-algebra-170-F02288	170	1.000	$1, \dots, \min(m, n)$	$1, \dots, \min(m, n)$	$1, \dots, \min(m, n)$,
lyche-numerical-linear-algebra-170-F02289	170	0.999	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$	$\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$	3. $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A}\boldsymbol{v}_j$ for $j = 1, \dots, r$ and
lyche-numerical-linear-algebra-170-F02290	170	0.999	$\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A}\boldsymbol{v}_j$	$\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A}\boldsymbol{v}_j$	3. $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A}\boldsymbol{v}_j$ for $j = 1, \dots, r$ and
lyche-numerical-linear-algebra-170-F02291	170	0.999	$j = 1, \dots, r$	$j = 1, \dots, r$	3. $\boldsymbol{U} := [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \in \mathbb{C}^{m \times m}$, where $\boldsymbol{u}_j = \sigma_j^{-1} \boldsymbol{A}\boldsymbol{v}_j$ for $j = 1, \dots, r$ and
lyche-numerical-linear-algebra-170-F02292	170	1.000	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ is any extension of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra-170-F02293	170	1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ is any extension of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra-170-F02294	170	1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$	$\boldsymbol{u}_{r+1}, \dots, \boldsymbol{u}_m$ is any extension of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ to an orthonormal basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_m$
lyche-numerical-linear-algebra-170-F02295	170	1.000	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$	$\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$	Proof Let $\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}$ be as in the theorem. The vectors $\boldsymbol{u}_1, \dots, \boldsymbol{u}_r$ are orthonormal

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lyche-numerical-linear-algebra-F02296	170	■ 1.000	$\sigma_j=\left \left \boldsymbol{A} \boldsymbol{v}_j\right \right _{2}>0, j=1, \ldots, r$	$\sigma_j=\left\ \boldsymbol{A} \boldsymbol{v}_j\right\ _2>0, j=1, \ldots, r$	since $\boldsymbol{A} \boldsymbol{v}_1, \ldots, \boldsymbol{A} \boldsymbol{v}_r$ are orthogonal and $\sigma_j=\left\ \boldsymbol{A} \boldsymbol{v}_j\right\ _2>0, j=1, \ldots, r$ by (7.3).
lyche-numerical-linear-algebra-F02297	171	■ 1.000	$\boldsymbol{U} \boldsymbol{\Sigma}=\boldsymbol{A} \boldsymbol{V}$	$\boldsymbol{U} \boldsymbol{\Sigma}=\boldsymbol{A} \boldsymbol{V}$	Thus $\boldsymbol{U} \boldsymbol{\Sigma}=\boldsymbol{A} \boldsymbol{V}$ and since $\boldsymbol{V}$ is square and unitary we find $\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*=\boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{A}$
lyche-numerical-linear-algebra-F02298	171	■ 1.000	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*=\boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{A}$	$\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*=\boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{A}$	Thus $\boldsymbol{U} \boldsymbol{\Sigma}=\boldsymbol{A} \boldsymbol{V}$ and since $\boldsymbol{V}$ is square and unitary we find $\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*=\boldsymbol{A} \boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{A}$
lyche-numerical-linear-algebra-F02299	171	■ 1.000	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$	and we have an SVD of $\boldsymbol{A}$ with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$.
lyche-numerical-linear-algebra-F02300	171	■ 1.000	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]$	$\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]$	<i>Example 7.2 (Find SVD)</i> To derive the SVD in (7.2) where $\boldsymbol{A}=\frac{1}{15}\left[\begin{array}{cc} 14 & 2 \\ 4 & 22 \\ 16 & 13 \end{array}\right]$, we
lyche-numerical-linear-algebra-F02301	171	■ 0.990	$\sigma_1=2, \sigma_2=1$	$\sigma_1=2, \sigma_2=1$	Thus $\sigma_1=2, \sigma_2=1$, and $\boldsymbol{V}=\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]$. Now $\boldsymbol{u}_1=\boldsymbol{A} \boldsymbol{v}_1 / \sigma_1=[1, 2, 2]^T / 3$,
lyche-numerical-linear-algebra-F02302	171	■ 0.990	$\boldsymbol{V}=\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]$	$\boldsymbol{V}=\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]$	Thus $\sigma_1=2, \sigma_2=1$, and $\boldsymbol{V}=\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]$. Now $\boldsymbol{u}_1=\boldsymbol{A} \boldsymbol{v}_1 / \sigma_1=[1, 2, 2]^T / 3$,
lyche-numerical-linear-algebra-F02303	171	■ 0.990	$\boldsymbol{u}_1=\boldsymbol{A} \boldsymbol{v}_1 / \sigma_1=[1, 2, 2]^T / 3$	$\boldsymbol{u}_1=\boldsymbol{A} \boldsymbol{v}_1 / \sigma_1=[1, 2, 2]^T / 3$	Thus $\sigma_1=2, \sigma_2=1$, and $\boldsymbol{V}=\frac{1}{5}\left[\begin{array}{cc} 3 & 4 \\ 4 & -3 \end{array}\right]$. Now $\boldsymbol{u}_1=\boldsymbol{A} \boldsymbol{v}_1 / \sigma_1=[1, 2, 2]^T / 3$,
lyche-numerical-linear-algebra-F02304	171	■ 0.998	$\boldsymbol{u}_2=\boldsymbol{A} \boldsymbol{v}_2 / \sigma_2=[2,-2,1]^T / 3$	$\boldsymbol{u}_2=\boldsymbol{A} \boldsymbol{v}_2 / \sigma_2=[2,-2,1]^T / 3$	$\boldsymbol{u}_2=\boldsymbol{A} \boldsymbol{v}_2 / \sigma_2=[2,-2,1]^T / 3$. For an SVD we also need $\boldsymbol{u}_3$ which is any vector
lyche-numerical-linear-algebra-F02305	171	■ 0.998	$\boldsymbol{u}_3$	$\boldsymbol{u}_3$	$\boldsymbol{u}_2=\boldsymbol{A} \boldsymbol{v}_2 / \sigma_2=[2,-2,1]^T / 3$. For an SVD we also need $\boldsymbol{u}_3$ which is any vector
lyche-numerical-linear-algebra-F02306	171	■ 1.000	$\boldsymbol{u}_1$	$\boldsymbol{u}_1$	of length one orthogonal to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$. $\boldsymbol{u}_3=[2,1,-2]^T / 3$ is such a vector and we
lyche-numerical-linear-algebra-F02307	171	■ 1.000	$\boldsymbol{u}_2 \cdot \boldsymbol{u}_3=[2,1,-2]^T / 3$	$\boldsymbol{u}_2 \cdot \boldsymbol{u}_3=[2,1,-2]^T / 3$	of length one orthogonal to $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$. $\boldsymbol{u}_3=[2,1,-2]^T / 3$ is such a vector and we
lyche-numerical-linear-algebra-F02308	171	■ 1.000	$\boldsymbol{\Sigma}_1$	$\boldsymbol{\Sigma}_1$	Thus $\boldsymbol{\Sigma}_1$ contains the r positive singular values on the diagonal and for $k, l \geq 0$ the
lyche-numerical-linear-algebra-F02309	171	■ 1.000	$k, l \geq 0$	$k, l \geq 0$	Thus $\boldsymbol{\Sigma}_1$ contains the r positive singular values on the diagonal and for $k, l \geq 0$ the
lyche-numerical-linear-algebra-F02310	171	■ 1.000	$\mathbf{0}_{k, l}=[]$	$\mathbf{0}_{k, l}=[]$	symbol $\mathbf{0}_{k, l}=[]$ denotes the empty matrix if $k=0$ or $l=0$, and the zero matrix
lyche-numerical-linear-algebra-F02311	171	■ 1.000	$k=0$	$k=0$	symbol $\mathbf{0}_{k, l}=[]$ denotes the empty matrix if $k=0$ or $l=0$, and the zero matrix
lyche-numerical-linear-algebra-F02312	171	■ 1.000	$l=0$	$l=0$	symbol $\mathbf{0}_{k, l}=[]$ denotes the empty matrix if $k=0$ or $l=0$, and the zero matrix
lyche-numerical-linear-algebra-F02313	172	■ 1.000	l	l	with k rows and l columns otherwise. We obtain by block multiplication a reduced
lyche-numerical-linear-algebra-F02314	172	■ 1.000	$\boldsymbol{A}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$	$\boldsymbol{A}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$	factorization of $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ of the form $\boldsymbol{A}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$, where $\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$ and
lyche-numerical-linear-algebra-F02315	172	■ 1.000	$\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$	$\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$	factorization of $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ of the form $\boldsymbol{A}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$, where $\boldsymbol{U}_1 \in \mathbb{C}^{m \times r}$ and
lyche-numerical-linear-algebra-F02316	172	■ 1.000	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$ have orthonormal columns, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a diagonal matrix with
lyche-numerical-linear-algebra-F02317	172	■ 1.000	$\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$	$\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$	$\boldsymbol{V}_1 \in \mathbb{C}^{n \times r}$ have orthonormal columns, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a diagonal matrix with
lyche-numerical-linear-algebra-F02318	172	■ 1.000	$\sigma_1 \geq \cdots \geq \sigma_r>0$	$\sigma_1 \geq \cdots \geq \sigma_r>0$	$\sigma_1 \geq \cdots \geq \sigma_r>0$.
lyche-numerical-linear-algebra-F02319	172	■ 1.000	$\boldsymbol{U}_1, \boldsymbol{V}_1$	$\boldsymbol{U}_1, \boldsymbol{V}_1$	$\boldsymbol{U}_1, \boldsymbol{V}_1$ contain the first r columns of $\boldsymbol{U}, \boldsymbol{V}$ respectively, and $\boldsymbol{\Sigma}_1 \in \mathbb{R}^{r \times r}$ is a

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lyche-numerical-linear-algebra- F02320	172	■ 1.000	$\backslash\boldsymbol{U}, \backslash\boldsymbol{V}$	U, V	U_1, V_1 contain the first r columns of U, V respectively, and $\Sigma_1 \in \mathbb{R}^{r \times r}$ is a
lyche-numerical-linear-algebra- F02321	172	■ 1.000	$\backslash\Sigma_{\{1\}} \in \mathbb{R}^{r \times r}$	$\Sigma_1 \in \mathbb{R}^{r \times r}$	U_1, V_1 contain the first r columns of U, V respectively, and $\Sigma_1 \in \mathbb{R}^{r \times r}$ is a
lyche-numerical-linear-algebra- F02322	172	■ 1.000	$\backslash\boldsymbol{U}_{\{1\}}$	U_1	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02323	172	■ 1.000	$\backslash\boldsymbol{V}_{\{1\}}$	V_1	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02324	172	■ 1.000	$\backslash\boldsymbol{U} \in \mathbb{C}^{m \times m}$	$U \in \mathbb{C}^{m \times m}$	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02325	172	■ 1.000	$\backslash\boldsymbol{V} \in \mathbb{C}^{n \times n}$	$V \in \mathbb{C}^{n \times n}$	U_1 and V_1 in any way to unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$, and let Σ
lyche-numerical-linear-algebra- F02326	172	■ 1.000	$\backslash\boldsymbol{A}=\backslashleft[\backslash\boldsymbol{u}_{\{1\}}, \backslashdots, \backslash\boldsymbol{u}_{\{r\}}\backslashright] \operatorname{diag} \backslashleft(\backslash\sigma_{\{1\}}, \backslashdots, \backslash\sigma_{\{r\}}\backslashright)\backslashleft[\backslash\boldsymbol{v}_{\{1\}}, \backslashdots, \backslash\boldsymbol{v}_{\{r\}}\backslashright]^{*}$	$A = [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r] \operatorname{diag}(\sigma_1, \dots, \sigma_r) [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r]^*$	3. If $A = [\boldsymbol{u}_1, \dots, \boldsymbol{u}_r] \operatorname{diag}(\sigma_1, \dots, \sigma_r) [\boldsymbol{v}_1, \dots, \boldsymbol{v}_r]^*$ is a singular value factoriza-
lyche-numerical-linear-algebra- F02327	172	■ 1.000	7.3($r < n < m$)	7.3($r < n < m$)	Example 7.3($r < n < m$) To find the SVF and SVD of
lyche-numerical-linear-algebra- F02328	173	■ 1.000	$\backslash\sigma_{\{1\}}=2, \backslash\sigma_{\{2\}}=0$	$\sigma_1 = 2, \sigma_2 = 0$	and we find $\sigma_1 = 2, \sigma_2 = 0$, Thus $r = 1, m = 3, n = 2$ and
lyche-numerical-linear-algebra- F02329	173	■ 1.000	$r=1, m=3, n=2$	$r = 1, m = 3, n = 2$	and we find $\sigma_1 = 2, \sigma_2 = 0$, Thus $r = 1, m = 3, n = 2$ and
lyche-numerical-linear-algebra- F02330	173	■ 1.000	$\backslash\boldsymbol{u}_{\{1\}}=\backslash\boldsymbol{A} \backslash\boldsymbol{v}_{\{1\}} / \backslash\sigma_{\{1\}}=\backslash\boldsymbol{A} \backslash\boldsymbol{v}_{\{1\}} / \backslash\sqrt{2}$	$\boldsymbol{u}_1 = A \boldsymbol{v}_1 / \sigma_1 = s_1 / \sqrt{2}$	We find $\boldsymbol{u}_1 = A \boldsymbol{v}_1 / \sigma_1 = s_1 / \sqrt{2}$, where $s_1 = [1, 1, 0]^T$, and the SVF of A is given
lyche-numerical-linear-algebra- F02331	173	■ 1.000	$\backslash\boldsymbol{s}_{\{1\}}=[1,1,0]^T$	$\boldsymbol{s}_1 = [1, 1, 0]^T$	We find $\boldsymbol{u}_1 = A \boldsymbol{v}_1 / \sigma_1 = s_1 / \sqrt{2}$, where $s_1 = [1, 1, 0]^T$, and the SVF of A is given
lyche-numerical-linear-algebra- F02332	173	■ 1.000	$\backslash\boldsymbol{s}_{\{1\}}$	$\boldsymbol{s}_1$	$\boldsymbol{s}_1$ to a basis $\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\}$ for $\mathbb{R}^3$, apply the Gram-Schmidt orthogonalization process
lyche-numerical-linear-algebra- F02333	173	■ 1.000	$\backslash\left\{\backslash\boldsymbol{s}_{\{1\}}, \backslash\boldsymbol{s}_{\{2\}}, \backslash\boldsymbol{s}_{\{3\}}\backslashright\}$	$\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\}$	$\boldsymbol{s}_1$ to a basis $\{\boldsymbol{s}_1, \boldsymbol{s}_2, \boldsymbol{s}_3\}$ for $\mathbb{R}^3$, apply the Gram-Schmidt orthogonalization process
lyche-numerical-linear-algebra- F02334	173	■ 0.979	$\backslash\boldsymbol{w}_{\{1\}}=\backslash\boldsymbol{s}_{\{1\}}, \backslash\boldsymbol{w}_{\{2\}}=\backslash\boldsymbol{s}_{\{2\}}-\backslash\frac{\backslash\boldsymbol{s}_{\{2\}}^T \backslash\boldsymbol{w}_{\{1\}}}{\backslash\boldsymbol{s}_{\{2\}}^T \backslash\boldsymbol{s}_{\{2\}}} \backslash\boldsymbol{s}_{\{2\}}, \backslash\boldsymbol{w}_{\{3\}}=\backslash\frac{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{w}_{\{1\}}}{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{s}_{\{3\}}} \backslash\boldsymbol{s}_{\{3\}}-\backslash\frac{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{w}_{\{2\}}}{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{s}_{\{3\}}} \backslash\boldsymbol{s}_{\{3\}}$	$\boldsymbol{w}_1 = \boldsymbol{s}_1, \boldsymbol{w}_2 = \boldsymbol{s}_2 - \frac{\boldsymbol{s}_2^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \end{bmatrix}, \boldsymbol{w}_3 = \boldsymbol{s}_3 -$	we find from (5.8) $\boldsymbol{w}_1 = \boldsymbol{s}_1, \boldsymbol{w}_2 = \boldsymbol{s}_2 - \frac{\boldsymbol{s}_2^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 = \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \end{bmatrix}, \boldsymbol{w}_3 = \boldsymbol{s}_3 -$
lyche-numerical-linear-algebra- F02335	173	■ 0.763	$\backslash\frac{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{w}_{\{1\}}}{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{s}_{\{3\}}} \backslash\boldsymbol{s}_{\{3\}}-\backslash\frac{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{w}_{\{2\}}}{\backslash\boldsymbol{s}_{\{3\}}^T \backslash\boldsymbol{s}_{\{3\}}} \backslash\boldsymbol{s}_{\{3\}}$	$\frac{\boldsymbol{s}_3^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 - \frac{\boldsymbol{s}_3^T \boldsymbol{w}_2}{\boldsymbol{w}_2^T \boldsymbol{w}_2} \boldsymbol{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	$\frac{\boldsymbol{s}_3^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 - \frac{\boldsymbol{s}_3^T \boldsymbol{w}_2}{\boldsymbol{w}_2^T \boldsymbol{w}_2} \boldsymbol{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the $\boldsymbol{w}_i$'s we obtain $\boldsymbol{u}_1 = \boldsymbol{w}_1 / \ \boldsymbol{w}_1\ _2 =$
lyche-numerical-linear-algebra- F02336	173	■ 0.763	$\backslash\boldsymbol{w}_{\{i\}}$	$\boldsymbol{w}_i$	$\frac{\boldsymbol{s}_3^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 - \frac{\boldsymbol{s}_3^T \boldsymbol{w}_2}{\boldsymbol{w}_2^T \boldsymbol{w}_2} \boldsymbol{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the $\boldsymbol{w}_i$'s we obtain $\boldsymbol{u}_1 = \boldsymbol{w}_1 / \ \boldsymbol{w}_1\ _2 =$
lyche-numerical-linear-algebra- F02337	173	■ 0.763	$\backslash\boldsymbol{u}_{\{1\}}=\backslash\boldsymbol{w}_{\{1\}} / \backslash\left\ \backslash\boldsymbol{w}_{\{1\}}\backslashright\ _2$	$\boldsymbol{u}_1 = \boldsymbol{w}_1 / \ \boldsymbol{w}_1\ _2 =$	$\frac{\boldsymbol{s}_3^T \boldsymbol{w}_1}{\boldsymbol{w}_1^T \boldsymbol{w}_1} \boldsymbol{w}_1 - \frac{\boldsymbol{s}_3^T \boldsymbol{w}_2}{\boldsymbol{w}_2^T \boldsymbol{w}_2} \boldsymbol{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Normalizing the $\boldsymbol{w}_i$'s we obtain $\boldsymbol{u}_1 = \boldsymbol{w}_1 / \ \boldsymbol{w}_1\ _2 =$
lyche-numerical-linear-algebra- F02338	173	■ 0.999	$[1 / \backslash\sqrt{2}, 1 / \backslash\sqrt{2}, 0]^T, \backslash\boldsymbol{u}_{\{2\}}=\backslash\boldsymbol{w}_{\{2\}} / \backslash\left\ \backslash\boldsymbol{w}_{\{2\}}\backslashright\ _2$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T, \boldsymbol{u}_2 = \boldsymbol{w}_2 / \ \boldsymbol{w}_2\ _2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T, \boldsymbol{u}_2 = \boldsymbol{w}_2 / \ \boldsymbol{w}_2\ _2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, and $\boldsymbol{u}_3 = \boldsymbol{s}_3 / \ \boldsymbol{s}_3\ _2 =$
lyche-numerical-linear-algebra- F02339	173	■ 0.999	$\backslash\boldsymbol{u}_{\{3\}}=\backslash\boldsymbol{s}_{\{3\}} / \backslash\left\ \backslash\boldsymbol{s}_{\{3\}}\backslashright\ _2$	$\boldsymbol{u}_3 = \boldsymbol{s}_3 / \ \boldsymbol{s}_3\ _2 =$	$[1/\sqrt{2}, 1/\sqrt{2}, 0]^T, \boldsymbol{u}_2 = \boldsymbol{w}_2 / \ \boldsymbol{w}_2\ _2 = [-1/\sqrt{2}, 1/\sqrt{2}, 0]^T$, and $\boldsymbol{u}_3 = \boldsymbol{s}_3 / \ \boldsymbol{s}_3\ _2 =$

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lyche-numerical-linear-algebra- F02340	173	■ 1.000	$[0,0,1]^T$	$[0,0,1]^T$	$[0,0,1]^T$. Therefore, $A = U \Sigma V^T$, is an SVD, where
lyche-numerical-linear-algebra- F02341	173	■ 1.000	$\boldsymbol{A}=\boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T$	$A = U \Sigma V^T$	$[0,0,1]^T$. Therefore, $A = U \Sigma V^T$, is an SVD, where
lyche-numerical-linear-algebra- F02342	174	■ 1.000	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^*), \mathcal{N}(\boldsymbol{A}^*)$	$\mathcal{R}(A), \mathcal{N}(A), \mathcal{R}(A^*)$	$\mathcal{R}(A), \mathcal{N}(A), \mathcal{R}(A^*)$, and $\mathcal{N}(A^*)$.
lyche-numerical-linear-algebra- F02343	174	■ 1.000	m, n	m, n	m, n let $A \in \mathbb{C}^{m \times n}$ have rank r and a singular value decomposition $A =$
lyche-numerical-linear-algebra- F02344	174	■ 0.985	$\left[\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\right] \boldsymbol{\Sigma} \left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right]^* = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^*$	$[\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \Sigma [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n]^* = U \Sigma V^*$	$[\boldsymbol{u}_1, \dots, \boldsymbol{u}_m] \Sigma [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n]^* = U \Sigma V^*$. Then the singular vectors satisfy
lyche-numerical-linear-algebra- F02345	174	■ 1.000	$\left\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\right\}$	$\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\}$	1. $\{\boldsymbol{u}_1, \dots, \boldsymbol{u}_r\}$ is an orthonormal basis for $\mathcal{R}(A)$,
lyche-numerical-linear-algebra- F02346	174	■ 1.000	$\left\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r\right\}$	$\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_r\}$	3. $\{\boldsymbol{v}_1, \dots, \boldsymbol{v}_r\}$ is an orthonormal basis for $\mathcal{R}(A^*)$,
lyche-numerical-linear-algebra- F02347	174	■ 1.000	$\boldsymbol{A} \boldsymbol{V} = \boldsymbol{U} \boldsymbol{\Sigma}$	$AV = U \Sigma$	Proof If $A = U \Sigma V^*$ then $AV = U \Sigma$, or in terms of the block partition (7.4)
lyche-numerical-linear-algebra- F02348	174	■ 0.980	$\boldsymbol{A} \left[\boldsymbol{V}_1, \boldsymbol{V}_2\right] = \left[\boldsymbol{U}_1, \boldsymbol{U}_2\right] \begin{bmatrix} \Sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	$A[V_1, V_2] = [U_1, U_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$	$A[V_1, V_2] = [U_1, U_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$. But then $AV_1 = U_1 \Sigma_1$, $AV_2 = \mathbf{0}$, and this
lyche-numerical-linear-algebra- F02349	174	■ 0.980	$\boldsymbol{A} \boldsymbol{V}_1 = \boldsymbol{U}_1 \Sigma_1, \boldsymbol{A} \boldsymbol{V}_2 = \mathbf{0}$	$AV_1 = U_1 \Sigma_1, AV_2 = \mathbf{0}$	$A[V_1, V_2] = [U_1, U_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$. But then $AV_1 = U_1 \Sigma_1$, $AV_2 = \mathbf{0}$, and this
lyche-numerical-linear-algebra- F02350	174	■ 1.000	$\boldsymbol{A}^* = \boldsymbol{V} \boldsymbol{\Sigma}^* \boldsymbol{U}^*$	$A^* = V \Sigma^* U^*$	$A^* = V \Sigma^* U^*$ or $A^* U = V \Sigma^*$. Using the block partition as before we obtain the
lyche-numerical-linear-algebra- F02351	174	■ 1.000	$\boldsymbol{A}^* \boldsymbol{U} = \boldsymbol{V} \boldsymbol{\Sigma}^*$	$A^* U = V \Sigma^*$	$A^* = V \Sigma^* U^*$ or $A^* U = V \Sigma^*$. Using the block partition as before we obtain the
lyche-numerical-linear-algebra- F02352	174	■ 1.000	$\mathcal{R}(\boldsymbol{A}), \{\boldsymbol{A}^* \boldsymbol{u}_1, \ldots, \boldsymbol{A}^* \boldsymbol{u}_r\}$	$\mathcal{R}(A), \{A^* \boldsymbol{u}_1, \dots, A^* \boldsymbol{u}_r\}$	$\mathcal{R}(A), \{A^* \boldsymbol{u}_1, \dots, A^* \boldsymbol{u}_r\}$ is an orthogonal basis for $\mathcal{R}(A^*)$, $\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_m\}$ is an
lyche-numerical-linear-algebra- F02353	174	■ 1.000	$\mathcal{R}(\boldsymbol{A}^*), \{\boldsymbol{v}_{r+1}, \ldots, \boldsymbol{v}_m\}$	$\mathcal{R}(A^*), \{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_m\}$	$\mathcal{R}(A), \{A^* \boldsymbol{u}_1, \dots, A^* \boldsymbol{u}_r\}$ is an orthogonal basis for $\mathcal{R}(A^*)$, $\{\boldsymbol{v}_{r+1}, \dots, \boldsymbol{v}_m\}$ is an
lyche-numerical-linear-algebra- F02354	174	■ 1.000	$\boldsymbol{T}: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	a linear transformation. Consider the linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by
lyche-numerical-linear-algebra- F02355	174	■ 0.825	$\boldsymbol{T} \boldsymbol{z} := \boldsymbol{A} \boldsymbol{z}$	$Tz := Az$	$Tz := Az$ where $A \in \mathbb{R}^{m \times n}$. Assume that $\text{rank}(A) = n$. In the following theorem
lyche-numerical-linear-algebra- F02356	174	■ 0.825	$(\boldsymbol{A}) = n$	$(A) = n$	$Tz := Az$ where $A \in \mathbb{R}^{m \times n}$. Assume that $\text{rank}(A) = n$. In the following theorem
lyche-numerical-linear-algebra- F02357	174	■ 0.672	$\mathcal{S} := \{z \in \mathbb{R}^n\}$	$\mathcal{S} := \{z \in \mathbb{R}^n$	we show that the function T maps the unit sphere in $\mathbb{R}^n$ given by $\mathcal{S} := \{z \in \mathbb{R}^n :$
lyche-numerical-linear-algebra- F02358	174	■ 0.943	$\ \boldsymbol{z}\ _2 = 1\}$	$\ z\ _2 = 1\}$	$\ z\ _2 = 1\}$ onto an ellipsoid $\mathcal{E} := A \mathcal{S} = \{Az : z \in \mathcal{S}\}$ in $\mathbb{R}^m$.
lyche-numerical-linear-algebra- F02359	174	■ 0.943	$\mathcal{E} := \boldsymbol{A} \boldsymbol{\mathcal{S}} = \{\boldsymbol{A} \boldsymbol{z} : \boldsymbol{z} \in \boldsymbol{\mathcal{S}}\}$	$\mathcal{E} := A \mathcal{S} = \{Az : z \in \mathcal{S}\}$	$\ z\ _2 = 1\}$ onto an ellipsoid $\mathcal{E} := A \mathcal{S} = \{Az : z \in \mathcal{S}\}$ in $\mathbb{R}^m$.
lyche-numerical-linear-algebra- F02360	174	■ 1.000	$r=n$	$r = n$	Theorem 7.4 (SVF Ellipse) Suppose $A \in \mathbb{R}^{m \times n}$ has rank $r = n$, and let $A =$
lyche-numerical-linear-algebra- F02361	174	■ 0.977	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T$	$U_1 \Sigma_1 V_1^T$	$U_1 \Sigma_1 V_1^T$ be a singular value factorization of A . Then
lyche-numerical-linear-algebra- F02362	175	■ 0.407	$z \in \mathcal{S}$	$z \in \mathcal{S}$	Proof Suppose $z \in \mathcal{S}$. Now $Az = U_1 \Sigma_1 V_1^T z = U_1 y$, where $y := \Sigma_1 V_1^T z$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F02363	175	■ 0.407	$\boldsymbol{z}=\boldsymbol{\Sigma}_1^{-1} \boldsymbol{z}=\boldsymbol{\Sigma}_1^{-1} \boldsymbol{z}$	$\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$	<i>Proof</i> Suppose $\boldsymbol{z} \in \mathcal{S}$. Now $\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y}:=\boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$.
lyche-numerical-linear-algebra- F02364	175	■ 0.407	$\boldsymbol{y}:=\boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$	$\boldsymbol{y}:=\boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$	<i>Proof</i> Suppose $\boldsymbol{z} \in \mathcal{S}$. Now $\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y}:=\boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}$.
lyche-numerical-linear-algebra- F02365	175	■ 0.997	$\operatorname{rank}(\boldsymbol{A})=n$	$\operatorname{rank}(\boldsymbol{A})=n$	Since $\operatorname{rank}(\boldsymbol{A})=n$ it follows that $\boldsymbol{V}_1=\boldsymbol{V}$ is square so that $\boldsymbol{V}_1 \boldsymbol{V}_1^T=\boldsymbol{I}$. But then
lyche-numerical-linear-algebra- F02366	175	■ 0.997	$\boldsymbol{V}_1=\boldsymbol{V}$	$\boldsymbol{V}_1=\boldsymbol{V}$	Since $\operatorname{rank}(\boldsymbol{A})=n$ it follows that $\boldsymbol{V}_1=\boldsymbol{V}$ is square so that $\boldsymbol{V}_1 \boldsymbol{V}_1^T=\boldsymbol{I}$. But then
lyche-numerical-linear-algebra- F02367	175	■ 0.997	$\boldsymbol{V}_1 \boldsymbol{V}_1^T=\boldsymbol{I}$	$\boldsymbol{V}_1 \boldsymbol{V}_1^T=\boldsymbol{I}$	Since $\operatorname{rank}(\boldsymbol{A})=n$ it follows that $\boldsymbol{V}_1=\boldsymbol{V}$ is square so that $\boldsymbol{V}_1 \boldsymbol{V}_1^T=\boldsymbol{I}$. But then
lyche-numerical-linear-algebra- F02368	175	■ 0.984	$\boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}=\boldsymbol{z}$	$\boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}=\boldsymbol{z}$	$\boldsymbol{V}_1 \boldsymbol{\Sigma}_1^{-1} \boldsymbol{y}=\boldsymbol{z}$ and we obtain
lyche-numerical-linear-algebra- F02369	175	■ 1.000	$\boldsymbol{y} \in \tilde{\mathcal{E}}$	$\boldsymbol{y} \in \tilde{\mathcal{E}}$	This implies that $\boldsymbol{y} \in \tilde{\mathcal{E}}$. Finally, $\boldsymbol{x}=\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y} \in \tilde{\mathcal{E}}$
lyche-numerical-linear-algebra- F02370	175	■ 1.000	$\boldsymbol{x}=\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$	$\boldsymbol{x}=\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$	This implies that $\boldsymbol{y} \in \tilde{\mathcal{E}}$. Finally, $\boldsymbol{x}=\boldsymbol{A} \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^T \boldsymbol{z}=\boldsymbol{U}_1 \boldsymbol{y}$, where $\boldsymbol{y} \in \tilde{\mathcal{E}}$
lyche-numerical-linear-algebra- F02371	175	■ 1.000	$\mathcal{E}=\boldsymbol{U}_1 \tilde{\mathcal{E}}$	$\mathcal{E}=\boldsymbol{U}_1 \tilde{\mathcal{E}}$	implies that $\mathcal{E}=\boldsymbol{U}_1 \tilde{\mathcal{E}}$.
lyche-numerical-linear-algebra- F02372	175	■ 1.000	$1=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}$	$1=\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}$	The equation $\frac{y_1^2}{\sigma_1^2}+\cdots+\frac{y_n^2}{\sigma_n^2}$ describes an ellipsoid in $\mathbb{R}^n$ with semiaxes
lyche-numerical-linear-algebra- F02373	175	■ 1.000	σ_j	σ_j	of length σ_j along the unit vectors $\boldsymbol{e}_j$ for $j=1, \dots, n$. Since the orthonormal
lyche-numerical-linear-algebra- F02374	175	■ 1.000	$\boldsymbol{e}_j$	$\boldsymbol{e}_j$	of length σ_j along the unit vectors $\boldsymbol{e}_j$ for $j=1, \dots, n$. Since the orthonormal
lyche-numerical-linear-algebra- F02375	175	■ 0.992	$\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$	$\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$	transformation $\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$ preserves length, the image $\mathcal{E}=\boldsymbol{A} \mathcal{S}$ is a rotated ellipsoid
lyche-numerical-linear-algebra- F02376	175	■ 0.992	$\mathcal{E}=\boldsymbol{A} \mathcal{S}$	$\mathcal{E}=\boldsymbol{A} \mathcal{S}$	transformation $\boldsymbol{U}_1 \boldsymbol{y} \rightarrow \boldsymbol{x}$ preserves length, the image $\mathcal{E}=\boldsymbol{A} \mathcal{S}$ is a rotated ellipsoid
lyche-numerical-linear-algebra- F02377	175	■ 1.000	$\boldsymbol{u}_j=\boldsymbol{U} \boldsymbol{e}_j$	$\boldsymbol{u}_j=\boldsymbol{U} \boldsymbol{e}_j$	with semiaxes along the left singular vectors $\boldsymbol{u}_j=\boldsymbol{U} \boldsymbol{e}_j$, of length $\sigma_j, j=1, \dots, n$.
lyche-numerical-linear-algebra- F02378	175	■ 1.000	$\sigma_j, j=1, \dots, n$	$\sigma_j, j=1, \dots, n$	with semiaxes along the left singular vectors $\boldsymbol{u}_j=\boldsymbol{U} \boldsymbol{e}_j$, of length $\sigma_j, j=1, \dots, n$.
lyche-numerical-linear-algebra- F02379	175	■ 1.000	$\boldsymbol{A} \boldsymbol{v}_j=\sigma_j \boldsymbol{u}_j$	$\boldsymbol{A} \boldsymbol{v}_j=\sigma_j \boldsymbol{u}_j$	Since $\boldsymbol{A} \boldsymbol{v}_j=\sigma_j \boldsymbol{u}_j$, for $j=1, \dots, n$ the right singular vectors defines points in $\mathcal{S}$
lyche-numerical-linear-algebra- F02380	175	■ 1.000	$\mathcal{E}$	$\mathcal{E}$	that are mapped onto the semiaxes of $\mathcal{E}$.
lyche-numerical-linear-algebra- F02381	175	■ 1.000	$\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	$\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$	<i>Example 7.4 (Ellipse)</i> Consider the transformation $\boldsymbol{A}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by the
lyche-numerical-linear-algebra- F02382	175	■ 1.000	$\sigma_1=3, \sigma_2=1, \boldsymbol{u}_1=[3, 4]^T / 5$	$\sigma_1=3, \sigma_2=1, \boldsymbol{u}_1=[3, 4]^T / 5$	in Example 7.8. Recall that $\sigma_1=3, \sigma_2=1, \boldsymbol{u}_1=[3, 4]^T / 5$ and $\boldsymbol{u}_2=[-4, 3]^T / 5$.
lyche-numerical-linear-algebra- F02383	175	■ 1.000	$\boldsymbol{u}_2=[-4, 3]^T / 5$	$\boldsymbol{u}_2=[-4, 3]^T / 5$	in Example 7.8. Recall that $\sigma_1=3, \sigma_2=1, \boldsymbol{u}_1=[3, 4]^T / 5$ and $\boldsymbol{u}_2=[-4, 3]^T / 5$.
lyche-numerical-linear-algebra- F02384	175	■ 1.000	$y_1^2 / \sigma_1^2+y_2^2 / \sigma_2^2=1$	$y_1^2 / \sigma_1^2+y_2^2 / \sigma_2^2=1$	The ellipses $y_1^2 / \sigma_1^2+y_2^2 / \sigma_2^2=1$ and $\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_1 \tilde{\mathcal{E}}$ are shown in Fig. 7.1. Since
lyche-numerical-linear-algebra- F02385	175	■ 1.000	$\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_1 \tilde{\mathcal{E}}$	$\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_1 \tilde{\mathcal{E}}$	The ellipses $y_1^2 / \sigma_1^2+y_2^2 / \sigma_2^2=1$ and $\mathcal{E}=\boldsymbol{A} \mathcal{S}=\boldsymbol{U}_1 \tilde{\mathcal{E}}$ are shown in Fig. 7.1. Since
lyche-numerical-linear-algebra- F02386	175	■ 1.000	$\boldsymbol{A} \mathcal{S}$	$\boldsymbol{A} \mathcal{S}$	Fig. 7.1 The ellipse $y_1^2 / 9+y_2^2=1$ (left) and the rotated ellipse $\boldsymbol{A} \mathcal{S}$ (right)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F02387	176	0.997	$\boldsymbol{y}=\boldsymbol{U}_1^T \boldsymbol{x}=\left[3 / 5 x_1+4 / 5 x_2,-4 / 5 x_1+3 / 5 x_2\right]^T$	$\boldsymbol{y}=\boldsymbol{U}_1^T \boldsymbol{x}=\left[3 / 5 x_1+4 / 5 x_2,-4 / 5 x_1+3 / 5 x_2\right]^T$	$\boldsymbol{y}=\boldsymbol{U}_1^T \boldsymbol{x}=\left[3 / 5 x_1+4 / 5 x_2,-4 / 5 x_1+3 / 5 x_2\right]^T$, the equation for the ellipse on the
lyche-numerical-linear-algebra-F02388	176	1.000	$\left\ \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}\right\ _F$	$\left\ \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}\right\ _F$	1. $\left\ \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}\right\ _F$,
lyche-numerical-linear-algebra-F02389	176	1.000	$\left\ \boldsymbol{A}\right\ _F^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{: j}\right\ _2^2$	$\left\ \boldsymbol{A}\right\ _F^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{: j}\right\ _2^2$	2. $\left\ \boldsymbol{A}\right\ _F^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{: j}\right\ _2^2$,
lyche-numerical-linear-algebra-F02390	177	1.000	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F=\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F=\left\ \boldsymbol{A}\right\ _F$	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F=\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F=\left\ \boldsymbol{A}\right\ _F$	3. $\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F=\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F=\left\ \boldsymbol{A}\right\ _F$ for any unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m \times m}$ and $\boldsymbol{V} \in$
lyche-numerical-linear-algebra-F02391	177	1.000	$\boldsymbol{V} \in$	$\boldsymbol{V} \in$	3. $\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F=\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F=\left\ \boldsymbol{A}\right\ _F$ for any unitary matrices $\boldsymbol{U} \in \mathbb{C}^{m \times m}$ and $\boldsymbol{V} \in$
lyche-numerical-linear-algebra-F02392	177	0.999	$\left\ \boldsymbol{A} \boldsymbol{B}\right\ _F \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{B}\right\ _F$	$\left\ \boldsymbol{A} \boldsymbol{B}\right\ _F \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{B}\right\ _F$	4. $\left\ \boldsymbol{A} \boldsymbol{B}\right\ _F<\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{B}\right\ _F$ for any $\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}$,
lyche-numerical-linear-algebra-F02393	177	0.999	$\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}$	$\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}$	4. $\left\ \boldsymbol{A} \boldsymbol{B}\right\ _F \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{B}\right\ _F$ for any $\boldsymbol{B} \in \mathbb{C}^{n, k}, \quad k \in \mathbb{N}$,
lyche-numerical-linear-algebra-F02394	177	0.980	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2 \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{x}\right\ _2$	$\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2 \leq\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{x}\right\ _2$	5. $\left\ \boldsymbol{A} \boldsymbol{x}\right\ _2<\left\ \boldsymbol{A}\right\ _F\left\ \boldsymbol{x}\right\ _2$, for all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-F02395	177	1.000	$\left\ \boldsymbol{A}^*\right\ _F^2=\sum_{j=1}^n \sum_{i=1}^m\left \bar{a}_{i j}\right ^2=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2=\left\ \boldsymbol{A}\right\ _F^2$	$\left\ \boldsymbol{A}^*\right\ _F^2=\sum_{j=1}^n \sum_{i=1}^m\left \bar{a}_{i j}\right ^2=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2=\left\ \boldsymbol{A}\right\ _F^2$	1. $\left\ \boldsymbol{A}^*\right\ _F^2=\sum_{j=1}^n \sum_{i=1}^m\left \bar{a}_{i j}\right ^2=\sum_{i=1}^m \sum_{j=1}^n\left a_{i j}\right ^2=\left\ \boldsymbol{A}\right\ _F^2$.
lyche-numerical-linear-algebra-F02396	177	1.000	$\left\ \boldsymbol{A}\right\ _F:=\left\ \operatorname{vec}(\boldsymbol{A})\right\ _2$	$\left\ \boldsymbol{A}\right\ _F:=\left\ \operatorname{vec}(\boldsymbol{A})\right\ _2$	$\left\ \boldsymbol{A}\right\ _F:=\left\ \operatorname{vec}(\boldsymbol{A})\right\ _2$, where $\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$ is the vector obtained by stacking
lyche-numerical-linear-algebra-F02397	177	1.000	$\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$	$\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$	$\left\ \boldsymbol{A}\right\ _F:=\left\ \operatorname{vec}(\boldsymbol{A})\right\ _2$, where $\operatorname{vec}(\boldsymbol{A}) \in \mathbb{C}^{mn}$ is the vector obtained by stacking
lyche-numerical-linear-algebra-F02398	177	1.000	$\boldsymbol{U}^* \boldsymbol{U}=\boldsymbol{I}$	$\boldsymbol{U}^* \boldsymbol{U}=\boldsymbol{I}$	3. Recall that if $\boldsymbol{U}^* \boldsymbol{U}=\boldsymbol{I}$ then $\left\ \boldsymbol{U} \boldsymbol{x}\right\ _2=\left\ \boldsymbol{x}\right\ _2$ for all $\boldsymbol{x} \in \mathbb{C}^n$. Applying this to each
lyche-numerical-linear-algebra-F02399	177	1.000	$\left\ \boldsymbol{U} \boldsymbol{x}\right\ _2=\left\ \boldsymbol{x}\right\ _2$	$\left\ \boldsymbol{U} \boldsymbol{x}\right\ _2=\left\ \boldsymbol{x}\right\ _2$	3. Recall that if $\boldsymbol{U}^* \boldsymbol{U}=\boldsymbol{I}$ then $\left\ \boldsymbol{U} \boldsymbol{x}\right\ _2=\left\ \boldsymbol{x}\right\ _2$ for all $\boldsymbol{x} \in \mathbb{C}^n$. Applying this to each
lyche-numerical-linear-algebra-F02400	177	0.997	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F^2 \stackrel{2}{=} \sum_{j=1}^n\left\ \boldsymbol{U} \boldsymbol{a}_{: j}\right\ _2^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{: j}\right\ _2^2 \stackrel{2}{=} \left\ \boldsymbol{A}\right\ _F^2$	$\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F^2 \stackrel{2}{=} \sum_{j=1}^n\left\ \boldsymbol{U} \boldsymbol{a}_{: j}\right\ _2^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{: j}\right\ _2^2 \stackrel{2}{=} \left\ \boldsymbol{A}\right\ _F^2$	column $\boldsymbol{a}_{: j}$ of $\boldsymbol{A}$ we find $\left\ \boldsymbol{U} \boldsymbol{A}\right\ _F^2 \stackrel{2}{=} \sum_{j=1}^n\left\ \boldsymbol{U} \boldsymbol{a}_{: j}\right\ _2^2=\sum_{j=1}^n\left\ \boldsymbol{a}_{: j}\right\ _2^2 \stackrel{2}{=} \left\ \boldsymbol{A}\right\ _F^2$.
lyche-numerical-linear-algebra-F02401	177	0.933	$\boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{I}$	$\boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{I}$	Similarly, since $\boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{I}$ we find $\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F \stackrel{1}{=} \left\ \boldsymbol{V}^* \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}^*\right\ _F \stackrel{1}{=} \left\ \boldsymbol{A}\right\ _F$.
lyche-numerical-linear-algebra-F02402	177	0.933	$\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F \stackrel{1}{=} \left\ \boldsymbol{V}^* \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}^*\right\ _F \stackrel{1}{=} \left\ \boldsymbol{A}\right\ _F$	$\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F \stackrel{1}{=} \left\ \boldsymbol{V}^* \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}^*\right\ _F \stackrel{1}{=} \left\ \boldsymbol{A}\right\ _F$	Similarly, since $\boldsymbol{V} \boldsymbol{V}^*=\boldsymbol{I}$ we find $\left\ \boldsymbol{A} \boldsymbol{V}\right\ _F \stackrel{1}{=} \left\ \boldsymbol{V}^* \boldsymbol{A}^*\right\ _F=\left\ \boldsymbol{A}^*\right\ _F \stackrel{1}{=} \left\ \boldsymbol{A}\right\ _F$.
lyche-numerical-linear-algebra-F02403	177	0.986	$\left\ \boldsymbol{v}\right\ _F=\left\ \boldsymbol{v}\right\ _2$	$\left\ \boldsymbol{v}\right\ _F=\left\ \boldsymbol{v}\right\ _2$	5. Since $\left\ \boldsymbol{v}\right\ _F=\left\ \boldsymbol{v}\right\ _2$ for a vector this follows by taking $k=1$ and $\boldsymbol{B}=\boldsymbol{x}$ in 4.
lyche-numerical-linear-algebra-F02404	177	0.986	$\boldsymbol{B}=\boldsymbol{x}$	$\boldsymbol{B}=\boldsymbol{x}$	5. Since $\left\ \boldsymbol{v}\right\ _F=\left\ \boldsymbol{v}\right\ _2$ for a vector this follows by taking $k=1$ and $\boldsymbol{B}=\boldsymbol{x}$ in 4.
lyche-numerical-linear-algebra-F02405	177	0.996	$\left\ \boldsymbol{A}\right\ _F=$	$\left\ \boldsymbol{A}\right\ _F=$	Theorem 7.5 (Frobenius Norm and Singular Values) We have $\left\ \boldsymbol{A}\right\ _F=$
lyche-numerical-linear-algebra-F02406	177	0.999	$\sqrt{\sigma_1^2+\cdots+\sigma_n^2}$	$\sqrt{\sigma_1^2+\cdots+\sigma_n^2}$	$\sqrt{\sigma_1^2+\cdots+\sigma_n^2}$, where $\sigma_1, \ldots, \sigma_n$ are the singular values of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02407	177	0.999	$\sigma_1, \ldots, \sigma_n$	$\sigma_1, \ldots, \sigma_n$	$\sqrt{\sigma_1^2+\cdots+\sigma_n^2}$, where $\sigma_1, \ldots, \sigma_n$ are the singular values of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02408	177	1.000	$m \geq n \geq 1$	$m \geq n \geq 1$	Suppose $m \geq n \geq 1$ and $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has a singular value decomposition $\boldsymbol{A}=$

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lyche-numerical-linear-algebra-F02409	177	■ 1.000	$\boldsymbol{U}\left[\begin{array}{c}\boldsymbol{D} \\ \mathbf{0}\end{array}\right]\boldsymbol{V}^*$	$U \begin{bmatrix} D \\ 0 \end{bmatrix} V^*$	$U \begin{bmatrix} D \\ 0 \end{bmatrix} V^*$, where $D = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02410	177	■ 1.000	$\boldsymbol{D}=\operatorname{diag}\left(\sigma_1,\ldots,\sigma_n\right)$	$D = \text{diag}(\sigma_1, \dots, \sigma_n)$	$U \begin{bmatrix} D \\ 0 \end{bmatrix} V^*$, where $D = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02411	177	■ 1.000	$\epsilon>0$	$\epsilon > 0$	$U \begin{bmatrix} D \\ 0 \end{bmatrix} V^*$, where $D = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02412	177	■ 1.000	$1\leq r\leq n$	$1 \leq r \leq n$	$U \begin{bmatrix} D \\ 0 \end{bmatrix} V^*$, where $D = \text{diag}(\sigma_1, \dots, \sigma_n)$. We choose $\epsilon > 0$ and let $1 \leq r \leq n$ be
lyche-numerical-linear-algebra-F02413	177	■ 1.000	$\sigma_{r+1}^2+\cdots+\sigma_n^2<\epsilon^2$	$\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$	the smallest integer such that $\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$. Define $A' := U \begin{bmatrix} D' \\ 0 \end{bmatrix} V^*$, where
lyche-numerical-linear-algebra-F02414	177	■ 1.000	$\boldsymbol{A}^{\prime}:=\boldsymbol{U}\left[\begin{array}{c}\boldsymbol{D}^{\prime} \\ \mathbf{0}\end{array}\right]\boldsymbol{V}^*$	$A' := U \begin{bmatrix} D' \\ 0 \end{bmatrix} V^*$	the smallest integer such that $\sigma_{r+1}^2 + \cdots + \sigma_n^2 < \epsilon^2$. Define $A' := U \begin{bmatrix} D' \\ 0 \end{bmatrix} V^*$, where
lyche-numerical-linear-algebra-F02415	177	■ 1.000	$\boldsymbol{D}^{\prime}:=\operatorname{diag}\left(\sigma_1,\ldots,\sigma_r,0,\ldots,0\right)\in\mathbb{R}^{n\times n}$	$D' := \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0) \in \mathbb{R}^{n \times n}$	$D' := \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0) \in \mathbb{R}^{n \times n}$. By Lemma 7.1
lyche-numerical-linear-algebra-F02416	178	■ 1.000	$\boldsymbol{A}^{\prime}$	A'	Thus, if ϵ is small then A is near a matrix A' of rank r . This can be used to determine
lyche-numerical-linear-algebra-F02417	178	■ 1.000	$\sqrt{\sigma_{r+1}^2+\cdots+\sigma_n^2}$	$\sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2}$	rank numerically. We choose an r such that $\sqrt{\sigma_{r+1}^2 + \cdots + \sigma_n^2}$ is “small”. Then we
lyche-numerical-linear-algebra-F02418	178	■ 0.972	$\operatorname{rank}(\boldsymbol{A})=r$	$\text{rank}(A) = r$	postulate that $\text{rank}(A) = r$ since A is close to a matrix of rank r .
lyche-numerical-linear-algebra-F02419	178	■ 1.000	$r,\boldsymbol{A}^{\prime}$	r, A'	The following theorem shows that of all $m \times n$ matrices of rank r , A' is closest
lyche-numerical-linear-algebra-F02420	178	■ 1.000	$\sigma_1\geq\cdots\geq\sigma_n\geq 0$	$\sigma_1 \geq \cdots \geq \sigma_n \geq 0$	values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. For any $r \leq \text{rank}(A)$ we have
lyche-numerical-linear-algebra-F02421	178	■ 1.000	$r\leq\operatorname{rank}(\boldsymbol{A})$	$r \leq \text{rank}(A)$	values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. For any $r \leq \text{rank}(A)$ we have
lyche-numerical-linear-algebra-F02422	178	■ 1.000	$\boldsymbol{A}=\left[\begin{array}{l}3\\4\end{array}\right]$	$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$	a) $A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
lyche-numerical-linear-algebra-F02423	178	■ 1.000	$\boldsymbol{A}=\left[\begin{array}{ll}1&1\\2&2\\2&2\end{array}\right]$	$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$	b) $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$.
lyche-numerical-linear-algebra-F02424	179	■ 1.000	$\boldsymbol{A}=\boldsymbol{e}_1$	$A = e_1$	a) $A = e_1$ the first unit vector in $\mathbb{R}^m$.
lyche-numerical-linear-algebra-F02425	179	■ 1.000	$\boldsymbol{A}=\boldsymbol{e}_n^T$	$A = e_n^T$	b) $A = e_n^T$ the last unit vector in $\mathbb{R}^n$.
lyche-numerical-linear-algebra-F02426	179	■ 1.000	$\boldsymbol{A}=\left[\begin{array}{rr}-1&0\\0&3\end{array}\right]$	$A = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$	c) $A = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$.
lyche-numerical-linear-algebra-F02427	179	■ 1.000	$\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}$	$U \Sigma V$	$U \Sigma V$ is a singular value decomposition of $A \in \mathbb{C}^{m \times n}$ then
lyche-numerical-linear-algebra-F02428	179	■ 1.000	$\boldsymbol{A}^*\boldsymbol{A}=\boldsymbol{V}\operatorname{diag}\left(\sigma_1^2,\ldots,\sigma_n^2\right)\boldsymbol{V}^*$	$A^* A = V \text{diag}(\sigma_1^2, \dots, \sigma_n^2) V^*$	a) $A^* A = V \text{diag}(\sigma_1^2, \dots, \sigma_n^2) V^*$ is a spectral decomposition of $A^* A$.
lyche-numerical-linear-algebra-F02429	179	■ 1.000	$\boldsymbol{A}\boldsymbol{A}^*=\boldsymbol{U}\operatorname{diag}\left(\sigma_1^2,\ldots,\sigma_m^2\right)\boldsymbol{U}^*$	$AA^* = U \text{diag}(\sigma_1^2, \dots, \sigma_m^2) U^*$	b) $AA^* = U \text{diag}(\sigma_1^2, \dots, \sigma_m^2) U^*$ is a spectral decomposition of AA^* .
lyche-numerical-linear-algebra-F02430	179	■ 0.803	$\left(\sigma_1,\ldots,\sigma_n\right)$	$(\sigma_1, \dots, \sigma_n)$	“If $A \in \mathbb{R}^{n \times n}$ has singular values $\sigma_1, \dots, \sigma_n$ then A^2 has singular values
lyche-numerical-linear-algebra-F02431	179	■ 0.803	$\boldsymbol{A}^2$	A^2	“If $A \in \mathbb{R}^{n \times n}$ has singular values $(\sigma_1, \dots, \sigma_n)$ then A^2 has singular values

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lyche-numerical-linear-algebra- F02432	179	1.000	$\left(\sigma_1^2, \dots, \sigma_n^2\right)$	$(\sigma_1^2, \dots, \sigma_n^2)$	$\sigma_1^2, \dots, \sigma_n^2$ ". Find a class of matrices for which the statement is true. Show that
lyche-numerical-linear-algebra- F02433	179	0.884	$\frac{1}{25}\begin{bmatrix} 11 & 48 \\ 48 & 39 \end{bmatrix}$	$A = \frac{1}{25} \begin{bmatrix} 11 & 48 \\ 48 & 39 \end{bmatrix}$	$A = \frac{1}{25} \begin{bmatrix} 11 & 48 \\ 48 & 39 \end{bmatrix}$. Also, using possibly a computer, find its spectral decomposition
lyche-numerical-linear-algebra- F02434	179	0.998	$\boldsymbol{U} \boldsymbol{D} \boldsymbol{U}^T$	UDU^T	UDU^T . The matrix A is normal, but the spectral decomposition is not an SVD.
lyche-numerical-linear-algebra- F02435	179	1.000	$\frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}\frac{1}{5}\begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	$A = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$	¹ Answer: $A = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$.
lyche-numerical-linear-algebra- F02436	180	1.000	$\operatorname{rank}(\boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A}^*)$	$\operatorname{rank}(A) = \operatorname{rank}(A^*)$	a) $\operatorname{rank}(A) = \operatorname{rank}(A^*)$.
lyche-numerical-linear-algebra- F02437	180	1.000	$\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}) = n$	$\operatorname{rank}(A) + \operatorname{null}(A) = n$	b) $\operatorname{rank}(A) + \operatorname{null}(A) = n$,
lyche-numerical-linear-algebra- F02438	180	1.000	$\operatorname{rank}(\boldsymbol{A}) + \operatorname{null}(\boldsymbol{A}^*) = m$	$\operatorname{rank}(A) + \operatorname{null}(A^*) = m$	c) $\operatorname{rank}(A) + \operatorname{null}(A^*) = m$,
lyche-numerical-linear-algebra- F02439	180	1.000	$\operatorname{null}(\boldsymbol{A})$	$\operatorname{null}(A)$	where $\operatorname{null}(A)$ is defined as the dimension of $\mathcal{N}(A)$.
lyche-numerical-linear-algebra- F02440	180	1.000	$\operatorname{rank} \boldsymbol{A} = \operatorname{rank}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{rank}(\boldsymbol{A} \boldsymbol{A}^*)$	$\operatorname{rank} A = \operatorname{rank}(A^* A) = \operatorname{rank}(A A^*)$	a) $\operatorname{rank} A = \operatorname{rank}(A^* A) = \operatorname{rank}(A A^*)$,
lyche-numerical-linear-algebra- F02441	180	0.997	$\operatorname{null}(\boldsymbol{A}^* \boldsymbol{A}) = \operatorname{null} \boldsymbol{A}$	$\operatorname{null}(A^* A) = \operatorname{null} A$	b) $\operatorname{null}(A^* A) = \operatorname{null} A$, and $\operatorname{null}(A A^*) = \operatorname{null}(A^*)$.
lyche-numerical-linear-algebra- F02442	180	0.997	$\operatorname{null}(\boldsymbol{A} \boldsymbol{A}^*) = \operatorname{null}(\boldsymbol{A}^*)$	$\operatorname{null}(A A^*) = \operatorname{null}(A^*)$	b) $\operatorname{null}(A^* A) = \operatorname{null} A$, and $\operatorname{null}(A A^*) = \operatorname{null}(A^*)$.
lyche-numerical-linear-algebra- F02443	180	1.000	$\mathcal{R}(\boldsymbol{B})$	$\mathcal{R}(B)$	Give orthonormal bases for $\mathcal{R}(B)$ and $\mathcal{N}(B)$.
lyche-numerical-linear-algebra- F02444	180	1.000	$\mathcal{N}(\boldsymbol{B})$	$\mathcal{N}(B)$	Give orthonormal bases for $\mathcal{R}(B)$ and $\mathcal{N}(B)$.
lyche-numerical-linear-algebra- F02445	180	1.000	$\mathcal{R}(\boldsymbol{A}^* \boldsymbol{A}) =$	$\mathcal{R}(A^* A) =$	Exercise 7.13 (Some Spanning Sets) Show for any $A \in \mathbb{C}^{m \times n}$ that $\mathcal{R}(A^* A) =$
lyche-numerical-linear-algebra- F02446	180	1.000	$\mathcal{R}(\boldsymbol{V}_1) = \mathcal{R}(\boldsymbol{A}^*)$	$\mathcal{R}(V_1) = \mathcal{R}(A^*)$	$\mathcal{R}(V_1) = \mathcal{R}(A^*)$
lyche-numerical-linear-algebra- F02447	180	1.000	$\boldsymbol{u}_1, \dots, \boldsymbol{u}_m \in \mathbb{C}^m$	$u_1, \dots, u_m \in \mathbb{C}^m$	$u_1, \dots, u_m \in \mathbb{C}^m$, and right singular vectors $v_1, \dots, v_n \in \mathbb{C}^n$. Show that the matrix
lyche-numerical-linear-algebra- F02448	180	1.000	$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \in \mathbb{C}^n$	$v_1, \dots, v_n \in \mathbb{C}^n$	$u_1, \dots, u_m \in \mathbb{C}^m$, and right singular vectors $v_1, \dots, v_n \in \mathbb{C}^n$. Show that the matrix
lyche-numerical-linear-algebra- F02449	181	1.000	$\boldsymbol{B} :=$	$B :=$	c) Use the singular value decomposition of A to give a suitable definition of $B :=$
lyche-numerical-linear-algebra- F02450	181	0.981	$\sqrt{\boldsymbol{A}^T \boldsymbol{A}}$	$\sqrt{A^T A}$	$\sqrt{A^T A}$ so that $P = B$.
lyche-numerical-linear-algebra- F02451	181	0.981	$\boldsymbol{P} = \boldsymbol{B}$	$P = B$	$\sqrt{A^T A}$ so that $P = B$.
lyche-numerical-linear-algebra- F02452	181	1.000	$\boldsymbol{X}_k = \boldsymbol{U} \boldsymbol{\Sigma}_k \boldsymbol{V}^T$	$X_k = U \Sigma_k V^T$	d) Show that the iteration (7.14) is well defined by showing that $X_k = U \Sigma_k V^T$,
lyche-numerical-linear-algebra- F02453	181	0.985	$\boldsymbol{\Sigma}_k$	Σ_k	for a diagonal matrix Σ_k with positive diagonal elements, $k = 0, 1, 2, \dots$
lyche-numerical-linear-algebra- F02454	181	0.985	$k=0, 1, 2, \dots$	$k = 0, 1, 2, \dots$	for a diagonal matrix Σ_k with positive diagonal elements, $k = 0, 1, 2, \dots$.
lyche-numerical-linear-algebra- F02455	181	0.749	$[\boldsymbol{Q}, \boldsymbol{P}, \boldsymbol{k}] =$	$[Q, P, k] =$	function <code>[Q, P, k] = polardecomp(A, tol, K)</code> to carry out the iter-

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lyche-numerical-linear-algebra-F02456	181	0.749	$(\mathrm{A}, \mathrm{tol}, \mathrm{K})$	(A, tol, K)	function $[Q, P, k] = \text{polardecomp}(A, \mathrm{tol}, K)$ to carry out the iter-
lyche-numerical-linear-algebra-F02457	181	1.000	$\boldsymbol{P} = \boldsymbol{Q}^T \boldsymbol{A}$	$\boldsymbol{P} = \boldsymbol{Q}^T \boldsymbol{A}$	ation in (7.14). The output is approximations $\boldsymbol{Q}$ and $\boldsymbol{P} = \boldsymbol{Q}^T \boldsymbol{A}$ to the polar
lyche-numerical-linear-algebra-F02458	181	0.977	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$	$\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$	decomposition $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{P}$ of $\boldsymbol{A}$ and the number of iterations k such that
lyche-numerical-linear-algebra-F02459	181	0.919	$\left\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k \right\ _F <$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F <$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F < \mathrm{tol} * \ \boldsymbol{X}_{k+1}\ _F$. Set $k = K + 1$ if convergence is not achieved
lyche-numerical-linear-algebra-F02460	181	0.919	$* \left\ \boldsymbol{X}_{k+1} \right\ _F$	$* \ \boldsymbol{X}_{k+1}\ _F$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F < \mathrm{tol} * \ \boldsymbol{X}_{k+1}\ _F$. Set $k = K + 1$ if convergence is not achieved
lyche-numerical-linear-algebra-F02461	181	0.919	$k = K + 1$	$k = K + 1$	$\ \boldsymbol{X}_{k+1} - \boldsymbol{X}_k\ _F < \mathrm{tol} * \ \boldsymbol{X}_{k+1}\ _F$. Set $k = K + 1$ if convergence is not achieved
lyche-numerical-linear-algebra-F02462	181	1.000	$\ \boldsymbol{A}\ _1$	$\ \boldsymbol{A}\ _1$	Compute $\ \boldsymbol{A}\ _1$ and $\ \boldsymbol{A}\ _\infty$.
lyche-numerical-linear-algebra-F02463	181	1.000	$\ \boldsymbol{A}\ _\infty$	$\ \boldsymbol{A}\ _\infty$	Compute $\ \boldsymbol{A}\ _1$ and $\ \boldsymbol{A}\ _\infty$.
lyche-numerical-linear-algebra-F02464	181	0.972	$\left(\boldsymbol{B}^T \right)$	$(\boldsymbol{B}^T)$	Find the spaces $\text{span}(\boldsymbol{B}^T)$ and $\ker(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F02465	181	0.972	$\ker(\boldsymbol{B})$	$\ker(\boldsymbol{B})$	Find the spaces $\text{span}(\boldsymbol{B}^T)$ and $\ker(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F02466	182	1.000	$\boldsymbol{x} \in \mathbb{R}^3$	$\boldsymbol{x} \in \mathbb{R}^3$	Find the solution $\boldsymbol{x} \in \mathbb{R}^3$ with $\ \boldsymbol{x}\ _2$ as small as possible.
lyche-numerical-linear-algebra-F02467	182	1.000	$\boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{b} \in \mathbb{R}^m$	d) Let $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ be a matrix with linearly independent columns, and $\boldsymbol{b} \in \mathbb{R}^m$
lyche-numerical-linear-algebra-F02468	182	1.000	$\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$	$\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$	the normal equations $\boldsymbol{A}^T \boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}^T \boldsymbol{b}$. Will the method converge? Justify your
lyche-numerical-linear-algebra-F02469	182	1.000	$\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$	$\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$	a) Give orthonormal bases for $\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A}^T)$.
lyche-numerical-linear-algebra-F02470	182	1.000	$\mathcal{N}(\boldsymbol{A}^T)$	$\mathcal{N}(\boldsymbol{A}^T)$	a) Give orthonormal bases for $\mathcal{R}(\boldsymbol{A}), \mathcal{R}(\boldsymbol{A}^T), \mathcal{N}(\boldsymbol{A})$ and $\mathcal{N}(\boldsymbol{A}^T)$.
lyche-numerical-linear-algebra-F02471	182	0.999	$\boldsymbol{B} \in \mathbb{R}^{4,3}$	$\boldsymbol{B} \in \mathbb{R}^{4,3}$	b) Explain why for all matrices $\boldsymbol{B} \in \mathbb{R}^{4,3}$ of rank one we have $\ \boldsymbol{A} - \boldsymbol{B}\ _F \geq 6$.
lyche-numerical-linear-algebra-F02472	182	0.999	$\ \boldsymbol{A} - \boldsymbol{B}\ _F \geq 6$	$\ \boldsymbol{A} - \boldsymbol{B}\ _F \geq 6$	b) Explain why for all matrices $\boldsymbol{B} \in \mathbb{R}^{4,3}$ of rank one we have $\ \boldsymbol{A} - \boldsymbol{B}\ _F > 6$.
lyche-numerical-linear-algebra-F02473	182	1.000	$\ \boldsymbol{A} - \boldsymbol{A}_1\ _F = 6$	$\ \boldsymbol{A} - \boldsymbol{A}_1\ _F = 6$	c) Give a matrix $\boldsymbol{A}_1$ of rank one such that $\ \boldsymbol{A} - \boldsymbol{A}_1\ _F = 6$.
lyche-numerical-linear-algebra-F02474	182	0.980	$(n, 1)$	$(n, 1)$	diagonal equal to -1 . Let $\boldsymbol{B}$ be the matrix obtained from $\boldsymbol{A}$ by changing the $(n, 1)$
lyche-numerical-linear-algebra-F02475	182	1.000	-2^{2-n}	-2^{2-n}	element from zero to -2^{2-n} .
lyche-numerical-linear-algebra-F02476	182	1.000	$\boldsymbol{x} := [2^{n-2}, 2^{n-3}, \dots, 2^0, 1]^T$	$\boldsymbol{x} := [2^{n-2}, 2^{n-3}, \dots, 2^0, 1]^T$	a) Show that $\boldsymbol{B} \boldsymbol{x} = \mathbf{0}$, where $\boldsymbol{x} := [2^{n-2}, 2^{n-3}, \dots, 2^0, 1]^T$. Conclude that $\boldsymbol{B}$ is
lyche-numerical-linear-algebra-F02477	182	1.000	$\det(\boldsymbol{A}) = 1$	$\det(\boldsymbol{A}) = 1$	singular, $\det(\boldsymbol{A}) = 1$, and $\ \boldsymbol{A} - \boldsymbol{B}\ _F = 2^{2-n}$. Thus even if $\det(\boldsymbol{A})$ is not small
lyche-numerical-linear-algebra-F02478	182	1.000	$\ \boldsymbol{A} - \boldsymbol{B}\ _F = 2^{2-n}$	$\ \boldsymbol{A} - \boldsymbol{B}\ _F = 2^{2-n}$	singular, $\det(\boldsymbol{A}) = 1$, and $\ \boldsymbol{A} - \boldsymbol{B}\ _F = 2^{2-n}$. Thus even if $\det(\boldsymbol{A})$ is not small
lyche-numerical-linear-algebra-F02479	182	1.000	$\boldsymbol{A} - \boldsymbol{B}$	$\boldsymbol{A} - \boldsymbol{B}$	the Frobenius norm of $\boldsymbol{A} - \boldsymbol{B}$ is small for large n , and the matrix $\boldsymbol{A}$ is very close
lyche-numerical-linear-algebra-F02480	182	1.000	σ_n	σ_n	b) Use Theorem 7.6 to show that the smallest singular vale σ_n of $\boldsymbol{A}$ is bounded
lyche-numerical-linear-algebra-F02481	182	1.000	2^{2-n}	2^{2-n}	above by 2^{2-n} .
lyche-numerical-linear-algebra-F02482	183	1.000	$\ \boldsymbol{A}\ _1, \ \boldsymbol{A}\ _\infty$	$\ \boldsymbol{A}\ _1, \ \boldsymbol{A}\ _\infty$	Compute $\ \boldsymbol{A}\ _1, \ \boldsymbol{A}\ _\infty$ and $\ \boldsymbol{A}\ _F$.

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lyche-numerical-linear-algebra-183-F02483	183	■ 1.000	$\ \boldsymbol{A}\ _F$	$\ \boldsymbol{A}\ _F$	Compute $\ \boldsymbol{A}\ _1$, $\ \boldsymbol{A}\ _\infty$ and $\ \boldsymbol{A}\ _F$.
lyche-numerical-linear-algebra-183-F02484	183	■ 1.000	$\boldsymbol{T}=\boldsymbol{L} \boldsymbol{L}^T$	$\boldsymbol{T} = \boldsymbol{L}\boldsymbol{L}^T$	$\boldsymbol{T} = \boldsymbol{L}\boldsymbol{L}^T$ of $\boldsymbol{T}$.
lyche-numerical-linear-algebra-183-F02485	183	■ 1.000	$\boldsymbol{A}^{\prime}=\sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$	$\boldsymbol{A}' = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$	$m \geq n$, and let $\boldsymbol{A}' = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$, where $1 \leq r \leq n$, σ_i are the singular values
lyche-numerical-linear-algebra-183-F02486	183	■ 1.000	$1 \leq r \leq n, \sigma_i$	$1 \leq r \leq n, \sigma_i$	$m \geq n$, and let $\boldsymbol{A}' = \sum_{i=1}^r \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^*$, where $1 \leq r \leq n$, σ_i are the singular values
lyche-numerical-linear-algebra-183-F02487	183	■ 1.000	$\boldsymbol{u}_i, \boldsymbol{v}_i$	$\boldsymbol{u}_i, \boldsymbol{v}_i$	of $\boldsymbol{A}$, and where $\boldsymbol{u}_i, \boldsymbol{v}_i$ are the columns of $\boldsymbol{U}$ and $\boldsymbol{V}$. Prove that
lyche-numerical-linear-algebra-185-F02488	185	■ 1.000	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$	$\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$	space $\mathcal{V}$ is a function $\ \cdot\ : \mathcal{V} \rightarrow \mathbb{R}$ that satisfies for all $\boldsymbol{x}, \boldsymbol{y}$ in $\mathcal{V}$ and all a in $\mathbb{R}$
lyche-numerical-linear-algebra-185-F02489	185	■ 0.728	$(\mathcal{V}, \mathbb{R}, \ \cdot\)$	$(\mathcal{V}, \mathbb{R}, \ \cdot\)$	The triple $(\mathcal{V}, \mathbb{R}, \ \cdot\)$ (resp. $(\mathcal{V}, \mathbb{C}, \ \cdot\)$) is called a normed vector space and the
lyche-numerical-linear-algebra-185-F02490	185	■ 0.728	$(\mathcal{V}, \mathbb{C}, \ \cdot\)$	$(\mathcal{V}, \mathbb{C}, \ \cdot\)$	The triple $(\mathcal{V}, \mathbb{R}, \ \cdot\)$ (resp. $(\mathcal{V}, \mathbb{C}, \ \cdot\)$) is called a normed vector space and the
lyche-numerical-linear-algebra-185-F02491	185	■ 1.000	$\mathbb{R}^n, \mathbb{C}^n$	$\mathbb{R}^n, \mathbb{C}^n$	In this book the vector space will be one of $\mathbb{R}^n, \mathbb{C}^n$ or one of the matrix spaces
lyche-numerical-linear-algebra-185-F02492	185	■ 1.000	$\mathcal{V}=\mathbb{C}^n$	$\mathcal{V} = \mathbb{C}^n$	book we will use the following family of vector norms on $\mathcal{V} = \mathbb{C}^n$ and $\mathcal{V} = \mathbb{R}^n$.
lyche-numerical-linear-algebra-185-F02493	185	■ 1.000	$\mathcal{V}=\mathbb{R}^n$	$\mathcal{V} = \mathbb{R}^n$	book we will use the following family of vector norms on $\mathcal{V} = \mathbb{C}^n$ and $\mathcal{V} = \mathbb{R}^n$.
lyche-numerical-linear-algebra-186-F02494	186	■ 0.987	$p \geq 1$	$p \geq 1$	Definition 8.2 (Vector p-Norms) We define for $p \geq 1$ and $\boldsymbol{x} \in \mathbb{R}^n$ or $\boldsymbol{x} \in \mathbb{C}^n$ the
lyche-numerical-linear-algebra-186-F02495	186	■ 1.000	$p=1,2, \infty$	$p = 1, 2, \infty$	The most important cases are $p = 1, 2, \infty$:
lyche-numerical-linear-algebra-186-F02496	186	■ 1.000	$\ \boldsymbol{x}\ _1:=\sum_{j=1}^n\left x_{j}\right $	$\ \boldsymbol{x}\ _1 := \sum_{j=1}^n x_j $	1. $\ \boldsymbol{x}\ _1 := \sum_{j=1}^n x_j $,
lyche-numerical-linear-algebra-186-F02497	186	■ 1.000	$\ \boldsymbol{x}\ _2:=\sqrt{\sum_{j=1}^n\left x_{j}\right ^2}$	$\ \boldsymbol{x}\ _2 := \sqrt{\sum_{j=1}^n x_j ^2}$	2. $\ \boldsymbol{x}\ _2 := \sqrt{\sum_{j=1}^n x_j ^2}$,
lyche-numerical-linear-algebra-186-F02498	186	■ 1.000	l_2	l_2	(the two-norm, l_2 -norm, or Euclidian norm)
lyche-numerical-linear-algebra-186-F02499	186	■ 1.000	$\ \boldsymbol{x}\ _{\infty}:=\max _{1 \leq j \leq n}\left x_{j}\right $	$\ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j $	3. $\ \boldsymbol{x}\ _\infty := \max_{1 \leq j \leq n} x_j $,
lyche-numerical-linear-algebra-186-F02500	186	■ 1.000	l_{∞}	l_∞	(the infinity-norm, l_∞ -norm, or max norm)
lyche-numerical-linear-algebra-186-F02501	186	■ 1.000	$1 \leq p \leq \infty$	$1 \leq p \leq \infty$	1. In Sect. 8.4, we show that the p -norms are vector norms for $1 < p < \infty$.
lyche-numerical-linear-algebra-186-F02502	186	■ 1.000	$\ \boldsymbol{x}+\boldsymbol{y}\ _p \leq\left\ \boldsymbol{x}\right\ _p+\left\ \boldsymbol{y}\right\ _p$	$\ \boldsymbol{x} + \boldsymbol{y}\ _p \leq \ \boldsymbol{x}\ _p + \ \boldsymbol{y}\ _p$	2. The triangle inequality $\ \boldsymbol{x} + \boldsymbol{y}\ _p \leq \ \boldsymbol{x}\ _p + \ \boldsymbol{y}\ _p$ is called Minkowski's
lyche-numerical-linear-algebra-186-F02503	186	■ 1.000	$\frac{1}{p}+\frac{1}{q}=1$	$\frac{1}{p} + \frac{1}{q} = 1$	The relation $\frac{1}{p} + \frac{1}{q} = 1$ means that if $p = 1$ then $q = \infty$ and vice versa.
lyche-numerical-linear-algebra-186-F02504	186	■ 1.000	$p=1$	$p = 1$	The relation $\frac{1}{p} + \frac{1}{q} = 1$ means that if $p = 1$ then $q = \infty$ and vice versa.
lyche-numerical-linear-algebra-186-F02505	186	■ 1.000	$q=\infty$	$q = \infty$	The relation $\frac{1}{p} + \frac{1}{q} = 1$ means that if $p = 1$ then $q = \infty$ and vice versa.
lyche-numerical-linear-algebra-186-F02506	186	■ 1.000	$p=2$	$p = 2$	Theorem 5.1) for the Euclidian norm $p = 2$.
lyche-numerical-linear-algebra-186-F02507	186	■ 1.000	$\lim _{p \rightarrow \infty} n^{1 / p}=1$	$\lim_{p \rightarrow \infty} n^{1/p} = 1$	Since $\lim_{p \rightarrow \infty} n^{1/p} = 1$ for any fixed $n \in \mathbb{N}$, we see that (8.4) follows.

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lyche-numerical-linear-algebra-F02508	187	■ 0.987	$\ \cdot\ $	$\ \cdot\ $	Definition 8.3 (Equivalent Norms) We say that two norms $\ \cdot\ $ and $\ \cdot\ '$ on $\mathcal{V}$ are
lyche-numerical-linear-algebra-F02509	187	■ 0.987	$\ \cdot\ '$	$\ \cdot\ '$	Definition 8.3 (Equivalent Norms) We say that two norms $\ \cdot\ $ and $\ \cdot\ '$ on $\mathcal{V}$ are
lyche-numerical-linear-algebra-F02510	187	■ 1.000	M	M	equivalent if there are positive constants m and M such that for all vectors $\mathbf{x} \in \mathcal{V}$
lyche-numerical-linear-algebra-F02511	187	■ 1.000	$\mathbf{x} \in \mathcal{V}$	$\mathbf{x} \in \mathcal{V}$	equivalent if there are positive constants m and M such that for all vectors $\mathbf{x} \in \mathcal{V}$
lyche-numerical-linear-algebra-F02512	187	■ 1.000	∞	∞	By (8.5) the p - and ∞ -norms are equivalent for any $p \geq 1$. This result is
lyche-numerical-linear-algebra-F02513	187	■ 1.000	$\ \mathbf{x} - \mathbf{y}\ \geq \left \ \mathbf{x}\ - \ \mathbf{y}\ \right $	$\ \mathbf{x} - \mathbf{y}\ \geq \left \ \mathbf{x}\ - \ \mathbf{y}\ \right $	1. $\ \mathbf{x} - \mathbf{y}\ \geq \left \ \mathbf{x}\ - \ \mathbf{y}\ \right $, for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ (inverse triangle inequality).
lyche-numerical-linear-algebra-F02514	187	■ 1.000	$\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$	$\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$	1. $\ \mathbf{x} - \mathbf{y}\ \geq \left \ \mathbf{x}\ - \ \mathbf{y}\ \right $, for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$ (inverse triangle inequality).
lyche-numerical-linear-algebra-F02515	187	■ 1.000	$\mathcal{V} \rightarrow \mathbb{R}$	$\mathcal{V} \rightarrow \mathbb{R}$	2. The vector norm is a continuous function $\mathcal{V} \rightarrow \mathbb{R}$.
lyche-numerical-linear-algebra-F02516	187	■ 1.000	$\ \mathbf{x}\ = \ \mathbf{x} - \mathbf{y} + \mathbf{y}\ \leq \ \mathbf{x} - \mathbf{y}\ + \ \mathbf{y}\ $	$\ \mathbf{x}\ = \ \mathbf{x} - \mathbf{y} + \mathbf{y}\ \leq \ \mathbf{x} - \mathbf{y}\ + \ \mathbf{y}\ $	1. Since $\ \mathbf{x}\ = \ \mathbf{x} - \mathbf{y} + \mathbf{y}\ \leq \ \mathbf{x} - \mathbf{y}\ + \ \mathbf{y}\ $ we obtain $\ \mathbf{x} - \mathbf{y}\ \geq \ \mathbf{x}\ - \ \mathbf{y}\ $. By
lyche-numerical-linear-algebra-F02517	187	■ 1.000	$\ \mathbf{x} - \mathbf{y}\ \geq \ \mathbf{x}\ - \ \mathbf{y}\ $	$\ \mathbf{x} - \mathbf{y}\ \geq \ \mathbf{x}\ - \ \mathbf{y}\ $	1. Since $\ \mathbf{x}\ = \ \mathbf{x} - \mathbf{y} + \mathbf{y}\ \leq \ \mathbf{x} - \mathbf{y}\ + \ \mathbf{y}\ $ we obtain $\ \mathbf{x} - \mathbf{y}\ \geq \ \mathbf{x}\ - \ \mathbf{y}\ $. By
lyche-numerical-linear-algebra-F02518	187	■ 1.000	$\ \mathbf{x} - \mathbf{y}\ = \ \mathbf{y} - \mathbf{x}\ \geq \ \mathbf{y}\ - \ \mathbf{x}\ $	$\ \mathbf{x} - \mathbf{y}\ = \ \mathbf{y} - \mathbf{x}\ \geq \ \mathbf{y}\ - \ \mathbf{x}\ $	symmetry $\ \mathbf{x} - \mathbf{y}\ = \ \mathbf{y} - \mathbf{x}\ \geq \ \mathbf{y}\ - \ \mathbf{x}\ $ and we obtain the inverse triangle
lyche-numerical-linear-algebra-F02519	187	■ 1.000	$f: \mathcal{S} \rightarrow \mathbb{R}$	$f: \mathcal{S} \rightarrow \mathbb{R}$	The set $\mathcal{S}$ is a closed and bounded set and the function $f: \mathcal{S} \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra-F02520	187	■ 1.000	$f(\mathbf{y}) = \ \mathbf{y}\ $	$f(\mathbf{y}) = \ \mathbf{y}\ $	$f(\mathbf{y}) = \ \mathbf{y}\ $ is continuous by what we just showed. Therefore f attains its
lyche-numerical-linear-algebra-F02521	187	■ 1.000	$\mathbf{y} := \mathbf{x} / \ \mathbf{x}\ ' \in \mathcal{S}$	$\mathbf{y} := \mathbf{x} / \ \mathbf{x}\ ' \in \mathcal{S}$	For any $\mathbf{x} \in \mathcal{V}$ we have $\mathbf{y} := \mathbf{x} / \ \mathbf{x}\ ' \in \mathcal{S}$, and (8.7) follows if we apply (8.8) to
lyche-numerical-linear-algebra-F02522	188	■ 0.997	$(\mathbb{C}^{m \times n}, \mathbb{C})$	$(\mathbb{C}^{m \times n}, \mathbb{C})$	For simplicity we consider only norms on the vector space $(\mathbb{C}^{m \times n}, \mathbb{C})$. All results
lyche-numerical-linear-algebra-F02523	188	■ 1.000	$(\mathbb{R}^{m \times n}, \mathbb{R})$	$(\mathbb{R}^{m \times n}, \mathbb{R})$	also holds for $(\mathbb{R}^{m \times n}, \mathbb{R})$. A matrix norm $\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is simply a vector norm
lyche-numerical-linear-algebra-F02524	188	■ 1.000	$\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$	$\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$	also holds for $(\mathbb{R}^{m \times n}, \mathbb{R})$. A matrix norm $\ \cdot\ : \mathbb{C}^{m \times n} \rightarrow \mathbb{R}$ is simply a vector norm
lyche-numerical-linear-algebra-F02525	188	■ 1.000	mn	mn	is a matrix norm. Indeed, writing all elements in $\mathbf{A}$ in a string of length mn we see
lyche-numerical-linear-algebra-F02526	188	■ 1.000	$\mathbb{C}^{mn}$	$\mathbb{C}^{mn}$	that the Frobenius norm is the Euclidian norm on the space $\mathbb{C}^{mn}$.
lyche-numerical-linear-algebra-F02527	188	■ 1.000	$\ \cdot\ _V$	$\ \cdot\ _V$	Any vector norm $\ \cdot\ _V$ on $\mathbb{C}^{mn}$ defines a matrix norm on $\mathbb{C}^{m \times n}$ given by $\ \mathbf{A}\ :=$
lyche-numerical-linear-algebra-F02528	188	■ 1.000	$\ \mathbf{A}\ :=$	$\ \mathbf{A}\ :=$	Any vector norm $\ \cdot\ _V$ on $\mathbb{C}^{mn}$ defines a matrix norm on $\mathbb{C}^{m \times n}$ given by $\ \mathbf{A}\ :=$
lyche-numerical-linear-algebra-F02529	188	■ 1.000	$\ \text{vec}(\mathbf{A})\ _V$	$\ \text{vec}(\mathbf{A})\ _V$	$\ \text{vec}(\mathbf{A})\ _V$, where $\text{vec}(\mathbf{A}) \in \mathbb{C}^{mn}$ is the vector obtained by stacking the columns
lyche-numerical-linear-algebra-F02530	189	■ 1.000	$\ \mathbf{AB}\ \leq \ \mathbf{A}\ \ \mathbf{B}\ $	$\ \mathbf{AB}\ \leq \ \mathbf{A}\ \ \mathbf{B}\ $	4. $\ \mathbf{AB}\ \leq \ \mathbf{A}\ \ \mathbf{B}\ $
lyche-numerical-linear-algebra-F02531	189	■ 0.998	$\ \cdot\ $	$\ \cdot\ $	Definition 8.5 (Subordinate Matrix Norms) Suppose $m, n \in \mathbb{N}$ are given, let $\ \cdot\ $
lyche-numerical-linear-algebra-F02532	189	■ 1.000	$\ \cdot\ _\beta$	$\ \cdot\ _\beta$	on $\mathbb{C}^m$ and $\ \cdot\ _\beta$ on $\mathbb{C}^n$ be vector norms, and let $\ \cdot\ $ be a matrix norm on $\mathbb{C}^{m \times n}$. We
lyche-numerical-linear-algebra-F02533	189	■ 1.000	$\ \cdot\ $	$\ \cdot\ $	on $\mathbb{C}^m$ and $\ \cdot\ _\beta$ on $\mathbb{C}^n$ be vector norms, and let $\ \cdot\ $ be a matrix norm on $\mathbb{C}^{m \times n}$. We

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lyche-numerical-linear-algebra-F02534	189	■ 0.999	$\ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _\beta$	$\ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _\beta$	$\ \boldsymbol{Ax}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ _\beta$ for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-F02535	189	■ 1.000	$m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}$	$m, n \in \mathbb{N}, \boldsymbol{A} \in \mathbb{C}^{m \times n}$	Proposition 8.1 For $m, n \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{C}^{m \times n}$, all $\boldsymbol{x} \in \mathbb{C}^n$ and any consistent matrix
lyche-numerical-linear-algebra-F02536	189	■ 1.000	$\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$	$\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$	consistency implies that (8.11) holds for $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and $\boldsymbol{B} := \boldsymbol{x} \in \mathbb{C}^{n \times 1}$. The last
lyche-numerical-linear-algebra-F02537	190	■ 0.969	$n \in \mathbb{N}.$	$n \in \mathbb{N}.$	defined on $\mathbb{C}^n$ for all $n \in \mathbb{N}$.
lyche-numerical-linear-algebra-F02538	190	■ 1.000	$\left\ \right\ $	$\ \ $	Definition 8.6 (Operator Norm) Let $\ \ $ be a vector norm defined on $\mathbb{C}^n$ for all
lyche-numerical-linear-algebra-F02539	190	■ 1.000	$\ \ : \mathcal{S} \rightarrow \mathbb{R}$	$\ \ : \mathcal{S} \rightarrow \mathbb{R}$	norm $\ \ : \mathcal{S} \rightarrow \mathbb{R}$ is a continuous function, it follows that the function
lyche-numerical-linear-algebra-F02540	190	■ 0.995	$f(\boldsymbol{x}) = \ \boldsymbol{Ax}\ $	$f(\boldsymbol{x}) = \ \boldsymbol{Ax}\ $	$f : \mathcal{S} \rightarrow \mathbb{R}$ given by $f(\boldsymbol{x}) = \ \boldsymbol{Ax}\ $ is continuous. But then f attains its max and
lyche-numerical-linear-algebra-F02541	190	■ 1.000	$\ \ _\beta$	$\ \ _\beta$	¹ In the case of one vector norm $\ \ $ on $\mathbb{C}^m$ and another vector norm $\ \ _\beta$ on $\mathbb{C}^n$ we would define
lyche-numerical-linear-algebra-F02542	190	■ 0.587	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{Ax}\ }{\ \boldsymbol{x}\ _\beta}$	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{Ax}\ }{\ \boldsymbol{x}\ _\beta}$	$\ \boldsymbol{A}\ := \max_{\boldsymbol{x} \neq 0} \frac{\ \boldsymbol{Ax}\ }{\ \boldsymbol{x}\ _\beta}$.
lyche-numerical-linear-algebra-F02543	191	■ 1.000	$\ \boldsymbol{I}\ = 1$	$\ \boldsymbol{I}\ = 1$	consistent matrix norm. Moreover, $\ \boldsymbol{I}\ = 1$.
lyche-numerical-linear-algebra-F02544	191	■ 1.000	$\ \boldsymbol{A}\ = 0$	$\ \boldsymbol{A}\ = 0$	1. Nonnegativity is obvious. If $\ \boldsymbol{A}\ = 0$ then $\ \boldsymbol{Ay}\ = 0$ for each $\boldsymbol{y} \in \mathbb{C}^n$. In
lyche-numerical-linear-algebra-F02545	191	■ 1.000	$\ \boldsymbol{Ay}\ = 0$	$\ \boldsymbol{Ay}\ = 0$	1. Nonnegativity is obvious. If $\ \boldsymbol{A}\ = 0$ then $\ \boldsymbol{Ay}\ = 0$ for each $\boldsymbol{y} \in \mathbb{C}^n$. In
lyche-numerical-linear-algebra-F02546	191	■ 1.000	$\boldsymbol{A} = 0$	$\boldsymbol{A} = 0$	particular, each column $\boldsymbol{Ae}_j$ in $\boldsymbol{A}$ is zero. Hence $\boldsymbol{A} = 0$.
lyche-numerical-linear-algebra-F02547	191	■ 1.000	$\ c\boldsymbol{A}\ = \max_{\boldsymbol{x}} \ c\boldsymbol{Ax}\ = \max_{\boldsymbol{x}} c \ \boldsymbol{Ax}\ = c \ \boldsymbol{A}\ $	$\ c\boldsymbol{A}\ = \max_{\boldsymbol{x}} \ c\boldsymbol{Ax}\ = \max_{\boldsymbol{x}} c \ \boldsymbol{Ax}\ = c \ \boldsymbol{A}\ $	2. $\ c\boldsymbol{A}\ = \max_{\boldsymbol{x}} \ c\boldsymbol{Ax}\ = \max_{\boldsymbol{x}} c \ \boldsymbol{Ax}\ = c \ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F02548	191	■ 1.000	$\ \boldsymbol{A} + \boldsymbol{B}\ = \max_{\boldsymbol{x}} \ (\boldsymbol{A} + \boldsymbol{B})\boldsymbol{x}\ \leq \max_{\boldsymbol{x}} \ \boldsymbol{Ax}\ + \max_{\boldsymbol{x}} \ \boldsymbol{Bx}\ = \ \boldsymbol{A}\ + \ \boldsymbol{B}\ $	$\ \boldsymbol{A} + \boldsymbol{B}\ = \max_{\boldsymbol{x}} \ (\boldsymbol{A} + \boldsymbol{B})\boldsymbol{x}\ \leq \max_{\boldsymbol{x}} \ \boldsymbol{Ax}\ + \max_{\boldsymbol{x}} \ \boldsymbol{Bx}\ = \ \boldsymbol{A}\ + \ \boldsymbol{B}\ $	3. $\ \boldsymbol{A} + \boldsymbol{B}\ = \max_{\boldsymbol{x}} \ (\boldsymbol{A} + \boldsymbol{B})\boldsymbol{x}\ \leq \max_{\boldsymbol{x}} \ \boldsymbol{Ax}\ + \max_{\boldsymbol{x}} \ \boldsymbol{Bx}\ = \ \boldsymbol{A}\ + \ \boldsymbol{B}\ $.
lyche-numerical-linear-algebra-F02549	191	■ 1.000	$\ \boldsymbol{I}\ _F = \sqrt{n}$	$\ \boldsymbol{I}\ _F = \sqrt{n}$	Since $\ \boldsymbol{I}\ _F = \sqrt{n}$, we see that the Frobenius norm is not an operator norm for
lyche-numerical-linear-algebra-F02550	191	■ 1.000	$n > 1$	$n > 1$	$n > 1$.
lyche-numerical-linear-algebra-F02551	191	■ 0.985	p	p	8.2.3 The Operator p-Norms
lyche-numerical-linear-algebra-F02552	191	■ 1.000	ℓ_p	ℓ_p	Recall that the p or ℓ_p vector norms (8.1) are given by
lyche-numerical-linear-algebra-F02553	191	■ 1.000	$\ \ _p$	$\ \ _p$	The operator norms $\ \ _p$ defined from these p -vector norms are used quite
lyche-numerical-linear-algebra-F02554	192	■ 1.000	$\ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\ _2$	The two-norm $\ \boldsymbol{A}\ _2$ is also called the spectral norm of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02555	192	■ 0.998	K_p	K_p	(a) We derive a constant K_p such that $\ \boldsymbol{Ax}\ _p \leq K_p$ for any $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _p = 1$.
lyche-numerical-linear-algebra-F02556	192	■ 0.998	$\ \boldsymbol{Ax}\ _p \leq K_p$	$\ \boldsymbol{Ax}\ _p \leq K_p$	(a) We derive a constant K_p such that $\ \boldsymbol{Ax}\ _p \leq K_p$ for any $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _p = 1$.
lyche-numerical-linear-algebra-F02557	192	■ 0.998	$\ \boldsymbol{x}\ _p = 1$	$\ \boldsymbol{x}\ _p = 1$	(a) We derive a constant K_p such that $\ \boldsymbol{Ax}\ _p \leq K_p$ for any $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _p = 1$.
lyche-numerical-linear-algebra-F02558	192	■ 1.000	$\boldsymbol{y}_e \in \mathbb{C}^n$	$\boldsymbol{y}_e \in \mathbb{C}^n$	(b) We give an extremal vector $\boldsymbol{y}_e \in \mathbb{C}^n$ with $\ \boldsymbol{y}_e\ _p = 1$ so that $\ \boldsymbol{Ay}_e\ _p = K_p$.

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lyche-numerical-linear-algebra-192-F02559	192	1.000	$\left\ \boldsymbol{y}_e\right\ _p=1$	$\ \boldsymbol{y}_e\ _p = 1$	(b) We give an extremal vector $\boldsymbol{y}_e \in \mathbb{C}^n$ with $\ \boldsymbol{y}_e\ _p = 1$ so that $\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$.
lyche-numerical-linear-algebra-192-F02560	192	1.000	$\left\ \boldsymbol{A}\boldsymbol{y}_e\right\ _p=K_p$	$\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$	(b) We give an extremal vector $\boldsymbol{y}_e \in \mathbb{C}^n$ with $\ \boldsymbol{y}_e\ _p = 1$ so that $\ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$.
lyche-numerical-linear-algebra-192-F02561	192	1.000	$\left\ \boldsymbol{A}\right\ _p=\left\ \boldsymbol{A}\boldsymbol{y}_e\right\ _p=K_p$	$\ \boldsymbol{A}\ _p = \ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$	It then follows from (8.17) that $\ \boldsymbol{A}\ _p = \ \boldsymbol{A}\boldsymbol{y}_e\ _p = K_p$.
lyche-numerical-linear-algebra-192-F02562	192	1.000	$K_2=$	$K_2 =$	2-norm: Let $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*$ be a singular value decomposition of $\boldsymbol{A}$, define $K_2 =$
lyche-numerical-linear-algebra-192-F02563	192	0.998	$\sigma_1, \boldsymbol{c}:=\boldsymbol{V}^*\boldsymbol{x}$	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^*\boldsymbol{x}$	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^*\boldsymbol{x}$, and $\boldsymbol{y}_e = \boldsymbol{v}_1$ the singular vector corresponding to σ_1 . Then $\boldsymbol{x} =$
lyche-numerical-linear-algebra-192-F02564	192	0.998	$\boldsymbol{y}_e=\boldsymbol{v}_1$	$\boldsymbol{y}_e = \boldsymbol{v}_1$	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^*\boldsymbol{x}$, and $\boldsymbol{y}_e = \boldsymbol{v}_1$ the singular vector corresponding to σ_1 . Then $\boldsymbol{x} =$
lyche-numerical-linear-algebra-192-F02565	192	0.998	σ_1	σ_1	$\sigma_1, \boldsymbol{c} := \boldsymbol{V}^*\boldsymbol{x}$, and $\boldsymbol{y}_e = \boldsymbol{v}_1$ the singular vector corresponding to σ_1 . Then $\boldsymbol{x} =$
lyche-numerical-linear-algebra-192-F02566	192	1.000	$\boldsymbol{V}\boldsymbol{c},\ \boldsymbol{c}\ _2=\ \boldsymbol{x}\ _2=1$	$\boldsymbol{V}\boldsymbol{c}, \ \boldsymbol{c}\ _2 = \ \boldsymbol{x}\ _2 = 1$	$\boldsymbol{V}\boldsymbol{c}, \ \boldsymbol{c}\ _2 = \ \boldsymbol{x}\ _2 = 1$, and using (7.7) in (b) we find
lyche-numerical-linear-algebra-192-F02567	192	0.998	$\ \boldsymbol{A}\boldsymbol{x}\ _2^2=\ \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*\boldsymbol{x}\ _2^2=\ \boldsymbol{\Sigma}\boldsymbol{c}\ _2^2=\sum_{j=1}^n\sigma_j^2 c_j ^2\leq\sigma_1^2\sum_{j=1}^n c_j ^2=\sigma_1^2$	$\ \boldsymbol{A}\boldsymbol{x}\ _2^2 = \ \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*\boldsymbol{x}\ _2^2 = \ \boldsymbol{\Sigma}\boldsymbol{c}\ _2^2 = \sum_{j=1}^n \sigma_j^2 c_j ^2 \leq \sigma_1^2 \sum_{j=1}^n c_j ^2 = \sigma_1^2$	(a) $\ \boldsymbol{A}\boldsymbol{x}\ _2^2 = \ \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^*\boldsymbol{x}\ _2^2 = \ \boldsymbol{\Sigma}\boldsymbol{c}\ _2^2 = \sum_{j=1}^n \sigma_j^2 c_j ^2 \leq \sigma_1^2 \sum_{j=1}^n c_j ^2 = \sigma_1^2$.
lyche-numerical-linear-algebra-192-F02568	192	1.000	$\left\ \boldsymbol{A}\boldsymbol{v}_1\right\ _2=\left\ \sigma_1\boldsymbol{u}_1\right\ _2=\sigma_1$	$\ \boldsymbol{A}\boldsymbol{v}_1\ _2 = \ \sigma_1\boldsymbol{u}_1\ _2 = \sigma_1$	(b) $\ \boldsymbol{A}\boldsymbol{v}_1\ _2 = \ \sigma_1\boldsymbol{u}_1\ _2 = \sigma_1$.
lyche-numerical-linear-algebra-192-F02569	192	1.000	K_1, c	K_1, c	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02570	192	1.000	$\boldsymbol{y}_e$	$\boldsymbol{y}_e$	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02571	192	1.000	$K_1:=\left\ \boldsymbol{A}\boldsymbol{e}_c\right\ _1=\max_{1 \leq j \leq n}\left\ \boldsymbol{A}\boldsymbol{e}_j\right\ _1$	$K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02572	192	1.000	$\boldsymbol{y}_e:=$	$\boldsymbol{y}_e :=$	1-norm: Define K_1, c and $\boldsymbol{y}_e$ by $K_1 := \ \boldsymbol{A}\boldsymbol{e}_c\ _1 = \max_{1 \leq j \leq n} \ \boldsymbol{A}\boldsymbol{e}_j\ _1$ and $\boldsymbol{y}_e :=$
lyche-numerical-linear-algebra-192-F02573	192	1.000	$\boldsymbol{e}_c$	$\boldsymbol{e}_c$	$\boldsymbol{e}_c$, a unit vector. Then $\ \boldsymbol{y}_e\ _1 = 1$ and we obtain
lyche-numerical-linear-algebra-192-F02574	192	1.000	$\left\ \boldsymbol{y}_e\right\ _1=1$	$\ \boldsymbol{y}_e\ _1 = 1$	$\boldsymbol{e}_c$, a unit vector. Then $\ \boldsymbol{y}_e\ _1 = 1$ and we obtain
lyche-numerical-linear-algebra-192-F02575	192	1.000	$\left\ \boldsymbol{A}\boldsymbol{y}_e\right\ _1=K_1$	$\ \boldsymbol{A}\boldsymbol{y}_e\ _1 = K_1$	(b) $\ \boldsymbol{A}\boldsymbol{y}_e\ _1 = K_1$.
lyche-numerical-linear-algebra-192-F02576	192	0.997	K_∞, r	K_∞, r	∞-norm: Define K_∞, r and $\boldsymbol{y}_e$ by $K_\infty := \ \boldsymbol{e}_r^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1$ and
lyche-numerical-linear-algebra-192-F02577	192	0.997	$K_\infty:=\left\ \boldsymbol{e}_r^T\boldsymbol{A}\right\ _1=\max_{1 \leq k \leq m}\left\ \boldsymbol{e}_k^T\boldsymbol{A}\right\ _1$	$K_\infty := \ \boldsymbol{e}_r^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1$	∞-norm: Define K_∞, r and $\boldsymbol{y}_e$ by $K_\infty := \ \boldsymbol{e}_r^T \boldsymbol{A}\ _1 = \max_{1 \leq k \leq m} \ \boldsymbol{e}_k^T \boldsymbol{A}\ _1$ and
lyche-numerical-linear-algebra-192-F02578	192	0.998	$\boldsymbol{y}_e:=\left[e^{-i \theta_1}, \ldots, e^{-i \theta_n}\right]^T$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \dots, e^{-i\theta_n}]^T$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \dots, e^{-i\theta_n}]^T$, where $a_{rj} = a_{rj} e^{i\theta_j}$ for $j = 1, \dots, n$.
lyche-numerical-linear-algebra-192-F02579	192	0.998	$a_{r j}=\left a_{r j}\right e^{i \theta_j}$	$a_{rj} = a_{rj} e^{i\theta_j}$	$\boldsymbol{y}_e := [e^{-i\theta_1}, \dots, e^{-i\theta_n}]^T$, where $a_{rj} = a_{rj} e^{i\theta_j}$ for $j = 1, \dots, n$.
lyche-numerical-linear-algebra-192-F02580	192	1.000	$\left\ \boldsymbol{A}\boldsymbol{y}^*\right\ _\infty=\max _{1 \leq k \leq m}\left \sum_{j=1}^n a_{k j} e^{-i \theta_j}\right =K_\infty$	$\ \boldsymbol{A}\boldsymbol{y}^*\ _\infty = \max_{1 \leq k \leq m} \left \sum_{j=1}^n a_{kj} e^{-i\theta_j} \right = K_\infty$	(b) $\ \boldsymbol{A}\boldsymbol{y}^*\ _\infty = \max_{1 \leq k \leq m} \left \sum_{j=1}^n a_{kj} e^{-i\theta_j} \right = K_\infty$.

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lyche-numerical-linear-algebra- F02581	192	■ 1.000	$\left \sum_{j=1}^n a_{kj} e^{-i\theta_j}\right \leq \sum_{j=1}^n a_{kj} \leq K_{\infty}$	$\sum_{j=1}^n a_{kj} e^{-i\theta_j} \leq \sum_{j=1}^n a_{kj} \leq K_{\infty}$	The last equality is correct because $ \sum_{j=1}^n a_{kj} e^{-i\theta_j} \leq \sum_{j=1}^n a_{kj} \leq K_{\infty}$ with
lyche-numerical-linear-algebra- F02582	192	■ 1.000	$k=r$	$k = r$	equality for $k=r$.
lyche-numerical-linear-algebra- F02583	192	■ 0.999	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix}$	$\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix}$	matrix $\boldsymbol{A} := \frac{1}{15} \begin{bmatrix} 14 & 4 & 16 \\ 2 & 22 & 13 \end{bmatrix}$, is $\sigma_1 = 2$ (cf. Example 7.9). We find
lyche-numerical-linear-algebra- F02584	193	■ 1.000	$\sigma_1 \geq \sigma_2 \geq$	$\sigma_1 \geq \sigma_2 \geq$	Theorem 8.4 (Spectral Norm) Suppose $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ has singular values $\sigma_1 \geq \sigma_2 \geq$
lyche-numerical-linear-algebra- F02585	193	■ 1.000	$\cdots \geq \sigma_n$	$\cdots \geq \sigma_n$	$\cdots \geq \sigma_n$ and eigenvalues $ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $. Then
lyche-numerical-linear-algebra- F02586	193	■ 1.000	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $	$\cdots \geq \sigma_n$ and eigenvalues $ \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n $. Then
lyche-numerical-linear-algebra- F02587	193	■ 1.000	$1/\sigma_n$	$1/\sigma_n$	Proof Since $1/\sigma_n$ is the largest singular value of $\boldsymbol{A}^{-1}$, (8.19) follows. By Exer-
lyche-numerical-linear-algebra- F02588	193	■ 1.000	$\ \boldsymbol{A}\ _2^2 \leq$	$\ \boldsymbol{A}\ _2^2 \leq$	Theorem 8.5 (Spectral Norm Bound) For any $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ we have $\ \boldsymbol{A}\ _2^2 \leq$
lyche-numerical-linear-algebra- F02589	193	■ 0.994	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}\ _1 \ \boldsymbol{A}\ _{\infty}$.
lyche-numerical-linear-algebra- F02590	193	■ 0.994	$(\sigma^2, \boldsymbol{v})$	$(\sigma^2, \boldsymbol{v})$	Proof Let $(\sigma^2, \boldsymbol{v})$ be an eigenpair for $\boldsymbol{A}^* \boldsymbol{A}$ corresponding to the largest singular
lyche-numerical-linear-algebra- F02591	193	■ 1.000	$\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$	$\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$	Observing that $\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$ by Theorem 8.3 and canceling $\ \boldsymbol{v}\ _1$ proves the
lyche-numerical-linear-algebra- F02592	193	■ 1.000	$\ \boldsymbol{v}\ _1$	$\ \boldsymbol{v}\ _1$	Observing that $\ \boldsymbol{A}^*\ _1 = \ \boldsymbol{A}\ _{\infty}$ by Theorem 8.3 and canceling $\ \boldsymbol{v}\ _1$ proves the
lyche-numerical-linear-algebra- F02593	193	■ 1.000	$\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $	$\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $	unitary invariant if $\ \boldsymbol{U} \boldsymbol{A} \boldsymbol{V}\ = \ \boldsymbol{A}\ $ for any $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and any unitary matrices
lyche-numerical-linear-algebra- F02594	193	■ 1.000	$\boldsymbol{U}(\boldsymbol{A} +$	$\boldsymbol{U}(\boldsymbol{A} +$	increased by a unitary transformation. Thus if $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary then $\boldsymbol{U}(\boldsymbol{A} +$
lyche-numerical-linear-algebra- F02595	193	■ 1.000	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$, where $\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $.
lyche-numerical-linear-algebra- F02596	193	■ 1.000	$\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $	$\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $	$\boldsymbol{E}) \boldsymbol{V} = \boldsymbol{U} \boldsymbol{A} \boldsymbol{V} + \boldsymbol{F}$, where $\ \boldsymbol{F}\ = \ \boldsymbol{E}\ $.
lyche-numerical-linear-algebra- F02597	194	■ 1.000	$\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$	Theorem 8.3 that $\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$. Moreover $\boldsymbol{V}^*$ is unitary. Using these facts we
lyche-numerical-linear-algebra- F02598	194	■ 1.000	$\boldsymbol{V}^*$	$\boldsymbol{V}^*$	Theorem 8.3 that $\ \boldsymbol{A}^*\ _2 = \ \boldsymbol{A}\ _2$. Moreover $\boldsymbol{V}^*$ is unitary. Using these facts we
lyche-numerical-linear-algebra- F02599	194	■ 1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $	A vector norm on $\mathbb{C}^n$ is an absolute norm if $\ \boldsymbol{x}\ = \ \boldsymbol{x}\ $ for all $\boldsymbol{x} \in \mathbb{C}^n$. Here
lyche-numerical-linear-algebra- F02600	194	■ 1.000	$ \boldsymbol{x} := [x_1 , \dots, x_n]^T$	$ \boldsymbol{x} := [x_1 , \dots, x_n]^T$	$ \boldsymbol{x} := [x_1 , \dots, x_n]^T$, the absolute values of the components of $\boldsymbol{x}$. Clearly the
lyche-numerical-linear-algebra- F02601	195	■ 1.000	$x_1 = x_2 = 10$	$x_1 = x_2 = 10$	whose exact solution is $x_1 = x_2 = 10$. If we replace the second equation by
lyche-numerical-linear-algebra- F02602	195	■ 1.000	$x_1 = 30, x_2 = -10$	$x_1 = 30, x_2 = -10$	the exact solution changes to $x_1 = 30, x_2 = -10$. Here a small change in one of
lyche-numerical-linear-algebra- F02603	195	■ 1.000	$1 - 10^{-16}$	$1 - 10^{-16}$	the coefficients, from $1 - 10^{-16}$ to $1 + 10^{-16}$, changed the exact solution by a large
lyche-numerical-linear-algebra- F02604	195	■ 1.000	$1 + 10^{-16}$	$1 + 10^{-16}$	the coefficients, from $1 - 10^{-16}$ to $1 + 10^{-16}$, changed the exact solution by a large
lyche-numerical-linear-algebra- F02605	195	■ 0.994	$\boldsymbol{A}, \boldsymbol{b}$	$\boldsymbol{A}, \boldsymbol{b}$	$\boldsymbol{A}, \boldsymbol{b}$ has on the inverse of $\boldsymbol{A}$ and on the solution $\boldsymbol{x}$ of a linear system $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$. To
lyche-numerical-linear-algebra- F02606	195	■ 1.000	$\boldsymbol{I} \in \mathbb{R}^{n \times n}$	$\boldsymbol{I} \in \mathbb{R}^{n \times n}$	vector norm. This follows from Lemma 7.1. We recall that if $\boldsymbol{I} \in \mathbb{R}^{n \times n}$ then $\ \boldsymbol{I}\ = 1$
lyche-numerical-linear-algebra- F02607	195	■ 0.996	$(\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	$(\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	Suppose $\boldsymbol{x}, \boldsymbol{y}$ solve $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ and $(\boldsymbol{A} + \boldsymbol{E}) \boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$, respectively. where $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$

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lyche-numerical-linear-algebra- F02608	195	■ 0.996	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$	Suppose $\boldsymbol{x}, \boldsymbol{y}$ solve $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$, respectively, where $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E} \in$
lyche-numerical-linear-algebra- F02609	195	■ 1.000	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$	$\mathbb{C}^{n \times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$. How large can $\boldsymbol{y} - \boldsymbol{x}$ be? The difference $\ \boldsymbol{y} - \boldsymbol{x}\ $
lyche-numerical-linear-algebra- F02610	195	■ 1.000	$\boldsymbol{y} - \boldsymbol{x}$	$\boldsymbol{y} - \boldsymbol{x}$	$\mathbb{C}^{n \times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$. How large can $\boldsymbol{y} - \boldsymbol{x}$ be? The difference $\ \boldsymbol{y} - \boldsymbol{x}\ $
lyche-numerical-linear-algebra- F02611	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ $	$\mathbb{C}^{n \times n}$ are nonsingular and $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n$. How large can $\boldsymbol{y} - \boldsymbol{x}$ be? The difference $\ \boldsymbol{y} - \boldsymbol{x}\ $
lyche-numerical-linear-algebra- F02612	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $	measures the absolute error in $\boldsymbol{y}$ as an approximation to $\boldsymbol{x}$, while $\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ $ and
lyche-numerical-linear-algebra- F02613	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{y}\ $ are measures for the relative error .
lyche-numerical-linear-algebra- F02614	195	■ 0.546	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$	nonsingular, $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$ and $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$. Then
lyche-numerical-linear-algebra- F02615	195	■ 0.546	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	nonsingular, $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^n, \boldsymbol{b} \neq \mathbf{0}$ and $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$ Then
lyche-numerical-linear-algebra- F02616	195	■ 1.000	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	$K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	where $K(\boldsymbol{A}) = \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $ is the condition number of $\boldsymbol{A}$.
lyche-numerical-linear-algebra- F02617	195	■ 1.000	$\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	$\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$	Proof Subtracting $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$ or $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra- F02618	195	■ 1.000	$\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$	$\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$	Proof Subtracting $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$ or $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra- F02619	195	■ 1.000	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$	Proof Subtracting $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ from $\boldsymbol{A}\boldsymbol{y} = \boldsymbol{b} + \boldsymbol{e}$ we have $\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{e}$ or $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{e}$.
lyche-numerical-linear-algebra- F02620	195	■ 1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $	Combining $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $ and $\ \boldsymbol{b}\ = \ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $ we
lyche-numerical-linear-algebra- F02621	195	■ 1.000	$\ \boldsymbol{b}\ = \ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{b}\ = \ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	Combining $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{e}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{e}\ $ and $\ \boldsymbol{b}\ = \ \boldsymbol{A}\boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $ we
lyche-numerical-linear-algebra- F02622	195	■ 1.000	$\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $	$\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $	obtain the upper bound in (8.22). Combining $\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $ and $\ \boldsymbol{x}\ \leq$
lyche-numerical-linear-algebra- F02623	195	■ 1.000	$\ \boldsymbol{x}\ \leq$	$\ \boldsymbol{x}\ \leq$	obtain the upper bound in (8.22). Combining $\ \boldsymbol{e}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{y} - \boldsymbol{x}\ $ and $\ \boldsymbol{x}\ \leq$
lyche-numerical-linear-algebra- F02624	195	■ 1.000	$\ \boldsymbol{A}^{-1}\ \ \boldsymbol{b}\ $	$\ \boldsymbol{A}^{-1}\ \ \boldsymbol{b}\ $	$\ \boldsymbol{A}^{-1}\ \ \boldsymbol{b}\ $ we obtain the lower bound.
lyche-numerical-linear-algebra- F02625	196	■ 1.000	$\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $	$\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $	Consider (8.22). $\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $ is a measure of the size of the perturbation $\boldsymbol{e}$ relative
lyche-numerical-linear-algebra- F02626	196	■ 1.000	$\boldsymbol{e}$	$\boldsymbol{e}$	Consider (8.22). $\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $ is a measure of the size of the perturbation $\boldsymbol{e}$ relative
lyche-numerical-linear-algebra- F02627	196	■ 0.999	$K(\boldsymbol{A})$	$K(\boldsymbol{A})$	$K(\boldsymbol{A})$ times as large as $\ \boldsymbol{e}\ / \ \boldsymbol{b}\ $.
lyche-numerical-linear-algebra- F02628	196	■ 1.000	$K(\boldsymbol{A}) \geq 1$	$K(\boldsymbol{A}) \geq 1$	well-conditioned (with respect to inversion). We always have $K(\boldsymbol{A}) \geq 1$. For since
lyche-numerical-linear-algebra- F02629	196	■ 1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $ for any $\boldsymbol{x}$ we have $\ \boldsymbol{I}\ \geq 1$ and therefore $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$
lyche-numerical-linear-algebra- F02630	196	■ 1.000	$\ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $ for any $\boldsymbol{x}$ we have $\ \boldsymbol{I}\ \geq 1$ and therefore $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$
lyche-numerical-linear-algebra- F02631	196	■ 1.000	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$	$\ \boldsymbol{x}\ = \ \boldsymbol{I}\boldsymbol{x}\ \leq \ \boldsymbol{I}\ \ \boldsymbol{x}\ $ for any $\boldsymbol{x}$ we have $\ \boldsymbol{I}\ \geq 1$ and therefore $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ \geq$
lyche-numerical-linear-algebra- F02632	196	■ 1.000	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ \geq 1$	$\ \boldsymbol{A}\boldsymbol{A}^{-1}\ = \ \boldsymbol{I}\ > 1$.
lyche-numerical-linear-algebra- F02633	196	■ 1.000	ℓ_1, ℓ_∞	ℓ_1, ℓ_∞	when discussing properties of the condition number, and the ℓ_1, ℓ_∞ , or Frobenius

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lyche-numerical-linear-algebra-F02634	196	0.874	$\boldsymbol{Ax}=\boldsymbol{b}$	$\boldsymbol{Ax} = \boldsymbol{b}$	Suppose we have computed an approximate solution $\boldsymbol{y}$ to $\boldsymbol{Ax} = \boldsymbol{b}$. The vector
lyche-numerical-linear-algebra-F02635	196	1.000	$\boldsymbol{r}(\boldsymbol{y}):=\boldsymbol{Ay}-\boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{Ay} - \boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{Ay} - \boldsymbol{b}$ is called the residual vector , or just the residual. We can bound
lyche-numerical-linear-algebra-F02636	196	1.000	$\boldsymbol{r}$	$\boldsymbol{r}$	$\boldsymbol{x} - \boldsymbol{y}$ in terms of $\boldsymbol{r}$.
lyche-numerical-linear-algebra-F02637	196	0.987	$\boldsymbol{b} \neq \mathbf{0}$	$\boldsymbol{b} \neq \mathbf{0}$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. Let $\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{Ay} - \boldsymbol{b}$ for $\boldsymbol{y} \in \mathbb{C}^n$. If $\boldsymbol{Ax} = \boldsymbol{b}$ then
lyche-numerical-linear-algebra-F02638	196	0.987	$\boldsymbol{r}(\boldsymbol{y})=\boldsymbol{Ay}-\boldsymbol{b}$	$\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{Ay} - \boldsymbol{b}$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. Let $\boldsymbol{r}(\boldsymbol{y}) = \boldsymbol{Ay} - \boldsymbol{b}$ for $\boldsymbol{y} \in \mathbb{C}^n$. If $\boldsymbol{Ax} = \boldsymbol{b}$ then
lyche-numerical-linear-algebra-F02639	196	1.000	$\boldsymbol{e}=\boldsymbol{r}(\boldsymbol{y})$	$\boldsymbol{e} = \boldsymbol{r}(\boldsymbol{y})$	Proof We simply take $\boldsymbol{e} = \boldsymbol{r}(\boldsymbol{y})$ in Theorem 8.7.
lyche-numerical-linear-algebra-F02640	196	0.999	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$ nonsingular. We like to compare the solution $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra-F02641	196	0.999	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$ with $\boldsymbol{A}, \boldsymbol{A} + \boldsymbol{E}$ nonsingular. We like to compare the solution $\boldsymbol{x}$ and $\boldsymbol{y}$
lyche-numerical-linear-algebra-F02642	196	0.999	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$	of the systems $\boldsymbol{Ax} = \boldsymbol{b}$ and $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$.
lyche-numerical-linear-algebra-F02643	196	0.615	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$	Theorem 8.9 (Perturbation in Matrix) Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}, \boldsymbol{b} \in \mathbb{C}^n$ with $\boldsymbol{A}$
lyche-numerical-linear-algebra-F02644	196	0.999	$r:=\ \boldsymbol{A}^{-1}\boldsymbol{E}\ <1$	$r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. If $r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$ then $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular. If $\boldsymbol{Ax} = \boldsymbol{b}$
lyche-numerical-linear-algebra-F02645	196	0.999	$\boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{A} + \boldsymbol{E}$	nonsingular and $\boldsymbol{b} \neq \mathbf{0}$. If $r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$ then $\boldsymbol{A} + \boldsymbol{E}$ is nonsingular. If $\boldsymbol{Ax} = \boldsymbol{b}$
lyche-numerical-linear-algebra-F02646	196	1.000	$r \geq 1$	$r \geq 1$	Proof We show $\boldsymbol{A} + \boldsymbol{E}$ singular implies $r \geq 1$. Suppose $\boldsymbol{A} + \boldsymbol{E}$ is singular. Then
lyche-numerical-linear-algebra-F02647	196	1.000	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ for some nonzero $\boldsymbol{x} \in \mathbb{C}^n$. Multiplying by $\boldsymbol{A}^{-1}$ it follows that
lyche-numerical-linear-algebra-F02648	196	1.000	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ and this implies that $\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ \leq r\ \boldsymbol{x}\ $. But then $r \geq 1$.
lyche-numerical-linear-algebra-F02649	196	1.000	$\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ \leq r\ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ \leq r\ \boldsymbol{x}\ $	$(\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})\boldsymbol{x} = \mathbf{0}$ and this implies that $\ \boldsymbol{x}\ = \ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{x}\ < r\ \boldsymbol{x}\ $. But then $r \geq 1$.
lyche-numerical-linear-algebra-F02650	196	0.967	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{E}\boldsymbol{y}$	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{E}\boldsymbol{y}$	Subtracting $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{y} = \boldsymbol{b}$ from $\boldsymbol{Ax} = \boldsymbol{b}$ gives $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{y}) = \boldsymbol{E}\boldsymbol{y}$ or
lyche-numerical-linear-algebra-F02651	196	1.000	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$. Taking norms and dividing by $\ \boldsymbol{y}\ $ proves (8.24). Solving
lyche-numerical-linear-algebra-F02652	196	1.000	$\ \boldsymbol{y}\ $	$\ \boldsymbol{y}\ $	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$. Taking norms and dividing by $\ \boldsymbol{y}\ $ proves (8.24). Solving
lyche-numerical-linear-algebra-F02653	197	1.000	$\boldsymbol{y} = (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\boldsymbol{x}$	$\boldsymbol{y} = (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\boldsymbol{x}$	$\boldsymbol{x} - \boldsymbol{y} = \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}$ for $\boldsymbol{y}$ we obtain $\boldsymbol{y} = (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\boldsymbol{x}$. By Theorem 12.14 we
lyche-numerical-linear-algebra-F02654	197	0.995	$\ \boldsymbol{y}\ \leq \ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$	$\ \boldsymbol{y}\ \leq \left\ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1} \right\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$	have $\ \boldsymbol{y}\ \leq \ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$ and (8.24) implies $\ \boldsymbol{y} - \boldsymbol{x}\ \leq r\ \boldsymbol{y}\ \leq$
lyche-numerical-linear-algebra-F02655	197	0.995	$\ \boldsymbol{y} - \boldsymbol{x}\ \leq r\ \boldsymbol{y}\ \leq$	$\ \boldsymbol{y} - \boldsymbol{x}\ \leq r\ \boldsymbol{y}\ \leq$	have $\ \boldsymbol{y}\ \leq \ (\boldsymbol{I} + \boldsymbol{A}^{-1}\boldsymbol{E})^{-1}\ \ \boldsymbol{x}\ \leq \frac{\ \boldsymbol{x}\ }{1-r}$ and (8.24) implies $\ \boldsymbol{y} - \boldsymbol{x}\ < r\ \boldsymbol{y}\ <$
lyche-numerical-linear-algebra-F02656	197	0.669	$\frac{r}{1-r}\ \boldsymbol{x}\ \leq \frac{K(\boldsymbol{A})}{1-r}\ \boldsymbol{E}\ \ \boldsymbol{A}\ $	$\frac{r}{1-r}\ \boldsymbol{x}\ \leq \frac{K(\boldsymbol{A})}{1-r}\ \boldsymbol{E}\ \ \boldsymbol{A}\ $	$\frac{r}{1-r}\ \boldsymbol{x}\ \leq \frac{K(\boldsymbol{A})}{1-r}\ \boldsymbol{E}\ \ \boldsymbol{A}\ $. Dividing by $\ \boldsymbol{x}\ $ gives (8.25).
lyche-numerical-linear-algebra-F02657	197	0.669	$\ \boldsymbol{x}\ $	$\ \boldsymbol{x}\ $	$\frac{r}{1-r}\ \boldsymbol{x}\ \leq \frac{K(\boldsymbol{A})}{1-r}\ \boldsymbol{E}\ \ \boldsymbol{A}\ $. Dividing by $\ \boldsymbol{x}\ $ gives (8.25).
lyche-numerical-linear-algebra-F02658	197	0.986	$\boldsymbol{x}.\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $	$\boldsymbol{x}.\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $	to $\boldsymbol{y}$ and the relative error in $\boldsymbol{y}$ as an approximation to $\boldsymbol{x}$. $\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $ is a measure for

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lyche-numerical-linear-algebra-F02659	197	1.000	$\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $	$\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $	again plays a crucial role. $\ \boldsymbol{y} - \boldsymbol{x}\ /\ \boldsymbol{y}\ $ can be as large as $K(\boldsymbol{A})$ times $\ \boldsymbol{E}\ /\ \boldsymbol{A}\ $. It
lyche-numerical-linear-algebra-F02660	197	1.000	$\ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{E}\ \ \boldsymbol{y}\ $	$\ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{E}\ \ \boldsymbol{y}\ $	the upper bound we used the inequality $\ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\ \leq \ \boldsymbol{A}^{-1}\ \ \boldsymbol{E}\ \ \boldsymbol{y}\ $. For a more
lyche-numerical-linear-algebra-F02661	197	1.000	$\ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\ $	$\ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\ $	or less random perturbation $\boldsymbol{E}$ this is not a severe overestimate for $\ \boldsymbol{A}^{-1}\boldsymbol{E}\boldsymbol{y}\ $. In the
lyche-numerical-linear-algebra-F02662	197	1.000	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$	$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$	with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$ and eigenvalues $ \lambda_1 \geq \lambda_2 \geq \cdots \geq$
lyche-numerical-linear-algebra-F02663	197	1.000	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq$	$ \lambda_1 \geq \lambda_2 \geq \cdots \geq$	with singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$ and eigenvalues $ \lambda_1 \geq \lambda_2 \geq \cdots \geq$
lyche-numerical-linear-algebra-F02664	197	1.000	$ \lambda_n > 0$	$ \lambda_n > 0$	$ \lambda_n > 0$. Then
lyche-numerical-linear-algebra-F02665	197	0.999	σ_1/σ_n	σ_1/σ_n	It follows that $\boldsymbol{A}$ is ill-conditioned with respect to inversion if and only if σ_1/σ_n
lyche-numerical-linear-algebra-F02666	197	1.000	λ_1/λ_n	λ_1/λ_n	is large, or λ_1/λ_n is large when $\boldsymbol{A}$ is positive definite.
lyche-numerical-linear-algebra-F02667	197	1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ /\ \boldsymbol{x}\ \approx \ \boldsymbol{r}(\boldsymbol{y})\ /\ \boldsymbol{b}\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ /\ \boldsymbol{x}\ \approx \ \boldsymbol{r}(\boldsymbol{y})\ /\ \boldsymbol{b}\ $	If $\boldsymbol{A}$ is well-conditioned, (8.23) says that $\ \boldsymbol{y} - \boldsymbol{x}\ /\ \boldsymbol{x}\ \approx \ \boldsymbol{r}(\boldsymbol{y})\ /\ \boldsymbol{b}\ $. In other
lyche-numerical-linear-algebra-F02668	197	1.000	$\ \boldsymbol{b}\ \approx 1$	$\ \boldsymbol{b}\ \approx 1$	long as $\ \boldsymbol{b}\ \approx 1$. If $\boldsymbol{A}$ is ill-conditioned, anything can happen. We can for example
lyche-numerical-linear-algebra-F02669	197	1.000	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{E}$	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{E}$	Suppose $\boldsymbol{A}$ is nonsingular and $\boldsymbol{E}$ a perturbation of $\boldsymbol{A}$. We expect $\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{E}$ to be
lyche-numerical-linear-algebra-F02670	197	1.000	$\ \boldsymbol{B}^{-1}\ $	$\ \boldsymbol{B}^{-1}\ $	to have bounds on $\ \boldsymbol{B}^{-1}\ $ in terms of $\ \boldsymbol{A}^{-1}\ $ and the difference $\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ $.
lyche-numerical-linear-algebra-F02671	197	1.000	$\ \boldsymbol{A}^{-1}\ $	$\ \boldsymbol{A}^{-1}\ $	to have bounds on $\ \boldsymbol{B}^{-1}\ $ in terms of $\ \boldsymbol{A}^{-1}\ $ and the difference $\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ $.
lyche-numerical-linear-algebra-F02672	197	1.000	$\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ $	$\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ $	to have bounds on $\ \boldsymbol{B}^{-1}\ $ in terms of $\ \boldsymbol{A}^{-1}\ $ and the difference $\ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ $.
lyche-numerical-linear-algebra-F02673	197	1.000	$\ \boldsymbol{B}^{-1}\ /\ \boldsymbol{A}^{-1}\ , \ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ /\ \boldsymbol{B}^{-1}\ $	$\ \boldsymbol{B}^{-1}\ /\ \boldsymbol{A}^{-1}\ , \ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ /\ \boldsymbol{B}^{-1}\ $	We consider the relative errors $\ \boldsymbol{B}^{-1}\ /\ \boldsymbol{A}^{-1}\ , \ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ /\ \boldsymbol{B}^{-1}\ $ and $\ \boldsymbol{B}^{-1} -$
lyche-numerical-linear-algebra-F02674	197	1.000	$\ \boldsymbol{B}^{-1} -$	$\ \boldsymbol{B}^{-1} -$	We consider the relative errors $\ \boldsymbol{B}^{-1}\ /\ \boldsymbol{A}^{-1}\ , \ \boldsymbol{B}^{-1} - \boldsymbol{A}^{-1}\ /\ \boldsymbol{B}^{-1}\ $ and $\ \boldsymbol{B}^{-1} -$
lyche-numerical-linear-algebra-F02675	197	1.000	$\boldsymbol{A}^{-1}\ /\ \boldsymbol{A}^{-1}\ $	$\boldsymbol{A}^{-1}\ /\ \boldsymbol{A}^{-1}\ $	$\boldsymbol{A}^{-1}\ /\ \boldsymbol{A}^{-1}\ $.
lyche-numerical-linear-algebra-F02676	198	1.000	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{E} \in \mathbb{C}^{n \times n}$	$\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{E} \in \mathbb{C}^{n \times n}$	gular and let $\boldsymbol{B} := \boldsymbol{A} + \boldsymbol{E} \in \mathbb{C}^{n \times n}$ be nonsingular. For any consistent matrix norm
lyche-numerical-linear-algebra-F02677	198	0.998	$K(\boldsymbol{A}) := \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	$K(\boldsymbol{A}) := \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	where $K(\boldsymbol{A}) := \ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $. If $r := \ \boldsymbol{A}^{-1}\boldsymbol{E}\ < 1$ then $\boldsymbol{B}$ is nonsingular and
lyche-numerical-linear-algebra-F02678	198	0.922	$\ \boldsymbol{A}^{-1}\boldsymbol{E}\ $	$\ \boldsymbol{A}^{-1}\boldsymbol{E}\ $	We can replace $\ \boldsymbol{A}^{-1}\boldsymbol{E}\ $ by $\ \boldsymbol{E}\boldsymbol{A}^{-1}\ $ everywhere.
lyche-numerical-linear-algebra-F02679	198	0.922	$\ \boldsymbol{E}\boldsymbol{A}^{-1}\ $	$\ \boldsymbol{E}\boldsymbol{A}^{-1}\ $	We can replace $\ \boldsymbol{A}^{-1}\boldsymbol{E}\ $ by $\ \boldsymbol{E}\boldsymbol{A}^{-1}\ $ everywhere.
lyche-numerical-linear-algebra-F02680	198	0.999	$r < 1$	$r < 1$	Proof That $\boldsymbol{B}$ is nonsingular if $r < 1$ follows from Theorem 8.9. We have $-\boldsymbol{E} =$
lyche-numerical-linear-algebra-F02681	198	0.999	$-\boldsymbol{E} =$	$-\boldsymbol{E} =$	Proof That $\boldsymbol{B}$ is nonsingular if $r < 1$ follows from Theorem 8.9. We have $-\boldsymbol{E} =$
lyche-numerical-linear-algebra-F02682	198	1.000	$\boldsymbol{A} - \boldsymbol{B} = \boldsymbol{A}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{B} = \boldsymbol{B}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{A}$	$\boldsymbol{A} - \boldsymbol{B} = \boldsymbol{A}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{B} = \boldsymbol{B}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{A}$	$\boldsymbol{A} - \boldsymbol{B} = \boldsymbol{A}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{B} = \boldsymbol{B}(\boldsymbol{B}^{-1} - \boldsymbol{A}^{-1})\boldsymbol{A}$ so that

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lyche-numerical-linear-algebra- F02683	198	1.000	$\left\ \boldsymbol{A}^{-1}\right\ \leq\left\ \boldsymbol{B}^{-1}\right\ +r\left\ \boldsymbol{B}^{-1}\right\ $	$\ \boldsymbol{A}^{-1}\ \leq\ \boldsymbol{B}^{-1}\ +r\ \boldsymbol{B}^{-1}\ $	Similarly we obtain the lower bound in (8.28) from $\ \boldsymbol{A}^{-1}\ \leq\ \boldsymbol{B}^{-1}\ +r\ \boldsymbol{B}^{-1}\ $.
lyche-numerical-linear-algebra- F02684	198	1.000	$\left\ \boldsymbol{B}^{-1}\right\ /\left\ \boldsymbol{A}^{-1}\right\ $	$\ \boldsymbol{B}^{-1}\ /\ \boldsymbol{A}^{-1}\ $	The bound in (8.29) follows by multiplying (8.27) by $\ \boldsymbol{B}^{-1}\ /\ \boldsymbol{A}^{-1}\ $ and using
lyche-numerical-linear-algebra- F02685	199	1.000	$1 \leq p \leq \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	$1 \leq p \leq \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$	Then for all $1 < p < \infty, \boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ and all $a \in \mathbb{C}$
lyche-numerical-linear-algebra- F02686	199	1.000	$\left\ \boldsymbol{x}\right\ _p \geq 0$	$\ \boldsymbol{x}\ _p \geq 0$	1. $\ \boldsymbol{x}\ _p \geq 0$ with equality if and only if $\boldsymbol{x}=\mathbf{0}$.
lyche-numerical-linear-algebra- F02687	199	1.000	$\left\ a \boldsymbol{x}\right\ _p=\left a\right \left\ \boldsymbol{x}\right\ _p$	$\ a \boldsymbol{x}\ _p= a \ \boldsymbol{x}\ _p$	2. $\ a \boldsymbol{x}\ _p= a \ \boldsymbol{x}\ _p$.
lyche-numerical-linear-algebra- F02688	199	1.000	$I \subset \mathbb{R}$	$I \subset \mathbb{R}$	Definition 8.8 (Convex Function) Let $I \subset \mathbb{R}$ be an interval. A function $f: I \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra- F02689	199	1.000	$f: I \rightarrow \mathbb{R}$	$f: I \rightarrow \mathbb{R}$	Definition 8.8 (Convex Function) Let $I \subset \mathbb{R}$ be an interval. A function $f: I \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra- F02690	199	1.000	$x_1, x_2 \in I$	$x_1, x_2 \in I$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in[0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra- F02691	199	1.000	$x_1 < x_2$	$x_1 < x_2$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in[0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra- F02692	199	1.000	$\lambda \in[0,1]$	$\lambda \in[0,1]$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in[0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra- F02693	199	1.000	$\sum_{j=1}^n \lambda_j x_j$	$\sum_{j=1}^n \lambda_j x_j$	for all $x_1, x_2 \in I$ with $x_1 < x_2$ and all $\lambda \in[0,1]$. The sum $\sum_{j=1}^n \lambda_j x_j$ is called a
lyche-numerical-linear-algebra- F02694	199	1.000	$x_1, \ldots, x_n$	$x_1, \ldots, x_n$	convex combination of $x_1, \ldots, x_n$ if $\lambda_j \geq 0$ for $j=1, \ldots, n$ and $\sum_{j=1}^n \lambda_j=1$.
lyche-numerical-linear-algebra- F02695	199	1.000	$\lambda_j \geq 0$	$\lambda_j \geq 0$	convex combination of $x_1, \ldots, x_n$ if $\lambda_j \geq 0$ for $j=1, \ldots, n$ and $\sum_{j=1}^n \lambda_j=1$.
lyche-numerical-linear-algebra- F02696	199	1.000	$\sum_{j=1}^n \lambda_j=1$	$\sum_{j=1}^n \lambda_j=1$	convex combination of $x_1, \ldots, x_n$ if $\lambda_j \geq 0$ for $j=1, \ldots, n$ and $\sum_{j=1}^n \lambda_j=1$.
lyche-numerical-linear-algebra- F02697	200	1.000	$f \in C^2[a, b]$	$f \in C^2[a, b]$	Lemma 8.2 (A Sufficient Condition for Convexity) If $f \in C^2[a, b]$ and $f''(x) \geq$
lyche-numerical-linear-algebra- F02698	200	1.000	$f'(\prime)(x) \geq$	$f''(x) \geq$	Lemma 8.2 (A Sufficient Condition for Convexity) If $f \in C^2[a, b]$ and $f''(x) \geq$
lyche-numerical-linear-algebra- F02699	200	1.000	$a \leq x_1 \leq x \leq x_2 \leq b$	$a \leq x_1 \leq x \leq x_2 \leq b$	numerical methods) For any $a \leq x_1 \leq x \leq x_2 \leq b$ there is a $c \in\left[x_1, x_2\right]$ such that
lyche-numerical-linear-algebra- F02700	200	1.000	$c \in\left[x_1, x_2\right]$	$c \in\left[x_1, x_2\right]$	numerical methods) For any $a \leq x_1 \leq x \leq x_2 \leq b$ there is a $c \in\left[x_1, x_2\right]$ such that
lyche-numerical-linear-algebra- F02701	200	1.000	$f(x) \leq(1-\lambda) f\left(x_1\right)+\lambda f\left(x_2\right)$	$f(x) \leq(1-\lambda) f\left(x_1\right)+\lambda f\left(x_2\right)$	Since $\lambda \in[0,1]$ we have $f(x) \leq(1-\lambda) f\left(x_1\right)+\lambda f\left(x_2\right)$. Moreover,
lyche-numerical-linear-algebra- F02702	200	0.999	$I \in \mathbb{R}$	$I \in \mathbb{R}$	Theorem 8.13 (Jensen's Inequality) Suppose $I \in \mathbb{R}$ is an interval and $f: I \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra- F02703	200	1.000	$z_1, \ldots, z_n \in I$	$z_1, \ldots, z_n \in I$	$\sum_{j=1}^n \lambda_j=1$, and all $z_1, \ldots, z_n \in I$
lyche-numerical-linear-algebra- F02704	200	1.000	$n \geq 2$	$n \geq 2$	Proof We use induction on n . The result is trivial for $n=1$. Let $n \geq 2$, assume
lyche-numerical-linear-algebra- F02705	200	0.999	λ_j, z_j	λ_j, z_j	the inequality holds for $n-1$, and let λ_j, z_j for $j=1, \ldots, n$ be given as in the
lyche-numerical-linear-algebra- F02706	200	1.000	$\lambda_i < 1$	$\lambda_i < 1$	theorem. Since $n \geq 2$ we have $\lambda_i < 1$ for at least one i so assume without loss of
lyche-numerical-linear-algebra- F02707	200	1.000	$\lambda_1 < 1$	$\lambda_1 < 1$	generality that $\lambda_1 < 1$, and define $u:=\sum_{j=2}^n \frac{\lambda_j}{1-\lambda_1} z_j$. Since $\sum_{j=2}^n \lambda_j=1-\lambda_1$

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lyche-numerical-linear-algebra-F02788	200	■ 1.000	$u := \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} z_j$	$u := \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} z_j$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} z_j$. Since $\sum_{j=2}^n \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-algebra-F02789	200	■ 1.000	$\sum_{j=2}^n \lambda_j = 1 - \lambda_1$	$\sum_{j=2}^n \lambda_j = 1 - \lambda_1$	generality that $\lambda_1 < 1$, and define $u := \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} z_j$. Since $\sum_{j=2}^n \lambda_j = 1 - \lambda_1$
lyche-numerical-linear-algebra-F02710	200	■ 1.000	$f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} f(z_j)$	$f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} f(z_j)$	that $f(u) \leq \sum_{j=2}^n \frac{\lambda_j}{1 - \lambda_1} f(z_j)$. But then by the convexity of f
lyche-numerical-linear-algebra-F02711	201	■ 1.000	$\sum_{j=1}^n \lambda_j a_j$	$\sum_{j=1}^n \lambda_j a_j$	$\sum_{j=1}^n \lambda_j a_j$ is a convex combination of nonnegative numbers $a_1, \dots, a_n$. Then
lyche-numerical-linear-algebra-F02712	201	■ 1.000	$a_1, \dots, a_n$	$a_1, \dots, a_n$	$\sum_{j=1}^n \lambda_j a_j$ is a convex combination of nonnegative numbers <u>$a_1, \dots, a_n$</u> . Then
lyche-numerical-linear-algebra-F02713	201	■ 1.000	$0^0 := 0$	$0^0 := 0$	where $0^0 := 0$.
lyche-numerical-linear-algebra-F02714	201	■ 1.000	a_j	a_j	Proof The result is trivial if one or more of the a_j 's are zero so assume $a_j > 0$
lyche-numerical-linear-algebra-F02715	201	■ 1.000	$a_j > 0$	$a_j > 0$	Proof The result is trivial if one or more of the a_j 's are zero so assume $a_j > 0$
lyche-numerical-linear-algebra-F02716	201	■ 1.000	$f: (0, \infty) \rightarrow \mathbb{R}$	$f: (0, \infty) \rightarrow \mathbb{R}$	for all j . Consider the function <u>$f: (0, \infty) \rightarrow \mathbb{R}$</u> given by $f(x) = -\log x$. Since
lyche-numerical-linear-algebra-F02717	201	■ 1.000	$f(x) = -\log x$	$f(x) = -\log x$	for all j . Consider the function $f: (0, \infty) \rightarrow \mathbb{R}$ given by $f(x) = -\log x$. Since
lyche-numerical-linear-algebra-F02718	201	■ 1.000	$f'(x) = 1/x^2 > 0$	$f''(x) = 1/x^2 > 0$	<u>$f''(x) = 1/x^2 > 0$</u> for $x \in (0, \infty)$, it follows from Lemma 8.2 that this function is
lyche-numerical-linear-algebra-F02719	201	■ 1.000	$x \in (0, \infty)$	$x \in (0, \infty)$	$f''(x) = 1/x^2 > 0$ for $x \in (0, \infty)$, it follows from Lemma 8.2 that this function is
lyche-numerical-linear-algebra-F02720	201	■ 1.000	$\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$	$\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$	or $\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$. The inequality follows since $\exp(\log x)$
lyche-numerical-linear-algebra-F02721	201	■ 1.000	$\exp(\log x)$	$\exp(\log x)$	or $\log(a_1^{\lambda_1} \cdots a_n^{\lambda_n}) \leq \log(\sum_{j=1}^n \lambda_j a_j)$. The inequality follows since $\exp(\log x)$
lyche-numerical-linear-algebra-F02722	201	■ 1.000	$= x$	$= x$	$= x$ for $x > 0$ and the exponential function is monotone increasing.
lyche-numerical-linear-algebra-F02723	201	■ 1.000	$x > 0$	$x > 0$	$= x$ for $x > 0$ and the exponential function is monotone increasing.
lyche-numerical-linear-algebra-F02724	201	■ 1.000	$\lambda_j = \frac{1}{n}$	$\lambda_j = \frac{1}{n}$	Taking $\lambda_j = \frac{1}{n}$ for all j in (8.32) we obtain the classical geometric/arithmetic
lyche-numerical-linear-algebra-F02725	201	■ 1.000	$p = \infty$	$p = \infty$	Proof We leave the proof for $p = 1$ and $p = \infty$ as an exercise so assume $1 < p <$
lyche-numerical-linear-algebra-F02726	201	■ 1.000	$1 < p <$	$1 < p <$	Proof We leave the proof for $p = 1$ and $p = \infty$ as an exercise so assume $1 < p <$
lyche-numerical-linear-algebra-F02727	201	■ 1.000	$a, b \geq 0$	$a, b \geq 0$	∞ . For any $a, b \geq 0$ the weighted arithmetic/geometric mean inequality implies
lyche-numerical-linear-algebra-F02728	202	■ 1.000	$(p-1)q = p$	$(p-1)q = p$	relation $(p-1)q = p$ the result is
lyche-numerical-linear-algebra-F02729	202	■ 1.000	$a = \pm 1$	$a = \pm 1$	Proof We set $a = \pm 1$ in (5.5) and add the two equations.
lyche-numerical-linear-algebra-F02730	202	■ 1.000	$\langle \mathbf{x}, \mathbf{x} \rangle = \ \mathbf{x}\ ^2$	$\langle \mathbf{x}, \mathbf{x} \rangle = \ \mathbf{x}\ ^2$	<u>$\langle \mathbf{x}, \mathbf{x} \rangle = \ \mathbf{x}\ ^2$</u> if and only if the parallelogram identity (8.35) holds for all $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.
lyche-numerical-linear-algebra-F02731	202	■ 1.000	$\mathbf{x}, \mathbf{y} \in \mathcal{V}$	$\mathbf{x}, \mathbf{y} \in \mathcal{V}$	$\langle \mathbf{x}, \mathbf{x} \rangle = \ \mathbf{x}\ ^2$ if and only if the parallelogram identity (8.35) holds for all $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.
lyche-numerical-linear-algebra-F02732	203	■ 0.670	$\langle \mathbf{y}, \mathbf{z} \rangle = 0$	$\langle \mathbf{y}, \mathbf{z} \rangle = 0$	In particular, since $\mathbf{y} = \mathbf{0}$ implies $\langle \mathbf{y}, \mathbf{z} \rangle = 0$ we obtain $\langle \mathbf{x}, \mathbf{z} \rangle = 2\langle \frac{\mathbf{x}}{2}, \mathbf{z} \rangle$ for all

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F02733	203	0.670	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2 \langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \rangle$	$\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2 \left\langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \right\rangle$	In particular, since $\boldsymbol{y} = \mathbf{0}$ implies $\langle \boldsymbol{y}, \boldsymbol{z} \rangle = 0$ we obtain $\langle \boldsymbol{x}, \boldsymbol{z} \rangle = 2 \langle \frac{\boldsymbol{x}}{2}, \boldsymbol{z} \rangle$ for all
lyche-numerical-linear-algebra- F02734	203	1.000	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$. This means that $2 \langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and (8.37)
lyche-numerical-linear-algebra- F02735	203	1.000	$2 \langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$	$2 \left\langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \right\rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$	$\boldsymbol{x}, \boldsymbol{z} \in \mathcal{V}$. This means that $2 \langle \frac{\boldsymbol{x} + \boldsymbol{y}}{2}, \boldsymbol{z} \rangle = \langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle$ for all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathcal{V}$ and (8.37)
lyche-numerical-linear-algebra- F02736	203	1.000	$a = n$	$a = n$	We first show (8.38) when $a = n$ is a positive integer. By induction
lyche-numerical-linear-algebra- F02737	203	1.000	$a > 0$	$a > 0$	Now if $a > 0$ there is a sequence $\{a_n\}$ of positive rational numbers converging to a .
lyche-numerical-linear-algebra- F02738	203	1.000	$\{a_n\}$	$\{a_n\}$	Now if $a > 0$ there is a sequence $\{a_n\}$ of positive rational numbers converging to a .
lyche-numerical-linear-algebra- F02739	204	1.000	$a \langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle a \boldsymbol{x}, \boldsymbol{y} \rangle$	$a \langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle a \boldsymbol{x}, \boldsymbol{y} \rangle$	Taking limits and using continuity of norms we obtain $\langle a \boldsymbol{x}, \boldsymbol{y} \rangle = \langle a \boldsymbol{x}, \boldsymbol{y} \rangle$. This also
lyche-numerical-linear-algebra- F02740	204	0.651	$a < 0$	$a < 0$	holds for $a = 0$. Finally, if $a < 0$ then $(-a) > 0$ and from what we just showed
lyche-numerical-linear-algebra- F02741	204	0.651	$(-a) > 0$	$(-a) > 0$	holds for $a = 0$. Finally, if $a < 0$ then $(-a) > 0$ and from what we just showed
lyche-numerical-linear-algebra- F02742	204	1.000	$\mathcal{V} = \mathbb{C}^n, 1 \leq p \leq \infty, n \geq 2$	$\mathcal{V} = \mathbb{C}^n, 1 \leq p \leq \infty, n \geq 2$	on $\mathcal{V} = \mathbb{R}^n$ or $\mathcal{V} = \mathbb{C}^n, 1 < p < \infty, n \geq 2$, there is an inner product on $\mathcal{V}$ such that
lyche-numerical-linear-algebra- F02743	204	1.000	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _p^2$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _p^2$	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle = \ \boldsymbol{x}\ _p^2$ for all $\boldsymbol{x} \in \mathcal{V}$ if and only if $p = 2$.
lyche-numerical-linear-algebra- F02744	204	1.000	$p \neq 2$	$p \neq 2$	standard inner product. If $p \neq 2$ then the parallelogram identity (8.35) does not
lyche-numerical-linear-algebra- F02745	204	1.000	$\boldsymbol{x} := \boldsymbol{e}_1$	$\boldsymbol{x} := \boldsymbol{e}_1$	hold for say $\boldsymbol{x} := \boldsymbol{e}_1$ and $\boldsymbol{y} := \boldsymbol{e}_2$.
lyche-numerical-linear-algebra- F02746	204	1.000	$\boldsymbol{y} := \boldsymbol{e}_2$	$\boldsymbol{y} := \boldsymbol{e}_2$	hold for say $\boldsymbol{x} := \boldsymbol{e}_1$ and $\boldsymbol{y} := \boldsymbol{e}_2$.
lyche-numerical-linear-algebra- F02747	204	1.000	$0 < \lambda_n \leq \dots \leq \lambda_1$	$0 < \lambda_n \leq \dots \leq \lambda_1$	metric positive definite matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ with eigenvalues $0 < \lambda_n < \dots < \lambda_1$.
lyche-numerical-linear-algebra- F02748	204	1.000	$\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$	$\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$	symmetric and positive definite matrix and assume $\boldsymbol{b}_1, \dots, \boldsymbol{b}_n$ is a basis for $\mathbb{R}^n$. We
lyche-numerical-linear-algebra- F02749	204	1.000	$\boldsymbol{B}_k := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_k] \in \mathbb{R}^{n \times k}$	$\boldsymbol{B}_k := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_k] \in \mathbb{R}^{n \times k}$	define $\boldsymbol{B}_k := [\boldsymbol{b}_1, \dots, \boldsymbol{b}_k] \in \mathbb{R}^{n \times k}$ for $k = 1, \dots, n$. We consider in this exercise the
lyche-numerical-linear-algebra- F02750	204	0.973	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$	inner product $\langle \cdot, \cdot \rangle$ defined by $\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y}$ for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and the corresponding
lyche-numerical-linear-algebra- F02751	204	1.000	$\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$	$\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$	norm $\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$. We define $\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$ and
lyche-numerical-linear-algebra- F02752	204	1.000	$\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$	$\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$	norm $\ \boldsymbol{x}\ _A := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$. We define $\tilde{\boldsymbol{b}}_1 := \boldsymbol{b}_1$ and
lyche-numerical-linear-algebra- F02753	204	1.000	$\boldsymbol{B}_k^T \boldsymbol{A} \boldsymbol{B}_k$	$\boldsymbol{B}_k^T \boldsymbol{A} \boldsymbol{B}_k$	a) Show that $\boldsymbol{B}_k^T \boldsymbol{A} \boldsymbol{B}_k$ is positive definite for $k = 1, \dots, n$.
lyche-numerical-linear-algebra- F02754	204	0.999	$\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$	$\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$	b) Show that for $k = 2, \dots, n$ we have (i) $\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$ for $j = 1, \dots, k - 1$ and
lyche-numerical-linear-algebra- F02755	204	0.999	$j = 1, \dots, k - 1$	$j = 1, \dots, k - 1$	b) Show that for $k = 2, \dots, n$ we have (i) $\langle \tilde{\boldsymbol{b}}_k, \boldsymbol{b}_j \rangle = 0$ for $j = 1, \dots, k - 1$ and
lyche-numerical-linear-algebra- F02756	204	1.000	$\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \text{span}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{k-1})$	$\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \text{span}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{k-1})$	(ii) $\tilde{\boldsymbol{b}}_k - \boldsymbol{b}_k \in \text{span}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{k-1})$.
lyche-numerical-linear-algebra- F02757	205	0.985	$\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n$	$\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n$	c) Explain why $\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n$ is a basis for $\mathbb{R}^n$ which in addition is $\boldsymbol{A}$ -orthogonal, i.e.,

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lyche-numerical-linear-algebra-205-F02758	205	■ 0.984	$\left\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \right\rangle = 0$	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$ for all $i, j \leq n, i \neq j$.
lyche-numerical-linear-algebra-205-F02759	205	■ 0.984	$i, j \leq n, i \neq j$	$i, j \leq n, i \neq j$	$\langle \tilde{\boldsymbol{b}}_i, \tilde{\boldsymbol{b}}_j \rangle = 0$ for all $i, j \leq n, i \neq j$.
lyche-numerical-linear-algebra-205-F02760	205	■ 1.000	$\tilde{\boldsymbol{B}}_n := \left[\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n \right]$	$\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n]$	d) Define $\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n]$. Show that there is an upper triangular matrix $\boldsymbol{T} \in$
lyche-numerical-linear-algebra-205-F02761	205	■ 1.000	$\boldsymbol{T} \in$	$\boldsymbol{T} \in$	d) Define $\tilde{\boldsymbol{B}}_n := [\tilde{\boldsymbol{b}}_1, \dots, \tilde{\boldsymbol{b}}_n]$. Show that there is an upper triangular matrix $\boldsymbol{T} \in$
lyche-numerical-linear-algebra-205-F02762	205	■ 1.000	$\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n \boldsymbol{T}$	$\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n \boldsymbol{T}$	$\mathbb{R}^{n \times n}$ with ones on the diagonal and satisfies $\boldsymbol{B}_n = \tilde{\boldsymbol{B}}_n \boldsymbol{T}$.
lyche-numerical-linear-algebra-205-F02763	205	■ 1.000	$ t_{ij} \leq \frac{1}{2}$	$ t_{ij} \leq \frac{1}{2}$	e) Assume that the matrix $\boldsymbol{T}$ in d) is such that $ t_{ij} \leq \frac{1}{2}$ for all $i, j \leq n, i \neq j$.
lyche-numerical-linear-algebra-205-F02764	205	■ 0.947	$\left\ \tilde{\boldsymbol{b}}_k \right\ _{\boldsymbol{A}}^2 \leq 2 \left\ \tilde{\boldsymbol{b}}_{k+1} \right\ _{\boldsymbol{A}}^2$	$\left\ \tilde{\boldsymbol{b}}_k \right\ _{\boldsymbol{A}}^2 \leq 2 \left\ \tilde{\boldsymbol{b}}_{k+1} \right\ _{\boldsymbol{A}}^2$	Assume also that $\left\ \tilde{\boldsymbol{b}}_k \right\ _{\boldsymbol{A}}^2 \leq 2 \left\ \tilde{\boldsymbol{b}}_{k+1} \right\ _{\boldsymbol{A}}^2$ for $k = 1, \dots, n-1$ and that $\det(\boldsymbol{B}_n) = 1$.
lyche-numerical-linear-algebra-205-F02765	205	■ 0.947	$\operatorname{det}(\boldsymbol{B}_n) = 1$	$\det(\boldsymbol{B}_n) = 1$	Assume also that $\left\ \tilde{\boldsymbol{b}}_k \right\ _{\boldsymbol{A}}^2 \leq 2 \left\ \tilde{\boldsymbol{b}}_{k+1} \right\ _{\boldsymbol{A}}^2$ for $k = 1, \dots, n-1$ and that $\det(\boldsymbol{B}_n) = 1$.
lyche-numerical-linear-algebra-205-F02766	205	■ 0.948	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	consistent by considering $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-205-F02767	205	■ 1.000	$\sqrt{mn}$	$\sqrt{mn}$	b) Show that the constant $\sqrt{mn}$ can be replaced by m and by n .
lyche-numerical-linear-algebra-205-F02768	205	■ 0.985	$\ \boldsymbol{Ax}\ _{\infty} \leq$	$\ \boldsymbol{Ax}\ _{\infty} \leq$	a) Show that the max norm is subordinate to the ∞ and 1 norm, i.e., $\ \boldsymbol{Ax}\ _{\infty} \leq$
lyche-numerical-linear-algebra-205-F02769	205	■ 1.000	$\ \boldsymbol{A}\ _M \ \boldsymbol{x}\ _1$	$\ \boldsymbol{A}\ _M \ \boldsymbol{x}\ _1$	$\ \boldsymbol{A}\ _M \ \boldsymbol{x}\ _1$ holds for all $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ and all $\boldsymbol{x} \in \mathbb{C}^n$.
lyche-numerical-linear-algebra-205-F02770	205	■ 1.000	$\ \boldsymbol{A}\ _M = a_{kl} $	$\ \boldsymbol{A}\ _M = a_{kl} $	b) Show that if $\ \boldsymbol{A}\ _M = a_{kl} $, then $\ \boldsymbol{Ae}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$.
lyche-numerical-linear-algebra-205-F02771	205	■ 1.000	$\ \boldsymbol{Ae}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$	$\ \boldsymbol{Ae}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$	b) Show that if $\ \boldsymbol{A}\ _M = a_{kl} $, then $\ \boldsymbol{Ae}_l\ _{\infty} = \ \boldsymbol{A}\ _M \ \boldsymbol{e}_l\ _1$.
lyche-numerical-linear-algebra-205-F02772	205	■ 1.000	$\ \boldsymbol{A}\ _M = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{Ax}\ _{\infty}}{\ \boldsymbol{x}\ _1}$	$\ \boldsymbol{A}\ _M = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{Ax}\ _{\infty}}{\ \boldsymbol{x}\ _1}$	c) Show that $\ \boldsymbol{A}\ _M = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{Ax}\ _{\infty}}{\ \boldsymbol{x}\ _1}$.
lyche-numerical-linear-algebra-205-F02773	205	■ 0.480	$\left\ \tilde{\boldsymbol{b}}_1 \right\ _{\boldsymbol{A}}^2 \cdots \left\ \tilde{\boldsymbol{b}}_n \right\ _{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$	$\left\ \tilde{\boldsymbol{b}}_1 \right\ _{\boldsymbol{A}}^2 \cdots \left\ \tilde{\boldsymbol{b}}_n \right\ _{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$	² Hint: Show that $\left\ \tilde{\boldsymbol{b}}_1 \right\ _{\boldsymbol{A}}^2 \cdots \left\ \tilde{\boldsymbol{b}}_n \right\ _{\boldsymbol{A}}^2 = \det(\boldsymbol{A})$.
lyche-numerical-linear-algebra-206-F02774	206	■ 0.991	$\ \boldsymbol{Ax}\ _2 \geq \sigma_n$	$\ \boldsymbol{Ax}\ _2 \geq \sigma_n$	Show that $\ \boldsymbol{Ax}\ _2 > \sigma_r$ for all $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _2 = 1$. Show that
lyche-numerical-linear-algebra-206-F02775	206	■ 1.000	$\ \boldsymbol{A}\ _p$	$\ \boldsymbol{A}\ _p$	Compute $\ \boldsymbol{A}\ _p$ and $\ \boldsymbol{A}^{-1}\ _p$ for $p = 1, 2, \infty$.
lyche-numerical-linear-algebra-206-F02776	206	■ 1.000	$\ \boldsymbol{A}^{-1}\ _p$	$\ \boldsymbol{A}^{-1}\ _p$	Compute $\ \boldsymbol{A}\ _p$ and $\ \boldsymbol{A}^{-1}\ _p$ for $p = 1, 2, \infty$.
lyche-numerical-linear-algebra-206-F02777	206	■ 1.000	$\ \boldsymbol{VA}\ _2 =$	$\ \boldsymbol{VA}\ _2 =$	Exercise 8.11 (Unitary Invariance of the Spectral Norm) Show that $\ \boldsymbol{VA}\ _2 =$
lyche-numerical-linear-algebra-206-F02778	206	■ 1.000	$\boldsymbol{V}^* \boldsymbol{V} = \boldsymbol{I}$	$\boldsymbol{V}^* \boldsymbol{V} = \boldsymbol{I}$	$\ \boldsymbol{A}\ _2$ holds even for a rectangular $\boldsymbol{V}$ as long as $\boldsymbol{V}^* \boldsymbol{V} = \boldsymbol{I}$.
lyche-numerical-linear-algebra-206-F02779	206	■ 1.000	$\ \boldsymbol{AU}\ _2$	$\ \boldsymbol{AU}\ _2$	Exercise 8.12 ($\ \boldsymbol{AU}\ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with
lyche-numerical-linear-algebra-206-F02780	206	■ 1.000	$\boldsymbol{A}$	$\boldsymbol{A}$	Exercise 8.12 ($\ \boldsymbol{AU}\ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with

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lyche-numerical-linear-algebra-206-F02781	206	1.000	$\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$	$\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$	Exercise 8.12 ($\ \boldsymbol{A}\boldsymbol{U}\ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with
lyche-numerical-linear-algebra-206-F02782	206	1.000	$\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$	$\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$	Exercise 8.12 ($\ \boldsymbol{A}\boldsymbol{U}\ _2$ Rectangular $\boldsymbol{A}$) Find $\boldsymbol{A} \in \mathbb{R}^{2 \times 2}$ and $\boldsymbol{U} \in \mathbb{R}^{2 \times 1}$ with
lyche-numerical-linear-algebra-206-F02783	206	1.000	$\ \boldsymbol{A}\boldsymbol{U}\ _2 < \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\boldsymbol{U}\ _2 < \ \boldsymbol{A}\ _2$	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$ such that $\ \boldsymbol{A}\boldsymbol{U}\ _2 < \ \boldsymbol{A}\ _2$. Thus, in general, $\ \boldsymbol{A}\boldsymbol{U}\ _2 = \ \boldsymbol{A}\ _2$ does
lyche-numerical-linear-algebra-206-F02784	206	1.000	$\ \boldsymbol{A}\boldsymbol{U}\ _2 = \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\boldsymbol{U}\ _2 = \ \boldsymbol{A}\ _2$	$\boldsymbol{U}^T \boldsymbol{U} = \boldsymbol{I}$ such that $\ \boldsymbol{A}\boldsymbol{U}\ _2 < \ \boldsymbol{A}\ _2$. Thus, in general, $\ \boldsymbol{A}\boldsymbol{U}\ _2 = \ \boldsymbol{A}\ _2$ does
lyche-numerical-linear-algebra-206-F02785	206	0.999	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A}) :=$	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A}) :=$	Exercise 8.13 (p-Norm of Diagonal Matrix) Show that $\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A}) :=$
lyche-numerical-linear-algebra-206-F02786	206	0.972	$\max \lambda_i $	$\max \lambda_i $	$\max \lambda_i $ (the largest eigenvalue of $\boldsymbol{A}$), $1 \leq p \leq \infty$, when $\boldsymbol{A}$ is a diagonal matrix.
lyche-numerical-linear-algebra-206-F02787	206	1.000	$\boldsymbol{a} \in \mathbb{C}^m$	$\boldsymbol{a} \in \mathbb{C}^m$	Exercise 8.14 (Spectral Norm of a Column Vector) A vector $\boldsymbol{a} \in \mathbb{C}^m$ can also be
lyche-numerical-linear-algebra-206-F02788	206	1.000	$\boldsymbol{A} \in \mathbb{C}^{m, 1}$	$\boldsymbol{A} \in \mathbb{C}^{m, 1}$	considered as a matrix $\boldsymbol{A} \in \mathbb{C}^{m, 1}$.
lyche-numerical-linear-algebra-206-F02789	206	1.000	$\ \boldsymbol{A}\ _p = \ \boldsymbol{a}\ _p$	$\ \boldsymbol{A}\ _p = \ \boldsymbol{a}\ _p$	b) Show that $\ \boldsymbol{A}\ _p = \ \boldsymbol{a}\ _p$ for $1 \leq p \leq \infty$.
lyche-numerical-linear-algebra-206-F02790	206	1.000	$ \boldsymbol{A} \in \mathbb{R}^{m \times n}$	$ \boldsymbol{A} \in \mathbb{R}^{m \times n}$	let $ \boldsymbol{A} \in \mathbb{R}^{m \times n}$ be the matrix with elements $ a_{ij} $.
lyche-numerical-linear-algebra-206-F02791	206	1.000	$ a_{ij} $	$ a_{ij} $	let $ \boldsymbol{A} \in \mathbb{R}^{m \times n}$ be the matrix with elements $ a_{ij} $.
lyche-numerical-linear-algebra-206-F02792	206	0.999	$ \boldsymbol{A} $	$ \boldsymbol{A} $	a) Compute $ \boldsymbol{A} $ if $\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$.
lyche-numerical-linear-algebra-206-F02793	206	0.999	$\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$	$\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$	a) Compute $ \boldsymbol{A} $ if $\boldsymbol{A} = \begin{bmatrix} 1+i & -2 \\ 1 & 1-i \end{bmatrix}$, $i = \sqrt{-1}$.
lyche-numerical-linear-algebra-206-F02794	206	1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F, \ \boldsymbol{A}\ _p = \ \boldsymbol{A}\ _p$	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F, \ \boldsymbol{A}\ _p = \ \boldsymbol{A}\ _p$	b) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F, \ \boldsymbol{A}\ _p = \ \boldsymbol{A}\ _p$ for $p = 1, \infty$.
lyche-numerical-linear-algebra-206-F02795	206	1.000	$p=1, \infty$	$p = 1, \infty$	b) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _F = \ \boldsymbol{A}\ _F, \ \boldsymbol{A}\ _p = \ \boldsymbol{A}\ _p$ for $p = 1, \infty$.
lyche-numerical-linear-algebra-206-F02796	206	1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _2$	$\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _2 \leq \ \boldsymbol{A}\ _2$	c) Show that for any $\boldsymbol{A} \in \mathbb{C}^{m \times n} \ \boldsymbol{A}\ _2 < \ \boldsymbol{A}\ _2$.
lyche-numerical-linear-algebra-206-F02797	206	0.806	$\ \boldsymbol{A}\ _2 < \ \boldsymbol{A}\ _2$	$\ \boldsymbol{A}\ _2 < \ \boldsymbol{A}\ _2$	d) Find a real symmetric 2×2 matrix $\boldsymbol{A}$ such that $\ \boldsymbol{A}\ _2 < \ \boldsymbol{A}\ _2$.
lyche-numerical-linear-algebra-206-F02798	206	1.000	$\{\lambda_1, \lambda_2, \dots, \lambda_m\}$	$\{\lambda_1, \lambda_2, \dots, \lambda_m\}$	$\{\lambda_1, \lambda_2, \dots, \lambda_m\}$ in no particular order. We have that $m \leq n$, since some of the
lyche-numerical-linear-algebra-206-F02799	206	1.000	$\boldsymbol{x}_0 \in \mathbb{C}^n$	$\boldsymbol{x}_0 \in \mathbb{C}^n$	eigenvalues may have multiplicity larger than one. Given $\boldsymbol{x}_0 \in \mathbb{C}^n$, and $k \geq 0$, we
lyche-numerical-linear-algebra-207-F02800	207	1.000	$\{\boldsymbol{x}_k\}_{k=0}^{m-1}$	$\{\boldsymbol{x}_k\}_{k=0}^{m-1}$	define the sequence $\{\boldsymbol{x}_k\}_{k=0}^{m-1}$ by
lyche-numerical-linear-algebra-207-F02801	207	1.000	c_{ik}	c_{ik}	a) Let the coefficients c_{ik} be defined by
lyche-numerical-linear-algebra-207-F02802	207	1.000	$\{(\sigma_i, \boldsymbol{u}_i)\}_{i=1}^n$	$\{(\sigma_i, \boldsymbol{u}_i)\}_{i=1}^n$	where $\{(\sigma_i, \boldsymbol{u}_i)\}_{i=1}^n$ are the eigenpairs of $\boldsymbol{A}$. Show that
lyche-numerical-linear-algebra-207-F02803	207	1.000	$l \leq m$	$l \leq m$	b) Show that for some $l \leq m$, we have that $\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \dots = \boldsymbol{x}_m = \boldsymbol{x}$, where
lyche-numerical-linear-algebra-207-F02804	207	1.000	$\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \dots = \boldsymbol{x}_m = \boldsymbol{x}$	$\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \dots = \boldsymbol{x}_m = \boldsymbol{x}$	b) Show that for some $l \leq m$, we have that $\boldsymbol{x}_l = \boldsymbol{x}_{l+1} = \dots = \boldsymbol{x}_m = \boldsymbol{x}$, where
lyche-numerical-linear-algebra-207-F02805	207	1.000	$\boldsymbol{T} = \text{tridiag}(c, d, c)$	$\boldsymbol{T} = \text{tridiag}(c, d, c)$	c) Consider this iteration for the $n \times n$ matrix $\boldsymbol{T} = \text{tridiag}(c, d, c)$, where d and c
lyche-numerical-linear-algebra-207-F02806	207	1.000	$d > 2c$	$d > 2c$	are positive real numbers and $d > 2c$. The eigenvalues of $\boldsymbol{T}$ are

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F02807	207	■ 1.000	$\boldsymbol{T} \ \boldsymbol{x}=\boldsymbol{b}$	$\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$	What is the operation count for solving $\boldsymbol{T}\boldsymbol{x} = \boldsymbol{b}$ using the iterative algorithm
lyche-numerical-linear-algebra-F02808	207	■ 1.000	$b_{ij}=0$	$b_{ij} = 0$	$b_{ij} = 0$ if $ i - j \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.
lyche-numerical-linear-algebra-F02809	207	■ 1.000	$ i-j \leq 1$	$ i - j \leq 1$	$b_{ij} = 0$ if $ i - j \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.
lyche-numerical-linear-algebra-F02810	207	■ 1.000	$\boldsymbol{A}=\boldsymbol{T}+\boldsymbol{B}$	$\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$	$b_{ij} = 0$ if $ i - j \leq 1$. Set $\boldsymbol{A} = \boldsymbol{T} + \boldsymbol{B}$, where $\boldsymbol{T}$ is the tridiagonal matrix above.
lyche-numerical-linear-algebra-F02811	207	■ 0.991	$\rho(\boldsymbol{E}):=\max_i \lambda_i $	$\rho(\boldsymbol{E}) := \max_i \lambda_i $	Recall that if $\boldsymbol{E} \in \mathbb{R}^{n \times n}$ has eigenvalues $\lambda_1, \dots, \lambda_n$ then $\rho(\boldsymbol{E}) := \max_i \lambda_i $ is
lyche-numerical-linear-algebra-F02812	207	■ 1.000	$\rho(\boldsymbol{T}^{-1}\boldsymbol{B}) \leq \rho(\boldsymbol{T}^{-1}) \rho(\boldsymbol{B})$	$\rho(\boldsymbol{T}^{-1}\boldsymbol{B}) \leq \rho(\boldsymbol{T}^{-1}) \rho(\boldsymbol{B})$	the spectral radius of $\boldsymbol{E}$. Show that $\rho(\boldsymbol{T}^{-1}\boldsymbol{B}) < \rho(\boldsymbol{T}^{-1})\rho(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F02813	208	■ 1.000	$\delta:=\ \boldsymbol{x}-\boldsymbol{y}\ _2/\ \boldsymbol{x}\ _2$	$\delta := \ \boldsymbol{x} - \boldsymbol{y}\ _2 / \ \boldsymbol{x}\ _2$	We define $\delta := \ \boldsymbol{x} - \boldsymbol{y}\ _2 / \ \boldsymbol{x}\ _2$.
lyche-numerical-linear-algebra-F02814	208	■ 1.000	$K_2(\boldsymbol{A})=\ \boldsymbol{A}\ _2\ \boldsymbol{A}^{-1}\ _2$	$K_2(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2$	a) Determine $K_2(\boldsymbol{A}) = \ \boldsymbol{A}\ _2 \ \boldsymbol{A}^{-1}\ _2$. Give an upper bound and a positive lower
lyche-numerical-linear-algebra-F02815	208	■ 1.000	δ	δ	bound for δ without computing $\boldsymbol{x}$ and $\boldsymbol{y}$.
lyche-numerical-linear-algebra-F02816	208	■ 1.000	$b_2=2.0$	$b_2 = 2.0$	b) Suppose as before that $b_2 = 2.0$ and $\ \boldsymbol{e}\ _2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which
lyche-numerical-linear-algebra-F02817	208	■ 1.000	$\ \boldsymbol{e}\ _2=0.1$	$\ \boldsymbol{e}\ _2 = 0.1$	b) Suppose as before that $b_2 = 2.0$ and $\ \boldsymbol{e}\ _2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which
lyche-numerical-linear-algebra-F02818	208	■ 1.000	b_1	b_1	b) Suppose as before that $b_2 = 2.0$ and $\ \boldsymbol{e}\ _2 = 0.1$. Determine b_1 and $\boldsymbol{e}$ which
lyche-numerical-linear-algebra-F02819	208	■ 1.000	$\boldsymbol{y}_A$	$\boldsymbol{y}_A$	special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_A$ and $\boldsymbol{y}_{A^{-1}}$ are vectors with $\ \boldsymbol{y}_A\ = \ \boldsymbol{y}_{A^{-1}}\ = 1$
lyche-numerical-linear-algebra-F02820	208	■ 1.000	$\boldsymbol{y}_{A^{-1}}$	$\boldsymbol{y}_{A^{-1}}$	special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_A$ and $\boldsymbol{y}_{A^{-1}}$ are vectors with $\ \boldsymbol{y}_A\ = \ \boldsymbol{y}_{A^{-1}}\ = 1$
lyche-numerical-linear-algebra-F02821	208	■ 1.000	$\ \boldsymbol{y}_A\ = \ \boldsymbol{y}_{A^{-1}}\ = 1$	$\ \boldsymbol{y}_A\ = \ \boldsymbol{y}_{A^{-1}}\ = 1$	special choices of $\boldsymbol{b}$. Suppose $\boldsymbol{y}_A$ and $\boldsymbol{y}_{A^{-1}}$ are vectors with $\ \boldsymbol{y}_A\ = \ \boldsymbol{y}_{A^{-1}}\ = 1$
lyche-numerical-linear-algebra-F02822	208	■ 0.968	$\ \boldsymbol{A}\ = \ \boldsymbol{A}\boldsymbol{y}_A\ $	$\ \boldsymbol{A}\ = \ \boldsymbol{A}\boldsymbol{y}_A\ $	and $\ \boldsymbol{A}\ = \ \boldsymbol{A}\boldsymbol{y}_A\ $ and $\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1}\boldsymbol{y}_{A^{-1}}\ $.
lyche-numerical-linear-algebra-F02823	208	■ 0.968	$\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1}\boldsymbol{y}_{A^{-1}}\ $	$\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1}\boldsymbol{y}_{A^{-1}}\ $	and $\ \boldsymbol{A}\ = \ \boldsymbol{A}\boldsymbol{y}_A\ $ and $\ \boldsymbol{A}^{-1}\ = \ \boldsymbol{A}^{-1}\boldsymbol{y}_{A^{-1}}\ $.
lyche-numerical-linear-algebra-F02824	208	■ 1.000	$\boldsymbol{b}=\boldsymbol{A}\boldsymbol{y}_A$	$\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}_A$	a) Show that the upper bound in (8.22) is attained if $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}_A$ and $\boldsymbol{e} = \boldsymbol{y}_{A^{-1}}$.
lyche-numerical-linear-algebra-F02825	208	■ 1.000	$\boldsymbol{e}=\boldsymbol{y}_{A^{-1}}$	$\boldsymbol{e} = \boldsymbol{y}_{A^{-1}}$	a) Show that the upper bound in (8.22) is attained if $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{y}_A$ and $\boldsymbol{e} = \boldsymbol{y}_{A^{-1}}$.
lyche-numerical-linear-algebra-F02826	208	■ 0.980	$\boldsymbol{b}=\boldsymbol{y}_{A^{-1}}$	$\boldsymbol{b} = \boldsymbol{y}_{A^{-1}}$	b) Show that the lower bound is attained if $\boldsymbol{b} = \boldsymbol{y}_{A^{-1}}$ and $\boldsymbol{e} = \boldsymbol{A}\boldsymbol{y}_A$.
lyche-numerical-linear-algebra-F02827	208	■ 0.980	$\boldsymbol{e}=\boldsymbol{A}\boldsymbol{y}_A$	$\boldsymbol{e} = \boldsymbol{A}\boldsymbol{y}_A$	b) Show that the lower bound is attained if $\boldsymbol{b} = \boldsymbol{y}_{A^{-1}}$ and $\boldsymbol{e} = \boldsymbol{A}\boldsymbol{y}_A$.
lyche-numerical-linear-algebra-F02828	208	■ 1.000	$m \geq 1$	$m \geq 1$	will show that for $m \geq 1$
lyche-numerical-linear-algebra-F02829	208	■ 1.000	$\boldsymbol{T}:=\text{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}$	$\boldsymbol{T} := \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$	where $\boldsymbol{T} := \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ and $\text{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$ is the p -
lyche-numerical-linear-algebra-F02830	208	■ 1.000	$\text{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$	$\text{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$	where $\boldsymbol{T} := \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ and $\text{cond}_p(\boldsymbol{T}) := \ \boldsymbol{T}\ _p \ \boldsymbol{T}^{-1}\ _p$ is the p -
lyche-numerical-linear-algebra-F02831	208	■ 1.000	$m \geq 3$	$m \geq 3$	a) Show that for $m \geq 3$
lyche-numerical-linear-algebra-F02832	208	■ 1.000	$\text{cond}_1(\boldsymbol{T}) = \text{cond}_\infty(\boldsymbol{T}) = 3$	$\text{cond}_1(\boldsymbol{T}) = \text{cond}_\infty(\boldsymbol{T}) = 3$	and that $\text{cond}_1(\boldsymbol{T}) = \text{cond}_\infty(\boldsymbol{T}) = 3$ for $m = 2$.

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lyche-numerical-linear-algebra-F02833	208	■ 1.000	m=2	$m = 2$	and that $\text{cond}_1(T) = \text{cond}_\infty(T) = 3$ for $m = 2$.
lyche-numerical-linear-algebra-F02834	209	■ 1.000	\tan x>x	$\tan x > x$	Hint: For the upper bound use the inequality $\tan x > x$ valid for $0 < x < \pi/2$.
lyche-numerical-linear-algebra-F02835	209	■ 1.000	0<x<\pi / 2	$0 < x < \pi/2$	Hint: For the upper bound use the inequality $\tan x > x$ valid for $0 < x < \pi/2$.
lyche-numerical-linear-algebra-F02836	209	■ 1.000	\cot ^{2} x>\frac{1}{x^{2}}-\frac{2}{3}	$\cot^2 x > \frac{1}{x^2} - \frac{2}{3}$	For the lower bound we use (without proof) the inequality $\cot^2 x > \frac{1}{x^2} - \frac{2}{3}$ for
lyche-numerical-linear-algebra-F02837	209	■ 1.000	\boldsymbol{I}-\boldsymbol{E}	$\boldsymbol{I} - \boldsymbol{E}$	a) Show that if $\boldsymbol{I} - \boldsymbol{E}$ is nonsingular then
lyche-numerical-linear-algebra-F02838	209	■ 1.000	\ \boldsymbol{E}\ <1	$\ \boldsymbol{E}\ < 1$	b) If $\ \boldsymbol{E}\ < 1$ then $(\boldsymbol{I} - \boldsymbol{E})^{-1}$ is nonsingular by exists and
lyche-numerical-linear-algebra-F02839	209	■ 1.000	(\boldsymbol{I}-\boldsymbol{E})^{-1}	$(\boldsymbol{I} - \boldsymbol{E})^{-1}$	b) If $\ \boldsymbol{E}\ < 1$ then $(\boldsymbol{I} - \boldsymbol{E})^{-1}$ is nonsingular by exists and
lyche-numerical-linear-algebra-F02840	209	■ 1.000	\boldsymbol{x}=\left[x_{0}, \ldots, x_{n}\right]^T \in \mathbb{R}^{n+1}	$\boldsymbol{x} = [x_0, \dots, x_n]^T \in \mathbb{R}^{n+1}$	components of $\boldsymbol{x} = [x_0, \dots, x_n]^T \in \mathbb{R}^{n+1}$ define a partition of the interval $[a, b]$,
lyche-numerical-linear-algebra-F02841	210	■ 0.999	\boldsymbol{y}:=\left[y_{0}, \ldots, y_{n}\right]^T \in \mathbb{R}^{n+1}	$\boldsymbol{y} := [y_0, \dots, y_n]^T \in \mathbb{R}^{n+1}$	and given a dataset $\boldsymbol{y} := [y_0, \dots, y_n]^T \in \mathbb{R}^{n+1}$ where we assume $y_0 = y_n$. The
lyche-numerical-linear-algebra-F02842	210	■ 0.999	y_{0}=y_{n}	$y_0 = y_n$	and given a dataset $\boldsymbol{y} := [y_0, \dots, y_n]^T \in \mathbb{R}^{n+1}$, where we assume $y_0 = y_n$. The
lyche-numerical-linear-algebra-F02843	210	■ 0.902	\left(x_{i-1}, x_{i}\right)	(x_{i-1}, x_i)	(Recall that g is a cubic polynomial on each interval (x_{i-1}, x_i) , for $i = 1, \dots, n$
lyche-numerical-linear-algebra-F02844	210	■ 1.000	C^2[a, b]	$C^2[a, b]$	with smoothness $C^2[a, b]$.)
lyche-numerical-linear-algebra-F02845	210	■ 0.664	s_{i}:=g^{\prime}\left(x_{i}\right), i=0, \ldots, n	$s_i := g'(x_i), i = 0, \dots, n$	We define $s_i := g'(x_i), i = 0, \dots, n$. It can be shown that the vector $\boldsymbol{s} :=$
lyche-numerical-linear-algebra-F02846	210	■ 1.000	\left[s_{1}, \ldots, s_{n}\right]^T	$[s_1, \dots, s_n]^T$	$[s_1, \dots, s_n]^T$ is determined from a linear system
lyche-numerical-linear-algebra-F02847	211	■ 1.000	\left\ \boldsymbol{A}^{-1}\right\ _{\infty} \leq 1	$\ \boldsymbol{A}^{-1}\ _\infty \leq 1$	b) Show that $\ \boldsymbol{A}^{-1}\ _\infty < 1$.
lyche-numerical-linear-algebra-F02848	211	■ 1.000	\left\ \boldsymbol{A}^{-1}\right\ _1 \leq \frac{3}{2}	$\ \boldsymbol{A}^{-1}\ _1 \leq \frac{3}{2}$	c) Show that $\ \boldsymbol{A}^{-1}\ _1 \leq \frac{3}{2}$.
lyche-numerical-linear-algebra-F02849	211	■ 1.000	\boldsymbol{e} \in \mathbb{R}^n	$\boldsymbol{e} \in \mathbb{R}^n$	d) Let $\boldsymbol{s}$ and $\boldsymbol{b}$ be as in (8.44), where we assume $\boldsymbol{b} \neq \mathbf{0}$. Let $\boldsymbol{e} \in \mathbb{R}^n$ be such that
lyche-numerical-linear-algebra-F02850	211	■ 1.000	\ \boldsymbol{e}\ _p / \ \boldsymbol{b}\ _p \leq 0.01	$\ \boldsymbol{e}\ _p / \ \boldsymbol{b}\ _p \leq 0.01$	$\ \boldsymbol{e}\ _p / \ \boldsymbol{b}\ _p \leq 0.01$. Suppose $\hat{\boldsymbol{s}}$ satisfies
lyche-numerical-linear-algebra-F02851	211	■ 1.000	\hat{\boldsymbol{s}}	$\hat{\boldsymbol{s}}$	$\ \boldsymbol{e}\ _p / \ \boldsymbol{b}\ _p \leq 0.01$. Suppose $\hat{\boldsymbol{s}}$ satisfies
lyche-numerical-linear-algebra-F02852	211	■ 1.000	\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m \times n}, \boldsymbol{b} \in \mathbb{R}^m$, where $\boldsymbol{A}$ has rank n and let $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^T$ be a singular
lyche-numerical-linear-algebra-F02853	211	■ 1.000	\boldsymbol{U} \in \mathbb{R}^{m \times n}	$\boldsymbol{U} \in \mathbb{R}^{m \times n}$	value factorization of $\boldsymbol{A}$. Thus $\boldsymbol{U} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$. Write a MATLAB
lyche-numerical-linear-algebra-F02854	211	■ 1.000	\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}	$\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$	value factorization of $\boldsymbol{A}$. Thus $\boldsymbol{U} \in \mathbb{R}^{m \times n}$ and $\boldsymbol{\Sigma}, \boldsymbol{V} \in \mathbb{R}^{n \times n}$. Write a MATLAB
lyche-numerical-linear-algebra-F02855	211	■ 0.932	[x, K]=lsq(A, b)	$[x, K] = \text{lsq}(\boldsymbol{A}, \boldsymbol{b})$	function <code>[x, K]=lsq(A, b)</code> that uses the singular value factorization of $\boldsymbol{A}$
lyche-numerical-linear-algebra-F02856	211	■ 1.000	\boldsymbol{x}=\boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}	$\boldsymbol{x} = \boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$	to calculate a least squares solution $\boldsymbol{x} = \boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$ to the system $\boldsymbol{A} \boldsymbol{x} =$
lyche-numerical-linear-algebra-F02857	211	■ 1.000	\boldsymbol{A} \boldsymbol{x} =	$\boldsymbol{A} \boldsymbol{x} =$	to calculate a least squares solution $\boldsymbol{x} = \boldsymbol{V} \boldsymbol{\Sigma}^{-1} \boldsymbol{U}^T \boldsymbol{b}$ to the system $\boldsymbol{A} \boldsymbol{x} =$
lyche-numerical-linear-algebra-F02858	211	■ 0.909	[U, Sigma, V]=svd(A, 0)	$[\boldsymbol{U}, \boldsymbol{\Sigma}, \boldsymbol{V}] = \text{svd}(\boldsymbol{A}, 0)$	<code>[U, Sigma, V]=svd(A, 0)</code> computes the singular value factorization of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02859	211	■ 0.998	\ \cdot\ _p	$\ \cdot\ _p$	Exercise 8.25 (p Norm for $p = 1$ and $p = \infty$) Show that $\ \cdot\ _p$ is a vector norm

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lyche-numerical-linear-algebra-F02886	214	■ 1.000	$p(t)=x_{\{1\}}+x_{\{2\}} \ t$	$p(t) = x_1 + x_2 t$	<i>Example 9.2 (Linear Regression)</i> We want to fit a straight line $p(t) = x_1 + x_2 t$ to
lyche-numerical-linear-algebra-F02887	214	■ 1.000	$\left(t_{\{k\}}, y_{\{k\}}\right) \in \mathbb{R}^2, k=1, \ldots, m$	$(t_k, y_k) \in \mathbb{R}^2, k = 1, \dots, m$	$m \geq 2$ given data $(t_k, y_k) \in \mathbb{R}^2, k = 1, \dots, m$. This is part of the linear regression
lyche-numerical-linear-algebra-F02888	214	■ 1.000	$m>2$	$m > 2$	This is square for $m = 2$ and overdetermined for $m > 2$. The matrix A has linearly
lyche-numerical-linear-algebra-F02889	214	■ 1.000	$\left\{t_{\{1\}}, \ldots, t_{\{m\}}\right\}$	$\{t_1, \dots, t_m\}$	independent columns if and only if the set $\{t_1, \dots, t_m\}$ of sites contains at least two
lyche-numerical-linear-algebra-F02890	214	■ 1.000	$t_{\{i\}} \neq t_{\{j\}}$	$t_i \neq t_j$	distinct elements. For if say $t_i \neq t_j$ then
lyche-numerical-linear-algebra-F02891	215	■ 1.000	$t_{\{1\}}=\cdots=t_{\{m\}}$	$t_1 = \dots = t_m$	Conversely, if $t_1 = \dots = t_m$ then the columns of A are linearly dependent. The
lyche-numerical-linear-algebra-F02892	215	■ 0.970	$t_{\{1\}} \neq t_{\{2\}}$	$t_1 \neq t_2$	$m = 2$ and $t_1 \neq t_2$ then both systems $Ax = b$ and $A^*Ax = A^*b$ are square, and p
lyche-numerical-linear-algebra-F02893	215	■ 1.000	$p\left(t_{\{k\}}\right)=y_{\{k\}}, k=1,2$	$p(t_k) = y_k, k = 1, 2$	is the linear interpolant to the data. Indeed, p is linear and $p(t_k) = y_k, k = 1, 2$.
lyche-numerical-linear-algebra-F02894	215	■ 1.000	$\left[\begin{array}{cc} 4 & 10 \\ 10 & 30 \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right] = \left[\begin{array}{c} 6 \\ 10.1 \end{array}\right]$	$\left[\begin{array}{cc} 4 & 10 \\ 10 & 30 \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right] = \left[\begin{array}{c} 6 \\ 10.1 \end{array}\right]$	the normal equations become $\left[\begin{array}{cc} 4 & 10 \\ 10 & 30 \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \end{array}\right] = \left[\begin{array}{c} 6 \\ 10.1 \end{array}\right]$. The data and the least
lyche-numerical-linear-algebra-F02895	215	■ 1.000	$p(t)=x_{\{1\}}+x_{\{2\}} \ t=3.95-0.98 \ t$	$p(t) = x_1 + x_2 t = 3.95 - 0.98 t$	squares polynomial $p(t) = x_1 + x_2 t = 3.95 - 0.98 t$ are shown in Fig. 9.1.
lyche-numerical-linear-algebra-F02896	215	■ 1.000	$y \in \mathbb{R}$	$y \in \mathbb{R}$	To every input $u \in \mathbb{R}^n$ we obtain an output $y \in \mathbb{R}$. Assuming we have a linear
lyche-numerical-linear-algebra-F02897	216	■ 0.994	$\min \ \boldsymbol{A} \boldsymbol{x}-\boldsymbol{b}\ _2^2$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$	We can estimate x by solving the least squares problem $\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$.
lyche-numerical-linear-algebra-F02898	216	■ 1.000	$1 \leq n \leq m$	$1 \leq n \leq m$	• size: $1 \leq n \leq m$,
lyche-numerical-linear-algebra-F02899	216	■ 1.000	$\mathcal{S}:=\left\{t_{\{1\}}, t_{\{2\}}, \ldots, t_{\{m\}}\right\} \subset[a, b]$	$\mathcal{S} := \{t_1, t_2, \dots, t_m\} \subset [a, b]$	• sites: $\mathcal{S} := \{t_1, t_2, \dots, t_m\} \subset [a, b]$,
lyche-numerical-linear-algebra-F02900	216	■ 1.000	$\boldsymbol{y}=\left[y_{\{1\}}, y_{\{2\}}, \ldots, y_{\{m\}}\right]^* \in \mathbb{R}^m$	$\boldsymbol{y} = [y_1, y_2, \dots, y_m]^* \in \mathbb{R}^m$	• y-values: $\boldsymbol{y} = [y_1, y_2, \dots, y_m]^* \in \mathbb{R}^m$
lyche-numerical-linear-algebra-F02901	216	■ 1.000	$\phi_{\{j\}}:[a, b] \rightarrow \mathbb{R}, j=1, \ldots, n$	$\phi_j : [a, b] \rightarrow \mathbb{R}, j = 1, \dots, n$	• functions: $\phi_j : [a, b] \rightarrow \mathbb{R}, j = 1, \dots, n$.
lyche-numerical-linear-algebra-F02902	216	■ 0.999	$p:[a, b] \rightarrow \mathbb{R}$	$p : [a, b] \rightarrow \mathbb{R}$	Find a function (curve fit) $p : [a, b] \rightarrow \mathbb{R}$ given by $p := \sum_{j=1}^n x_j \phi_j$ such that
lyche-numerical-linear-algebra-F02903	216	■ 0.999	$p:=\sum_{j=1}^n x_{\{j\}} \ \phi_{\{j\}}$	$p := \sum_{j=1}^n x_j \phi_j$	Find a function (curve fit) $p : [a, b] \rightarrow \mathbb{R}$ given by $p := \sum_{j=1}^n x_j \phi_j$ such that
lyche-numerical-linear-algebra-F02904	216	■ 1.000	$p\left(t_{\{k\}}\right) \approx y_{\{k\}}$	$p(t_k) \approx y_k$	$p(t_k) \approx y_k$ for $k = 1, \dots, m$.
lyche-numerical-linear-algebra-F02905	217	■ 1.000	$\phi_{\{j\}}$	ϕ_j	Typical examples of functions ϕ_j are polynomials, trigonometric functions, expo-
lyche-numerical-linear-algebra-F02906	217	■ 1.000	$w_{\{k\}}>0$	$w_k > 0$	In (9.2) one can also include weights $w_k > 0$ for $k = 1, \dots, m$ and minimize
lyche-numerical-linear-algebra-F02907	217	■ 1.000	$y_{\{k\}}$	y_k	If y_k is an accurate observation, we can choose a large weight w_k . This will force
lyche-numerical-linear-algebra-F02908	217	■ 1.000	$w_{\{k\}}$	w_k	If y_k is an accurate observation, we can choose a large weight w_k . This will force
lyche-numerical-linear-algebra-F02909	217	■ 1.000	$p\left(t_{\{k\}}\right)-y_{\{k\}}$	$p(t_k) - y_k$	$p(t_k) - y_k$ to be small. Similarly, a small w_k will allow $p(t_k) - y_k$ to be large. If
lyche-numerical-linear-algebra-F02910	217	■ 1.000	$\delta y_{\{k\}}$	δy_k	an estimate for the standard deviation δy_k in y_k is known for each k , we can choose
lyche-numerical-linear-algebra-F02911	217	■ 1.000	$w_{\{k\}}=1 /\left(\delta y_{\{k\}}\right)^2, k=1,2, \ldots, m$	$w_k = 1 / (\delta y_k)^2, k = 1, 2, \dots, m$	$w_k = 1 / (\delta y_k)^2, k = 1, 2, \dots, m$. For simplicity we will assume in the following

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lyche-numerical-linear-algebra- F02912	217	■ 1.000	$w_{\{k\}}=1$	$w_k = 1$	that $w_k = 1$ for all k .
lyche-numerical-linear-algebra- F02913	217	■ 1.000	4.2 $\boldsymbol{A}^* \boldsymbol{A}$	$4.2 \boldsymbol{A}^* \boldsymbol{A}$	Proof By Lemma 4.2 $\boldsymbol{A}^* \boldsymbol{A}$ is positive definite if and only if $\boldsymbol{A}$ has linearly
lyche-numerical-linear-algebra- F02914	217	■ 0.989	$(\boldsymbol{A} \boldsymbol{x})_k = \sum_{j=1}^n x_j \phi_j(t_k), k=1, \ldots, m$	$(\boldsymbol{A} \boldsymbol{x})_k = \sum_{j=1}^n x_j \phi_j(t_k), k=1, \ldots, m$	independent columns. Since $(\boldsymbol{A} \boldsymbol{x})_k = \sum_{j=1}^n x_j \phi_j(t_k), k=1, \ldots, m$ this is
lyche-numerical-linear-algebra- F02915	217	■ 1.000	$\phi_{\{j\}}(t) := t^{j-1}$	$\phi_j(t) := t^{j-1}$	mials $\phi_j(t) := t^{j-1}$, for $j=1, \ldots, n$ and equidistant sites $t_k = (k-1)/(m-1)$
lyche-numerical-linear-algebra- F02916	217	■ 1.000	$t_{\{k\}} = (k-1)/(m-1)$	$t_k = (k-1)/(m-1)$	mials $\phi_j(t) := t^{j-1}$, for $j=1, \ldots, n$ and equidistant sites $t_k = (k-1)/(m-1)$
lyche-numerical-linear-algebra- F02917	217	■ 1.000	$\boldsymbol{B}_n \boldsymbol{x} = \boldsymbol{c}_n$	$\boldsymbol{B}_n \boldsymbol{x} = \boldsymbol{c}_n$	for $k=1, \ldots, m$. The normal equations are $\boldsymbol{B}_n \boldsymbol{x} = \boldsymbol{c}_n$, where for $n=3$
lyche-numerical-linear-algebra- F02918	217	■ 1.000	$\boldsymbol{B}_n$	$\boldsymbol{B}_n$	$\boldsymbol{B}_n$ is positive definite if at least n of the t 's are distinct. However $\boldsymbol{B}_n$ is extremely
lyche-numerical-linear-algebra- F02919	217	■ 1.000	t	t	$\boldsymbol{B}_n$ is positive definite if at least n of the t 's are distinct. However $\boldsymbol{B}_n$ is extremely
lyche-numerical-linear-algebra- F02920	217	■ 1.000	$\frac{1}{m} \boldsymbol{B}_n \approx \boldsymbol{H}_n$	$\frac{1}{m} \boldsymbol{B}_n \approx \boldsymbol{H}_n$	ill-conditioned even for moderate n . Indeed, $\frac{1}{m} \boldsymbol{B}_n \approx \boldsymbol{H}_n$, where $\boldsymbol{H}_n \in \mathbb{R}^{n \times n}$ is the
lyche-numerical-linear-algebra- F02921	217	■ 1.000	$\boldsymbol{H}_n \in \mathbb{R}^{n \times n}$	$\boldsymbol{H}_n \in \mathbb{R}^{n \times n}$	ill-conditioned even for moderate n . Indeed, $\frac{1}{m} \boldsymbol{B}_n \approx \boldsymbol{H}_n$, where $\boldsymbol{H}_n \in \mathbb{R}^{n \times n}$ is the
lyche-numerical-linear-algebra- F02922	217	■ 1.000	$1/(i+j-1)$	$1/(i+j-1)$	Hilbert Matrix with i, j element $1/(i+j-1)$. Thus, for $n=3$
lyche-numerical-linear-algebra- F02923	218	■ 1.000	$\frac{1}{m} \boldsymbol{B}_n$	$\frac{1}{m} \boldsymbol{B}_n$	The elements of $\frac{1}{m} \boldsymbol{B}_n$ are Riemann sums approximations to the elements of $\boldsymbol{H}_n$. In
lyche-numerical-linear-algebra- F02924	218	■ 1.000	$\boldsymbol{H}_n$	$\boldsymbol{H}_n$	The elements of $\frac{1}{m} \boldsymbol{B}_n$ are Riemann sums approximations to the elements of $\boldsymbol{H}_n$. In
lyche-numerical-linear-algebra- F02925	218	■ 1.000	$\boldsymbol{B}_n = [b_{i,j}]_{i,j=1}^n$	$\boldsymbol{B}_n = [b_{i,j}]_{i,j=1}^n$	fact, if $\boldsymbol{B}_n = [b_{i,j}]_{i,j=1}^n$ then
lyche-numerical-linear-algebra- F02926	218	■ 1.000	$\boldsymbol{H}_n^{-1}$	$\boldsymbol{H}_n^{-1}$	The elements of $\boldsymbol{H}_n^{-1}$ are determined in Exercise 1.13. We find $K_1(\boldsymbol{H}_6) :=$
lyche-numerical-linear-algebra- F02927	218	■ 1.000	$K_1(\boldsymbol{H}_6) :=$	$K_1(\boldsymbol{H}_6) :=$	The elements of $\boldsymbol{H}_n^{-1}$ are determined in Exercise 1.13. We find $K_1(\boldsymbol{H}_6) :=$
lyche-numerical-linear-algebra- F02928	218	■ 0.998	$\ \boldsymbol{H}_6\ _1 \ \boldsymbol{H}_6^{-1}\ _1 \approx 3 \cdot 10^7$	$\ \boldsymbol{H}_6\ _1 \ \boldsymbol{H}_6^{-1}\ _1 \approx 3 \cdot 10^7$	$\ \boldsymbol{H}_6\ _1 \ \boldsymbol{H}_6^{-1}\ _1 \approx 3 \cdot 10^7$. It appears that $\frac{1}{m} \boldsymbol{B}_n$ and hence $\boldsymbol{B}_n$ is ill-conditioned
lyche-numerical-linear-algebra- F02929	218	■ 1.000	$(t - \tilde{t})^{j-1}, j=1, \ldots, n$	$(t - \tilde{t})^{j-1}, j=1, \ldots, n$	possibility is to use the shifted power basis $(t - \tilde{t})^{j-1}, j=1, \ldots, n$, for a suitable $\tilde{t}$.
lyche-numerical-linear-algebra- F02930	218	■ 1.000	$\tilde{t}$	$\tilde{t}$	possibility is to use the shifted power basis $(t - \tilde{t})^{j-1}, j=1, \ldots, n$, for a suitable $\tilde{t}$.
lyche-numerical-linear-algebra- F02931	218	■ 1.000	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{y}^* \boldsymbol{x}$	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{y}^* \boldsymbol{x}$	With the usual inner product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{y}^* \boldsymbol{x}$, orthogonal sums and projections we
lyche-numerical-linear-algebra- F02932	218	■ 1.000	$\mathcal{S} := \mathcal{R}(\boldsymbol{A})$	$\mathcal{S} := \mathcal{R}(\boldsymbol{A})$	problems. For $\boldsymbol{A} \in \mathbb{C}^{m \times n}$ we consider the column space $\mathcal{S} := \mathcal{R}(\boldsymbol{A})$ of $\boldsymbol{A}$ and the
lyche-numerical-linear-algebra- F02933	218	■ 1.000	$\mathcal{T} := \mathcal{N}(\boldsymbol{A}^*)$	$\mathcal{T} := \mathcal{N}(\boldsymbol{A}^*)$	null space $\mathcal{T} := \mathcal{N}(\boldsymbol{A}^*)$ of $\boldsymbol{A}^*$. These are subspaces of $\mathbb{C}^m$ and by Theorem 7.3 we
lyche-numerical-linear-algebra- F02934	218	■ 1.000	$\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$	$\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$	uniquely as $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$, where $\boldsymbol{b}_1$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$
lyche-numerical-linear-algebra- F02935	218	■ 1.000	$\boldsymbol{b}_1$	$\boldsymbol{b}_1$	uniquely as $\boldsymbol{b} = \boldsymbol{b}_1 + \boldsymbol{b}_2$, where $\boldsymbol{b}_1$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$
lyche-numerical-linear-algebra- F02936	218	■ 1.000	$\boldsymbol{b}_2$	$\boldsymbol{b}_2$	and $\boldsymbol{b}_2$ is the orthogonal projection of $\boldsymbol{b}$ into $\mathcal{N}(\boldsymbol{A}^*)$. Suppose $\boldsymbol{x} \in \mathbb{C}^n$. Clearly
lyche-numerical-linear-algebra- F02937	218	■ 1.000	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$

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lyche-numerical-linear-algebra-2019-02-29-19-F02938	218	1.000	$\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{Ax} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$
lyche-numerical-linear-algebra-2019-02-29-19-F02939	218	1.000	$\langle \boldsymbol{Ax} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$	$\langle \boldsymbol{Ax} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$	$\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$ since it is a subspace and $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. But then $\langle \boldsymbol{Ax} - \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle = 0$
lyche-numerical-linear-algebra-2019-02-29-19-F02940	219	1.000	$\boldsymbol{Ax} = \boldsymbol{b}_1$	$\boldsymbol{Ax} = \boldsymbol{b}_1$	with equality if and only if $\boldsymbol{Ax} = \boldsymbol{b}_1$. It follows that the set of all least squares
lyche-numerical-linear-algebra-2019-02-29-19-F02941	219	1.000	$\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$	This set is nonempty since $\boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A})$.
lyche-numerical-linear-algebra-2019-02-29-19-F02942	219	1.000	$\boldsymbol{Ax} - \boldsymbol{b}_1 = \mathbf{0}$	$\boldsymbol{Ax} - \boldsymbol{b}_1 = \mathbf{0}$	Proof of Theorem 9.3 If $\boldsymbol{x}$ solves the least squares problem then $\boldsymbol{Ax} - \boldsymbol{b}_1 = \mathbf{0}$ and
lyche-numerical-linear-algebra-2019-02-29-19-F02943	219	0.699	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$	it follows that $\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}) = \boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$ since $\boldsymbol{b}_2 \in \mathcal{N}(\boldsymbol{A}^*)$. This shows that
lyche-numerical-linear-algebra-2019-02-29-19-F02944	219	1.000	$\boldsymbol{A}^*\boldsymbol{b}_2 = \mathbf{0}$	$\boldsymbol{A}^*\boldsymbol{b}_2 = \mathbf{0}$	the normal equations hold. Conversely, if $\boldsymbol{A}^*\boldsymbol{Ax} = \boldsymbol{A}^*\boldsymbol{b}$ then $\boldsymbol{A}^*\boldsymbol{b}_2 = \mathbf{0}$ implies that
lyche-numerical-linear-algebra-2019-02-29-19-F02945	219	0.995	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$. But then $\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$ showing that $\boldsymbol{Ax} - \boldsymbol{b}_1 = \mathbf{0}$,
lyche-numerical-linear-algebra-2019-02-29-19-F02946	219	0.995	$\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{A}^*(\boldsymbol{Ax} - \boldsymbol{b}_1) = \mathbf{0}$. But then $\boldsymbol{Ax} - \boldsymbol{b}_1 \in \mathcal{R}(\boldsymbol{A}) \cap \mathcal{N}(\boldsymbol{A}^*)$ showing that $\boldsymbol{Ax} - \boldsymbol{b}_1 = \mathbf{0}$,
lyche-numerical-linear-algebra-2019-02-29-19-F02947	219	1.000	$m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$	$m \geq n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$	We assume that $m > n, \boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^m$. We consider numerical methods based
lyche-numerical-linear-algebra-2019-02-29-19-F02948	219	1.000	$\boldsymbol{B} := \boldsymbol{A}^*\boldsymbol{A}$	$\boldsymbol{B} := \boldsymbol{A}^*\boldsymbol{A}$	coefficient matrix $\boldsymbol{B} := \boldsymbol{A}^*\boldsymbol{A}$ in the normal equations is positive definite, and we
lyche-numerical-linear-algebra-2019-02-29-19-F02949	219	0.998	$(\boldsymbol{A}^*\boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \dots, n$	$(\boldsymbol{A}^*\boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \dots, n$	1. inner product: $(\boldsymbol{A}^*\boldsymbol{A})_{i,j} = \sum_{k=1}^m \bar{a}_{k,i} a_{k,j}, i, j = 1, \dots, n$,
lyche-numerical-linear-algebra-2019-02-29-19-F02950	219	1.000	$\boldsymbol{A}^*\boldsymbol{A} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} [a_{k1} \cdots a_{kn}], \boldsymbol{A}^*\boldsymbol{b} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} b_k$	$\boldsymbol{A}^*\boldsymbol{A} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} [a_{k1} \cdots a_{kn}], \boldsymbol{A}^*\boldsymbol{b} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} b_k$	2. outer product: $\boldsymbol{A}^*\boldsymbol{A} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} [a_{k1} \cdots a_{kn}], \boldsymbol{A}^*\boldsymbol{b} = \sum_{k=1}^m \begin{bmatrix} \bar{a}_{k,1} \\ \vdots \\ \bar{a}_{k,n} \end{bmatrix} b_k$.
lyche-numerical-linear-algebra-2019-02-29-19-F02951	219	1.000	$n(n+1)/2$	$n(n+1)/2$	we only need to compute $n(n+1)/2$ such inner products. It follows that $\boldsymbol{B}$ can be
lyche-numerical-linear-algebra-2019-02-29-19-F02952	219	1.000	mn^2	mn^2	computed in approximately mn^2 arithmetic operations. In conclusion the number
lyche-numerical-linear-algebra-2019-02-29-19-F02953	220	1.000	$\boldsymbol{B}, 2mn$	$\boldsymbol{B}, 2mn$	of operations are mn^2 to find $\boldsymbol{B}$, $2mn$ to find $\boldsymbol{c} := \boldsymbol{A}^*\boldsymbol{b}$, $n^3/3$ to find $\boldsymbol{L}$ such that
lyche-numerical-linear-algebra-2019-02-29-19-F02954	220	1.000	$\boldsymbol{c} := \boldsymbol{A}^*\boldsymbol{b}, n^3/3$	$\boldsymbol{c} := \boldsymbol{A}^*\boldsymbol{b}, n^3/3$	of operations are mn^2 to find $\boldsymbol{B}$, $2mn$ to find $\boldsymbol{c} := \boldsymbol{A}^*\boldsymbol{b}$, $n^3/3$ to find $\boldsymbol{L}$ such that
lyche-numerical-linear-algebra-2019-02-29-19-F02955	220	1.000	$\boldsymbol{B} = \boldsymbol{LL}^*, n^2$	$\boldsymbol{B} = \boldsymbol{LL}^*, n^2$	$\boldsymbol{B} = \boldsymbol{LL}^*, n^2$ to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra-2019-02-29-19-F02956	220	1.000	$\boldsymbol{Ly} = \boldsymbol{c}$	$\boldsymbol{Ly} = \boldsymbol{c}$	$\boldsymbol{B} = \boldsymbol{LL}^*, n^2$ to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra-2019-02-29-19-F02957	220	1.000	$\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$	$\boldsymbol{B} = \boldsymbol{LL}^*, n^2$ to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra-2019-02-29-19-F02958	220	1.000	$m \approx n$	$m \approx n$	$\boldsymbol{B} = \boldsymbol{LL}^*, n^2$ to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra-2019-02-29-19-F02959	220	1.000	$\frac{4}{3}n^3 =$	$\frac{4}{3}n^3 =$	$\boldsymbol{B} = \boldsymbol{LL}^*, n^2$ to solve $\boldsymbol{Ly} = \boldsymbol{c}$ and n^2 to solve $\boldsymbol{L}^*\boldsymbol{x} = \boldsymbol{y}$. If $m \approx n$ it takes $\frac{4}{3}n^3 =$
lyche-numerical-linear-algebra-2019-02-29-19-F02960	220	1.000	$\lambda_1 \geq \cdots \geq \lambda_n > 0$	$\lambda_1 \geq \cdots \geq \lambda_n > 0$	where $\lambda_1 \geq \cdots \geq \lambda_n > 0$ are the eigenvalues of $\boldsymbol{A}^*\boldsymbol{A}$, and $\sigma_1 \geq \cdots \geq \sigma_n > 0$ are

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lyche-numerical-linear-algebra-F02961	220	1.000	$\sigma_1 \geq \dots \geq \sigma_n > 0$	$\sigma_1 \geq \dots \geq \sigma_n > 0$	where $\lambda_1 \geq \dots \geq \lambda_n > 0$ are the eigenvalues of A^*A , and $\sigma_1 \geq \dots \geq \sigma_n > 0$ are
lyche-numerical-linear-algebra-F02962	220	1.000	$K_2(\boldsymbol{A}) = \frac{\sigma_1}{\sigma_n}$	$K_2(\boldsymbol{A}) = \frac{\sigma_1}{\sigma_n}$	Proof Since A^*A is Hermitian it follows from Theorem 8.10 that $K_2(\boldsymbol{A}) = \frac{\sigma_1}{\sigma_n}$ and
lyche-numerical-linear-algebra-F02963	220	0.999	$K_2(\boldsymbol{A}^* \boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$	$K_2(\boldsymbol{A}^* \boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$	$K_2(\boldsymbol{A}^* \boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$. But $\lambda_i = \sigma_i^2$ by Theorem 7.2 and the proof is complete.
lyche-numerical-linear-algebra-F02964	220	0.999	$\lambda_i = \sigma_i^2$	$\lambda_i = \sigma_i^2$	$K_2(\boldsymbol{A}^* \boldsymbol{A}) = \frac{\lambda_1}{\lambda_n}$. But $\lambda_i = \sigma_i^2$ by Theorem 7.2 and the proof is complete.
lyche-numerical-linear-algebra-F02965	220	1.000	$\boldsymbol{R}_1^*$	$\boldsymbol{R}_1^*$	Since $\boldsymbol{A}$ has rank n the matrix $\boldsymbol{R}_1^*$ is nonsingular and can be canceled. Thus
lyche-numerical-linear-algebra-F02966	220	1.000	$\boldsymbol{c}_1$	$\boldsymbol{c}_1$	We can use Householder transformations or Givens rotations to find $\boldsymbol{R}_1$ and $\boldsymbol{c}_1$.
lyche-numerical-linear-algebra-F02967	220	0.997	$\boldsymbol{R} = \boldsymbol{Q}^* \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{Q}^* \boldsymbol{A}$	$\boldsymbol{R} = \boldsymbol{Q}^* \boldsymbol{A}$ and $\boldsymbol{c} = \boldsymbol{Q}^* \boldsymbol{b}$, where $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$ is the QR decomposition of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02968	220	0.997	$\boldsymbol{c} = \boldsymbol{Q}^* \boldsymbol{b}$	$\boldsymbol{c} = \boldsymbol{Q}^* \boldsymbol{b}$	$\boldsymbol{R} = \boldsymbol{Q}^* \boldsymbol{A}$ and $\boldsymbol{c} = \boldsymbol{Q}^* \boldsymbol{b}$, where $\boldsymbol{A} = \boldsymbol{Q} \boldsymbol{R}$ is the QR decomposition of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F02969	220	1.000	$\boldsymbol{c}$	$\boldsymbol{c}$	The matrices $\boldsymbol{R}_1$ and $\boldsymbol{c}_1$ are located in the first n rows of $\boldsymbol{R}$ and $\boldsymbol{c}$. Using also
lyche-numerical-linear-algebra-F02970	220	0.769	$[\boldsymbol{R}, \boldsymbol{c}] =$	$[\boldsymbol{R}, \boldsymbol{c}] =$	1. $[\boldsymbol{R}, \boldsymbol{c}] = \text{housetriang}(\boldsymbol{A}, \boldsymbol{b})$.
lyche-numerical-linear-algebra-F02971	221	0.958	$\boldsymbol{x} = [1, 0, 0]^*$	$\boldsymbol{x} = [1, 0, 0]^*$	and we find $\boldsymbol{x} = [1, 0, 0]^*$.
lyche-numerical-linear-algebra-F02972	221	1.000	$\boldsymbol{R}_1 \boldsymbol{x} = \boldsymbol{c}_1$	$\boldsymbol{R}_1 \boldsymbol{x} = \boldsymbol{c}_1$	problems, see [2]. The 2 norm condition number for the system $\boldsymbol{R}_1 \boldsymbol{x} = \boldsymbol{c}_1$ is
lyche-numerical-linear-algebra-F02973	221	1.000	$K_2(\boldsymbol{R}_1) = K_2(\boldsymbol{Q}_1 \boldsymbol{R}_1) = K_2(\boldsymbol{A})$	$K_2(\boldsymbol{R}_1) = K_2(\boldsymbol{Q}_1 \boldsymbol{R}_1) = K_2(\boldsymbol{A})$	$K_2(\boldsymbol{R}_1) = K_2(\boldsymbol{Q}_1 \boldsymbol{R}_1) = K_2(\boldsymbol{A})$, and as discussed in the previous section this
lyche-numerical-linear-algebra-F02974	221	1.000	$K_2(\boldsymbol{A}^* \boldsymbol{A})$	$K_2(\boldsymbol{A}^* \boldsymbol{A})$	is the square root of $K_2(\boldsymbol{A}^* \boldsymbol{A})$, the condition number for the normal equations.
lyche-numerical-linear-algebra-F02975	221	1.000	$2mn^2 -$	$2mn^2 -$	the number of arithmetic operations in Algorithm 5.2 is approximately $2mn^2 -$
lyche-numerical-linear-algebra-F02976	221	1.000	$\boldsymbol{x} = \boldsymbol{A} \backslash \boldsymbol{b}$	$\boldsymbol{x} = \boldsymbol{A} \backslash \boldsymbol{b}$	Using MATLAB a least squares solution can be found using $\boldsymbol{x} = \boldsymbol{A} \backslash \boldsymbol{b}$ if $\boldsymbol{A}$ has
lyche-numerical-linear-algebra-F02977	221	0.996	$\boldsymbol{x} = \text{lscov}(\boldsymbol{A}, \boldsymbol{b})$	$\boldsymbol{x} = \text{lscov}(\boldsymbol{A}, \boldsymbol{b})$	full rank. For rank deficient problems the function $\boldsymbol{x} = \text{lscov}(\boldsymbol{A}, \boldsymbol{b})$ finds a least
lyche-numerical-linear-algebra-F02978	222	1.000	$\boldsymbol{U}_2$	$\boldsymbol{U}_2$	• the set of columns of $\boldsymbol{U}_2$ is an orthonormal basis for $\mathcal{N}(\boldsymbol{A}^*)$,
lyche-numerical-linear-algebra-F02979	222	1.000	$\boldsymbol{V}_2$	$\boldsymbol{V}_2$	• the set of columns of $\boldsymbol{V}_2$ is an orthonormal basis for $\mathcal{N}(\boldsymbol{A})$.
lyche-numerical-linear-algebra-F02980	222	1.000	$\boldsymbol{A}^\dagger \in \mathbb{C}^{n \times m}$	$\boldsymbol{A}^\dagger \in \mathbb{C}^{n \times m}$	there is a unique matrix $\boldsymbol{A}^\dagger \in \mathbb{C}^{n \times m}$ such that
lyche-numerical-linear-algebra-F02981	222	1.000	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$	$\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$	If $\boldsymbol{U}_1 \boldsymbol{\Sigma}_1 \boldsymbol{V}_1^*$ is a singular value factorization of $\boldsymbol{A}$ then
lyche-numerical-linear-algebra-F02982	222	1.000	$(\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*$	$(\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*$	A similar calculation shows that $(\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*$ and $(\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*$ showing
lyche-numerical-linear-algebra-F02983	222	1.000	$(\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*$	$(\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*$	A similar calculation shows that $(\boldsymbol{A}^\dagger \boldsymbol{A})^* = \boldsymbol{V}_1 \boldsymbol{V}_1^*$ and $(\boldsymbol{A} \boldsymbol{A}^\dagger)^* = \boldsymbol{U}_1 \boldsymbol{U}_1^*$ showing
lyche-numerical-linear-algebra-F02984	222	1.000	$\boldsymbol{A}^\dagger \boldsymbol{A}$	$\boldsymbol{A}^\dagger \boldsymbol{A}$	that $\boldsymbol{A}^\dagger \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^\dagger$ are Hermitian. Moreover, by (9.10)
lyche-numerical-linear-algebra-F02985	222	1.000	$\boldsymbol{A} \boldsymbol{A}^\dagger$	$\boldsymbol{A} \boldsymbol{A}^\dagger$	that $\boldsymbol{A}^\dagger \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^\dagger$ are Hermitian. Moreover, by (9.10)
lyche-numerical-linear-algebra-F02986	223	1.000	$\boldsymbol{A}^\dagger$	$\boldsymbol{A}^\dagger$	The matrix $\boldsymbol{A}^\dagger$ is called the generalized inverse of $\boldsymbol{A}$. We note that

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lyche-numerical-linear-algebra-223-F02987	223	■ 0.999	$\boldsymbol{A}^{-1}=\boldsymbol{A}^{\dagger}$	$\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$	1. If $\boldsymbol{A}$ is square and nonsingular then $\boldsymbol{A}^{-1}$ satisfies (9.8) so that $\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$. Indeed,
lyche-numerical-linear-algebra-223-F02988	223	■ 1.000	$\boldsymbol{A}^{-1} \boldsymbol{A}$	$\boldsymbol{A}^{-1} \boldsymbol{A}$	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$ implies that $\boldsymbol{A}^{-1} \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^{-1}$ are Hermitian. Moreover,
lyche-numerical-linear-algebra-223-F02989	223	■ 1.000	$\boldsymbol{A} \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1}$	$\boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{I}$ implies that $\boldsymbol{A}^{-1} \boldsymbol{A}$ and $\boldsymbol{A} \boldsymbol{A}^{-1}$ are Hermitian. Moreover,
lyche-numerical-linear-algebra-223-F02990	223	■ 1.000	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$	$\boldsymbol{A} \boldsymbol{A}^{-1} \boldsymbol{A} = \boldsymbol{A}, \boldsymbol{A}^{-1} \boldsymbol{A} \boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}$. By uniqueness $\boldsymbol{A}^{-1} = \boldsymbol{A}^{\dagger}$.
lyche-numerical-linear-algebra-223-F02991	223	■ 1.000	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$	$\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$	and let $\mathcal{S}$ be one of the subspaces $\mathcal{R}(\boldsymbol{A}), \mathcal{N}(\boldsymbol{A}^*)$. The orthogonal projection of
lyche-numerical-linear-algebra-223-F02992	223	■ 1.000	$\boldsymbol{v} \in \mathbb{C}^m$	$\boldsymbol{v} \in \mathbb{C}^m$	$\boldsymbol{v} \in \mathbb{C}^m$ into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$,
lyche-numerical-linear-algebra-223-F02993	223	■ 1.000	$\boldsymbol{P}_{\mathcal{S}}$	$\boldsymbol{P}_{\mathcal{S}}$	$\boldsymbol{v} \in \mathbb{C}^m$ into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$,
lyche-numerical-linear-algebra-223-F02994	223	■ 1.000	$\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$	$\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$	$\boldsymbol{v} \in \mathbb{C}^m$ into $\mathcal{S}$ can be written as a matrix $\boldsymbol{P}_{\mathcal{S}}$ times the vector $\boldsymbol{v}$ in the form $\boldsymbol{P}_{\mathcal{S}} \boldsymbol{v}$,
lyche-numerical-linear-algebra-223-F02995	223	■ 0.998	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$	$\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$	where $\boldsymbol{A}^{\dagger}$ is the generalized inverse of $\boldsymbol{A}$, and $\boldsymbol{A} = \boldsymbol{U} \boldsymbol{\Sigma} \boldsymbol{V}^* \in \mathbb{C}^{m \times n}$ is a singular
lyche-numerical-linear-algebra-223-F02996	223	■ 1.000	$\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$	$\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$	where $\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$ and $\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$. By uniqueness and (9.10)
lyche-numerical-linear-algebra-223-F02997	223	■ 1.000	$\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$	$\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$	where $\boldsymbol{s} = \boldsymbol{U}_1 \boldsymbol{U}_1^* \boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$ and $\boldsymbol{t} = \boldsymbol{U}_2 \boldsymbol{U}_2^* \boldsymbol{v} \in \mathcal{N}(\boldsymbol{A}^*)$. By uniqueness and (9.10)
lyche-numerical-linear-algebra-223-F02998	223	■ 1.000	$\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$	$\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$	we obtain the first equation in (9.12). Since $\boldsymbol{v} = (\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^*) \boldsymbol{v}$ for any $\boldsymbol{v} \in \mathbb{C}^m$
lyche-numerical-linear-algebra-223-F02999	223	■ 0.999	$\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$	$\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$	we have $\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$, and hence $\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$ and the
lyche-numerical-linear-algebra-223-F03000	223	■ 0.999	$\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$	$\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$	we have $\boldsymbol{U}_1 \boldsymbol{U}_1^* + \boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I}$, and hence $\boldsymbol{U}_2 \boldsymbol{U}_2^* = \boldsymbol{I} - \boldsymbol{U}_1 \boldsymbol{U}_1^* = \boldsymbol{I} - \boldsymbol{A} \boldsymbol{A}^{\dagger}$ and the
lyche-numerical-linear-algebra-223-F03001	223	■ 0.848	$\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$	$\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$	solves the least squares problem $\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$ if and only if $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$,
lyche-numerical-linear-algebra-223-F03002	223	■ 0.848	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$	solves the least squares problem $\min_x \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2^2$ if and only if $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$,
lyche-numerical-linear-algebra-223-F03003	223	■ 0.893	$\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})$	$\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})$	where $\boldsymbol{A}^{\dagger}$ is the generalized inverse of $\boldsymbol{A}$ and $\boldsymbol{z} \in \mathcal{N}(\boldsymbol{A})$.
lyche-numerical-linear-algebra-224-F03004	224	■ 1.000	$\boldsymbol{b}_1 := \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}$	$\boldsymbol{b}_1 := \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}$	Proof It follows from Theorem 9.6 that $\boldsymbol{b}_1 := \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b}$ is the orthogonal projection
lyche-numerical-linear-algebra-224-F03005	224	■ 0.998	$\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$	$\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$	Let $\boldsymbol{x}$ be a least squares solution, i.e., $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}_1$. If $\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$ then $\boldsymbol{A} \boldsymbol{z} =$
lyche-numerical-linear-algebra-224-F03006	224	■ 0.998	$\boldsymbol{A} \boldsymbol{z} =$	$\boldsymbol{A} \boldsymbol{z} =$	Let $\boldsymbol{x}$ be a least squares solution, i.e., $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}_1$. If $\boldsymbol{z} := \boldsymbol{x} - \boldsymbol{A}^{\dagger} \boldsymbol{b}$ then $\boldsymbol{A} \boldsymbol{z} =$
lyche-numerical-linear-algebra-224-F03007	224	■ 1.000	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{b}_1 - \boldsymbol{b}_1 = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{b}_1 - \boldsymbol{b}_1 = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{x} - \boldsymbol{A} \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{b}_1 - \boldsymbol{b}_1 = \mathbf{0}$ and $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$.
lyche-numerical-linear-algebra-224-F03008	224	■ 0.997	$\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$	$\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$	Conversely, if $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$ with $\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$ then $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$ and $\boldsymbol{x}$ is
lyche-numerical-linear-algebra-224-F03009	224	■ 0.997	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$	Conversely, if $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}$ with $\boldsymbol{A} \boldsymbol{z} = \mathbf{0}$ then $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{A}(\boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}) = \boldsymbol{b}_1$ and $\boldsymbol{x}$ is

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lyche-numerical-linear-algebra-224 F03010	224	1.000	$\boldsymbol{A}^{\dagger} \boldsymbol{b}$	$\boldsymbol{A}^{\dagger} \boldsymbol{b}$	The least squares solution $\boldsymbol{A}^{\dagger} \boldsymbol{b}$ has an interesting property.
lyche-numerical-linear-algebra-224 F03011	224	1.000	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}$	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}$	Euclidian norm is $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}$ corresponding to $\boldsymbol{z} = \mathbf{0}$.
lyche-numerical-linear-algebra-224 F03012	224	0.999	$\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$	$\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$	$\mathcal{N}(\boldsymbol{A})$ we have $\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$ for some $\boldsymbol{y}$. Moreover, $\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$ since $\boldsymbol{V}$ has orthonormal
lyche-numerical-linear-algebra-224 F03013	224	0.999	$\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$	$\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$	$\mathcal{N}(\boldsymbol{A})$ we have $\boldsymbol{z} = \boldsymbol{V}_2 \boldsymbol{y}$ for some $\boldsymbol{y}$. Moreover, $\boldsymbol{V}_2^* \boldsymbol{V}_1 = \mathbf{0}$ since $\boldsymbol{V}$ has orthonormal
lyche-numerical-linear-algebra-224 F03014	224	1.000	$\boldsymbol{z}^* \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{y}^* \boldsymbol{V}_2^* \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* \boldsymbol{b} = \mathbf{0}$	$\boldsymbol{z}^* \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{y}^* \boldsymbol{V}_2^* \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* \boldsymbol{b} = \mathbf{0}$	columns. But then $\boldsymbol{z}^* \boldsymbol{A}^{\dagger} \boldsymbol{b} = \boldsymbol{y}^* \boldsymbol{V}_2^* \boldsymbol{V}_1 \boldsymbol{\Sigma}^{-1} \boldsymbol{U}_1^* \boldsymbol{b} = \mathbf{0}$. Thus $\boldsymbol{z}$ and $\boldsymbol{A}^{\dagger} \boldsymbol{b}$ are orthogonal
lyche-numerical-linear-algebra-224 F03015	224	1.000	$\ \boldsymbol{x}\ _2^2 = \ \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}\ _2^2 = \ \boldsymbol{A}^{\dagger} \boldsymbol{b}\ _2^2 + \ \boldsymbol{z}\ _2^2 \geq \ \boldsymbol{A}^{\dagger} \boldsymbol{b}\ _2^2$	$\ \boldsymbol{x}\ _2^2 = \ \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}\ _2^2 = \ \boldsymbol{A}^{\dagger} \boldsymbol{b}\ _2^2 + \ \boldsymbol{z}\ _2^2 \geq \ \boldsymbol{A}^{\dagger} \boldsymbol{b}\ _2^2$	so that by Pythagoras $\ \boldsymbol{x}\ _2^2 = \ \boldsymbol{A}^{\dagger} \boldsymbol{b} + \boldsymbol{z}\ _2^2 = \ \boldsymbol{A}^{\dagger} \boldsymbol{b}\ _2^2 + \ \boldsymbol{z}\ _2^2 \geq \ \boldsymbol{A}^{\dagger} \boldsymbol{b}\ _2^2$ with
lyche-numerical-linear-algebra-224 F03016	224	0.997	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	problem with $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\boldsymbol{b} = [1, 1]^*$. The singular value factorization, $\boldsymbol{A}^{\dagger}$ and
lyche-numerical-linear-algebra-224 F03017	224	0.997	$\boldsymbol{b} = [1, 1]^*$	$\boldsymbol{b} = [1, 1]^*$	problem with $\boldsymbol{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\boldsymbol{b} = [1, 1]^*$. The singular value factorization, $\boldsymbol{A}^{\dagger}$ and
lyche-numerical-linear-algebra-224 F03018	224	1.000	$[1/2, 1/2] + [a, -a]$	$[1/2, 1/2] + [a, -a]$	Using Corollary 9.1 we find the general solution $[1/2, 1/2] + [a, -a]$ for any $a \in \mathbb{C}$.
lyche-numerical-linear-algebra-224 F03019	224	0.653	$a = 1/2$	$a = 1/2$	The MATLAB function <code>lsqcov</code> gives the solution $[1, 0]^*$ corresponding to $a = 1/2$,
lyche-numerical-linear-algebra-224 F03020	224	0.761	$[1/2, 1/2]$	$[1/2, 1/2]$	while the minimal norm solution is $[1/2, 1/2]$ obtained for $a = 0$.
lyche-numerical-linear-algebra-224 F03021	224	0.933	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$	solution $\boldsymbol{x}$ of the least squares problem $\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$.
lyche-numerical-linear-algebra-225 F03022	225	1.000	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$	$\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$	independent columns, and let $\boldsymbol{b}, \boldsymbol{e} \in \mathbb{C}^m$. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^n$ be the solutions of
lyche-numerical-linear-algebra-225 F03023	225	0.994	$\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$	$\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$ and $\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$. Finally, let $\boldsymbol{b}_1, \boldsymbol{e}_1$ be the orthogonal
lyche-numerical-linear-algebra-225 F03024	225	0.994	$\boldsymbol{b}_1, \boldsymbol{e}_1$	$\boldsymbol{b}_1, \boldsymbol{e}_1$	$\min \ \boldsymbol{A} \boldsymbol{x} - \boldsymbol{b}\ _2$ and $\min \ \boldsymbol{A} \boldsymbol{y} - \boldsymbol{b} - \boldsymbol{e}\ _2$. Finally, let $\boldsymbol{b}_1, \boldsymbol{e}_1$ be the orthogonal
lyche-numerical-linear-algebra-225 F03025	225	1.000	$\boldsymbol{b}_1 \neq \mathbf{0}$	$\boldsymbol{b}_1 \neq \mathbf{0}$	projections of $\boldsymbol{b}$ and $\boldsymbol{e}$ into $\mathcal{R}(\boldsymbol{A})$. If $\boldsymbol{b}_1 \neq \mathbf{0}$, we have for any operator norm
lyche-numerical-linear-algebra-225 F03026	225	1.000	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1$	$\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1$	Proof Subtracting $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1 + \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$.
lyche-numerical-linear-algebra-225 F03027	225	1.000	$\boldsymbol{y} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1 + \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$	$\boldsymbol{y} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1 + \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$	Proof Subtracting $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1 + \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$.
lyche-numerical-linear-algebra-225 F03028	225	1.000	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$	$\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$	Proof Subtracting $\boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1$ from $\boldsymbol{y} = \boldsymbol{A}^{\dagger} \boldsymbol{b}_1 + \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$ we have $\boldsymbol{y} - \boldsymbol{x} = \boldsymbol{A}^{\dagger} \boldsymbol{e}_1$.
lyche-numerical-linear-algebra-225 F03029	225	1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{\dagger} \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{\dagger} \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ $	Thus $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{\dagger} \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ $. Moreover, $\ \boldsymbol{b}_1\ = \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $.
lyche-numerical-linear-algebra-225 F03030	225	1.000	$\ \boldsymbol{b}_1\ = \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	$\ \boldsymbol{b}_1\ = \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $	Thus $\ \boldsymbol{y} - \boldsymbol{x}\ = \ \boldsymbol{A}^{\dagger} \boldsymbol{e}_1\ \leq \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ $. Moreover, $\ \boldsymbol{b}_1\ = \ \boldsymbol{A} \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{x}\ $.
lyche-numerical-linear-algebra-225 F03031	225	1.000	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $	$\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $	Therefore $\ \boldsymbol{y} - \boldsymbol{x}\ / \ \boldsymbol{x}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{A}^{\dagger}\ \ \boldsymbol{e}_1\ / \ \boldsymbol{b}_1\ $ proving the rightmost inequality.

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lyche-numerical-linear-algebra-2025-F03032	225	■ 1.000	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\boldsymbol{e}_1$	$\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\boldsymbol{e}_1$	From $\boldsymbol{A}(\boldsymbol{x}-\boldsymbol{y})=\boldsymbol{e}$ and $\boldsymbol{x}=\boldsymbol{A}^\dagger\boldsymbol{b}_1$ we obtain the leftmost inequality.
lyche-numerical-linear-algebra-2025-F03033	225	■ 1.000	$K(\boldsymbol{A})=\ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $	$K(\boldsymbol{A})=\ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $	the number $K(\boldsymbol{A})=\ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $ generalizes the condition number $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $ for
lyche-numerical-linear-algebra-2025-F03034	225	■ 1.000	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	$\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $	the number $K(\boldsymbol{A})=\ \boldsymbol{A}\ \ \boldsymbol{A}^\dagger\ $ generalizes the condition number $\ \boldsymbol{A}\ \ \boldsymbol{A}^{-1}\ $ for
lyche-numerical-linear-algebra-2025-F03035	225	■ 0.997	$\ \boldsymbol{e}_1\ /\ \boldsymbol{b}_1\ $	$\ \boldsymbol{e}_1\ /\ \boldsymbol{b}_1\ $	$\ \boldsymbol{e}\ /\ \boldsymbol{b}\ $ in (8.22) has been replaced by $\ \boldsymbol{e}_1\ /\ \boldsymbol{b}_1\ $, the orthogonal projections of
lyche-numerical-linear-algebra-2025-F03036	225	■ 1.000	$\ \boldsymbol{b}\ /\ \boldsymbol{b}_1\ $	$\ \boldsymbol{b}\ /\ \boldsymbol{b}_1\ $	$\boldsymbol{e}$ and $\boldsymbol{b}$ into $\mathcal{R}(\boldsymbol{A})$. If $\boldsymbol{b}$ lies almost entirely in $\mathcal{N}(\boldsymbol{A}^*)$, i.e. $\ \boldsymbol{b}\ /\ \boldsymbol{b}_1\ $ is large, then
lyche-numerical-linear-algebra-2025-F03037	226	■ 1.000	$K_\infty(\boldsymbol{A})=\ \boldsymbol{A}\ _\infty\ \boldsymbol{A}^\dagger\ _\infty=2\cdot 2=4$	$K_\infty(\boldsymbol{A})=\ \boldsymbol{A}\ _\infty\ \boldsymbol{A}^\dagger\ _\infty=2\cdot 2=4$	Thus $K_\infty(\boldsymbol{A})=\ \boldsymbol{A}\ _\infty\ \boldsymbol{A}^\dagger\ _\infty=2\cdot 2=4$ is quite small.
lyche-numerical-linear-algebra-2025-F03038	226	■ 0.999	$\boldsymbol{A}\boldsymbol{A}^\dagger=\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$	$\boldsymbol{A}\boldsymbol{A}^\dagger=\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$	Consider now the projections $\boldsymbol{b}_1$ and $\boldsymbol{e}_1$. We find $\boldsymbol{A}\boldsymbol{A}^\dagger=\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$. Hence
lyche-numerical-linear-algebra-2025-F03039	226	■ 1.000	$\ \boldsymbol{e}_1\ _\infty/\ \boldsymbol{b}_1\ _\infty=10^{-2}$	$\ \boldsymbol{e}_1\ _\infty/\ \boldsymbol{b}_1\ _\infty=10^{-2}$	Thus $\ \boldsymbol{e}_1\ _\infty/\ \boldsymbol{b}_1\ _\infty=10^{-2}$ and (9.13) takes the form
lyche-numerical-linear-algebra-2025-F03040	226	■ 1.000	$\boldsymbol{x}=\boldsymbol{A}^\dagger\boldsymbol{b}=[10^{-4},0]^*$	$\boldsymbol{x}=\boldsymbol{A}^\dagger\boldsymbol{b}=[10^{-4},0]^*$	To verify the bounds we compute the solutions as $\boldsymbol{x}=\boldsymbol{A}^\dagger\boldsymbol{b}=[10^{-4},0]^*$ and
lyche-numerical-linear-algebra-2025-F03041	226	■ 0.999	$\boldsymbol{y}=\boldsymbol{A}^\dagger(\boldsymbol{b}+\boldsymbol{e})=[10^{-4}+10^{-6},0]^*$	$\boldsymbol{y}=\boldsymbol{A}^\dagger(\boldsymbol{b}+\boldsymbol{e})=[10^{-4}+10^{-6},0]^*$	$\boldsymbol{y}=\boldsymbol{A}^\dagger(\boldsymbol{b}+\boldsymbol{e})=[10^{-4}+10^{-6},0]^*$. Hence
lyche-numerical-linear-algebra-2025-F03042	226	■ 1.000	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m > n$	$\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m > n$	Theorem 9.9 (Perturbing the Matrix) Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{m \times n}, m > n$, where $\boldsymbol{A}$
lyche-numerical-linear-algebra-2025-F03043	226	■ 1.000	$\alpha:=1-\ \boldsymbol{E}\ _2\ \boldsymbol{A}^\dagger\ _2>0$	$\alpha:=1-\ \boldsymbol{E}\ _2\ \boldsymbol{A}^\dagger\ _2>0$	has linearly independent columns and $\alpha:=1-\ \boldsymbol{E}\ _2\ \boldsymbol{A}^\dagger\ _2>0$. Then $\boldsymbol{A}+\boldsymbol{E}$
lyche-numerical-linear-algebra-2025-F03044	226	■ 1.000	$\boldsymbol{b}=\boldsymbol{b}_1+\boldsymbol{b}_2 \in \mathbb{C}^m$	$\boldsymbol{b}=\boldsymbol{b}_1+\boldsymbol{b}_2 \in \mathbb{C}^m$	has linearly independent columns. Let $\boldsymbol{b}=\boldsymbol{b}_1+\boldsymbol{b}_2 \in \mathbb{C}^m$ where $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are the
lyche-numerical-linear-algebra-2025-F03045	226	■ 0.755	$\min\ (\boldsymbol{A}+\boldsymbol{E})\boldsymbol{y}-\boldsymbol{b}\ _2$	$\min\ (\boldsymbol{A}+\boldsymbol{E})\boldsymbol{y}-\boldsymbol{b}\ _2$	and $\boldsymbol{y}$ be the solutions of $\min\ \boldsymbol{A}\boldsymbol{x}-\boldsymbol{b}\ _2$ and $\min\ (\boldsymbol{A}+\boldsymbol{E})\boldsymbol{y}-\boldsymbol{b}\ _2$. Then
lyche-numerical-linear-algebra-2025-F03046	226	■ 1.000	$K(1+\beta K)/\alpha$	$K(1+\beta K)/\alpha$	most $K(1+\beta K)/\alpha$ times as large as the size $\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$ of the relative perturbation
lyche-numerical-linear-algebra-2025-F03047	226	■ 1.000	$\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$	$\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$	most $K(1+\beta K)/\alpha$ times as large as the size $\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$ of the relative perturbation
lyche-numerical-linear-algebra-2025-F03048	227	■ 0.993	$\boldsymbol{A}.\beta$	$\boldsymbol{A}.\beta$	in $\boldsymbol{A}$. β will be small if $\boldsymbol{b}$ lies almost entirely in $\mathcal{R}(\boldsymbol{A})$, and we have approximately
lyche-numerical-linear-algebra-2025-F03049	227	■ 1.000	$\rho \leq \frac{1}{\alpha}K\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$	$\rho \leq \frac{1}{\alpha}K\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$	$\rho \leq \frac{1}{\alpha}K\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$. This corresponds to the estimate (8.25) for linear systems.
lyche-numerical-linear-algebra-2025-F03050	227	■ 1.000	β	β	If β is not small, the term $\frac{1}{\alpha}K^2\beta\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$ will dominate. In other words, the
lyche-numerical-linear-algebra-2025-F03051	227	■ 1.000	$\frac{1}{\alpha}K^2\beta\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$	$\frac{1}{\alpha}K^2\beta\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$	If β is not small, the term $\frac{1}{\alpha}K^2\beta\ \boldsymbol{E}\ _2/\ \boldsymbol{A}\ _2$ will dominate. In other words, the
lyche-numerical-linear-algebra-2025-F03052	227	■ 1.000	$K(\boldsymbol{A})^2\beta$	$K(\boldsymbol{A})^2\beta$	condition number is roughly $K(\boldsymbol{A})$ if β is small and $K(\boldsymbol{A})^2\beta$ if β is not small. Note
lyche-numerical-linear-algebra-2025-F03053	227	■ 0.999	$\boldsymbol{b}_2=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}$	$\boldsymbol{b}_2=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}$	that β is large if $\boldsymbol{b}$ is almost orthogonal to $\mathcal{R}(\boldsymbol{A})$ and that $\boldsymbol{b}_2=\boldsymbol{b}-\boldsymbol{A}\boldsymbol{x}$ is the residual
lyche-numerical-linear-algebra-2025-F03054	227	■ 0.999	$\sigma_1, \sigma_2, \dots, \sigma_n$	$\sigma_1, \sigma_2, \dots, \sigma_n$	$\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has singular values $\sigma_1, \sigma_2, \dots, \sigma_n$ ordered so that $\sigma_1 \geq \dots \geq \sigma_n$. Then
lyche-numerical-linear-algebra-2025-F03055	227	■ 0.999	$\sigma_1 \geq \dots \geq \sigma_n$	$\sigma_1 \geq \dots \geq \sigma_n$	$\boldsymbol{A} \in \mathbb{C}^{m \times n}$ has singular values $\sigma_1, \sigma_2, \dots, \sigma_n$ ordered so that $\sigma_1 \geq \dots \geq \sigma_n$. Then
lyche-numerical-linear-algebra-2025-F03056	228	■ 1.000	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$	$\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$	Theorem 9.11 (Perturbation of Singular Values) Let $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m \times n}$ be rectangu-

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lyche-numerical-linear-algebra-2020-F03057	228	1.000	n-j+1	$n - j + 1$	Proof Fix j and let S be the $n - j + 1$ dimensional subspace for which the minimum
lyche-numerical-linear-algebra-2020-F03058	228	1.000	$\beta_j \leq \alpha_j + \ \mathbf{A} - \mathbf{B}\ _2$	$\beta_j \leq \alpha_j + \ \mathbf{A} - \mathbf{B}\ _2$	By symmetry we obtain $\beta_j \leq \alpha_j + \ \mathbf{A} - \mathbf{B}\ _2$ and the proof is complete.
lyche-numerical-linear-algebra-2020-F03059	228	1.000	$\mathbf{A}, \mathbf{E} \in$	$\mathbf{A}, \mathbf{E} \in$	Theorem 9.12 (Generalized Inverse When Perturbing the Matrix) Let $\mathbf{A}, \mathbf{E} \in$
lyche-numerical-linear-algebra-2020-F03060	228	1.000	$\alpha_1 \geq \cdots \geq \alpha_n$	$\alpha_1 \geq \cdots \geq \alpha_n$	$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \cdots \geq \alpha_n$ and $\epsilon_1 \geq \cdots \geq \epsilon_n$. If $\text{rank}(\mathbf{A} + \mathbf{E}) \leq$
lyche-numerical-linear-algebra-2020-F03061	228	1.000	$\epsilon_1 \geq \cdots \geq \epsilon_n$	$\epsilon_1 \geq \cdots \geq \epsilon_n$	$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \cdots \geq \alpha_n$ and $\epsilon_1 \geq \cdots \geq \epsilon_n$. If $\text{rank}(\mathbf{A} + \mathbf{E}) \leq$
lyche-numerical-linear-algebra-2020-F03062	228	1.000	$\text{rank}(\mathbf{A} + \mathbf{E}) \leq$	$\text{rank}(\mathbf{A} + \mathbf{E}) \leq$	$\mathbb{R}^{m \times n}$ have singular values $\alpha_1 \geq \cdots \geq \alpha_n$ and $\epsilon_1 \geq \cdots \geq \epsilon_n$. If $\text{rank}(\mathbf{A} + \mathbf{E}) \leq$
lyche-numerical-linear-algebra-2020-F03063	228	0.893	$\ \mathbf{A}^\dagger\ _2 \ \mathbf{E}\ _2 < 1$	$\ \mathbf{A}^\dagger\ _2 \ \mathbf{E}\ _2 < 1$	$\text{rank}(\mathbf{A}) = r$ and $\ \mathbf{A}^\dagger\ _2 \ \mathbf{E}\ _2 < 1$ then
lyche-numerical-linear-algebra-2020-F03064	228	1.000	$\text{rank}(\mathbf{A} + \mathbf{E}) = \text{rank}(\mathbf{A})$	$\text{rank}(\mathbf{A} + \mathbf{E}) = \text{rank}(\mathbf{A})$	1. $\text{rank}(\mathbf{A} + \mathbf{E}) = \text{rank}(\mathbf{A})$,
lyche-numerical-linear-algebra-2020-F03065	228	1.000	$\ (\mathbf{A} + \mathbf{E})^\dagger \ _2 \leq \frac{\ \mathbf{A}^\dagger \ _2}{1 - \ \mathbf{A}^\dagger \ _2 \ \mathbf{E} \ _2} = \frac{1}{\alpha_r - \epsilon_1}$	$\ (\mathbf{A} + \mathbf{E})^\dagger \ _2 \leq \frac{\ \mathbf{A}^\dagger \ _2}{1 - \ \mathbf{A}^\dagger \ _2 \ \mathbf{E} \ _2} = \frac{1}{\alpha_r - \epsilon_1}$	2. $\ (\mathbf{A} + \mathbf{E})^\dagger \ _2 \leq \frac{\ \mathbf{A}^\dagger \ _2}{1 - \ \mathbf{A}^\dagger \ _2 \ \mathbf{E} \ _2} = \frac{1}{\alpha_r - \epsilon_1}$,
lyche-numerical-linear-algebra-2020-F03066	228	1.000	$\beta_1 \geq$	$\beta_1 \geq$	Proof Suppose $\mathbf{A}$ has rank r and let $\mathbf{B} := \mathbf{A} + \mathbf{E}$ have singular values $\beta_1 \geq$
lyche-numerical-linear-algebra-2020-F03067	228	1.000	$\cdots \geq \beta_n$	$\cdots \geq \beta_n$	$\cdots > \beta_r$. In terms of singular values the inequality $\ \mathbf{A}^\dagger\ _2 \ \mathbf{E}\ _2 < 1$ can be written
lyche-numerical-linear-algebra-2020-F03068	228	1.000	$\epsilon_1 / \alpha_r < 1$	$\epsilon_1 / \alpha_r < 1$	$\epsilon_1 / \alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \ \mathbf{E}\ _2 = \epsilon_1$, which
lyche-numerical-linear-algebra-2020-F03069	228	1.000	$\alpha_r > \epsilon_1$	$\alpha_r > \epsilon_1$	$\epsilon_1 / \alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \ \mathbf{E}\ _2 = \epsilon_1$, which
lyche-numerical-linear-algebra-2020-F03070	228	1.000	$\alpha_r - \beta_r \leq \ \mathbf{E}\ _2 = \epsilon_1$	$\alpha_r - \beta_r \leq \ \mathbf{E}\ _2 = \epsilon_1$	$\epsilon_1 / \alpha_r < 1$ or $\alpha_r > \epsilon_1$. By Theorem 9.11 we have $\alpha_r - \beta_r \leq \ \mathbf{E}\ _2 = \epsilon_1$, which
lyche-numerical-linear-algebra-2020-F03071	228	0.703	$\beta_r \geq \alpha_r - \epsilon_1 > 0$	$\beta_r \geq \alpha_r - \epsilon_1 > 0$	implies $\beta_r \geq \alpha_r - \epsilon_1 > 0$, and this shows that $\text{rank}(\mathbf{A} + \mathbf{E}) \geq r$. Thus 1. follows.
lyche-numerical-linear-algebra-2020-F03072	228	0.703	$\text{rank}(\mathbf{A} + \mathbf{E}) \geq r$	$\text{rank}(\mathbf{A} + \mathbf{E}) \geq r$	implies $\beta_r \geq \alpha_r - \epsilon_1 > 0$, and this shows that $\text{rank}(\mathbf{A} + \mathbf{E}) \geq r$. Thus 1. follows.
lyche-numerical-linear-algebra-2020-F03073	228	0.960	$\beta_r \geq \alpha_r - \epsilon_1$	$\beta_r \geq \alpha_r - \epsilon_1$	To prove 2., the inequality $\beta_r \geq \alpha_r - \epsilon_1$ implies that
lyche-numerical-linear-algebra-2020-F03074	228	1.000	$\mu_1 \geq \cdots \geq \mu_n$	$\mu_1 \geq \cdots \geq \mu_n$	and $\mu_1 \geq \cdots \geq \mu_r$, respectively.
lyche-numerical-linear-algebra-2020-F03075	229	1.000	$\mathbf{A}, \mathbf{B} \in \mathbb{C}^{m \times n}$	$\mathbf{A}, \mathbf{B} \in \mathbb{C}^{m \times n}$	$m, n \in \mathbb{N}$ and $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{m \times n}$ we have
lyche-numerical-linear-algebra-2020-F03076	229	1.000	$\beta_1 \geq \cdots \geq \beta_n$	$\beta_1 \geq \cdots \geq \beta_n$	where $\alpha_1 \geq \cdots \geq \alpha_n$ and $\beta_1 \geq \cdots \geq \beta_n$ are the singular values of $\mathbf{A}$ and $\mathbf{B}$,
lyche-numerical-linear-algebra-2020-F03077	229	1.000	$\lambda_1 \geq \cdots \geq \lambda_{m+n}$	$\lambda_1 \geq \cdots \geq \lambda_{m+n}$	If $\mathbf{C}$ and $\mathbf{D}$ have eigenvalues $\lambda_1 \geq \cdots \geq \lambda_{m+n}$ and $\mu_1 \geq \cdots \geq \mu_{m+n}$, respectively
lyche-numerical-linear-algebra-2020-F03078	229	1.000	$\mu_1 \geq \cdots \geq \mu_{m+n}$	$\mu_1 \geq \cdots \geq \mu_{m+n}$	If $\mathbf{C}$ and $\mathbf{D}$ have eigenvalues $\lambda_1 \geq \cdots \geq \lambda_{m+n}$ and $\mu_1 \geq \cdots \geq \mu_{m+n}$, respectively
lyche-numerical-linear-algebra-2020-F03079	229	0.651	$\text{SVD}[\mathbf{u}_1, \dots, \mathbf{u}_m] \mathbf{\Sigma} [\mathbf{v}_1, \dots, \mathbf{v}_n]^*$	$\text{SVD}[\mathbf{u}_1, \dots, \mathbf{u}_m] \mathbf{\Sigma} [\mathbf{v}_1, \dots, \mathbf{v}_n]^*$	Suppose $\mathbf{A}$ has rank r and $\text{SVD}[\mathbf{u}_1, \dots, \mathbf{u}_m] \mathbf{\Sigma} [\mathbf{v}_1, \dots, \mathbf{v}_n]^*$. We use (7.7) and
lyche-numerical-linear-algebra-2020-F03080	229	1.000	$2r$	$2r$	Thus $\mathbf{C}$ has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$ and $m + n - 2r$ additional
lyche-numerical-linear-algebra-2020-F03081	229	1.000	$\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$	$\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$	Thus $\mathbf{C}$ has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$ and $m + n - 2r$ additional
lyche-numerical-linear-algebra-2020-F03082	229	1.000	$m + n - 2r$	$m + n - 2r$	Thus $\mathbf{C}$ has the $2r$ eigenvalues $\alpha_1, -\alpha_1, \dots, \alpha_r, -\alpha_r$ and $m + n - 2r$ additional

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lyche-numerical-linear-algebra-F03083	229	■ 1.000	$2 \ s$	$2s$	zero eigenvalues. Similarly, if $\mathbf{B}$ has rank s then $\mathbf{D}$ has the $2s$ eigenvalues
lyche-numerical-linear-algebra-F03084	229	■ 0.999	$\backslash\mathrm{beta}_{\{1\}},\backslash\mathrm{beta}_{\{1\}}, \ \backslash\mathrm{dots}, \ \backslash\mathrm{beta}_{\{s\}},\backslash\mathrm{beta}_{\{s\}}$	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$ and $m+n-2s$ additional zero eigenvalues. Let
lyche-numerical-linear-algebra-F03085	229	■ 0.999	$m+n-2 \ s$	$m+n-2s$	$\beta_1, -\beta_1, \dots, \beta_s, -\beta_s$ and $m+n-2s$ additional zero eigenvalues. Let
lyche-numerical-linear-algebra-F03086	230	■ 1.000	$\backslash\mathrm{sum}_{\{i=1\}^{\{t\}}\backslash\mathrm{left} \backslash\mathrm{alpha}_{\{i\}}-\backslash\mathrm{beta}_{\{i\}}\backslash\mathrm{right} ^{\{2\}} \ \backslash\mathrm{leq} \ \backslash \backslash\mathrm{boldsymbol{B}}-\backslash\mathrm{boldsymbol{A}}\backslash _{\{F\}}^{\{2\}}$	$\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ \mathbf{B} - \mathbf{A}\ _F^2$	But then (9.21) implies $\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ \mathbf{B} - \mathbf{A}\ _F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra-F03087	230	■ 1.000	$t \ \backslash\mathrm{leq} \ n$	$t \leq n$	But then (9.21) implies $\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ \mathbf{B} - \mathbf{A}\ _F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra-F03088	230	■ 1.000	$\backslash\mathrm{alpha}_{\{i\}}=\backslash\mathrm{beta}_{\{i\}}=0$	$\alpha_i = \beta_i = 0$	But then (9.21) implies $\sum_{i=1}^t \alpha_i - \beta_i ^2 \leq \ \mathbf{B} - \mathbf{A}\ _F^2$. Since $t \leq n$ and $\alpha_i = \beta_i = 0$
lyche-numerical-linear-algebra-F03089	230	■ 1.000	$i=t+1, \ \backslash\mathrm{dots}, \ n$	$i = t + 1, \dots, n$	for $i = t + 1, \dots, n$ we obtain (9.20).
lyche-numerical-linear-algebra-F03090	230	■ 1.000	$\backslash\mathrm{left}(t\text{-}c_{\{1\}}\backslash\mathrm{right})^{\{2\}}+\backslash\mathrm{left}(y\text{-}c_{\{2\}}\backslash\mathrm{right})^{\{2\}}=r^{\{2\}}$	$(t - c_1)^2 + (y - c_2)^2 = r^2$	to fit a circle $(t - c_1)^2 + (y - c_2)^2 = r^2$ to $m \geq 3$ given points $(t_i, y_i)_{i=1}^m$ in the
lyche-numerical-linear-algebra-F03091	230	■ 1.000	$\backslash\mathrm{left}(t_{\{i\}}, \ y_{\{i\}}\backslash\mathrm{right})_{\{i=1\}^{\{m\}}}$	$(t_i, y_i)_{i=1}^m$	to fit a circle $(t - c_1)^2 + (y - c_2)^2 = r^2$ to $m \geq 3$ given points $(t_i, y_i)_{i=1}^m$ in the
lyche-numerical-linear-algebra-F03092	230	■ 0.999	$(t, \ y)$	(t, y)	(t, y) -plane. We obtain the overdetermined system
lyche-numerical-linear-algebra-F03093	230	■ 1.000	$c_{\{1\}}, \ c_{\{2\}}$	c_1, c_2	of m equations in the three unknowns c_1, c_2 and r . This system is nonlinear, but it
lyche-numerical-linear-algebra-F03094	230	■ 1.000	$c_{\{1\}}=x_{\{1\}} \ / \ 2, \ c_{\{2\}}=x_{\{2\}} \ / \ 2$	$c_1 = x_1/2, c_2 = x_2/2$	and then setting $c_1 = x_1/2, c_2 = x_2/2$ and $r^2 = c_1^2 + c_2^2 + x_3$,
lyche-numerical-linear-algebra-F03095	230	■ 1.000	$r^{\{2\}}=c_{\{1\}}^{\{2\}}+c_{\{2\}}^{\{2\}}+x_{\{3\}}$	$r^2 = c_1^2 + c_2^2 + x_3$	and then setting $c_1 = x_1/2, c_2 = x_2/2$ and $r^2 = c_1^2 + c_2^2 + x_3$.
lyche-numerical-linear-algebra-F03096	230	■ 1.000	$c_{\{1\}}, \ c_{\{2\}}, \ r$	c_1, c_2, r	a) Derive (9.23) from (9.22). Explain how we can find c_1, c_2, r once $[x_1, x_2, x_3]$ is
lyche-numerical-linear-algebra-F03097	230	■ 1.000	$\backslash\mathrm{left}[x_{\{1\}}, \ x_{\{2\}}, \ x_{\{3\}}\backslash\mathrm{right}]$	$[x_1, x_2, x_3]$	a) Derive (9.23) from (9.22). Explain how we can find c_1, c_2, r once $[x_1, x_2, x_3]$ is
lyche-numerical-linear-algebra-F03098	230	■ 0.593	$(1,4), (3,2), (1,0)$	$(1, 4), (3, 2), (1, 0)$	d) Use (9.23) to find the circle passing through the three points $(1, 4), (3, 2), (1, 0)$.
lyche-numerical-linear-algebra-F03099	231	■ 1.000	$\backslash\mathrm{left}[\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}\backslash\mathrm{sqrt}\{2\} \ \& \ \backslash\mathrm{sqrt}\{2\} \ \backslash\backslash \ 0 \ \& \ \backslash\mathrm{sqrt}\{3\}\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}]$	$\begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$	a) Let $\mathbf{A}$ be the matrix $\begin{bmatrix} \sqrt{2} & \sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$. Find the singular values of $\mathbf{A}$, and compute
lyche-numerical-linear-algebra-F03100	231	■ 0.993	$\backslash\mathrm{boldsymbol{A}}=\backslash\mathrm{left}[\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cc}\}3 \ \backslash\mathrm{alpha} \ \backslash\backslash \ \backslash\mathrm{alpha} \ \& \ 1\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}]$	$\mathbf{A} = \begin{bmatrix} 3 & \alpha \\ \alpha & 1 \end{bmatrix}$	b) Consider the matrix $\mathbf{A} = \begin{bmatrix} 3 & \alpha \\ \alpha & 1 \end{bmatrix}$, where α is a real number. For which values of
lyche-numerical-linear-algebra-F03101	231	■ 1.000	$\backslash\mathrm{boldsymbol{p}}_{\{1\}}=(0,1), \ \backslash\mathrm{boldsymbol{p}}_{\{2\}}=(1,0), \ \backslash\mathrm{boldsymbol{p}}_{\{3\}}=(2,1)$	$\mathbf{p}_1 = (0, 1), \mathbf{p}_2 = (1, 0), \mathbf{p}_3 = (2, 1)$	c) We would like to fit the points $\mathbf{p}_1 = (0, 1), \mathbf{p}_2 = (1, 0), \mathbf{p}_3 = (2, 1)$ to a straight
lyche-numerical-linear-algebra-F03102	231	■ 1.000	$p(x)=m \ x+b$	$p(x) = mx + b$	line in the plane. Find a line $p(x) = mx + b$ which minimizes
lyche-numerical-linear-algebra-F03103	231	■ 1.000	$\backslash\mathrm{boldsymbol{p}}_{\{i\}}=\backslash\mathrm{left}(x_{\{i\}}, \ y_{\{i\}}\backslash\mathrm{right})$	$\mathbf{p}_i = (x_i, y_i)$	where $\mathbf{p}_i = (x_i, y_i)$. Is this solution unique?
lyche-numerical-linear-algebra-F03104	231	■ 1.000	$F: \ \backslash\mathrm{mathbb{R}}^{\{n\}} \ \rightarrowtail \ \backslash\mathrm{mathbb{R}}^{\{n\}}$	$F : \mathbb{R}^n \rightarrow \mathbb{R}^n$	$\mathbb{R}^{m \times n}$ and let $\mathbf{I} \in \mathbb{R}^{n \times n}$ be the identity matrix. We define $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by
lyche-numerical-linear-algebra-F03105	231	■ 1.000	$\backslash\mathrm{boldsymbol{B}}:=\backslash\mathrm{boldsymbol{I}}+\backslash\mathrm{boldsymbol{A}}^{\{T\}} \ \backslash\mathrm{boldsymbol{A}}$	$\mathbf{B} := \mathbf{I} + \mathbf{A}^T \mathbf{A}$	a) Show that the matrix $\mathbf{B} := \mathbf{I} + \mathbf{A}^T \mathbf{A}$ is symmetric and positive definite.
lyche-numerical-linear-algebra-F03106	231	■ 1.000	$\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{I}}+\backslash\mathrm{boldsymbol{A}}^{\{T\}} \ \backslash\mathrm{boldsymbol{A}} \ \backslash\mathrm{right}) \ \backslash\mathrm{boldsymbol{x}}=\backslash\mathrm{boldsymbol{A}}^{\{T\}} \ \backslash\mathrm{boldsymbol{b}}$	$(\mathbf{I} + \mathbf{A}^T \mathbf{A}) \mathbf{x} = \mathbf{A}^T \mathbf{b}$	is the unique solution of the linear system $(\mathbf{I} + \mathbf{A}^T \mathbf{A}) \mathbf{x} = \mathbf{A}^T \mathbf{b}$.
lyche-numerical-linear-algebra-F03107	231	■ 1.000	$\backslash\mathrm{operatorname{diag}}\backslash\mathrm{left}(d_{\{1\}}, \ d_{\{2\}}, \ \backslash\mathrm{dots}, \ d_{\{m\}} \ \backslash\mathrm{right}) \ \backslash\mathrm{in} \ \backslash\mathrm{mathbb{R}}^{\{m\}} \ \backslash\mathrm{times} \ m$	$\mathrm{diag}(d_1, d_2, \dots, d_m) \in \mathbb{R}^{m \times m}$	$\mathrm{diag}(d_1, d_2, \dots, d_m) \in \mathbb{R}^{m \times m}$, where $d_i > 0, i = 1, 2, \dots, m$. We want to minimize

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lyche-numerical-linear-algebra-2021-F03108	231	1.000	$d_i > 0, i = 1, 2, \dots, m$	$d_i > 0, i = 1, 2, \dots, m$	$\text{diag}(d_1, d_2, \dots, d_m) \in \mathbb{R}^{m \times m}$, where $d_i > 0, i = 1, 2, \dots, m$. We want to minimize
lyche-numerical-linear-algebra-2021-F03109	232	1.000	$r_i = r_i(\mathbf{x}), i = 1, 2, \dots, m$	$r_i = r_i(\mathbf{x}), i = 1, 2, \dots, m$	where $r_i = r_i(\mathbf{x}), i = 1, 2, \dots, m$ are the components of the vector
lyche-numerical-linear-algebra-2021-F03110	232	1.000	$\ \mathbf{r}(\mathbf{x})\ _D^2$	$\ \mathbf{r}(\mathbf{x})\ _D^2$	a) Show that $\ \mathbf{r}(\mathbf{x})\ _D^2$ in (9.24) obtains a unique minimum when $\mathbf{x} = \mathbf{x}_{\min}$ is the
lyche-numerical-linear-algebra-2021-F03111	232	1.000	$\mathbf{x} = \mathbf{x}_{\min}$	$\mathbf{x} = \mathbf{x}_{\min}$	a) Show that $\ \mathbf{r}(\mathbf{x})\ _D^2$ in (9.24) obtains a unique minimum when $\mathbf{x} = \mathbf{x}_{\min}$ is the
lyche-numerical-linear-algebra-2021-F03112	232	1.000	$K_2(\mathbf{B}) := \ \mathbf{B}\ _2 \ \mathbf{B}^{-1}\ _2$	$K_2(\mathbf{B}) := \ \mathbf{B}\ _2 \ \mathbf{B}^{-1}\ _2$	where for any nonsingular matrix $K_2(\mathbf{B}) := \ \mathbf{B}\ _2 \ \mathbf{B}^{-1}\ _2$.
lyche-numerical-linear-algebra-2021-F03113	232	0.999	$\mathbf{B}, \mathbf{C} \in \mathbb{C}^{n \times m}$	$\mathbf{B}, \mathbf{C} \in \mathbb{C}^{n \times m}$	pose $\mathbf{B}, \mathbf{C} \in \mathbb{C}^{n \times m}$ satisfy
lyche-numerical-linear-algebra-2021-F03114	232	1.000	$\mathbf{B} = \mathbf{C}$	$\mathbf{B} = \mathbf{C}$	Verify the following proof that $\mathbf{B} = \mathbf{C}$.
lyche-numerical-linear-algebra-2021-F03115	232	1.000	$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$	$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$	generalized inverse of $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ is $\mathbf{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ without using the singular
lyche-numerical-linear-algebra-2021-F03116	232	1.000	$\mathbf{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$	$\mathbf{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$	generalized inverse of $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$ is $\mathbf{A}^\dagger = \frac{1}{4} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ without using the singular
lyche-numerical-linear-algebra-2021-F03117	232	1.000	$\mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^*$	$\mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^*$	and $\mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^*$. If $\mathbf{A}$ has linearly independent rows, then show that $\mathbf{A} \mathbf{A}^*$ is
lyche-numerical-linear-algebra-2021-F03118	232	1.000	$\mathbf{A}^\dagger = \mathbf{A}^* (\mathbf{A} \mathbf{A}^*)^{-1}$	$\mathbf{A}^\dagger = \mathbf{A}^* (\mathbf{A} \mathbf{A}^*)^{-1}$	nonsingular and $\mathbf{A}^\dagger = \mathbf{A}^* (\mathbf{A} \mathbf{A}^*)^{-1}$.
lyche-numerical-linear-algebra-2021-F03119	233	1.000	$\mathcal{R}(\mathbf{A}^*), \mathcal{N}(\mathbf{A})$	$\mathcal{R}(\mathbf{A}^*), \mathcal{N}(\mathbf{A})$	rank r , and let $\mathcal{S}$ be one of the subspaces $\mathcal{R}(\mathbf{A}^*), \mathcal{N}(\mathbf{A})$. Show that the orthogonal
lyche-numerical-linear-algebra-2021-F03120	233	0.996	$\mathbb{C}^n = \mathcal{R}(\mathbf{A}^*) \oplus \mathcal{N}(\mathbf{A})$	$\mathbb{C}^n = \mathcal{R}(\mathbf{A}^*) \oplus \mathcal{N}(\mathbf{A})$	orthogonal sum $\mathbb{C}^n = \mathcal{R}(\mathbf{A}^*) \oplus \mathcal{N}(\mathbf{A})$
lyche-numerical-linear-algebra-2021-F03121	233	1.000	$\mathbf{u}^\dagger = (\mathbf{u}^* \mathbf{u})^{-1} \mathbf{u}^*$	$\mathbf{u}^\dagger = (\mathbf{u}^* \mathbf{u})^{-1} \mathbf{u}^*$	Exercise 9.9 (The Generalized Inverse of a Vector) Show that $\mathbf{u}^\dagger = (\mathbf{u}^* \mathbf{u})^{-1} \mathbf{u}^*$
lyche-numerical-linear-algebra-2021-F03122	233	1.000	$\mathbf{u} \in \mathbb{C}^{n,1}$	$\mathbf{u} \in \mathbb{C}^{n,1}$	if $\mathbf{u} \in \mathbb{C}^{n,1}$ is nonzero.
lyche-numerical-linear-algebra-2021-F03123	233	1.000	$\mathbf{A} = \mathbf{u} \mathbf{v}^*$	$\mathbf{A} = \mathbf{u} \mathbf{v}^*$	Exercise 9.10 (The Generalized Inverse of an Outer Product) If $\mathbf{A} = \mathbf{u} \mathbf{v}^*$ where
lyche-numerical-linear-algebra-2021-F03124	233	1.000	$\mathbf{u} \in \mathbb{C}^m, \mathbf{v} \in \mathbb{C}^n$	$\mathbf{u} \in \mathbb{C}^m, \mathbf{v} \in \mathbb{C}^n$	$\mathbf{u} \in \mathbb{C}^m, \mathbf{v} \in \mathbb{C}^n$ are nonzero, show that
lyche-numerical-linear-algebra-2021-F03125	233	0.990	$\text{diag}(\lambda_1, \dots, \lambda_n)^\dagger = \text{diag}(\lambda_1^\dagger, \dots, \lambda_n^\dagger)$	$\text{diag}(\lambda_1, \dots, \lambda_n)^\dagger = \text{diag}(\lambda_1^\dagger, \dots, \lambda_n^\dagger)$	$\text{diag}(\lambda_1, \dots, \lambda_n)^\dagger = \text{diag}(\lambda_1^\dagger, \dots, \lambda_n^\dagger)$ where
lyche-numerical-linear-algebra-2021-F03126	233	1.000	$(\mathbf{A}^*)^\dagger = (\mathbf{A}^\dagger)^*$	$(\mathbf{A}^*)^\dagger = (\mathbf{A}^\dagger)^*$	a) $(\mathbf{A}^*)^\dagger = (\mathbf{A}^\dagger)^*$.
lyche-numerical-linear-algebra-2021-F03127	233	0.999	$(\mathbf{A}^\dagger)^\dagger = \mathbf{A}$	$(\mathbf{A}^\dagger)^\dagger = \mathbf{A}$	b) $(\mathbf{A}^\dagger)^\dagger = \mathbf{A}$.
lyche-numerical-linear-algebra-2021-F03128	233	0.995	$(\alpha \mathbf{A})^\dagger = \frac{1}{\alpha} \mathbf{A}^\dagger, \alpha \neq 0$	$(\alpha \mathbf{A})^\dagger = \frac{1}{\alpha} \mathbf{A}^\dagger, \alpha \neq 0$	c) $(\alpha \mathbf{A})^\dagger = \frac{1}{\alpha} \mathbf{A}^\dagger, \alpha \neq 0$.
lyche-numerical-linear-algebra-2021-F03129	233	1.000	$k, m, n \in \mathbb{N}, \mathbf{A} \in \mathbb{C}^{m \times n}, \mathbf{B} \in \mathbb{C}^{n \times k}$	$k, m, n \in \mathbb{N}, \mathbf{A} \in \mathbb{C}^{m \times n}, \mathbf{B} \in \mathbb{C}^{n \times k}$	Exercise 9.13 (The Generalized Inverse of a Product) Suppose $k, m, n \in \mathbb{N}, \mathbf{A} \in \mathbb{C}^{m \times n}, \mathbf{B} \in \mathbb{C}^{n \times k}$. Suppose $\mathbf{A}$ has linearly independent columns and $\mathbf{B}$ has linearly
lyche-numerical-linear-algebra-2021-F03130	233	0.623	$(\mathbf{A} \mathbf{B})^\dagger = \mathbf{B}^\dagger \mathbf{A}^\dagger$	$(\mathbf{A} \mathbf{B})^\dagger = \mathbf{B}^\dagger \mathbf{A}^\dagger$	a) Show that $(\mathbf{A} \mathbf{B})^\dagger = \mathbf{B}^\dagger \mathbf{A}^\dagger$.
lyche-numerical-linear-algebra-2021-F03131	233	0.993	$(\mathbf{A} \mathbf{B})^\dagger = \mathbf{B}^\dagger \mathbf{A}^\dagger$	$(\mathbf{A} \mathbf{B})^\dagger = \mathbf{B}^\dagger \mathbf{A}^\dagger$	a) Show that $(\mathbf{A} \mathbf{B})^\dagger = \mathbf{B}^\dagger \mathbf{A}^\dagger$.

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lyche-numerical-linear-algebra-233 F03132	233	1.000	$\boldsymbol{E}=\boldsymbol{A} \boldsymbol{B}, \boldsymbol{F}=\boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$	$E = AB, F = B^{\dagger} A^{\dagger}$	Hint: Let $E = AB, F = B^{\dagger} A^{\dagger}$. Show by using $A^{\dagger} A = BB^{\dagger} = I$ that F is the
lyche-numerical-linear-algebra-233 F03133	233	1.000	$\boldsymbol{A}^{\dagger} \boldsymbol{A}=\boldsymbol{B} \boldsymbol{B}^{\dagger}=\boldsymbol{I}$	$A^{\dagger} A = BB^{\dagger} = I$	Hint: Let $E = AB, F = B^{\dagger} A^{\dagger}$. Show by using $A^{\dagger} A = BB^{\dagger} = I$ that F is the
lyche-numerical-linear-algebra-233 F03134	233	0.998	$\boldsymbol{A} \in \mathbb{R}^{1,2}, \boldsymbol{B} \in \mathbb{R}^{2,1}$	$A \in \mathbb{R}^{1,2}, B \in \mathbb{R}^{2,1}$	b) Find $A \in \mathbb{R}^{1,2}, B \in \mathbb{R}^{2,1}$ such that $(AB)^{\dagger} \neq B^{\dagger} A^{\dagger}$.
lyche-numerical-linear-algebra-233 F03135	233	0.998	$(\boldsymbol{A} \boldsymbol{B})^{\dagger} \neq \boldsymbol{B}^{\dagger} \boldsymbol{A}^{\dagger}$	$(AB)^{\dagger} \neq B^{\dagger} A^{\dagger}$	b) Find $A \in \mathbb{R}^{1,2}, B \in \mathbb{R}^{2,1}$ such that $(AB)^{\dagger} \neq B^{\dagger} A^{\dagger}$.
lyche-numerical-linear-algebra-234 F03136	234	1.000	$\boldsymbol{A}^*=\boldsymbol{A}^{\dagger}$	$A^* = A^{\dagger}$	$A^* = A^{\dagger}$ if and only if all singular values of A are either zero or one.
lyche-numerical-linear-algebra-234 F03137	234	0.998	$\boldsymbol{A}(\boldsymbol{A}^* \boldsymbol{A})^{-1} \boldsymbol{A}^* \boldsymbol{b}$	$A(A^* A)^{-1} A^* b$	$A(A^* A)^{-1} A^* b$ is the projection of b into $\mathcal{R}(A)$. (Cf. Exercise 9.8.)
lyche-numerical-linear-algebra-234 F03138	234	1.000	$r>0$	$r > 0$	system $Ax = b$ where $A \in \mathbb{C}^{n \times n}$ has rank $r > 0$ and $b \in \mathbb{C}^n$. Let
lyche-numerical-linear-algebra-234 F03139	234	1.000	$\boldsymbol{c}=\left[c_1, \ldots, c_n\right]^*=\boldsymbol{U}^* \boldsymbol{b}$	$c = [c_1, \dots, c_n]^* = U^* b$	a) Let $c = [c_1, \dots, c_n]^* = U^* b$ and $y = [y_1, \dots, y_n]^* = V^* x$. Show that $Ax = b$
lyche-numerical-linear-algebra-234 F03140	234	1.000	$\boldsymbol{y}=\left[y_1, \ldots, y_n\right]^*=\boldsymbol{V}^* \boldsymbol{x}$	$y = [y_1, \dots, y_n]^* = V^* x$	a) Let $c = [c_1, \dots, c_n]^* = U^* b$ and $y = [y_1, \dots, y_n]^* = V^* x$. Show that $Ax = b$
lyche-numerical-linear-algebra-234 F03141	234	0.999	$c_{r+1}=\cdots=c_n=0$	$c_{r+1} = \dots = c_n = 0$	b) Show that $Ax = b$ has a solution x if and only if $c_{r+1} = \dots = c_n = 0$.
lyche-numerical-linear-algebra-234 F03142	234	1.000	$\boldsymbol{A} \in \mathbb{C}^{m \times n}, \boldsymbol{b} \in \mathbb{C}^n$	$A \in \mathbb{C}^{m \times n}, b \in \mathbb{C}^n$	Exercise 9.17 (Fredholm's Alternative) For any $A \in \mathbb{C}^{m \times n}, b \in \mathbb{C}^n$ show that
lyche-numerical-linear-algebra-234 F03143	234	1.000	$\boldsymbol{b} \in \mathcal{R}(\boldsymbol{A})$	$b \in \mathcal{R}(A)$	In other words either $b \in \mathcal{R}(A)$, or we can find $y \in \mathcal{N}(A^*)$ such that $y^* b \neq 0$. This
lyche-numerical-linear-algebra-234 F03144	234	1.000	$\boldsymbol{y} \in \mathcal{N}\left(\boldsymbol{A}^*\right)$	$y \in \mathcal{N}(A^*)$	In other words either $b \in \mathcal{R}(A)$, or we can find $y \in \mathcal{N}(A^*)$ such that $y^* b \neq 0$. This
lyche-numerical-linear-algebra-234 F03145	234	1.000	$\boldsymbol{y}^* \boldsymbol{b} \neq 0$	$y^* b \neq 0$	In other words either $b \in \mathcal{R}(A)$, or we can find $y \in \mathcal{N}(A^*)$ such that $y^* b \neq 0$. This
lyche-numerical-linear-algebra-234 F03146	234	1.000	$\mathbb{C}^{(m-n) \times n}$	$\mathbb{C}^{(m-n) \times n}$	where B is a non-singular $n \times n$ matrix and C is in $\mathbb{C}^{(m-n) \times n}$. Let $A^{\dagger}$ denote the
lyche-numerical-linear-algebra-234 F03147	234	1.000	$\left\ \boldsymbol{A}^{\dagger}\right\ _2 \leq\left\ \boldsymbol{B}^{-1}\right\ _2$	$\ A^{\dagger}\ _2 \leq \ B^{-1}\ _2$	pseudoinverse of A . Show that $\ A^{\dagger}\ _2 < \ B^{-1}\ _2$.
lyche-numerical-linear-algebra-235 F03148	235	1.000	$K(\boldsymbol{A})=\left\ \boldsymbol{A}\right\ _2\left\ \boldsymbol{A}^{\dagger}\right\ _2$	$K(A) = \ A\ _2 \ A^{\dagger}\ _2$	b) Compute $K(A) = \ A\ _2 \ A^{\dagger}\ _2$.
lyche-numerical-linear-algebra-235 F03149	235	0.999	$\boldsymbol{y}_{\boldsymbol{A}^{\dagger}}$	$y_{A^{\dagger}}$	and $y_{A^{\dagger}}$ are vectors with $\ y_A\ = \ y_{A^{\dagger}}\ = 1$ and $\ A\ = \ Ay_A\ $ and $\ A^{\dagger}\ =$
lyche-numerical-linear-algebra-235 F03150	235	0.999	$\left\ \boldsymbol{y}_A\right\ =\left\ \boldsymbol{y}_{\boldsymbol{A}^{\dagger}}\right\ =1$	$\ y_A\ = \ y_{A^{\dagger}}\ = 1$	and $y_{A^{\dagger}}$ are vectors with $\ y_A\ = \ y_{A^{\dagger}}\ = 1$ and $\ A\ = \ Ay_A\ $ and $\ A^{\dagger}\ =$
lyche-numerical-linear-algebra-235 F03151	235	0.999	$\left\ \boldsymbol{A}^{\dagger}\right\ $	$\ A^{\dagger}\ =$	and $y_{A^{\dagger}}$ are vectors with $\ y_A\ = \ y_{A^{\dagger}}\ = 1$ and $\ A\ = \ Ay_A\ $ and $\ A^{\dagger}\ =$
lyche-numerical-linear-algebra-235 F03152	235	0.999	$\left\ \boldsymbol{A}^{\dagger} \boldsymbol{y}_{\boldsymbol{A}^{\dagger}}\right\ $	$\ A^{\dagger} y_{A^{\dagger}}\ $	$\ A^{\dagger} y_{A^{\dagger}}\ $.
lyche-numerical-linear-algebra-235 F03153	235	1.000	$\boldsymbol{b}=\boldsymbol{A} \boldsymbol{y}_A, \boldsymbol{e}_1=\boldsymbol{y}_{\boldsymbol{A}^{\dagger}}$	$b = Ay_A, e_1 = y_{A^{\dagger}}$	a) Show that we have equality to the right in (9.13) if $b = Ay_A, e_1 = y_{A^{\dagger}}$.
lyche-numerical-linear-algebra-235 F03154	235	1.000	$\boldsymbol{A}^{\dagger}=\boldsymbol{A}^{-1}$	$A^{\dagger} = A^{-1}$	nonsingular then $A^{\dagger} = A^{-1}$ and $e_1 = e$.
lyche-numerical-linear-algebra-235 F03155	235	1.000	$\boldsymbol{e}_1=\boldsymbol{e}$	$e_1 = e$	nonsingular then $A^{\dagger} = A^{-1}$ and $e_1 = e$.
lyche-numerical-linear-algebra-235 F03156	235	1.000	$(2,2)$	$(2, 2)$	b) Suppose ϵ is small and we replace the (2, 2) entry $3+2\epsilon+\epsilon^2$ in $A^* A$ by $3+2\epsilon$.
lyche-numerical-linear-algebra-235 F03157	235	1.000	$3+2 \epsilon+\epsilon^2$	$3 + 2\epsilon + \epsilon^2$	b) Suppose ϵ is small and we replace the (2, 2) entry $3+2\epsilon+\epsilon^2$ in $A^* A$ by $3+2\epsilon$.

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lyche-numerical-linear-algebra-235-F03158	235	1.000	$3+2 \ \epsilon$	$3 + 2\epsilon$	b) Suppose ϵ is small and we replace the (2, 2) entry $3+2\epsilon+\epsilon^2$ in A^*A by $3+2\epsilon$.
lyche-numerical-linear-algebra-235-F03159	235	1.000	$\epsilon<\sqrt{u}, u$	$\epsilon < \sqrt{u}, u$	(This will be done in a computer if $\epsilon < \sqrt{u}$, u being the round-off unit). For
lyche-numerical-linear-algebra-235-F03160	235	1.000	$u=10^{-16}$	$u = 10^{-16}$	example, if $u = 10^{-16}$ then $\sqrt{u} = 10^{-8}$. Solve $A^*Ax = A^*b$ for x and compare
lyche-numerical-linear-algebra-235-F03161	235	1.000	$\sqrt{u}=10^{-8}$	$\sqrt{u} = 10^{-8}$	example, if $u = 10^{-16}$ then $\sqrt{u} = 10^{-8}$. Solve $A^*Ax = A^*b$ for x and compare
lyche-numerical-linear-algebra-235-F03162	235	1.000	$\boldsymbol{A}(\epsilon) \in \mathbb{R}^{n \times n}$	$A(\epsilon) \in \mathbb{R}^{n \times n}$	$A(\epsilon) \in \mathbb{R}^{n \times n}$ be bidiagonal with $a_{i,j} = 0$ for $i, j = 1, \dots, n$ and $j \neq i, i + 1$.
lyche-numerical-linear-algebra-235-F03163	235	1.000	$j \neq i, i+1$	$j \neq i, i + 1$	$A(\epsilon) \in \mathbb{R}^{n \times n}$ be bidiagonal with $a_{i,j} = 0$ for $i, j = 1, \dots, n$ and $j \neq i, i + 1$.
lyche-numerical-linear-algebra-235-F03164	235	1.000	$a_{k, k+1}=\epsilon \in \mathbb{R}$	$a_{k,k+1} = \epsilon \in \mathbb{R}$	Moreover, for some $1 \leq k \leq n - 1$ we have $a_{k,k+1} = \epsilon \in \mathbb{R}$. Show that
lyche-numerical-linear-algebra-235-F03165	235	1.000	$\sigma_i(\epsilon), i=1, \ldots, n$	$\sigma_i(\epsilon), i = 1, \dots, n$	where $\sigma_i(\epsilon), i = 1, \dots, n$ are the singular values of $A(\epsilon)$.
lyche-numerical-linear-algebra-235-F03166	235	1.000	$\boldsymbol{A}(\epsilon)$	$A(\epsilon)$	where $\sigma_i(\epsilon), i = 1, \dots, n$ are the singular values of $A(\epsilon)$.
lyche-numerical-linear-algebra-238-F03167	238	1.000	$\Omega:=(0,1)^2=\{(x, y): 0<x, y<1\}$	$\Omega := (0, 1)^2 = \{(x, y) : 0 < x, y < 1\}$	Let $\Omega := (0, 1)^2 = \{(x, y) : 0 < x, y < 1\}$ be the open unit square with boundary
lyche-numerical-linear-algebra-238-F03168	238	1.000	$\partial \Omega$	$\partial \Omega$	$\partial \Omega$. Consider the problem
lyche-numerical-linear-algebra-238-F03169	238	1.000	$u=$	$u =$	Here the function f is given and continuous on Ω , and we seek a function $u =$
lyche-numerical-linear-algebra-238-F03170	238	1.000	$u(x, y)$	$u(x, y)$	$u(x, y)$ such that (10.1) holds and which is zero on $\partial \Omega$.
lyche-numerical-linear-algebra-238-F03171	238	1.000	$v_{j, k} \approx u(j h, k h)$	$v_{j,k} \approx u(jh, kh)$	approximations $v_{j,k} \approx u(jh, kh)$ on a grid of points given by
lyche-numerical-linear-algebra-238-F03172	238	1.000	$\Omega_h:=\{(j h, k h): j, k=1, \ldots, m\}$	$\Omega_h := \{(jh, kh) : j, k = 1, \dots, m\}$	The points $\Omega_h := \{(jh, kh) : j, k = 1, \dots, m\}$ are called the interior points, while
lyche-numerical-linear-algebra-238-F03173	238	1.000	$\bar{\Omega}_h \backslash \Omega_h$	$\bar{\Omega}_h \backslash \Omega_h$	$\bar{\Omega}_h \backslash \Omega_h$ are the boundary points. The solution is zero at the boundary points. Using
lyche-numerical-linear-algebra-239-F03174	239	1.000	$f_{j, k}:=f(j h, k h)$	$f_{j,k} := f(jh, kh)$	where $f_{j,k} := f(jh, kh)$ and
lyche-numerical-linear-algebra-239-F03175	239	1.000	$n:=m^2$	$n := m^2$	Let us take a closer look at (10.2). It consists of $n := m^2$ linear equations. Since the
lyche-numerical-linear-algebra-239-F03176	239	1.000	$v_{j, k}$	$v_{j,k}$	values at the boundary points are known, the unknowns are the n numbers $v_{j,k}$ at
lyche-numerical-linear-algebra-239-F03177	239	1.000	$\boldsymbol{T}=\operatorname{tridiag}(-1,2,-1) \in \mathbb{R}^{m \times m}$	$T = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$	where $T = \operatorname{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ is the second derivative matrix given by
lyche-numerical-linear-algebra-239-F03178	239	0.913	j, k	j, k	Indeed, the (j, k) element in $TV + VT$ is given by
lyche-numerical-linear-algebra-239-F03179	239	0.913	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}$	$TV + VT$	Indeed, the (j, k) element in $TV + VT$ is given by
lyche-numerical-linear-algebra-239-F03180	239	1.000	$\boldsymbol{B} \in \mathbb{R}^{m \times n}$	$B \in \mathbb{R}^{m \times n}$	Definition 10.1 (vec Operation) For any $B \in \mathbb{R}^{m \times n}$ we define the vector
lyche-numerical-linear-algebra-240-F03181	240	1.000	$\boldsymbol{x}:=\operatorname{vec}(\boldsymbol{V}) \in \mathbb{R}^n$	$x := \operatorname{vec}(V) \in \mathbb{R}^n$	Let $x := \operatorname{vec}(V) \in \mathbb{R}^n$, where $n = m^2$. Note that forming x by stacking the
lyche-numerical-linear-algebra-240-F03182	240	1.000	$n=m^2$	$n = m^2$	Let $x := \operatorname{vec}(V) \in \mathbb{R}^n$, where $n = m^2$. Note that forming x by stacking the
lyche-numerical-linear-algebra-240-F03183	240	1.000	$x_i=v_{j, k}$	$x_i = v_{j,k}$	where $x_i = v_{j,k}$ and $b_i = h^2 f_{j,k}$. This must be modified close to the boundary. We
lyche-numerical-linear-algebra-240-F03184	240	1.000	$b_i=h^2 f_{j, k}$	$b_i = h^2 f_{j,k}$	where $x_i = v_{j,k}$ and $b_i = h^2 f_{j,k}$. This must be modified close to the boundary. We

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lyche-numerical-linear-algebra-F03185	240	1.000	$\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}),$ $\boldsymbol{b}=h^2\operatorname{vec}(\boldsymbol{F})$	$\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{b} = h^2 \operatorname{vec}(\boldsymbol{F})$	where $\boldsymbol{x} = \operatorname{vec}(\boldsymbol{V}), \boldsymbol{b} = h^2 \operatorname{vec}(\boldsymbol{F})$ and $\boldsymbol{A}$ is the Poisson matrix given by
lyche-numerical-linear-algebra-F03186	241	0.905	$\boldsymbol{T}_1 \in \mathbb{R}^{m \times m}$	$\boldsymbol{T}_1 \in \mathbb{R}^{m \times m}$	In Sect. 2.4 we encountered the 1-dimensional test matrix $\boldsymbol{T}_1 \in \mathbb{R}^{m \times m}$ defined for
lyche-numerical-linear-algebra-F03187	241	1.000	a, d	a, d	any real numbers a, d by
lyche-numerical-linear-algebra-F03188	241	1.000	$\boldsymbol{T}_2=\left[\boldsymbol{a}_{ij}\right] \in$	$\boldsymbol{T}_2 = [a_{ij}] \in$	The (2-dimensional) Poisson matrix is a special case of the matrix $\boldsymbol{T}_2 = [a_{ij}] \in$
lyche-numerical-linear-algebra-F03189	242	0.956	$\boldsymbol{T}_2$	$\boldsymbol{T}_2$	The partition into 3×3 sub matrices shows that $\boldsymbol{T}_2$ is block tridiagonal.
lyche-numerical-linear-algebra-F03190	242	1.000	p, q, r, s	p, q, r, s	Definition 10.2 (Kronecker Product) For any positive integers p, q, r, s we
lyche-numerical-linear-algebra-F03191	242	1.000	$\boldsymbol{A} \in \mathbb{R}^{p \times q}$	$\boldsymbol{A} \in \mathbb{R}^{p \times q}$	define the Kronecker product of two matrices $\boldsymbol{A} \in \mathbb{R}^{p \times q}$ and $\boldsymbol{B} \in \mathbb{R}^{r \times s}$ as a
lyche-numerical-linear-algebra-F03192	242	1.000	$\boldsymbol{B} \in \mathbb{R}^{r \times s}$	$\boldsymbol{B} \in \mathbb{R}^{r \times s}$	define the Kronecker product of two matrices $\boldsymbol{A} \in \mathbb{R}^{p \times q}$ and $\boldsymbol{B} \in \mathbb{R}^{r \times s}$ as a
lyche-numerical-linear-algebra-F03193	242	1.000	$\boldsymbol{C} \in \mathbb{R}^{pr \times qs}$	$\boldsymbol{C} \in \mathbb{R}^{pr \times qs}$	matrix $\boldsymbol{C} \in \mathbb{R}^{pr \times qs}$ given in block form as
lyche-numerical-linear-algebra-F03194	242	1.000	$\boldsymbol{C}=\boldsymbol{A} \otimes \boldsymbol{B}$	$\boldsymbol{C} = \boldsymbol{A} \otimes \boldsymbol{B}$	We denote the Kronecker product of $\boldsymbol{A}$ and $\boldsymbol{B}$ by $\boldsymbol{C} = \boldsymbol{A} \otimes \boldsymbol{B}$.
lyche-numerical-linear-algebra-F03195	242	1.000	$\boldsymbol{B} \otimes \boldsymbol{A}$	$\boldsymbol{B} \otimes \boldsymbol{A}$	which in our notation is given by $\boldsymbol{B} \otimes \boldsymbol{A}$.
lyche-numerical-linear-algebra-F03196	242	1.000	$\boldsymbol{u} \otimes \boldsymbol{v}=\left[\boldsymbol{u}^T \boldsymbol{v}_1, \ldots, \boldsymbol{u}^T \boldsymbol{v}_r\right]^T$	$\boldsymbol{u} \otimes \boldsymbol{v} = \left[\boldsymbol{u}^T v_1, \dots, \boldsymbol{u}^T v_r\right]^T$	The Kronecker product $\boldsymbol{u} \otimes \boldsymbol{v} = \left[\boldsymbol{u}^T v_1, \dots, \boldsymbol{u}^T v_r\right]^T$ of two column vectors
lyche-numerical-linear-algebra-F03197	242	1.000	$\boldsymbol{u} \in \mathbb{R}^p$	$\boldsymbol{u} \in \mathbb{R}^p$	$\boldsymbol{u} \in \mathbb{R}^p$ and $\boldsymbol{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra-F03198	242	1.000	$\boldsymbol{v} \in \mathbb{R}^r$	$\boldsymbol{v} \in \mathbb{R}^r$	$\boldsymbol{u} \in \mathbb{R}^p$ and $\boldsymbol{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra-F03199	242	1.000	$p \cdot r$	$p \cdot r$	$\boldsymbol{u} \in \mathbb{R}^p$ and $\boldsymbol{v} \in \mathbb{R}^r$ is a column vector of length $p \cdot r$.
lyche-numerical-linear-algebra-F03200	243	1.000	r, s, k	r, s, k	Definition 10.3 (Kronecker Sum) For positive integers r, s, k , let $\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$
lyche-numerical-linear-algebra-F03201	243	1.000	$\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$	$\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$	Definition 10.3 (Kronecker Sum) For positive integers r, s, k , let $\boldsymbol{A} \in \mathbb{R}^{r \times r}, \boldsymbol{B} \in$
lyche-numerical-linear-algebra-F03202	243	1.000	$\mathbb{R}^{s \times s}$	$\mathbb{R}^{s \times s}$	$\mathbb{R}^{s \times s}$, and $\boldsymbol{I}_k$ be the identity matrix of order k . The sum $\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$ is known
lyche-numerical-linear-algebra-F03203	243	1.000	$\boldsymbol{I}_k$	$\boldsymbol{I}_k$	$\mathbb{R}^{s \times s}$, and $\boldsymbol{I}_k$ be the identity matrix of order k . The sum $\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$ is known
lyche-numerical-linear-algebra-F03204	243	1.000	$\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$	$\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$	$\mathbb{R}^{s \times s}$, and $\boldsymbol{I}_k$ be the identity matrix of order k . The sum $\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}$ is known
lyche-numerical-linear-algebra-F03205	243	1.000	λ, μ	λ, μ	λ, μ and matrices $\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$ of dimensions such that the operations
lyche-numerical-linear-algebra-F03206	243	1.000	$\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$	$\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$	λ, μ and matrices $\boldsymbol{A}, \boldsymbol{A}_1, \boldsymbol{A}_2, \boldsymbol{B}, \boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{C}$ of dimensions such that the operations
lyche-numerical-linear-algebra-F03207	243	1.000	$\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$	$\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$	Note however that in general we have $\boldsymbol{A} \otimes \boldsymbol{B} \neq \boldsymbol{B} \otimes \boldsymbol{A}$, but it can be shown that
lyche-numerical-linear-algebra-F03208	243	1.000	$\boldsymbol{P}, \boldsymbol{Q}$	$\boldsymbol{P}, \boldsymbol{Q}$	there are permutation matrices $\boldsymbol{P}, \boldsymbol{Q}$ such that $\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$, see [9].
lyche-numerical-linear-algebra-F03209	243	1.000	$\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$	$\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$	there are permutation matrices $\boldsymbol{P}, \boldsymbol{Q}$ such that $\boldsymbol{B} \otimes \boldsymbol{A} = \boldsymbol{P}(\boldsymbol{A} \otimes \boldsymbol{B})\boldsymbol{Q}$, see [9].
lyche-numerical-linear-algebra-F03210	243	0.989	$\boldsymbol{B} \boldsymbol{D}$	$\boldsymbol{B} \boldsymbol{D}$	ces with dimensions so that the products $\boldsymbol{A} \boldsymbol{C}$ and $\boldsymbol{B} \boldsymbol{D}$ are well defined. Then the
lyche-numerical-linear-algebra-F03211	243	1.000	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$	product $\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{C} \otimes \boldsymbol{D})$ is defined and

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lyche-numerical-linear-algebra F03212	243	1.000	$\boldsymbol{B} \in \mathbb{R}^{r, t}$	$B \in \mathbb{R}^{r, t}$	Proof If $B \in \mathbb{R}^{r, t}$ and $D \in \mathbb{R}^{t, s}$ for some integers r, s, t , then
lyche-numerical-linear-algebra F03213	243	1.000	$\boldsymbol{D} \in \mathbb{R}^{t, s}$	$D \in \mathbb{R}^{t, s}$	Proof If $B \in \mathbb{R}^{r, t}$ and $D \in \mathbb{R}^{t, s}$ for some integers r, s, t , then
lyche-numerical-linear-algebra F03214	243	1.000	r, s, t	r, s, t	Proof If $B \in \mathbb{R}^{r, t}$ and $D \in \mathbb{R}^{t, s}$ for some integers r, s, t , then
lyche-numerical-linear-algebra F03215	244	1.000	$r, s \in \mathbb{N}$	$r, s \in \mathbb{N}$	Theorem 10.1 (Properties of Kronecker Products) Suppose for $r, s \in \mathbb{N}$ that
lyche-numerical-linear-algebra F03216	244	1.000	$\boldsymbol{A} \in \mathbb{R}^{r \times r}$	$A \in \mathbb{R}^{r \times r}$	$A \in \mathbb{R}^{r \times r}$ and $B \in \mathbb{R}^{s \times s}$ are square matrices with eigenpairs $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$
lyche-numerical-linear-algebra F03217	244	1.000	$\boldsymbol{B} \in \mathbb{R}^{s \times s}$	$B \in \mathbb{R}^{s \times s}$	$A \in \mathbb{R}^{r \times r}$ and $B \in \mathbb{R}^{s \times s}$ are square matrices with eigenpairs $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$
lyche-numerical-linear-algebra F03218	244	1.000	$(\lambda_i, \boldsymbol{u}_i)$ $i=1, \dots, r$	$(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$	$A \in \mathbb{R}^{r \times r}$ and $B \in \mathbb{R}^{s \times s}$ are square matrices with eigenpairs $(\lambda_i, \boldsymbol{u}_i)$ $i = 1, \dots, r$
lyche-numerical-linear-algebra F03219	244	1.000	$(\mu_j, \boldsymbol{v}_j)$ $j=1, \dots, s$	$(\mu_j, \boldsymbol{v}_j)$ $j = 1, \dots, s$	and $(\mu_j, \boldsymbol{v}_j)$ $j = 1, \dots, s$. Moreover, let $F, V \in \mathbb{R}^{r \times s}$. Then
lyche-numerical-linear-algebra F03220	244	1.000	$F, V \in \mathbb{R}^{r \times s}$	$F, V \in \mathbb{R}^{r \times s}$	and $(\mu_j, \boldsymbol{v}_j)$ $j = 1, \dots, s$. Moreover, let $F, V \in \mathbb{R}^{r \times s}$. Then
lyche-numerical-linear-algebra F03221	244	0.999	$(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T$	$(A \otimes B)^T = A^T \otimes B^T$	1. $(A \otimes B)^T = A^T \otimes B^T$, (this also holds for rectangular matrices).
lyche-numerical-linear-algebra F03222	244	0.999	$\boldsymbol{A} \otimes \boldsymbol{B}$	$A \otimes B$	2. If A and B are nonsingular then $A \otimes B$ is nonsingular. with $(A \otimes B)^{-1} =$
lyche-numerical-linear-algebra F03223	244	0.999	$(\boldsymbol{A} \otimes \boldsymbol{B})^{-1} =$	$(A \otimes B)^{-1} =$	2. If A and B are nonsingular then $A \otimes B$ is nonsingular. with $(A \otimes B)^{-1} =$
lyche-numerical-linear-algebra F03224	244	1.000	$\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}$	$A^{-1} \otimes B^{-1}$	$A^{-1} \otimes B^{-1}$.
lyche-numerical-linear-algebra F03225	244	1.000	$(\boldsymbol{A} \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	$(A \otimes B)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	4. $(A \otimes B)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$,
lyche-numerical-linear-algebra F03226	244	0.999	$(\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B})(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\lambda_i + \mu_j)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	$(A \otimes I_s + I_r \otimes B)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\lambda_i + \mu_j)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$	5. $(A \otimes I_s + I_r \otimes B)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\lambda_i + \mu_j)(\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, $i = 1, \dots, r$, $j = 1, \dots, s$,
lyche-numerical-linear-algebra F03227	244	1.000	$\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	$A \otimes I + I \otimes B$	$A \otimes I + I \otimes B$ is positive definite.
lyche-numerical-linear-algebra F03228	244	0.689	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	$AVB^T = F \Leftrightarrow (A \otimes B) \text{vec}(V) = \text{vec}(F)$	7. $AVB^T = F \Leftrightarrow (A \otimes B) \text{vec}(V) = \text{vec}(F)$,
lyche-numerical-linear-algebra F03229	244	1.000	$\boldsymbol{A} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{B}^T = \boldsymbol{F} \Leftrightarrow (\boldsymbol{A} \otimes \boldsymbol{I}_s + \boldsymbol{I}_r \otimes \boldsymbol{B}) \text{vec}(\boldsymbol{V}) = \text{vec}(\boldsymbol{F})$	$AV + VB^T = F \Leftrightarrow (A \otimes I_s + I_r \otimes B) \text{vec}(V) = \text{vec}(F)$	8. $AV + VB^T = F \Leftrightarrow (A \otimes I_s + I_r \otimes B) \text{vec}(V) = \text{vec}(F)$.
lyche-numerical-linear-algebra F03230	244	1.000	$\boldsymbol{A} = \boldsymbol{T} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}$	$A = T \otimes I + I \otimes T$	Poisson matrix $A = T \otimes I + I \otimes T$ is also positive definite.
lyche-numerical-linear-algebra F03231	244	1.000	$\boldsymbol{A} \boldsymbol{B}^T = \boldsymbol{F}$	$AVB^T = F$	4. The system $AVB^T = F$ in part 7 can be solved by first finding W from $AW =$
lyche-numerical-linear-algebra F03232	244	1.000	$\boldsymbol{A} \boldsymbol{W} =$	$AW =$	4. The system $AVB^T = F$ in part 7 can be solved by first finding W from $AW =$

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lyche-numerical-linear-algebra- F03233	244	■ 1.000	$\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$	$\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$	$\boldsymbol{F}$, and then finding $\boldsymbol{V}$ from $\boldsymbol{B} \boldsymbol{V}^T = \boldsymbol{W}^T$. This is preferable to solving the much
lyche-numerical-linear-algebra- F03234	244	■ 0.544	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{A} \boldsymbol{V} \boldsymbol{B}^T)$	$(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$	larger linear system $(\boldsymbol{A} \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) = \operatorname{vec}(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03235	244	■ 0.965	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$	$\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$	5. A fast way to solve the 2D Poisson problem in the form $\boldsymbol{T} \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T} = \boldsymbol{F}$ will be
lyche-numerical-linear-algebra- F03236	245	■ 0.998	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) = \boldsymbol{I}$	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) =$	2. By the mixed product rule $(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{A}^{-1} \otimes \boldsymbol{B}^{-1}) = (\boldsymbol{A} \boldsymbol{A}^{-1}) \otimes (\boldsymbol{B} \boldsymbol{B}^{-1}) =$
lyche-numerical-linear-algebra- F03237	245	■ 0.997	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$. Thus $(\boldsymbol{A} \otimes \boldsymbol{B})$ is nonsingular with the indicated inverse.
lyche-numerical-linear-algebra- F03238	245	■ 0.997	$(\boldsymbol{A} \otimes \boldsymbol{B})$	$(\boldsymbol{A} \otimes \boldsymbol{B})$	$\boldsymbol{I}_r \otimes \boldsymbol{I}_s = \boldsymbol{I}_{rs}$. Thus $(\boldsymbol{A} \otimes \boldsymbol{B})$ is nonsingular with the indicated inverse.
lyche-numerical-linear-algebra- F03239	245	■ 1.000	$1, (\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$	$1, (\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra- F03240	245	■ 1.000	$\boldsymbol{A} \otimes \boldsymbol{I}$	$\boldsymbol{A} \otimes \boldsymbol{I}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra- F03241	245	■ 1.000	$\boldsymbol{I} \otimes \boldsymbol{B}$	$\boldsymbol{I} \otimes \boldsymbol{B}$	3. By 1, $(\boldsymbol{A} \otimes \boldsymbol{B})^T = \boldsymbol{A}^T \otimes \boldsymbol{B}^T = \boldsymbol{A} \otimes \boldsymbol{B}$. Moreover, since then $\boldsymbol{A} \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}$
lyche-numerical-linear-algebra- F03242	245	■ 1.000	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\boldsymbol{A} \boldsymbol{u}_i) \otimes (\boldsymbol{B} \boldsymbol{v}_j) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	$(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\boldsymbol{A} \boldsymbol{u}_i) \otimes (\boldsymbol{B} \boldsymbol{v}_j) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	4. $(\boldsymbol{A} \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = (\boldsymbol{A} \boldsymbol{u}_i) \otimes (\boldsymbol{B} \boldsymbol{v}_j) = (\lambda_i \boldsymbol{u}_i) \otimes (\mu_j \boldsymbol{v}_j) = (\lambda_i \mu_j) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$, for
lyche-numerical-linear-algebra- F03243	245	■ 0.475	$(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$	5. $(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j).$
lyche-numerical-linear-algebra- F03244	245	■ 0.475	$(\boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	$(\boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j)$	5. $(\boldsymbol{A} \otimes \boldsymbol{I}_s) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \lambda_i (\boldsymbol{u}_i \otimes \boldsymbol{v}_j),$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) (\boldsymbol{u}_i \otimes \boldsymbol{v}_j) = \mu_j (\boldsymbol{u}_i \otimes \boldsymbol{v}_j).$
lyche-numerical-linear-algebra- F03245	245	■ 0.892	$1, \boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	$1, \boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$	6. By 1, $\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$ is symmetric. Moreover, the eigenvalues $\lambda_i + \mu_j$ are positive
lyche-numerical-linear-algebra- F03246	245	■ 0.892	$\lambda_i + \mu_j$	$\lambda_i + \mu_j$	6. By 1, $\boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$ is symmetric. Moreover, the eigenvalues $\lambda_i + \mu_j$ are positive
lyche-numerical-linear-algebra- F03247	245	■ 0.500	μ_j	μ_j	since for all i, j , both λ_i and μ_j are nonnegative and one of them is positive. It
lyche-numerical-linear-algebra- F03248	245	■ 1.000	$\boldsymbol{V}, \boldsymbol{F}$	$\boldsymbol{V}, \boldsymbol{F}$	7. We partition $\boldsymbol{V}, \boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra- F03249	245	■ 1.000	$\boldsymbol{B}^T$	$\boldsymbol{B}^T$	7. We partition $\boldsymbol{V}, \boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra- F03250	245	■ 1.000	$\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$	$\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$	7. We partition $\boldsymbol{V}, \boldsymbol{F}$, and $\boldsymbol{B}^T$ by columns as $\boldsymbol{V} = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_s], \boldsymbol{F} = [\boldsymbol{f}_1, \dots, \boldsymbol{f}_s]$
lyche-numerical-linear-algebra- F03251	245	■ 1.000	$\boldsymbol{B}^T = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_s]$	$\boldsymbol{B}^T = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_s]$	and $\boldsymbol{B}^T = [\boldsymbol{b}_1, \dots, \boldsymbol{b}_s]$. Then we have

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lyche-numerical-linear-algebra- F03252	245	0.890	$\left(\boldsymbol{A} \otimes \boldsymbol{I}_s\right) \operatorname{vec}(\boldsymbol{V})=\boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$	8. From the proof of 7. we have $(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$
lyche-numerical-linear-algebra- F03253	245	0.890	$\left(\boldsymbol{I}_r \otimes \boldsymbol{B}\right) \operatorname{vec}(\boldsymbol{V})=$	$(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$	8. From the proof of 7. we have $(\boldsymbol{A} \otimes \boldsymbol{I}_s) \operatorname{vec}(\boldsymbol{V}) = \boldsymbol{A} \boldsymbol{V} \boldsymbol{I}_s^T$ and $(\boldsymbol{I}_r \otimes \boldsymbol{B}) \operatorname{vec}(\boldsymbol{V}) =$
lyche-numerical-linear-algebra- F03254	245	1.000	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$	$\boldsymbol{I}_r \boldsymbol{V} \boldsymbol{B}^T$. But then
lyche-numerical-linear-algebra- F03255	246	0.527	$\left(\lambda_j, \boldsymbol{s}_j\right)$	$(\lambda_j, \boldsymbol{s}_j)$	where $(\lambda_j, \boldsymbol{s}_j)$ are the eigenpairs of $\boldsymbol{T}_1$ given by (10.15) and (10.16). The
lyche-numerical-linear-algebra- F03256	246	0.975	$\boldsymbol{s}_j \otimes \boldsymbol{s}_k$	$\boldsymbol{s}_j \otimes \boldsymbol{s}_k$	eigenvectors $\boldsymbol{s}_j \otimes \boldsymbol{s}_k$ are orthogonal
lyche-numerical-linear-algebra- F03257	246	1.000	$d > 0$	$d > 0$	and $\boldsymbol{T}_2$ is positive definite if $d > 0$ and $d \geq 2 a $.
lyche-numerical-linear-algebra- F03258	246	1.000	$d \geq 2 a $	$d \geq 2 a $	and $\boldsymbol{T}_2$ is positive definite if $d > 0$ and $d \geq 2 a $.
lyche-numerical-linear-algebra- F03259	246	1.000	$j, k, p, q=1, \ldots, m$	$j, k, p, q = 1, \dots, m$	rule, the mixed product rule and (2.32) we find for $j, k, p, q = 1, \dots, m$
lyche-numerical-linear-algebra- F03260	246	1.000	λ_j	λ_j	and (10.19) follows. Since $\boldsymbol{T}_2$ is symmetric, $\boldsymbol{T}_2$ is positive definite if the λ_j given
lyche-numerical-linear-algebra- F03261	246	0.990	$\mathbb{R}^{m^2 \times m^2}$	$\mathbb{R}^{m^2 \times m^2}$	given by (10.8) is given by
lyche-numerical-linear-algebra- F03262	246	1.000	$d=2, a=-1$	$d = 2, a = -1$	Proof Recall that by (10.15) with $d = 2, a = -1$, and (10.18), the eigenvalues $\lambda_{j,k}$
lyche-numerical-linear-algebra- F03263	246	1.000	$\lambda_{j, k}$	$\lambda_{j,k}$	Proof Recall that by (10.15) with $d = 2, a = -1$, and (10.18), the eigenvalues $\lambda_{j,k}$
lyche-numerical-linear-algebra- F03264	247	0.967	$\ \boldsymbol{A}\ _2\left\ \boldsymbol{A}^{-1}\right\ _2=\frac{\lambda_{\max }}{\lambda_{\min }}$	$\ \boldsymbol{A}\ _2\left\ \boldsymbol{A}^{-1}\right\ _2 = \frac{\lambda_{\max}}{\lambda_{\min}}$	$\ \boldsymbol{A}\ _2\ \boldsymbol{A}^{-1}\ _2 = \frac{\lambda_{\max}}{\lambda_{\min}}$ and (10.20) follows.
lyche-numerical-linear-algebra- F03265	247	1.000	$\lambda=2 a+2 d$	$\lambda = 2a + 2d$	Show that $\lambda = 2a + 2d$ is an eigenvalue corresponding to the eigenvector $\boldsymbol{x} =$
lyche-numerical-linear-algebra- F03266	247	1.000	$[1,1,1,1]^T$	$[1, 1, 1, 1]^T$	$[1, 1, 1, 1]^T$. Verify that apart from a scaling of the eigenvector this agrees with
lyche-numerical-linear-algebra- F03267	247	1.000	$j=k=1$	$j = k = 1$	(10.15) and (10.16) for $j = k = 1$ and $m = 2$.
lyche-numerical-linear-algebra- F03268	247	1.000	$-\Delta u=f, u=0$	$-\Delta u = f, u = 0$	9 point difference approximation to the Poisson problem $-\Delta u = f, u = 0$ on the
lyche-numerical-linear-algebra- F03269	248	1.000	$-\left(\square_h v\right)_{j, k}=(\mu f)_{j, k}$	$-(\square_h v)_{j,k} = (\mu f)_{j,k}$	(a) $-(\square_h v)_{j,k} = (\mu f)_{j,k}$
lyche-numerical-linear-algebra- F03270	248	1.000	$j, k=1, \ldots, m$	$j, k = 1, \dots, m$	$j, k = 1, \dots, m$
lyche-numerical-linear-algebra- F03271	248	1.000	$0=v_{0, k}=v_{m+1, k}=v_{j, 0}=v_{j, m+1}, j, k=0,1, \ldots, m+1$	$0 = v_{0,k} = v_{m+1,k} = v_{j,0} = v_{j,m+1}, j, k = 0, 1, \dots, m + 1$	$0 = v_{0,k} = v_{m+1,k} = v_{j,0} = v_{j,m+1}, j, k = 0, 1, \dots, m + 1$,
lyche-numerical-linear-algebra- F03272	248	1.000	$-\left(\square_h v\right)_{j, k}=\left[20 v_{j, k}-4 v_{j-1, k}-4 v_{j, k-1}-4 v_{j+1, k}-4 v_{j, k+1}\right.$	$-(\square_h v)_{j,k} = [20v_{j,k} - 4v_{j-1,k} - 4v_{j,k-1} - 4v_{j+1,k} - 4v_{j,k+1}$	$-(\square_h v)_{j,k} = [20v_{j,k} - 4v_{j-1,k} - 4v_{j,k-1} - 4v_{j+1,k} - 4v_{j,k+1}$
lyche-numerical-linear-algebra- F03273	248	1.000	$\left.-v_{j-1, k-1}-v_{j+1, k-1}-v_{j-1, k+1}-v_{j+1, k+1}\right] /\left(6 h^2\right)$	$-v_{j-1,k-1} - v_{j+1,k-1} - v_{j-1,k+1} - v_{j+1,k+1}] / (6h^2)$	$-v_{j-1,k-1} - v_{j+1,k-1} - v_{j-1,k+1} - v_{j+1,k+1}]/(6h^2)$.
lyche-numerical-linear-algebra- F03274	248	0.877	$(\mu f)_{j, k}=\left[8 f_{j, k}+f_{j-1, k}+f_{j, k-1}+f_{j+1, k}+f_{j, k+1}\right] / 12$	$(\mu f)_{j,k} = [8f_{j,k} + f_{j-1,k} + f_{j,k-1} + f_{j+1,k} + f_{j,k+1}] / 12$	$(\mu f)_{j,k} = [8f_{j,k} + f_{j-1,k} + f_{j,k-1} + f_{j+1,k} + f_{j,k+1}] / 12$
lyche-numerical-linear-algebra- F03275	248	1.000	$j, k=1,2$	$j, k = 1, 2$	b) Find $v_{j,k}$ for $j, k = 1, 2$, if $f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y)$ and $m = 2$.
lyche-numerical-linear-algebra- F03276	248	1.000	$f(x, y)=2 \pi^2 \sin (\pi x) \sin (\pi y)$	$f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y)$	b) Find $v_{j,k}$ for $j, k = 1, 2$, if $f(x, y) = 2\pi^2 \sin(\pi x) \sin(\pi y)$ and $m = 2$.

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lyche-numerical-linear-algebra- F03277	248	■ 1.000	$v_{\{j, k\}=5 \pi^2} / 66$	$v_{j,k} = 5\pi^2/66$	Answer: $v_{i,k} = 5\pi^2/66$.
lyche-numerical-linear-algebra- F03278	248	■ 1.000	$O\left(h^4\right)$	$O\left(h^4\right)$	It can be shown that (10.21) defines an $O(h^4)$ approximation to (10.1).
lyche-numerical-linear-algebra- F03279	248	■ 0.997	$\mu \boldsymbol{F}$	$\mu \boldsymbol{F}$	Here $\mu \boldsymbol{F}$ has elements $(\mu f)_{j,k}$ given by (10.21d) and $\boldsymbol{T} = \text{tridiag}(-1, 2, -1)$.
lyche-numerical-linear-algebra- F03280	248	■ 0.997	$(\mu f)_{\{j, k\}}$	$(\mu f)_{j,k}$	Here $\mu \boldsymbol{F}$ has elements $(\mu f)_{j,k}$ given by (10.21d) and $\boldsymbol{T} = \text{tridiag}(-1, 2, -1)$.
lyche-numerical-linear-algebra- F03281	248	■ 1.000	$\boldsymbol{A}=\boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\text { vec }(\boldsymbol{V})$	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\text { vec }(\boldsymbol{V})$	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\text { vec }(\boldsymbol{V})$, and $\boldsymbol{b}=h^2 \text { vec }(\mu \boldsymbol{F})$.
lyche-numerical-linear-algebra- F03282	248	■ 1.000	$\boldsymbol{b}=h^2 \text { vec }(\mu \boldsymbol{F})$	$\boldsymbol{b}=h^2 \text { vec }(\mu \boldsymbol{F})$	$\boldsymbol{A}=\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}-\frac{1}{6} \boldsymbol{T} \otimes \boldsymbol{T}, \boldsymbol{x}=\text { vec }(\boldsymbol{V})$, and $\boldsymbol{b}=h^2 \text { vec }(\mu \boldsymbol{F})$.
lyche-numerical-linear-algebra- F03283	248	■ 1.000	$\Delta u=0$	$\Delta u=0$	Here Ω is the open unit square. The condition $\Delta u=0$ is called the <i>Navier boundary</i>
lyche-numerical-linear-algebra- F03284	248	■ 0.994	$\Delta^2 u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}$	$\Delta^2 u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}$	condition. Moreover, $\Delta^2 u=u_{x x x x}+2 u_{x x y y}+u_{y y y y}$.
lyche-numerical-linear-algebra- F03285	248	■ 1.000	$v=-\Delta u$	$v=-\Delta u$	a) Let $v=-\Delta u$. Show that (10.23) can be written as a system
lyche-numerical-linear-algebra- F03286	248	■ 1.000	$\boldsymbol{T}=\text { tridiag }(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+$	$\boldsymbol{T}=\text { tridiag }(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+$	b) Discretizing, using (10.4), with $\boldsymbol{T}=\text { tridiag }(-1,2,-1) \in \mathbb{R}^{m \times m}, h=1 /(m+$
lyche-numerical-linear-algebra- F03287	248	■ 0.584	$\boldsymbol{F}=(f(j h, k h))_{j, k=1}^m$	$\boldsymbol{F}=(f(j h, k h))_{j, k=1}^m$	1), and $\boldsymbol{F}=(f(j h, k h))_{j, k=1}^m$ we get two matrix equations
lyche-numerical-linear-algebra- F03288	249	■ 1.000	$\boldsymbol{A}=(\boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T})^2$	$\boldsymbol{A}=(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T})^2$	and hence $\boldsymbol{A}=(\boldsymbol{T} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T})^2$ is the matrix for the standard form of the
lyche-numerical-linear-algebra- F03289	249	■ 0.999	$\boldsymbol{A}=\boldsymbol{A} \otimes \boldsymbol{I}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\text { vec }(\boldsymbol{U})$	$\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\text { vec }(\boldsymbol{U})$	where $\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\text { vec }(\boldsymbol{U})$, and $\boldsymbol{b}=h^4 \text { vec }(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03290	249	■ 0.999	$\boldsymbol{b}=h^4 \text { vec }(\boldsymbol{F})$	$\boldsymbol{b}=h^4 \text { vec }(\boldsymbol{F})$	where $\boldsymbol{A}=\boldsymbol{T}^2 \otimes \boldsymbol{I}+2 \boldsymbol{T} \otimes \boldsymbol{T}+\boldsymbol{I} \otimes \boldsymbol{T}^2 \in \mathbb{R}^{n \times n}, \boldsymbol{x}=\text { vec }(\boldsymbol{U})$, and $\boldsymbol{b}=h^4 \text { vec }(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03291	249	■ 0.831	$\mathbf{b}$	$\mathbf{b}$	representations for the matrix $\boldsymbol{A}$, the product one in b) and the one in c) . Which
lyche-numerical-linear-algebra- F03292	249	■ 0.831	$\mathbf{c}$	$\mathbf{c}$	representations for the matrix $\boldsymbol{A}$, the product one in b) and the one in c) . Which
lyche-numerical-linear-algebra- F03293	249	■ 1.000	$\boldsymbol{T}_1:=\text { tridiagonal }(a, d, a)$	$\boldsymbol{T}_1:=\text { tridiagonal }(a, d, a)$	10.5.2 What are the eigenpairs of $\boldsymbol{T}_1:=\text { tridiagonal }(a, d, a)$?
lyche-numerical-linear-algebra- F03294	249	■ 1.000	$(\boldsymbol{A} \otimes$	$(\boldsymbol{A} \otimes$	• give an economical general way to solve the linear system $(\boldsymbol{A} \otimes$
lyche-numerical-linear-algebra- F03295	249	■ 1.000	$\boldsymbol{B}) \text { vec }(\boldsymbol{V})=\text { vec }(\boldsymbol{F})$	$\boldsymbol{B}) \text { vec }(\boldsymbol{V})=\text { vec }(\boldsymbol{F})$	$\boldsymbol{B}) \text { vec }(\boldsymbol{V})=\text { vec }(\boldsymbol{F})$?
lyche-numerical-linear-algebra- F03296	249	■ 0.999	$(\boldsymbol{A} \otimes \boldsymbol{I}_s+\boldsymbol{I}_r \otimes \boldsymbol{B}) \text { vec }(\boldsymbol{V})=\text { vec }(\boldsymbol{F})$	$(\boldsymbol{A} \otimes \boldsymbol{I}_s+\boldsymbol{I}_r \otimes \boldsymbol{B}) \text { vec }(\boldsymbol{V})=\text { vec }(\boldsymbol{F})$	• Same for $(\boldsymbol{A} \otimes \boldsymbol{I}_s+\boldsymbol{I}_r \otimes \boldsymbol{B}) \text { vec }(\boldsymbol{V})=\text { vec }(\boldsymbol{F})$.
lyche-numerical-linear-algebra- F03297	250	■ 1.000	$n=9$	$n=9$	matrix (10.8). Thus, for $n=9$
lyche-numerical-linear-algebra- F03298	251	■ 1.000	$d=\sqrt{n}$	$d=\sqrt{n}$	lower triangular, to solve $\boldsymbol{A} \boldsymbol{x}=\boldsymbol{b}$. Since $\boldsymbol{A}$ and $\boldsymbol{L}$ has the same bandwidth $d=\sqrt{n}$
lyche-numerical-linear-algebra- F03299	251	■ 1.000	$O\left(n d^2\right)=O\left(n^2\right)$	$O\left(n d^2\right)=O\left(n^2\right)$	the complexity of this factorization is $O\left(n d^2\right)=O\left(n^2\right)$, cf. Algorithm 4.2. We need

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lyche-numerical-linear-algebra- F03300	251	■ 0.929	(n=100)	$(n = 100)$	Fig. 11.1 Fill-inn in the Cholesky factor of the Poisson matrix ($n = 100$)
lyche-numerical-linear-algebra- F03301	252	■ 1.000	$\boldsymbol{D}_{\{1\}}, \ldots, \boldsymbol{D}_{\{m\}}$	$\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$	Here $\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1,$
lyche-numerical-linear-algebra- F03302	252	■ 1.000	$\boldsymbol{U}_{\{1\}}, \ldots, \boldsymbol{U}_{\{m\}}$	$\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$	Here $\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1,$
lyche-numerical-linear-algebra- F03303	252	■ 1.000	$\boldsymbol{A}_{\{1\}}, \ldots, \boldsymbol{A}_{\{m-1\}},$ $\boldsymbol{L}_{\{1\}}$	$\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1$	Here $\boldsymbol{D}_1, \dots, \boldsymbol{D}_m$ and $\boldsymbol{U}_1, \dots, \boldsymbol{U}_m$ are square matrices while $\boldsymbol{A}_1, \dots, \boldsymbol{A}_{m-1}, \boldsymbol{L}_1,$
lyche-numerical-linear-algebra- F03304	252	■ 1.000	$\ldots, \boldsymbol{L}_{\{m-1\}}$	$\dots, \boldsymbol{L}_{m-1}$	$\dots, \boldsymbol{L}_{m-1}$ and $\boldsymbol{C}_1, \dots, \boldsymbol{C}_{m-1}$ can be rectangular.
lyche-numerical-linear-algebra- F03305	252	■ 1.000	$\boldsymbol{C}_{\{1\}}, \ldots, \boldsymbol{C}_{\{m-1\}}$	$\boldsymbol{C}_1, \dots, \boldsymbol{C}_{m-1}$	$\dots, \boldsymbol{L}_{m-1}$ and $\boldsymbol{C}_1, \dots, \boldsymbol{C}_{m-1}$ can be rectangular.
lyche-numerical-linear-algebra- F03306	252	■ 0.995	$\boldsymbol{b}^{\mathrm{T}} =$	$\boldsymbol{b}^T =$	To solve the system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ we partition $\boldsymbol{b}$ conformally with $\boldsymbol{A}$ in the form $\boldsymbol{b}^T =$
lyche-numerical-linear-algebra- F03307	252	■ 1.000	$\left[\boldsymbol{b}_{\{1\}}^{\mathrm{T}}, \ldots, \boldsymbol{b}_{\{m\}}^{\mathrm{T}}\right]$	$[\boldsymbol{b}_1^T, \dots, \boldsymbol{b}_m^T]$	$[\boldsymbol{b}_1^T, \dots, \boldsymbol{b}_m^T]$. The formulas for solving $\boldsymbol{L}\boldsymbol{y} = \boldsymbol{b}$ and $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{y}$ are as follows:
lyche-numerical-linear-algebra- F03308	252	■ 1.000	$\boldsymbol{x}^{\mathrm{T}} = \left[\boldsymbol{x}_{\{1\}}^{\mathrm{T}}, \ldots, \boldsymbol{x}_{\{m\}}^{\mathrm{T}}\right]$	$\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \dots, \boldsymbol{x}_m^T]$	The solution is then $\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \dots, \boldsymbol{x}_m^T]$. To find $\boldsymbol{L}_k$ in (11.2) we solve the linear
lyche-numerical-linear-algebra- F03309	252	■ 1.000	$\boldsymbol{L}_{\{k\}}$	$\boldsymbol{L}_k$	The solution is then $\boldsymbol{x}^T = [\boldsymbol{x}_1^T, \dots, \boldsymbol{x}_m^T]$. To find $\boldsymbol{L}_k$ in (11.2) we solve the linear
lyche-numerical-linear-algebra- F03310	252	■ 1.000	$\boldsymbol{L}_{\{k\}} \boldsymbol{U}_{\{k\}} = \boldsymbol{A}_{\{k\}}$	$\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$	systems $\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$. Similarly we need to solve a linear system to find $\boldsymbol{x}_k$ in (11.3).
lyche-numerical-linear-algebra- F03311	252	■ 1.000	$\boldsymbol{x}_{\{k\}}$	$\boldsymbol{x}_k$	systems $\boldsymbol{L}_k \boldsymbol{U}_k = \boldsymbol{A}_k$. Similarly we need to solve a linear system to find $\boldsymbol{x}_k$ in (11.3).
lyche-numerical-linear-algebra- F03312	252	■ 1.000	$O(n^{\frac{3}{2}})$	$O\left(n^{3/2}\right)$	The algorithm we now derive will only require $O(n^{3/2})$ arithmetic operations and
lyche-numerical-linear-algebra- F03313	252	■ 1.000	$O(n \log n)$	$O(n \log n)$	number of arithmetic operations can be reduced further to $O(n \log n)$.
lyche-numerical-linear-algebra- F03314	252	■ 1.000	$\boldsymbol{V} =$	$\boldsymbol{V} =$	where $\boldsymbol{T} = \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{m \times m}$ is the second derivative matrix, $\boldsymbol{V} =$
lyche-numerical-linear-algebra- F03315	252	■ 1.000	$\left(v_{\{j, k\}}\right) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m \times m}$ are the unknowns, and $\boldsymbol{F} = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$ contains
lyche-numerical-linear-algebra- F03316	252	■ 1.000	$\boldsymbol{F} = \left(f_{\{j, k\}}\right) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$\boldsymbol{F} = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$(v_{j,k}) \in \mathbb{R}^{m \times m}$ are the unknowns, and $\boldsymbol{F} = (f_{j,k}) = (f(jh, kh)) \in \mathbb{R}^{m \times m}$ contains
lyche-numerical-linear-algebra- F03317	253	■ 0.999	$\boldsymbol{X} \in \mathbb{R}^{m \times m}$	$\boldsymbol{X} \in \mathbb{R}^{m \times m}$	Define $\boldsymbol{X} \in \mathbb{R}^{m \times m}$ by $\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$, where $\boldsymbol{V}$ is the solution of $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F}$.
lyche-numerical-linear-algebra- F03318	253	■ 0.999	$\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$	$\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$	Define $\boldsymbol{X} \in \mathbb{R}^{m \times m}$ by $\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$, where $\boldsymbol{V}$ is the solution of $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F}$.
lyche-numerical-linear-algebra- F03319	253	■ 0.999	$\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F}$	$\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F}$	Define $\boldsymbol{X} \in \mathbb{R}^{m \times m}$ by $\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$, where $\boldsymbol{V}$ is the solution of $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = \boldsymbol{h}^2 \boldsymbol{F}$.
lyche-numerical-linear-algebra- F03320	253	■ 1.000	$\boldsymbol{D}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{D} = 4\boldsymbol{h}^4 \boldsymbol{G}$	$\boldsymbol{D}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{D} = 4\boldsymbol{h}^4 \boldsymbol{G}$	Since $\boldsymbol{D}$ is diagonal, the equation $\boldsymbol{D}\boldsymbol{X} + \boldsymbol{X}\boldsymbol{D} = 4\boldsymbol{h}^4 \boldsymbol{G}$, is easy to solve. For the j, k
lyche-numerical-linear-algebra- F03321	254	■ 1.000	$\boldsymbol{G} = \boldsymbol{S}\boldsymbol{F}\boldsymbol{S}$	$\boldsymbol{G} = \boldsymbol{S}\boldsymbol{F}\boldsymbol{S}$	1. $\boldsymbol{G} = \boldsymbol{S}\boldsymbol{F}\boldsymbol{S}$,
lyche-numerical-linear-algebra- F03322	254	■ 1.000	$x_{\{j, k\}} = \boldsymbol{h}^4 g_{\{j, k\}} / (\sigma_j + \sigma_k), \quad j, k = 1, \dots, m$	$x_{j,k} = \boldsymbol{h}^4 g_{j,k} / (\sigma_j + \sigma_k), \quad j, k = 1, \dots, m$	2. $x_{j,k} = \boldsymbol{h}^4 g_{j,k} / (\sigma_j + \sigma_k), \quad j, k = 1, \dots, m$,
lyche-numerical-linear-algebra- F03323	254	■ 1.000	$\boldsymbol{X}, \boldsymbol{S}$	$\boldsymbol{X}, \boldsymbol{S}$	We can compute $\boldsymbol{X}, \boldsymbol{S}$ and the σ 's without using loops. Using outer products,
lyche-numerical-linear-algebra- F03324	254	■ 0.999	$-\Delta u = f$	$-\Delta u = f$	$-\Delta u = f$ on $\Omega = (0, 1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-algebra- F03325	254	■ 0.999	$\Omega = (0, 1)^2$	$\Omega = (0, 1)^2$	$-\Delta u = f$ on $\Omega = (0, 1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$
lyche-numerical-linear-algebra- F03326	254	■ 0.999	$m \in$	$m \in$	$-\Delta u = f$ on $\Omega = (0, 1)^2$ and $u = 0$ on $\partial\Omega$ using the 5-point scheme, i.e., let $m \in$

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lyche-numerical-linear-algebra-254-4 F03327	254	■ 1.000	$\mathbb{N}, h=1/(m+1)$	$\mathbb{N}, h = 1/(m+1)$	$\mathbb{N}, h = 1/(m+1)$, and $\mathbf{F} = (f(jh, kh)) \in \mathbb{R}^{m \times m}$. We compute $\mathbf{V} \in \mathbb{R}^{(m+2) \times (m+2)}$
lyche-numerical-linear-algebra-254-4 F03328	254	■ 1.000	$\mathbf{F} = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$\mathbf{F} = (f(jh, kh)) \in \mathbb{R}^{m \times m}$	$\mathbb{N}, h = 1/(m+1)$, and $\mathbf{F} = (f(jh, kh)) \in \mathbb{R}^{m \times m}$. We compute $\mathbf{V} \in \mathbb{R}^{(m+2) \times (m+2)}$
lyche-numerical-linear-algebra-254-4 F03329	254	■ 1.000	$\mathbf{V} \in \mathbb{R}^{(m+2) \times (m+2)}$	$\mathbf{V} \in \mathbb{R}^{(m+2) \times (m+2)}$	$\mathbb{N}, h = 1/(m+1)$, and $\mathbf{F} = (f(jh, kh)) \in \mathbb{R}^{m \times m}$. We compute $\mathbf{V} \in \mathbb{R}^{(m+2) \times (m+2)}$
lyche-numerical-linear-algebra-254-4 F03330	254	■ 1.000	$O(m^3)$	$O(m^3)$	requires $O(m^3)$ arithmetic operations, the complexity of this algorithm is for large
lyche-numerical-linear-algebra-254-4 F03331	254	■ 1.000	$O(4 \times 2m^3)$	$O(4 \times 2m^3) =$	m determined by 4 m -by- m matrix multiplications and is given by $O(4 \times 2m^3) =$
lyche-numerical-linear-algebra-254-4 F03332	254	■ 0.943	$O(8n^{3/2})$	$O(8n^{3/2})$	$O(8n^{3/2})$. ¹ The method is very fast and will be used as a preconditioner for a more
lyche-numerical-linear-algebra-254-4 F03333	254	■ 1.000	10^6	10^6	find the 10^6 unknowns $v_{j,k}$ on a 1000×1000 grid.
lyche-numerical-linear-algebra-254-4 F03334	254	■ 1.000	1000×1000	1000×1000	find the 10^6 unknowns $v_{j,k}$ on a 1000×1000 grid.
lyche-numerical-linear-algebra-254-4 F03335	254	■ 1.000	$O(4n^{3/2})$	$O(4n^{3/2})$	to $O(4n^{3/2})$. This is detailed in Problem 11.4.
lyche-numerical-linear-algebra-255-4 F03336	255	■ 1.000	$\mathbf{A} \in \mathbb{R}^{m \times m}$	$\mathbf{A} \in \mathbb{R}^{m \times m}$	given by (11.4) and a matrix $\mathbf{A} \in \mathbb{R}^{m \times m}$. Since the matrices are m -by- m this will
lyche-numerical-linear-algebra-255-4 F03337	255	■ 1.000	$\mathbf{S} \mathbf{A}$	$\mathbf{S} \mathbf{A}$	calculate the products $\mathbf{S} \mathbf{A}$ and $\mathbf{A} \mathbf{S}$ in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-algebra-255-4 F03338	255	■ 1.000	$\mathbf{A} \mathbf{S}$	$\mathbf{A} \mathbf{S}$	calculate the products $\mathbf{S} \mathbf{A}$ and $\mathbf{A} \mathbf{S}$ in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-algebra-255-4 F03339	255	■ 1.000	$O(m^2 \log_2 m)$	$O(m^2 \log_2 m)$	calculate the products $\mathbf{S} \mathbf{A}$ and $\mathbf{A} \mathbf{S}$ in $O(m^2 \log_2 m)$ operations.
lyche-numerical-linear-algebra-255-4 F03340	255	■ 1.000	$\mathbf{v} = [v_1, \dots, v_m]^T \in \mathbb{R}^m$	$\mathbf{v} = [v_1, \dots, v_m]^T \in \mathbb{R}^m$	Given $\mathbf{v} = [v_1, \dots, v_m]^T \in \mathbb{R}^m$ we say that the vector $\mathbf{w} = [w_1, \dots, w_m]^T$ given
lyche-numerical-linear-algebra-255-4 F03341	255	■ 1.000	$\mathbf{w} = [w_1, \dots, w_m]^T$	$\mathbf{w} = [w_1, \dots, w_m]^T$	Given $\mathbf{v} = [v_1, \dots, v_m]^T \in \mathbb{R}^m$ we say that the vector $\mathbf{w} = [w_1, \dots, w_m]^T$ given
lyche-numerical-linear-algebra-255-4 F03342	255	■ 1.000	$\mathbf{w} = \mathbf{S} \mathbf{v}$	$\mathbf{w} = \mathbf{S} \mathbf{v}$	as the matrix times vector $\mathbf{w} = \mathbf{S} \mathbf{v}$, where $\mathbf{S}$ is the sine matrix given by (11.4). We
lyche-numerical-linear-algebra-255-4 F03343	255	■ 1.000	$\mathbf{B} = \mathbf{S} \mathbf{A}$	$\mathbf{B} = \mathbf{S} \mathbf{A}$	can then identify the matrix $\mathbf{B} = \mathbf{S} \mathbf{A}$ as the DST of $\mathbf{A} \in \mathbb{R}^{m,n}$, i.e. as the DST of
lyche-numerical-linear-algebra-255-4 F03344	255	■ 1.000	$\mathbf{A} \in \mathbb{R}^{m,n}$	$\mathbf{A} \in \mathbb{R}^{m,n}$	can then identify the matrix $\mathbf{B} = \mathbf{S} \mathbf{A}$ as the DST of $\mathbf{A} \in \mathbb{R}^{m,n}$, i.e. as the DST of
lyche-numerical-linear-algebra-255-4 F03345	255	■ 1.000	$\mathbf{B} = \mathbf{A} \mathbf{S}$	$\mathbf{B} = \mathbf{A} \mathbf{S}$	the columns of $\mathbf{A}$. The product $\mathbf{B} = \mathbf{A} \mathbf{S}$ can also be interpreted as a DST. Indeed,
lyche-numerical-linear-algebra-255-4 F03346	255	■ 1.000	$\mathbf{B} = (\mathbf{S} \mathbf{A})^T$	$\mathbf{B} = (\mathbf{S} \mathbf{A})^T$	since $\mathbf{S}$ is symmetric we have $\mathbf{B} = (\mathbf{S} \mathbf{A})^T$ which means that $\mathbf{B}$ is the transpose
lyche-numerical-linear-algebra-255-4 F03347	255	■ 1.000	$N \in \mathbb{N}$	$N \in \mathbb{N}$	as the fast Fourier transform (FFT). To define the DFT let for $N \in \mathbb{N}$
lyche-numerical-linear-algebra-256-4 F03348	256	■ 1.000	$\mathbf{y} = [y_1, \dots, y_N]^T \in \mathbb{R}^N$	$\mathbf{y} = [y_1, \dots, y_N]^T \in \mathbb{R}^N$	where $i = \sqrt{-1}$ is the imaginary unit. Given $\mathbf{y} = [y_1, \dots, y_N]^T \in \mathbb{R}^N$ we say that
lyche-numerical-linear-algebra-256-4 F03349	256	■ 1.000	$\mathbf{z} = [z_1, \dots, z_N]^T$	$\mathbf{z} = [z_1, \dots, z_N]^T$	$\mathbf{z} = [z_1, \dots, z_N]^T$ given by
lyche-numerical-linear-algebra-256-4 F03350	256	■ 0.991	$\mathbf{z} = \mathbf{F}_N \mathbf{y}$	$\mathbf{z} = \mathbf{F}_N \mathbf{y}$	vector product $\mathbf{z} = \mathbf{F}_N \mathbf{y}$, where the Fourier matrix $\mathbf{F}_N \in \mathbb{C}^{N \times N}$ has elements
lyche-numerical-linear-algebra-256-4 F03351	256	■ 0.991	$\mathbf{F}_N \in \mathbb{C}^{N \times N}$	$\mathbf{F}_N \in \mathbb{C}^{N \times N}$	vector product $\mathbf{z} = \mathbf{F}_N \mathbf{y}$, where the Fourier matrix $\mathbf{F}_N \in \mathbb{C}^{N \times N}$ has elements
lyche-numerical-linear-algebra-256-4 F03352	256	■ 1.000	$\omega_N^{jk}, j, k = 0, 1, \dots, N-1$	$\omega_N^{jk}, j, k = 0, 1, \dots, N-1$	$\omega_N^{jk}, j, k = 0, 1, \dots, N-1$. For a matrix we say that $\mathbf{B} = \mathbf{F}_N \mathbf{A}$ is the DFT of $\mathbf{A}$.

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lyche-numerical-linear-algebra-F03353	256	1.000	$\boldsymbol{B}=\boldsymbol{F}_N\boldsymbol{A}$	$\boldsymbol{B} = \boldsymbol{F}_N \boldsymbol{A}$	ω_N^{jk} , $j, k = 0, 1, \dots, N - 1$. For a matrix we say that $\boldsymbol{B} = \boldsymbol{F}_N \boldsymbol{A}$ is the DFT of $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F03354	256	1.000	$\omega_4^2=(-i)^2=-1$, $\omega_4^3=(-i)(-1)=i$, $\omega_4^4=(-1)^2=1$, $\omega_4^6=i^2=-1$	$\omega_4^2 = (-i)^2 = -1, \omega_4^3 = (-i)(-1) = i, \omega_4^4 = (-1)^2 = 1, \omega_4^6 = i^2 = -1$	we find $\omega_4^2 = (-i)^2 = -1, \omega_4^3 = (-i)(-1) = i, \omega_4^4 = (-1)^2 = 1, \omega_4^6 = i^2 = -1$,
lyche-numerical-linear-algebra-F03355	256	1.000	$\omega_4^9=i^3=-i$	$\omega_4^9 = i^3 = -i$	$\omega_4^9 = i^3 = -i$, and so
lyche-numerical-linear-algebra-F03356	256	1.000	$2m+2$	$2m + 2$	computed from the discrete Fourier transform of order $2m + 2$. We recall that for
lyche-numerical-linear-algebra-F03357	256	1.000	w	w	any complex number w
lyche-numerical-linear-algebra-F03358	256	1.000	$\boldsymbol{S}\boldsymbol{x}$	$\boldsymbol{S}\boldsymbol{x}$	and a vector $\boldsymbol{x} \in \mathbb{R}^m$. Component k of $\boldsymbol{S}\boldsymbol{x}$ is equal to $i/2$ times component $k + 1$ of
lyche-numerical-linear-algebra-F03359	256	1.000	$i / 2$	$i/2$	and a vector $\boldsymbol{x} \in \mathbb{R}^m$. Component k of $\boldsymbol{S}\boldsymbol{x}$ is equal to $i/2$ times component $k + 1$ of
lyche-numerical-linear-algebra-F03360	256	1.000	$\boldsymbol{F}_{2m+2}\boldsymbol{z}$	$\boldsymbol{F}_{2m+2}\boldsymbol{z}$	$\boldsymbol{F}_{2m+2}$ where
lyche-numerical-linear-algebra-F03361	256	1.000	$\omega=\omega_{2m+2}=e^{-2\pi i/(2m+2)}=e^{-\pi i/(m+1)}$	$\omega = \omega_{2m+2} = e^{-2\pi i/(2m+2)} = e^{-\pi i/(m+1)}$	Proof Let $\omega = \omega_{2m+2} = e^{-2\pi i/(2m+2)} = e^{-\pi i/(m+1)}$. We note that
lyche-numerical-linear-algebra-F03362	257	1.000	$-2i$	$-2i$	Dividing both sides by $-2i$ and noting $-1/(2i) = -i/(2i^2) = i/2$, proves the
lyche-numerical-linear-algebra-F03363	257	1.000	$-1/(2i)=-i/\left(2i^2\right)=i/2$	$-1/(2i) = -i/(2i^2) = i/2$	Dividing both sides by $-2i$ and noting $-1/(2i) = -i/(2i^2) = i/2$, proves the
lyche-numerical-linear-algebra-F03364	257	1.000	$N=2m+2$	$N = 2m + 2$	from the DFT of length $N = 2m + 2$.
lyche-numerical-linear-algebra-F03365	257	1.000	$\boldsymbol{F}_N\boldsymbol{y}$	$\boldsymbol{F}_N\boldsymbol{y}$	the matrix- vector product $\boldsymbol{F}_N\boldsymbol{y}$. Suppose N is even. The key to the FFT is a
lyche-numerical-linear-algebra-F03366	257	1.000	$\boldsymbol{F}_N$	$\boldsymbol{F}_N$	connection between $\boldsymbol{F}_N$ and $\boldsymbol{F}_{N/2}$ which makes it possible to compute the FFT
lyche-numerical-linear-algebra-F03367	257	1.000	$\boldsymbol{F}_{N/2}$	$\boldsymbol{F}_{N/2}$	connection between $\boldsymbol{F}_N$ and $\boldsymbol{F}_{N/2}$ which makes it possible to compute the FFT
lyche-numerical-linear-algebra-F03368	257	1.000	$N/2$	$N/2$	of order N as two FFT's of order $N/2$. By repeating this process we can reduce the
lyche-numerical-linear-algebra-F03369	257	1.000	$O\left(N^2\right)$	$O\left(N^2\right)$	number of arithmetic operations to compute a DFT from $O(N^2)$ to $O(N \log_2 N)$.
lyche-numerical-linear-algebra-F03370	257	1.000	$O\left(N \log _2 N\right)$	$O\left(N \log _2 N\right)$	number of arithmetic operations to compute a DFT from $O(N^2)$ to $O(N \log_2 N)$.
lyche-numerical-linear-algebra-F03371	257	1.000	$\boldsymbol{P}_N \in \mathbb{R}^{N \times N}$	$\boldsymbol{P}_N \in \mathbb{R}^{N \times N}$	tion matrix $\boldsymbol{P}_N \in \mathbb{R}^{N \times N}$ given by
lyche-numerical-linear-algebra-F03372	257	0.998	$\boldsymbol{e}_k=\left(\delta_{j, k}\right)$	$\boldsymbol{e}_k = (\delta_{j,k})$	where the $\boldsymbol{e}_k = (\delta_{j,k})$ are unit vectors. If $\boldsymbol{A}$ is a matrix with N columns $[\boldsymbol{a}_1, \dots, \boldsymbol{a}_N]$
lyche-numerical-linear-algebra-F03373	257	0.998	$\left[\boldsymbol{a}_1, \ldots, \boldsymbol{a}_N\right]$	$[\boldsymbol{a}_1, \dots, \boldsymbol{a}_N]$	where the $\boldsymbol{e}_k = (\delta_{j,k})$ are unit vectors. If $\boldsymbol{A}$ is a matrix with N columns $[\boldsymbol{a}_1, \dots, \boldsymbol{a}_N]$
lyche-numerical-linear-algebra-F03374	257	0.965	$\boldsymbol{P}_N$	$\boldsymbol{P}_N$	i.e. post multiplying $\boldsymbol{A}$ by $\boldsymbol{P}_N$ permutes the columns of $\boldsymbol{A}$ so that all the odd-indexed
lyche-numerical-linear-algebra-F03375	258	1.000	$\boldsymbol{F}_4\boldsymbol{P}_4$	$\boldsymbol{F}_4\boldsymbol{P}_4$	where we have indicated a certain block structure of $\boldsymbol{F}_4\boldsymbol{P}_4$. These blocks can be
lyche-numerical-linear-algebra-F03376	258	1.000	$\boldsymbol{F}_2$	$\boldsymbol{F}_2$	related to the 2-by-2 matrix $\boldsymbol{F}_2$. We define the diagonal scaling matrix $\boldsymbol{D}_2$ by
lyche-numerical-linear-algebra-F03377	258	1.000	$\omega_2=\exp ^{-2 \pi i / 2}=-1$	$\omega_2 = \exp^{-2\pi i/2} = -1$	Since $\omega_2 = \exp^{-2\pi i/2} = -1$ we find
lyche-numerical-linear-algebra-F03378	258	1.000	$N=2m$	$N = 2m$	Theorem 11.1 (Fast Fourier Transform) If $N = 2m$ is even then

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lyche-numerical-linear-algebra-258 F03379	258	1.000	p, q	p, q	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-258 F03380	258	1.000	$1 \leq p, q \leq m$	$1 \leq p, q \leq m$	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-258 F03381	258	1.000	$j := p - 1$	$j := p - 1$	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-258 F03382	258	1.000	$k := q - 1$	$k := q - 1$	Proof Fix integers p, q with $1 \leq p, q \leq m$ and set $j := p - 1$ and $k := q - 1$.
lyche-numerical-linear-algebra-258 F03383	258	1.000	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}$	$\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}$	It follows that the four m -by- m blocks of $\boldsymbol{F}_{2m} \boldsymbol{P}_{2m}$ have the required structure.
lyche-numerical-linear-algebra-258 F03384	258	1.000	$\boldsymbol{y} \in \mathbb{R}^{2m}$	$\boldsymbol{y} \in \mathbb{R}^{2m}$	$\boldsymbol{y} \in \mathbb{R}^{2m}$ and set $\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$, where
lyche-numerical-linear-algebra-258 F03385	258	1.000	$\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$	$\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$	$\boldsymbol{y} \in \mathbb{R}^{2m}$ and set $\boldsymbol{w} = \boldsymbol{P}_{2m}^T \boldsymbol{y} = [\boldsymbol{w}_1^T, \boldsymbol{w}_2^T]^T$, where
lyche-numerical-linear-algebra-259 F03386	259	1.000	$\boldsymbol{F}_{2m} \boldsymbol{y}$	$\boldsymbol{F}_{2m} \boldsymbol{y}$	In order to compute $\boldsymbol{F}_{2m} \boldsymbol{y}$ we need to compute $\boldsymbol{F}_m \boldsymbol{w}_1$ and $\boldsymbol{F}_m \boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-algebra-259 F03387	259	1.000	$\boldsymbol{F}_m \boldsymbol{w}_1$	$\boldsymbol{F}_m \boldsymbol{w}_1$	In order to compute $\boldsymbol{F}_{2m} \boldsymbol{y}$ we need to compute $\boldsymbol{F}_m \boldsymbol{w}_1$ and $\boldsymbol{F}_m \boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-algebra-259 F03388	259	1.000	$\boldsymbol{F}_m \boldsymbol{w}_2$	$\boldsymbol{F}_m \boldsymbol{w}_2$	In order to compute $\boldsymbol{F}_{2m} \boldsymbol{y}$ we need to compute $\boldsymbol{F}_m \boldsymbol{w}_1$ and $\boldsymbol{F}_m \boldsymbol{w}_2$. Thus, by
lyche-numerical-linear-algebra-259 F03389	259	1.000	$n = 2^k$	$n = 2^k$	combining two FFT's of order m we obtain an FFT of order $2m$. If $n = 2^k$ then
lyche-numerical-linear-algebra-259 F03390	259	1.000	$\boldsymbol{y} \in \mathbb{R}^n$	$\boldsymbol{y} \in \mathbb{R}^n$	Statement 3 is included so that the input $\boldsymbol{y} \in \mathbb{R}^n$ can be either a row or column
lyche-numerical-linear-algebra-259 F03391	259	1.000	$\gamma N \log_2 N$	$\gamma N \log_2 N$	The complexity of the FFT is given by $\gamma N \log_2 N$ for some constant γ
lyche-numerical-linear-algebra-259 F03392	259	1.000	γ	γ	The complexity of the FFT is given by $\gamma N \log_2 N$ for some constant γ
lyche-numerical-linear-algebra-259 F03393	259	1.000	$N = 2^k$	$N = 2^k$	let x_k be the complexity (the number of arithmetic operations) when $N = 2^k$.
lyche-numerical-linear-algebra-259 F03394	259	1.000	$N/2 = 2^{k-1}$	$N/2 = 2^{k-1}$	Since we need two FFT's of order $N/2 = 2^{k-1}$ and a multiplication with the
lyche-numerical-linear-algebra-259 F03395	259	1.000	$\boldsymbol{D}_{N/2}$	$\boldsymbol{D}_{N/2}$	diagonal matrix $\boldsymbol{D}_{N/2}$, it is reasonable to assume that $x_k = 2x_{k-1} + \gamma 2^k$ for
lyche-numerical-linear-algebra-259 F03396	259	1.000	$x_k = 2x_{k-1} + \gamma 2^k$	$x_k = 2x_{k-1} + \gamma 2^k$	diagonal matrix $\boldsymbol{D}_{N/2}$, it is reasonable to assume that $x_k = 2x_{k-1} + \gamma 2^k$ for
lyche-numerical-linear-algebra-259 F03397	259	1.000	$x_0 = 0$	$x_0 = 0$	some constant γ independent of k . Since $x_0 = 0$ we obtain by induction on k
lyche-numerical-linear-algebra-259 F03398	259	1.000	$x_k = \gamma k 2^k$	$x_k = \gamma k 2^k$	that $x_k = \gamma k 2^k$. Indeed, this holds for $k = 0$ and if $x_{k-1} = \gamma(k-1)2^{k-1}$ then
lyche-numerical-linear-algebra-259 F03399	259	1.000	$x_{k-1} = \gamma(k-1)2^{k-1}$	$x_{k-1} = \gamma(k-1)2^{k-1}$	that $x_k = \gamma k 2^k$. Indeed, this holds for $k = 0$ and if $x_{k-1} = \gamma(k-1)2^{k-1}$ then
lyche-numerical-linear-algebra-259 F03400	259	1.000	$x_k = 2x_{k-1} + \gamma 2^k = 2\gamma(k-1)2^{k-1} + \gamma 2^k = \gamma k 2^k$	$x_k = 2x_{k-1} + \gamma 2^k = 2\gamma(k-1)2^{k-1} + \gamma 2^k = \gamma k 2^k$	$x_k = 2x_{k-1} + \gamma 2^k = 2\gamma(k-1)2^{k-1} + \gamma 2^k = \gamma k 2^k$. Reasonable implementations
lyche-numerical-linear-algebra-259 F03401	259	1.000	$\gamma \approx 5$	$\gamma \approx 5$	of FFT typically have $\gamma \approx 5$, see [14].
lyche-numerical-linear-algebra-259 F03402	259	1.000	$O(8n^2)$	$O(8n^2)$	large N . The direct multiplication $\boldsymbol{F}_N \boldsymbol{y}$ requires $O(8n^2)$ arithmetic operations since
lyche-numerical-linear-algebra-260 F03403	260	1.000	$5N \log_2 N$	$5N \log_2 N$	complex arithmetic is involved. Assuming that the FFT uses $5N \log_2 N$ arithmetic
lyche-numerical-linear-algebra-260 F03404	260	1.000	$N = 2^{20} \approx 10^6$	$N = 2^{20} \approx 10^6$	operations we find for $N = 2^{20} \approx 10^6$ the ratio

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lyche-numerical-linear-algebra-260 F03405	260	■ 1.000	$\boldsymbol{H}=\boldsymbol{S} \boldsymbol{F}$	$\boldsymbol{H} = \boldsymbol{S}\boldsymbol{F}$	Algorithm 11.1. We first compute $\boldsymbol{H} = \boldsymbol{S}\boldsymbol{F}$ using Lemma 11.1 and m FFT's, one
lyche-numerical-linear-algebra-260 F03406	260	■ 0.998	$\boldsymbol{G}=\boldsymbol{H} \boldsymbol{S}$	$\boldsymbol{G} = \boldsymbol{H}\boldsymbol{S}$	for each of the m columns of $\boldsymbol{F}$. We then compute $\boldsymbol{G} = \boldsymbol{H}\boldsymbol{S}$ by m FFT's, one for
lyche-numerical-linear-algebra-260 F03407	260	■ 1.000	$\boldsymbol{Z}=\boldsymbol{S} \boldsymbol{X}$	$\boldsymbol{Z} = \boldsymbol{S}\boldsymbol{X}$	each of the rows of $\boldsymbol{H}$. After $\boldsymbol{X}$ is determined we compute $\boldsymbol{Z} = \boldsymbol{S}\boldsymbol{X}$ and $\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{S}$
lyche-numerical-linear-algebra-260 F03408	260	■ 1.000	$\boldsymbol{V}=\boldsymbol{Z} \boldsymbol{S}$	$\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{S}$	each of the rows of $\boldsymbol{H}$. After $\boldsymbol{X}$ is determined we compute $\boldsymbol{Z} = \boldsymbol{S}\boldsymbol{X}$ and $\boldsymbol{V} = \boldsymbol{Z}\boldsymbol{S}$
lyche-numerical-linear-algebra-260 F03409	260	■ 1.000	$O\left(\gamma(2m+2)\log _{2}(2m+2)\right)$	$O\left(\gamma(2m+2)\log _2(2m+2)\right)$	one FFT requires $O(\gamma(2m+2)\log _2(2m+2))$ arithmetic operations the $4m$ FFT's
lyche-numerical-linear-algebra-260 F03410	260	■ 1.000	$O\left(8 n^{3 / 2}\right)$	$O\left(8 n^{3 / 2}\right)$	if Cholesky factorization was used. This should be compared to the $O(8 n^{3 / 2})$
lyche-numerical-linear-algebra-260 F03411	260	■ 1.000	$\boldsymbol{F}_{4}$	$\boldsymbol{F}_4$	Exercise 11.1 (Fourier Matrix) Show that the Fourier matrix $\boldsymbol{F}_4$ is symmetric,
lyche-numerical-linear-algebra-260 F03412	261	■ 1.000	$\left(v_{i, j}\right)_{i, j=1}^m$	$\left(v_{i, j}\right)_{i, j=1}^m$	$\left(v_{i, j}\right)_{i, j=1}^m$, where
lyche-numerical-linear-algebra-260 F03413	261	■ 0.608	$\boldsymbol{T} \boldsymbol{V}+\boldsymbol{V} \boldsymbol{T}=h^2 \boldsymbol{F}$	$\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$	in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$, where $\lambda_j = 4 \sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra-260 F03414	261	■ 0.608	$\boldsymbol{D}=\operatorname{diag}\left(\lambda_1,\ldots,\lambda_m\right)$	$\boldsymbol{D} = \text{diag}\left(\lambda_1, \dots, \lambda_m\right)$	in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$, where $\lambda_j = 4 \sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra-260 F03415	261	■ 0.608	$\lambda_j=4 \sin ^2(j \pi h / 2)$	$\lambda_j = 4 \sin ^2(j \pi h / 2)$	in $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2\boldsymbol{F}$. Let $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$, where $\lambda_j = 4 \sin^2(j\pi h/2)$,
lyche-numerical-linear-algebra-260 F03416	261	■ 1.000	$\boldsymbol{X}=\left[\boldsymbol{x}_1,\ldots,\boldsymbol{x}_m\right]$	$\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_m]$	where $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_m]$ and $\boldsymbol{C} = [\boldsymbol{c}_1, \dots, \boldsymbol{c}_m]$. Thus we can find $\boldsymbol{X}$ by solving
lyche-numerical-linear-algebra-260 F03417	261	■ 1.000	$\boldsymbol{C}=\left[\boldsymbol{c}_1,\ldots,\boldsymbol{c}_m\right]$	$\boldsymbol{C} = [\boldsymbol{c}_1, \dots, \boldsymbol{c}_m]$	where $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_m]$ and $\boldsymbol{C} = [\boldsymbol{c}_1, \dots, \boldsymbol{c}_m]$. Thus we can find $\boldsymbol{X}$ by solving
lyche-numerical-linear-algebra-260 F03418	261	■ 1.000	$8 m-7$	$8m - 7$	$m \times m$ system can be solved by Algorithms 2.1 and 2.2 in $8m - 7$ arithmetic
lyche-numerical-linear-algebra-260 F03419	261	■ 1.000	$O\left(\delta m^2\right)$	$O\left(\delta m^2\right)$	operations. Give an algorithm to find $\boldsymbol{X}$ which only requires $O(\delta m^2)$ arithmetic
lyche-numerical-linear-algebra-260 F03420	261	■ 1.000	$O\left(4 m^3\right)=O\left(4 n^{3 / 2}\right)$	$O\left(4 m^3\right)=O\left(4 n^{3 / 2}\right)$	c) Describe a method to compute $\boldsymbol{V}$ which only requires $O(4 m^3) = O(4 n^{3 / 2})$
lyche-numerical-linear-algebra-260 F03421	261	■ 1.000	$O\left(2 \gamma n \log _2 n\right)$	$O\left(2 \gamma n \log _2 n\right)$	$O(2 \gamma n \log _2 n)$ where γ is the same constant as mentioned at the end of the
lyche-numerical-linear-algebra-260 F03422	261	■ 1.000	$\boldsymbol{V}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}=\left(x_{j, k}\right)$	$\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S} = (x_{j,k})$	$\boldsymbol{V} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S} = (x_{j,k})$ where $\boldsymbol{V}$ is the solution of (10.22). Show that
lyche-numerical-linear-algebra-260 F03423	262	■ 1.000	$\lambda_j=4 \sin ^2(j \pi h / 2), j=1,\ldots, m$	$\lambda_j = 4 \sin ^2(j \pi h / 2), j = 1, \dots, m$	where $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_m)$, with $\lambda_j = 4 \sin^2(j\pi h/2), j = 1, \dots, m$, and that
lyche-numerical-linear-algebra-260 F03424	262	■ 1.000	$\sigma_j+\sigma_k-\frac{2}{3} \sigma_j \sigma_k>0$	$\sigma_j + \sigma_k - \frac{2}{3} \sigma_j \sigma_k > 0$	Show that $\sigma_j + \sigma_k - \frac{2}{3} \sigma_j \sigma_k > 0$ for $j, k = 1, 2, \dots, m$. Conclude that the matrix
lyche-numerical-linear-algebra-260 F03425	262	■ 1.000	$j, k=1,2,\ldots, m$	$j, k = 1, 2, \dots, m$	Show that $\sigma_j + \sigma_k - \frac{2}{3} \sigma_j \sigma_k > 0$ for $j, k = 1, 2, \dots, m$. Conclude that the matrix
lyche-numerical-linear-algebra-260 F03426	262	■ 1.000	$\boldsymbol{X}=\left(x_{j, k}\right)$	$\boldsymbol{X} = (x_{j,k})$	Define the matrix $\boldsymbol{X} = (x_{j,k})$ by $\boldsymbol{U} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$ where $\boldsymbol{U}$ is the solution of (10.25).
lyche-numerical-linear-algebra-260 F03427	262	■ 1.000	$\boldsymbol{U}=\boldsymbol{S} \boldsymbol{X} \boldsymbol{S}$	$\boldsymbol{U} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$	Define the matrix $\boldsymbol{X} = (x_{j,k})$ by $\boldsymbol{U} = \boldsymbol{S}\boldsymbol{X}\boldsymbol{S}$ where $\boldsymbol{U}$ is the solution of (10.25).
lyche-numerical-linear-algebra-260 F03428	262	■ 1.000	$O\left(\delta n^{3 / 2}\right)$	$O\left(\delta n^{3 / 2}\right)$	which requires only $O(\delta n^{3 / 2})$ operations to find $\boldsymbol{U}$ in Problem 10.9. Here δ is some
lyche-numerical-linear-algebra-260 F03429	262	■ 0.713	$\boldsymbol{F} =$	$\boldsymbol{F} =$	Exercise 11.8 compute the solution $\boldsymbol{U}$ corresponding to $\boldsymbol{F} = \text{ones}(m, m)$. For
lyche-numerical-linear-algebra-260 F03430	262	■ 1.000	$\boldsymbol{x}=\boldsymbol{A} \backslash \boldsymbol{b}$	$\boldsymbol{x} = \boldsymbol{A} \backslash \boldsymbol{b}$	standard form $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ in (10.25). You can use $\boldsymbol{x} = \boldsymbol{A} \backslash \boldsymbol{b}$ for solving $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-262-F03431	262	0.581	$\boldsymbol{x} \in \mathbb{R}^{m^2}$	$\boldsymbol{x} \in \mathbb{R}^{m^2}$	Use <code>F(:)</code> to vectorize a matrix and <code>reshape(x,m,m)</code> to turn a vector $\boldsymbol{x} \in \mathbb{R}^{m^2}$
lyche-numerical-linear-algebra-262-F03432	262	1.000	U	U	into an $m \times m$ matrix. Use the MATLAB command <code>surf(U)</code> for plotting U for, say,
lyche-numerical-linear-algebra-262-F03433	262	1.000	$m=50$	$m = 50$	$m = 50$. Compare the result with Exercise 11.8 by plotting the difference between
lyche-numerical-linear-algebra-264-F03434	264	1.000	$\boldsymbol{x}_0$	$\boldsymbol{x}_0$	approximation $\boldsymbol{x}_0$ to the exact solution $\boldsymbol{x}$ and then compute a sequence $\{\boldsymbol{x}_k\}$ such that
lyche-numerical-linear-algebra-264-F03435	264	1.000	$\{\boldsymbol{x}_k\}$	$\{\boldsymbol{x}_k\}$	approximation $\boldsymbol{x}_0$ to the exact solution $\boldsymbol{x}$ and then compute a sequence $\{\boldsymbol{x}_k\}$ such that
lyche-numerical-linear-algebra-264-F03436	264	1.000	$\boldsymbol{x}_k \rightarrow \boldsymbol{x}$	$\boldsymbol{x}_k \rightarrow \boldsymbol{x}$	hopefully $\boldsymbol{x}_k \rightarrow \boldsymbol{x}$. Iterative methods are mainly used for large sparse systems, i.e.,
lyche-numerical-linear-algebra-265-F03437	265	1.000	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	onal elements, the linear system $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ can be written in component
lyche-numerical-linear-algebra-265-F03438	265	1.000	$y=(z+1) / 2$	$y = (z + 1)/2$	form as $y = (z + 1)/2$ and $z = (y + 1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-algebra-265-F03439	265	1.000	$z=(y+1) / 2$	$z = (y + 1)/2$	form as $y = (z + 1)/2$ and $z = (y + 1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-algebra-265-F03440	265	1.000	$y_{\{0\}}, z_{\{0\}}$	y_0, z_0	form as $y = (z + 1)/2$ and $z = (y + 1)/2$. Starting with y_0, z_0 we generate two
lyche-numerical-linear-algebra-265-F03441	265	0.996	$\{y_k\}$	$\{y_k\}$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k + 1)/2$ and
lyche-numerical-linear-algebra-265-F03442	265	0.996	$\{z_k\}$	$\{z_k\}$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k + 1)/2$ and
lyche-numerical-linear-algebra-265-F03443	265	0.996	$y_{k+1}=\left(z_{k+1}+1\right) / 2$	$y_{k+1} = (z_k + 1) / 2$	sequences $\{y_k\}$ and $\{z_k\}$ using the difference equations $y_{k+1} = (z_k + 1)/2$ and
lyche-numerical-linear-algebra-265-F03444	265	1.000	$z_{k+1}=\left(y_{k+1}+1\right) / 2$	$z_{k+1} = (y_k + 1) / 2$	$z_{k+1} = (y_k + 1)/2$. This is an example of Jacobi's method. If $y_0 = z_0 = 0$ then
lyche-numerical-linear-algebra-265-F03445	265	1.000	$y_{\{0\}}=z_{\{0\}}=0$	$y_0 = z_0 = 0$	$z_{k+1} = (y_k + 1)/2$. This is an example of Jacobi's method. If $y_0 = z_0 = 0$ then
lyche-numerical-linear-algebra-265-F03446	265	1.000	$y_{\{1\}}=z_{\{1\}}=1 / 2$	$y_1 = z_1 = 1/2$	we find $y_1 = z_1 = 1/2$ and in general $y_k = z_k = 1 - 2^{-k}$ for $k = 0, 1, 2, 3, \dots$
lyche-numerical-linear-algebra-265-F03447	265	1.000	$y_{\{k\}}=z_{\{k\}}=1-2^{\{-k\}}$	$y_k = z_k = 1 - 2^{-k}$	we find $y_1 = z_1 = 1/2$ and in general $y_k = z_k = 1 - 2^{-k}$ for $k = 0, 1, 2, 3, \dots$
lyche-numerical-linear-algebra-265-F03448	265	1.000	$k=0,1,2,3, \ldots$	$k = 0, 1, 2, 3, \dots$	we find $y_1 = z_1 = 1/2$ and in general $y_k = z_k = 1 - 2^{-k}$ for $k = 0, 1, 2, 3, \dots$.
lyche-numerical-linear-algebra-265-F03449	265	1.000	$[1,1]^T$	$[1,1]^T$	The iteration converges to the exact solution $[1,1]^T$, and the error is halved in each
lyche-numerical-linear-algebra-266-F03450	266	1.000	$z_{k+1}=\left(y_{k+1}+1\right) / 2$	$z_{k+1} = (y_{k+1} + 1) / 2$	$z_{k+1} = (y_{k+1} + 1)/2$. If $y_0 = z_0 = 0$ then we find $y_1 = 1/2, z_1 = 3/4, y_2 = 7/8$,
lyche-numerical-linear-algebra-266-F03451	266	1.000	$y_{\{1\}}=1 / 2, z_{\{1\}}=3 / 4, y_{\{2\}}=7 / 8$	$y_1 = 1/2, z_1 = 3/4, y_2 = 7/8$	$z_{k+1} = (y_{k+1} + 1)/2$. If $y_0 = z_0 = 0$ then we find $y_1 = 1/2, z_1 = 3/4, y_2 = 7/8$,
lyche-numerical-linear-algebra-266-F03452	266	1.000	$z_{\{2\}}=15 / 16$	$z_2 = 15/16$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \dots$
lyche-numerical-linear-algebra-266-F03453	266	1.000	$y_{\{k\}}=1-2 \cdot 4^{\{-k\}}$	$y_k = 1 - 2 \cdot 4^{-k}$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \dots$
lyche-numerical-linear-algebra-266-F03454	266	1.000	$z_{\{k\}}=1-4^{\{-k\}}$	$z_k = 1 - 4^{-k}$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \dots$
lyche-numerical-linear-algebra-266-F03455	266	1.000	$k=1,2,3, \ldots$	$k = 1, 2, 3, \dots$	$z_2 = 15/16$, and in general $y_k = 1 - 2 \cdot 4^{-k}$ and $z_k = 1 - 4^{-k}$ for $k = 1, 2, 3, \dots$.
lyche-numerical-linear-algebra-266-F03456	266	1.000	$\boldsymbol{x}_{\{k\}}=\left[\boldsymbol{x}_{\{k\}}(1), \ldots, \boldsymbol{x}_{\{k\}}(n)\right]^T$	$\boldsymbol{x}_k = [\boldsymbol{x}_k(1), \dots, \boldsymbol{x}_k(n)]^T$	Suppose we know an approximation $\boldsymbol{x}_k = [\boldsymbol{x}_k(1), \dots, \boldsymbol{x}_k(n)]^T$ to the exact solution
lyche-numerical-linear-algebra-266-F03457	266	1.000	$\boldsymbol{x}(i)$	$\boldsymbol{x}(i)$	diagonal elements. Solving the i th equation of $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ for $\boldsymbol{x}(i)$, we obtain a fixed-

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lyche-numerical-linear-a2466 F03458	266	■ 1.000	$\boldsymbol{x}_{k+1}(i)$	$\boldsymbol{x}_{k+1}(i)$	where we use the new $\boldsymbol{x}_{k+1}(i)$ immediately after it has been computed.																								
lyche-numerical-linear-a2466 F03459	266	■ 1.000	$0 < \omega < 2$	$0 < \omega < 2$	ducing an acceleration parameter $0 < \omega < 2$ in the GS method. We write																								
lyche-numerical-linear-a2466 F03460	266	■ 1.000	$\boldsymbol{x}(i) = \omega \boldsymbol{x}(i) + (1 - \omega) \boldsymbol{x}(i)$	$\boldsymbol{x}(i) = \omega \boldsymbol{x}(i) + (1 - \omega) \boldsymbol{x}(i)$	$\boldsymbol{x}(i) = \omega \boldsymbol{x}(i) + (1 - \omega) \boldsymbol{x}(i)$ and this leads to the method																								
lyche-numerical-linear-a2466 F03461	266	■ 1.000	$\omega = 1$	$\omega = 1$	The SOR method reduces to the Gauss-Seidel method for $\omega = 1$. Denoting the																								
lyche-numerical-linear-a2466 F03462	266	■ 1.000	$\boldsymbol{x}_{k+1}^{gs}$	$\boldsymbol{x}_{k+1}^{gs}$	right hand side of (12.3) by $\boldsymbol{x}_{k+1}^{gs}$ we can write (12.4) as $\boldsymbol{x}_{k+1} = \omega \boldsymbol{x}_{k+1}^{gs} + (1 -$																								
lyche-numerical-linear-a2466 F03463	266	■ 1.000	$\boldsymbol{x}_{k+1} = \omega \boldsymbol{x}_{k+1}^{gs} + (1 -$	$\boldsymbol{x}_{k+1} = \omega \boldsymbol{x}_{k+1}^{gs} + (1 -$	right hand side of (12.3) by $\boldsymbol{x}_{k+1}^{gs}$ we can write (12.4) as $\boldsymbol{x}_{k+1} = \omega \boldsymbol{x}_{k+1}^{gs} + (1 -$																								
lyche-numerical-linear-a2466 F03464	266	■ 1.000	$\omega) \boldsymbol{x}_k$	$\omega) \boldsymbol{x}_k$	$\omega) \boldsymbol{x}_k$, and we see that $\boldsymbol{x}_{k+1}$ is located on the straight line passing through the																								
lyche-numerical-linear-a2466 F03465	266	■ 1.000	$\boldsymbol{x}_{k+1}$	$\boldsymbol{x}_{k+1}$	$\omega) \boldsymbol{x}_k$, and we see that $\boldsymbol{x}_{k+1}$ is located on the straight line passing through the																								
lyche-numerical-linear-a2467 F03466	267	■ 1.000	$1 \leq \omega < 2$	$1 \leq \omega < 2$	parameter ω in the range $1 \leq \omega < 2$ and then $\boldsymbol{x}_{k+1}$ is computed by linear																								
lyche-numerical-linear-a2467 F03467	267	■ 1.000	$\boldsymbol{x}_{k+1/2}$	$\boldsymbol{x}_{k+1/2}$	(12.4), computing an approximation denoted $\boldsymbol{x}_{k+1/2}$ instead of $\boldsymbol{x}_{k+1}$, is followed																								
lyche-numerical-linear-a2467 F03468	267	■ 1.000	$i = n, n-1, \dots, 1$	$i = n, n-1, \dots, 1$	in the order $i = n, n-1, \dots, 1$. The method is slower and more complicated																								
lyche-numerical-linear-a2467 F03469	267	■ 0.999	$\boldsymbol{v}(i, j)$	$\boldsymbol{v}(i, j)$	with homogenous boundary conditions also given in (10.4). Solving for $\boldsymbol{v}(i, j)$ we																								
lyche-numerical-linear-a2467 F03470	267	■ 0.570	$e_{i,j} := f_{i,j} / (m+1)^2$	$e_{i,j} := f_{i,j} / (m+1)^2$	where $e_{i,j} := f_{i,j} / (m+1)^2$. The J, GS , and SOR methods take the form																								
lyche-numerical-linear-a2468 F03471	268	■ 0.651	k_{100}	k_{100}	<table> <tr><td></td><td>k_{100}</td><td>k_{2500}</td><td>$k_{10\,000}$</td><td>$k_{40\,000}$</td><td>$k_{160\,000}$</td></tr> <tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr> <tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr> <tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr> </table>		k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$	J	385	8386				GS	194	4194				SOR	35	164	324	645	1286
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lyche-numerical-linear-a2468 F03472	268	■ 0.651	k_{2500}	k_{2500}	<table> <tr><td></td><td>k_{100}</td><td>k_{2500}</td><td>$k_{10\,000}$</td><td>$k_{40\,000}$</td><td>$k_{160\,000}$</td></tr> <tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr> <tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr> <tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr> </table>		k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$	J	385	8386				GS	194	4194				SOR	35	164	324	645	1286
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lyche-numerical-linear-a2468 F03473	268	■ 0.651	$k_{10\,000}$	$k_{10\,000}$	<table> <tr><td></td><td>k_{100}</td><td>k_{2500}</td><td>$k_{10\,000}$</td><td>$k_{40\,000}$</td><td>$k_{160\,000}$</td></tr> <tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr> <tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr> <tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr> </table>		k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$	J	385	8386				GS	194	4194				SOR	35	164	324	645	1286
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lyche-numerical-linear-a2468 F03474	268	■ 0.651	k_{40000}	k_{40000}	<table> <tr><td></td><td>k_{100}</td><td>k_{2500}</td><td>$k_{10\,000}$</td><td>$k_{40\,000}$</td><td>$k_{160\,000}$</td></tr> <tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr> <tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr> <tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr> </table>		k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$	J	385	8386				GS	194	4194				SOR	35	164	324	645	1286
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lyche-numerical-linear-a2468 F03475	268	■ 0.651	$k_{160\,000}$	$k_{160\,000}$	<table> <tr><td></td><td>k_{100}</td><td>k_{2500}</td><td>$k_{10\,000}$</td><td>$k_{40\,000}$</td><td>$k_{160\,000}$</td></tr> <tr><td>J</td><td>385</td><td>8386</td><td></td><td></td><td></td></tr> <tr><td>GS</td><td>194</td><td>4194</td><td></td><td></td><td></td></tr> <tr><td>SOR</td><td>35</td><td>164</td><td>324</td><td>645</td><td>1286</td></tr> </table>		k_{100}	k_{2500}	$k_{10\,000}$	$k_{40\,000}$	$k_{160\,000}$	J	385	8386				GS	194	4194				SOR	35	164	324	645	1286
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lyche-numerical-linear-a2468 F03476	268	■ 0.995	$\left(i_1, j_1\right) <$	$(i_1, j_1) <$	We note that for GS and SOR we have used the natural ordering , i.e., $(i_1, j_1) <$																								
lyche-numerical-linear-a2468 F03477	268	■ 0.960	i_2, j_2	i_2, j_2	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering																								

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lyche-numerical-linear-algebra- F03478	268	■ 0.960	$j_{\{1\}} \leq j_{\{2\}}$	$j_1 \leq j_2$	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering
lyche-numerical-linear-algebra- F03479	268	■ 0.960	$i_{\{1\}} < i_{\{2\}}$	$i_1 < i_2$	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering
lyche-numerical-linear-algebra- F03480	268	■ 0.960	$j_{\{1\}} = j_{\{2\}}$	$j_1 = j_2$	(i_2, j_2) if and only if $j_1 \leq j_2$ and $i_1 < i_2$ if $j_1 = j_2$. For the J method any ordering
lyche-numerical-linear-algebra- F03481	268	■ 1.000	$\boldsymbol{F} = \left(f_{\{i, j\}} \right) \in \mathbb{R}^{m \times m}$	$\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m \times m}$	on the linear system (12.6) with $\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m \times m}$, starting with $\boldsymbol{V}_0 = \mathbf{0} \in$
lyche-numerical-linear-algebra- F03482	268	■ 1.000	$\boldsymbol{V}_{\{0\}} = \mathbf{0}$	$\boldsymbol{V}_0 = \mathbf{0} \in$	on the linear system (12.6) with $\boldsymbol{F} = (f_{i,j}) \in \mathbb{R}^{m \times m}$, starting with $\boldsymbol{V}_0 = \mathbf{0} \in$
lyche-numerical-linear-algebra- F03483	268	■ 1.000	$\mathbb{R}^{(m+2) \times (m+2)}$	$\mathbb{R}^{(m+2) \times (m+2)}$	$\mathbb{R}^{(m+2) \times (m+2)}$. The output is the number of iterations k , to obtain $\ \boldsymbol{V}^{(k)} - \boldsymbol{U}\ _M :=$
lyche-numerical-linear-algebra- F03484	268	■ 1.000	$\left\ \boldsymbol{V}^{(k)} - \boldsymbol{U} \right\ _M :=$	$\left\ \boldsymbol{V}^{(k)} - \boldsymbol{U} \right\ _M :=$	$\mathbb{R}^{(m+2) \times (m+2)}$. The output is the number of iterations k , to obtain $\ \boldsymbol{V}^{(k)} - \boldsymbol{U}\ _M :=$
lyche-numerical-linear-algebra- F03485	268	■ 0.911	$\max_{\{i, j\}} v_{\{i j\}} - u_{\{i j\}} <$	$\max_{i,j} v_{ij} - u_{ij} <$	$\max_{i,j} v_{ij} - u_{ij} < tol$. Here $[u_{ij}] \in \mathbb{R}^{(m+2) \times (m+2)}$ is the "exact" solution of (12.6)
lyche-numerical-linear-algebra- F03486	268	■ 0.911	$\left[u_{\{i j\}} \right] \in \mathbb{R}^{(m+2) \times (m+2)}$	$[u_{ij}] \in \mathbb{R}^{(m+2) \times (m+2)}$	$\max_{i,j} v_{ij} - u_{ij} < tol$. Here $[u_{ij}] \in \mathbb{R}^{(m+2) \times (m+2)}$ is the "exact" solution of (12.6)
lyche-numerical-linear-algebra- F03487	268	■ 0.843	$k = k_{\{n\}}$	$k = k_n$	$k = k_n$ from this algorithm using $\boldsymbol{F} = \text{ones}(m, m)$ for $m = 10, 50, K = 10^4$, and
lyche-numerical-linear-algebra- F03488	268	■ 0.843	$\boldsymbol{F} = \text{operatorname{ones}}(m, m)$	$\boldsymbol{F} = \text{ones}(m, m)$	$k = k_n$ from this algorithm using $\boldsymbol{F} = \text{ones}(m, m)$ for $m = 10, 50, K = 10^4$, and
lyche-numerical-linear-algebra- F03489	268	■ 0.843	$m=10, 50, K=10^4$	$m = 10, 50, K = 10^4$	$k = k_n$ from this algorithm using $\boldsymbol{F} = \text{ones}(m, m)$ for $m = 10, 50, K = 10^4$, and
lyche-numerical-linear-algebra- F03490	268	■ 0.825	$= 10^{-8}$	$= 10^{-8}$	$tol = 10^{-8}$. We also show the number of iterations for Gauss-Seidel and SOR with
lyche-numerical-linear-algebra- F03491	268	■ 1.000	$\omega^* := 2 / \left(1 + \sin \left(\frac{\pi}{m+1} \right) \right)$	$\omega^* := 2 / \left(1 + \sin \left(\frac{\pi}{m+1} \right) \right)$	a value of ω known as the optimal acceleration parameter $\omega^* := 2 / (1 + \sin(\frac{\pi}{m+1}))$.
lyche-numerical-linear-algebra- F03492	269	■ 1.000	$0 < w < 2$	$0 < w < 2$	is an acceleration parameter with $0 < w < 2$. For $w = 1$ we obtain Gauss-Seidel's
lyche-numerical-linear-algebra- F03493	269	■ 1.000	$w=1$	$w = 1$	is an acceleration parameter with $0 < w < 2$. For $w = 1$ we obtain Gauss-Seidel's
lyche-numerical-linear-algebra- F03494	269	■ 1.000	$k_{\{n\}} = O(n)$	$k_n = O(n)$	$k_n = O(n)$ for the J and GS method and $k_n = O(\sqrt{n})$ for SOR with optimal ω .
lyche-numerical-linear-algebra- F03495	269	■ 1.000	$k_{\{n\}} = O(\sqrt{n})$	$k_n = O(\sqrt{n})$	$k_n = O(n)$ for the J and GS method and $k_n = O(\sqrt{n})$ for SOR with optimal ω .
lyche-numerical-linear-algebra- F03496	269	■ 1.000	$\sqrt{n}$	$\sqrt{n}$	The choice of tol will only influence the constants multiplying n or $\sqrt{n}$.
lyche-numerical-linear-algebra- F03497	269	■ 1.000	$O(k_{\{n\}} \times n)$	$O(k_n \times n)$	$O(k_n \times n)$. Therefore the number of arithmetic operations for the J and GS
lyche-numerical-linear-algebra- F03498	270	■ 1.000	$\boldsymbol{G} \in \mathbb{C}^{n \times n}$	$\boldsymbol{G} \in \mathbb{C}^{n \times n}$	$\{\boldsymbol{x}_k\}$ of vectors in $\mathbb{C}^n$. For a general $\boldsymbol{G} \in \mathbb{C}^{n \times n}$ and $\boldsymbol{c} \in \mathbb{C}^n$ a solution of $\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$
lyche-numerical-linear-algebra- F03499	270	■ 1.000	$\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$	$\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$	$\{\boldsymbol{x}_k\}$ of vectors in $\mathbb{C}^n$. For a general $\boldsymbol{G} \in \mathbb{C}^{n \times n}$ and $\boldsymbol{c} \in \mathbb{C}^n$ a solution of $\boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$
lyche-numerical-linear-algebra- F03500	270	■ 1.000	$\boldsymbol{I} - \boldsymbol{G}$	$\boldsymbol{I} - \boldsymbol{G}$	is called a fixed-point . The fixed-point is unique if $\boldsymbol{I} - \boldsymbol{G}$ is nonsingular.
lyche-numerical-linear-algebra- F03501	270	■ 1.000	$\lim_{k \rightarrow \infty} \boldsymbol{x}_{\{k\}} = \boldsymbol{x}$	$\lim_{k \rightarrow \infty} \boldsymbol{x}_k = \boldsymbol{x}$	If $\lim_{k \rightarrow \infty} \boldsymbol{x}_k = \boldsymbol{x}$ for some $\boldsymbol{x} \in \mathbb{C}^n$ then $\boldsymbol{x}$ is a fixed point since
lyche-numerical-linear-algebra- F03502	270	■ 1.000	$\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ in the equivalent form $\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{M}^{-1} \boldsymbol{b}$ or $\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} + \boldsymbol{M}^{-1} \boldsymbol{b}$.
lyche-numerical-linear-algebra- F03503	270	■ 1.000	$\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} + \boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} + \boldsymbol{M}^{-1} \boldsymbol{b}$	$\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$ in the equivalent form $\boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} = \boldsymbol{M}^{-1} \boldsymbol{b}$ or $\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{M}^{-1} \boldsymbol{A} \boldsymbol{x} + \boldsymbol{M}^{-1} \boldsymbol{b}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F03504	270	1.000	$\boldsymbol{A}=\boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}-\boldsymbol{A}_{\boldsymbol{R}}$	$\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$, where $-\boldsymbol{A}_L$, $\boldsymbol{D}$, and $-\boldsymbol{A}_R$ are the lower,
lyche-numerical-linear-algebra- F03505	270	1.000	$-\boldsymbol{A}_{\boldsymbol{L}}, \boldsymbol{D}$	$-\boldsymbol{A}_L, \boldsymbol{D}$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$, where $-\boldsymbol{A}_L$, $\boldsymbol{D}$, and $-\boldsymbol{A}_R$ are the lower,
lyche-numerical-linear-algebra- F03506	270	1.000	$-\boldsymbol{A}_{\boldsymbol{R}}$	$-\boldsymbol{A}_R$	of three matrices, $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{A}_L - \boldsymbol{A}_R$, where $-\boldsymbol{A}_L$, $\boldsymbol{D}$, and $-\boldsymbol{A}_R$ are the lower,
lyche-numerical-linear-algebra- F03507	270	1.000	$\boldsymbol{D}:=\operatorname{diag}\left(a_{11}, \ldots, a_{nn}\right)$	$\boldsymbol{D} := \operatorname{diag}\left(a_{11}, \ldots, a_{nn}\right)$	diagonal, and upper part of $\boldsymbol{A}$, respectively. Thus $\boldsymbol{D} := \operatorname{diag}(a_{11}, \ldots, a_{nn})$,
lyche-numerical-linear-algebra- F03508	271	0.753	$\boldsymbol{M}_{\boldsymbol{J}}, \boldsymbol{M}_{\boldsymbol{1}}$	$\boldsymbol{M}_J, \boldsymbol{M}_1$	$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J , GS and SOR methods are given by
lyche-numerical-linear-algebra- F03509	271	0.753	$\boldsymbol{M}_{\omega}$	$\boldsymbol{M}_\omega$	$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J , GS and SOR methods are given by
lyche-numerical-linear-algebra- F03510	271	0.753	J, G S	J, GS	$\boldsymbol{M}_J$, $\boldsymbol{M}_1$ and $\boldsymbol{M}_\omega$ for the J , GS and SOR methods are given by
lyche-numerical-linear-algebra- F03511	271	0.955	$\boldsymbol{D} \boldsymbol{x}_{k+1}=(\boldsymbol{D}-\boldsymbol{A}) \boldsymbol{x}_k+\boldsymbol{b}$	$\boldsymbol{D} \boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$	obtain the matrix form $\boldsymbol{D} \boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$ showing that $\boldsymbol{M}_J = \boldsymbol{D}$.
lyche-numerical-linear-algebra- F03512	271	0.955	$\boldsymbol{M}_{\boldsymbol{J}}=\boldsymbol{D}$	$\boldsymbol{M}_J = \boldsymbol{D}$	obtain the matrix form $\boldsymbol{D} \boldsymbol{x}_{k+1} = (\boldsymbol{D} - \boldsymbol{A}) \boldsymbol{x}_k + \boldsymbol{b}$ showing that $\boldsymbol{M}_J = \boldsymbol{D}$.
lyche-numerical-linear-algebra- F03513	271	1.000	$\boldsymbol{A}_{\boldsymbol{L}} \boldsymbol{x}_{k+1}$	$\boldsymbol{A}_L \boldsymbol{x}_{k+1}$	Dividing both sides by ω and moving $\boldsymbol{A}_L \boldsymbol{x}_{k+1}$ to the left hand side this takes the
lyche-numerical-linear-algebra- F03514	271	1.000	$\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right) \boldsymbol{x}_{k+1}=\boldsymbol{A}_{\boldsymbol{R}} \boldsymbol{x}_k+\boldsymbol{b}+\left(\omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{x}_k$	$\left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L\right) \boldsymbol{x}_{k+1} = \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b} + \left(\omega^{-1} - 1\right) \boldsymbol{D} \boldsymbol{x}_k$	form $\left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L\right) \boldsymbol{x}_{k+1} = \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b} + \left(\omega^{-1} - 1\right) \boldsymbol{D} \boldsymbol{x}_k$ showing that $\boldsymbol{M}_\omega =$
lyche-numerical-linear-algebra- F03515	271	1.000	$\boldsymbol{M}_{\omega}=\left(\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}\right) \boldsymbol{x}_{k+1}=\boldsymbol{A}_{\boldsymbol{R}} \boldsymbol{x}_k+\boldsymbol{b}+\left(\omega^{-1}-1\right) \boldsymbol{D} \boldsymbol{x}_k$	$\boldsymbol{M}_\omega = \left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L\right) \boldsymbol{x}_{k+1} = \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b} + \left(\omega^{-1} - 1\right) \boldsymbol{D} \boldsymbol{x}_k$	form $\left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L\right) \boldsymbol{x}_{k+1} = \boldsymbol{A}_R \boldsymbol{x}_k + \boldsymbol{b} + \left(\omega^{-1} - 1\right) \boldsymbol{D} \boldsymbol{x}_k$ showing that $\boldsymbol{M}_\omega =$
lyche-numerical-linear-algebra- F03516	271	1.000	$\omega^{-1} \boldsymbol{D}-\boldsymbol{A}_{\boldsymbol{L}}$	$\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L$	$\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L$. We obtain $\boldsymbol{M}_1$ by letting $\omega = 1$ in $\boldsymbol{M}_\omega$.
lyche-numerical-linear-algebra- F03517	271	1.000	$\boldsymbol{M}_{\boldsymbol{1}}$	$\boldsymbol{M}_1$	$\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L$. We obtain $\boldsymbol{M}_1$ by letting $\omega = 1$ in $\boldsymbol{M}_\omega$.
lyche-numerical-linear-algebra- F03518	271	1.000	$\boldsymbol{G}_{\omega}=\boldsymbol{I}-\boldsymbol{M}_{\omega}^{-1} \boldsymbol{A}$	$\boldsymbol{G}_\omega = \boldsymbol{I} - \boldsymbol{M}_\omega^{-1} \boldsymbol{A}$	The iteration matrix $\boldsymbol{G}_\omega = \boldsymbol{I} - \boldsymbol{M}_\omega^{-1} \boldsymbol{A}$ is given by
lyche-numerical-linear-algebra- F03519	272	1.000	$\boldsymbol{x}_{k+1}(1)$	$\boldsymbol{x}_{k+1}(1)$	Substituting the value of $\boldsymbol{x}_{k+1}(1)$ from the first equation into the second equation
lyche-numerical-linear-algebra- F03520	272	1.000	$\boldsymbol{x}_{k+1}:=\boldsymbol{G} \boldsymbol{x}_k+\boldsymbol{c}$	$\boldsymbol{x}_{k+1} := \boldsymbol{G} \boldsymbol{x}_k + \boldsymbol{c}$	for methods of the form $\boldsymbol{x}_{k+1} := \boldsymbol{G} \boldsymbol{x}_k + \boldsymbol{c}$.
lyche-numerical-linear-algebra- F03521	272	1.000	$\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$	$\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$	In the following we assume that $\boldsymbol{G} \in \mathbb{C}^{n \times n}, \boldsymbol{c} \in \mathbb{C}^n$ and $\boldsymbol{I} - \boldsymbol{G}$ is nonsingular.
lyche-numerical-linear-algebra- F03522	272	1.000	$\lim _{k \rightarrow \infty} \boldsymbol{G}^k=\mathbf{0}$	$\lim_{k \rightarrow \infty} \boldsymbol{G}^k = \mathbf{0}$	$\boldsymbol{x}_{k+1} := \boldsymbol{G} \boldsymbol{x}_k + \boldsymbol{c}$ converges if and only if $\lim_{k \rightarrow \infty} \boldsymbol{G}^k = \mathbf{0}$.
lyche-numerical-linear-algebra- F03523	272	0.991	$\boldsymbol{x}_{k+1}=\boldsymbol{G} \boldsymbol{x}_k+\boldsymbol{c}$	$\boldsymbol{x}_{k+1} = \boldsymbol{G} \boldsymbol{x}_k + \boldsymbol{c}$	$\boldsymbol{x} = \boldsymbol{G} \boldsymbol{x} + \boldsymbol{c}$ from $\boldsymbol{x}_{k+1} = \boldsymbol{G} \boldsymbol{x}_k + \boldsymbol{c}$ The vector $\boldsymbol{c}$ cancels and we obtain $\boldsymbol{x}_{k+1} - \boldsymbol{x} =$
lyche-numerical-linear-algebra- F03524	272	0.991	$\boldsymbol{x}_{k+1}-\boldsymbol{x}=\mathbf{0}$	$\boldsymbol{x}_{k+1} - \boldsymbol{x} =$	$\boldsymbol{x} = \boldsymbol{G} \boldsymbol{x} + \boldsymbol{c}$ from $\boldsymbol{x}_{k+1} = \boldsymbol{G} \boldsymbol{x}_k + \boldsymbol{c}$. The vector $\boldsymbol{c}$ cancels and we obtain $\boldsymbol{x}_{k+1} - \boldsymbol{x} =$
lyche-numerical-linear-algebra- F03525	272	1.000	$\boldsymbol{G}\left(\boldsymbol{x}_k-\boldsymbol{x}\right)$	$\boldsymbol{G}\left(\boldsymbol{x}_k - \boldsymbol{x}\right)$	$\boldsymbol{G}\left(\boldsymbol{x}_k - \boldsymbol{x}\right)$. By induction on k
lyche-numerical-linear-algebra- F03526	272	0.999	$\boldsymbol{x}_k-\boldsymbol{x} \rightarrow \mathbf{0}$	$\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$	Clearly $\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$ if $\boldsymbol{G}^k \rightarrow \mathbf{0}$ and the method converges. For the converse we let
lyche-numerical-linear-algebra- F03527	272	0.999	$\boldsymbol{G}^k \rightarrow \mathbf{0}$	$\boldsymbol{G}^k \rightarrow \mathbf{0}$	Clearly $\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$ if $\boldsymbol{G}^k \rightarrow \mathbf{0}$ and the method converges. For the converse we let
lyche-numerical-linear-algebra- F03528	272	1.000	$\boldsymbol{x}_0-\boldsymbol{x}=\boldsymbol{e}_j$	$\boldsymbol{x}_0 - \boldsymbol{x} = \boldsymbol{e}_j$	$\boldsymbol{x}$ be as before and choose $\boldsymbol{x}_0 - \boldsymbol{x} = \boldsymbol{e}_j$, the j th unit vector for $j = 1, \ldots, n$. Since
lyche-numerical-linear-algebra- F03529	272	1.000	$\boldsymbol{G}^k \boldsymbol{e}_j \rightarrow \mathbf{0}$	$\boldsymbol{G}^k \boldsymbol{e}_j \rightarrow \mathbf{0}$	$\boldsymbol{x}_k - \boldsymbol{x} \rightarrow \mathbf{0}$ for any $\boldsymbol{x}_0$ we have $\boldsymbol{G}^k \boldsymbol{e}_j \rightarrow \mathbf{0}$ for $j = 1, \ldots, n$ which implies that

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lyche-numerical-linear-algebra- F03530	272	■ 1.000	$\ \boldsymbol{G}\ < 1$	$\ \boldsymbol{G}\ < 1$	Theorem 12.3 (Sufficient Condition for Convergence) <i>If $\ \boldsymbol{G}\ < 1$ then the</i>
lyche-numerical-linear-algebra- F03531	273	■ 1.000	$\rho(\boldsymbol{G}) < 1$	$\rho(\boldsymbol{G}) < 1$	<i>and only if $\rho(\boldsymbol{G}) < 1$.</i>
lyche-numerical-linear-algebra- F03532	273	■ 1.000	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$	$\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$	adding a multiple of the residual vector $\boldsymbol{r}_k := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$. Note that we do not need
lyche-numerical-linear-algebra- F03533	273	■ 0.992	$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$	$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$	values $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$ then the R method given by $\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$
lyche-numerical-linear-algebra- F03534	273	■ 0.992	$\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$	$\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$	values $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$ then the R method given by $\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$
lyche-numerical-linear-algebra- F03535	273	■ 1.000	$0 < \alpha < 2 / \lambda_1$	$0 < \alpha < 2 / \lambda_1$	<i>converges if and only if $0 < \alpha < 2 / \lambda_1$. Moreover,</i>
lyche-numerical-linear-algebra- F03536	274	■ 1.000	$\boldsymbol{I} - \alpha \boldsymbol{A}$	$\boldsymbol{I} - \alpha \boldsymbol{A}$	Proof The eigenvalues of $\boldsymbol{I} - \alpha \boldsymbol{A}$ are $1 - \alpha \lambda_j$, $j = 1, \dots, n$. We have
lyche-numerical-linear-algebra- F03537	274	■ 1.000	$1 - \alpha \lambda_j$, $j = 1, \dots, n$	$1 - \alpha \lambda_j$, $j = 1, \dots, n$	Proof The eigenvalues of $\boldsymbol{I} - \alpha \boldsymbol{A}$ are $1 - \alpha \lambda_j$, $j = 1, \dots, n$. We have
lyche-numerical-linear-algebra- F03538	274	■ 1.000	$1 - \alpha \lambda_n = \alpha \lambda_1 - 1$	$1 - \alpha \lambda_n = \alpha \lambda_1 - 1$	see Fig. 12.1. Clearly $1 - \alpha \lambda_n = \alpha \lambda_1 - 1$ for $\alpha = \alpha_o$ and
lyche-numerical-linear-algebra- F03539	274	■ 1.000	$\alpha = \alpha_o$	$\alpha = \alpha_o$	see Fig. 12.1. Clearly $1 - \alpha \lambda_n = \alpha \lambda_1 - 1$ for $\alpha = \alpha_o$ and
lyche-numerical-linear-algebra- F03540	274	■ 1.000	$\rho_\alpha < 1$	$\rho_\alpha < 1$	We have $\rho_\alpha < 1$ if and only if $\alpha > 0$ and $\alpha \lambda_1 - 1 < 1$ showing convergence if
lyche-numerical-linear-algebra- F03541	274	■ 1.000	$\alpha \lambda_1 - 1 < 1$	$\alpha \lambda_1 - 1 < 1$	We have $\rho_\alpha < 1$ if and only if $\alpha > 0$ and $\alpha \lambda_1 - 1 < 1$ showing convergence if
lyche-numerical-linear-algebra- F03542	274	■ 1.000	$\rho_\alpha > \rho_{\alpha_o}$	$\rho_\alpha > \rho_{\alpha_o}$	and only if $0 < \alpha < 2 / \lambda_1$ and $\rho_\alpha > \rho_{\alpha_o}$ for $\alpha \leq 0$ and $\alpha \geq 2 / \lambda_1$. Finally, if
lyche-numerical-linear-algebra- F03543	274	■ 1.000	$\alpha \leq 0$	$\alpha \leq 0$	and only if $0 < \alpha < 2 / \lambda_1$ and $\rho_\alpha > \rho_{\alpha_o}$ for $\alpha \leq 0$ and $\alpha \geq 2 / \lambda_1$. Finally, if
lyche-numerical-linear-algebra- F03544	274	■ 1.000	$\alpha \geq 2 / \lambda_1$	$\alpha \geq 2 / \lambda_1$	and only if $0 < \alpha < 2 / \lambda_1$ and $\rho_\alpha > \rho_{\alpha_o}$ for $\alpha \leq 0$ and $\alpha \geq 2 / \lambda_1$. Finally, if
lyche-numerical-linear-algebra- F03545	274	■ 1.000	$0 < \alpha < \alpha_o$	$0 < \alpha < \alpha_o$	$0 < \alpha < \alpha_o$ then $\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$ and if $\alpha_o < \alpha < 2 / \lambda_1$ then
lyche-numerical-linear-algebra- F03546	274	■ 1.000	$\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$	$\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$	$0 < \alpha < \alpha_o$ then $\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$ and if $\alpha_o < \alpha < 2 / \lambda_1$ then
lyche-numerical-linear-algebra- F03547	274	■ 1.000	$\alpha_o < \alpha < 2 / \lambda_1$	$\alpha_o < \alpha < 2 / \lambda_1$	$0 < \alpha < \alpha_o$ then $\rho_\alpha = 1 - \alpha \lambda_n > 1 - \alpha_o \lambda_n = \rho_{\alpha_o}$ and if $\alpha_o < \alpha < 2 / \lambda_1$ then
lyche-numerical-linear-algebra- F03548	274	■ 0.998	$\rho_\alpha = \alpha \lambda_1 - 1 > \alpha_o \lambda_1 - 1 = \rho_{\alpha_o}$	$\rho_\alpha = \alpha \lambda_1 - 1 > \alpha_o \lambda_1 - 1 = \rho_{\alpha_o}$	$\rho_\alpha = \alpha \lambda_1 - 1 > \alpha_o \lambda_1 - 1 = \rho_{\alpha_o}$.
lyche-numerical-linear-algebra- F03549	274	■ 0.964	$\mathbf{R}$	$\mathbf{R}$	Corollary 12.1 (Rate of Convergence for the R Method) <i>Suppose $\boldsymbol{A}$ is positive</i>
lyche-numerical-linear-algebra- F03550	274	■ 0.991	$\lambda_{\max}$	$\lambda_{\max}$	<i>definite with largest and smallest eigenvalue $\lambda_{\max}$ and $\lambda_{\min}$, respectively. Richard-</i>
lyche-numerical-linear-algebra- F03551	274	■ 0.999	$0 < \alpha < 2 / \lambda_{\max}$	$0 < \alpha < 2 / \lambda_{\max}$	<i>son's method $\boldsymbol{x}_{k+1} = (\boldsymbol{I} - \alpha \boldsymbol{A})\boldsymbol{x}_k + \boldsymbol{b}$ converges if and only if $0 < \alpha < 2 / \lambda_{\max}$,</i>
lyche-numerical-linear-algebra- F03552	274	■ 1.000	$\alpha = \alpha_o := \frac{2}{\lambda_{\max} + \lambda_{\min}}$	$\alpha = \alpha_o := \frac{2}{\lambda_{\max} + \lambda_{\min}}$	<i>With $\alpha = \alpha_o := \frac{2}{\lambda_{\max} + \lambda_{\min}}$ we have the error estimate</i>
lyche-numerical-linear-algebra- F03553	274	■ 0.960	$\kappa := \lambda_{\max} / \lambda_{\min}$	$\kappa := \lambda_{\max} / \lambda_{\min}$	<i>where $\kappa := \lambda_{\max} / \lambda_{\min}$ is the spectral condition number of $\boldsymbol{A}$.</i>
lyche-numerical-linear-algebra- F03554	274	■ 0.988	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} -$	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} -$	Proof The spectral norm $\ \cdot\ _2$ is consistent and therefore $\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} -$
lyche-numerical-linear-algebra- F03555	274	■ 0.988	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} -$	$\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} -$	Proof The spectral norm $\ \cdot\ _2$ is consistent and therefore $\ \boldsymbol{x}_k - \boldsymbol{x}\ _2 \leq \ \boldsymbol{I} -$
lyche-numerical-linear-algebra- F03556	274	■ 0.994	$\alpha_o \ \boldsymbol{A}\ _2^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2$	$\alpha_o \ \boldsymbol{A}\ _2^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2$	$\alpha_o \ \boldsymbol{A}\ _2^k \ \boldsymbol{x}_0 - \boldsymbol{x}\ _2$. But for a positive definite matrix the spectral norm is equal to

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lyche-numerical-linear-algebra-275 F03557	275	1.000	$\omega \in (0, 2)$	$\omega \in (0, 2)$	The condition $\omega \in (0, 2)$ is necessary for convergence of the SOR method.
lyche-numerical-linear-algebra-275 F03558	275	1.000	$\mathbf{x}_{k+1} = \omega \left(\mathbf{A}_L \mathbf{x}_{k+1} + \mathbf{A}_R \mathbf{x}_k + \mathbf{b} \right) + (1 - \omega) \mathbf{x}_k$	$\mathbf{D} \mathbf{x}_{k+1} = \omega (\mathbf{A}_L \mathbf{x}_{k+1} + \mathbf{A}_R \mathbf{x}_k + \mathbf{b}) + (1 - \omega) \mathbf{D} \mathbf{x}_k$	Proof We have (cf. (12.13)) $\mathbf{D} \mathbf{x}_{k+1} = \omega (\mathbf{A}_L \mathbf{x}_{k+1} + \mathbf{A}_R \mathbf{x}_k + \mathbf{b}) + (1 - \omega) \mathbf{D} \mathbf{x}_k$
lyche-numerical-linear-algebra-275 F03559	275	1.000	$\mathbf{x}_{k+1} = \omega \left(\mathbf{L} \mathbf{x}_{k+1} + \mathbf{R} \mathbf{x}_k + \mathbf{D}^{-1} \mathbf{b} \right) + (1 - \omega) \mathbf{x}_k$	$\mathbf{x}_{k+1} = \omega (\mathbf{L} \mathbf{x}_{k+1} + \mathbf{R} \mathbf{x}_k + \mathbf{D}^{-1} \mathbf{b}) + (1 - \omega) \mathbf{x}_k$	or $\mathbf{x}_{k+1} = \omega (\mathbf{L} \mathbf{x}_{k+1} + \mathbf{R} \mathbf{x}_k + \mathbf{D}^{-1} \mathbf{b}) + (1 - \omega) \mathbf{x}_k$, where $\mathbf{L} := \mathbf{D}^{-1} \mathbf{A}_L$ and
lyche-numerical-linear-algebra-275 F03560	275	1.000	$\mathbf{L} := \mathbf{D}^{-1} \mathbf{A}_L$	$\mathbf{L} := \mathbf{D}^{-1} \mathbf{A}_L$	or $\mathbf{x}_{k+1} = \omega (\mathbf{L} \mathbf{x}_{k+1} + \mathbf{R} \mathbf{x}_k + \mathbf{D}^{-1} \mathbf{b}) + (1 - \omega) \mathbf{x}_k$, where $\mathbf{L} := \mathbf{D}^{-1} \mathbf{A}_L$ and
lyche-numerical-linear-algebra-275 F03561	275	1.000	$\mathbf{R} := \mathbf{D}^{-1} \mathbf{A}_R$	$\mathbf{R} := \mathbf{D}^{-1} \mathbf{A}_R$	$\mathbf{R} := \mathbf{D}^{-1} \mathbf{A}_R$. Thus $(\mathbf{I} - \omega \mathbf{L}) \mathbf{x}_{k+1} = (\omega \mathbf{R} + (1 - \omega) \mathbf{I}) \mathbf{x}_k + \omega \mathbf{D}^{-1} \mathbf{b}$ so the
lyche-numerical-linear-algebra-275 F03562	275	1.000	$(\mathbf{I} - \omega \mathbf{L}) \mathbf{x}_{k+1} = (\omega \mathbf{R} + (1 - \omega) \mathbf{I}) \mathbf{x}_k + \omega \mathbf{D}^{-1} \mathbf{b}$	$(\mathbf{I} - \omega \mathbf{L}) \mathbf{x}_{k+1} = (\omega \mathbf{R} + (1 - \omega) \mathbf{I}) \mathbf{x}_k + \omega \mathbf{D}^{-1} \mathbf{b}$	$\mathbf{R} := \mathbf{D}^{-1} \mathbf{A}_R$. Thus $(\mathbf{I} - \omega \mathbf{L}) \mathbf{x}_{k+1} = (\omega \mathbf{R} + (1 - \omega) \mathbf{I}) \mathbf{x}_k + \omega \mathbf{D}^{-1} \mathbf{b}$ so the
lyche-numerical-linear-algebra-275 F03563	275	1.000	$\mathbf{G}_\omega$	$\mathbf{G}_\omega$	We next compute the determinant of $\mathbf{G}_\omega$. Since $\mathbf{I} - \omega \mathbf{L}$ is lower triangular with
lyche-numerical-linear-algebra-275 F03564	275	1.000	$\mathbf{I} - \omega \mathbf{L}$	$\mathbf{I} - \omega \mathbf{L}$	We next compute the determinant of $\mathbf{G}_\omega$. Since $\mathbf{I} - \omega \mathbf{L}$ is lower triangular with
lyche-numerical-linear-algebra-275 F03565	275	1.000	$\omega \mathbf{R} + (1 - \omega) \mathbf{I}$	$\omega \mathbf{R} + (1 - \omega) \mathbf{I}$	the determinant of this matrix is equal to one. The matrix $\omega \mathbf{R} + (1 - \omega) \mathbf{I}$ is upper
lyche-numerical-linear-algebra-275 F03566	275	1.000	$1 - \omega$	$1 - \omega$	triangular with $1 - \omega$ on the diagonal and therefore its determinant equals $(1 - \omega)^n$.
lyche-numerical-linear-algebra-275 F03567	275	1.000	$(1 - \omega)^n$	$(1 - \omega)^n$	triangular with $1 - \omega$ on the diagonal and therefore its determinant equals $(1 - \omega)^n$.
lyche-numerical-linear-algebra-275 F03568	275	0.998	$\det(\mathbf{G}_\omega) = (1 - \omega)^n$	$\det(\mathbf{G}_\omega) = (1 - \omega)^n$	It follows that $\det(\mathbf{G}_\omega) = (1 - \omega)^n$. Since the determinant of a matrix equals the
lyche-numerical-linear-algebra-275 F03569	275	1.000	$ \lambda \geq 1 - \omega $	$ \lambda \geq 1 - \omega $	product of its eigenvalues we must have $ \lambda \geq 1 - \omega $ for at least one eigenvalue λ
lyche-numerical-linear-algebra-275 F03570	275	1.000	$\rho(\mathbf{G}_\omega) \geq \omega - 1 $	$\rho(\mathbf{G}_\omega) \geq \omega - 1 $	of $\mathbf{G}_\omega$ and we conclude that $\rho(\mathbf{G}_\omega) \geq \omega - 1 $. But then $\rho(\mathbf{G}_\omega) \geq 1$ if ω is not in
lyche-numerical-linear-algebra-275 F03571	275	1.000	$\rho(\mathbf{G}_\omega) \geq 1$	$\rho(\mathbf{G}_\omega) \geq 1$	of $\mathbf{G}_\omega$ and we conclude that $\rho(\mathbf{G}_\omega) \geq \omega - 1 $. But then $\rho(\mathbf{G}_\omega) \geq 1$ if ω is not in
lyche-numerical-linear-algebra-275 F03572	275	1.000	$ \lambda < 1$	$ \lambda < 1$	to show that $ \lambda < 1$. The following identity will be shown below:
lyche-numerical-linear-algebra-275 F03573	275	1.000	$\mathbf{x}^* \mathbf{A} \mathbf{x}$	$\mathbf{x}^* \mathbf{A} \mathbf{x}$	where $\mathbf{D} := \text{diag}(a_{11}, \dots, a_{nn})$. Now $\mathbf{x}^* \mathbf{A} \mathbf{x}$ and $\mathbf{x}^* \mathbf{D} \mathbf{x}$ are positive for all nonzero
lyche-numerical-linear-algebra-275 F03574	275	1.000	$\mathbf{x}^* \mathbf{D} \mathbf{x}$	$\mathbf{x}^* \mathbf{D} \mathbf{x}$	where $\mathbf{D} := \text{diag}(a_{11}, \dots, a_{nn})$. Now $\mathbf{x}^* \mathbf{A} \mathbf{x}$ and $\mathbf{x}^* \mathbf{D} \mathbf{x}$ are positive for all nonzero
lyche-numerical-linear-algebra-275 F03575	275	1.000	$\mathbf{x} \in \mathbb{C}^n$	$\mathbf{x} \in \mathbb{C}^n$	$\mathbf{x} \in \mathbb{C}^n$ since a positive definite matrix has positive diagonal elements $a_{ii} =$
lyche-numerical-linear-algebra-275 F03576	275	1.000	$a_{ii} =$	$a_{ii} =$	$\mathbf{x} \in \mathbb{C}^n$ since a positive definite matrix has positive diagonal elements $a_{ii} =$
lyche-numerical-linear-algebra-275 F03577	275	1.000	$\mathbf{e}_i^T \mathbf{A} \mathbf{e}_i > 0$	$\mathbf{e}_i^T \mathbf{A} \mathbf{e}_i > 0$	$\mathbf{e}_i^T \mathbf{A} \mathbf{e}_i > 0$. It follows that the left hand side of (12.22) is nonnegative and then
lyche-numerical-linear-algebra-275 F03578	275	1.000	$ \lambda \leq 1$	$ \lambda \leq 1$	the right hand side must be nonnegative as well. This implies $ \lambda \leq 1$. If $ \lambda = 1$
lyche-numerical-linear-algebra-276 F03579	276	1.000	$\mathbf{G}_\omega \mathbf{x} = \lambda \mathbf{x}$	$\mathbf{G}_\omega \mathbf{x} = \lambda \mathbf{x}$	and the eigenpair equation $\mathbf{G}_\omega \mathbf{x} = \lambda \mathbf{x}$ can be written $\mathbf{x} - (\omega^{-1} \mathbf{D} - \mathbf{A}_L)^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$
lyche-numerical-linear-algebra-276 F03580	276	1.000	$\mathbf{x} - (\omega^{-1} \mathbf{D} - \mathbf{A}_L)^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$	$\mathbf{x} - (\omega^{-1} \mathbf{D} - \mathbf{A}_L)^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$	and the eigenpair equation $\mathbf{G}_\omega \mathbf{x} = \lambda \mathbf{x}$ can be written $\mathbf{x} - (\omega^{-1} \mathbf{D} - \mathbf{A}_L)^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$

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lyche-numerical-linear-algebra- F03581	276	■ 0.976	$\boldsymbol{A} \boldsymbol{x} \neq \mathbf{0}$	$\boldsymbol{Ax} \neq \mathbf{0}$	Now $\boldsymbol{Ax} \neq \mathbf{0}$ implies that $\lambda \neq 1$.
lyche-numerical-linear-algebra- F03582	276	■ 0.976	$\lambda \neq 1$	$\lambda \neq 1$	Now $\boldsymbol{Ax} \neq \mathbf{0}$ implies that $\lambda \neq 1$.
lyche-numerical-linear-algebra- F03583	276	■ 1.000	$(\boldsymbol{A} \boldsymbol{x})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} = \boldsymbol{y}^* \boldsymbol{y}^* \left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L \right)^* \boldsymbol{y} =$	$(\boldsymbol{Ax})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} =$	Again using (12.23), (12.24) and adding $(\boldsymbol{Ax})^* \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_L)^* \boldsymbol{y} =$
lyche-numerical-linear-algebra- F03584	276	■ 1.000	$\boldsymbol{y}^* \left(\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R \right) \boldsymbol{y}$	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$ and $\boldsymbol{y}^* (\lambda \boldsymbol{Ax}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$ we find
lyche-numerical-linear-algebra- F03585	276	■ 1.000	$\boldsymbol{y}^* (\lambda \boldsymbol{Ax}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$	$\boldsymbol{y}^* (\lambda \boldsymbol{Ax}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$	$\boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} - \boldsymbol{A}_R) \boldsymbol{y}$ and $\boldsymbol{y}^* (\lambda \boldsymbol{Ax}) = \boldsymbol{y}^* \boldsymbol{E} \boldsymbol{y} = \boldsymbol{y}^* (\omega^{-1} \boldsymbol{D} + \boldsymbol{A}_R - \boldsymbol{D}) \boldsymbol{y}$ we find
lyche-numerical-linear-algebra- F03586	276	■ 1.000	$(\boldsymbol{A} \boldsymbol{x})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$	$(\boldsymbol{Ax})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$	Since $(\boldsymbol{Ax})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$ and $\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$ this also equals
lyche-numerical-linear-algebra- F03587	276	■ 1.000	$\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$	$\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$	Since $(\boldsymbol{Ax})^* = \boldsymbol{x}^* \boldsymbol{A}^* = \boldsymbol{x}^* \boldsymbol{A}, \boldsymbol{y} := (1 - \lambda) \boldsymbol{x}$ and $\boldsymbol{y}^* = (1 - \bar{\lambda}) \boldsymbol{x}^*$ this also equals
lyche-numerical-linear-algebra- F03588	276	■ 0.997	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A}/4$	$\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A}/4$	Consider first the J method. The matrix $\boldsymbol{G}_J = \boldsymbol{I} - \boldsymbol{D}^{-1} \boldsymbol{A} = \boldsymbol{I} - \boldsymbol{A}/4$ has eigenvalues
lyche-numerical-linear-algebra- F03589	277	■ 1.000	$\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$	$\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$	It follows that $\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$. Since $\boldsymbol{G}_J$ is symmetric it is normal, and the
lyche-numerical-linear-algebra- F03590	277	■ 1.000	$\boldsymbol{G}_J$	$\boldsymbol{G}_J$	It follows that $\rho(\boldsymbol{G}_J) = \cos(\pi h) < 1$. Since $\boldsymbol{G}_J$ is symmetric it is normal, and the
lyche-numerical-linear-algebra- F03591	277	■ 0.932	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$	The R method given by $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$ with $\alpha = 2/(\lambda_{\max} + \lambda_{\min}) = 1/4$
lyche-numerical-linear-algebra- F03592	277	■ 0.932	$\alpha = 2/(\lambda_{\max} + \lambda_{\min}) = 1/4$	$\alpha = 2/(\lambda_{\max} + \lambda_{\min}) = 1/4$	The R method given by $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{r}_k$ with $\alpha = 2/(\lambda_{\max} + \lambda_{\min}) = 1/4$
lyche-numerical-linear-algebra- F03593	277	■ 1.000	κ	κ	from Corollary 12.1 with κ given by (10.20).
lyche-numerical-linear-algebra- F03594	277	■ 1.000	$\rho(\boldsymbol{G}_\omega)$	$\rho(\boldsymbol{G}_\omega)$	For the SOR method it is possible to explicitly determine $\rho(\boldsymbol{G}_\omega)$ for any $\omega \in$
lyche-numerical-linear-algebra- F03595	277	■ 1.000	$\omega \in$	$\omega \in$	For the SOR method it is possible to explicitly determine $\rho(\boldsymbol{G}_\omega)$ for any $\omega \in$
lyche-numerical-linear-algebra- F03596	277	■ 1.000	$\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$	$\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$	where $\beta := \rho(\boldsymbol{G}_J) = \cos(\pi h)$ and
lyche-numerical-linear-algebra- F03597	277	■ 1.000	ω^*	ω^*	decreases monotonically to the minimum ω^* . Then it increases linearly to the value
lyche-numerical-linear-algebra- F03598	277	■ 1.000	$\omega = 2$	$\omega = 2$	one for $\omega = 2$. We call ω^* the optimal relaxation parameter .
lyche-numerical-linear-algebra- F03599	277	■ 1.000	$\beta = \cos(\pi h)$	$\beta = \cos(\pi h)$	For the discrete Poisson problem we have $\beta = \cos(\pi h)$ and it follows from
lyche-numerical-linear-algebra- F03600	277	■ 0.998	$\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$	$\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$	Letting $\omega = 1$ in (12.27) we find $\rho(\boldsymbol{G}_1) = \beta^2 = \rho(\boldsymbol{G}_J)^2 = \cos^2(\pi h)$ for the GS
lyche-numerical-linear-algebra- F03601	277	■ 0.999	$\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$	$\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$	The values of $\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$, and $\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$ are shown in Table 12.2 for
lyche-numerical-linear-algebra- F03602	277	■ 0.999	$\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$	$\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$	The values of $\rho(\boldsymbol{G}_J), \rho(\boldsymbol{G}_1)$, and $\rho(\boldsymbol{G}_{\omega^*}) = \omega^* - 1$ are shown in Table 12.2 for
lyche-numerical-linear-algebra- F03603	277	■ 1.000	$\rho(\boldsymbol{G})^{k_n} \leq$	$\rho(\boldsymbol{G})^{k_n} \leq$	$n = 100$ and $n = 2500$. We also show the smallest integer k_n such that $\rho(\boldsymbol{G})^{k_n} \leq$

Identifier	Page	Conf.	LaTeX source	Rendered	Image																				
lyche-numerical-linear-algebra F03604	278	■ 1.000	$\mathrm{n}=100$	$n = 100$	<table><tr><td></td><td>n=100</td><td>n = 2500</td><td>k_{100}</td><td>k_{2500}</td></tr><tr><td>J</td><td>0.959493</td><td>0.998103</td><td>446</td><td>9703</td></tr><tr><td>GS</td><td>0.920627</td><td>0.99621</td><td>223</td><td>4852</td></tr><tr><td>SOR</td><td>0.56039</td><td>0.88402</td><td>32</td><td>150</td></tr></table>		n=100	n = 2500	k_{100}	k_{2500}	J	0.959493	0.998103	446	9703	GS	0.920627	0.99621	223	4852	SOR	0.56039	0.88402	32	150
	n=100	n = 2500	k_{100}	k_{2500}																					
J	0.959493	0.998103	446	9703																					
GS	0.920627	0.99621	223	4852																					
SOR	0.56039	0.88402	32	150																					
lyche-numerical-linear-algebra F03605	278	■ 1.000	$\mathrm{n}=2500$	$n = 2500$	<table><tr><td></td><td>n=100</td><td>n = 2500</td><td>k_{100}</td><td>k_{2500}</td></tr><tr><td>J</td><td>0.959493</td><td>0.998103</td><td>446</td><td>9703</td></tr><tr><td>GS</td><td>0.920627</td><td>0.99621</td><td>223</td><td>4852</td></tr><tr><td>SOR</td><td>0.56039</td><td>0.88402</td><td>32</td><td>150</td></tr></table>		n=100	n = 2500	k_{100}	k_{2500}	J	0.959493	0.998103	446	9703	GS	0.920627	0.99621	223	4852	SOR	0.56039	0.88402	32	150
	n=100	n = 2500	k_{100}	k_{2500}																					
J	0.959493	0.998103	446	9703																					
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SOR	0.56039	0.88402	32	150																					
lyche-numerical-linear-algebra F03606	278	■ 1.000	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x})$	$\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x})$	know how fast the iterative method converges. Recall that $\boldsymbol{x}_k - \boldsymbol{x} = \boldsymbol{G}^k (\boldsymbol{x}_0 - \boldsymbol{x})$.																				
lyche-numerical-linear-algebra F03607	278	■ 1.000	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} =$	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} =$	For the last formula we apply Theorem 12.13 which says that $\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} =$																				
lyche-numerical-linear-algebra F03608	278	■ 1.000	$\rho(\boldsymbol{G})$	$\rho(\boldsymbol{G})$	$\rho(\boldsymbol{G})$. For Jacobi's method and the spectral norm we have $\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$ (cf.																				
lyche-numerical-linear-algebra F03609	278	■ 1.000	$\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$	$\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$	$\rho(\boldsymbol{G})$. For Jacobi's method and the spectral norm we have $\ \boldsymbol{G}_J^k\ _2 = \rho(\boldsymbol{G}_J)^k$ (cf.																				
lyche-numerical-linear-algebra F03610	278	■ 1.000	$\rho(\boldsymbol{G}) = 1 - \eta$	$\rho(\boldsymbol{G}) = 1 - \eta$	Lemma 12.1 (Number of Iterations) Suppose $\rho(\boldsymbol{G}) = 1 - \eta$ for some $0 < \eta < 1$																				
lyche-numerical-linear-algebra F03611	278	■ 1.000	$0 < \eta < 1$	$0 < \eta < 1$	Lemma 12.1 (Number of Iterations) Suppose $\rho(\boldsymbol{G}) = 1 - \eta$ for some $0 < \eta < 1$																				
lyche-numerical-linear-algebra F03612	278	■ 1.000	$s \in \mathbb{N}$	$s \in \mathbb{N}$	and let $s \in \mathbb{N}$. Then																				
lyche-numerical-linear-algebra F03613	278	■ 1.000	$\rho(\boldsymbol{G})^k \leq 10^{-s}$	$\rho(\boldsymbol{G})^k \leq 10^{-s}$	is an estimate for the smallest number of iterations k so that $\rho(\boldsymbol{G})^k \leq 10^{-s}$.																				
lyche-numerical-linear-algebra F03614	279	■ 1.000	$\tilde{k}$	$\tilde{k}$	Proof The estimate $\tilde{k}$ is an approximate solution of the equation $\rho(\boldsymbol{G})^{\tilde{k}} = 10^{-s}$.																				
lyche-numerical-linear-algebra F03615	279	■ 1.000	$\rho(\boldsymbol{G})^k = 10^{-s}$	$\rho(\boldsymbol{G})^k = 10^{-s}$	Proof The estimate $\tilde{k}$ is an approximate solution of the equation $\rho(\boldsymbol{G})^{\tilde{k}} = 10^{-s}$.																				
lyche-numerical-linear-algebra F03616	279	■ 1.000	$-\log(1 - \eta) \approx \eta$	$-\log(1 - \eta) \approx \eta$	Thus, since $-\log(1 - \eta) \approx \eta$ when η is small																				
lyche-numerical-linear-algebra F03617	279	■ 1.000	η	η	Thus, since $-\log(1 - \eta) \approx \eta$ when η is small																				
lyche-numerical-linear-algebra F03618	279	■ 0.996	$\rho(\boldsymbol{G}_J) = \cos(\pi h) = 1 - \eta, \eta = 1 - \cos(\pi h) = \frac{1}{2}\pi^2 h^2 + O(h^4) =$	$\rho(\boldsymbol{G}_J) = \cos(\pi h) = 1 - \eta, \eta = 1 - \cos(\pi h) = \frac{1}{2}\pi^2 h^2 + O(h^4) =$	• R and J: $\rho(\boldsymbol{G}_J) = \cos(\pi h) = 1 - \eta, \eta = 1 - \cos(\pi h) = \frac{1}{2}\pi^2 h^2 + O(h^4) =$																				
lyche-numerical-linear-algebra F03619	279	■ 1.000	$\frac{\pi^2}{2}/n + O(n^{-2})$	$\frac{\pi^2}{2}/n + O(n^{-2})$	$\frac{\pi^2}{2}/n + O(n^{-2})$. Thus,																				
lyche-numerical-linear-algebra F03620	279	■ 0.512	$\rho(\boldsymbol{G}_1) = \cos^2(\pi h) = 1 - \eta, \eta = 1 - \cos^2(\pi h) = \sin^2 \pi h = \pi^2 h^2 +$	$\rho(\boldsymbol{G}_1) = \cos^2(\pi h) = 1 - \eta, \eta = 1 - \cos^2(\pi h) = \sin^2 \pi h = \pi^2 h^2 +$	• GS: $\rho(\boldsymbol{G}_1) = \cos^2(\pi h) = 1 - \eta, \eta = 1 - \cos^2(\pi h) = \sin^2 \pi h = \pi^2 h^2 +$																				
lyche-numerical-linear-algebra F03621	279	■ 1.000	$O(h^4) = \pi^2/n + O(n^{-2})$	$O(h^4) = \pi^2/n + O(n^{-2})$	$O(h^4) = \pi^2/n + O(n^{-2})$. Thus,																				
lyche-numerical-linear-algebra F03622	279	■ 0.906	$\rho(\boldsymbol{G}_{\omega^*}) = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)} = 1 - 2\pi h + O(h^2)$	$\rho(\boldsymbol{G}_{\omega^*}) = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)} = 1 - 2\pi h + O(h^2)$	• SOR: $\rho(\boldsymbol{G}_{\omega^*}) = \frac{1 - \sin(\pi h)}{1 + \sin(\pi h)} = 1 - 2\pi h + O(h^2)$. Thus,																				
lyche-numerical-linear-algebra F03623	279	■ 1.000	$\boldsymbol{G}^k$	$\boldsymbol{G}^k$	1. The convergence depends on the behavior of the powers $\boldsymbol{G}^k$ as k increases.																				
lyche-numerical-linear-algebra F03624	279	■ 0.998	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} = \rho(\boldsymbol{G})$	$\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} = \rho(\boldsymbol{G})$	2. The convergence $\lim_{k \rightarrow \infty} \ \boldsymbol{G}^k\ ^{1/k} = \rho(\boldsymbol{G})$ can be quite slow (cf. Exer-																				
lyche-numerical-linear-algebra F03625	279	■ 1.000	$\ \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ $	$\ \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ $	One possibility is to choose a vector norm, keep track of $\ \boldsymbol{x}_{k+1} - \boldsymbol{x}_k\ $, and stop																				
lyche-numerical-linear-algebra F03626	280	■ 1.000	$\ \boldsymbol{x}_k - \boldsymbol{x}\ $	$\ \boldsymbol{x}_k - \boldsymbol{x}\ $	when this number is sufficiently small. The following result indicates that $\ \boldsymbol{x}_k - \boldsymbol{x}\ $																				

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lyche-numerical-linear-algebra-F03627	280	■ 1.000	$\ \boldsymbol{G}\ $	$\ \boldsymbol{G}\ $	can be quite large if $\ \boldsymbol{G}\ $ is close to one.
lyche-numerical-linear-algebra-F03628	280	■ 1.000	$\boldsymbol{x}_k = \boldsymbol{G}\boldsymbol{x}_{k-1} + \boldsymbol{c}, \boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$	$\boldsymbol{x}_k = \boldsymbol{G}\boldsymbol{x}_{k-1} + \boldsymbol{c}, \boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$	Lemma 12.2 (Be Careful When Stopping) If $\boldsymbol{x}_k = \boldsymbol{G}\boldsymbol{x}_{k-1} + \boldsymbol{c}, \boldsymbol{x} = \boldsymbol{G}\boldsymbol{x} + \boldsymbol{c}$ and
lyche-numerical-linear-algebra-F03629	280	■ 0.998	$(1 - \ \boldsymbol{G}\) \ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}\ \ \boldsymbol{x}_{k-1} - \boldsymbol{x}_k\ $	$(1 - \ \boldsymbol{G}\) \ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}\ \ \boldsymbol{x}_{k-1} - \boldsymbol{x}_k\ $	Thus $(1 - \ \boldsymbol{G}\) \ \boldsymbol{x}_k - \boldsymbol{x}\ \leq \ \boldsymbol{G}\ \ \boldsymbol{x}_{k-1} - \boldsymbol{x}_k\ $ which implies (12.32).
lyche-numerical-linear-algebra-F03630	280	■ 0.997	$\boldsymbol{M}\boldsymbol{x}_{k+1} = \boldsymbol{M}\boldsymbol{x}_k - \boldsymbol{A}\boldsymbol{x}_k + \boldsymbol{b}$	$\boldsymbol{M}\boldsymbol{x}_{k+1} = \boldsymbol{M}\boldsymbol{x}_k - \boldsymbol{A}\boldsymbol{x}_k + \boldsymbol{b}$	splitting matrix of the method then by (12.9) we have $\boldsymbol{M}\boldsymbol{x}_{k+1} = \boldsymbol{M}\boldsymbol{x}_k - \boldsymbol{A}\boldsymbol{x}_k + \boldsymbol{b}$.
lyche-numerical-linear-algebra-F03631	280	■ 1.000	$\{\boldsymbol{A}^k\}$	$\{\boldsymbol{A}^k\}$	sequence $\{\boldsymbol{A}^k\}$ of powers of $\boldsymbol{A}$. We want to know when this sequence converges to
lyche-numerical-linear-algebra-F03632	281	■ 1.000	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$	$\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$	Theorem 12.10 (When Is $\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$?) For any $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ we have
lyche-numerical-linear-algebra-F03633	281	■ 1.000	$\rho(\boldsymbol{A})$	$\rho(\boldsymbol{A})$	where $\rho(\boldsymbol{A})$ is the spectral radius of $\boldsymbol{A}$ given by (12.17).
lyche-numerical-linear-algebra-F03634	281	■ 0.913	$\rho(\boldsymbol{A}) < 1$	$\rho(\boldsymbol{A}) < 1$	Clearly $\rho(\boldsymbol{A}) < 1$ is a necessary condition for $\lim_{k \rightarrow \infty} \boldsymbol{A}^k = \mathbf{0}$. For if $(\lambda, \boldsymbol{x})$ is
lyche-numerical-linear-algebra-F03635	281	■ 1.000	$ \lambda \geq 1$	$ \lambda \geq 1$	an eigenpair of $\boldsymbol{A}$ with $ \lambda \geq 1$ and $\ \boldsymbol{x}\ _2 = 1$ then $\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$, and this implies
lyche-numerical-linear-algebra-F03636	281	■ 1.000	$\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$	$\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$	an eigenpair of $\boldsymbol{A}$ with $ \lambda \geq 1$ and $\ \boldsymbol{x}\ _2 = 1$ then $\boldsymbol{A}^k \boldsymbol{x} = \lambda^k \boldsymbol{x}$, and this implies
lyche-numerical-linear-algebra-F03637	281	■ 1.000	$\ \boldsymbol{A}^k\ _2 \geq \ \boldsymbol{A}^k \boldsymbol{x}\ _2 = \ \lambda^k \boldsymbol{x}\ _2 = \lambda ^k$	$\ \boldsymbol{A}^k\ _2 \geq \ \boldsymbol{A}^k \boldsymbol{x}\ _2 = \ \lambda^k \boldsymbol{x}\ _2 = \lambda ^k$	$\ \boldsymbol{A}^k\ _2 \geq \ \boldsymbol{A}^k \boldsymbol{x}\ _2 = \ \lambda^k \boldsymbol{x}\ _2 = \lambda ^k$, and it follows that $\boldsymbol{A}^k$ does not tend to zero.
lyche-numerical-linear-algebra-F03638	281	■ 1.000	$\ \boldsymbol{A}\ < 1$	$\ \boldsymbol{A}\ < 1$	norm on $\mathbb{C}^{n \times n}$ such that $\ \boldsymbol{A}\ < 1$ and then use Theorems 12.2 and 12.3.
lyche-numerical-linear-algebra-F03639	281	■ 1.000	$\rho(\boldsymbol{A}) \leq$	$\rho(\boldsymbol{A}) \leq$	matrix norm $\ \cdot\ $ that is consistent on $\mathbb{C}^{n \times n}$ and any $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ we have $\rho(\boldsymbol{A}) \leq$
lyche-numerical-linear-algebra-F03640	281	■ 1.000	$\ \boldsymbol{A}\ $	$\ \boldsymbol{A}\ $	$\ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F03641	281	■ 0.679	$\boldsymbol{A}, \ \cdot\ $	$\boldsymbol{A}, \ \cdot\ $	Proof Let $(\lambda, \boldsymbol{x})$ be an eigenpair for $\boldsymbol{A}$, $\ \cdot\ $ a consistent matrix norm on $\mathbb{C}^{n \times n}$
lyche-numerical-linear-algebra-F03642	281	■ 1.000	$\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$	$\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$	and define $\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$. Then $\lambda \boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$, which implies $ \lambda \ \boldsymbol{X}\ =$
lyche-numerical-linear-algebra-F03643	281	■ 1.000	$\lambda \boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$	$\lambda \boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$	and define $\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$. Then $\lambda \boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$, which implies $ \lambda \ \boldsymbol{X}\ =$
lyche-numerical-linear-algebra-F03644	281	■ 1.000	$ \lambda \ \boldsymbol{X}\ =$	$ \lambda \ \boldsymbol{X}\ =$	and define $\boldsymbol{X} := [\boldsymbol{x}, \dots, \boldsymbol{x}] \in \mathbb{C}^{n \times n}$. Then $\lambda \boldsymbol{X} = \boldsymbol{A}\boldsymbol{X}$, which implies $ \lambda \ \boldsymbol{X}\ =$
lyche-numerical-linear-algebra-F03645	281	■ 0.999	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A}\boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A}\boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A}\boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $. Since $\ \boldsymbol{X}\ \neq 0$ we obtain $ \lambda \leq \ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F03646	281	■ 0.999	$\ \boldsymbol{X}\ \neq 0$	$\ \boldsymbol{X}\ \neq 0$	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A}\boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $. Since $\ \boldsymbol{X}\ \neq 0$ we obtain $ \lambda \leq \ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F03647	281	■ 0.999	$ \lambda \leq \ \boldsymbol{A}\ $	$ \lambda \leq \ \boldsymbol{A}\ $	$\ \lambda \boldsymbol{X}\ = \ \boldsymbol{A}\boldsymbol{X}\ \leq \ \boldsymbol{A}\ \ \boldsymbol{X}\ $. Since $\ \boldsymbol{X}\ \neq 0$ we obtain $ \lambda \leq \ \boldsymbol{A}\ $.
lyche-numerical-linear-algebra-F03648	281	■ 1.000	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ \leq \rho(\boldsymbol{A}) + \epsilon$	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ \leq \rho(\boldsymbol{A}) + \epsilon$	that $\rho(\boldsymbol{A}) < \ \boldsymbol{A}\ < \rho(\boldsymbol{A}) + \epsilon$.
lyche-numerical-linear-algebra-F03649	281	■ 1.000	$\boldsymbol{R} = [r_{ij}]$	$\boldsymbol{R} = [r_{ij}]$	there is a unitary matrix $\boldsymbol{U}$ and an upper triangular matrix $\boldsymbol{R} = [r_{ij}]$ such that
lyche-numerical-linear-algebra-F03650	281	■ 1.000	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$. For $t > 0$ we define $\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}$, and note that
lyche-numerical-linear-algebra-F03651	281	■ 1.000	$\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}$	$\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}$	$\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{R}$. For $t > 0$ we define $\boldsymbol{D}_t := \text{diag}(t, t^2, \dots, t^n) \in \mathbb{R}^{n \times n}$, and note that

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lyche-numerical-linear-algebra- F03652	281	0.718	(i, j)	(i, j)	the (i, j) element in $\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$ is given by $t^{i-j} r_{ij}$ for all i, j . For $n = 3$
lyche-numerical-linear-algebra- F03653	281	0.718	$\boldsymbol{D}_{\boldsymbol{t}} \boldsymbol{R} \boldsymbol{D}_{\boldsymbol{t}}^{-1}$	$\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$	the (i, j) element in $\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$ is given by $t^{i-j} r_{ij}$ for all i, j . For $n = 3$
lyche-numerical-linear-algebra- F03654	281	0.718	$t^{i-j} r_{ij}$	$t^{i-j} r_{ij}$	the (i, j) element in $\boldsymbol{D}_t \boldsymbol{R} \boldsymbol{D}_t^{-1}$ is given by $t^{i-j} r_{ij}$ for all i, j . For $n = 3$
lyche-numerical-linear-algebra- F03655	281	0.877	$\ \boldsymbol{B}\ _t := \ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1}\ _1$	$\ \boldsymbol{B}\ _t := \ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1}\ _1$	$\ \boldsymbol{B}\ _t := \ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1}\ _1$. We leave it as an exercise to show that $\ \cdot\ _t$ is a
lyche-numerical-linear-algebra- F03656	281	0.877	$\ \cdot\ _t$	$\ \cdot\ _t$	$\ \boldsymbol{B}\ _t := \ \boldsymbol{D}_t \boldsymbol{U}^* \boldsymbol{B} \boldsymbol{U} \boldsymbol{D}_t^{-1}\ _1$. We leave it as an exercise to show that $\ \cdot\ _t$ is a
lyche-numerical-linear-algebra- F03657	281	1.000	$\ \boldsymbol{B}\ := \ \boldsymbol{B}\ _t$	$\ \boldsymbol{B}\ := \ \boldsymbol{B}\ _t$	consistent matrix norm on $\mathbb{C}^{n \times n}$. We define $\ \boldsymbol{B}\ := \ \boldsymbol{B}\ _t$, where t is chosen so
lyche-numerical-linear-algebra- F03658	282	1.000	λ^k	λ^k	then λ^k is an eigenvalue of $\boldsymbol{A}^k$ for any $k \in \mathbb{N}$. By Theorem 12.11 we then obtain
lyche-numerical-linear-algebra- F03659	282	1.000	$\rho(\boldsymbol{A}^k) = \rho(\boldsymbol{A})^{1/k}$	$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) \leq \ \boldsymbol{A}^k\ $	$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) < \ \boldsymbol{A}^k\ $ for any $k \in \mathbb{N}$ so that $\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ ^{1/k}$. Let $\epsilon > 0$ and
lyche-numerical-linear-algebra- F03660	282	1.000	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ ^{1/k}$	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}^k\ ^{1/k}$	$\rho(\boldsymbol{A})^k = \rho(\boldsymbol{A}^k) \leq \ \boldsymbol{A}^k\ $ for any $k \in \mathbb{N}$ so that $\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ ^{1/k}$. Let $\epsilon > 0$ and
lyche-numerical-linear-algebra- F03661	282	1.000	$\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$	$\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$	consider the matrix $\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$. Then $\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$
lyche-numerical-linear-algebra- F03662	282	1.000	$\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$	$\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$	consider the matrix $\boldsymbol{B} := (\rho(\boldsymbol{A}) + \epsilon)^{-1} \boldsymbol{A}$. Then $\rho(\boldsymbol{B}) = \rho(\boldsymbol{A})/(\rho(\boldsymbol{A}) + \epsilon) < 1$
lyche-numerical-linear-algebra- F03663	282	1.000	$\ \boldsymbol{B}^k\ \rightarrow 0$	$\ \boldsymbol{B}^k\ \rightarrow 0$	and $\ \boldsymbol{B}^k\ \rightarrow 0$ by Theorem 12.10 as $k \rightarrow \infty$. Choose $N \in \mathbb{N}$ such that $\ \boldsymbol{B}^k\ < 1$
lyche-numerical-linear-algebra- F03664	282	1.000	$k \rightarrow \infty$	$k \rightarrow \infty$	and $\ \boldsymbol{B}^k\ \rightarrow 0$ by Theorem 12.10 as $k \rightarrow \infty$. Choose $N \in \mathbb{N}$ such that $\ \boldsymbol{B}^k\ < 1$
lyche-numerical-linear-algebra- F03665	282	1.000	$\ \boldsymbol{B}^k\ < 1$	$\ \boldsymbol{B}^k\ < 1$	and $\ \boldsymbol{B}^k\ \rightarrow 0$ by Theorem 12.10 as $k \rightarrow \infty$. Choose $N \in \mathbb{N}$ such that $\ \boldsymbol{B}^k\ < 1$
lyche-numerical-linear-algebra- F03666	282	1.000	$k \geq N$	$k \geq N$	for all $k \geq N$. Then for $k \geq N$
lyche-numerical-linear-algebra- F03667	282	1.000	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ ^{1/k} \leq \rho(\boldsymbol{A}) + \epsilon$	$\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}^k\ ^{1/k} \leq \rho(\boldsymbol{A}) + \epsilon$	We have shown that $\rho(\boldsymbol{A}) \leq \ \boldsymbol{A}\ ^{1/k} \leq \rho(\boldsymbol{A}) + \epsilon$ for $k \geq N$. Since ϵ is arbitrary
lyche-numerical-linear-algebra- F03668	282	1.000	$\sum_{k=0}^{\infty} \boldsymbol{A}_k$	$\sum_{k=0}^{\infty} \boldsymbol{A}_k$	Consider an infinite series $\sum_{k=0}^{\infty} \boldsymbol{A}_k$ of matrices in $\mathbb{C}^{n \times n}$. We say that the series
lyche-numerical-linear-algebra- F03669	282	1.000	$\{\boldsymbol{S}_m\}$	$\{\boldsymbol{S}_m\}$	converges if the sequence of partial sums $\{\boldsymbol{S}_m\}$ given by $\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$ converges.
lyche-numerical-linear-algebra- F03670	282	1.000	$\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$	$\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$	converges if the sequence of partial sums $\{\boldsymbol{S}_m\}$ given by $\boldsymbol{S}_m = \sum_{k=0}^m \boldsymbol{A}_k$ converges.
lyche-numerical-linear-algebra- F03671	282	1.000	$\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$	$\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$	there exists an integer N so that $\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$ for all $l > m \geq N$.
lyche-numerical-linear-algebra- F03672	282	1.000	$l > m \geq N$	$l > m \geq N$	there exists an integer N so that $\ \boldsymbol{S}_l - \boldsymbol{S}_m\ < \epsilon$ for all $l > m \geq N$.
lyche-numerical-linear-algebra- F03673	282	1.000	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	$\sum_{k=0}^{\infty} \boldsymbol{B}^k$	1. The series $\sum_{k=0}^{\infty} \boldsymbol{B}^k$ converges if and only if $\rho(\boldsymbol{B}) < 1$.
lyche-numerical-linear-algebra- F03674	282	1.000	$\rho(\boldsymbol{B}) < 1$	$\rho(\boldsymbol{B}) < 1$	1. The series $\sum_{k=0}^{\infty} \boldsymbol{B}^k$ converges if and only if $\rho(\boldsymbol{B}) < 1$.
lyche-numerical-linear-algebra- F03675	282	1.000	$(\boldsymbol{I} - \boldsymbol{B})$	$(\boldsymbol{I} - \boldsymbol{B})$	2. If $\rho(\boldsymbol{B}) < 1$ then $(\boldsymbol{I} - \boldsymbol{B})$ is nonsingular and $(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$.
lyche-numerical-linear-algebra- F03676	282	1.000	$(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$	$(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$	2. If $\rho(\boldsymbol{B}) < 1$ then $(\boldsymbol{I} - \boldsymbol{B})$ is nonsingular and $(\boldsymbol{I} - \boldsymbol{B})^{-1} = \sum_{k=0}^{\infty} \boldsymbol{B}^k$.

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lyche-numerical-linear-algebra-282-F03677	282	■ 1.000	$\ \boldsymbol{B}\ < 1$	$\ \boldsymbol{B}\ < 1$	3. If $\ \boldsymbol{B}\ < 1$ for some consistent matrix norm $\ \cdot\ $ on $\mathbb{C}^{n \times n}$ then
lyche-numerical-linear-algebra-283-F03678	283	■ 1.000	$\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$	$\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$	1. Suppose $\rho(\boldsymbol{B}) < 1$. We show that $\boldsymbol{S}_m := \sum_{k=0}^m \boldsymbol{B}^k$ is a Cauchy sequence and
lyche-numerical-linear-algebra-283-F03679	283	■ 1.000	$l > m$	$l > m$	norm $\ \cdot\ $ on $\mathbb{C}^{n \times n}$ such that $\ \boldsymbol{B}\ < 1$. Then for $l > m$
lyche-numerical-linear-algebra-283-F03680	283	■ 1.000	$\frac{\ \boldsymbol{B}\ ^{N+1}}{1 - \ \boldsymbol{B}\ } < \epsilon$	$\frac{\ \boldsymbol{B}\ ^{N+1}}{1 - \ \boldsymbol{B}\ } < \epsilon$	But then $\{\boldsymbol{S}_m\}$ is a Cauchy sequence provided N is such that $\frac{\ \boldsymbol{B}\ ^{N+1}}{1 - \ \boldsymbol{B}\ } < \epsilon$.
lyche-numerical-linear-algebra-283-F03681	283	■ 0.876	$\boldsymbol{S}_m \boldsymbol{x} =$	$\boldsymbol{S}_m \boldsymbol{x} =$	Conversely, suppose $(\lambda, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{B}$ with $ \lambda \geq 1$. We find $\boldsymbol{S}_m \boldsymbol{x} =$
lyche-numerical-linear-algebra-283-F03682	283	■ 1.000	$\sum_{k=0}^m \boldsymbol{B}^k \boldsymbol{x} = \left(\sum_{k=0}^m \lambda^k \right) \boldsymbol{x}$	$\sum_{k=0}^m \boldsymbol{B}^k \boldsymbol{x} = \left(\sum_{k=0}^m \lambda^k \right) \boldsymbol{x}$	$\sum_{k=0}^m \boldsymbol{B}^k \boldsymbol{x} = \left(\sum_{k=0}^m \lambda^k \right) \boldsymbol{x}$. Since λ^k does not tend to zero the series $\sum_{k=0}^{\infty} \lambda^k$ is
lyche-numerical-linear-algebra-283-F03683	283	■ 1.000	$\sum_{k=0}^{\infty} \lambda^k$	$\sum_{k=0}^{\infty} \lambda^k$	$\sum_{k=0}^m \boldsymbol{B}^k \boldsymbol{x} = \left(\sum_{k=0}^m \lambda^k \right) \boldsymbol{x}$. Since λ^k does not tend to zero the series $\sum_{k=0}^{\infty} \lambda^k$ is
lyche-numerical-linear-algebra-283-F03684	283	■ 1.000	$\{\boldsymbol{S}_m \boldsymbol{x}\}$	$\{\boldsymbol{S}_m \boldsymbol{x}\}$	not convergent and therefore $\{\boldsymbol{S}_m \boldsymbol{x}\}$ and hence $\{\boldsymbol{S}_m\}$ does not converge.
lyche-numerical-linear-algebra-283-F03685	283	■ 1.000	$\boldsymbol{B}^{m+1} \rightarrow 0$	$\boldsymbol{B}^{m+1} \rightarrow 0$	Since $\rho(\boldsymbol{B}) < 1$ we conclude that $\boldsymbol{B}^{m+1} \rightarrow 0$ and hence taking limits in (12.36)
lyche-numerical-linear-algebra-283-F03686	283	■ 0.947	$\left(\sum_{k=0}^{\infty} \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I}$	$\left(\sum_{k=0}^{\infty} \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I}$	we obtain $\left(\sum_{k=0}^{\infty} \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) = \boldsymbol{I}$ which completes the proof of 2.
lyche-numerical-linear-algebra-283-F03687	283	■ 0.999	$\left\ \left(\sum_{k=0}^{\infty} \boldsymbol{B}^k \right) (\boldsymbol{I} - \boldsymbol{B}) - \boldsymbol{I} \right\ \leq \sum_{k=0}^{\infty} \ \boldsymbol{B}\ ^k = \frac{1}{1 - \ \boldsymbol{B}\ }$	$\ (I - B)^{-1}\ = \left\ \sum_{k=0}^{\infty} B^k \right\ \leq \sum_{k=0}^{\infty} \ B\ ^k = \frac{1}{1 - \ B\ }$	3. By 2: $\ (I - B)^{-1}\ = \ \sum_{k=0}^{\infty} B^k\ \leq \sum_{k=0}^{\infty} \ B\ ^k = \frac{1}{1 - \ B\ }$.
lyche-numerical-linear-algebra-283-F03688	283	■ 1.000	$\boldsymbol{\omega}$	$\boldsymbol{\omega}$	12.5 The Optimal SOR Parameter $\boldsymbol{\omega}$
lyche-numerical-linear-algebra-283-F03689	283	■ 0.546	$\boldsymbol{G}_J \boldsymbol{v} = \mu \boldsymbol{v}$	$\boldsymbol{G}_J \boldsymbol{v} = \mu \boldsymbol{v}$	write these equations using the matrix formulation $\boldsymbol{T}\boldsymbol{V} + \boldsymbol{V}\boldsymbol{T} = h^2 \boldsymbol{F}$. If $\boldsymbol{G}_J \boldsymbol{v} = \mu \boldsymbol{v}$
lyche-numerical-linear-algebra-283-F03690	283	■ 1.000	$\boldsymbol{v} := \text{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$	$\boldsymbol{v} := \text{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$	where $\boldsymbol{v} := \text{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03691	283	■ 1.000	$v_{i,j} = 0$	$v_{i,j} = 0$	where $\boldsymbol{v} := \text{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03692	283	■ 1.000	$i \in \{0, m+1\}$	$i \in \{0, m+1\}$	where $\boldsymbol{v} := \text{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03693	283	■ 1.000	$j \in \{0, m+1\}$	$j \in \{0, m+1\}$	where $\boldsymbol{v} := \text{vec}(\boldsymbol{V}) \in \mathbb{R}^{m^2}$ and $v_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-283-F03694	283	■ 0.588	$(\lambda, \boldsymbol{w})$	$(\lambda, \boldsymbol{w})$	Suppose $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_{\omega}$. By (12.21) $(\boldsymbol{I} - \omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R} + (1 -$
lyche-numerical-linear-algebra-283-F03695	283	■ 0.588	$(12.21)(\boldsymbol{I} - \omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R} + (1 -$	$(12.21)(\boldsymbol{I} - \omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R} + (1 -$	Suppose $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_{\omega}$. By (12.21) $(\boldsymbol{I} - \omega \boldsymbol{L})^{-1}(\omega \boldsymbol{R} + (1 -$
lyche-numerical-linear-algebra-283-F03696	283	■ 1.000	$\omega) \boldsymbol{I}) \boldsymbol{w} = \lambda \boldsymbol{w}$	$\omega) \boldsymbol{I}) \boldsymbol{w} = \lambda \boldsymbol{w}$	$\omega) \boldsymbol{I}) \boldsymbol{w} = \lambda \boldsymbol{w}$ or
lyche-numerical-linear-algebra-284-F03697	284	■ 1.000	$\boldsymbol{w} = \text{vec}(\boldsymbol{W})$	$\boldsymbol{w} = \text{vec}(\boldsymbol{W})$	Let $\boldsymbol{w} = \text{vec}(\boldsymbol{W})$, where $\boldsymbol{W} \in \mathbb{C}^{m \times m}$. Then (12.38) can be written
lyche-numerical-linear-algebra-284-F03698	284	■ 1.000	$\boldsymbol{W} \in \mathbb{C}^{m \times m}$	$\boldsymbol{W} \in \mathbb{C}^{m \times m}$	Let $\boldsymbol{w} = \text{vec}(\boldsymbol{W})$, where $\boldsymbol{W} \in \mathbb{C}^{m \times m}$. Then (12.38) can be written
lyche-numerical-linear-algebra-284-F03699	284	■ 1.000	$w_{i,j} = 0$	$w_{i,j} = 0$	where $w_{i,j} = 0$ if $i \in \{0, m+1\}$ or $j \in \{0, m+1\}$.
lyche-numerical-linear-algebra-284-F03700	284	■ 0.430	$(\mu, \boldsymbol{v})$	$(\mu, \boldsymbol{v})$	Proof Suppose $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_{\omega}$. We claim that $(\mu, \boldsymbol{v})$ is an eigenpair

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lyche-numerical-linear-algebra- F03701	284	1.000	$\boldsymbol{v}=(\boldsymbol{V})$	$\boldsymbol{v} = (\boldsymbol{V})$	for $\boldsymbol{G}_J$, where μ is given by (12.40) and $\boldsymbol{v} = (\boldsymbol{V})$ with $v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$.
lyche-numerical-linear-algebra- F03702	284	1.000	$v_{\{i, j\}}:=\lambda^{\{-(i+j) / 2\}} w_{\{i, j\}}$	$v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$	for $\boldsymbol{G}_J$, where μ is given by (12.40) and $\boldsymbol{v} = (\boldsymbol{V})$ with $v_{i,j} := \lambda^{-(i+j)/2} w_{i,j}$.
lyche-numerical-linear-algebra- F03703	284	1.000	$w_{\{i, j\}}$	$w_{i,j}$	Indeed, replacing $w_{i,j}$ by $\lambda^{(i+j)/2} v_{i,j}$ in (12.39) and cancelling the common factor
lyche-numerical-linear-algebra- F03704	284	1.000	$\lambda^{\{i+j\} / 2} v_{\{i, j\}}$	$\lambda^{(i+j)/2} v_{i,j}$	Indeed, replacing $w_{i,j}$ by $\lambda^{(i+j)/2} v_{i,j}$ in (12.39) and cancelling the common factor
lyche-numerical-linear-algebra- F03705	284	1.000	$\lambda^{\{i+j\} / 2}$	$\lambda^{(i+j)/2}$	$\lambda^{(i+j)/2}$ we obtain
lyche-numerical-linear-algebra- F03706	284	0.999	$\boldsymbol{v}=: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W}=\operatorname{vec}(\boldsymbol{W})$	$\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$	(12.41). We define as before $\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$ with $w_{i,j} := \lambda^{(i+j)/2} v_{i,j}$.
lyche-numerical-linear-algebra- F03707	284	0.999	$w_{\{i, j\}}:=\lambda^{\{i+j\} / 2} v_{\{i, j\}}$	$w_{i,j} := \lambda^{(i+j)/2} v_{i,j}$	(12.41). We define as before $\boldsymbol{v} =: \operatorname{vec}(\boldsymbol{V}), \boldsymbol{W} = \operatorname{vec}(\boldsymbol{W})$ with $w_{i,j} := \lambda^{(i+j)/2} v_{i,j}$.
lyche-numerical-linear-algebra- F03708	284	1.000	$\lambda^{\{-(i+j) / 2\}}$	$\lambda^{-(i+j)/2}$	Inserting this in (12.37) and canceling $\lambda^{-(i+j)/2}$ we obtain
lyche-numerical-linear-algebra- F03709	284	1.000	$\omega \lambda^{1 / 2}$	$\omega \lambda^{1/2}$	Multiplying by $\omega \lambda^{1/2}$ we obtain
lyche-numerical-linear-algebra- F03710	284	0.952	$\omega \mu \lambda^{1 / 2}=\lambda+\omega-1$	$\omega \mu \lambda^{1/2} = \lambda + \omega - 1$	Thus, if $\omega \mu \lambda^{1/2} = \lambda + \omega - 1$ then by (12.39) $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$.
lyche-numerical-linear-algebra- F03711	284	0.952	$\lambda, \boldsymbol{w}$	$\lambda, \boldsymbol{w}$	Thus, if $\omega \mu \lambda^{1/2} = \lambda + \omega - 1$ then by (12.39) $(\lambda, \boldsymbol{w})$ is an eigenpair for $\boldsymbol{G}_\omega$.
lyche-numerical-linear-algebra- F03712	285	1.000	$\rho\left(\boldsymbol{G}_{\omega}\right)= \lambda(\mu) $	$\rho\left(\boldsymbol{G}_\omega\right)= \lambda(\mu) $	that $\rho\left(\boldsymbol{G}_\omega\right)= \lambda(\mu) $, where $\lambda(\mu)$ is an eigenvalue of $\boldsymbol{G}_\omega$ satisfying (12.41) for
lyche-numerical-linear-algebra- F03713	285	1.000	$\lambda(\mu)$	$\lambda(\mu)$	that $\rho\left(\boldsymbol{G}_\omega\right)= \lambda(\mu) $, where $\lambda(\mu)$ is an eigenvalue of $\boldsymbol{G}_\omega$ satisfying (12.41) for
lyche-numerical-linear-algebra- F03714	285	1.000	$\frac{1}{2} \cos (j \pi h)+\frac{1}{2} \cos (k \pi h)$	$\frac{1}{2} \cos (j \pi h) + \frac{1}{2} \cos (k \pi h)$	some eigenvalue μ of $\boldsymbol{G}_J$. The eigenvalues of $\boldsymbol{G}_J$ are $\frac{1}{2} \cos (j \pi h) + \frac{1}{2} \cos (k \pi h)$,
lyche-numerical-linear-algebra- F03715	285	1.000	$-\mu$	$-\mu$	$j, k=1, \dots, m$, so μ is real and both μ and $-\mu$ are eigenvalues. Thus, to compute
lyche-numerical-linear-algebra- F03716	285	1.000	$\lambda=\lambda(\mu)$	$\lambda = \lambda(\mu)$	(12.41) for $\lambda = \lambda(\mu)$ gives
lyche-numerical-linear-algebra- F03717	285	0.998	$d(0)=4>0$	$d(0)=4>0$	Moreover $d(0)=4>0$ and $d(2)=4 \mu^2-4<0$. As a function of ω , $\lambda(\mu)$ changes
lyche-numerical-linear-algebra- F03718	285	0.998	$d(2)=4 \mu^2-4<0$	$d(2)=4 \mu^2-4<0$	Moreover $d(0)=4>0$ and $d(2)=4 \mu^2-4<0$. As a function of ω , $\lambda(\mu)$ changes
lyche-numerical-linear-algebra- F03719	285	0.998	$\omega, \lambda(\mu)$	$\omega, \lambda(\mu)$	Moreover $d(0)=4>0$ and $d(2)=4 \mu^2-4<0$. As a function of ω , $\lambda(\mu)$ changes
lyche-numerical-linear-algebra- F03720	285	0.618	$d(\omega)=0$	$d(\omega)=0$	from real to complex when $d(\omega)=0$. The root in $(0, 2)$ is
lyche-numerical-linear-algebra- F03721	285	1.000	$\tilde{\omega}(\mu)$	$\tilde{\omega}(\mu)$	Both $\lambda(\mu)$ and $\tilde{\omega}(\mu)$ are strictly increasing as functions of μ . It follows that $ \lambda(\mu) $
lyche-numerical-linear-algebra- F03722	285	1.000	$ \lambda(\mu) $	$ \lambda(\mu) $	Both $\lambda(\mu)$ and $\tilde{\omega}(\mu)$ are strictly increasing as functions of μ . It follows that $ \lambda(\mu) $
lyche-numerical-linear-algebra- F03723	285	1.000	$\mu=\rho\left(\boldsymbol{G}_J\right)=: \beta$	$\mu = \rho\left(\boldsymbol{G}_J\right) =: \beta$	is maximized for $\mu = \rho\left(\boldsymbol{G}_J\right) =: \beta$ and for this value of μ we obtain (12.27) for
lyche-numerical-linear-algebra- F03724	285	0.995	$0<\omega \leq \tilde{\omega}(\beta)=\omega^*$	$0<\omega \leq \tilde{\omega}(\beta) = \omega^*$	$0<\omega \leq \omega(\beta) = \omega^*$.
lyche-numerical-linear-algebra- F03725	285	1.000	$\rho\left(\boldsymbol{G}_{\omega}\right)=\omega-1$	$\rho\left(\boldsymbol{G}_\omega\right)=\omega-1$	Evidently $\rho\left(\boldsymbol{G}_\omega\right)=\omega-1$ is strictly increasing in $\omega^* < \omega < 2$. Equation (12.29)
lyche-numerical-linear-algebra- F03726	285	1.000	$\omega^*<\omega<2$	$\omega^* < \omega < 2$	Evidently $\rho\left(\boldsymbol{G}_\omega\right)=\omega-1$ is strictly increasing in $\omega^* < \omega < 2$. Equation (12.29)

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lyche-numerical-linear-algebra-285-F03727	285	■ 1.000	$0 < \omega < \omega^*$	$0 < \omega < \omega^*$	will follow if we can show that $\rho(\mathbf{G}_\omega)$ is strictly decreasing in $0 < \omega < \omega^*$. By
lyche-numerical-linear-algebra-285-F03728	285	■ 1.000	$\beta^2 (\omega^2 \beta^2 - 4\omega + 4) < (2 - \omega \beta^2)^2$	$\beta^2 (\omega^2 \beta^2 - 4\omega + 4) < (2 - \omega \beta^2)^2$	Since $\beta^2 (\omega^2 \beta^2 - 4\omega + 4) < (2 - \omega \beta^2)^2$ the numerator is negative and the strict
lyche-numerical-linear-algebra-286-F03729	286	■ 1.000	$a_{ii} = d \neq 0$	$a_{ii} = d \neq 0$	Exercise 12.1 (Richardson and Jacobi) Show that if $a_{ii} = d \neq 0$ for all i then
lyche-numerical-linear-algebra-286-F03730	286	■ 1.000	$\alpha := 1/d$	$\alpha := 1/d$	Richardson's method with $\alpha := 1/d$ is the same as Jacobi's method.
lyche-numerical-linear-algebra-286-F03731	286	■ 1.000	u_j	u_j	all eigenvalues λ_j of $\mathbf{A}$ have positive real parts u_j for $j = 1, \dots, n$ and that α is
lyche-numerical-linear-algebra-286-F03732	286	■ 0.766	$0 < \alpha < \min_j \left(2u_j / \lambda_j ^2 \right)$	$0 < \alpha < \min_j \left(2u_j / \lambda_j ^2 \right)$	real. Show that the R method converges if and only if $0 < \alpha < \min_j (2u_j / \lambda_j ^2)$.
lyche-numerical-linear-algebra-286-F03733	286	■ 1.000	$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	method and Gauss-Seidel's method diverge for $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.
lyche-numerical-linear-algebra-286-F03734	286	■ 0.945	$\mathbf{A} := \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$	$\mathbf{A} := \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$	and only if Jacobi converges for a 2 by 2 matrix $\mathbf{A} := \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}$.
lyche-numerical-linear-algebra-286-F03735	286	■ 1.000	$\begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$	values) that the matrix $\begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$ is positive definite for $-1/2 < a < 1$. Thus, GS
lyche-numerical-linear-algebra-286-F03736	286	■ 1.000	$-1/2 < a < 1$	$-1/2 < a < 1$	values) that the matrix $\begin{bmatrix} 1 & a & a \\ a & 1 & a \\ a & a & 1 \end{bmatrix}$ is positive definite for $-1/2 < a < 1$. Thus, GS
lyche-numerical-linear-algebra-286-F03737	286	■ 1.000	$1/2 < a < 1$	$1/2 < a < 1$	$1/2 < a < 1$.
lyche-numerical-linear-algebra-286-F03738	286	■ 0.995	$\mathbf{G}_1$	$\mathbf{G}_1$	Exercise 12.6 (Example: GS Diverges, J Converges) Let $\mathbf{G}_J$ and $\mathbf{G}_1$ be the
lyche-numerical-linear-algebra-286-F03739	286	■ 0.924	$\mathbf{A} := \begin{bmatrix} 1 & 0 & 1/2 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$	$\mathbf{A} := \begin{bmatrix} 1 & 0 & 1/2 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$	$\mathbf{A} := \begin{bmatrix} 1 & 0 & 1/2 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$.
lyche-numerical-linear-algebra-286-F03740	286	■ 0.996	$\mathbf{G}_1 := \begin{bmatrix} 0 & 0 & -1/2 \\ 0 & 0 & 1/2 \\ 0 & 0 & -1 \end{bmatrix}$	$\mathbf{G}_1 := \begin{bmatrix} 0 & 0 & -1/2 \\ 0 & 0 & 1/2 \\ 0 & 0 & -1 \end{bmatrix}$	a) Show that $\mathbf{G}_1 := \begin{bmatrix} 0 & 0 & -1/2 \\ 0 & 0 & 1/2 \\ 0 & 0 & -1 \end{bmatrix}$ and conclude that GS diverges.
lyche-numerical-linear-algebra-286-F03741	286	■ 1.000	$p(\lambda) := \det(\lambda \mathbf{I} - \mathbf{G}_J) = \lambda^3 + \frac{1}{2}\lambda + \frac{1}{2}$	$p(\lambda) := \det(\lambda \mathbf{I} - \mathbf{G}_J) = \lambda^3 + \frac{1}{2}\lambda + \frac{1}{2}$	b) Show that $p(\lambda) := \det(\lambda \mathbf{I} - \mathbf{G}_J) = \lambda^3 + \frac{1}{2}\lambda + \frac{1}{2}$.
lyche-numerical-linear-algebra-286-F03742	286	■ 0.997	$p(\lambda) \neq 0$	$p(\lambda) \neq 0$	c) Show that if $ \lambda \geq 1$ then $p(\lambda) \neq 0$. Conclude that J converges.
lyche-numerical-linear-algebra-286-F03743	286	■ 1.000	$ a_{ii} > \sum_{j \neq i} a_{ij} $	$ a_{ii} > \sum_{j \neq i} a_{ij} $	method converges if $ a_{ii} > \sum_{j \neq i} a_{ij} $ for $i = 1, \dots, n$.
lyche-numerical-linear-algebra-287-F03744	287	■ 1.000	$r := \max_i r_i < 1$	$r := \max_i r_i < 1$	GS method. Suppose $r := \max_i r_i < 1$, where $r_i = \sum_{j \neq i} \frac{ a_{ij} }{ a_{ii} }$. Show using
lyche-numerical-linear-algebra-287-F03745	287	■ 1.000	$r_i = \sum_{j \neq i} \frac{ a_{ij} }{ a_{ii} }$	$r_i = \sum_{j \neq i} \frac{ a_{ij} }{ a_{ii} }$	GS method. Suppose $r := \max_i r_i < 1$, where $r_i = \sum_{j \neq i} \frac{ a_{ij} }{ a_{ii} }$. Show using
lyche-numerical-linear-algebra-287-F03746	287	■ 1.000	$ \epsilon_{k+1}(j) \leq r \ \epsilon_k\ _\infty$	$ \epsilon_{k+1}(j) \leq r \ \epsilon_k\ _\infty$	induction on i that $ \epsilon_{k+1}(j) \leq r \ \epsilon_k\ _\infty$ for $j = 1, \dots, i$. Conclude that Gauss-
lyche-numerical-linear-algebra-287-F03747	287	■ 1.000	$j = 1, \dots, i$	$j = 1, \dots, i$	induction on i that $ \epsilon_{k+1}(j) \leq r \ \epsilon_k\ _\infty$ for $j = 1, \dots, i$. Conclude that Gauss-
lyche-numerical-linear-algebra-287-F03748	287	■ 1.000	$a \in$	$a \in$	Exercise 12.9 (Convergence Example for Fix Point Iteration) Consider for $a \in$
lyche-numerical-linear-algebra-287-F03749	287	■ 1.000	$\mathbf{x}_0 = \mathbf{0}$	$\mathbf{x}_0 = \mathbf{0}$	Starting with $\mathbf{x}_0 = \mathbf{0}$ show by induction
lyche-numerical-linear-algebra-287-F03750	287	■ 1.000	$\mathbf{x} = [1, 1]^T$	$\mathbf{x} = [1, 1]^T$	and conclude that the iteration converges to the fixed-point $\mathbf{x} = [1, 1]^T$ for $ a < 1$

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lyche-numerical-linear-algebra-F03751	287	1.000	$ a < 1$	$ a < 1$	and conclude that the iteration converges to the fixed-point $\mathbf{x} = [1, 1]^T$ for $ a < 1$
lyche-numerical-linear-algebra-F03752	287	1.000	$ a > 1$	$ a > 1$	and diverges for $ a > 1$. Show that $\rho(G) = 1 - \eta$ with $\eta = 1 - a $. Compute the
lyche-numerical-linear-algebra-F03753	287	1.000	$\eta = 1 - a $	$\eta = 1 - a $	and diverges for $ a > 1$. Show that $\rho(G) = 1 - \eta$ with $\eta = 1 - a $. Compute the
lyche-numerical-linear-algebra-F03754	287	0.999	$a = 0.9$	$a = 0.9$	estimate (12.31) for the rate of convergence for $a = 0.9$ and $s = 16$ and compare
lyche-numerical-linear-algebra-F03755	287	0.999	$s = 16$	$s = 16$	estimate (12.31) for the rate of convergence for $a = 0.9$ and $s = 16$ and compare
lyche-numerical-linear-algebra-F03756	287	1.000	$ a ^k \leq 10^{-16}$	$ a ^k \leq 10^{-16}$	with the true number of iterations determined from $ a ^k < 10^{-16}$.
lyche-numerical-linear-algebra-F03757	287	1.000	$\rho(G_J) = 1/2$	$\rho(G_J) = 1/2$	in Example 12.2. Show that $\rho(G_J) = 1/2$. Then show that $\mathbf{x}_k(1) = \mathbf{x}_k(2) = 1 - 2^{-k}$
lyche-numerical-linear-algebra-F03758	287	1.000	$\mathbf{x}_k(1) = \mathbf{x}_k(2) = 1 - 2^{-k}$	$\mathbf{x}_k(1) = \mathbf{x}_k(2) = 1 - 2^{-k}$	in Example 12.2. Show that $\rho(G_J) = 1/2$. Then show that $\mathbf{x}_k(1) = \mathbf{x}_k(2) = 1 - 2^{-k}$
lyche-numerical-linear-algebra-F03759	287	1.000	$A = I - L - L^T$	$A = I - L - L^T$	be written $A = I - L - L^T$, where L is lower triangular with zero's on the diagonal,
lyche-numerical-linear-algebra-F03760	287	1.000	$l_{i,j} = 0$	$l_{i,j} = 0$	$l_{i,j} = 0$, when $j \geq i$. The method is defined by
lyche-numerical-linear-algebra-F03761	287	1.000	$j \geq i$	$j \geq i$	$l_{i,j} = 0$, when $j \geq i$. The method is defined by
lyche-numerical-linear-algebra-F03762	287	1.000	N	N	where M and N are given by the splitting
lyche-numerical-linear-algebra-F03763	287	1.000	$M^{-1}N$	$M^{-1}N$	a) Let $\mathbf{x} \neq \mathbf{0}$ be an eigenvector of $M^{-1}N$ with eigenvalue λ . Show that
lyche-numerical-linear-algebra-F03764	288	1.000	$\mathbf{x} = [x(1), x(2), \dots, x(n)]^T$	$\mathbf{x} = [x(1), x(2), \dots, x(n)]^T$	1. Choose $\mathbf{x} = [x(1), x(2), \dots, x(n)]^T$.
lyche-numerical-linear-algebra-F03765	288	0.996	$i = 1, 2, \dots, n-1, n, n, n-1, n-2, \dots, 1$	$i = 1, 2, \dots, n-1, n, n, n-1, n-2, \dots, 1$	for $i = 1, 2, \dots, n-1, n, n, n-1, n-2, \dots, 1$
lyche-numerical-linear-algebra-F03766	288	0.999	$\mathbf{b} := [1, 9, -2]^T$	$\mathbf{b} := [1, 9, -2]^T$	and $\mathbf{b} := [1, 9, -2]^T$.
lyche-numerical-linear-algebra-F03767	288	1.000	$\mathbf{x}_0 = [1, 1, 1]^t$	$\mathbf{x}_0 = [1, 1, 1]^t$	a) With $\mathbf{x}_0 = [1, 1, 1]^t$, carry out one iteration of the Gauss-Seidel method to find
lyche-numerical-linear-algebra-F03768	288	1.000	$\mathbf{x}_1 \in \mathbb{R}^3$	$\mathbf{x}_1 \in \mathbb{R}^3$	$\mathbf{x}_1 \in \mathbb{R}^3$.
lyche-numerical-linear-algebra-F03769	288	1.000	$E \in \mathbb{C}^{n \times n}$	$E \in \mathbb{C}^{n \times n}$	$E \in \mathbb{C}^{n \times n}$. Show that $A + E$ is nonsingular if $\rho(A^{-1}E) < 1$.
lyche-numerical-linear-algebra-F03770	288	1.000	$\rho(A^{-1}E) < 1$	$\rho(A^{-1}E) < 1$	$E \in \mathbb{C}^{n \times n}$. Show that $A + E$ is nonsingular if $\rho(A^{-1}E) < 1$.
lyche-numerical-linear-algebra-F03771	289	0.998	$\lim_{k \rightarrow \infty} \ \mathbf{A}^k\ ^{1/k} = \rho(A)$	$\lim_{k \rightarrow \infty} \ \mathbf{A}^k\ ^{1/k} = \rho(A)$	Exercise 12.15 (Slow Spectral Radius Convergence) The convergence $\lim_{k \rightarrow \infty} \ \mathbf{A}^k\ ^{1/k} = \rho(A)$ can be quite slow. Consider
lyche-numerical-linear-algebra-F03772	289	1.000	$\ \mathbf{A}^k\ ^{1/k} = \rho(A)$	$\ \mathbf{A}^k\ ^{1/k} = \rho(A)$	$\ \mathbf{A}^k\ ^{1/k} = \rho(A)$ can be quite slow. Consider
lyche-numerical-linear-algebra-F03773	289	1.000	$ \lambda = \rho(A) < 1$	$ \lambda = \rho(A) < 1$	If $ \lambda = \rho(A) < 1$ then $\lim_{k \rightarrow \infty} \mathbf{A}^k = \mathbf{0}$ for any $a \in \mathbb{R}$. We show below that the
lyche-numerical-linear-algebra-F03774	289	1.000	$(1, n)$	$(1, n)$	$(1, n)$ element of $\mathbf{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-algebra-F03775	289	1.000	$f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$	$f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$	$(1, n)$ element of $\mathbf{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$ for $k \geq n-1$.
lyche-numerical-linear-algebra-F03776	289	1.000	$k \geq n-1$	$k \geq n-1$	$(1, n)$ element of $\mathbf{A}^k$ is given by $f(k) := \binom{k}{n-1} a^{n-1} \lambda^{k-n+1}$ for $k \geq n-1$.

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lyche-numerical-linear-algebra-F03777	289	1.000	n=5	$n = 5$	a) Pick an n , e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and
lyche-numerical-linear-algebra-F03778	289	1.000	f(k)	$f(k)$	a) Pick an n , e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and
lyche-numerical-linear-algebra-F03779	289	1.000	\lambda=0.9, a=10	$\lambda = 0.9, a = 10$	a) Pick an n , e.g. $n = 5$, and make a plot of $f(k)$ for $\lambda = 0.9$, $a = 10$, and
lyche-numerical-linear-algebra-F03780	289	1.000	n-1 \leq k \leq 200	$n - 1 \leq k \leq 200$	$n - 1 \leq k \leq 200$. Your program should also compute $\max_k f(k)$. Use your
lyche-numerical-linear-algebra-F03781	289	1.000	\max_{k} f(k)	$\max_k f(k)$	$n - 1 \leq k \leq 200$. Your program should also compute $\max_k f(k)$. Use your
lyche-numerical-linear-algebra-F03782	289	1.000	f(k)<10^{-8}	$f(k) < 10^{-8}$	program to determine how large k must be before $f(k) < 10^{-8}$.
lyche-numerical-linear-algebra-F03783	289	1.000	\boldsymbol{E}:= (\boldsymbol{A}-\lambda \boldsymbol{I}) / a	$\boldsymbol{E} := (\boldsymbol{A} - \lambda \boldsymbol{I})/a$	b) We can determine the elements of $\boldsymbol{A}^k$ explicitly for any k . Let $\boldsymbol{E} := (\boldsymbol{A} - \lambda \boldsymbol{I})/a$.
lyche-numerical-linear-algebra-F03784	289	1.000	\boldsymbol{E}^k=\left[\begin{array}{cc}\mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0}\end{array}\right] & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0}\end{array}\right]	$\boldsymbol{E}^k = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$	Show by induction that $\boldsymbol{E}^k = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ for $1 \leq k \leq n - 1$ and that $\boldsymbol{E}^n = \mathbf{0}$.
lyche-numerical-linear-algebra-F03785	289	1.000	\boldsymbol{E}^n=\mathbf{0}	$\boldsymbol{E}^n = \mathbf{0}$	Show by induction that $\boldsymbol{E}^k = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_{n-k} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$ for $1 \leq k \leq n - 1$ and that $\boldsymbol{E}^n = \mathbf{0}$.
lyche-numerical-linear-algebra-F03786	289	0.997	\boldsymbol{A}^k=(a \boldsymbol{E}+\lambda \boldsymbol{I})^k=\sum_{j=0}^{\min \{k, n-1\}} \binom{k}{j} a^j \lambda^{k-j} \boldsymbol{E}^j	$\boldsymbol{A}^k = (a\boldsymbol{E} + \lambda \boldsymbol{I})^k = \sum_{j=0}^{\min\{k, n-1\}} \binom{k}{j} a^j \lambda^{k-j} \boldsymbol{E}^j$	c) We have $\boldsymbol{A}^k = (a\boldsymbol{E} + \lambda \boldsymbol{I})^k = \sum_{j=0}^{\min\{k, n-1\}} \binom{k}{j} a^j \lambda^{k-j} \boldsymbol{E}^j$ and conclude that the
lyche-numerical-linear-algebra-F03787	289	1.000	\boldsymbol{A}^n \rightarrow \mathbf{0}	$\boldsymbol{A}^n \rightarrow \mathbf{0}$	• Give a necessary and sufficient condition that $\boldsymbol{A}^n \rightarrow \mathbf{0}$.
lyche-numerical-linear-algebra-F03788	289	1.000	\ \boldsymbol{A}\ < \rho(\boldsymbol{A})	$\ \boldsymbol{A}\ < \rho(\boldsymbol{A})$	• Is there a matrix norm $\ \cdot\ $ consistent on $\mathbb{C}^{n \times n}$ such that $\ \boldsymbol{A}\ < \rho(\boldsymbol{A})$?
lyche-numerical-linear-algebra-F03789	290	0.969	\kappa:=\frac{\lambda_{\text {max }}}{\lambda_{\text {min }}}}	$\kappa := \frac{\lambda_{\max}}{\lambda_{\min}}$	condition number of $\boldsymbol{A}$ is given by $\kappa := \frac{\lambda_{\max}}{\lambda_{\min}}$, where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest
lyche-numerical-linear-algebra-F03790	290	1.000	\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^T \boldsymbol{y}	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^T \boldsymbol{y}$	on $\mathbb{R}^n$, the usual inner product $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{x}^T \boldsymbol{y}$ with the associated Euclidian norm
lyche-numerical-linear-algebra-F03791	290	1.000	\ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^T \boldsymbol{x}}	$\ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^T \boldsymbol{x}}$	$\ \boldsymbol{x}\ _2 = \sqrt{\boldsymbol{x}^T \boldsymbol{x}}$, and the A-inner product and the corresponding A-norm given by
lyche-numerical-linear-algebra-F03792	291	1.000	\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in$	We note that the A-inner product is an inner product on $\mathbb{R}^n$. Indeed, for any $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in$
lyche-numerical-linear-algebra-F03793	291	1.000	\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \geq 0	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \geq 0$	1. $\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} > 0$ and $\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = 0$ if and only if $\boldsymbol{x} = \mathbf{0}$, since $\boldsymbol{A}$ is positive
lyche-numerical-linear-algebra-F03794	291	1.000	\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = 0	$\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = 0$	1. $\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} \geq 0$ and $\langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\boldsymbol{A}} = 0$ if and only if $\boldsymbol{x} = \mathbf{0}$, since $\boldsymbol{A}$ is positive
lyche-numerical-linear-algebra-F03795	291	0.983	\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} = (\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y})^T = \boldsymbol{y}^T \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{y}, \boldsymbol{x} \rangle_{\boldsymbol{A}}	$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} = (\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y})^T = \boldsymbol{y}^T \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{y}, \boldsymbol{x} \rangle_{\boldsymbol{A}}$	2. $\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y} = (\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{y})^T = \boldsymbol{y}^T \boldsymbol{A}^T \boldsymbol{x} = \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{x} = \langle \boldsymbol{y}, \boldsymbol{x} \rangle_{\boldsymbol{A}}$ by symmetry of
lyche-numerical-linear-algebra-F03796	291	1.000	\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{z} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{z} = \langle \boldsymbol{x}, \boldsymbol{z} \rangle_{\boldsymbol{A}} + \langle \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}}	$\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{z} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{z} = \langle \boldsymbol{x}, \boldsymbol{z} \rangle_{\boldsymbol{A}} + \langle \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}}$	3. $\langle \boldsymbol{x} + \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}} := \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{z} + \boldsymbol{y}^T \boldsymbol{A} \boldsymbol{z} = \langle \boldsymbol{x}, \boldsymbol{z} \rangle_{\boldsymbol{A}} + \langle \boldsymbol{y}, \boldsymbol{z} \rangle_{\boldsymbol{A}}$, true for any $\boldsymbol{A}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F03797	291	1.000	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{b} \in \mathbb{R}^n$	$\boldsymbol{A} \in \mathbb{R}^{n \times n}, \boldsymbol{b} \in \mathbb{R}^n$	Consider for a positive definite $\boldsymbol{A} \in \mathbb{R}^{n \times n}$, $\boldsymbol{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$ the quadratic
lyche-numerical-linear-algebra-F03798	291	1.000	$c \in \mathbb{R}$	$c \in \mathbb{R}$	Consider for a positive definite $\boldsymbol{A} \in \mathbb{R}^{n \times n}$, $\boldsymbol{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$ the quadratic
lyche-numerical-linear-algebra-F03799	291	1.000	$Q: \mathbb{R}^n \rightarrow \mathbb{R}$	$Q: \mathbb{R}^n \rightarrow \mathbb{R}$	function $Q: \mathbb{R}^n \rightarrow \mathbb{R}$ given by
lyche-numerical-linear-algebra-F03800	291	1.000	$\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$	$\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$	The following expansion will be used repeatedly. For $\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}$
lyche-numerical-linear-algebra-F03801	291	1.000	$\varepsilon \in \mathbb{R}$	$\varepsilon \in \mathbb{R}$	The following expansion will be used repeatedly. For $\boldsymbol{y}, \boldsymbol{h} \in \mathbb{R}^n$ and $\varepsilon \in \mathbb{R}$
lyche-numerical-linear-algebra-F03802	291	0.996	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{y}$	$\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{y}$	if and only if $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. Moreover, the residual $\boldsymbol{r}(\boldsymbol{y}) := \boldsymbol{b} - \boldsymbol{A}\boldsymbol{y}$ for any $\boldsymbol{y} \in \mathbb{R}^n$ is
lyche-numerical-linear-algebra-F03803	291	1.000	$\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$	$\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$	equal to the negative gradient, i.e., $\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$, where $\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$.
lyche-numerical-linear-algebra-F03804	291	1.000	$\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$	$\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$	equal to the negative gradient, i.e., $\boldsymbol{r}(\boldsymbol{y}) = -\nabla Q(\boldsymbol{y})$, where $\nabla := \left[\frac{\partial}{\partial y_1}, \dots, \frac{\partial}{\partial y_n} \right]^T$.
lyche-numerical-linear-algebra-F03805	291	0.998	$\boldsymbol{y} = \boldsymbol{x}, \varepsilon = 1$	$\boldsymbol{y} = \boldsymbol{x}, \varepsilon = 1$	Proof If $\boldsymbol{y} = \boldsymbol{x}$, $\varepsilon = 1$, and $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, then (13.5) simplifies to $Q(\boldsymbol{x} + \boldsymbol{h}) =$
lyche-numerical-linear-algebra-F03806	291	0.998	$Q(\boldsymbol{x} + \boldsymbol{h}) =$	$Q(\boldsymbol{x} + \boldsymbol{h}) =$	Proof If $\boldsymbol{y} = \boldsymbol{x}$, $\varepsilon = 1$, and $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$, then (13.5) simplifies to $Q(\boldsymbol{x} + \boldsymbol{h}) =$
lyche-numerical-linear-algebra-F03807	291	1.000	$Q(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}$	$Q(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}$	$Q(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}$, and since $\boldsymbol{A}$ is positive definite $Q(\boldsymbol{x} + \boldsymbol{h}) > Q(\boldsymbol{x})$ for all nonzero
lyche-numerical-linear-algebra-F03808	291	1.000	$Q(\boldsymbol{x} + \boldsymbol{h}) > Q(\boldsymbol{x})$	$Q(\boldsymbol{x} + \boldsymbol{h}) > Q(\boldsymbol{x})$	$Q(\boldsymbol{x}) + \frac{1}{2} \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}$, and since $\boldsymbol{A}$ is positive definite $Q(\boldsymbol{x} + \boldsymbol{h}) > Q(\boldsymbol{x})$ for all nonzero
lyche-numerical-linear-algebra-F03809	291	0.982	$\boldsymbol{h} \in \mathbb{R}^n$	$\boldsymbol{h} \in \mathbb{R}^n$	$\boldsymbol{h} \in \mathbb{R}^n$. It follows that $\boldsymbol{x}$ is the unique minimum of Q . Conversely, if $\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$ and
lyche-numerical-linear-algebra-F03810	291	0.982	$\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$	$\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$	$\boldsymbol{h} \in \mathbb{R}^n$. It follows that $\boldsymbol{x}$ is the unique minimum of Q . Conversely, if $\boldsymbol{A}\boldsymbol{x} \neq \boldsymbol{b}$ and
lyche-numerical-linear-algebra-F03811	292	1.000	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$, then by (13.5), $Q(\boldsymbol{x} + \varepsilon \boldsymbol{h}) - Q(\boldsymbol{x}) = -\varepsilon(\boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{x}) - \frac{1}{2} \varepsilon \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}) < 0$ for
lyche-numerical-linear-algebra-F03812	292	1.000	$Q(\boldsymbol{x} + \varepsilon \boldsymbol{h}) - Q(\boldsymbol{x}) = -\varepsilon \left(\boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{x}) - \frac{1}{2} \varepsilon \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h} \right) < 0$	$Q(\boldsymbol{x} + \varepsilon \boldsymbol{h}) - Q(\boldsymbol{x}) = -\varepsilon \left(\boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{x}) - \frac{1}{2} \varepsilon \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h} \right) < 0$	$\boldsymbol{h} := \boldsymbol{r}(\boldsymbol{x})$, then by (13.5), $Q(\boldsymbol{x} + \varepsilon \boldsymbol{h}) - Q(\boldsymbol{x}) = -\varepsilon(\boldsymbol{h}^T \boldsymbol{r}(\boldsymbol{x}) - \frac{1}{2} \varepsilon \boldsymbol{h}^T \boldsymbol{A} \boldsymbol{h}) < 0$ for
lyche-numerical-linear-algebra-F03813	292	1.000	$\boldsymbol{x}_0 \in \mathbb{R}^n$	$\boldsymbol{x}_0 \in \mathbb{R}^n$	1. Choose $\boldsymbol{x}_0 \in \mathbb{R}^n$.
lyche-numerical-linear-algebra-F03814	292	1.000	α_k	α_k	Choose a “step length” α_k ,
lyche-numerical-linear-algebra-F03815	292	1.000	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$	$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$	Compute $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$.
lyche-numerical-linear-algebra-F03816	293	0.999	$Q(\boldsymbol{x}_{k+1})$	$Q(\boldsymbol{x}_{k+1})$	For a fixed direction $\boldsymbol{p}_k$ we say that α_k is optimal if $Q(\boldsymbol{x}_{k+1})$ is as small as
lyche-numerical-linear-algebra-F03817	293	1.000	$Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = Q(\boldsymbol{x}_k) - \alpha \boldsymbol{p}_k^T \boldsymbol{r}_k + \frac{1}{2} \alpha^2 \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k$	$Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = Q(\boldsymbol{x}_k) - \alpha \boldsymbol{p}_k^T \boldsymbol{r}_k + \frac{1}{2} \alpha^2 \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k$	By (13.5) we have $Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = Q(\boldsymbol{x}_k) - \alpha \boldsymbol{p}_k^T \boldsymbol{r}_k + \frac{1}{2} \alpha^2 \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k$, where $\boldsymbol{r}_k :=$
lyche-numerical-linear-algebra-F03818	293	1.000	$\boldsymbol{r}_k :=$	$\boldsymbol{r}_k :=$	By (13.5) we have $Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = Q(\boldsymbol{x}_k) - \alpha \boldsymbol{p}_k^T \boldsymbol{r}_k + \frac{1}{2} \alpha^2 \boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k$, where $\boldsymbol{r}_k :=$
lyche-numerical-linear-algebra-F03819	293	1.000	$\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$	$\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$	$\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k$. Since $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k \geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial \alpha} Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = 0$.

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lyche-numerical-linear-algebra F03820	293	■ 1.000	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k \geq 0$	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k \geq 0$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$. Since $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k \geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial \alpha} Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = 0$.
lyche-numerical-linear-algebra F03821	293	■ 1.000	$\frac{\partial}{\partial \alpha} Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = 0$	$\frac{\partial}{\partial \alpha} Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = 0$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_k$. Since $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k \geq 0$ we find a minimum α_k by solving $\frac{\partial}{\partial \alpha} Q(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) = 0$.
lyche-numerical-linear-algebra F03822	293	■ 1.000	$\boldsymbol{p}_k = \boldsymbol{r}_k$	$\boldsymbol{p}_k = \boldsymbol{r}_k$	choose $\boldsymbol{p}_k = \boldsymbol{r}_k$ the negative gradient, and the optimal α_k . Starting from $\boldsymbol{x}_0$ we
lyche-numerical-linear-algebra F03823	293	■ 1.000	$k=0,1,2 \dots$	$k = 0, 1, 2 \dots$	compute for $k=0,1,2\dots$
lyche-numerical-linear-algebra F03824	293	■ 1.000	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$	are orthogonal. Indeed, by (13.10), $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$ since $\alpha_k =$
lyche-numerical-linear-algebra F03825	293	■ 1.000	$\alpha_k =$	$\alpha_k =$	are orthogonal. Indeed, by (13.10), $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_k = (\boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{r}_k)^T \boldsymbol{r}_k = 0$ since $\alpha_k =$
lyche-numerical-linear-algebra F03826	293	■ 0.887	$\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$	$\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$	$\frac{\boldsymbol{r}_k^T \boldsymbol{r}_k}{\boldsymbol{r}_k^T \boldsymbol{A} \boldsymbol{r}_k}$ and $\boldsymbol{A}$ is symmetric.
lyche-numerical-linear-algebra F03827	293	■ 1.000	$\boldsymbol{x}_0 = [-1, -1/2]^T$	$\boldsymbol{x}_0 = [-1, -1/2]^T$	Starting with $\boldsymbol{x}_0 = [-1, -1/2]^T$ and $\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$ we find
lyche-numerical-linear-algebra F03828	293	■ 1.000	$\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$	$\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$	Starting with $\boldsymbol{x}_0 = [-1, -1/2]^T$ and $\boldsymbol{r}_0 = -\boldsymbol{A} \boldsymbol{x}_0 = [3/2, 0]^T$ we find
lyche-numerical-linear-algebra F03829	294	■ 1.000	$k \geq 1$	$k \geq 1$	and in general for $k \geq 1$
lyche-numerical-linear-algebra F03830	294	■ 1.000	$\alpha_k = 1/2$	$\alpha_k = 1/2$	Since $\alpha_k = 1/2$ is constant for all k the methods of Richardson, Jacobi and steepest
lyche-numerical-linear-algebra F03831	294	■ 1.000	$\ \boldsymbol{x}_{j+1}\ _2 / \ \boldsymbol{x}_j\ = \ \boldsymbol{r}_{j+1}\ _2 / \ \boldsymbol{r}_j\ _2 = 1/2$	$\ \boldsymbol{x}_{j+1}\ _2 / \ \boldsymbol{x}_j\ = \ \boldsymbol{r}_{j+1}\ _2 / \ \boldsymbol{r}_j\ _2 = 1/2$	of convergence is determined from $\ \boldsymbol{x}_{j+1}\ _2 / \ \boldsymbol{x}_j\ = \ \boldsymbol{r}_{j+1}\ _2 / \ \boldsymbol{r}_j\ _2 = 1/2$ for
lyche-numerical-linear-algebra F03832	294	■ 1.000	$\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$	$\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$	A-orthogonal search directions i.e., $\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0$ for all $i \neq j$.
lyche-numerical-linear-algebra F03833	294	■ 1.000	$\boldsymbol{r}_0 =$	$\boldsymbol{r}_0 =$	As in the steepest descent method we choose a starting vector $\boldsymbol{x}_0 \in \mathbb{R}^n$. If $\boldsymbol{r}_0 =$
lyche-numerical-linear-algebra F03834	294	■ 1.000	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0 = \mathbf{0}$ then $\boldsymbol{x}_0$ is the exact solution and we are finished, otherwise we initially
lyche-numerical-linear-algebra F03835	294	■ 1.000	$\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$	$\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$	make a steepest descent step. It follows that $\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$ and $\boldsymbol{p}_0 := \boldsymbol{r}_0$.
lyche-numerical-linear-algebra F03836	294	■ 1.000	$\boldsymbol{p}_0 := \boldsymbol{r}_0$	$\boldsymbol{p}_0 := \boldsymbol{r}_0$	make a steepest descent step. It follows that $\boldsymbol{r}_1^T \boldsymbol{r}_0 = 0$ and $\boldsymbol{p}_0 := \boldsymbol{r}_0$.
lyche-numerical-linear-algebra F03837	294	■ 1.000	$j \geq 0$	$j \geq 0$	For the general case we define for $j \geq 0$
lyche-numerical-linear-algebra F03838	294	■ 1.000	$\boldsymbol{p}_j$	$\boldsymbol{p}_j$	1. $\boldsymbol{p}_j$ is computed by the Gram-Schmidt orthogonalization process applied to the
lyche-numerical-linear-algebra F03839	294	■ 0.999	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_j$	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_j$	residuals $\boldsymbol{r}_0, \dots, \boldsymbol{r}_j$ using the $\boldsymbol{A}$ -inner product. The search directions are there-
lyche-numerical-linear-algebra F03840	294	■ 1.000	α_j	α_j	3. It can be shown that the step length α_j is optimal for all j (cf. Exercise 13.7).
lyche-numerical-linear-algebra F03841	295	■ 1.000	$\boldsymbol{x}_j$	$\boldsymbol{x}_j$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-algebra F03842	295	■ 1.000	$\boldsymbol{r}_j \neq 0$	$\boldsymbol{r}_j \neq 0$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-algebra F03843	295	■ 1.000	$\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$	$\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$
lyche-numerical-linear-algebra F03844	295	■ 1.000	$i, j=0,1, \dots, k, i \neq j$	$i, j = 0, 1, \dots, k, i \neq j$	$\boldsymbol{x}_j$ is well defined, $\boldsymbol{r}_j \neq 0$, and $\boldsymbol{r}_i^T \boldsymbol{r}_j = 0$ for $i, j = 0, 1, \dots, k, i \neq j$. Then $\boldsymbol{x}_{k+1}$

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lyche-numerical-linear-algebra-F03845	295	■ 1.000	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$	$\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$	is well defined and $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$ for $j = 0, 1, \dots, k$.
lyche-numerical-linear-algebra-F03846	295	■ 1.000	$j=0,1, \ldots, k$	$j = 0, 1, \dots, k$	is well defined and $\boldsymbol{r}_{k+1}^T \boldsymbol{r}_j = 0$ for $j = 0, 1, \dots, k$.
lyche-numerical-linear-algebra-F03847	295	■ 1.000	$\boldsymbol{r}_j$	$\boldsymbol{r}_j$	Proof Since the residuals $\boldsymbol{r}_j$ are orthogonal and nonzero for $j \leq k$, they are linearly
lyche-numerical-linear-algebra-F03848	295	■ 1.000	$j \leq k$	$j \leq k$	Proof Since the residuals $\boldsymbol{r}_j$ are orthogonal and nonzero for $j \leq k$, they are linearly
lyche-numerical-linear-algebra-F03849	295	■ 1.000	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$	$\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-algebra-F03850	295	■ 1.000	$i < k$	$i < k$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-algebra-F03851	295	■ 1.000	$\boldsymbol{r}_{k+1}$	$\boldsymbol{r}_{k+1}$	and $\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_i = 0$ for $i < k$. But then $\boldsymbol{x}_{k+1}$ and $\boldsymbol{r}_{k+1}$ are well defined. Now
lyche-numerical-linear-algebra-F03852	295	■ 1.000	$j < k$	$j < k$	orthogonality for $j < k$ and by the definition of α_k for $j = k$. This completes
lyche-numerical-linear-algebra-F03853	295	■ 1.000	$\operatorname{dim} \mathbb{R}^n = n$	$\dim \mathbb{R}^n = n$	orthogonal and therefore linearly independent if they are nonzero. Since $\dim \mathbb{R}^n = n$
lyche-numerical-linear-algebra-F03854	295	■ 1.000	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$	$\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$	the $n + 1$ residuals $\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$ cannot all be nonzero and we must have $\boldsymbol{r}_k = 0$ for
lyche-numerical-linear-algebra-F03855	295	■ 1.000	$\boldsymbol{r}_k = 0$	$\boldsymbol{r}_k = 0$	the $n + 1$ residuals $\boldsymbol{r}_0, \dots, \boldsymbol{r}_n$ cannot all be nonzero and we must have $\boldsymbol{r}_k = 0$ for
lyche-numerical-linear-algebra-F03856	295	■ 1.000	$j = k + 1$	$j = k + 1$	With $j = k + 1$ (13.11) therefore takes the simple form $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$ and
lyche-numerical-linear-algebra-F03857	295	■ 1.000	$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$	$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$	With $j = k + 1$ (13.11) therefore takes the simple form $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k$ and
lyche-numerical-linear-algebra-F03858	295	■ 1.000	$\boldsymbol{x}_0, \boldsymbol{p}_0 = \boldsymbol{r}_0 =$	$\boldsymbol{x}_0, \boldsymbol{p}_0 = \boldsymbol{r}_0 =$	To summarize, in the conjugate gradient method we start with $\boldsymbol{x}_0, \boldsymbol{p}_0 = \boldsymbol{r}_0 =$
lyche-numerical-linear-algebra-F03859	295	■ 1.000	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$ and then generate a sequence of vectors $\{\boldsymbol{x}_k\}$ as follows:
lyche-numerical-linear-algebra-F03860	296	■ 1.000	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$	itive definite linear system $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Starting as in Example 13.1 with
lyche-numerical-linear-algebra-F03861	296	■ 1.000	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$ we find $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$ and then
lyche-numerical-linear-algebra-F03862	296	■ 1.000	$\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$	$\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$	$\boldsymbol{x}_0 = \begin{bmatrix} -1 \\ -1/2 \end{bmatrix}$ we find $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \begin{bmatrix} 3/2 \\ 0 \end{bmatrix}$ and then
lyche-numerical-linear-algebra-F03863	296	■ 1.000	$\boldsymbol{x}_2$	$\boldsymbol{x}_2$	Thus $\boldsymbol{x}_2$ is the exact solution as illustrated in the right part of Fig. 13.1.
lyche-numerical-linear-algebra-F03864	296	■ 0.969	$\ \boldsymbol{r}_k\ _2 / \ \boldsymbol{b}\ _2 \leq$	$\ \boldsymbol{r}_k\ _2 / \ \boldsymbol{b}\ _2 \leq$	starting vector for the iteration. The iteration is stopped when $\ \boldsymbol{r}_k\ _2 / \ \boldsymbol{b}\ _2 \leq \text{tol}$ or
lyche-numerical-linear-algebra-F03865	297	■ 0.803	$k >$	$k >$	$k > \text{itmax}$. K is the number of iterations used:
lyche-numerical-linear-algebra-F03866	297	■ 0.996	$(\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p})$	$(\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p})$	1. one matrix times vector ($\boldsymbol{t} = \boldsymbol{A} \boldsymbol{p}$),
lyche-numerical-linear-algebra-F03867	297	■ 1.000	$(\boldsymbol{p}^T \boldsymbol{t})$	$(\boldsymbol{p}^T \boldsymbol{t})$	2. two inner products ($(\boldsymbol{p}^T \boldsymbol{t})$ and $\boldsymbol{r}^T \boldsymbol{r}$),
lyche-numerical-linear-algebra-F03868	297	■ 1.000	$(\boldsymbol{r}^T \boldsymbol{r})$	$\boldsymbol{r}^T \boldsymbol{r}$	2. two inner products ($(\boldsymbol{p}^T \boldsymbol{t})$ and $\boldsymbol{r}^T \boldsymbol{r}$),
lyche-numerical-linear-algebra-F03869	297	■ 0.758	$\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{r}$	$\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{r}$	3. three vector-plus-scalar-times-vector ($\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{r}$) and $\boldsymbol{p} = \boldsymbol{r} +$
lyche-numerical-linear-algebra-F03870	297	■ 0.758	$\boldsymbol{p} = \boldsymbol{r} +$	$\boldsymbol{p} = \boldsymbol{r} +$	3. three vector-plus-scalar-times-vector ($\boldsymbol{x} = \boldsymbol{x} + \alpha \boldsymbol{p}, \boldsymbol{r} = \boldsymbol{r} - \alpha \boldsymbol{r}$ and $\boldsymbol{p} = \boldsymbol{r} +$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F03871	297	■ 1.000	$\boldsymbol{t}=\boldsymbol{A}\boldsymbol{p}$	$\boldsymbol{t} = \boldsymbol{A}\boldsymbol{p}$	The dominating part is the computation of $\boldsymbol{t} = \boldsymbol{A}\boldsymbol{p}$.
lyche-numerical-linear-algebra-F03872	297	■ 1.000	$\boldsymbol{T}_2:=\boldsymbol{T}_1\otimes\boldsymbol{I}+\boldsymbol{I}\otimes\boldsymbol{T}_1$	$\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$	Kronecker sum $\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$, where $\boldsymbol{T}_1 := \text{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$
lyche-numerical-linear-algebra-F03873	297	■ 1.000	$\boldsymbol{T}_1:=\operatorname{tridiag}_m(a, d, a)$ $\in \mathbb{R}^{m \times m}$	$\boldsymbol{T}_1 := \text{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$	Kronecker sum $\boldsymbol{T}_2 := \boldsymbol{T}_1 \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{T}_1$, where $\boldsymbol{T}_1 := \text{tridiag}_m(a, d, a) \in \mathbb{R}^{m \times m}$
lyche-numerical-linear-algebra-F03874	297	■ 1.000	$\boldsymbol{f}=[1, \ldots, 1]^T \in \mathbb{R}^n$	$\boldsymbol{f} = [1, \dots, 1]^T \in \mathbb{R}^n$	(cf. Theorem 10.2). We set $h = 1/(m + 1)$ and $\boldsymbol{f} = [1, \dots, 1]^T \in \mathbb{R}^n$
lyche-numerical-linear-algebra-F03875	297	■ 1.000	$a=1 / 9, d=5 / 18$	$a = 1/9, d = 5/18$	1. $a = 1/9, d = 5/18$, the Averaging matrix.
lyche-numerical-linear-algebra-F03876	297	■ 1.000	$a=-1, d=2$	$a = -1, d = 2$	2. $a = -1, d = 2$, the Poisson matrix.
lyche-numerical-linear-algebra-F03877	297	■ 1.000	$O(5 \ n)$	$O(5n)$	Note that for our test problems $\boldsymbol{T}_2$ only has $O(5n)$ nonzero elements. Therefore,
lyche-numerical-linear-algebra-F03878	297	■ 1.000	$\boldsymbol{t}$	$\boldsymbol{t}$	taking advantage of the sparseness of $\boldsymbol{T}_2$ we can compute $\boldsymbol{t}$ in Algorithm 13.1
lyche-numerical-linear-algebra-F03879	298	■ 1.000	$\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$ $\mathbb{R}^{m \times m}$	$\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$	tageous to use a matrix equation formulation. We define matrices $\boldsymbol{V}, \boldsymbol{R}, \boldsymbol{P}, \boldsymbol{B}, \boldsymbol{T} \in$
lyche-numerical-linear-algebra-F03880	298	■ 1.000	$\boldsymbol{x}=\operatorname{vec}(\boldsymbol{V}),$ $\boldsymbol{r}=\operatorname{vec}(\boldsymbol{R}),$ $\boldsymbol{p}=\operatorname{vec}(\boldsymbol{P}),$ $\boldsymbol{t}=\operatorname{vec}(\boldsymbol{T})$	$\boldsymbol{x} = \text{vec}(\boldsymbol{V}), \boldsymbol{r} = \text{vec}(\boldsymbol{R}), \boldsymbol{p} = \text{vec}(\boldsymbol{P}), \boldsymbol{t} = \text{vec}(\boldsymbol{T})$	$\mathbb{R}^{m \times m}$ by $\boldsymbol{x} = \text{vec}(\boldsymbol{V}), \boldsymbol{r} = \text{vec}(\boldsymbol{R}), \boldsymbol{p} = \text{vec}(\boldsymbol{P}), \boldsymbol{t} = \text{vec}(\boldsymbol{T})$, and $h^2 \boldsymbol{f} = \text{vec}(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F03881	298	■ 1.000	$h^2 \boldsymbol{f}=\operatorname{vec}(\boldsymbol{B})$	$h^2 \boldsymbol{f} = \text{vec}(\boldsymbol{B})$	$\mathbb{R}^{m \times m}$ by $\boldsymbol{x} = \text{vec}(\boldsymbol{V}), \boldsymbol{r} = \text{vec}(\boldsymbol{R}), \boldsymbol{p} = \text{vec}(\boldsymbol{P}), \boldsymbol{t} = \text{vec}(\boldsymbol{T})$, and $h^2 \boldsymbol{f} = \text{vec}(\boldsymbol{B})$.
lyche-numerical-linear-algebra-F03882	298	■ 1.000	$\boldsymbol{T}_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff \boldsymbol{T}_1 \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T}_1 = \boldsymbol{B}$	$\boldsymbol{T}_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff \boldsymbol{T}_1 \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T}_1 = \boldsymbol{B}$	Then $\boldsymbol{T}_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff \boldsymbol{T}_1 \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T}_1 = \boldsymbol{B}$, and $\boldsymbol{t} = \boldsymbol{T}_2 \boldsymbol{p} \iff \boldsymbol{T} = \boldsymbol{T}_1 \boldsymbol{P} + \boldsymbol{P} \boldsymbol{T}_1$.
lyche-numerical-linear-algebra-F03883	298	■ 1.000	$\boldsymbol{t} = \boldsymbol{T}_2 \boldsymbol{p} \iff \boldsymbol{T} = \boldsymbol{T}_1 \boldsymbol{P} + \boldsymbol{P} \boldsymbol{T}_1$	$\boldsymbol{t} = \boldsymbol{T}_2 \boldsymbol{p} \iff \boldsymbol{T} = \boldsymbol{T}_1 \boldsymbol{P} + \boldsymbol{P} \boldsymbol{T}_1$	Then $\boldsymbol{T}_2 \boldsymbol{x} = h^2 \boldsymbol{f} \iff \boldsymbol{T}_1 \boldsymbol{V} + \boldsymbol{V} \boldsymbol{T}_1 = \boldsymbol{B}$, and $\boldsymbol{t} = \boldsymbol{T}_2 \boldsymbol{p} \iff \boldsymbol{T} = \boldsymbol{T}_1 \boldsymbol{P} + \boldsymbol{P} \boldsymbol{T}_1$.
lyche-numerical-linear-algebra-F03884	298	■ 0.958	$tol=10^{-8}$	$tol = 10^{-8}$	For both the averaging- and Poisson matrix we use $tol = 10^{-8}$.
lyche-numerical-linear-algebra-F03885	298	■ 1.000	$K / \sqrt{n}$	$K / \sqrt{n}$	of iterations, and $K / \sqrt{n}$.
lyche-numerical-linear-algebra-F03886	299	■ 1.000	$\ \boldsymbol{x}\ _A := \sqrt{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}$	$\ \boldsymbol{x}\ _A := \sqrt{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}$	Recall that the A -norm of a vector $\boldsymbol{x} \in \mathbb{R}^n$ is given by $\ \boldsymbol{x}\ _A := \sqrt{\boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x}}$. The
lyche-numerical-linear-algebra-F03887	299	■ 0.995	$\kappa=\operatorname{cond}_2(\boldsymbol{A}):=\lambda_{\max} / \lambda_{\min}$	$\kappa = \text{cond}_2(\boldsymbol{A}) := \lambda_{\max} / \lambda_{\min}$	Here $\kappa = \text{cond}_2(\boldsymbol{A}) := \lambda_{\max} / \lambda_{\min}$ is the spectral condition number of $\boldsymbol{A}$, where
lyche-numerical-linear-algebra-F03888	299	■ 1.000	$\frac{\kappa-1}{\kappa+1}<1$	$\frac{\kappa-1}{\kappa+1} < 1$	1. Since $\frac{\kappa-1}{\kappa+1} < 1$ the steepest descent method always converges for a positive
lyche-numerical-linear-algebra-F03889	299	■ 1.000	$\frac{\kappa-1}{\kappa+1}$	$\frac{\kappa-1}{\kappa+1}$	definite matrix. The convergence can be slow when $\frac{\kappa-1}{\kappa+1}$ is close to one, and this
lyche-numerical-linear-algebra-F03890	300	■ 1.000	$d=5 / 18, a=1 / 9$	$d = 5/18, a = 1/9$	For the averaging problem given by $d = 5/18, a = 1/9$, the largest and smallest
lyche-numerical-linear-algebra-F03891	300	■ 0.999	$\lambda_{\max}=\frac{5}{9}+\frac{4}{9} \cos (\pi h)$	$\lambda_{\max} = \frac{5}{9} + \frac{4}{9} \cos(\pi h)$	eigenvalue of $\boldsymbol{T}_2$ are given by $\lambda_{\max} = \frac{5}{9} + \frac{4}{9} \cos(\pi h)$ and $\lambda_{\min} = \frac{5}{9} - \frac{4}{9} \cos(\pi h)$.
lyche-numerical-linear-algebra-F03892	300	■ 0.999	$\lambda_{\min}=\frac{5}{9}-\frac{4}{9} \cos (\pi h)$	$\lambda_{\min} = \frac{5}{9} - \frac{4}{9} \cos(\pi h)$	eigenvalue of $\boldsymbol{T}_2$ are given by $\lambda_{\max} = \frac{5}{9} + \frac{4}{9} \cos(\pi h)$ and $\lambda_{\min} = \frac{5}{9} - \frac{4}{9} \cos(\pi h)$.
lyche-numerical-linear-algebra-F03893	300	■ 1.000	$\mathbb{W}_0=\{\mathbf{0}\}$	$\mathbb{W}_0 = \{\mathbf{0}\}$	The Krylov spaces are defined by $\mathbb{W}_0 = \{\mathbf{0}\}$ and
lyche-numerical-linear-algebra-F03894	301	■ 1.000	$\operatorname{dim}\left(\mathbb{W}_k\right) \leq k$	$\dim(\mathbb{W}_k) \leq k$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra-F03895	301	■ 1.000	$\boldsymbol{v} \in \mathbb{W}_k$	$\boldsymbol{v} \in \mathbb{W}_k$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$.

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lyche-numerical-linear-algebra-F03896	301	1.000	$\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$	$\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$	with $\dim(\mathbb{W}_k) \leq k$ for all $k \geq 0$. Moreover, If $\boldsymbol{v} \in \mathbb{W}_k$ then $\boldsymbol{A} \boldsymbol{v} \in \mathbb{W}_{k+1}$.
lyche-numerical-linear-algebra-F03897	301	1.000	$\boldsymbol{p}_{\{0\}} = \boldsymbol{r}_{\{0\}}$	$\boldsymbol{p}_0 = \boldsymbol{r}_0$	Proof Equation (13.22) clearly holds for $k = 0$ since $\boldsymbol{p}_0 = \boldsymbol{r}_0$. Suppose it holds
lyche-numerical-linear-algebra-F03898	301	1.000	$\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k \in \mathbb{W}_{k+2}, \boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k \in \mathbb{W}_{k+2},$ $\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k \in \mathbb{W}_{k+2}$	$\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k \in \mathbb{W}_{k+2}, \boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k \in \mathbb{W}_{k+2}$	for some $k \geq 0$. Then $\boldsymbol{r}_{k+1} = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k \in \mathbb{W}_{k+2}, \boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k \in \mathbb{W}_{k+2}$
lyche-numerical-linear-algebra-F03899	301	0.995	$\mathbb{W}_{k+2}$	$\mathbb{W}_{k+2}$	$\mathbb{W}_{k+2}$ and $\boldsymbol{x}_{k+1} - \boldsymbol{x}_0 \stackrel{(13.12)}{=} \boldsymbol{x}_k - \boldsymbol{x}_0 + \alpha_k \boldsymbol{p}_k \in \mathbb{W}_{k+1}$. Thus (13.22) follows by
lyche-numerical-linear-algebra-F03900	301	0.995	$\boldsymbol{x}_{k+1} - \boldsymbol{x}_0 \stackrel{(13.12)}{=} \boldsymbol{x}_k - \boldsymbol{x}_0 + \alpha_k \boldsymbol{p}_k \in \mathbb{W}_{k+1}$	$\boldsymbol{x}_{k+1} - \boldsymbol{x}_0 \stackrel{(13.12)}{=} \boldsymbol{x}_k - \boldsymbol{x}_0 + \alpha_k \boldsymbol{p}_k \in \mathbb{W}_{k+1}$	$\mathbb{W}_{k+2}$ and $\boldsymbol{x}_{k+1} - \boldsymbol{x}_0 \stackrel{(13.12)}{=} \boldsymbol{x}_k - \boldsymbol{x}_0 + \alpha_k \boldsymbol{p}_k \in \mathbb{W}_{k+1}$. Thus (13.22) follows by
lyche-numerical-linear-algebra-F03901	301	0.984	$\boldsymbol{w} \in \mathbb{W}_k$	$\boldsymbol{w} \in \mathbb{W}_k$	induction. The equation (13.23) follows since any $\boldsymbol{w} \in \mathbb{W}_k$ is a linear combination
lyche-numerical-linear-algebra-F03902	301	1.000	$\{\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1}\}$	$\{\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1}\}$	of $\{\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1}\}$ and also $\{\boldsymbol{p}_0, \boldsymbol{p}_1, \dots, \boldsymbol{p}_{k-1}\}$.
lyche-numerical-linear-algebra-F03903	301	1.000	$\{\boldsymbol{p}_0, \boldsymbol{p}_1, \dots, \boldsymbol{p}_{k-1}\}$	$\{\boldsymbol{p}_0, \boldsymbol{p}_1, \dots, \boldsymbol{p}_{k-1}\}$	of $\{\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1}\}$ and also $\{\boldsymbol{p}_0, \boldsymbol{p}_1, \dots, \boldsymbol{p}_{k-1}\}$
lyche-numerical-linear-algebra-F03904	301	0.995	$\boldsymbol{u} := \boldsymbol{x}_k - \boldsymbol{x}_0 - \boldsymbol{w}$	$\boldsymbol{u} := \boldsymbol{x}_k - \boldsymbol{x}_0 - \boldsymbol{w}$	Proof Fix k , let $\boldsymbol{w} \in \mathbb{W}_k$ and $\boldsymbol{u} := \boldsymbol{x}_k - \boldsymbol{x}_0 - \boldsymbol{w}$. By (13.22) $\boldsymbol{u} \in \mathbb{W}_k$ and then (13.23)
lyche-numerical-linear-algebra-F03905	301	0.995	$\boldsymbol{u} \in \mathbb{W}_k$	$\boldsymbol{u} \in \mathbb{W}_k$	Proof Fix k , let $\boldsymbol{w} \in \mathbb{W}_k$ and $\boldsymbol{u} := \boldsymbol{x}_k - \boldsymbol{x}_0 - \boldsymbol{w}$. By (13.22) $\boldsymbol{u} \in \mathbb{W}_k$ and then (13.23)
lyche-numerical-linear-algebra-F03906	301	1.000	$\langle \boldsymbol{x} - \boldsymbol{x}_k, \boldsymbol{u} \rangle = \boldsymbol{r}_k^T \boldsymbol{u} = 0$	$\langle \boldsymbol{x} - \boldsymbol{x}_k, \boldsymbol{u} \rangle = \boldsymbol{r}_k^T \boldsymbol{u} = 0$	implies that $\langle \boldsymbol{x} - \boldsymbol{x}_k, \boldsymbol{u} \rangle = \boldsymbol{r}_k^T \boldsymbol{u} = 0$. Using Corollary 5.2 we obtain
lyche-numerical-linear-algebra-F03907	301	1.000	$\boldsymbol{u} = \mathbf{0}$	$\boldsymbol{u} = \mathbf{0}$	with equality for $\boldsymbol{u} = \mathbf{0}$.
lyche-numerical-linear-algebra-F03908	301	1.000	$\boldsymbol{x}_0 \neq \mathbf{0}$	$\boldsymbol{x}_0 \neq \mathbf{0}$	solution $\boldsymbol{x}$ in the $\boldsymbol{A}$ -norm. More generally, if $\boldsymbol{x}_0 \neq \mathbf{0}$ then $\boldsymbol{x} - \boldsymbol{x}_k = (\boldsymbol{x} - \boldsymbol{x}_0) -$
lyche-numerical-linear-algebra-F03909	301	1.000	$\boldsymbol{x} - \boldsymbol{x}_k = (\boldsymbol{x} - \boldsymbol{x}_0) -$	$\boldsymbol{x} - \boldsymbol{x}_k = (\boldsymbol{x} - \boldsymbol{x}_0) -$	solution $\boldsymbol{x}$ in the $\boldsymbol{A}$ -norm. More generally, if $\boldsymbol{x}_0 \neq \mathbf{0}$ then $\boldsymbol{x} - \boldsymbol{x}_k = (\boldsymbol{x} - \boldsymbol{x}_0) -$
lyche-numerical-linear-algebra-F03910	301	1.000	$(\boldsymbol{x}_k - \boldsymbol{x}_0)$	$(\boldsymbol{x}_k - \boldsymbol{x}_0)$	$(\boldsymbol{x}_k - \boldsymbol{x}_0)$ and $\boldsymbol{x}_k - \boldsymbol{x}_0$ is the element in $\mathbb{W}_k$ that is closest to $\boldsymbol{x} - \boldsymbol{x}_0$ in the $\boldsymbol{A}$ -norm.
lyche-numerical-linear-algebra-F03911	301	1.000	$\boldsymbol{x}_k - \boldsymbol{x}_0$	$\boldsymbol{x}_k - \boldsymbol{x}_0$	$(\boldsymbol{x}_k - \boldsymbol{x}_0)$ and $\boldsymbol{x}_k - \boldsymbol{x}_0$ is the element in $\mathbb{W}_k$ that is closest to $\boldsymbol{x} - \boldsymbol{x}_0$ in the $\boldsymbol{A}$ -norm.
lyche-numerical-linear-algebra-F03912	301	1.000	$p(t) := \sum_{j=0}^m a_j t^j$	$p(t) := \sum_{j=0}^m a_j t^j$	Recall that to each polynomial $p(t) := \sum_{j=0}^m a_j t^j$ there corresponds a matrix
lyche-numerical-linear-algebra-F03913	301	0.808	$p(\boldsymbol{A}) := a_0 \boldsymbol{I} + a_1 \boldsymbol{A} + \dots + a_m \boldsymbol{A}^m$	$p(\boldsymbol{A}) := a_0 \boldsymbol{I} + a_1 \boldsymbol{A} + \dots + a_m \boldsymbol{A}^m$	polynomial $p(\boldsymbol{A}) := a_0 \boldsymbol{I} + a_1 \boldsymbol{A} + \dots + a_m \boldsymbol{A}^m$. Moreover, if $(\lambda_j, \boldsymbol{u}_j)$ are eigenpairs
lyche-numerical-linear-algebra-F03914	301	0.979	$(p(\lambda_j), \boldsymbol{u}_j)$	$(p(\lambda_j), \boldsymbol{u}_j)$	of $\boldsymbol{A}$ then $(p(\lambda_j), \boldsymbol{u}_j)$ are eigenpairs of $p(\boldsymbol{A})$ for $j = 1, \dots, n$.
lyche-numerical-linear-algebra-F03915	301	0.979	$p(\boldsymbol{A})$	$p(\boldsymbol{A})$	of $\boldsymbol{A}$ then $(p(\lambda_j), \boldsymbol{u}_j)$ are eigenpairs of $p(\boldsymbol{A})$ for $j = 1, \dots, n$.
lyche-numerical-linear-algebra-F03916	301	0.991	$(\lambda_j, \boldsymbol{u}_j), j = 1, 2, \dots, n$	$(\lambda_j, \boldsymbol{u}_j), j = 1, 2, \dots, n$	$\mathbb{R}^{n \times n}$ is positive definite with orthonormal eigenpairs $(\lambda_j, \boldsymbol{u}_j), j = 1, 2, \dots, n$,
lyche-numerical-linear-algebra-F03917	301	1.000	$\boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	and let $\boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$ for some $\boldsymbol{x}_0 \in \mathbb{R}^n$. To each $\boldsymbol{w} \in \mathbb{W}_k$ there corresponds
lyche-numerical-linear-algebra-F03918	302	1.000	$P(t) := \sum_{j=0}^{k-1} a_j t^{k-1-j}$	$P(t) := \sum_{j=0}^{k-1} a_j t^{k-1-j}$	a polynomial $P(t) := \sum_{j=0}^{k-1} a_j t^{k-1-j}$ such that $\boldsymbol{w} = P(\boldsymbol{A}) \boldsymbol{r}_0$. Moreover, if $\boldsymbol{r}_0 =$
lyche-numerical-linear-algebra-F03919	302	1.000	$\boldsymbol{w} = P(\boldsymbol{A}) \boldsymbol{r}_0$	$\boldsymbol{w} = P(\boldsymbol{A}) \boldsymbol{r}_0$	a polynomial $P(t) := \sum_{j=0}^{k-1} a_j t^{k-1-j}$ such that $\boldsymbol{w} = P(\boldsymbol{A}) \boldsymbol{r}_0$. Moreover, if $\boldsymbol{r}_0 =$

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lyche-numerical-linear-algebra- F03920	302	1.000	$\sum_{j=1}^n \sigma_j \boldsymbol{u}_j$	$\sum_{j=1}^n \sigma_j \boldsymbol{u}_j$	$\sum_{j=1}^n \sigma_j \boldsymbol{u}_j$ then
lyche-numerical-linear-algebra- F03921	302	1.000	$\boldsymbol{w} = \boldsymbol{a}_0 + \boldsymbol{A} \boldsymbol{r}_0 + \cdots + \boldsymbol{A}^{k-1} \boldsymbol{r}_0$	$\boldsymbol{w} = a_0 \boldsymbol{r}_0 + a_1 \boldsymbol{A} \boldsymbol{r}_0 + \cdots + a_{k-1} \boldsymbol{A}^{k-1} \boldsymbol{r}_0$	Proof If $\boldsymbol{w} \in \mathbb{W}_k$ then $\boldsymbol{w} = a_0 \boldsymbol{r}_0 + a_1 \boldsymbol{A} \boldsymbol{r}_0 + \cdots + a_{k-1} \boldsymbol{A}^{k-1} \boldsymbol{r}_0$ for some scalars
lyche-numerical-linear-algebra- F03922	302	1.000	$\boldsymbol{a}_0, \ldots, \boldsymbol{a}_{k-1}$	$a_0, \ldots, a_{k-1}$	$a_0, \ldots, a_{k-1}$. But then $\boldsymbol{w} = P(\boldsymbol{A}) \boldsymbol{r}_0$. We find $\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1}(\boldsymbol{r}_0 -$
lyche-numerical-linear-algebra- F03923	302	1.000	$\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1}(\boldsymbol{r}_0 -$	$\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1}(\boldsymbol{r}_0 -$	$a_0, \ldots, a_{k-1}$. But then $\boldsymbol{w} = P(\boldsymbol{A}) \boldsymbol{r}_0$. We find $\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1}(\boldsymbol{r}_0 -$
lyche-numerical-linear-algebra- F03924	302	1.000	$\boldsymbol{A} P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_0$	$\boldsymbol{A} P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_0$	$\boldsymbol{A} P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_0$ and $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0) = Q(\boldsymbol{A}) \boldsymbol{r}_0$. Therefore,
lyche-numerical-linear-algebra- F03925	302	1.000	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0) = Q(\boldsymbol{A}) \boldsymbol{r}_0$	$\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0) = Q(\boldsymbol{A}) \boldsymbol{r}_0$	$\boldsymbol{A} P(\boldsymbol{A}) \boldsymbol{r}_0 = \boldsymbol{A}^{-1} Q(\boldsymbol{A}) \boldsymbol{r}_0$ and $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}_0 - P(\boldsymbol{A}) \boldsymbol{r}_0) = Q(\boldsymbol{A}) \boldsymbol{r}_0$. Therefore,
lyche-numerical-linear-algebra- F03926	302	1.000	$\boldsymbol{r}_0$	$\boldsymbol{r}_0$	Using the eigenvector expansion for $\boldsymbol{r}_0$ we obtain
lyche-numerical-linear-algebra- F03927	303	0.978	$0 < a < b$	$0 < a < b$	$0 < a < b$ is an interval containing all the eigenvalues of $\boldsymbol{A}$. Then in the conjugate
lyche-numerical-linear-algebra- F03928	303	0.978	$\boldsymbol{A}$	$\boldsymbol{A}$	$0 < a < b$ is an interval containing all the eigenvalues of $\boldsymbol{A}$. Then in the conjugate
lyche-numerical-linear-algebra- F03929	303	1.000	Π_k	Π_k	where Π_k denotes the class of univariate polynomials of degree $\leq k$ with real
lyche-numerical-linear-algebra- F03930	303	1.000	$\leq k$	$\leq k$	where Π_k denotes the class of univariate polynomials of degree $\leq k$ with real
lyche-numerical-linear-algebra- F03931	303	0.998	$Q(t) = 1$	$Q(t) = 1$	Proof By (13.25) with $Q(t) = 1$ (corresponding to $P(\boldsymbol{A}) = \mathbf{0}$) we find $\ \boldsymbol{x} -$
lyche-numerical-linear-algebra- F03932	303	0.998	$P(\boldsymbol{A}) = \mathbf{0}$	$P(\boldsymbol{A}) = \mathbf{0}$	Proof By (13.25) with $Q(t) = 1$ (corresponding to $P(\boldsymbol{A}) = \mathbf{0}$) we find $\ \boldsymbol{x} -$
lyche-numerical-linear-algebra- F03933	303	0.998	$\ \boldsymbol{x} -$	$\ \boldsymbol{x} -$	Proof By (13.25) with $Q(t) = 1$ (corresponding to $P(\boldsymbol{A}) = \mathbf{0}$) we find $\ \boldsymbol{x} -$
lyche-numerical-linear-algebra- F03934	303	0.969	$\ \boldsymbol{x}_0\ _{\boldsymbol{A}}^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}$	$\ \boldsymbol{x}_0\ _{\boldsymbol{A}}^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}$	$\ \boldsymbol{x}_0\ _{\boldsymbol{A}}^2 = \sum_{j=1}^n \frac{\sigma_j^2}{\lambda_j}$. Therefore, by the best approximation property Theorem 13.4
lyche-numerical-linear-algebra- F03935	303	1.000	$Q \in \Pi_k$	$Q \in \Pi_k$	where $Q \in \Pi_k$ and $Q(0) = 1$. Minimizing over such polynomials Q and taking
lyche-numerical-linear-algebra- F03936	303	1.000	$Q(0) = 1$	$Q(0) = 1$	where $Q \in \Pi_k$ and $Q(0) = 1$. Minimizing over such polynomials Q and taking
lyche-numerical-linear-algebra- F03937	303	0.999	$\kappa = \lambda_{\max} / \lambda_{\min}$	$\kappa = \lambda_{\max} / \lambda_{\min}$	where $\kappa = \lambda_{\max} / \lambda_{\min}$ is the spectral condition number of $\boldsymbol{A}$. Ignoring the second
lyche-numerical-linear-algebra- F03938	303	1.000	$y = 1/x$	$y = 1/x$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03939	303	1.000	$2^k/k! = 2, k = 1, 2$	$2^k/k! = 2, k = 1, 2$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03940	303	1.000	$2^k/k! < 2$	$2^k/k! < 2$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03941	303	1.000	$k > 2$	$k > 2$	$y = 1/x$, using $2^k/k! = 2, k = 1, 2$, and $2^k/k! < 2$ for $k > 2$, we find
lyche-numerical-linear-algebra- F03942	304	1.000	$c \notin [a, b]$	$c \notin [a, b]$	$c \notin [a, b]$ and $k \in \mathbb{N}$. Consider the set $\mathcal{S}_k$ of all polynomials Q of degree $\leq k$ such
lyche-numerical-linear-algebra- F03943	304	1.000	$\mathcal{S}_k$	$\mathcal{S}_k$	$c \notin [a, b]$ and $k \in \mathbb{N}$. Consider the set $\mathcal{S}_k$ of all polynomials Q of degree $\leq k$ such
lyche-numerical-linear-algebra- F03944	304	1.000	$Q(c) = 1$	$Q(c) = 1$	that $Q(c) = 1$. For any continuous function f on $[a, b]$ we define

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lyche-numerical-linear-algebra-F03945	304	■ 1.000	$Q^*\in\mathcal{S}_k$	$Q^*\in\mathcal{S}_k$	We want to find a polynomial $Q^*\in\mathcal{S}_k$ such that
lyche-numerical-linear-algebra-F03946	304	■ 1.000	Q^*	Q^*	We will show that Q^* is uniquely given as a suitably shifted and normalized
lyche-numerical-linear-algebra-F03947	304	■ 1.000	$T_{\{n\}}$	T_n	version of the Chebyshev polynomial . The Chebyshev polynomial T_n of degree n
lyche-numerical-linear-algebra-F03948	304	■ 0.999	$T_{\{0\}}(t)=1$	$T_0(t)=1$	starting with $T_0(t)=1$ and $T_1(t)=t$. Thus $T_2(t)=2t^2-1$, $T_3(t)=4t^3-3t$ etc.
lyche-numerical-linear-algebra-F03949	304	■ 0.999	$T_{\{1\}}(t)=t$	$T_1(t)=t$	starting with $T_0(t)=1$ and $T_1(t)=t$. Thus $T_2(t)=2t^2-1$, $T_3(t)=4t^3-3t$ etc.
lyche-numerical-linear-algebra-F03950	304	■ 0.999	$T_{\{2\}}(t)=2\,t^2-1$, $T_{\{3\}}(t)=4\,t^3-3\,t$	$T_2(t)=2t^2-1, T_3(t)=4t^3-3t$	starting with $T_0(t)=1$ and $T_1(t)=t$. Thus $T_2(t)=2t^2-1, T_3(t)=4t^3-3t$ etc.
lyche-numerical-linear-algebra-F03951	304	■ 1.000	$T_{\{n\}}(t)=\cos\left(n\arccos t\right)$	$T_n(t)=\cos(n\arccos t)$	1. $T_n(t)=\cos(n\arccos t)$ for $t\in[-1,1]$,
lyche-numerical-linear-algebra-F03952	304	■ 1.000	$t\in[-1,1]$	$t\in[-1,1]$	1. $T_n(t)=\cos(n\arccos t)$ for $t\in[-1,1]$,
lyche-numerical-linear-algebra-F03953	304	■ 1.000	$T_{\{n\}}(t)=\frac{1}{2}\left[\left(t+\sqrt{t^2-1}\right)^n+\left(t+\sqrt{t^2-1}\right)^{-n}\right]$	$T_n(t)=\frac{1}{2}\left[\left(t+\sqrt{t^2-1}\right)^n+\left(t+\sqrt{t^2-1}\right)^{-n}\right]$	2. $T_n(t)=\frac{1}{2}\left[\left(t+\sqrt{t^2-1}\right)^n+\left(t+\sqrt{t^2-1}\right)^{-n}\right]$ for $ t \geq 1$.
lyche-numerical-linear-algebra-F03954	304	■ 1.000	$ t \geq 1$	$ t \geq 1$	2. $T_n(t)=\frac{1}{2}\left[\left(t+\sqrt{t^2-1}\right)^n+\left(t+\sqrt{t^2-1}\right)^{-n}\right]$ for $ t \geq 1$.
lyche-numerical-linear-algebra-F03955	304	■ 1.000	$P_{\{n\}}(t)=\cos\left(n\arccos t\right)$	$P_n(t)=\cos(n\arccos t)$	1. With $P_n(t)=\cos(n\arccos t)$ we have $P_n(t)=\cos n\phi$, where $t=\cos\phi$.
lyche-numerical-linear-algebra-F03956	304	■ 1.000	$P_{\{n\}}(t)=\cos n\phi$	$P_n(t)=\cos n\phi$	1. With $P_n(t)=\cos(n\arccos t)$ we have $P_n(t)=\cos n\phi$, where $t=\cos\phi$.
lyche-numerical-linear-algebra-F03957	304	■ 1.000	$t=\cos\phi$	$t=\cos\phi$	1. With $P_n(t)=\cos(n\arccos t)$ we have $P_n(t)=\cos n\phi$, where $t=\cos\phi$.
lyche-numerical-linear-algebra-F03958	304	■ 1.000	$P_{\{n\}}$	P_n	and it follows that P_n satisfies the same recurrence relation as T_n . Since $P_0=T_0$
lyche-numerical-linear-algebra-F03959	304	■ 1.000	$P_{\{0\}}=T_{\{0\}}$	$P_0=T_0$	and it follows that P_n satisfies the same recurrence relation as T_n . Since $P_0=T_0$
lyche-numerical-linear-algebra-F03960	304	■ 1.000	$P_{\{1\}}=T_{\{1\}}$	$P_1=T_1$	and $P_1=T_1$ we have $P_n=T_n$ for all $n\geq 0$.
lyche-numerical-linear-algebra-F03961	304	■ 1.000	$P_{\{n\}}=T_{\{n\}}$	$P_n=T_n$	and $P_1=T_1$ we have $P_n=T_n$ for all $n\geq 0$.
lyche-numerical-linear-algebra-F03962	304	■ 1.000	$x_{\{n\}}:=T_{\{n\}}(t)$	$x_n:=T_n(t)$	2. Fix t with $ t \geq 1$ and let $x_n:=T_n(t)$ for $n\geq 0$. The recurrence relation for the
lyche-numerical-linear-algebra-F03963	305	■ 0.711	$x_{\{n\}}=z^{\{n\}}$	$x_n=z^n$	To solve this difference equation we insert $x_n=z^n$ into (13.31) and obtain $z^{n+1}-$
lyche-numerical-linear-algebra-F03964	305	■ 0.711	$z^{\{n+1\}}-$	$z^{n+1}-$	To solve this difference equation we insert $x_n=z^n$ into (13.31) and obtain $z^{n+1}-$
lyche-numerical-linear-algebra-F03965	305	■ 1.000	$2\,t\,z^{\{n\}}+z^{\{n-1\}}=0$	$2tz^n+z^{n-1}=0$	$2tz^n+z^{n-1}=0$ or $z^2-2tz+1=0$. The roots of this equation are
lyche-numerical-linear-algebra-F03966	305	■ 1.000	$z^{\{2\}}-2\,t\,z+1=0$	$z^2-2tz+1=0$	$2tz^n+z^{n-1}=0$ or $z^2-2tz+1=0$. The roots of this equation are
lyche-numerical-linear-algebra-F03967	305	■ 1.000	$z_{\{1\}}^{\{n\}},\,z_{\{2\}}^{\{n\}}$	z_1^n, z_2^n	Now z_1^n, z_2^n and more generally $c_1z_1^n+c_2z_2^n$ are solutions of (13.31) for any
lyche-numerical-linear-algebra-F03968	305	■ 1.000	$c_{\{1\}}\,z_{\{1\}}^{\{n\}}+c_{\{2\}}\,z_{\{2\}}^{\{n\}}$	$c_1z_1^n+c_2z_2^n$	Now z_1^n, z_2^n and more generally $c_1z_1^n+c_2z_2^n$ are solutions of (13.31) for any
lyche-numerical-linear-algebra-F03969	305	■ 1.000	$c_{\{1\}}$	c_1	constants c_1 and c_2 . We find these constants from the initial conditions $x_0=$
lyche-numerical-linear-algebra-F03970	305	■ 1.000	$c_{\{2\}}$	c_2	constants c_1 and c_2 . We find these constants from the initial conditions $x_0=$
lyche-numerical-linear-algebra-F03971	305	■ 1.000	$x_{\{0\}}=$	$x_0=$	constants c_1 and c_2 . We find these constants from the initial conditions $x_0=$

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lyche-numerical-linear-algebra- F03972	305	■ 1.000	$c_{\{1\}}+c_{\{2\}}=1$	$c_1 + c_2 = 1$	$c_1 + c_2 = 1$ and $x_1 = c_1 z_1 + c_2 z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra- F03973	305	■ 1.000	$x_{\{1\}}=c_{\{1\}} z_{\{1\}}+c_{\{2\}} z_{\{2\}}=t$	$x_1 = c_1 z_1 + c_2 z_2 = t$	$c_1 + c_2 = 1$ and $x_1 = c_1 z_1 + c_2 z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra- F03974	305	■ 1.000	$z_{\{1\}}+z_{\{2\}}=2 t$	$z_1 + z_2 = 2t$	$c_1 + c_2 = 1$ and $x_1 = c_1 z_1 + c_2 z_2 = t$. Since $z_1 + z_2 = 2t$ the solution is
lyche-numerical-linear-algebra- F03975	305	■ 1.000	$c_{\{1\}}=c_{\{2\}}=\frac{1}{2}$	$c_1 = c_2 = \frac{1}{2}$	$c_1 = c_2 = \frac{1}{2}$.
lyche-numerical-linear-algebra- F03976	305	■ 1.000	$a<b, c \notin[a, b]$	$a < b, c \notin [a, b]$	Theorem 13.6 (A Minimal Norm Problem) Suppose $a < b, c \notin [a, b]$ and $k \in$
lyche-numerical-linear-algebra- F03977	305	■ 1.000	$k \in$	$k \in$	Theorem 13.6 (A Minimal Norm Problem) Suppose $a < b, c \notin [a, b]$ and $k \in$
lyche-numerical-linear-algebra- F03978	305	■ 1.000	$\mathbb{N}$	$\mathbb{N}$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03979	305	■ 1.000	$Q \in \mathcal{S}_k$	$Q \in \mathcal{S}_k$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03980	305	■ 1.000	$Q \neq Q^*$	$Q \neq Q^*$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03981	305	■ 1.000	$\ Q\ _\infty > \ Q^*\ _\infty$	$\ Q\ _\infty > \ Q^*\ _\infty$	$\mathbb{N}$. If $Q \in \mathcal{S}_k$ and $Q \neq Q^*$ then $\ Q\ _\infty > \ Q^*\ _\infty$.
lyche-numerical-linear-algebra- F03982	305	■ 1.000	$p(z)=p'(z)=0$	$p(z) = p'(z) = 0$	$p(z) = p'(z) = 0$, we say that p has a double zero at z . Counting such a zero as
lyche-numerical-linear-algebra- F03983	305	■ 1.000	$ Q^* $	$ Q^* $	$ Q^* $ takes on its maximum $1/ T_k(u(c)) $ at the $k + 1$ points $\mu_0, \dots, \mu_k$ in $[a, b]$
lyche-numerical-linear-algebra- F03984	305	■ 1.000	$1/ T_k(u(c)) $	$1/ T_k(u(c)) $	$ Q^* $ takes on its maximum $1/ T_k(u(c)) $ at the $k + 1$ points $\mu_0, \dots, \mu_k$ in $[a, b]$
lyche-numerical-linear-algebra- F03985	305	■ 1.000	$\mu_0, \dots, \mu_k$	$\mu_0, \dots, \mu_k$	$ Q^* $ takes on its maximum $1/ T_k(u(c)) $ at the $k + 1$ points $\mu_0, \dots, \mu_k$ in $[a, b]$
lyche-numerical-linear-algebra- F03986	305	■ 1.000	$u(\mu_i) = \cos(i\pi/k)$	$u(\mu_i) = \cos(i\pi/k)$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \dots, k$. Suppose $Q \in \mathcal{S}_k$ and that
lyche-numerical-linear-algebra- F03987	305	■ 1.000	$i = 0, 1, \dots, k$	$i = 0, 1, \dots, k$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \dots, k$. Suppose $Q \in \mathcal{S}_k$ and that
lyche-numerical-linear-algebra- F03988	305	■ 1.000	$Q \in \mathcal{S}_k$	$Q \in \mathcal{S}_k$	such that $u(\mu_i) = \cos(i\pi/k)$ for $i = 0, 1, \dots, k$. Suppose $Q \in \mathcal{S}_k$ and that
lyche-numerical-linear-algebra- F03989	305	■ 1.000	$\ Q\ _\infty \leq \ Q^*\ _\infty$	$\ Q\ _\infty \leq \ Q^*\ _\infty$	$\ Q\ _\infty \leq \ Q^*\ _\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra- F03990	305	■ 1.000	$Q \equiv Q^*$	$Q \equiv Q^*$	$\ Q\ _\infty \leq \ Q^*\ _\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra- F03991	305	■ 1.000	$f \equiv Q - Q^*$	$f \equiv Q - Q^*$	$\ Q\ _\infty \leq \ Q^*\ _\infty$. We have to show that $Q \equiv Q^*$. Let $f \equiv Q - Q^*$. We show
lyche-numerical-linear-algebra- F03992	305	■ 1.000	$f(c)=0$	$f(c) = 0$	$f(c) = 0$, this means that $f \equiv 0$ or equivalently $Q \equiv Q^*$.
lyche-numerical-linear-algebra- F03993	305	■ 1.000	$f \equiv 0$	$f \equiv 0$	$f(c) = 0$, this means that $f \equiv 0$ or equivalently $Q \equiv Q^*$.
lyche-numerical-linear-algebra- F03994	305	■ 1.000	$I_j = [\mu_{j-1}, \mu_j]$	$I_j = [\mu_{j-1}, \mu_j]$	Consider $I_j = [\mu_{j-1}, \mu_j]$ for a fixed j . Let
lyche-numerical-linear-algebra- F03995	305	■ 1.000	$\sigma_j \leq 0$	$\sigma_j \leq 0$	We have $\sigma_j \leq 0$. For if say $Q^*(\mu_j) > 0$ then
lyche-numerical-linear-algebra- F03996	305	■ 1.000	$Q^*(\mu_j) > 0$	$Q^*(\mu_j) > 0$	We have $\sigma_j \leq 0$. For if say $Q^*(\mu_j) > 0$ then
lyche-numerical-linear-algebra- F03997	305	■ 1.000	$f(\mu_j) \leq 0$	$f(\mu_j) \leq 0$	so that $f(\mu_j) \leq 0$. Moreover,
lyche-numerical-linear-algebra- F03998	305	■ 1.000	$f(\mu_{j-1}) \geq 0$	$f(\mu_{j-1}) \geq 0$	Thus $f(\mu_{j-1}) \geq 0$ and it follows that $\sigma_j \leq 0$. Similarly, $\sigma_j \leq 0$ if $Q^*(\mu_j) < 0$.

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lyche-numerical-linear-algebra-F04025	307	■ 1.000	$f(x) := x + 1 / x$	$f(x) := x + 1/x$	where $f : [mc, Mc] \rightarrow \mathbb{R}$ is given by $f(x) := x + 1/x$. By (13.38)
lyche-numerical-linear-algebra-F04026	307	■ 1.000	$f \in C^2$	$f \in C^2$	Since $f \in C^2$ and f'' is positive it follows from Lemma 8.2 that f is a convex
lyche-numerical-linear-algebra-F04027	307	■ 1.000	$f^{\prime \prime}$	f''	Since $f \in C^2$ and f'' is positive it follows from Lemma 8.2 that f is a convex
lyche-numerical-linear-algebra-F04028	308	■ 1.000	$c := 1 / \sqrt{m M}$	$c := 1/\sqrt{mM}$	Choosing $c := 1/\sqrt{mM}$ we find $f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$.
lyche-numerical-linear-algebra-F04029	308	■ 1.000	$f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$	$f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$	Choosing $c := 1/\sqrt{mM}$ we find $f(mc) = f(Mc) = \sqrt{\frac{M}{m}} + \sqrt{\frac{m}{M}} = \frac{M+m}{\sqrt{mM}}$.
lyche-numerical-linear-algebra-F04030	308	■ 1.000	$\epsilon_j := x - x_j, j = 0, 1, \dots$	$\epsilon_j := x - x_j, j = 0, 1, \dots$	<i>Proof of (13.19)</i> Let $\epsilon_j := x - x_j, j = 0, 1, \dots$, where $Ax = b$. It is enough to
lyche-numerical-linear-algebra-F04031	308	■ 1.000	$\left\ \epsilon_k \right\ _A \leq \left(\frac{\kappa - 1}{\kappa + 1} \right) \left\ \epsilon_{k-1} \right\ \leq \dots \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k \left\ \epsilon_0 \right\ $	$\ \epsilon_k\ _A \leq \left(\frac{\kappa - 1}{\kappa + 1} \right) \ \epsilon_{k-1}\ \leq \dots \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k \ \epsilon_0\ $	for then $\ \epsilon_k\ _A \leq \left(\frac{\kappa-1}{\kappa+1} \right) \ \epsilon_{k-1}\ \leq \dots \leq \left(\frac{\kappa-1}{\kappa+1} \right)^k \ \epsilon_0\ $. It follows from (13.8) that
lyche-numerical-linear-algebra-F04032	309	■ 1.000	$r_{m+1} = \mathbf{0}$	$r_{m+1} = \mathbf{0}$	gradient method m be the smallest integer such that $r_{m+1} = \mathbf{0}$. For $k \leq m$ we
lyche-numerical-linear-algebra-F04033	309	■ 1.000	$k \leq m$	$k \leq m$	gradient method m be the smallest integer such that $r_{m+1} = \mathbf{0}$. For $k \leq m$ we
lyche-numerical-linear-algebra-F04034	309	■ 1.000	$\left\ \epsilon_{k+1} \right\ _2 < \left\ \epsilon_k \right\ _2$	$\ \epsilon_{k+1}\ _2 < \ \epsilon_k\ _2$	have $\ \epsilon_{k+1}\ _2 < \ \epsilon_k\ _2$. More precisely,
lyche-numerical-linear-algebra-F04035	309	■ 0.983	$\epsilon_j = x - x_j$	$\epsilon_j = x - x_j$	where $\epsilon_j = x - x_j$ and $Ax = b$.
lyche-numerical-linear-algebra-F04036	309	■ 1.000	$j \leq m$	$j \leq m$	Proof For $j \leq m$
lyche-numerical-linear-algebra-F04037	309	■ 1.000	$\beta_{i-1} \cdots \beta_k = (r_i^T r_i) / (r_k^T r_k)$	$\beta_{i-1} \cdots \beta_k = (r_i^T r_i) / (r_k^T r_k)$	and since $\beta_{i-1} \cdots \beta_k = (r_i^T r_i) / (r_k^T r_k)$ we find
lyche-numerical-linear-algebra-F04038	310	■ 1.000	$\operatorname{cond}_2(A)$	$\operatorname{cond}_2(A)$	For problems $Ax = b$ of size n , where both n and $\operatorname{cond}_2(A)$ are large, it is
lyche-numerical-linear-algebra-F04039	310	■ 1.000	$B Ax = B b$	$B Ax = B b$	equivalent system $B Ax = B b$, where B is nonsingular and $\operatorname{cond}_2(BA)$ is smaller
lyche-numerical-linear-algebra-F04040	310	■ 1.000	$\operatorname{cond}_2(BA)$	$\operatorname{cond}_2(BA)$	equivalent system $B Ax = B b$, where B is nonsingular and $\operatorname{cond}_2(BA)$ is smaller
lyche-numerical-linear-algebra-F04041	310	■ 0.578	$B = M^{-1}$	$B = M^{-1}$	$B = M^{-1}$. We cannot use CG on $B Ax = B b$ directly since BA in general is not
lyche-numerical-linear-algebra-F04042	310	■ 1.000	$B = C^T C$	$B = C^T C$	such that $B = C^T C$. (C is only needed for the derivation and will not appear in the
lyche-numerical-linear-algebra-F04043	310	■ 1.000	$y := C^{-T} x$	$y := C^{-T} x$	where $y := C^{-T} x$. We have 3 linear systems
lyche-numerical-linear-algebra-F04044	311	■ 1.000	CAC^T	CAC^T	being positive definite the matrix CAC^T is similar to BA . Indeed,
lyche-numerical-linear-algebra-F04045	311	■ 1.000	$(CAC^T)y = Cb$	$(CAC^T)y = Cb$	We apply the conjugate gradient method to $(CAC^T)y = Cb$. Denoting the
lyche-numerical-linear-algebra-F04046	311	■ 0.997	q_k	q_k	search direction by q_k and the residual by $z_k := Cb - CAC^T y_k$ we obtain the
lyche-numerical-linear-algebra-F04047	311	■ 0.997	$z_k := Cb - CAC^T y_k$	$z_k := Cb - CAC^T y_k$	search direction by q_k and the residual by $z_k := Cb - CAC^T y_k$ we obtain the

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lyche-numerical-linear-algebra- F04048	311	■ 0.998	$\backslash\boldsymbol{A} \ \backslash\boldsymbol{x}=\backslash\boldsymbol{b},$ $\backslash\boldsymbol{r}_{\{k\}}=\backslash\boldsymbol{b}-\backslash\boldsymbol{A}$ $\backslash\boldsymbol{x}_{\{k\}}$	$\boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{Ax}_k$	Here $\boldsymbol{x}_k$ will be an approximation to the solution $\boldsymbol{x}$ of $\boldsymbol{Ax} = \boldsymbol{b}, \boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{Ax}_k$ is
lyche-numerical-linear-algebra- F04049	311	■ 0.573	$\backslash\boldsymbol{s}_{\{k\}}=\backslash\boldsymbol{C}^{\wedge\{T\}} \ \backslash\boldsymbol{z}_{\{k\}}=$ $\backslash\boldsymbol{C}^{\wedge\{T\}}\backslash\left(\backslash\boldsymbol{C}-\backslash\boldsymbol{C}$ $\backslash\boldsymbol{A} \ \backslash\boldsymbol{C}^{\wedge\{T\}}\backslash\right)$ $\backslash\boldsymbol{y}_{\{k\}}=\backslash\boldsymbol{B} \ \backslash\boldsymbol{b}-$	$\boldsymbol{s}_k = \boldsymbol{C}^T \boldsymbol{z}_k = \boldsymbol{C}^T (\boldsymbol{C} - \boldsymbol{CAC}^T) \boldsymbol{y}_k = \boldsymbol{Bb}-$	the residual in the original system, and $\boldsymbol{s}_k = \boldsymbol{C}^T \boldsymbol{z}_k = \boldsymbol{C}^T (\boldsymbol{C} - \boldsymbol{CAC}^T) \boldsymbol{y}_k = \boldsymbol{Bb}-$
lyche-numerical-linear-algebra- F04050	311	■ 1.000	$\backslash\boldsymbol{B} \ \backslash\boldsymbol{A} \ \backslash\boldsymbol{x}_{\{k\}}$	$\boldsymbol{BAx}_k$	$\boldsymbol{BAx}_k$ is the residual in the preconditioned system. If we set $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{Ax}_0$, $\boldsymbol{p}_0 =$
lyche-numerical-linear-algebra- F04051	311	■ 1.000	$\backslash\boldsymbol{r}_{\{0\}}=\backslash\boldsymbol{b}-\backslash\boldsymbol{A}$ $\backslash\boldsymbol{x}_{\{0\}}, \ \backslash\boldsymbol{p}_{\{0\}}=$	$\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{Ax}_0, \boldsymbol{p}_0 =$	$\boldsymbol{BAx}_k$ is the residual in the preconditioned system. If we set $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{Ax}_0$, $\boldsymbol{p}_0 =$
lyche-numerical-linear-algebra- F04052	311	■ 0.835	$\backslash\boldsymbol{s}_{\{0\}}=\backslash\boldsymbol{B} \ \boldsymbol{r}_{\{0\}}$	$\boldsymbol{s}_0 = \boldsymbol{Br}_0$	$\boldsymbol{s}_0 = \boldsymbol{Br}_0$, we obtain the following preconditioned conjugate gradient algorithm for
lyche-numerical-linear-algebra- F04053	311	■ 0.995	$\mathrm{x}\backslash\left(=\backslash\boldsymbol{x}_{\{0\}}\backslash\right)$	$\boldsymbol{x}(= \boldsymbol{x}_0)$	and $\boldsymbol{x}(= \boldsymbol{x}_0)$ is the starting iteration.
lyche-numerical-linear-algebra- F04054	312	■ 1.000	$\backslash\boldsymbol{\rho}$	ρ	Apart from the calculation of ρ this algorithm is quite similar to Algorithm 13.1.
lyche-numerical-linear-algebra- F04055	312	■ 1.000	$\boldsymbol{w}=\boldsymbol{B} * \boldsymbol{t}$	$\boldsymbol{w} = \boldsymbol{B} * \boldsymbol{t}$	The main additional work is contained in $\boldsymbol{w} = \boldsymbol{B} * \boldsymbol{t}$. We'll discuss this further in
lyche-numerical-linear-algebra- F04056	312	■ 1.000	$\backslash\boldsymbol{y}_{\{k\}}$	$\boldsymbol{y}_k$	where $\boldsymbol{y}_k$ is the conjugate gradient approximation to the solution $\boldsymbol{y}$ of (13.46) and κ
lyche-numerical-linear-algebra- F04057	313	■ 0.998	$\backslash\boldsymbol{B} \ \backslash\boldsymbol{x}$	$\boldsymbol{Bx}$	2. The evaluation of $\boldsymbol{Bx}$ for a given vector $\boldsymbol{x}$ should not be expensive in storage and
lyche-numerical-linear-algebra- F04058	313	■ 1.000	$\boldsymbol{u}=\boldsymbol{u}(\boldsymbol{x}, \ \boldsymbol{y})$	$\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x}, \boldsymbol{y})$	and $\boldsymbol{c}$ are given and we seek a function $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x}, \boldsymbol{y})$ such that (13.52) holds. We
lyche-numerical-linear-algebra- F04059	313	■ 1.000	$\boldsymbol{c}(\boldsymbol{x}, \ \boldsymbol{y})>0$	$\boldsymbol{c}(\boldsymbol{x}, \boldsymbol{y}) > 0$	assume that $\boldsymbol{c}$ and $\boldsymbol{f}$ are defined and continuous on Ω and that $\boldsymbol{c}(\boldsymbol{x}, \boldsymbol{y}) > 0$ for
lyche-numerical-linear-algebra- F04060	313	■ 0.991	$(\boldsymbol{x}, \ \boldsymbol{y}) \ \backslash\mathrm{in} \ \backslash\Omega$	$(\boldsymbol{x}, \boldsymbol{y}) \in \Omega$	all $(\boldsymbol{x}, \boldsymbol{y}) \in \Omega$. The problem (13.52) reduces to the Poisson problem (10.1) in the
lyche-numerical-linear-algebra- F04061	313	■ 1.000	$\boldsymbol{c}(\boldsymbol{x}, \ \boldsymbol{y})=1$	$\boldsymbol{c}(\boldsymbol{x}, \boldsymbol{y}) = 1$	special case where $\boldsymbol{c}(\boldsymbol{x}, \boldsymbol{y}) = 1$ for $(\boldsymbol{x}, \boldsymbol{y}) \in \Omega$.
lyche-numerical-linear-algebra- F04062	313	■ 1.000	$\boldsymbol{m} \ \backslash\mathrm{in} \ \backslash\mathbb{N}$	$\boldsymbol{m} \in \mathbb{N}$	To solve (13.52) numerically, we choose $\boldsymbol{m} \in \mathbb{N}$, set $\boldsymbol{h} := 1/(\boldsymbol{m} + 1)$, and define
lyche-numerical-linear-algebra- F04063	313	■ 1.000	$\boldsymbol{v}_{\{j, \ k\}} \ \backslash\mathrm{approx} \ \boldsymbol{u}\backslash\left(\boldsymbol{x}_{\{j\}}, \ \boldsymbol{y}_{\{k\}}\backslash\right)$	$\boldsymbol{v}_{j,k} \approx \boldsymbol{u}(\boldsymbol{x}_j, \boldsymbol{y}_k)$	We compute approximations $\boldsymbol{v}_{j,k} \approx \boldsymbol{u}(\boldsymbol{x}_j, \boldsymbol{y}_k)$ on a grid of points
lyche-numerical-linear-algebra- F04064	314	■ 1.000	$\boldsymbol{f}, \ \boldsymbol{g}$	$\boldsymbol{f}, \boldsymbol{g}$	using a finite difference method. For univariate functions $\boldsymbol{f}, \boldsymbol{g}$ we approximate
lyche-numerical-linear-algebra- F04065	314	■ 1.000	$\boldsymbol{f}_{\{j, \ k\}}:=\boldsymbol{f}\backslash\left(\boldsymbol{x}_{\{j\}}, \ \boldsymbol{y}_{\{k\}}\backslash\right),(\boldsymbol{d} \ \boldsymbol{v})_{\{j, \ k\}}:=$ $\backslash\left(\boldsymbol{d}_{\{1\}} \ \boldsymbol{v}\backslash\right)_{\{j, \ k\}}+\backslash\left(\boldsymbol{d}_{\{2\}} \ \boldsymbol{v}\backslash\right)_{\{j, \ k\}}$	$\boldsymbol{f}_{j,k} := \boldsymbol{f}(\boldsymbol{x}_j, \boldsymbol{y}_k), (\boldsymbol{dv})_{j,k} := (\boldsymbol{d}_1 \boldsymbol{v})_{j,k} + (\boldsymbol{d}_2 \boldsymbol{v})_{j,k}$	where $\boldsymbol{f}_{j,k} := \boldsymbol{f}(\boldsymbol{x}_j, \boldsymbol{y}_k), (\boldsymbol{dv})_{j,k} := (\boldsymbol{d}_1 \boldsymbol{v})_{j,k} + (\boldsymbol{d}_2 \boldsymbol{v})_{j,k}$,
lyche-numerical-linear-algebra- F04066	314	■ 1.000	$\boldsymbol{c}_{\{p, \ q\}}=\boldsymbol{c}(\boldsymbol{p} \ \boldsymbol{h}, \ \boldsymbol{q} \ \boldsymbol{h})$	$\boldsymbol{c}_{p,q} = \boldsymbol{c}(\boldsymbol{ph}, \boldsymbol{qh})$	and where $\boldsymbol{c}_{p,q} = \boldsymbol{c}(\boldsymbol{ph}, \boldsymbol{qh})$ for $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}$. The equation (13.53) can be written in
lyche-numerical-linear-algebra- F04067	314	■ 1.000	$\boldsymbol{p}, \ \boldsymbol{q} \ \backslash\mathrm{in} \ \backslash\mathbb{R}$	$\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}$	and where $\boldsymbol{c}_{p,q} = \boldsymbol{c}(\boldsymbol{ph}, \boldsymbol{qh})$ for $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}$. The equation (13.53) can be written in
lyche-numerical-linear-algebra- F04068	314	■ 0.997	$\boldsymbol{h}^{\{2\}} \ \boldsymbol{L}_{\{h\}} \ \boldsymbol{v}=\boldsymbol{h}^{\{2\}} \ \backslash\boldsymbol{F}$	$\boldsymbol{h}^2 \boldsymbol{L}_h \boldsymbol{v} = \boldsymbol{h}^2 \boldsymbol{F}$	as unknowns. The system $\boldsymbol{h}^2 \boldsymbol{L}_h \boldsymbol{v} = \boldsymbol{h}^2 \boldsymbol{F}$ can be written in standard form $\boldsymbol{Ax} = \boldsymbol{b}$
lyche-numerical-linear-algebra- F04069	315	■ 1.000	$\backslash\boldsymbol{V}, \ \backslash\boldsymbol{W} \ \backslash\mathrm{in} \ \backslash\mathbb{R}^{\wedge\{\boldsymbol{m} \ \mathrm{times} \ \boldsymbol{m}\}}$	$\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$	Lemma 13.6 (Discrete Inner Product) If $\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$ and $\boldsymbol{v}_{j,k} = \boldsymbol{w}_{j,k} = 0$ for
lyche-numerical-linear-algebra- F04070	315	■ 1.000	$\boldsymbol{v}_{\{j, \ k\}}=\boldsymbol{w}_{\{j, \ k\}}=0$	$\boldsymbol{v}_{j,k} = \boldsymbol{w}_{j,k} = 0$	Lemma 13.6 (Discrete Inner Product) If $\boldsymbol{V}, \boldsymbol{W} \in \mathbb{R}^{m \times m}$ and $\boldsymbol{v}_{j,k} = \boldsymbol{w}_{j,k} = 0$ for
lyche-numerical-linear-algebra- F04071	315	■ 1.000	$(\boldsymbol{j}, \ \boldsymbol{k}) \ \backslash\mathrm{in} \ \backslash\partial \boldsymbol{I}_{\{\boldsymbol{m}\}}$	$(\boldsymbol{j}, \boldsymbol{k}) \in \partial \boldsymbol{I}_m$	$(\boldsymbol{j}, \boldsymbol{k}) \in \partial \boldsymbol{I}_m$, then
lyche-numerical-linear-algebra- F04072	315	■ 0.994	$\boldsymbol{m} \ \backslash\mathrm{in} \ \backslash\mathbb{N}, \ \boldsymbol{a}_{\{\boldsymbol{i}\}}, \ \boldsymbol{b}_{\{\boldsymbol{i}\}}, \ \boldsymbol{c}_{\{\boldsymbol{i}\}} \ \backslash\mathrm{in} \ \backslash\mathbb{R}$	$\boldsymbol{m} \in \mathbb{N}, \boldsymbol{a}_i, \boldsymbol{b}_i, \boldsymbol{c}_i \in \mathbb{R}$	Proof If $\boldsymbol{m} \in \mathbb{N}, \boldsymbol{a}_i, \boldsymbol{b}_i, \boldsymbol{c}_i \in \mathbb{R}$ for $\boldsymbol{i} = 0, \dots, \boldsymbol{m}$ and $\boldsymbol{b}_0 = \boldsymbol{c}_0 = \boldsymbol{b}_{\boldsymbol{m}+1} = \boldsymbol{c}_{\boldsymbol{m}+1} = 0$
lyche-numerical-linear-algebra- F04073	315	■ 0.994	$\boldsymbol{i}=0, \ \backslash\mathrm{ldots}, \ \boldsymbol{m}$	$\boldsymbol{i} = 0, \dots, \boldsymbol{m}$	Proof If $\boldsymbol{m} \in \mathbb{N}, \boldsymbol{a}_i, \boldsymbol{b}_i, \boldsymbol{c}_i \in \mathbb{R}$ for $\boldsymbol{i} = 0, \dots, \boldsymbol{m}$ and $\boldsymbol{b}_0 = \boldsymbol{c}_0 = \boldsymbol{b}_{\boldsymbol{m}+1} = \boldsymbol{c}_{\boldsymbol{m}+1} = 0$

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lyche-numerical-linear-algebra- F04874	315	■ 0.994	$b_{\{0\}}=c_{\{0\}}=b_{\{m+1\}}=c_{\{m+1\}}=0$	$b_0 = c_0 = b_{m+1} = c_{m+1} = 0$	Proof If $m \in \mathbb{N}$, $a_i, b_i, c_i \in \mathbb{R}$ for $i = 0, \dots, m$ and $b_0 = c_0 = b_{m+1} = c_{m+1} = 0$
lyche-numerical-linear-algebra- F04875	315	■ 1.000	$\left(d_{\{1\}} v\right)_{\{j, k\}} w_{\{j, k\}}$	$(d_1 v)_{j,k} w_{j,k}$	and the right hand side of (13.60) follows. We apply (13.60) to $(d_1 v)_{j,k} w_{j,k}$ and
lyche-numerical-linear-algebra- F04876	315	■ 1.000	$\left(d_{\{2\}} v\right)_{\{j, k\}} w_{\{j, k\}}$	$(d_2 v)_{j,k} w_{j,k}$	$(d_2 v)_{j,k} w_{j,k}$ given by (13.55) and (13.59) follows.
lyche-numerical-linear-algebra- F04877	315	■ 1.000	$\left\langle L_h v, \boldsymbol{W} \right\rangle = \left\langle \boldsymbol{W}, L_h v \right\rangle$	$\langle L_h v, \boldsymbol{W} \rangle = \langle \boldsymbol{W}, L_h v \rangle$	Proof By (13.59) $\langle L_h v, \boldsymbol{W} \rangle = \langle \boldsymbol{W}, L_h v \rangle$ and symmetry follows. We take $\boldsymbol{W} =$
lyche-numerical-linear-algebra- F04878	315	■ 1.000	$\boldsymbol{W} =$	$\boldsymbol{W} =$	Proof By (13.59) $\langle L_h v, \boldsymbol{W} \rangle = \langle \boldsymbol{W}, L_h v \rangle$ and symmetry follows. We take $\boldsymbol{W} =$
lyche-numerical-linear-algebra- F04879	315	■ 1.000	$c_{\{j+\frac{1}{2}\}, k}$	$c_{j+\frac{1}{2},k}$	$\boldsymbol{V}$ and obtain quadratic factors in (13.59). Since $c_{j+\frac{1}{2},k}$ and $c_{j,k+\frac{1}{2}}$ correspond to
lyche-numerical-linear-algebra- F04880	315	■ 1.000	$c_{\{j, k+\frac{1}{2}\}}$	$c_{j,k+\frac{1}{2}}$	$\boldsymbol{V}$ and obtain quadratic factors in (13.59). Since $c_{j+\frac{1}{2},k}$ and $c_{j,k+\frac{1}{2}}$ correspond to
lyche-numerical-linear-algebra- F04881	315	■ 1.000	$\left\langle L_h v, \boldsymbol{V} \right\rangle \geq 0$	$\langle L_h v, \boldsymbol{V} \rangle \geq 0$	and $\langle L_h v, \boldsymbol{V} \rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m \times m}$. If $\langle L_h v, \boldsymbol{V} \rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra- F04882	315	■ 1.000	$\boldsymbol{V} \in \mathbb{R}^{m \times m}$	$\boldsymbol{V} \in \mathbb{R}^{m \times m}$	and $\langle L_h v, \boldsymbol{V} \rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m \times m}$. If $\langle L_h v, \boldsymbol{V} \rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra- F04883	315	■ 1.000	$\left\langle L_h v, \boldsymbol{V} \right\rangle = 0$	$\langle L_h v, \boldsymbol{V} \rangle = 0$	and $\langle L_h v, \boldsymbol{V} \rangle \geq 0$ for all $\boldsymbol{V} \in \mathbb{R}^{m \times m}$. If $\langle L_h v, \boldsymbol{V} \rangle = 0$ then all the quadratic factors
lyche-numerical-linear-algebra- F04884	315	■ 1.000	$v_{\{j, k+1\}} = v_{\{j, k\}}$	$v_{j,k+1} = v_{j,k}$	must be zero, and $v_{j,k+1} = v_{j,k}$ for $k = 0, 1, \dots, m$ and $j = 1, \dots, m$. Now
lyche-numerical-linear-algebra- F04885	315	■ 1.000	$k=0,1, \ldots, m$	$k = 0, 1, \dots, m$	must be zero, and $v_{j,k+1} = v_{j,k}$ for $k = 0, 1, \dots, m$ and $j = 1, \dots, m$. Now
lyche-numerical-linear-algebra- F04886	315	■ 1.000	$v_{\{j, 0\}} = v_{\{j, m+1\}} = 0$	$v_{j,0} = v_{j,m+1} = 0$	$v_{j,0} = v_{j,m+1} = 0$ implies that $\boldsymbol{V} = \mathbf{0}$. It follows that the linear system (13.56) is
lyche-numerical-linear-algebra- F04887	315	■ 1.000	$\boldsymbol{V} = \mathbf{0}$	$\boldsymbol{V} = \mathbf{0}$	$v_{j,0} = v_{j,m+1} = 0$ implies that $\boldsymbol{V} = \mathbf{0}$. It follows that the linear system (13.56) is
lyche-numerical-linear-algebra- F04888	316	■ 1.000	$\boldsymbol{b} \in \mathbb{R}^n$	$\boldsymbol{b} \in \mathbb{R}^n$	Consider solving $\boldsymbol{A} \boldsymbol{x} = \boldsymbol{b}$, where $\boldsymbol{A}$ is given by (13.57) and $\boldsymbol{b} \in \mathbb{R}^n$. Since $\boldsymbol{A}$ is
lyche-numerical-linear-algebra- F04889	316	■ 1.000	$\boldsymbol{x} \in \mathbb{R}^n$	$\boldsymbol{x} \in \mathbb{R}^n$	positive definite it is nonsingular and the system has a unique solution $\boldsymbol{x} \in \mathbb{R}^n$.
lyche-numerical-linear-algebra- F04890	316	■ 1.000	$m = \sqrt{n}$	$m = \sqrt{n}$	to find $\boldsymbol{x}$. Since the bandwidth of $\boldsymbol{A}$ is $m = \sqrt{n}$ both of these methods require $O(n^2)$
lyche-numerical-linear-algebra- F04891	316	■ 1.000	$c(x, y) \equiv 1$	$c(x, y) \equiv 1$	If we choose $c(x, y) \equiv 1$ in (13.52), we get the Poisson problem. With this in
lyche-numerical-linear-algebra- F04892	316	■ 1.000	$\boldsymbol{A}_p$	$\boldsymbol{A}_p$	mind, we may think of the coefficient matrix $\boldsymbol{A}_p$ arising from the discretization of
lyche-numerical-linear-algebra- F04893	316	■ 1.000	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$	$\boldsymbol{B} = \boldsymbol{A}_p^{-1}$, the inverse of the discrete Poisson matrix as a preconditioner for the
lyche-numerical-linear-algebra- F04894	316	■ 0.928	$\boldsymbol{w} = \boldsymbol{B} \boldsymbol{t}$	$\boldsymbol{w} = \boldsymbol{B} \boldsymbol{t}$	Consider Algorithm 13.3. With this preconditioner the calculation $\boldsymbol{w} = \boldsymbol{B} \boldsymbol{t}$ takes
lyche-numerical-linear-algebra- F04895	316	■ 1.000	$\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$	$\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$	the form $\boldsymbol{A}_p \boldsymbol{w}_k = \boldsymbol{t}_k$.
lyche-numerical-linear-algebra- F04896	316	■ 1.000	$\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$	$\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$	This method can be utilized to solve $\boldsymbol{A}_p \boldsymbol{w} = \boldsymbol{t}$.
lyche-numerical-linear-algebra- F04897	316	■ 1.000	$\boldsymbol{x}_0 = 0$	$\boldsymbol{x}_0 = 0$	lem (13.52). We used $\boldsymbol{x}_0 = 0$ and $\epsilon = 10^{-8}$. The results are shown in Table 13.3.
lyche-numerical-linear-algebra- F04898	316	■ 1.000	$\epsilon = 10^{-8}$	$\epsilon = 10^{-8}$	lem (13.52). We used $\boldsymbol{x}_0 = 0$ and $\epsilon = 10^{-8}$. The results are shown in Table 13.3.
lyche-numerical-linear-algebra- F04899	316	■ 1.000	$O(n \log_2 n)$	$O(n \log_2 n)$	without preconditioning to $O(n^{3/2})$ or $O(n \log_2 n)$ with preconditioning. This is not

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lyche-numerical-linear-algebra F04100	316	1.000	K_{pre}	K_{pre}	<table> <tr> <td>n</td><td>2500</td><td>10000</td><td>22500</td><td>40000</td><td>62500</td></tr> <tr> <td>K</td><td>222</td><td>472</td><td>728</td><td>986</td><td>1246</td></tr> <tr> <td>$K/\sqrt{n}$</td><td>4.44</td><td>4.72</td><td>4.85</td><td>4.93</td><td>4.98</td></tr> <tr> <td>K_{pre}</td><td>22</td><td>23</td><td>23</td><td>23</td><td>23</td></tr> </table>	n	2500	10000	22500	40000	62500	K	222	472	728	986	1246	$K/\sqrt{n}$	4.44	4.72	4.85	4.93	4.98	K_{pre}	22	23	23	23	23
n	2500	10000	22500	40000	62500																								
K	222	472	728	986	1246																								
$K/\sqrt{n}$	4.44	4.72	4.85	4.93	4.98																								
K_{pre}	22	23	23	23	23																								
lyche-numerical-linear-algebra F04101	317	1.000	$0 < c_0 \leq$	$0 < c_0 \leq$	Theorem 13.12 (Eigenvalues of Preconditioned Matrix) Suppose $0 < c_0 \leq$																								
lyche-numerical-linear-algebra F04102	317	0.973	$c(x, y) \leq c_1$	$c(x, y) \leq c_1$	$c(x, y) \leq c_1$ for all $(x, y) \in [0, 1]^2$. For the eigenvalues of the matrix $\mathbf{BA} = \mathbf{A}_p^{-1}\mathbf{A}$																								
lyche-numerical-linear-algebra F04103	317	0.973	$(x, y) \in [0, 1]^2$	$(x, y) \in [0, 1]^2$	$c(x, y) \leq c_1$ for all $(x, y) \in [0, 1]^2$. For the eigenvalues of the matrix $\mathbf{BA} = \mathbf{A}_p^{-1}\mathbf{A}$																								
lyche-numerical-linear-algebra F04104	317	0.973	$\mathbf{B} \mathbf{A} = \mathbf{A}_p^{-1} \mathbf{A}$	$\mathbf{BA} = \mathbf{A}_p^{-1} \mathbf{A}$	$c(x, y) \leq c_1$ for all $(x, y) \in [0, 1]^2$. For the eigenvalues of the matrix $\mathbf{BA} = \mathbf{A}_p^{-1}\mathbf{A}$																								
lyche-numerical-linear-algebra F04105	317	1.000	$\mathbf{A}_p^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$	$\mathbf{A}_p^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$	Proof Suppose $\mathbf{A}_p^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\mathbf{A} \mathbf{x} = \lambda \mathbf{A}_p \mathbf{x}$.																								
lyche-numerical-linear-algebra F04106	317	1.000	$\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$	$\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$	Proof Suppose $\mathbf{A}_p^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\mathbf{A} \mathbf{x} = \lambda \mathbf{A}_p \mathbf{x}$.																								
lyche-numerical-linear-algebra F04107	317	1.000	$\mathbf{A} \mathbf{x} = \lambda \mathbf{A}_p \mathbf{x}$	$\mathbf{A} \mathbf{x} = \lambda \mathbf{A}_p \mathbf{x}$	Proof Suppose $\mathbf{A}_p^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$. Then $\mathbf{A} \mathbf{x} = \lambda \mathbf{A}_p \mathbf{x}$.																								
lyche-numerical-linear-algebra F04108	317	1.000	$\mathbf{x}^T \mathbf{A} \mathbf{x}$	$\mathbf{x}^T \mathbf{A} \mathbf{x}$	We computed $\mathbf{x}^T \mathbf{A} \mathbf{x}$ in (13.59) and we obtain $\mathbf{x}^T \mathbf{A}_p \mathbf{x}$ by setting all the c 's there																								
lyche-numerical-linear-algebra F04109	317	1.000	$\mathbf{x}^T \mathbf{A}_p \mathbf{x}$	$\mathbf{x}^T \mathbf{A}_p \mathbf{x}$	We computed $\mathbf{x}^T \mathbf{A} \mathbf{x}$ in (13.59) and we obtain $\mathbf{x}^T \mathbf{A}_p \mathbf{x}$ by setting all the c 's there																								
lyche-numerical-linear-algebra F04110	317	0.997	$\mathbf{x}^T \mathbf{A}_p \mathbf{x} > 0$	$\mathbf{x}^T \mathbf{A}_p \mathbf{x} > 0$	Thus $\mathbf{x}^T \mathbf{A}_p \mathbf{x} > 0$ and bounding all the c 's in (13.59) from below by c_0 and above																								
lyche-numerical-linear-algebra F04111	317	0.997	c_0	c_0	Thus $\mathbf{x}^T \mathbf{A}_p \mathbf{x} > 0$ and bounding all the c 's in (13.59) from below by c_0 and above																								
lyche-numerical-linear-algebra F04112	317	1.000	$c_0 \leq \lambda \leq c_1$	$c_0 \leq \lambda \leq c_1$	which implies that $c_0 < \lambda < c_1$ for all eigenvalues λ of $\mathbf{BA} = \mathbf{A}_p^{-1}\mathbf{A}$.																								
lyche-numerical-linear-algebra F04113	317	0.998	$c(x, y) = e^{-x+y}$	$c(x, y) = e^{-x+y}$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$																								
lyche-numerical-linear-algebra F04114	317	0.998	$c_0 = e^{-2}$	$c_0 = e^{-2}$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$																								
lyche-numerical-linear-algebra F04115	317	0.998	$c_1 = 1$	$c_1 = 1$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa \leq e^2 \approx$																								
lyche-numerical-linear-algebra F04116	317	0.998	$\kappa \leq e^2 \approx$	$\kappa \leq e^2 \approx$	Using $c(x, y) = e^{-x+y}$ as above, we find $c_0 = e^{-2}$ and $c_1 = 1$. Thus $\kappa < e^2 \approx$																								
lyche-numerical-linear-algebra F04117	317	1.000	$\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^T$	$\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^T$	Exercise 13.2 (Paraboloid) Let $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^T$ be the spectral decomposition of $\mathbf{A}$,																								
lyche-numerical-linear-algebra F04118	318	0.966	$\mathbf{v} = [v_1, \dots, v_n]^T := \mathbf{U}^T \mathbf{y}$	$\mathbf{v} = [v_1, \dots, v_n]^T := \mathbf{U}^T \mathbf{y}$	$\mathbf{v} = [v_1, \dots, v_n]^T := \mathbf{U}^T \mathbf{y}$, and set $\mathbf{c} := \mathbf{U}^T \mathbf{b} = [c_1, \dots, c_n]^T$. Show that																								
lyche-numerical-linear-algebra F04119	318	0.966	$\mathbf{c} := \mathbf{U}^T \mathbf{b} = [c_1, \dots, c_n]^T$	$\mathbf{c} := \mathbf{U}^T \mathbf{b} = [c_1, \dots, c_n]^T$	$\mathbf{v} = [v_1, \dots, v_n]^T := \mathbf{U}^T \mathbf{y}$, and set $\mathbf{c} := \mathbf{U}^T \mathbf{b} = [c_1, \dots, c_n]^T$. Show that																								
lyche-numerical-linear-algebra F04120	318	0.999	$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{r}_k$	$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{r}_k$	the iteration is $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{r}_k$, where $\mathbf{r}_k$ is the residual, $\mathbf{r}_k = \mathbf{b} - \mathbf{A} \mathbf{x}_k$, and																								
lyche-numerical-linear-algebra F04121	318	0.999	$\mathbf{r}_k = \mathbf{b} - \mathbf{A} \mathbf{x}_k$	$\mathbf{r}_k = \mathbf{b} - \mathbf{A} \mathbf{x}_k$	the iteration is $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{r}_k$, where $\mathbf{r}_k$ is the residual, $\mathbf{r}_k = \mathbf{b} - \mathbf{A} \mathbf{x}_k$, and																								
lyche-numerical-linear-algebra F04122	318	0.973	$\alpha_k = \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{r}_k^T \mathbf{A} \mathbf{r}_k}$	$\alpha_k = \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{r}_k^T \mathbf{A} \mathbf{r}_k}$	$\alpha_k = \frac{\mathbf{r}_k^T \mathbf{r}_k}{\mathbf{r}_k^T \mathbf{A} \mathbf{r}_k}$.																								

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra F04123	318	0.927	$\boldsymbol{A}=\left[\begin{array}{cc}2 & -1 \\ -1 & 2\end{array}\right], \boldsymbol{b}=\left[\begin{array}{c}1 \\ 1\end{array}\right]^T$	$\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$	a) Compute $\boldsymbol{x}_1$ if $\boldsymbol{A} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \boldsymbol{b} = [1 \ 1]^T$ and $\boldsymbol{x}_0 = \mathbf{0}$.
lyche-numerical-linear-algebra F04124	318	1.000	$\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}$	$\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}$	b) If the k -th error, $\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}$, is an eigenvector of $\boldsymbol{A}$, what can you say about
lyche-numerical-linear-algebra F04125	318	0.907	$\boldsymbol{x}_1 = \left(\frac{\boldsymbol{b}^T \boldsymbol{b}}{\boldsymbol{b}^T \boldsymbol{A} \boldsymbol{b}}\right) \boldsymbol{b}$	$\boldsymbol{x}_1 = \left(\frac{\boldsymbol{b}^T \boldsymbol{b}}{\boldsymbol{b}^T \boldsymbol{A} \boldsymbol{b}}\right) \boldsymbol{b}$	conjugate gradient method when $\boldsymbol{x}_0 = \mathbf{0}$. (Answer: $\boldsymbol{x}_1 = \left(\frac{\boldsymbol{b}^T \boldsymbol{b}}{\boldsymbol{b}^T \boldsymbol{A} \boldsymbol{b}}\right) \boldsymbol{b}$.)
lyche-numerical-linear-algebra F04126	318	0.990	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$	system $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$ starting with $\boldsymbol{y}_0 = \mathbf{0}$. ³
lyche-numerical-linear-algebra F04127	318	0.990	$\boldsymbol{y}_0 = \mathbf{0}$.	$\boldsymbol{y}_0 = \mathbf{0}$. ³	system $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0 := \boldsymbol{b} - \boldsymbol{A} \boldsymbol{x}_0$ starting with $\boldsymbol{y}_0 = \mathbf{0}$. ³
lyche-numerical-linear-algebra F04128	318	1.000	$\boldsymbol{p}_k = \boldsymbol{r}_k + \sum_{j=0}^{k-1} a_{k,j} \boldsymbol{r}_j$	$\boldsymbol{p}_k = \boldsymbol{r}_k + \sum_{j=0}^{k-1} a_{k,j} \boldsymbol{r}_j$	² Hint: use induction on k to show that $\boldsymbol{p}_k = \boldsymbol{r}_k + \sum_{j=0}^{k-1} a_{k,j} \boldsymbol{r}_j$ for some constants $a_{k,j}$.
lyche-numerical-linear-algebra F04129	318	0.881	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$	$\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$	³ Hint: The conjugate gradient method for $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$ can be written $\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k, \gamma_k :=$
lyche-numerical-linear-algebra F04130	318	0.881	$\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k, \gamma_k :=$	$\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k, \gamma_k :=$	³ Hint: The conjugate gradient method for $\boldsymbol{A} \boldsymbol{y} = \boldsymbol{r}_0$ can be written $\boldsymbol{y}_{k+1} := \boldsymbol{y}_k + \gamma_k \boldsymbol{q}_k, \gamma_k :=$
lyche-numerical-linear-algebra F04131	318	0.888	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}, \boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k, \boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k, \delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}, \boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k, \boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k, \delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}, \boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k, \boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k, \delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$. Show that $\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$,
lyche-numerical-linear-algebra F04132	318	0.888	$\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$	$\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$	$\frac{\boldsymbol{s}_k^T \boldsymbol{s}_k}{\boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k}, \boldsymbol{s}_{k+1} := \boldsymbol{s}_k - \gamma_k \boldsymbol{A} \boldsymbol{q}_k, \boldsymbol{q}_{k+1} := \boldsymbol{s}_{k+1} + \delta_k \boldsymbol{q}_k, \delta_k := \frac{\boldsymbol{s}_{k+1}^T \boldsymbol{s}_{k+1}}{\boldsymbol{s}_k^T \boldsymbol{s}_k}$. Show that $\boldsymbol{y}_k = \boldsymbol{x}_k - \boldsymbol{x}_0$,
lyche-numerical-linear-algebra F04133	318	0.977	$\boldsymbol{s}_k = \boldsymbol{r}_k$	$\boldsymbol{s}_k = \boldsymbol{r}_k$	$\boldsymbol{s}_k = \boldsymbol{r}_k$, and $\boldsymbol{q}_k = \boldsymbol{p}_k$, for $k = 0, 1, 2 \dots$
lyche-numerical-linear-algebra F04134	318	0.977	$\boldsymbol{q}_k = \boldsymbol{p}_k$	$\boldsymbol{q}_k = \boldsymbol{p}_k$	$\boldsymbol{s}_k = \boldsymbol{r}_k$, and $\boldsymbol{q}_k = \boldsymbol{p}_k$, for $k = 0, 1, 2 \dots$
lyche-numerical-linear-algebra F04135	319	1.000	$n = 100, 400, 1600, 2500$	$n = 100, 400, 1600, 2500$	to test for say $n = 100, 400, 1600, 2500$, and tabulate K/n instead of $K/\sqrt{n}$ in the
lyche-numerical-linear-algebra F04136	319	1.000	K / n	K/n	to test for say $n = 100, 400, 1600, 2500$, and tabulate K/n instead of $K/\sqrt{n}$ in the
lyche-numerical-linear-algebra F04137	319	1.000	$\boldsymbol{A} \in \mathbb{R}^{m,n}, \boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m,n}, \boldsymbol{b} \in \mathbb{R}^m$	$\boldsymbol{A} \in \mathbb{R}^{m,n}, \boldsymbol{b} \in \mathbb{R}^m$ and $\boldsymbol{A}^T \boldsymbol{A}$ is positive definite. ⁴ Explain why only the following
lyche-numerical-linear-algebra F04138	319	0.996	$\boldsymbol{r} = \boldsymbol{A}'(\boldsymbol{b} - \boldsymbol{A}^* \boldsymbol{x}) ; \boldsymbol{p} = \boldsymbol{r}$	$\boldsymbol{r} = \boldsymbol{A}'(\boldsymbol{b} - \boldsymbol{A}^* \boldsymbol{x}) ; \boldsymbol{p} = \boldsymbol{r}$	1. $\boldsymbol{r} = \boldsymbol{A}'(\boldsymbol{b} - \boldsymbol{A}^* \boldsymbol{x}) ; \boldsymbol{p} = \boldsymbol{r}$;
lyche-numerical-linear-algebra F04139	319	0.926	$\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{a}^* \boldsymbol{A}^* \boldsymbol{t}$	$\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{a}^* \boldsymbol{A}^* \boldsymbol{t}$	3. $\boldsymbol{r} = \boldsymbol{r} - \boldsymbol{a}^* \boldsymbol{A}^* \boldsymbol{t}$;
lyche-numerical-linear-algebra F04140	319	1.000	$\text{cond}_2(\boldsymbol{A})^2$	$\text{cond}_2(\boldsymbol{A})^2$	Note that the condition number of the normal equations is $\text{cond}_2(\boldsymbol{A})^2$, the square of
lyche-numerical-linear-algebra F04141	319	1.000	$\{\boldsymbol{v}_i\}_{i=1}^k$	$\{\boldsymbol{v}_i\}_{i=1}^k$	a) Let $\{\boldsymbol{v}_i\}_{i=1}^k$ be a set of linearly independent vectors in $\mathbb{R}^n$, and let $\langle \cdot, \cdot \rangle$ be an
lyche-numerical-linear-algebra F04142	319	1.000	$n_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$	$n_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$	inner product in $\mathbb{R}^n$. Explain that the $k \times k$ -matrix $\boldsymbol{N}$ with entries $n_{ij} = \langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle$
lyche-numerical-linear-algebra F04143	319	0.997	$\mathbb{W} \subset \mathbb{R}^n$	$\mathbb{W} \subset \mathbb{R}^n$	b) Let $\mathbb{W} \subset \mathbb{R}^n$ be any linear subspace. Show that there is one and only one vector

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lyche-numerical-linear-algebra-F04144	319	■ 1.000	$\hat{\mathbf{x}} \in \mathbb{W}$	$\hat{\mathbf{x}} \in \mathbb{W}$	$\hat{\mathbf{x}} \in \mathbb{W}$ so that
lyche-numerical-linear-algebra-F04145	319	■ 1.000	$\hat{\mathbf{x}}$	$\hat{\mathbf{x}}$	and that $\hat{\mathbf{x}}$ satisfies
lyche-numerical-linear-algebra-F04146	319	■ 1.000	$\mathbb{W}$	$\mathbb{W}$	space $\mathbb{W}$ is taken to be the Krylov space
lyche-numerical-linear-algebra-F04147	320	■ 1.000	$\mathbf{x}_k \in \mathbb{W}_k$	$\mathbf{x}_k \in \mathbb{W}_k$	$\mathbf{x}_k$. Assume that $\mathbf{x}_k \in \mathbb{W}_k$ is already determined. In addition, assume that we
lyche-numerical-linear-algebra-F04148	320	■ 1.000	$\mathbf{p}_k \in \mathbb{W}_{k+1}$	$\mathbf{p}_k \in \mathbb{W}_{k+1}$	already have computed a “search direction” $\mathbf{p}_k \in \mathbb{W}_{k+1}$ such that $\ \mathbf{A}\mathbf{p}_k\ _2 =$
lyche-numerical-linear-algebra-F04149	320	■ 1.000	$\ \mathbf{A}\mathbf{p}_k\ _2 =$	$\ \mathbf{A}\mathbf{p}_k\ _2 =$	already have computed a “search direction” $\mathbf{p}_k \in \mathbb{W}_{k+1}$ such that $\ \mathbf{A}\mathbf{p}_k\ _2 =$
lyche-numerical-linear-algebra-F04150	320	■ 1.000	$\ \mathbf{p}_k\ _{\mathbf{A}} = 1$	$\ \mathbf{p}_k\ _{\mathbf{A}} = 1$	$\ \mathbf{p}_k\ _{\mathbf{A}} = 1$, and such that
lyche-numerical-linear-algebra-F04151	320	■ 1.000	$\alpha_k \in \mathbb{R}$	$\alpha_k \in \mathbb{R}$	Show that $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$ for a suitable $\alpha_k \in \mathbb{R}$, and express α_k in terms of
lyche-numerical-linear-algebra-F04152	320	■ 1.000	$\mathbf{p}_{k-2}, \mathbf{p}_{k-1}$	$\mathbf{p}_{k-2}, \mathbf{p}_{k-1}$	further that the vectors $\mathbf{p}_{k-2}$, $\mathbf{p}_{k-1}$, and $\mathbf{p}_k$ are already known with properties as
lyche-numerical-linear-algebra-F04153	320	■ 1.000	$k \leq 3$	$k \leq 3$	a) Determine the vectors defining the Krylov spaces for $k \leq 3$ taking as initial
lyche-numerical-linear-algebra-F04154	320	■ 1.000	$\left[\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}\right] = \begin{bmatrix} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{bmatrix}$		approximation $\mathbf{x} = \mathbf{0}$. Answer: $[\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}] = \begin{bmatrix} 4 & 8 & 20 \\ 0 & -4 & -16 \\ 0 & 0 & 4 \end{bmatrix}$.
lyche-numerical-linear-algebra-F04155	321	■ 0.983	$\dim(\mathbb{W}_k) = k$	$\dim(\mathbb{W}_k) = k$	• $\dim(\mathbb{W}_k) = k$ for $k = 0, 1, 2, 3$.
lyche-numerical-linear-algebra-F04156	321	■ 0.983	$k = 0, 1, 2, 3$	$k = 0, 1, 2, 3$	• $\dim(\mathbb{W}_k) = k$ for $k = 0, 1, 2, 3$.
lyche-numerical-linear-algebra-F04157	321	■ 0.987	$\mathbf{x}_3$	$\mathbf{x}_3$	• $\mathbf{x}_3$ is the exact solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.
lyche-numerical-linear-algebra-F04158	321	■ 1.000	$\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$	$\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$	• $\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$ is an orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra-F04159	321	■ 1.000	$k = 1, 2, 3$	$k = 1, 2, 3$	• $\mathbf{r}_0, \dots, \mathbf{r}_{k-1}$ is an orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra-F04160	321	■ 1.000	$\mathbf{p}_0, \dots, \mathbf{p}_{k-1}$	$\mathbf{p}_0, \dots, \mathbf{p}_{k-1}$	• $\mathbf{p}_0, \dots, \mathbf{p}_{k-1}$ is an $\mathbf{A}$ -orthogonal basis for $\mathbb{W}_k$ for $k = 1, 2, 3$.
lyche-numerical-linear-algebra-F04161	321	■ 1.000	$\{\ \mathbf{r}_k\ \}$	$\{\ \mathbf{r}_k\ \}$	• $\{\ \mathbf{r}_k\ \}$ is monotonically decreasing.
lyche-numerical-linear-algebra-F04162	321	■ 0.991	$\{\ \mathbf{x}_k - \mathbf{x}\ \}$	$\{\ \mathbf{x}_k - \mathbf{x}\ \}$	• $\{\ \mathbf{x}_k - \mathbf{x}\ \}$ is monotonically decreasing.
lyche-numerical-linear-algebra-F04163	321	■ 1.000	$\mathbf{B}^T = -\mathbf{B}$	$\mathbf{B}^T = -\mathbf{B}$	Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be an antisymmetric matrix, i.e., $\mathbf{B}^T = -\mathbf{B}$, and let $\mathbf{A} := \mathbf{I} - \mathbf{B}$,
lyche-numerical-linear-algebra-F04164	321	■ 1.000	$\mathbf{A} := \mathbf{I} - \mathbf{B}$	$\mathbf{A} := \mathbf{I} - \mathbf{B}$	Let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be an antisymmetric matrix, i.e., $\mathbf{B}^T = -\mathbf{B}$, and let $\mathbf{A} := \mathbf{I} - \mathbf{B}$,
lyche-numerical-linear-algebra-F04165	321	■ 0.998	$\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$	$\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$	b) Show that $\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$ and that $\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$.
lyche-numerical-linear-algebra-F04166	321	■ 0.998	$\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$	$\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$	b) Show that $\ \mathbf{A}\mathbf{x}\ _2^2 = \ \mathbf{x}\ _2^2 + \ \mathbf{B}\mathbf{x}\ _2^2$ and that $\ \mathbf{A}\ _2 = \sqrt{1 + \ \mathbf{B}\ _2^2}$.
lyche-numerical-linear-algebra-F04167	321	■ 1.000	$\ \mathbf{A}\ _2 \leq 1$	$\ \mathbf{A}\ _2 \leq 1$	and $\ \mathbf{A}\ _2 \leq 1$.
lyche-numerical-linear-algebra-F04168	321	■ 1.000	$1 \leq k \leq n, \mathcal{W} = \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$	$1 \leq k \leq n, \mathcal{W} = \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$	d) Let $\mathcal{W} = \text{span}(\mathbf{w}_1, \dots, \mathbf{w}_k)$ a k -dimensional subspace of $\mathbb{R}^n$ and

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lyche-numerical-linear-algebra F04169	321	■ 1.000	$\boldsymbol{x} \in \mathcal{W}$	$\boldsymbol{x} \in \mathcal{W}$	$\boldsymbol{b} \in \mathbb{R}^n$. Show that if $\boldsymbol{x} \in \mathcal{W}$ is such that
lyche-numerical-linear-algebra F04170	321	■ 0.835	$\ \boldsymbol{x}\ _2 \leq \ \boldsymbol{b}\ _2$	$\ \boldsymbol{x}\ _2 \leq \ \boldsymbol{b}\ _2$	then $\ \boldsymbol{x}\ _2 \leq \ \boldsymbol{b}\ _2$.
lyche-numerical-linear-algebra F04171	322	■ 1.000	$\boldsymbol{x} := \sum_{j=1}^k x_j \boldsymbol{w}_j$	$\boldsymbol{x} := \sum_{j=1}^k x_j \boldsymbol{w}_j$	With $\boldsymbol{x} := \sum_{j=1}^k x_j \boldsymbol{w}_j$ the problem (13.65) is equivalent to finding real
lyche-numerical-linear-algebra F04172	322	■ 1.000	$x_1, \dots, x_k$	$x_1, \dots, x_k$	numbers $x_1, \dots, x_k$ solving the linear system
lyche-numerical-linear-algebra F04173	322	■ 0.996	$\boldsymbol{x}^* := \boldsymbol{A}^{-1} \boldsymbol{b}$	$\boldsymbol{x}^* := \boldsymbol{A}^{-1} \boldsymbol{b}$	e) Let $\boldsymbol{x}^* := \boldsymbol{A}^{-1} \boldsymbol{b}$. Show that
lyche-numerical-linear-algebra F04174	322	■ 1.000	$\boldsymbol{x}_k \in \mathcal{W}_k$	$\boldsymbol{x}_k \in \mathcal{W}_k$	For $k = 1, 2, \dots, n$ we define $\boldsymbol{x}_k \in \mathcal{W}_k$ by
lyche-numerical-linear-algebra F04175	322	■ 0.986	$\boldsymbol{x}^*$	$\boldsymbol{x}^*$	“good” approximation to $\boldsymbol{x}^*$ follows from (13.67). In this exercise we will derive a
lyche-numerical-linear-algebra F04176	322	■ 1.000	$k=0, \dots, n$	$k = 0, \dots, n$	For $k = 0, \dots, n$ we set
lyche-numerical-linear-algebra F04177	322	■ 1.000	$\omega_0, \omega_1, \dots, \omega_m$	$\omega_0, \omega_1, \dots, \omega_m$	Let $\omega_0, \omega_1, \dots, \omega_m$ be real numbers defined recursively for $k = 1, 2, \dots, m$ by
lyche-numerical-linear-algebra F04178	322	■ 1.000	$k=1, 2, \dots, m$	$k = 1, 2, \dots, m$	Let $\omega_0, \omega_1, \dots, \omega_m$ be real numbers defined recursively for $k = 1, 2, \dots, m$ by
lyche-numerical-linear-algebra F04179	323	■ 1.000	$k=0, 1, \dots, m-1$	$k = 0, 1, \dots, m-1$	$k = 0, 1, \dots, m-1$
lyche-numerical-linear-algebra F04180	323	■ 1.000	$\boldsymbol{x}_0 = \boldsymbol{x}_{-1} = 0$	$\boldsymbol{x}_0 = \boldsymbol{x}_{-1} = 0$	starting with $\boldsymbol{x}_0 = \boldsymbol{x}_{-1} = 0$ and $\boldsymbol{r}_0 = \boldsymbol{r}_{-1} = \boldsymbol{b}$.
lyche-numerical-linear-algebra F04181	323	■ 1.000	$\boldsymbol{r}_0 = \boldsymbol{r}_{-1} = \boldsymbol{b}$	$\boldsymbol{r}_0 = \boldsymbol{r}_{-1} = \boldsymbol{b}$	starting with $\boldsymbol{x}_0 = \boldsymbol{x}_{-1} = 0$ and $\boldsymbol{r}_0 = \boldsymbol{r}_{-1} = \boldsymbol{b}$.
lyche-numerical-linear-algebra F04182	323	■ 1.000	$0 < \omega_k < 1$	$0 < \omega_k < 1$	a) Show that $0 < \omega_k < 1$ for $k = 1, 2, \dots, m$.
lyche-numerical-linear-algebra F04183	323	■ 1.000	$\langle \boldsymbol{r}_k, \boldsymbol{r}_j \rangle = 0$	$\langle \boldsymbol{r}_k, \boldsymbol{r}_j \rangle = 0$	c) Show that $\langle \boldsymbol{r}_k, \boldsymbol{r}_j \rangle = 0$ for $j = 0, 1, \dots, k-1$.
lyche-numerical-linear-algebra F04184	323	■ 1.000	$j=0, 1, \dots, k-1$	$j = 0, 1, \dots, k-1$	c) Show that $\langle \boldsymbol{r}_k, \boldsymbol{r}_j \rangle = 0$ for $j = 0, 1, \dots, k-1$.
lyche-numerical-linear-algebra F04185	323	■ 1.000	$k \leq m+1$	$k \leq m+1$	d) Show that if $k \leq m+1$ then $\mathcal{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1})$ and $\dim \mathcal{W}_k = k$.
lyche-numerical-linear-algebra F04186	323	■ 1.000	$\mathcal{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1})$	$\mathcal{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1})$	d) Show that if $k \leq m+1$ then $\mathcal{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1})$ and $\dim \mathcal{W}_k = k$.
lyche-numerical-linear-algebra F04187	323	■ 1.000	$\dim \mathcal{W}_k = k$	$\dim \mathcal{W}_k = k$	d) Show that if $k \leq m+1$ then $\mathcal{W}_k = \text{span}(\boldsymbol{r}_0, \boldsymbol{r}_1, \dots, \boldsymbol{r}_{k-1})$ and $\dim \mathcal{W}_k = k$.
lyche-numerical-linear-algebra F04188	323	■ 1.000	$1 \leq k \leq m-1$	$1 \leq k \leq m-1$	e) Show that if $1 \leq k \leq m-1$ then
lyche-numerical-linear-algebra F04189	323	■ 0.605	$\alpha_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k+1} \rangle / \rho_{k+1}$	$\alpha_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k+1} \rangle / \rho_{k+1}$	where $\alpha_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k+1} \rangle / \rho_{k+1}$ and $\beta_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k-1} \rangle / \rho_{k-1}$.
lyche-numerical-linear-algebra F04190	323	■ 0.605	$\beta_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k-1} \rangle / \rho_{k-1}$	$\beta_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k-1} \rangle / \rho_{k-1}$	where $\alpha_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k+1} \rangle / \rho_{k+1}$ and $\beta_k := \langle \boldsymbol{B} \boldsymbol{r}_k, \boldsymbol{r}_{k-1} \rangle / \rho_{k-1}$.
lyche-numerical-linear-algebra F04191	323	■ 0.865	$\alpha_0 := \langle \boldsymbol{B} \boldsymbol{r}_0, \boldsymbol{r}_1 \rangle / \rho_1$	$\alpha_0 := \langle \boldsymbol{B} \boldsymbol{r}_0, \boldsymbol{r}_1 \rangle / \rho_1$	f) Define $\alpha_0 := \langle \boldsymbol{B} \boldsymbol{r}_0, \boldsymbol{r}_1 \rangle / \rho_1$ and show that $\alpha_0 = 1$.
lyche-numerical-linear-algebra F04192	323	■ 0.865	$\alpha_0 = 1$	$\alpha_0 = 1$	f) Define $\alpha_0 := \langle \boldsymbol{B} \boldsymbol{r}_0, \boldsymbol{r}_1 \rangle / \rho_1$ and show that $\alpha_0 = 1$.
lyche-numerical-linear-algebra F04193	323	■ 1.000	$\beta_k = -\alpha_{k-1} \rho_k / \rho_{k-1}$	$\beta_k = -\alpha_{k-1} \rho_k / \rho_{k-1}$	g) Show that if $1 \leq k \leq m-1$ then $\beta_k = -\alpha_{k-1} \rho_k / \rho_{k-1}$.
lyche-numerical-linear-algebra F04194	323	■ 1.000	$\alpha_k + \beta_k = 1$	$\alpha_k + \beta_k = 1$	i) Use (13.71) and (13.72) to show that $\alpha_k + \beta_k = 1$ for $k = 1, 2, \dots, m-1$.
lyche-numerical-linear-algebra F04195	323	■ 1.000	$k=1, 2, \dots, m-1$	$k = 1, 2, \dots, m-1$	i) Use (13.71) and (13.72) to show that $\alpha_k + \beta_k = 1$ for $k = 1, 2, \dots, m-1$.

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lyche-numerical-linear-algebra-F04196	323	1.000	$\alpha_k \geq 1$	$\alpha_k \geq 1$	j) Show that $\alpha_k \geq 1$ for $k = 1, 2, \dots, m - 1$.
lyche-numerical-linear-algebra-F04197	323	1.000	$\mathbf{x}_k, \mathbf{r}_k$	$\mathbf{x}_k, \mathbf{r}_k$	k) Show that $\mathbf{x}_k, \mathbf{r}_k$ and ω_k satisfy the recurrence relations (13.68), (13.69)
lyche-numerical-linear-algebra-F04198	323	1.000	ω_k	ω_k	k) Show that $\mathbf{x}_k, \mathbf{r}_k$ and ω_k satisfy the recurrence relations (13.68), (13.69)
lyche-numerical-linear-algebra-F04199	323	0.887	$\operatorname{arccosh}$	$\operatorname{arccosh}$	where $\operatorname{arccosh}$ is the inverse function of $\cosh x := (e^x + e^{-x})/2$.
lyche-numerical-linear-algebra-F04200	323	0.887	$\cosh x := \left(e^x + e^{-x} \right) / 2$	$\cosh x := (e^x + e^{-x})/2$	where $\operatorname{arccosh}$ is the inverse function of $\cosh x := (e^x + e^{-x})/2$
lyche-numerical-linear-algebra-F04201	323	1.000	$\mathbf{A}^{-1}(\mathbf{r}_{k+1} - \mathbf{r}_j) \in \mathcal{W}_{k+1}$	$\mathbf{A}^{-1}(\mathbf{r}_{k+1} - \mathbf{r}_j) \in \mathcal{W}_{k+1}$	⁵ Hint: Show that $\mathbf{A}^{-1}(\mathbf{r}_{k+1} - \mathbf{r}_j) \in \mathcal{W}_{k+1}$.
lyche-numerical-linear-algebra-F04202	324	1.000	$f: [a, b] \rightarrow \mathbb{R}$	$f: [a, b] \rightarrow \mathbb{R}$	Exercise 13.16 (Maximum of a Convex Function) Show that if $f: [a, b] \rightarrow \mathbb{R}$
lyche-numerical-linear-algebra-F04203	324	1.000	$\max_{a \leq x \leq b} f(x) \leq \max\{f(a), f(b)\}$	$\max_{a \leq x \leq b} f(x) \leq \max\{f(a), f(b)\}$	is convex then $\max_{a \leq x \leq b} f(x) \leq \max\{f(a), f(b)\}$.
lyche-numerical-linear-algebra-F04204	326	1.000	$\mathbf{A} \mathbf{e}_i = a_{ii} \mathbf{e}_i$	$\mathbf{A} \mathbf{e}_i = a_{ii} \mathbf{e}_i$	Diagonal Matrices The eigenpairs are easily determined. Since $\mathbf{A} \mathbf{e}_i = a_{ii} \mathbf{e}_i$
lyche-numerical-linear-algebra-F04205	326	0.965	$(\lambda_i, \mathbf{e}_i)$	$(\lambda_i, \mathbf{e}_i)$	the eigenpairs are $(\lambda_i, \mathbf{e}_i)$, where $\lambda_i = a_{ii}$ for $i = 1, \dots, n$. Moreover, the
lyche-numerical-linear-algebra-F04206	326	0.965	$\lambda_i = a_{ii}$	$\lambda_i = a_{ii}$	the eigenpairs are $(\lambda_i, \mathbf{e}_i)$, where $\lambda_i = a_{ii}$ for $i = 1, \dots, n$. Moreover, the
lyche-numerical-linear-algebra-F04207	326	1.000	$\det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{i=1}^n (a_{ii} - \lambda)$	$\det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{i=1}^n (a_{ii} - \lambda)$	the eigenvalues Since $\det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{i=1}^n (a_{ii} - \lambda)$ the eigenvalues are $\lambda_i = a_{ii}$
lyche-numerical-linear-algebra-F04208	326	1.000	$\mathbf{A}_i \mathbf{X}_i = \mathbf{X}_i \mathbf{D}_i$	$\mathbf{A}_i \mathbf{X}_i = \mathbf{X}_i \mathbf{D}_i$	Here the eigenpair problem reduces to r smaller problems. Let $\mathbf{A}_i \mathbf{X}_i = \mathbf{X}_i \mathbf{D}_i$
lyche-numerical-linear-algebra-F04209	326	1.000	$\mathbf{A}_i$	$\mathbf{A}_i$	define the eigenpairs of $\mathbf{A}_i$ for $i = 1, \dots, r$ and let $\mathbf{X} := \operatorname{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$,
lyche-numerical-linear-algebra-F04210	326	1.000	$\mathbf{X} := \operatorname{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$	$\mathbf{X} := \operatorname{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$	define the eigenpairs of $\mathbf{A}_i$ for $i = 1, \dots, r$ and let $\mathbf{X} := \operatorname{diag}(\mathbf{X}_1, \dots, \mathbf{X}_r)$,
lyche-numerical-linear-algebra-F04211	326	1.000	$\mathbf{D} := \operatorname{diag}(\mathbf{D}_1, \dots, \mathbf{D}_r)$	$\mathbf{D} := \operatorname{diag}(\mathbf{D}_1, \dots, \mathbf{D}_r)$	$\mathbf{D} := \operatorname{diag}(\mathbf{D}_1, \dots, \mathbf{D}_r)$. Then the eigenpairs for $\mathbf{A}$ are given by
lyche-numerical-linear-algebra-F04212	327	0.988	$\mathbf{A}_{11}, \mathbf{A}_{22}, \dots, \mathbf{A}_{rr}$	$\mathbf{A}_{11}, \mathbf{A}_{22}, \dots, \mathbf{A}_{rr}$	$\mathbf{A}_{11}, \mathbf{A}_{22}, \dots, \mathbf{A}_{rr}$ be the diagonal blocks of $\mathbf{A}$. By Property 8. of determinants
lyche-numerical-linear-algebra-F04213	327	1.000	$\pi_{\mathbf{A}}(\lambda)$	$\pi_{\mathbf{A}}(\lambda)$	a small change in one of the coefficients of $\pi_{\mathbf{A}}(\lambda)$ can lead to a large change in the
lyche-numerical-linear-algebra-F04214	327	0.405	$\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$	$\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$	roots of the polynomial. For example, if $\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$ and $q(\lambda) = \lambda^{16} - 10^{-16}$ then
lyche-numerical-linear-algebra-F04215	327	0.405	$q(\lambda) = \lambda^{16} - 10^{-16}$	$q(\lambda) = \lambda^{16} - 10^{-16}$	roots of the polynomial. For example, if $\pi_{\mathbf{A}}(\lambda) = \lambda^{16}$ and $q(\lambda) = \lambda^{16} - 10^{-16}$ then
lyche-numerical-linear-algebra-F04216	327	0.999	$\lambda_j = 10^{-1} e^{2\pi i j / 16}$	$\lambda_j = 10^{-1} e^{2\pi i j / 16}$	the roots of $\pi_{\mathbf{A}}$ are all equal to zero, while the roots of q are $\lambda_j = 10^{-1} e^{2\pi i j / 16}$,
lyche-numerical-linear-algebra-F04217	327	1.000	$j=1, \dots, 16$	$j = 1, \dots, 16$	$j = 1, \dots, 16$. The roots of q have absolute value 0.1 and a perturbation in one
lyche-numerical-linear-algebra-F04218	327	1.000	10^{-16}	10^{-16}	of the polynomial coefficients of magnitude 10^{-16} has led to an error in the roots
lyche-numerical-linear-algebra-F04219	328	1.000	$R \cap C$	$R \cap C$	Then any eigenvalue of $\mathbf{A}$ lies in $R \cap C$ where $R = R_1 \cup R_2 \cup \dots \cup R_n$ and
lyche-numerical-linear-algebra-F04220	328	1.000	$R = R_1 \cup R_2 \cup \dots \cup R_n$	$R = R_1 \cup R_2 \cup \dots \cup R_n$	Then any eigenvalue of $\mathbf{A}$ lies in $R \cap C$ where $R = R_1 \cup R_2 \cup \dots \cup R_n$ and
lyche-numerical-linear-algebra-F04221	328	1.000	$C = C_1 \cup C_2 \cup \dots \cup C_n$	$C = C_1 \cup C_2 \cup \dots \cup C_n$	$C = C_1 \cup C_2 \cup \dots \cup C_n$.
lyche-numerical-linear-algebra-F04222	328	0.621	$\lambda \in R_i$	$\lambda \in R_i$	Proof Suppose $(\lambda, \mathbf{x})$ is an eigenpair for $\mathbf{A}$. We claim that $\lambda \in R_i$, where i is such

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lyche-numerical-linear-algebra- F04223	328	■ 0.997	$\left x_i\right =\left \boldsymbol{x}\right _{\infty}$	$ x_i = \ \boldsymbol{x}\ _\infty$	that $ x_i = \ \boldsymbol{x}\ _\infty$. Indeed, $\boldsymbol{Ax} = \lambda\boldsymbol{x}$ implies that $\sum_j a_{ij}x_j = \lambda x_i$ or $(\lambda - a_{ii})x_i =$
lyche-numerical-linear-algebra- F04224	328	■ 0.997	$\sum_j a_{ij} x_j=\lambda x_i$	$\sum_j a_{ij} x_j = \lambda x_i$	that $ x_i = \ \boldsymbol{x}\ _\infty$. Indeed, $\boldsymbol{Ax} = \lambda\boldsymbol{x}$ implies that $\sum_j a_{ij}x_j = \lambda x_i$ or $(\lambda - a_{ii})x_i =$
lyche-numerical-linear-algebra- F04225	328	■ 0.997	$\left(\lambda-a_{i i}\right) x_i=$	$(\lambda - a_{ii}) x_i =$	that $ x_i = \ \boldsymbol{x}\ _\infty$. Indeed, $\boldsymbol{Ax} = \lambda\boldsymbol{x}$ implies that $\sum_j a_{ij}x_j = \lambda x_i$ or $(\lambda - a_{ii})x_i =$
lyche-numerical-linear-algebra- F04226	328	■ 1.000	$\sum_{j \neq i} a_{ij} x_j$	$\sum_{j \neq i} a_{ij} x_j$	$\sum_{j \neq i} a_{ij} x_j$. Dividing by x_i and taking absolute values we find
lyche-numerical-linear-algebra- F04227	328	■ 1.000	$\left x_j / x_i\right \leq 1$	$ x_j/x_i \leq 1$	since $ x_j/x_i \leq$ for all j . Thus $\lambda \in R_i$.
lyche-numerical-linear-algebra- F04228	328	■ 1.000	$C_{\{j\}}$	C_j	But these are the column disks C_j of $\boldsymbol{A}$. Hence $\lambda \in C_j$ for some j .
lyche-numerical-linear-algebra- F04229	328	■ 1.000	$\lambda \in C_{\{j\}}$	$\lambda \in C_j$	But these are the column disks C_j of $\boldsymbol{A}$. Hence $\lambda \in C_j$ for some j .
lyche-numerical-linear-algebra- F04230	328	■ 1.000	$r_{\{i\}}$	r_i	with center at a_{ii} and radius r_i , c.f. Fig. 14.1. R_i is called a (Gerschgorin) row disk.
lyche-numerical-linear-algebra- F04231	328	■ 1.000	$R_{\{1\}}, \ldots, R_{\{n\}}$	$R_1, \dots, R_n$	An eigenvalue λ lies in the union of the row disks $R_1, \dots, R_n$ and also in the
lyche-numerical-linear-algebra- F04232	328	■ 1.000	$C_{\{1\}}, \ldots, C_{\{n\}}$	$C_1, \dots, C_n$	union of the column disks $C_1, \dots, C_n$. If $\boldsymbol{A}$ is Hermitian then $R_i = C_i$ for $i =$
lyche-numerical-linear-algebra- F04233	328	■ 1.000	$R_{\{i\}}=C_{\{i\}}$	$R_i = C_i$	union of the column disks $C_1, \dots, C_n$. If $\boldsymbol{A}$ is Hermitian then $R_i = C_i$ for $i =$
lyche-numerical-linear-algebra- F04234	328	■ 1.000	$R_{\{1\}}=R_{\{m\}}=\left\{z \in \mathbb{R}: z-2 \leq 1\right\}$	$R_1 = R_m = \{z \in \mathbb{R} : z - 2 \leq 1\}$	eigenvalues are real. We find $R_1 = R_m = \{z \in \mathbb{R} : z - 2 \leq 1\}$ and
lyche-numerical-linear-algebra- F04235	328	■ 1.000	$\lambda \in[0,4]$	$\lambda \in [0, 4]$	We conclude that $\lambda \in [0, 4]$ for any eigenvalue λ of $\boldsymbol{T}$. To check this, we recall that
lyche-numerical-linear-algebra- F04236	329	■ 1.000	$4\left[\sin \frac{\pi}{2(m+1)}\right]^2$	$4\left[\sin \frac{\pi}{2(m+1)}\right]^2$	When m is large the smallest eigenvalue $4\left[\sin \frac{\pi}{2(m+1)}\right]^2$ is very close to zero and
lyche-numerical-linear-algebra- F04237	329	■ 0.969	$4\left[\sin \frac{m \pi}{2(m+1)}\right]^2$	$4\left[\sin \frac{m \pi}{2(m+1)}\right]^2$	the largest eigenvalue $4\left[\sin \frac{m \pi}{2(m+1)}\right]^2$ is very close to 4. Thus Gerschgorin's theorem
lyche-numerical-linear-algebra- F04238	329	■ 1.000	$\boldsymbol{A}(0)=\boldsymbol{D}$	$\boldsymbol{A}(0) = \boldsymbol{D}$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t , every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra- F04239	329	■ 1.000	$\boldsymbol{A}(1)=\boldsymbol{A}$	$\boldsymbol{A}(1) = \boldsymbol{A}$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t , every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra- F04240	329	■ 1.000	$\boldsymbol{A}(t)$	$\boldsymbol{A}(t)$	We have $\boldsymbol{A}(0) = \boldsymbol{D}$ and $\boldsymbol{A}(1) = \boldsymbol{A}$. As a function of t , every eigenvalue of $\boldsymbol{A}(t)$ is
lyche-numerical-linear-algebra- F04241	329	■ 1.000	$R_{\{i\}}(t)$	$R_i(t)$	row disks $R_i(t)$ of $\boldsymbol{A}(t)$ have radius proportional to t , indeed
lyche-numerical-linear-algebra- F04242	329	■ 1.000	$0 \leq t_1 < t_2 \leq 1$	$0 \leq t_1 < t_2 \leq 1$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of
lyche-numerical-linear-algebra- F04243	329	■ 1.000	$R_{\{i\}}\left(t_1\right) \subset R_{\{i\}}\left(t_2\right)$	$R_i\left(t_1\right) \subset R_i\left(t_2\right)$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of
lyche-numerical-linear-algebra- F04244	329	■ 1.000	$R_{\{i\}}(1)$	$R_i(1)$	Clearly $0 \leq t_1 < t_2 \leq 1$ implies $R_i(t_1) \subset R_i(t_2)$ and $R_i(1)$ is a row disk of
lyche-numerical-linear-algebra- F04245	329	■ 0.992	$\bigcup_{k=1}^p R_{\{i_k\}}(1)$	$\bigcup_{k=1}^p R_{i_k}(1)$	$\boldsymbol{A}$ for all i . Suppose $\bigcup_{k=1}^p R_{i_k}(1)$ are disjoint from the other disks of $\boldsymbol{A}$ and set
lyche-numerical-linear-algebra- F04246	329	■ 0.992	$R^p(t):=\bigcup_{k=1}^p R_{\{i_k\}}(t)$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$ for $t \in [0, 1]$. Now $R^p(0)$ contains only the p eigenvalues
lyche-numerical-linear-algebra- F04247	329	■ 0.992	$t \in[0,1]$	$t \in [0, 1]$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$ for $t \in [0, 1]$. Now $R^p(0)$ contains only the p eigenvalues
lyche-numerical-linear-algebra- F04248	329	■ 0.992	$R^p(0)$	$R^p(0)$	$R^p(t) := \bigcup_{k=1}^p R_{i_k}(t)$ for $t \in [0, 1]$. Now $R^p(0)$ contains only the p eigenvalues

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lyche-numerical-linear-algebra-F04249	329	1.000	$a_{i_1, i_1}, i_1\}, \ldots, a_{i_p, i_p}\}$	$a_{i_1, i_1}, \ldots, a_{i_p, i_p}$	$a_{i_1, i_1}, \ldots, a_{i_p, i_p}$ of $A(0) = D$. As t increases from zero to one the set $R^p(t)$ is
lyche-numerical-linear-algebra-F04250	329	1.000	$R^{\{p\}}(t)$	$R^p(t)$	$a_{i_1, i_1}, \ldots, a_{i_p, i_p}$ of $A(0) = D$. As t increases from zero to one the set $R^p(t)$ is
lyche-numerical-linear-algebra-F04251	329	1.000	$R^{\{p\}}(1)$	$R^p(1)$	cannot loose or gain eigenvalues. It follows that $R^p(1)$ must contain p eigenvalues
lyche-numerical-linear-algebra-F04252	329	1.000	$\boldsymbol{A}=\left[\begin{array}{ccc}1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3\end{array}\right]$	$\boldsymbol{A} = \begin{bmatrix} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{bmatrix}$	<i>Example 14.2</i> Consider the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{bmatrix}$, where $ \epsilon_i \leq 10^{-15}$ all i . By
lyche-numerical-linear-algebra-F04253	329	1.000	$ \epsilon_i \leq 10^{-15}$	$ \epsilon_i \leq 10^{-15}$	<i>Example 14.2</i> Consider the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & \epsilon_1 & \epsilon_2 \\ \epsilon_3 & 2 & \epsilon_4 \\ \epsilon_5 & \epsilon_6 & 3 \end{bmatrix}$, where $ \epsilon_i \leq 10^{-15}$ all i . By
lyche-numerical-linear-algebra-F04254	329	1.000	$\lambda_1, \lambda_2, \lambda_3$	$\lambda_1, \lambda_2, \lambda_3$	Corollary 14.1 the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $\boldsymbol{A}$ are distinct and satisfy $ \lambda_j - j \leq$
lyche-numerical-linear-algebra-F04255	329	1.000	$ \lambda_j - j \leq$	$ \lambda_j - j \leq$	Corollary 14.1 the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of $\boldsymbol{A}$ are distinct and satisfy $ \lambda_j - j \leq$
lyche-numerical-linear-algebra-F04256	329	1.000	2×10^{-15}	2×10^{-15}	2×10^{-15} for $j = 1, 2, 3$.
lyche-numerical-linear-algebra-F04257	329	1.000	$j=1,2,3$	$j = 1, 2, 3$	2×10^{-15} for $j = 1, 2, 3$.
lyche-numerical-linear-algebra-F04258	329	0.979	$\boldsymbol{A}+$	$\boldsymbol{A}+$	we think of $\boldsymbol{E}$ as a perturbation of $\boldsymbol{A}$. By how much do the eigenvalues of $\boldsymbol{A}$ and $\boldsymbol{A}+$
lyche-numerical-linear-algebra-F04259	329	1.000	$\boldsymbol{A}_0 := \mathbf{0}$	$\boldsymbol{A}_0 := \mathbf{0}$	We illustrate this by considering two examples. Suppose $\boldsymbol{A}_0 := \mathbf{0}$ is the zero
lyche-numerical-linear-algebra-F04260	329	0.998	$\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$	$\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$	matrix. If $\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$, then $ \lambda \leq \ \boldsymbol{E}\ _\infty$ by Theorem 12.11, and any
lyche-numerical-linear-algebra-F04261	329	0.998	$ \lambda \leq \ \boldsymbol{E}\ _\infty$	$ \lambda \leq \ \boldsymbol{E}\ _\infty$	matrix. If $\lambda \in \sigma(\boldsymbol{A}_0 + \boldsymbol{E}) = \sigma(\boldsymbol{E})$, then $ \lambda \leq \ \boldsymbol{E}\ _\infty$ by Theorem 12.11, and any
lyche-numerical-linear-algebra-F04262	330	0.998	$\boldsymbol{A}_0$	$\boldsymbol{A}_0$	zero eigenvalue of $\boldsymbol{A}_0$ is perturbed by at most $\ \boldsymbol{E}\ _\infty$. On the other hand consider
lyche-numerical-linear-algebra-F04263	330	0.998	$\ \boldsymbol{E}\ _\infty$	$\ \boldsymbol{E}\ _\infty$	zero eigenvalue of $\boldsymbol{A}_0$ is perturbed by at most $\ \boldsymbol{E}\ _\infty$. On the other hand consider
lyche-numerical-linear-algebra-F04264	330	1.000	$\boldsymbol{A}_1 + \boldsymbol{E}$	$\boldsymbol{A}_1 + \boldsymbol{E}$	The characteristic polynomial of $\boldsymbol{A}_1 + \boldsymbol{E}$ is $\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$, and the zero
lyche-numerical-linear-algebra-F04265	330	1.000	$\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$	$\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$	The characteristic polynomial of $\boldsymbol{A}_1 + \boldsymbol{E}$ is $\pi(\lambda) := (-1)^n(\lambda^n - \epsilon)$, and the zero
lyche-numerical-linear-algebra-F04266	330	1.000	$ \lambda = \ \boldsymbol{E}\ _\infty^{1/n}$	$ \lambda = \ \boldsymbol{E}\ _\infty^{1/n}$	eigenvalues of $\boldsymbol{A}_1$ are perturbed by the amount $ \lambda = \ \boldsymbol{E}\ _\infty^{1/n}$. Thus, for $n = 16$, a
lyche-numerical-linear-algebra-F04267	330	1.000	$n=16$	$n = 16$	eigenvalues of $\boldsymbol{A}_1$ are perturbed by the amount $ \lambda = \ \boldsymbol{E}\ _\infty^{1/n}$. Thus, for $n = 16$, a
lyche-numerical-linear-algebra-F04268	330	0.978	$\epsilon = 10^{-16}$	$\epsilon = 10^{-16}$	perturbation of say $\epsilon = 10^{-16}$ gives a change in eigenvalue of 0.1.
lyche-numerical-linear-algebra-F04269	330	1.000	$\ \boldsymbol{E}\ _\infty^{1/n}$	$\ \boldsymbol{E}\ _\infty^{1/n}$	The following theorem shows that a dependence $\ \boldsymbol{E}\ _\infty^{1/n}$ is the worst that can
lyche-numerical-linear-algebra-F04270	330	1.000	$\mu \in$	$\mu \in$	Theorem 14.2 (Elsner’s Theorem (1985)) Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$. To every $\mu \in$
lyche-numerical-linear-algebra-F04271	330	1.000	$\sigma(\boldsymbol{A} + \boldsymbol{E})$	$\sigma(\boldsymbol{A} + \boldsymbol{E})$	$\sigma(\boldsymbol{A} + \boldsymbol{E})$ there is a $\lambda \in \sigma(\boldsymbol{A})$ such that
lyche-numerical-linear-algebra-F04272	330	1.000	$\ \boldsymbol{u}_1\ _2 = 1$	$\ \boldsymbol{u}_1\ _2 = 1$	μ . Let $\boldsymbol{u}_1$ with $\ \boldsymbol{u}_1\ _2 = 1$ be an eigenvector corresponding to μ , and extend $\boldsymbol{u}_1$ to
lyche-numerical-linear-algebra-F04273	330	1.000	$\ \boldsymbol{E}\ _2^{1/n}$	$\ \boldsymbol{E}\ _2^{1/n}$	elements of the matrix. The factor $\ \boldsymbol{E}\ _2^{1/n}$ shows that this dependence is almost,

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lyche-numerical-linear-algebra-F04274	330	1.000	$\left[\begin{array}{cc} 1 & 1 \\ \epsilon & 1 \end{array}\right]$	$\begin{bmatrix} 1 & 1 \\ \epsilon & 1 \end{bmatrix}$	but not quite, differentiable. As an example, the eigenvalues of the matrix $\begin{bmatrix} 1 & 1 \\ \epsilon & 1 \end{bmatrix}$ are
lyche-numerical-linear-algebra-F04275	330	1.000	$1 \pm \sqrt{\epsilon}$	$1 \pm \sqrt{\epsilon}$	$1 \pm \sqrt{\epsilon}$ and these expressions are not differentiable at $\epsilon = 0$.
lyche-numerical-linear-algebra-F04276	330	1.000	$\epsilon=0$	$\epsilon = 0$	$1 \pm \sqrt{\epsilon}$ and these expressions are not differentiable at $\epsilon = 0$.
lyche-numerical-linear-algebra-F04277	331	1.000	$1 / n$	$1/n$	nondefective matrices we can get rid of the annoying exponent $1/n$ in $\ E\ _2$
lyche-numerical-linear-algebra-F04278	331	1.000	$\ \boldsymbol{E}\ _2$	$\ \boldsymbol{E}\ _2$	nondefective matrices we can get rid of the annoying exponent $1/n$ in $\ \boldsymbol{E}\ _2$
lyche-numerical-linear-algebra-F04279	331	1.000	$\mu \in \mathbb{C}$	$\mu \in \mathbb{C}$	any $\mu \in \mathbb{C}$ and $\boldsymbol{x} \in \mathbb{C}^n$ with $\ \boldsymbol{x}\ _p = 1$ we can find an eigenvalue λ of $\boldsymbol{A}$ such that
lyche-numerical-linear-algebra-F04280	331	0.999	$\boldsymbol{r} := \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x}$	$\boldsymbol{r} := \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x}$	where $\boldsymbol{r} := \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x}$ and $K_p(\boldsymbol{X}) := \ \boldsymbol{X}\ _p \ \boldsymbol{X}^{-1}\ _p$. If for some $\boldsymbol{E} \in \mathbb{C}^{n \times n}$ it holds
lyche-numerical-linear-algebra-F04281	331	0.999	$K_p(\boldsymbol{X}) := \ \boldsymbol{X}\ _p \ \boldsymbol{X}^{-1}\ _p$	$K_p(\boldsymbol{X}) := \ \boldsymbol{X}\ _p \ \boldsymbol{X}^{-1}\ _p$	where $\boldsymbol{r} := \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x}$ and $K_p(\boldsymbol{X}) := \ \boldsymbol{X}\ _p \ \boldsymbol{X}^{-1}\ _p$. If for some $\boldsymbol{E} \in \mathbb{C}^{n \times n}$ it holds
lyche-numerical-linear-algebra-F04282	331	0.902	$(\mu, \boldsymbol{x})$	$(\mu, \boldsymbol{x})$	that $(\mu, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{A} + \boldsymbol{E}$, then we can find an eigenvalue λ of $\boldsymbol{A}$ such
lyche-numerical-linear-algebra-F04283	331	1.000	$\mu \in \sigma(\boldsymbol{A})$	$\mu \in \sigma(\boldsymbol{A})$	Proof If $\mu \in \sigma(\boldsymbol{A})$ then we can take $\lambda = \mu$ and (14.2), (14.3) hold trivially. So
lyche-numerical-linear-algebra-F04284	331	1.000	$\lambda = \mu$	$\lambda = \mu$	Proof If $\mu \in \sigma(\boldsymbol{A})$ then we can take $\lambda = \mu$ and (14.2), (14.3) hold trivially. So
lyche-numerical-linear-algebra-F04285	331	1.000	$\mu \notin \sigma(\boldsymbol{A})$	$\mu \notin \sigma(\boldsymbol{A})$	assume $\mu \notin \sigma(\boldsymbol{A})$. Since $\boldsymbol{A}$ is nondefective it can be diagonalized, we have $\boldsymbol{A} =$
lyche-numerical-linear-algebra-F04286	331	1.000	$\boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^{-1}$	$\boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^{-1}$	$\boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^{-1}$, where $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $(\lambda_j, \boldsymbol{x}_j)$ are the eigenpairs of $\boldsymbol{A}$ for $j =$
lyche-numerical-linear-algebra-F04287	331	1.000	$(\lambda_j, \boldsymbol{x}_j)$	$(\lambda_j, \boldsymbol{x}_j)$	$\boldsymbol{X}\boldsymbol{D}\boldsymbol{X}^{-1}$, where $\boldsymbol{D} = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $(\lambda_j, \boldsymbol{x}_j)$ are the eigenpairs of $\boldsymbol{A}$ for $j =$
lyche-numerical-linear-algebra-F04288	331	1.000	$\boldsymbol{D}_1 := \boldsymbol{D} - \mu\boldsymbol{I}$	$\boldsymbol{D}_1 := \boldsymbol{D} - \mu\boldsymbol{I}$	$1, \dots, n$. Define $\boldsymbol{D}_1 := \boldsymbol{D} - \mu\boldsymbol{I}$. Then $\boldsymbol{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$
lyche-numerical-linear-algebra-F04289	331	1.000	$\boldsymbol{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$	$\boldsymbol{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$	$1, \dots, n$. Define $\boldsymbol{D}_1 := \boldsymbol{D} - \mu\boldsymbol{I}$. Then $\boldsymbol{D}_1^{-1} = \text{diag}((\lambda_1 - \mu)^{-1}, \dots, (\lambda_n - \mu)^{-1})$
lyche-numerical-linear-algebra-F04290	331	0.658	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mu\boldsymbol{x}$	$(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mu\boldsymbol{x}$	But then (14.2) follows. If $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mu\boldsymbol{x}$ then $\boldsymbol{0} = \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x} + \boldsymbol{E}\boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E}\boldsymbol{x}$.
lyche-numerical-linear-algebra-F04291	331	0.658	$\boldsymbol{0} = \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x} + \boldsymbol{E}\boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E}\boldsymbol{x}$	$\boldsymbol{0} = \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x} + \boldsymbol{E}\boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E}\boldsymbol{x}$	But then (14.2) follows. If $(\boldsymbol{A} + \boldsymbol{E})\boldsymbol{x} = \mu\boldsymbol{x}$ then $\boldsymbol{0} = \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x} + \boldsymbol{E}\boldsymbol{x} = \boldsymbol{r} + \boldsymbol{E}\boldsymbol{x}$.
lyche-numerical-linear-algebra-F04292	331	0.954	$\ \boldsymbol{r}\ _p = \ \boldsymbol{E}\boldsymbol{x}\ _p \leq \ \boldsymbol{E}\ _p \ \boldsymbol{x}\ _p$	$\ \boldsymbol{r}\ _p = \ \boldsymbol{E}\boldsymbol{x}\ _p \leq \ \boldsymbol{E}\ _p \ \boldsymbol{x}\ _p$	But then $\ \boldsymbol{r}\ _p = \ \boldsymbol{E}\boldsymbol{x}\ _p \leq \ \boldsymbol{E}\ _p$. Inserting this in (14.2) proves (14.3).
lyche-numerical-linear-algebra-F04293	331	1.000	$K_p(\boldsymbol{X})$	$K_p(\boldsymbol{X})$	be magnified by at most $K_p(\boldsymbol{X})$, the condition number of the eigenvector matrix
lyche-numerical-linear-algebra-F04294	332	0.593	$K_2(\boldsymbol{X}) = 1$	$K_2(\boldsymbol{X}) = 1$	we restrict attention to the 2-norm then $K_2(\boldsymbol{X}) = 1$ and (14.3) implies the following
lyche-numerical-linear-algebra-F04295	332	1.000	$ \lambda - \mu \leq \ \boldsymbol{E}\ _2$	$ \lambda - \mu \leq \ \boldsymbol{E}\ _2$	eigenvalue λ of $\boldsymbol{A}$ such that $ \lambda - \mu \leq \ \boldsymbol{E}\ _2$.
lyche-numerical-linear-algebra-F04296	332	0.995	$\boldsymbol{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$	$\boldsymbol{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$	Lemma 14.1 (p-Norm of a Diagonal Matrix) If $\boldsymbol{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$ is a
lyche-numerical-linear-algebra-F04297	332	1.000	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A})$	$\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A})$	diagonal matrix then $\ \boldsymbol{A}\ _p = \rho(\boldsymbol{A})$ for $1 \leq p \leq \infty$.
lyche-numerical-linear-algebra-F04298	332	1.000	$p < \infty$	$p < \infty$	Proof For $p = \infty$ the proof is left as an exercise. For any $\boldsymbol{x} \in \mathbb{C}^n$ and $p < \infty$ we

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lyche-numerical-linear-algebra- F04299	332	0.874	$\ \boldsymbol{A}\ _p = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _p}{\ \boldsymbol{x}\ _p} \leq \rho(\boldsymbol{A})$	$\ \boldsymbol{A}\ _p = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\ _p}{\ \boldsymbol{x}\ _p} \leq \rho(\boldsymbol{A})$	Thus $\ \boldsymbol{A}\ _p = \max_{\boldsymbol{x} \neq \mathbf{0}} \frac{\ \boldsymbol{A}\boldsymbol{x}\ _p}{\ \boldsymbol{x}\ _p} \leq \rho(\boldsymbol{A})$. But from Theorem 12.11 we have $\rho(\boldsymbol{A}) \leq$
lyche-numerical-linear-algebra- F04300	332	0.784	$\mu, \boldsymbol{x}$	$\mu, \boldsymbol{x}$	where $\boldsymbol{r} := \boldsymbol{A}\boldsymbol{x} - \mu\boldsymbol{x}$. If for some $\boldsymbol{E} \in \mathbb{C}^{n \times n}$ it holds that $(\mu, \boldsymbol{x})$ is an eigenpair for
lyche-numerical-linear-algebra- F04301	333	1.000	$\boldsymbol{B} := \mu \boldsymbol{A}^{-1}$	$\boldsymbol{B} := \mu \boldsymbol{A}^{-1}$	and (14.4) follows from (14.2). To prove (14.5) we define the matrices $\boldsymbol{B} := \mu \boldsymbol{A}^{-1}$
lyche-numerical-linear-algebra- F04302	333	0.858	$\boldsymbol{F} := -\boldsymbol{A}^{-1} \boldsymbol{E}$	$\boldsymbol{F} := -\boldsymbol{A}^{-1} \boldsymbol{E}$	and $\boldsymbol{F} := -\boldsymbol{A}^{-1} \boldsymbol{E}$. If $(\lambda_j, \boldsymbol{x})$ are the eigenpairs for $\boldsymbol{A}$ then $(\frac{\mu}{\lambda_j}, \boldsymbol{x})$ are the eigenpairs
lyche-numerical-linear-algebra- F04303	333	0.858	$(\lambda_j, \boldsymbol{x})$	$(\lambda_j, \boldsymbol{x})$	and $\boldsymbol{F} := -\boldsymbol{A}^{-1} \boldsymbol{E}$. If $(\lambda_j, \boldsymbol{x})$ are the eigenpairs for $\boldsymbol{A}$ then $(\frac{\mu}{\lambda_j}, \boldsymbol{x})$ are the eigenpairs
lyche-numerical-linear-algebra- F04304	333	0.858	$(\frac{\mu}{\lambda_j}, \boldsymbol{x})$	$(\frac{\mu}{\lambda_j}, \boldsymbol{x})$	and $\boldsymbol{F} := -\boldsymbol{A}^{-1} \boldsymbol{E}$. If $(\lambda_j, \boldsymbol{x})$ are the eigenpairs for $\boldsymbol{A}$ then $(\frac{\mu}{\lambda_j}, \boldsymbol{x})$ are the eigenpairs
lyche-numerical-linear-algebra- F04305	333	0.999	$(1, \boldsymbol{x})$	$(1, \boldsymbol{x})$	Thus $(1, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{B} + \boldsymbol{F}$. Applying Theorem 14.3 to this eigenvalue
lyche-numerical-linear-algebra- F04306	333	0.999	$\boldsymbol{B} + \boldsymbol{F}$	$\boldsymbol{B} + \boldsymbol{F}$	Thus $(1, \boldsymbol{x})$ is an eigenpair for $\boldsymbol{B} + \boldsymbol{F}$. Applying Theorem 14.3 to this eigenvalue
lyche-numerical-linear-algebra- F04307	333	1.000	$\left \frac{\mu}{\lambda} - 1 \right \leq K_p(\boldsymbol{X}) \ \boldsymbol{F}\ _p = K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p$	$\left \frac{\mu}{\lambda} - 1 \right \leq K_p(\boldsymbol{X}) \ \boldsymbol{F}\ _p = K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p$	we can find $\lambda \in \sigma(\bar{\boldsymbol{A}})$ such that $\left \frac{\mu}{\lambda} - 1 \right \leq K_p(\boldsymbol{X}) \ \boldsymbol{F}\ _p = K_p(\boldsymbol{X}) \ \boldsymbol{A}^{-1} \boldsymbol{E}\ _p$ which
lyche-numerical-linear-algebra- F04308	333	1.000	$1, 2, \dots, i-2, i=3, 4, \dots, n$	$1, 2, \dots, i-2, i=3, 4, \dots, n$	$1, 2, \dots, i-2, i=3, 4, \dots, n$. We will reduce $\boldsymbol{A} \in \mathbb{C}^{n \times n}$ to upper Hessenberg form
lyche-numerical-linear-algebra- F04309	333	1.000	$\boldsymbol{A}_1 = \boldsymbol{A}$	$\boldsymbol{A}_1 = \boldsymbol{A}$	by unitary similarity transformations. Let $\boldsymbol{A}_1 = \boldsymbol{A}$ and define $\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$ for
lyche-numerical-linear-algebra- F04310	333	1.000	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$	$\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$	by unitary similarity transformations. Let $\boldsymbol{A}_1 = \boldsymbol{A}$ and define $\boldsymbol{A}_{k+1} = \boldsymbol{H}_k \boldsymbol{A}_k \boldsymbol{H}_k$ for
lyche-numerical-linear-algebra- F04311	333	1.000	$k=1, 2, \dots, n-2$	$k=1, 2, \dots, n-2$	$k=1, 2, \dots, n-2$. Here $\boldsymbol{H}_k$ is a Householder transformation chosen to introduce
lyche-numerical-linear-algebra- F04312	333	1.000	$\boldsymbol{H}_k$	$\boldsymbol{H}_k$	$k=1, 2, \dots, n-2$. Here $\boldsymbol{H}_k$ is a Householder transformation chosen to introduce
lyche-numerical-linear-algebra- F04313	333	1.000	$\boldsymbol{A}_{n-1}$	$\boldsymbol{A}_{n-1}$	$\boldsymbol{A}_{n-1}$ will be upper Hessenberg. Householder transformations were used in Chap. 5
lyche-numerical-linear-algebra- F04314	333	1.000	$\boldsymbol{A}_k^* = \boldsymbol{A}_k$	$\boldsymbol{A}_k^* = \boldsymbol{A}_k$	if $\boldsymbol{A}_k^* = \boldsymbol{A}_k$ then
lyche-numerical-linear-algebra- F04315	334	1.000	$\boldsymbol{B}_k \in \mathbb{C}^{k,k}$	$\boldsymbol{B}_k \in \mathbb{C}^{k,k}$	Suppose $\boldsymbol{B}_k \in \mathbb{C}^{k,k}$ is upper Hessenberg, and the first $k-1$ columns of $\boldsymbol{D}_k \in$
lyche-numerical-linear-algebra- F04316	334	1.000	$\boldsymbol{D}_k \in$	$\boldsymbol{D}_k \in$	Suppose $\boldsymbol{B}_k \in \mathbb{C}^{k,k}$ is upper Hessenberg, and the first $k-1$ columns of $\boldsymbol{D}_k \in$
lyche-numerical-linear-algebra- F04317	334	0.999	$\mathbb{C}^{n-k,k}$	$\mathbb{C}^{n-k,k}$	$\mathbb{C}^{n-k,k}$ are zero, i.e. $\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \dots, \mathbf{0}, d_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$ be a
lyche-numerical-linear-algebra- F04318	334	0.999	$\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \dots, \mathbf{0}, d_k]$	$\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \dots, \mathbf{0}, d_k]$	$\mathbb{C}^{n-k,k}$ are zero, i.e. $\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \dots, \mathbf{0}, d_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$ be a
lyche-numerical-linear-algebra- F04319	334	0.999	$\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$	$\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$	$\mathbb{C}^{n-k,k}$ are zero, i.e. $\boldsymbol{D}_k = [\mathbf{0}, \mathbf{0}, \dots, \mathbf{0}, d_k]$. Let $\boldsymbol{V}_k = \boldsymbol{I} - \boldsymbol{v}_k \boldsymbol{v}_k^* \in \mathbb{C}^{n-k, n-k}$ be a
lyche-numerical-linear-algebra- F04320	334	0.999	$\boldsymbol{V}_k \boldsymbol{d}_k = \alpha_k \boldsymbol{e}_1$	$\boldsymbol{V}_k \boldsymbol{d}_k = \alpha_k \boldsymbol{e}_1$	Householder transformation such that $\boldsymbol{V}_k \boldsymbol{d}_k = \alpha_k \boldsymbol{e}_1$. Define
lyche-numerical-linear-algebra- F04321	334	1.000	$\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \mathbf{0}, \dots, \boldsymbol{V}_k \mathbf{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\mathbf{0}, \dots, \mathbf{0}, \alpha_k \boldsymbol{e}_1)$	$\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \mathbf{0}, \dots, \boldsymbol{V}_k \mathbf{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\mathbf{0}, \dots, \mathbf{0}, \alpha_k \boldsymbol{e}_1)$	Now $\boldsymbol{V}_k \boldsymbol{D}_k = [\boldsymbol{V}_k \mathbf{0}, \dots, \boldsymbol{V}_k \mathbf{0}, \boldsymbol{V}_k \boldsymbol{d}_k] = (\mathbf{0}, \dots, \mathbf{0}, \alpha_k \boldsymbol{e}_1)$. Moreover, the matrix

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lyche-numerical-linear-algebra-F04322	334	1.000	$(k+1) \times (k+1)$	$(k+1) \times (k+1)$	B_k is not affected by the H_k transformation. Therefore the upper left $(k+1) \times (k+1)$
lyche-numerical-linear-algebra-F04323	334	1.000	$\boldsymbol{v}_{\{k\}}$	v_k	To find A_{k+1} we use Algorithm 5.1 to find v_k and α_k . We store v_k in the k th
lyche-numerical-linear-algebra-F04324	334	1.000	$\boldsymbol{L}(k+1:n, k)=\boldsymbol{v}_{\{k\}}$	$L(k+1:n, k) = v_k$	column of a matrix L as $L(k+1:n, k) = v_k$. This leads to the following algorithm
lyche-numerical-linear-algebra-F04325	334	0.994	$\boldsymbol{B} \cdot \boldsymbol{B}$	$B.B$	transformations. The algorithm returns the reduced matrix B . B is tridiagonal if A
lyche-numerical-linear-algebra-F04326	334	1.000	$\boldsymbol{B}=\boldsymbol{Q}^*\boldsymbol{A}\boldsymbol{Q}$ $\boldsymbol{Q}$	$B = Q^*AQ$	unitary matrix Q such that $B = Q^*AQ$. Algorithm 5.1 is used in each step of the
lyche-numerical-linear-algebra-F04327	335	1.000	$\boldsymbol{Q}^*\boldsymbol{A}\boldsymbol{Q}$	Q^*AQ	Q is orthogonal and Q^*AQ is upper Hessenberg. We need to compute the product
lyche-numerical-linear-algebra-F04328	335	0.950	$\boldsymbol{Q}=\boldsymbol{H}_{\{1\}} \boldsymbol{H}_{\{2\}} \cdots \boldsymbol{H}_{\{n-2\}}$	$Q = H_1H_2 \cdots H_{n-2}$	$Q = H_1H_2 \cdots H_{n-2}$, where $H_k = \begin{bmatrix} I & \mathbf{0} \\ \mathbf{0} & I-v_kv_k^T \end{bmatrix}$ and $v_k \in \mathbb{R}^{n-k}$. Since $v_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra-F04329	335	0.950	$\boldsymbol{H}_{\{k\}}=\left[\begin{array}{cc}\boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I}-\boldsymbol{v}_{\{k\}}\boldsymbol{v}_{\{k\}}^T\end{array}\right]$ & $\mathbf{0} \setminus \mathbf{0}$ & $\boldsymbol{I}-\boldsymbol{v}_{\{k\}}\boldsymbol{v}_{\{k\}}^T$ $\boldsymbol{v}_{\{k\}}\boldsymbol{v}_{\{k\}}^T$ $\right]$	$H_k = \begin{bmatrix} I & \mathbf{0} \\ \mathbf{0} & I-v_kv_k^T \end{bmatrix}$	$Q = H_1H_2 \cdots H_{n-2}$, where $H_k = \begin{bmatrix} I & \mathbf{0} \\ \mathbf{0} & I-v_kv_k^T \end{bmatrix}$ and $v_k \in \mathbb{R}^{n-k}$. Since $v_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra-F04330	335	0.950	$\boldsymbol{v}_{\{k\}} \in \mathbb{R}^{n-k}$	$v_k \in \mathbb{R}^{n-k}$	$Q = H_1H_2 \cdots H_{n-2}$, where $H_k = \begin{bmatrix} I & \mathbf{0} \\ \mathbf{0} & I-v_kv_k^T \end{bmatrix}$ and $v_k \in \mathbb{R}^{n-k}$. Since $v_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra-F04331	335	0.950	$\boldsymbol{v}_{\{1\}} \in \mathbb{R}^{n-1}$	$v_1 \in \mathbb{R}^{n-1}$	$Q = H_1H_2 \cdots H_{n-2}$, where $H_k = \begin{bmatrix} I & \mathbf{0} \\ \mathbf{0} & I-v_kv_k^T \end{bmatrix}$ and $v_k \in \mathbb{R}^{n-k}$. Since $v_1 \in \mathbb{R}^{n-1}$
lyche-numerical-linear-algebra-F04332	335	1.000	$\boldsymbol{v}_{\{n-2\}} \in \mathbb{R}^2$	$v_{n-2} \in \mathbb{R}^2$	and $v_{n-2} \in \mathbb{R}^2$ it is most economical to assemble the product from right to left. We
lyche-numerical-linear-algebra-F04333	335	1.000	$\boldsymbol{Q}_{\{k+1\}}$	Q_{k+1}	Suppose Q_{k+1} has the form $\begin{bmatrix} I_k & \mathbf{0} \\ \mathbf{0} & U_k \end{bmatrix}$, where $U_k \in \mathbb{R}^{n-k, n-k}$. Then
lyche-numerical-linear-algebra-F04334	335	1.000	$\left[\begin{array}{cc}\boldsymbol{I}_{\{k\}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{U}_{\{k\}}\end{array}\right]$ & $\mathbf{0} \setminus \mathbf{0}$ & $\boldsymbol{U}_{\{k\}}$ $\mathbf{0} \setminus \mathbf{0}$ & $\boldsymbol{U}_{\{k\}}$	Suppose Q_{k+1} has the form $\begin{bmatrix} I_k & \mathbf{0} \\ \mathbf{0} & U_k \end{bmatrix}$, where $U_k \in \mathbb{R}^{n-k, n-k}$. Then	
lyche-numerical-linear-algebra-F04335	335	1.000	$\boldsymbol{U}_{\{k\}} \in \mathbb{R}^{n-k, n-k}$	$U_k \in \mathbb{R}^{n-k, n-k}$	Suppose Q_{k+1} has the form $\begin{bmatrix} I_k & \mathbf{0} \\ \mathbf{0} & U_k \end{bmatrix}$, where $U_k \in \mathbb{R}^{n-k, n-k}$. Then
lyche-numerical-linear-algebra-F04336	335	1.000	$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$	$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$	Let $A \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues $\lambda_1 > \lambda_2 > \cdots > \lambda_n$. In this section
lyche-numerical-linear-algebra-F04337	335	1.000	$1 \leq m \leq n$	$1 \leq m \leq n$	some $1 \leq m \leq n$. Using Householder similarity transformations as outlined in the
lyche-numerical-linear-algebra-F04338	336	1.000	$c_{\{i\}}=0$	$c_i = 0$	Suppose one of the off-diagonal elements is equal to zero, say $c_i = 0$. We then have
lyche-numerical-linear-algebra-F04339	336	1.000	$\boldsymbol{A}=\left[\begin{array}{cc}\boldsymbol{A}_{\{1\}} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{A}_{\{2\}}\end{array}\right]$ & $\mathbf{0} \setminus \mathbf{0}$ & $\boldsymbol{A}_{\{2\}}$ $\end{array}\right]$	$A = \begin{bmatrix} A_1 & \mathbf{0} \\ \mathbf{0} & A_2 \end{bmatrix}$, where	
lyche-numerical-linear-algebra-F04340	336	1.000	$c_{\{i\}} \neq 0$	$c_i \neq 0$	so that A is irreducible, i.e., $c_i \neq 0$ for $i = 1, \dots, n-1$.
lyche-numerical-linear-algebra-F04341	336	1.000	$x \in \mathbb{R}$	$x \in \mathbb{R}$	for $x \in \mathbb{R}$ the polynomial $p_k(x) := \det(xI_k - A_k)$ for $k = 1, \dots, n$, where A_k
lyche-numerical-linear-algebra-F04342	336	1.000	$p_{\{k\}}(x):=\operatorname{det}\left(x \boldsymbol{I}_{\{k\}}-\boldsymbol{A}_{\{k\}}\right)$	$p_k(x) := \det(xI_k - A_k)$	for $x \in \mathbb{R}$ the polynomial $p_k(x) := \det(xI_k - A_k)$ for $k = 1, \dots, n$, where A_k
lyche-numerical-linear-algebra-F04343	336	1.000	$p_{\{n\}}$	p_n	The eigenvalues of A are the roots of the polynomial p_n . Using the last column to
lyche-numerical-linear-algebra-F04344	336	1.000	$k \geq 2$	$k \geq 2$	expand for $k \geq 2$ the determinant $p_{k+1}(x)$ we find

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lyche-numerical-linear-algebra F04345	336	1.000	$p_{k+1}(x)$	$p_{k+1}(x)$	expand for $k \geq 2$ the determinant $p_{k+1}(x)$ we find
lyche-numerical-linear-algebra F04346	336	0.990	$p_1(x)=x-d_1$	$p_1(x) = x - d_1$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-algebra F04347	336	0.990	$p_2(x)=\left(x-d_2\right)\left(x-d_1\right)-c_1^2$	$p_2(x) = (x - d_2)(x - d_1) - c_1^2$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-algebra F04348	336	0.990	$k=0,1$	$k = 0, 1$	Since $p_1(x) = x - d_1$ and $p_2(x) = (x - d_2)(x - d_1) - c_1^2$ this also holds for $k = 0, 1$
lyche-numerical-linear-algebra F04349	336	1.000	$p_{-1}(x)=0$	$p_{-1}(x) = 0$	if we define $p_{-1}(x) = 0$ and $p_0(x) = 1$. For M sufficiently large we have
lyche-numerical-linear-algebra F04350	336	1.000	$p_0(x)=1$	$p_0(x) = 1$	if we define $p_{-1}(x) = 0$ and $p_0(x) = 1$. For M sufficiently large we have
lyche-numerical-linear-algebra F04351	336	1.000	$y_1 \in (-M, d_1)$	$y_1 \in (-M, d_1)$	Since p_2 is continuous there are $y_1 \in (-M, d_1)$ and $y_2 \in (d_1, M)$ such that
lyche-numerical-linear-algebra F04352	336	1.000	$y_2 \in (d_1, M)$	$y_2 \in (d_1, M)$	Since p_2 is continuous there are $y_1 \in (-M, d_1)$ and $y_2 \in (d_1, M)$ such that
lyche-numerical-linear-algebra F04353	336	1.000	$p_2(y_1)=p_2(y_2)=0$	$p_2(y_1) = p_2(y_2) = 0$	$p_2(y_1) = p_2(y_2) = 0$. It follows that the root d_1 of p_1 separates the roots of p_2 , so
lyche-numerical-linear-algebra F04354	336	1.000	y_1	y_1	y_1 and y_2 must be distinct. Consider next
lyche-numerical-linear-algebra F04355	336	1.000	y_2	y_2	y_1 and y_2 must be distinct. Consider next
lyche-numerical-linear-algebra F04356	337	1.000	$y_1 < d_1 < y_2$	$y_1 < d_1 < y_2$	Since $y_1 < d_1 < y_2$ we have for M sufficiently large
lyche-numerical-linear-algebra F04357	337	1.000	x_1, x_2, x_3	x_1, x_2, x_3	Thus the roots x_1, x_2, x_3 of p_3 are separated by the roots y_1, y_2 of p_2 . In the general
lyche-numerical-linear-algebra F04358	337	1.000	p_3	p_3	Thus the roots x_1, x_2, x_3 of p_3 are separated by the roots y_1, y_2 of p_2 . In the general
lyche-numerical-linear-algebra F04359	337	1.000	$z_1, \dots, z_{k-1}$	$z_1, \dots, z_{k-1}$	case suppose for $k \geq 2$ that the roots $z_1, \dots, z_{k-1}$ of p_{k-1} separate the roots
lyche-numerical-linear-algebra F04360	337	1.000	p_{k-1}	p_{k-1}	case suppose for $k \geq 2$ that the roots $z_1, \dots, z_{k-1}$ of p_{k-1} separate the roots
lyche-numerical-linear-algebra F04361	337	1.000	$y_1, \dots, y_k$	$y_1, \dots, y_k$	$y_1, \dots, y_k$ of p_k . Choose M so that $y_0 := -M < y_1, y_{k+1} := M > y_k$. Then
lyche-numerical-linear-algebra F04362	337	1.000	p_k	p_k	$y_1, \dots, y_k$ of p_k . Choose M so that $y_0 := -M < y_1, y_{k+1} := M > y_k$. Then
lyche-numerical-linear-algebra F04363	337	1.000	$y_0 := -M < y_1, y_{k+1} := M > y_k$	$y_0 := -M < y_1, y_{k+1} := M > y_k$	$y_1, \dots, y_k$ of p_k . Choose M so that $y_0 := -M < y_1, y_{k+1} := M > y_k$. Then
lyche-numerical-linear-algebra F04364	337	1.000	$j=0, k+1$	$j = 0, k + 1$	This holds for $j = 0, k + 1$, and for $j = 1, \dots, k$ since
lyche-numerical-linear-algebra F04365	337	0.832	$x_1, \dots, x_{k+1}$	$x_1, \dots, x_{k+1}$	It follows that the roots $x_1, \dots, x_{k+1}$ are separated by the roots $y_1, \dots, y_k$ of p_k and
lyche-numerical-linear-algebra F04366	337	1.000	$\boldsymbol{A}=\boldsymbol{E}^*\boldsymbol{B}\boldsymbol{E}$	$\boldsymbol{A} = \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E}$	We say that two matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{C}^{n \times n}$ are congruent if $\boldsymbol{A} = \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E}$ for some
lyche-numerical-linear-algebra F04367	337	1.000	$\boldsymbol{D}, \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$	$\boldsymbol{D}, \boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$	congruent and similar to a diagonal matrix $\boldsymbol{D}$, $\boldsymbol{U}^* \boldsymbol{A} \boldsymbol{U} = \boldsymbol{D}$ where $\boldsymbol{U}$ is unitary. The
lyche-numerical-linear-algebra F04368	337	1.000	$\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$	$\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$	eigenvalues of $\boldsymbol{A}$ are the diagonal elements of $\boldsymbol{D}$. Let $\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$ and $v(\boldsymbol{A})$ denote
lyche-numerical-linear-algebra F04369	337	1.000	$v(\boldsymbol{A})$	$v(\boldsymbol{A})$	eigenvalues of $\boldsymbol{A}$ are the diagonal elements of $\boldsymbol{D}$. Let $\pi(\boldsymbol{A}), \zeta(\boldsymbol{A})$ and $v(\boldsymbol{A})$ denote
lyche-numerical-linear-algebra F04370	337	1.000	$\pi(\boldsymbol{A})+\zeta(\boldsymbol{A})+v(\boldsymbol{A})=n$	$\pi(\boldsymbol{A}) + \zeta(\boldsymbol{A}) + v(\boldsymbol{A}) = n$	all eigenvalues are real and $\pi(\boldsymbol{A}) + \zeta(\boldsymbol{A}) + v(\boldsymbol{A}) = n$.
lyche-numerical-linear-algebra F04371	337	1.000	$\pi(\boldsymbol{A})=\pi(\boldsymbol{B}), \zeta(\boldsymbol{A})=\zeta(\boldsymbol{B})$	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B}), \zeta(\boldsymbol{A}) = \zeta(\boldsymbol{B})$	congruent then $\pi(\boldsymbol{A}) = \pi(\boldsymbol{B}), \zeta(\boldsymbol{A}) = \zeta(\boldsymbol{B})$ and $v(\boldsymbol{A}) = v(\boldsymbol{B})$.

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lyche-numerical-linear-algebra-F04372	337	1.000	$v(\boldsymbol{A})=v(\boldsymbol{B})$	$v(\boldsymbol{A}) = v(\boldsymbol{B})$	$\text{congruent then } \pi(\boldsymbol{A}) = \pi(\boldsymbol{B}), \zeta(\boldsymbol{A}) = \zeta(\boldsymbol{B}) \text{ and } v(\boldsymbol{A}) = v(\boldsymbol{B}).$
lyche-numerical-linear-algebra-F04373	337	1.000	$\pi(\boldsymbol{A})=k$	$\pi(\boldsymbol{A}) = k$	are diagonal matrices. Suppose $\pi(\boldsymbol{A}) = k$ and $\pi(\boldsymbol{B}) = m < k$. We shall show that
lyche-numerical-linear-algebra-F04374	337	1.000	$\pi(\boldsymbol{B})=m<k$	$\pi(\boldsymbol{B}) = m < k$	are diagonal matrices. Suppose $\pi(\boldsymbol{A}) = k$ and $\pi(\boldsymbol{B}) = m < k$. We shall show that
lyche-numerical-linear-algebra-F04375	337	1.000	$\boldsymbol{E}_1$	$\boldsymbol{E}_1$	this leads to a contradiction. Let $\boldsymbol{E}_1$ be the upper left $m \times k$ corner of $\boldsymbol{E}$. Since
lyche-numerical-linear-algebra-F04376	337	1.000	$m \times k$	$m \times k$	this leads to a contradiction. Let $\boldsymbol{E}_1$ be the upper left $m \times k$ corner of $\boldsymbol{E}$. Since
lyche-numerical-linear-algebra-F04377	337	1.000	$m < k$	$m < k$	$m < k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T =$
lyche-numerical-linear-algebra-F04378	337	1.000	$\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$	$\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$	$m < k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T =$
lyche-numerical-linear-algebra-F04379	337	1.000	$\boldsymbol{y}^T =$	$\boldsymbol{y}^T =$	$m < k$, we can find a nonzero $\boldsymbol{x}$ such that $\boldsymbol{E}_1 \boldsymbol{x} = \mathbf{0}$ (cf. Lemma 1.3). Let $\boldsymbol{y}^T =$
lyche-numerical-linear-algebra-F04380	337	0.996	$\left[\boldsymbol{x}^T, \mathbf{0}^T \right] \in \mathbb{C}^n$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04381	337	0.996	$\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$	$\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04382	337	0.996	$z_i = 0$	$z_i = 0$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04383	337	0.996	$i=1,2, \dots, m$	$i = 1, 2, \dots, m$	$[\boldsymbol{x}^T, \mathbf{0}^T] \in \mathbb{C}^n$, and $\boldsymbol{z} = [z_1, \dots, z_n]^T = \boldsymbol{E} \boldsymbol{y}$. Then $z_i = 0$ for $i = 1, 2, \dots, m$.
lyche-numerical-linear-algebra-F04384	337	1.000	$\mu_i \leq 0$	$\mu_i \leq 0$	$\mu_i \leq 0$ for $i \geq m + 1$ then
lyche-numerical-linear-algebra-F04385	337	1.000	$i \geq m+1$	$i \geq m + 1$	$\mu_i \leq 0$ for $i \geq m + 1$ then
lyche-numerical-linear-algebra-F04386	338	1.000	$\pi(\boldsymbol{A})=\pi(\boldsymbol{B})$	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$	We conclude that $\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$ if $\boldsymbol{A}$ and $\boldsymbol{B}$ are diagonal. Moreover, $v(\boldsymbol{A}) =$
lyche-numerical-linear-algebra-F04387	338	1.000	$v(\boldsymbol{A})=$	$v(\boldsymbol{A}) =$	We conclude that $\pi(\boldsymbol{A}) = \pi(\boldsymbol{B})$ if $\boldsymbol{A}$ and $\boldsymbol{B}$ are diagonal. Moreover, $v(\boldsymbol{A}) =$
lyche-numerical-linear-algebra-F04388	338	1.000	$\pi(-\boldsymbol{A})=\pi(-\boldsymbol{B})=v(\boldsymbol{B})$	$\pi(-\boldsymbol{A}) = \pi(-\boldsymbol{B}) = v(\boldsymbol{B})$	$\pi(-\boldsymbol{A}) = \pi(-\boldsymbol{B}) = v(\boldsymbol{B})$ and $\zeta(\boldsymbol{A}) = n - \pi(\boldsymbol{A}) - v(\boldsymbol{A}) = n - \pi(\boldsymbol{B}) - v(\boldsymbol{B}) =$
lyche-numerical-linear-algebra-F04389	338	1.000	$\zeta(\boldsymbol{A})=n-\pi(\boldsymbol{A})-v(\boldsymbol{A})=n-\pi(\boldsymbol{B})-v(\boldsymbol{B})=$	$\zeta(\boldsymbol{A}) = n - \pi(\boldsymbol{A}) - v(\boldsymbol{A}) = n - \pi(\boldsymbol{B}) - v(\boldsymbol{B}) =$	$\pi(-\boldsymbol{A}) = \pi(-\boldsymbol{B}) = v(\boldsymbol{B})$ and $\zeta(\boldsymbol{A}) = n - \pi(\boldsymbol{A}) - v(\boldsymbol{A}) = n - \pi(\boldsymbol{B}) - v(\boldsymbol{B}) =$
lyche-numerical-linear-algebra-F04390	338	1.000	$\zeta(\boldsymbol{B})$	$\zeta(\boldsymbol{B})$	$\zeta(\boldsymbol{B})$. This completes the proof for diagonal matrices.
lyche-numerical-linear-algebra-F04391	338	1.000	$\boldsymbol{U}_1^* \boldsymbol{A} \boldsymbol{U}_1 = \boldsymbol{D}_1$	$\boldsymbol{U}_1^* \boldsymbol{A} \boldsymbol{U}_1 = \boldsymbol{D}_1$	Let in the general case $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ be unitary matrices such that $\boldsymbol{U}_1^* \boldsymbol{A} \boldsymbol{U}_1 = \boldsymbol{D}_1$
lyche-numerical-linear-algebra-F04392	338	0.999	$\boldsymbol{U}_2^* \boldsymbol{B} \boldsymbol{U}_2 = \boldsymbol{D}_2$	$\boldsymbol{U}_2^* \boldsymbol{B} \boldsymbol{U}_2 = \boldsymbol{D}_2$	and $\boldsymbol{U}_2^* \boldsymbol{B} \boldsymbol{U}_2 = \boldsymbol{D}_2$ where $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ are diagonal matrices. Since $\boldsymbol{A} = \boldsymbol{E}^* \boldsymbol{B} \boldsymbol{E}$,
lyche-numerical-linear-algebra-F04393	338	0.999	$\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$	$\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$	we find $\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$ where $\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$ is nonsingular. Thus $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$
lyche-numerical-linear-algebra-F04394	338	0.999	$\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$	$\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$	we find $\boldsymbol{D}_1 = \boldsymbol{F}^* \boldsymbol{D}_2 \boldsymbol{F}$ where $\boldsymbol{F} = \boldsymbol{U}_2^* \boldsymbol{E} \boldsymbol{U}_1$ is nonsingular. Thus $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$
lyche-numerical-linear-algebra-F04395	338	1.000	$\boldsymbol{D}_1, \boldsymbol{B}$	$\boldsymbol{D}_1, \boldsymbol{B}$	are congruent diagonal matrices. But since $\boldsymbol{A}$ and $\boldsymbol{D}_1$, $\boldsymbol{B}$ and $\boldsymbol{D}_2$ have the same
lyche-numerical-linear-algebra-F04396	338	1.000	$\pi(\boldsymbol{A})=\pi(\boldsymbol{D}_1)=\pi(\boldsymbol{D}_2)=\pi(\boldsymbol{B})$	$\pi(\boldsymbol{A}) = \pi(\boldsymbol{D}_1) = \pi(\boldsymbol{D}_2) = \pi(\boldsymbol{B})$	eigenvalues, we find $\pi(\boldsymbol{A}) = \pi(\boldsymbol{D}_1) = \pi(\boldsymbol{D}_2) = \pi(\boldsymbol{B})$. Similar results hold for ζ
lyche-numerical-linear-algebra-F04397	338	1.000	ζ	ζ	eigenvalues, we find $\pi(\boldsymbol{A}) = \pi(\boldsymbol{D}_1) = \pi(\boldsymbol{D}_2) = \pi(\boldsymbol{B})$. Similar results hold for ζ
lyche-numerical-linear-algebra-F04398	338	1.000	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$	$\boldsymbol{A} = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$ is symmetric and that $\alpha \in \mathbb{R}$ is such that $\boldsymbol{A} - \alpha \boldsymbol{I}$

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lyche-numerical-linear-algebra- F04399	338	1.000	$\alpha \in \mathbb{R}$	$\alpha \in \mathbb{R}$	$A = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$ is symmetric and that $\alpha \in \mathbb{R}$ is such that $A - \alpha I$
lyche-numerical-linear-algebra- F04400	338	1.000	$\boldsymbol{A} - \alpha \boldsymbol{I}$	$A - \alpha I$	$A = \text{tridiag}(c_i, d_i, c_i) \in \mathbb{R}^{n \times n}$ is symmetric and that $\alpha \in \mathbb{R}$ is such that $A - \alpha I$
lyche-numerical-linear-algebra- F04401	338	0.999	$\boldsymbol{A} - \alpha \boldsymbol{I} = \boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$	$A - \alpha I = LDL^T$	has an symmetric LU factorization, i.e. $A - \alpha I = LDL^T$ where L is unit lower
lyche-numerical-linear-algebra- F04402	338	1.000	$d_1(\alpha), \dots, d_n(\alpha)$	$d_1(\alpha), \dots, d_n(\alpha)$	$d_1(\alpha), \dots, d_n(\alpha)$ in D can be computed recursively as follows
lyche-numerical-linear-algebra- F04403	338	1.000	$\boldsymbol{L} \boldsymbol{D} \boldsymbol{L}^T$	LDL^T	elements in D in an LDL^T factorization we see that the formulas in (14.8) follows
lyche-numerical-linear-algebra- F04404	338	1.000	$v(\boldsymbol{A} - \alpha \boldsymbol{I}) = v(\boldsymbol{D})$	$v(A - \alpha I) = v(D)$	By the previous theorem $v(A - \alpha I) = v(D)$, the number of negative diagonal
lyche-numerical-linear-algebra- F04405	338	0.988	$(\boldsymbol{A} - \alpha \boldsymbol{I}) \boldsymbol{x} = (\lambda - \alpha) \boldsymbol{x}$	$(A - \alpha I)x = (\lambda - \alpha)x$	elements in D . If $Ax = \lambda x$ then $(A - \alpha I)x = (\lambda - \alpha)x$, and $\lambda - \alpha$ is an eigenvalue
lyche-numerical-linear-algebra- F04406	338	0.988	$\lambda - \alpha$	$\lambda - \alpha$	elements in D . If $Ax = \lambda x$ then $(A - \alpha I)x = (\lambda - \alpha)x$, and $\lambda - \alpha$ is an eigenvalue
lyche-numerical-linear-algebra- F04407	338	1.000	$v(\boldsymbol{A} - \alpha \boldsymbol{I})$	$v(A - \alpha I)$	of $A - \alpha I$. But then $v(A - \alpha I)$ equals the number of eigenvalues of A which are
lyche-numerical-linear-algebra- F04408	338	1.000	$\lambda_1 \geq \lambda_2 \geq$	$\lambda_1 \geq \lambda_2 \geq$	Corollary 14.2 can be used to determine the m th eigenvalue of A , where $\lambda_1 \geq \lambda_2 \geq$
lyche-numerical-linear-algebra- F04409	338	0.999	$\dots \geq \lambda_n$	$\dots \geq \lambda_n$	$\dots \geq \lambda_n$. Using Gerschgorin's theorem we first find an interval $[a, b]$, such that
lyche-numerical-linear-algebra- F04410	338	0.999	(a, b)	(a, b)	(a, b) contains the eigenvalues of A . Let for $x \in [a, b]$
lyche-numerical-linear-algebra- F04411	339	1.000	$\rho(a) =$	$\rho(a) =$	be the number of eigenvalues of A which are strictly greater than x . Clearly $\rho(a) =$
lyche-numerical-linear-algebra- F04412	339	1.000	$n, \rho(b) = 0$	$n, \rho(b) = 0$	$n, \rho(b) = 0$. Choosing a tolerance ϵ and using bisection we proceed as follows:
lyche-numerical-linear-algebra- F04413	339	1.000	$\{[a_j, b_j]\}$	$\{[a_j, b_j]\}$	We generate a sequence $\{[a_j, b_j]\}$ of intervals, each containing λ_m and $b_j - a_j =$
lyche-numerical-linear-algebra- F04414	339	1.000	$b_j - a_j =$	$b_j - a_j =$	We generate a sequence $\{[a_j, b_j]\}$ of intervals, each containing λ_m and $b_j - a_j =$
lyche-numerical-linear-algebra- F04415	339	1.000	$2^{-j}(b - a)$	$2^{-j}(b - a)$	$2^{-j}(b - a)$.
lyche-numerical-linear-algebra- F04416	339	1.000	$d_k(\alpha)$	$d_k(\alpha)$	As it stands this method will fail if in (14.8) one of the $d_k(\alpha)$ is zero. One
lyche-numerical-linear-algebra- F04417	339	1.000	$\delta_k =$	$\delta_k =$	possibility is to replace such a $d_k(\alpha)$ by a suitable small number, say $\delta_k =$
lyche-numerical-linear-algebra- F04418	339	1.000	$c_k \epsilon_M$	$c_k \epsilon_M$	$c_k \epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra- F04419	339	1.000	ϵ_M	ϵ_M	$c_k \epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra- F04420	339	1.000	2×10^{-16}	2×10^{-16}	$c_k \epsilon_M$, where ϵ_M is the Machine epsilon, typically 2×10^{-16} for MATLAB. This
lyche-numerical-linear-algebra- F04421	339	1.000	$ d_k(\alpha) < \delta_k $	$ d_k(\alpha) < \delta_k $	replacement is done if $ d_k(\alpha) < \delta_k $.
lyche-numerical-linear-algebra- F04422	340	0.998	$\boldsymbol{A} (a_{i,i} >$	$A (a_{i,i} >$	Gershgorin's circle theorem that a strictly diagonally dominant matrix $A (a_{i,i} >$
lyche-numerical-linear-algebra- F04423	340	1.000	$\sum_{j \neq i} a_{i,j} $	$\sum_{j \neq i} a_{i,j} $	$\sum_{j \neq i} a_{i,j} $ for all i) is nonsingular.
lyche-numerical-linear-algebra- F04424	340	1.000	$R := R_1 \cup \dots \cup R_n$	$R := R_1 \cup \dots \cup R_n$	$A \in \mathbb{R}^{n,n}$ lie inside $R \cap C$, where $R := R_1 \cup \dots \cup R_n$ is the union of the row disks
lyche-numerical-linear-algebra- F04425	340	1.000	$C = C_1 \cup \dots \cup C_n$	$C = C_1 \cup \dots \cup C_n$	R_i of A , and $C := C_1 \cup \dots \cup C_n$ is the union of the column disks C_j . You do not

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lyche-numerical-linear-algebra-F04426	340	■ 0.987	$\backslash\boldsymbol{s}=\backslash\left[s_{\{1\}}, \backslash\ldots, s_{\{n\}}\backslash\right] \backslash\in \backslash\mathbb{R}^n$	$\boldsymbol{s} = [s_1, \dots, s_n] \in \mathbb{R}^n$	computes the centres $\boldsymbol{s} = [s_1, \dots, s_n] \in \mathbb{R}^n$ of the row and column disks, and their
lyche-numerical-linear-algebra-F04427	340	■ 1.000	$\backslash\boldsymbol{r}=\backslash\left[r_{\{1\}}, \backslash\ldots, r_{\{n\}}\backslash\right] \backslash\in \backslash\mathbb{R}^n$	$\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$	radii $\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$ and $\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$, respectively.
lyche-numerical-linear-algebra-F04428	340	■ 1.000	$\backslash\boldsymbol{c}=\backslash\left[c_{\{1\}}, \backslash\ldots, c_{\{n\}}\backslash\right] \backslash\in \backslash\mathbb{R}^n$	$\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$	radii $\boldsymbol{r} = [r_1, \dots, r_n] \in \mathbb{R}^n$ and $\boldsymbol{c} = [c_1, \dots, c_n] \in \mathbb{R}^n$, respectively.
lyche-numerical-linear-algebra-F04429	340	■ 1.000	$\backslash\boldsymbol{A} \backslash\left(t_{\{2\}}\backslash\right)$	$\boldsymbol{A}(t_2)$	$0 \leq t_1 < t_2 \leq 1$ and that μ is an eigenvalue of $\boldsymbol{A}(t_2)$. Show, using Theorem 14.2
lyche-numerical-linear-algebra-F04430	340	■ 1.000	$\backslash\boldsymbol{A}=\backslash\boldsymbol{A} \backslash\left(t_{\{1\}}\backslash\right)$	$\boldsymbol{A} = \boldsymbol{A}(t_1)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra-F04431	340	■ 1.000	$\backslash\boldsymbol{E}=\backslash\boldsymbol{A} \backslash\left(t_{\{2\}}\backslash\right)-\backslash\boldsymbol{A} \backslash\left(t_{\{1\}}\backslash\right)$	$\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra-F04432	340	■ 1.000	$\backslash\boldsymbol{A} \backslash\left(t_{\{1\}}\backslash\right)$	$\boldsymbol{A}(t_1)$	with $\boldsymbol{A} = \boldsymbol{A}(t_1)$ and $\boldsymbol{E} = \boldsymbol{A}(t_2) - \boldsymbol{A}(t_1)$, that $\boldsymbol{A}(t_1)$ has an eigenvalue λ such that
lyche-numerical-linear-algebra-F04433	340	■ 1.000	$\backslash\backslash\boldsymbol{A} \backslash\backslash_{\{ \infty \}}=$	$\ \boldsymbol{A}\ _\infty =$	Exercise 14.6 (∞-Norm of a Diagonal Matrix) Give a direct proof that $\ \boldsymbol{A}\ _\infty =$
lyche-numerical-linear-algebra-F04434	340	■ 1.000	$\backslash\left[a_{\{k\ j\}}\backslash\right], \backslash\boldsymbol{E}=\backslash\left[e_{\{k\ j\}}\backslash\right]$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra-F04435	340	■ 1.000	$\backslash\boldsymbol{B}=\backslash\left[b_{\{k\ j\}}\backslash\right]$	$\boldsymbol{B} = [b_{kj}]$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra-F04436	340	■ 1.000	$\backslash\mathbb{R}^{\{n, n\}}$	$\mathbb{R}^{n,n}$	$[a_{kj}], \boldsymbol{E} = [e_{kj}]$, and $\boldsymbol{B} = [b_{kj}]$ be matrices in $\mathbb{R}^{n,n}$ with
lyche-numerical-linear-algebra-F04437	341	■ 1.000	$\backslash\boldsymbol{B}=\backslash\boldsymbol{A}+\backslash\boldsymbol{E}$	$\boldsymbol{B} = \boldsymbol{A} + \boldsymbol{E}$	and $\boldsymbol{B} = \boldsymbol{A} + \boldsymbol{E}$, where $0 < \epsilon < 1$. Thus for $n = 4$,
lyche-numerical-linear-algebra-F04438	341	■ 1.000	$\backslash\backslash\boldsymbol{A} \backslash\backslash_{\{ 2 \}}=\backslash\backslash\boldsymbol{B} \backslash\backslash_{\{ 2 \}}=1$	$\ \boldsymbol{A}\ _2 = \ \boldsymbol{B}\ _2 = 1$	b) Show that $\ \boldsymbol{A}\ _2 = \ \boldsymbol{B}\ _2 =$ for arbitrary $n \in \mathbb{N}$.
lyche-numerical-linear-algebra-F04439	341	■ 1.000	$\backslash\boldsymbol{A}, \backslash\boldsymbol{E}, \backslash\boldsymbol{B}$	$\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$	c) Recall Elsner's Theorem (Theorem 14.2). Let $\boldsymbol{A}, \boldsymbol{E}, \boldsymbol{B}$ be given by (14.10).
lyche-numerical-linear-algebra-F04440	341	■ 1.000	$\backslash\mu=\backslash\epsilon^{\{1 / n\}}$	$\mu = \epsilon^{1/n}$	$\mu = \epsilon^{1/n}$ of $\boldsymbol{B}$? How sharp is this upper bound?
lyche-numerical-linear-algebra-F04441	341	■ 1.000	$\backslash\frac{\{10\}\{3\}}{n^{\{3\}}}=\backslash5 \ G_{\{n\}}$	$\frac{10}{3}n^3 = 5G_n$	Show that the number of arithmetic operations for Algorithm 14.1 is $\frac{10}{3}n^3 = 5G_n$.
lyche-numerical-linear-algebra-F04442	341	■ 1.000	$\backslash\frac{\{4\}\{3\}}{n^{\{3\}}}=\backslash2 \ G_{\{n\}}$	$\frac{4}{3}n^3 = 2G_n$	of arithmetic operations required by Algorithm 14.2 is $\frac{4}{3}n^3 = 2G_n$.
lyche-numerical-linear-algebra-F04443	341	■ 1.000	$\backslash\boldsymbol{V}_{\{k\}} \backslash\boldsymbol{E}_{\{k\}} \backslash\boldsymbol{V}_{\{k\}}$	$\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$	compute $\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$ where $\boldsymbol{E}_k$ is symmetric. Dropping subscripts we have to compute
lyche-numerical-linear-algebra-F04444	341	■ 1.000	$\backslash\boldsymbol{E}_{\{k\}}$	$\boldsymbol{E}_k$	compute $\boldsymbol{V}_k \boldsymbol{E}_k \boldsymbol{V}_k$ where $\boldsymbol{E}_k$ is symmetric. Dropping subscripts we have to compute
lyche-numerical-linear-algebra-F04445	341	■ 1.000	$\backslash\boldsymbol{G}=\backslash\left(\backslash\boldsymbol{I}-\backslash\boldsymbol{v} \backslash\boldsymbol{v}^{\{T\}}\backslash\right) \backslash\boldsymbol{E} \backslash\left(\backslash\boldsymbol{v} \backslash\boldsymbol{v}^{\{T\}}\backslash\right) \backslash\boldsymbol{E} \backslash\left(\backslash\boldsymbol{I}-\backslash\boldsymbol{v} \backslash\boldsymbol{v}^{\{T\}}\backslash\right)$	$\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{E} (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T)$	a product of the form $\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{E} (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T)$. Let $\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}$, $\beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$ and
lyche-numerical-linear-algebra-F04446	341	■ 1.000	$\backslash\boldsymbol{w}:=\backslash\boldsymbol{E} \backslash\boldsymbol{v}, \backslash\beta:=\backslash\frac{\{1\}\{2\}}{\backslash\boldsymbol{v}^{\{T\}} \backslash\boldsymbol{w}}$	$\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}, \beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$	a product of the form $\boldsymbol{G} = (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T) \boldsymbol{E} (\boldsymbol{I} - \boldsymbol{v} \boldsymbol{v}^T)$. Let $\boldsymbol{w} := \boldsymbol{E} \boldsymbol{v}$, $\beta := \frac{1}{2} \boldsymbol{v}^T \boldsymbol{w}$ and
lyche-numerical-linear-algebra-F04447	341	■ 0.999	$\backslash\boldsymbol{z}:=\backslash\boldsymbol{w}-\backslash\beta \boldsymbol{v}$	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$. Show that $\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$. Since $\boldsymbol{G}$ is symmetric, only the sub- or
lyche-numerical-linear-algebra-F04448	341	■ 0.999	$\backslash\boldsymbol{G}=\backslash\boldsymbol{E}-\backslash\boldsymbol{v} \boldsymbol{z}^{\{T\}} \backslash-\backslash\boldsymbol{z} \boldsymbol{v}^{\{T\}}$	$\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$	$\boldsymbol{z} := \boldsymbol{w} - \beta \boldsymbol{v}$. Show that $\boldsymbol{G} = \boldsymbol{E} - \boldsymbol{v} \boldsymbol{z}^T - \boldsymbol{z} \boldsymbol{v}^T$. Since $\boldsymbol{G}$ is symmetric, only the sub- or
lyche-numerical-linear-algebra-F04449	341	■ 1.000	$\backslash\left(4 \ n^{\{3\}} / 3\backslash\right)$	$O(4n^3/3)$	be shown that we need $O(4n^3/3)$ operations to tridiagonalize a symmetric matrix
lyche-numerical-linear-algebra-F04450	342	■ 0.999	$\backslash d_{\{k\}}$	d_k	a) Let d_k be the diagonal elements of $\boldsymbol{D}$ in an LDL* factorization of $\boldsymbol{A}_n$. Show that
lyche-numerical-linear-algebra-F04451	342	■ 0.999	$\backslash\boldsymbol{A}_{\{n\}}$	$\boldsymbol{A}_n$	a) Let d_k be the diagonal elements of $\boldsymbol{D}$ in an LDL* factorization of $\boldsymbol{A}_n$. Show that
lyche-numerical-linear-algebra-F04452	342	■ 1.000	$\backslash 5+\backslash\sqrt{\{24\}}<\backslash d_{\{k\}} \backslash\leq 10, k=1,2, \backslash\ldots, n$	$5 + \sqrt{24} < d_k \leq 10, k = 1, 2, \dots, n$	$5 + \sqrt{24} < d_k < 10, k = 1, 2, \dots, n$.

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lyche-numerical-linear-algebra- F04453	342	■ 1.000	$D_{\{n\}} := \operatorname{det}(\left(\boldsymbol{A}_{\{n\}} \right) > (5 + \sqrt{24})^n$	$D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$	b) Show that $D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$. Give $n_0 \in \mathbb{N}$ such that your computer
lyche-numerical-linear-algebra- F04454	342	■ 1.000	$n_{\{0\}} \in \mathbb{N}$	$n_0 \in \mathbb{N}$	b) Show that $D_n := \det(\boldsymbol{A}_n) > (5 + \sqrt{24})^n$. Give $n_0 \in \mathbb{N}$ such that your computer
lyche-numerical-linear-algebra- F04455	342	■ 1.000	$D_{\{n_{\{0\}}\}}$	D_{n_0}	gives an overflow when D_{n_0} is computed in floating point arithmetic.
lyche-numerical-linear-algebra- F04456	342	■ 0.935	$\boldsymbol{B} = \boldsymbol{U}^T \boldsymbol{D} \boldsymbol{U}$	$\boldsymbol{B} = \boldsymbol{U}^T \boldsymbol{D} \boldsymbol{U}$	where $\boldsymbol{A}^T = \boldsymbol{A}$ and $\boldsymbol{B}$ is symmetric positive definite. Then $\boldsymbol{B} = \boldsymbol{U}^T \boldsymbol{D} \boldsymbol{U}$ where
lyche-numerical-linear-algebra- F04457	342	■ 1.000	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^T \boldsymbol{D}^{-1/2}$	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^T \boldsymbol{D}^{-1/2}$	$\hat{\boldsymbol{A}} = \boldsymbol{D}^{-1/2} \boldsymbol{U} \boldsymbol{A} \boldsymbol{U}^T \boldsymbol{D}^{-1/2}$ where
lyche-numerical-linear-algebra- F04458	342	■ 1.000	$\hat{\boldsymbol{A}}$	$\hat{\boldsymbol{A}}$	a) Show that $\hat{\boldsymbol{A}}$ is symmetric.
lyche-numerical-linear-algebra- F04459	342	■ 1.000	$\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$	$\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$	Let $\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$ where $\hat{\boldsymbol{U}}$ is orthonormal and $\hat{\boldsymbol{D}}$ is diagonal. Set $\boldsymbol{E} =$
lyche-numerical-linear-algebra- F04460	342	■ 1.000	$\hat{\boldsymbol{D}}$	$\hat{\boldsymbol{D}}$	Let $\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$ where $\hat{\boldsymbol{U}}$ is orthonormal and $\hat{\boldsymbol{D}}$ is diagonal. Set $\boldsymbol{E} =$
lyche-numerical-linear-algebra- F04461	342	■ 1.000	$\boldsymbol{E} =$	$\boldsymbol{E} =$	Let $\hat{\boldsymbol{A}} = \hat{\boldsymbol{U}}^T \hat{\boldsymbol{D}} \hat{\boldsymbol{U}}$ where $\hat{\boldsymbol{U}}$ is orthonormal and $\hat{\boldsymbol{D}}$ is diagonal. Set $\boldsymbol{E} =$
lyche-numerical-linear-algebra- F04462	342	■ 1.000	$\boldsymbol{U}^T \boldsymbol{D}^{-1/2} \hat{\boldsymbol{U}}^T$	$\boldsymbol{U}^T \boldsymbol{D}^{-1/2} \hat{\boldsymbol{U}}^T$	$\boldsymbol{U}^T \boldsymbol{D}^{-1/2} \hat{\boldsymbol{U}}^T$.
lyche-numerical-linear-algebra- F04463	342	■ 0.999	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \hat{\boldsymbol{D}}, \boldsymbol{E}^T \boldsymbol{B} \boldsymbol{E} = \boldsymbol{I}$	$\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \hat{\boldsymbol{D}}, \boldsymbol{E}^T \boldsymbol{B} \boldsymbol{E} = \boldsymbol{I}$	b) Show that $\boldsymbol{E}$ is nonsingular and that $\boldsymbol{E}^T \boldsymbol{A} \boldsymbol{E} = \hat{\boldsymbol{D}}, \boldsymbol{E}^T \boldsymbol{B} \boldsymbol{E} = \boldsymbol{I}$.
lyche-numerical-linear-algebra- F04464	342	■ 1.000	$\boldsymbol{A} = \operatorname{tridiag}(\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{c})$	$\boldsymbol{A} = \operatorname{tridiag}(\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{c})$	Exercise 14.14 (Program Code for One Eigenvalue) Suppose $\boldsymbol{A} = \operatorname{tridiag}(\boldsymbol{c}, \boldsymbol{d}, \boldsymbol{c})$
lyche-numerical-linear-algebra- F04465	342	■ 1.000	$\boldsymbol{d}_{\{1\}}, \dots, \boldsymbol{d}_{\{n\}}$	$\boldsymbol{d}_1, \dots, \boldsymbol{d}_n$	is symmetric and tridiagonal with elements $\boldsymbol{d}_1, \dots, \boldsymbol{d}_n$ on the diagonal and
lyche-numerical-linear-algebra- F04466	342	■ 1.000	$\boldsymbol{c}_{\{1\}}, \dots, \boldsymbol{c}_{\{n-1\}}$	$\boldsymbol{c}_1, \dots, \boldsymbol{c}_{n-1}$	$\boldsymbol{c}_1, \dots, \boldsymbol{c}_{n-1}$ on the neighboring subdiagonals. Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be the
lyche-numerical-linear-algebra- F04467	342	■ 0.998	$\mathrm{k} = \operatorname{counting}(\mathrm{c}, \mathrm{d}, \mathrm{x})$	$\mathrm{k} = \operatorname{counting}(\mathrm{c}, \mathrm{d}, \mathrm{x})$	a) Write a function $\mathrm{k} = \operatorname{counting}(\mathrm{c}, \mathrm{d}, \mathrm{x})$ which for given x counts the
lyche-numerical-linear-algebra- F04468	342	■ 1.000	$\boldsymbol{d}_{\{j\}}(x)$	$\boldsymbol{d}_j(x)$	above if one of the $\boldsymbol{d}_j(x)$ is close to zero.
lyche-numerical-linear-algebra- F04469	342	■ 0.907	$(\boldsymbol{a}, \boldsymbol{b}]$	$(\boldsymbol{a}, \boldsymbol{b}]$	interval $(\boldsymbol{a}, \boldsymbol{b}]$ containing all eigenvalues of $\boldsymbol{A}$ and then generates a sequence
lyche-numerical-linear-algebra- F04470	342	■ 0.901	$\left\{ \left(\boldsymbol{a}_{\{j\}}, \boldsymbol{b}_{\{j\}} \right) \right\}$	$\{(\boldsymbol{a}_j, \boldsymbol{b}_j)\}$	$\{(\boldsymbol{a}_j, \boldsymbol{b}_j)\}$ of intervals each containing λ_m . Iterate until $\boldsymbol{b}_j - \boldsymbol{a}_j \leq (\boldsymbol{b} - \boldsymbol{a})\epsilon_M$,
lyche-numerical-linear-algebra- F04471	342	■ 0.901	$\boldsymbol{b}_{\{j\}} - \boldsymbol{a}_{\{j\}} \leq (\boldsymbol{b} - \boldsymbol{a}) \epsilon_M$	$\boldsymbol{b}_j - \boldsymbol{a}_j \leq (\boldsymbol{b} - \boldsymbol{a})\epsilon_M$	$\{(\boldsymbol{a}_j, \boldsymbol{b}_j)\}$ of intervals each containing λ_m . Iterate until $\boldsymbol{b}_j - \boldsymbol{a}_j \leq (\boldsymbol{b} - \boldsymbol{a})\epsilon_M$,
lyche-numerical-linear-algebra- F04472	342	■ 1.000	$\epsilon_M \approx 2.22 \times 10^{-16}$	$\epsilon_M \approx 2.22 \times 10^{-16}$	where ϵ_M is MATLAB's machine epsilon eps. Typically $\epsilon_M \approx 2.22 \times 10^{-16}$.
lyche-numerical-linear-algebra- F04473	342	■ 0.795	$\boldsymbol{T} := \operatorname{tridiag}(-1, 2, -1)$	$\boldsymbol{T} := \operatorname{tridiag}(-1, 2, -1)$	c) Test the program on $\boldsymbol{T} := \operatorname{tridiag}(-1, 2, -1)$ of size 100. Compare the exact
lyche-numerical-linear-algebra- F04474	342	■ 1.000	λ_5	λ_5	value of λ_5 with your result and the result obtained by using MATLAB's built-in
lyche-numerical-linear-algebra- F04475	343	■ 1.000	$x \in \mathbb{C}$	$x \in \mathbb{C}$	is upper Hessenberg and $x \in \mathbb{C}$. We will study two algorithms to compute $f(x) =$
lyche-numerical-linear-algebra- F04476	343	■ 1.000	$f(x) =$	$f(x) =$	is upper Hessenberg and $x \in \mathbb{C}$. We will study two algorithms to compute $f(x) =$
lyche-numerical-linear-algebra- F04477	343	■ 1.000	$\operatorname{det}(\boldsymbol{A} - x \boldsymbol{I})$	$\det(\boldsymbol{A} - x \boldsymbol{I})$	$\det(\boldsymbol{A} - x \boldsymbol{I})$.
lyche-numerical-linear-algebra- F04478	343	■ 1.000	$\mu \in \sigma(\boldsymbol{A} + \boldsymbol{E})$	$\mu \in \sigma(\boldsymbol{A} + \boldsymbol{E})$	14.7.1 Suppose $\boldsymbol{A}, \boldsymbol{E} \in \mathbb{C}^{n \times n}$. To every $\mu \in \sigma(\boldsymbol{A} + \boldsymbol{E})$ there is a $\lambda \in \sigma(\boldsymbol{A})$ which

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lyche-numerical-linear-algebra- F04479	345	0.604	$2 z_k / 3^k$	$2z_k/3^k$	It follows that $2z_k/3^k$ converges to $[1, -1]$, an eigenvector corresponding to the
lyche-numerical-linear-algebra- F04480	345	0.604	$[1, -1]$	$[1, -1]$	It follows that $2z_k/3^k$ converges to $[1, -1]$, an eigenvector corresponding to the
lyche-numerical-linear-algebra- F04481	345	0.954	$\lambda=3$	$\lambda = 3$	dominant eigenvalue $\lambda = 3$. The sequence of Rayleigh quotients $\{z_k^T A z_k / z_k^T z_k\}$
lyche-numerical-linear-algebra- F04482	345	0.954	$\left\{ z_k^T A z_k / z_k^T z_k \right\}$	$\{z_k^T A z_k / z_k^T z_k\}$	dominant eigenvalue $\lambda = 3$. The sequence of Rayleigh quotients $\{z_k^T A z_k / z_k^T z_k\}$
lyche-numerical-linear-algebra- F04483	345	0.628	z_0	z_0	To understand better what happens we expand z_0 in terms of the eigenvectors
lyche-numerical-linear-algebra- F04484	345	1.000	$(\lambda_j^k, v_j), j=1,2$	$(\lambda_j^k, v_j), j = 1, 2$	Since A^k has eigenpairs $(\lambda_j^k, v_j), j = 1, 2$ we find
lyche-numerical-linear-algebra- F04485	345	0.749	$3^{-k} z_k = c_1 v_1 + 3^{-k} c_2 v_2 \rightarrow c_1 v_1$	$3^{-k} z_k = c_1 v_1 + 3^{-k} c_2 v_2 \rightarrow c_1 v_1$	Thus $3^{-k} z_k = c_1 v_1 + 3^{-k} c_2 v_2 \rightarrow c_1 v_1$. Since $c_1 \neq 0$ we obtain convergence to the
lyche-numerical-linear-algebra- F04486	345	0.749	$c_1 \neq 0$	$c_1 \neq 0$	Thus $3^{-k} z_k = c_1 v_1 + 3^{-k} c_2 v_2 \rightarrow c_1 v_1$. Since $c_1 \neq 0$ we obtain convergence to the
lyche-numerical-linear-algebra- F04487	345	1.000	$(\lambda_j, v_j), j=1, \dots, n$	$(\lambda_j, v_j), j = 1, \dots, n$	Let $A \in \mathbb{C}^{n \times n}$ have eigenpairs $(\lambda_j, v_j), j = 1, \dots, n$ with $ \lambda_1 > \lambda_2 \geq \dots \geq$
lyche-numerical-linear-algebra- F04488	345	1.000	$ \lambda_1 > \lambda_2 \geq \dots \geq$	$ \lambda_1 > \lambda_2 \geq \dots \geq$	Let $A \in \mathbb{C}^{n \times n}$ have eigenpairs $(\lambda_j, v_j), j = 1, \dots, n$ with $ \lambda_1 > \lambda_2 \geq \dots \geq$
lyche-numerical-linear-algebra- F04489	345	1.000	$ \lambda_n $	$ \lambda_n $	$ \lambda_n $.
lyche-numerical-linear-algebra- F04490	345	0.905	$z_0 \in \mathbb{C}^n$	$z_0 \in \mathbb{C}^n$	Given $z_0 \in \mathbb{C}^n$ we assume that
lyche-numerical-linear-algebra- F04491	345	0.999	z_0	z_0	multiplicity one. The second assumption says that z_0 has a component in the
lyche-numerical-linear-algebra- F04492	345	0.522	$z_0 = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$	$z_0 = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$	To see what happens let $z_0 = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$ where by assumption
lyche-numerical-linear-algebra- F04493	345	1.000	$A^k v_j = \lambda_j^k v_j$	$A^k v_j = \lambda_j^k v_j$	(ii) of (15.2) we have $c_1 \neq 0$. Since $A^k v_j = \lambda_j^k v_j$ for all j we see that
lyche-numerical-linear-algebra- F04494	345	1.000	λ_1^k	λ_1^k	Dividing by λ_1^k we find
lyche-numerical-linear-algebra- F04495	346	0.772	$(\lambda_j / \lambda_1)^k \rightarrow 0$	$(\lambda_j / \lambda_1)^k \rightarrow 0$	Assumption (i) of (15.2) implies that $(\lambda_j / \lambda_1)^k \rightarrow 0$ as $k \rightarrow \infty$ for all $j \geq 2$ and
lyche-numerical-linear-algebra- F04496	346	0.534	(i)	(i)	matrices as long as (i) and (ii) of (15.2) hold, see for example page 58 of [18].
lyche-numerical-linear-algebra- F04497	346	0.989	z_k	z_k	In practice we need to scale the iterates z_k somehow and we normally do not
lyche-numerical-linear-algebra- F04498	346	0.766	$x_0 = z_0 / \ z_0\ $	$x_0 = z_0 / \ z_0\ $	know λ_1 . Instead we choose a norm on $\mathbb{C}^n$, set $x_0 = z_0 / \ z_0\ $ and generate for
lyche-numerical-linear-algebra- F04499	346	1.000	$k=1,2, \dots$	$k = 1, 2, \dots$	$k = 1, 2, \dots$ unit vectors as follows:
lyche-numerical-linear-algebra- F04500	346	1.000	$\lambda_1 > 0$	$\lambda_1 > 0$	In particular, if $\lambda_1 > 0$ and $c_1 > 0$ then the sequence $\{x_k\}$ will converge to the
lyche-numerical-linear-algebra- F04501	346	1.000	$c_1 > 0$	$c_1 > 0$	In particular, if $\lambda_1 > 0$ and $c_1 > 0$ then the sequence $\{x_k\}$ will converge to the
lyche-numerical-linear-algebra- F04502	346	1.000	$u_1 := v_1 / \ v_1\ $	$u_1 := v_1 / \ v_1\ $	eigenvector $u_1 := v_1 / \ v_1\ $ of unit length.
lyche-numerical-linear-algebra- F04503	346	0.980	$x_k = z_k / \ z_k\ $	$x_k = z_k / \ z_k\ $	Proof By induction on k it follows that $x_k = z_k / \ z_k\ $ for all $k \geq 0$, where $z_k =$
lyche-numerical-linear-algebra- F04504	346	0.980	$z_k =$	$z_k =$	Proof By induction on k it follows that $x_k = z_k / \ z_k\ $ for all $k \geq 0$, where $z_k =$
lyche-numerical-linear-algebra- F04505	346	0.999	$A^k z_0$	$A^k z_0$	$A^k z_0$. Indeed, this holds for $k = 1$, and if it holds for $k - 1$ then $y_k = A x_{k-1} =$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra-F04506	346	0.999	$\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1}$	$\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1}$	$\boldsymbol{A}^k \boldsymbol{z}_0$. Indeed, this holds for $k = 1$, and if it holds for $k - 1$ then $\boldsymbol{y}_k = \boldsymbol{A} \boldsymbol{x}_{k-1} =$
lyche-numerical-linear-algebra-F04507	346	1.000	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $ and $\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\)(\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $,
lyche-numerical-linear-algebra-F04508	346	1.000	$\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\)(\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $	$\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\)(\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $	$\boldsymbol{A} \boldsymbol{z}_{k-1} / \ \boldsymbol{z}_{k-1}\ = \boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\ $ and $\boldsymbol{x}_k = (\boldsymbol{z}_k / \ \boldsymbol{z}_{k-1}\)(\ \boldsymbol{z}_{k-1}\ / \ \boldsymbol{z}_k\) = \boldsymbol{z}_k / \ \boldsymbol{z}_k\ $.
lyche-numerical-linear-algebra-F04509	346	1.000	$r(\lambda) := \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}$	$r(\lambda) := \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}$	residual $r(\lambda) := \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}$.
lyche-numerical-linear-algebra-F04510	346	1.000	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \rightarrow \mathbb{R}$ be given by $\rho(\lambda) = \ \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}\ _2$. Then ρ is
lyche-numerical-linear-algebra-F04511	346	1.000	$\rho : \mathbb{C} \rightarrow \mathbb{R}$	$\rho : \mathbb{C} \rightarrow \mathbb{R}$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \rightarrow \mathbb{R}$ be given by $\rho(\lambda) = \ \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}\ _2$. Then ρ is
lyche-numerical-linear-algebra-F04512	346	1.000	$\rho(\lambda) = \ \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}\ _2$	$\rho(\lambda) = \ \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}\ _2$	$\boldsymbol{u} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, and let $\rho : \mathbb{C} \rightarrow \mathbb{R}$ be given by $\rho(\lambda) = \ \boldsymbol{A} \boldsymbol{u} - \lambda \boldsymbol{u}\ _2$. Then ρ is
lyche-numerical-linear-algebra-F04513	346	0.906	$\lambda := \frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$	$\lambda := \frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$	minimized when $\lambda := \frac{\boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}}{\boldsymbol{u}^* \boldsymbol{u}}$, the Rayleigh quotient for $\boldsymbol{A}$.
lyche-numerical-linear-algebra-F04514	347	1.000	$\{\boldsymbol{u}, \boldsymbol{U}\}$	$\{\boldsymbol{u}, \boldsymbol{U}\}$	Proof Assume $\boldsymbol{u}^* \boldsymbol{u} = 1$ and extend $\boldsymbol{u}$ to an orthonormal basis $\{\boldsymbol{u}, \boldsymbol{U}\}$ for $\mathbb{C}^n$. Then
lyche-numerical-linear-algebra-F04515	347	1.000	$\boldsymbol{U}^* \boldsymbol{u} = \mathbf{0}$	$\boldsymbol{U}^* \boldsymbol{u} = \mathbf{0}$	$\boldsymbol{U}^* \boldsymbol{u} = \mathbf{0}$ and
lyche-numerical-linear-algebra-F04516	347	1.000	$\lambda = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$	$\lambda = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$	and ρ has a global minimum at $\lambda = \boldsymbol{u}^* \boldsymbol{A} \boldsymbol{u}$.
lyche-numerical-linear-algebra-F04517	347	0.909	$(\mu, \boldsymbol{u})$	$(\mu, \boldsymbol{u})$	$(\mu, \boldsymbol{u})$ is an approximate eigenpair with $\ \boldsymbol{u}\ _2 = 1$, then by (14.4) we can find an
lyche-numerical-linear-algebra-F04518	347	0.909	$\ \boldsymbol{u}\ _2 = 1$	$\ \boldsymbol{u}\ _2 = 1$	$(\mu, \boldsymbol{u})$ is an approximate eigenpair with $\ \boldsymbol{u}\ _2 = 1$, then by (14.4) we can find an
lyche-numerical-linear-algebra-F04519	347	0.860	$l, \boldsymbol{x}$	$l, \boldsymbol{x}$	estimate for the eigenvalue is used to compute a dominant eigenpair $(l, \boldsymbol{x})$ of $\boldsymbol{A}$
lyche-numerical-linear-algebra-F04520	347	0.767	$\ \boldsymbol{A} \boldsymbol{x} - l \boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$	$\ \boldsymbol{A} \boldsymbol{x} - l \boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$	$\ \boldsymbol{A} \boldsymbol{x} - l \boldsymbol{x}\ _2 / \ \boldsymbol{A}\ _F < tol$. If no such eigenpair is found in K iterations the value
lyche-numerical-linear-algebra-F04521	347	0.877	$= K + 1$	$= K + 1$	it $\equiv K + 1$ is returned.
lyche-numerical-linear-algebra-F04522	348	0.900	$\boldsymbol{z} = [0.6602, 0.3420]$	$\boldsymbol{z} = [0.6602, 0.3420]$	In each case we start with the random vector $\boldsymbol{z} = [0.6602, 0.3420]$ and $tol = 10^{-6}$.
lyche-numerical-linear-algebra-F04523	348	0.900	$= 10^{-6}$	$= 10^{-6}$	In each case we start with the random vector $\boldsymbol{z} = [0.6602, 0.3420]$ and $tol = 10^{-6}$.
lyche-numerical-linear-algebra-F04524	348	1.000	$\frac{ \lambda_2 }{ \lambda_1 }$	$\frac{ \lambda_2 }{ \lambda_1 }$	convergence depends on the ratio $\frac{ \lambda_2 }{ \lambda_1 }$. If this ratio is small then the convergence
lyche-numerical-linear-algebra-F04525	348	1.000	$\lambda_1 = 5.3723$	$\lambda_1 = 5.3723$	$\boldsymbol{A}_1$ are $\lambda_1 = 5.3723$ and $\lambda_2 = -0.3723$ giving a quite small ratio of 0.07 and
lyche-numerical-linear-algebra-F04526	348	1.000	$\lambda_2 = -0.3723$	$\lambda_2 = -0.3723$	$\boldsymbol{A}_1$ are $\lambda_1 = 5.3723$ and $\lambda_2 = -0.3723$ giving a quite small ratio of 0.07 and
lyche-numerical-linear-algebra-F04527	348	1.000	$\lambda_2 = 1.9$	$\lambda_2 = 1.9$	$\lambda_2 = 1.9$ and the corresponding ratio is 0.95 resulting in slow convergence.
lyche-numerical-linear-algebra-F04528	348	1.000	$\boldsymbol{A} - s \boldsymbol{I}$	$\boldsymbol{A} - s \boldsymbol{I}$	choose a number s and apply the power method to the matrix $\boldsymbol{A} - s \boldsymbol{I}$. The number
lyche-numerical-linear-algebra-F04529	348	1.000	$\lambda - s$	$\lambda - s$	s is called a shift since it shifts an eigenvalue λ of $\boldsymbol{A}$ to $\lambda - s$ of $\boldsymbol{A} - s \boldsymbol{I}$. Sometimes
lyche-numerical-linear-algebra-F04530	348	1.000	$(\boldsymbol{A} - s \boldsymbol{I})^{-1}$	$(\boldsymbol{A} - s \boldsymbol{I})^{-1}$	method to the inverse matrix $(\boldsymbol{A} - s \boldsymbol{I})^{-1}$, where s is a shift. If $\boldsymbol{A}$ has eigenvalues

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lyche-numerical-linear-algebra-F04531	348	1.000	$\lim_{s \rightarrow \lambda_1} \mu_1(s) = \infty$	$\lim_{s \rightarrow \lambda_1} \mu_1(s) = \infty$	Suppose λ_1 is a simple eigenvalue of A . Then $\lim_{s \rightarrow \lambda_1} \mu_1(s) = \infty$, while																		
lyche-numerical-linear-algebra-F04532	348	1.000	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$ for $j = 2, \dots, n$. Hence, by choosing s																		
lyche-numerical-linear-algebra-F04533	348	1.000	$j=2, \dots, n$	$j = 2, \dots, n$	$\lim_{s \rightarrow \lambda_1} \mu_j(s) = (\lambda_j - \lambda_1)^{-1} < \infty$ for $j = 2, \dots, n$. Hence, by choosing s																		
lyche-numerical-linear-algebra-F04534	349	0.987	s_{k-1}	s_{k-1}	previous Rayleigh quotient s_{k-1} as the current shift. In each iteration we need to																		
lyche-numerical-linear-algebra-F04535	349	1.000	Ax_k	Ax_k	We can avoid the calculation of Ax_k in (iii) and (iv). Let																		
lyche-numerical-linear-algebra-F04536	350	0.913	ii	ii	this error disappears when we normalize y_k in (ii). Miraculously, the normalized																		
lyche-numerical-linear-algebra-F04537	350	0.995	(s, x)	(s, x)	Given an approximation (s, x) to an eigenpair (λ, v) of a matrix $A \in \mathbb{C}^{n \times n}$. The																		
lyche-numerical-linear-algebra-F04538	350	0.995	(λ, v)	(λ, v)	Given an approximation (s, x) to an eigenpair (λ, v) of a matrix $A \in \mathbb{C}^{n \times n}$. The																		
lyche-numerical-linear-algebra-F04539	350	1.000	nr	nr	one Rayleigh quotient iteration. The length nr of the new residual is also returned.																		
lyche-numerical-linear-algebra-F04540	350	1.000	$\lambda_1 = (5 - \sqrt{33})/2 \approx -0.37$	$\lambda_1 = (5 - \sqrt{33})/2 \approx -0.37$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is $\lambda_1 = (5 - \sqrt{33})/2 \approx -0.37$. Starting with $x = [1, 1]^T$ and $s =$																		
lyche-numerical-linear-algebra-F04541	350	1.000	$s =$	$s =$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is $\lambda_1 = (5 - \sqrt{33})/2 \approx -0.37$. Starting with $x = [1, 1]^T$ and $s =$																		
lyche-numerical-linear-algebra-F04542	350	0.798	$\ r\ _2$	$\ r\ _2$	<table> <tr><td>k</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr><td>$\ r\ _2$</td><td>1.0e+000</td><td>7.7e-002</td><td>1.6e-004</td><td>8.2e-010</td><td>2.0e-020</td></tr> <tr><td>$s - \lambda_1$</td><td>3.7e-001</td><td>-1.2e-002</td><td>-2.9e-005</td><td>-1.4e-010</td><td>-2.2e-016</td></tr> </table>	k	1	2	3	4	5	$\ r\ _2$	1.0e+000	7.7e-002	1.6e-004	8.2e-010	2.0e-020	$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016
k	1	2	3	4	5																		
$\ r\ _2$	1.0e+000	7.7e-002	1.6e-004	8.2e-010	2.0e-020																		
$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016																		
lyche-numerical-linear-algebra-F04543	350	0.798	1.0×10^0	$1.0e + 000$	<table> <tr><td>k</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr><td>$\ r\ _2$</td><td>1.0e+000</td><td>7.7e-002</td><td>1.6e-004</td><td>8.2e-010</td><td>2.0e-020</td></tr> <tr><td>$s - \lambda_1$</td><td>3.7e-001</td><td>-1.2e-002</td><td>-2.9e-005</td><td>-1.4e-010</td><td>-2.2e-016</td></tr> </table>	k	1	2	3	4	5	$\ r\ _2$	1.0e+000	7.7e-002	1.6e-004	8.2e-010	2.0e-020	$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016
k	1	2	3	4	5																		
$\ r\ _2$	1.0e+000	7.7e-002	1.6e-004	8.2e-010	2.0e-020																		
$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016																		
lyche-numerical-linear-algebra-F04544	350	0.798	$ s - \lambda_1 $	$ s - \lambda_1 $	<table> <tr><td>k</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr><td>$\ r\ _2$</td><td>1.0e+000</td><td>7.7e-002</td><td>1.6e-004</td><td>8.2e-010</td><td>2.0e-020</td></tr> <tr><td>$s - \lambda_1$</td><td>3.7e-001</td><td>-1.2e-002</td><td>-2.9e-005</td><td>-1.4e-010</td><td>-2.2e-016</td></tr> </table>	k	1	2	3	4	5	$\ r\ _2$	1.0e+000	7.7e-002	1.6e-004	8.2e-010	2.0e-020	$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016
k	1	2	3	4	5																		
$\ r\ _2$	1.0e+000	7.7e-002	1.6e-004	8.2e-010	2.0e-020																		
$ s - \lambda_1 $	3.7e-001	-1.2e-002	-2.9e-005	-1.4e-010	-2.2e-016																		
lyche-numerical-linear-algebra-F04545	351	0.960	$Q \in \mathbb{C}^{n \times n}$	$Q \in \mathbb{C}^{n \times n}$	same. If $A = QR$ is a QR factorization then $Q \in \mathbb{C}^{n \times n}$ is unitary, $Q^*Q = I$ and																		
lyche-numerical-linear-algebra-F04546	351	1.000	$R \in \mathbb{C}^{n \times n}$	$R \in \mathbb{C}^{n \times n}$	$R \in \mathbb{C}^{n \times n}$ is upper triangular.																		
lyche-numerical-linear-algebra-F04547	351	1.000	$R_k Q_k$	$R_k Q_k$	The determination of the QR factorization of A_k and the computation of $R_k Q_k$																		
lyche-numerical-linear-algebra-F04548	351	1.000	$R_k = Q_k^* A_k$	$R_k = Q_k^* A_k$	point, since $R_k = Q_k^* A_k$ we find																		
lyche-numerical-linear-algebra-F04549	352	1.000	A_{10}	A_{10}	A_{10} is almost diagonal and contains approximations to the eigenvalues $\lambda_1 = 3$ and																		
lyche-numerical-linear-algebra-F04550	352	1.000	$\lambda_1 = 3$	$\lambda_1 = 3$	A_{10} is almost diagonal and contains approximations to the eigenvalues $\lambda_1 = 3$ and																		
lyche-numerical-linear-algebra-F04551	352	0.999	$\lambda_2 = 1$	$\lambda_2 = 1$	$\lambda_2 = 1$ on the diagonal.																		
lyche-numerical-linear-algebra-F04552	352	1.000	$\lambda_1, \dots, \lambda_4$	$\lambda_1, \dots, \lambda_4$	This matrix is almost quasi-triangular and estimates for the eigenvalues $\lambda_1, \dots, \lambda_4$																		
lyche-numerical-linear-algebra-F04553	352	0.835	A_{14}	A_{14}	of A can now easily be determined from the diagonal blocks of A_{14} . The 1×1																		
lyche-numerical-linear-algebra-F04554	352	1.000	$\lambda_1 \approx 2.323$	$\lambda_1 \approx 2.323$	blocks give us two real eigenvalues $\lambda_1 \approx 2.323$ and $\lambda_4 \approx 0.2275$. The middle																		

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lyche-numerical-linear-algebra- F04555	352	■ 1.000	$\lambda_4 \approx 0.2275$	$\lambda_4 \approx 0.2275$	blocks give us two real eigenvalues $\lambda_1 \approx 2.323$ and $\lambda_4 \approx 0.2275$. The middle
lyche-numerical-linear-algebra- F04556	352	■ 0.813	$\lambda_2 \approx 0.0914 + 0.4586i$	$\lambda_2 \approx 0.0914 + 0.4586i$	2×2 block has complex eigenvalues resulting in $\lambda_2 \approx 0.0914 + 0.4586i$ and $\lambda_3 \approx$
lyche-numerical-linear-algebra- F04557	352	■ 0.813	$\lambda_3 \approx$	$\lambda_3 \approx$	2×2 block has complex eigenvalues resulting in $\lambda_2 \approx 0.0914 + 0.4586i$ and $\lambda_3 \approx$
lyche-numerical-linear-algebra- F04558	352	■ 1.000	$(\mathbf{A}_k)_k$	$(\mathbf{A}_k)_k$	$(\mathbf{A}_k)_k$ converges to the triangular Schur form (Cf. Theorem 6.5) if all the eigenvalues
lyche-numerical-linear-algebra- F04559	352	■ 0.674	$k=1,2,3 \dots$	$k = 1, 2, 3 \dots$	Theorem 15.3 (QR and Power) For $k = 1, 2, 3, \dots$, the QR factorization of $\mathbf{A}^k$ is
lyche-numerical-linear-algebra- F04560	352	■ 0.674	$\mathbf{A}^k$	$\mathbf{A}^k$	Theorem 15.3 (QR and Power) For $k = 1, 2, 3, \dots$, the QR factorization of $\mathbf{A}^k$ is
lyche-numerical-linear-algebra- F04561	352	■ 1.000	$\mathbf{A}^k = \tilde{\mathbf{Q}}_k \tilde{\mathbf{R}}_k$	$\mathbf{A}^k = \tilde{\mathbf{Q}}_k \tilde{\mathbf{R}}_k$	$\mathbf{A}^k = \tilde{\mathbf{Q}}_k \tilde{\mathbf{R}}_k$, where
lyche-numerical-linear-algebra- F04562	352	■ 1.000	$\mathbf{Q}_1, \dots, \mathbf{Q}_k, \mathbf{R}_1, \dots, \mathbf{R}_k$	$\mathbf{Q}_1, \dots, \mathbf{Q}_k, \mathbf{R}_1, \dots, \mathbf{R}_k$	and $\mathbf{Q}_1, \dots, \mathbf{Q}_k, \mathbf{R}_1, \dots, \mathbf{R}_k$ are the matrices generated by the basic QR algo-
lyche-numerical-linear-algebra- F04563	353	■ 1.000	$\tilde{\mathbf{Q}}_1 \tilde{\mathbf{R}}_1 = \mathbf{Q}_1 \mathbf{R}_1 = \mathbf{A}_1$	$\tilde{\mathbf{Q}}_1 \tilde{\mathbf{R}}_1 = \mathbf{Q}_1 \mathbf{R}_1 = \mathbf{A}_1$	The proof is by induction on k . Clearly $\tilde{\mathbf{Q}}_1 \tilde{\mathbf{R}}_1 = \mathbf{Q}_1 \mathbf{R}_1 = \mathbf{A}_1$. Suppose
lyche-numerical-linear-algebra- F04564	353	■ 1.000	$\tilde{\mathbf{Q}}_{k-1} \tilde{\mathbf{R}}_{k-1} = \mathbf{A}^{k-1}$	$\tilde{\mathbf{Q}}_{k-1} \tilde{\mathbf{R}}_{k-1} = \mathbf{A}^{k-1}$	$\tilde{\mathbf{Q}}_{k-1} \tilde{\mathbf{R}}_{k-1} = \mathbf{A}^{k-1}$ for some $k \geq 2$. Since $\mathbf{Q}_k \mathbf{R}_k = \mathbf{A}_k$ and using (15.11)
lyche-numerical-linear-algebra- F04565	353	■ 1.000	$\mathbf{Q}_k \mathbf{R}_k = \mathbf{A}_k$	$\mathbf{Q}_k \mathbf{R}_k = \mathbf{A}_k$	$\tilde{\mathbf{Q}}_{k-1} \tilde{\mathbf{R}}_{k-1} = \mathbf{A}^{k-1}$ for some $k \geq 2$. Since $\mathbf{Q}_k \mathbf{R}_k = \mathbf{A}_k$ and using (15.11)
lyche-numerical-linear-algebra- F04566	353	■ 1.000	$\tilde{\mathbf{R}}_k$	$\tilde{\mathbf{R}}_k$	Since $\tilde{\mathbf{R}}_k$ is upper triangular, its first column is a multiple of $\mathbf{e}_1$ so that
lyche-numerical-linear-algebra- F04567	353	■ 1.000	$\ \tilde{\mathbf{q}}_1^{(k)}\ _2 = 1$	$\ \tilde{\mathbf{q}}_1^{(k)}\ _2 = 1$	Since $\ \tilde{\mathbf{q}}_1^{(k)}\ _2 = 1$ the first column of $\tilde{\mathbf{Q}}_k$ is the result of applying the normalized
lyche-numerical-linear-algebra- F04568	353	■ 1.000	$\tilde{\mathbf{Q}}_k$	$\tilde{\mathbf{Q}}_k$	Since $\ \tilde{\mathbf{q}}_1^{(k)}\ _2 = 1$ the first column of $\tilde{\mathbf{Q}}_k$ is the result of applying the normalized
lyche-numerical-linear-algebra- F04569	353	■ 1.000	$\mathbf{x}_0 = \mathbf{e}_1$	$\mathbf{x}_0 = \mathbf{e}_1$	power iteration (15.6) to the starting vector $\mathbf{x}_0 = \mathbf{e}_1$. If this iteration converges we
lyche-numerical-linear-algebra- F04570	353	■ 1.000	$\lambda_1 \mathbf{e}_1$	$\lambda_1 \mathbf{e}_1$	It can be shown that the first column of $\mathbf{A}_k$ must then converge to $\lambda_1 \mathbf{e}_1$, where λ_1 is a
lyche-numerical-linear-algebra- F04571	353	■ 1.000	$(\tilde{\mathbf{Q}}_k^* \mathbf{A} \tilde{\mathbf{Q}}_k)_k$	$(\tilde{\mathbf{Q}}_k^* \mathbf{A} \tilde{\mathbf{Q}}_k)_k$	Indeed, what is observed in practice is that the sequence $(\tilde{\mathbf{Q}}_k^* \mathbf{A} \tilde{\mathbf{Q}}_k)_k$ converges to a
lyche-numerical-linear-algebra- F04572	353	■ 1.000	$\mathbf{H}_1$	$\mathbf{H}_1$	an initial reduction of $\mathbf{A}$ to upper Hessenberg form $\mathbf{H}_1$ using Algorithm 14.1, the
lyche-numerical-linear-algebra- F04573	353	■ 1.000	$\mathbf{P}_{i,i+1}, i = 1, \dots, n-1$	$\mathbf{P}_{i,i+1}, i = 1, \dots, n-1$	determine plane rotations $\mathbf{P}_{i,i+1}, i = 1, \dots, n-1$ so that $\mathbf{P}_{n-1,n} \cdots \mathbf{P}_{1,2} \mathbf{H}_1 = \mathbf{R}_1$
lyche-numerical-linear-algebra- F04574	353	■ 1.000	$\mathbf{P}_{n-1,n} \cdots \mathbf{P}_{1,2} \mathbf{H}_1 = \mathbf{R}_1$	$\mathbf{P}_{n-1,n} \cdots \mathbf{P}_{1,2} \mathbf{H}_1 = \mathbf{R}_1$	determine plane rotations $\mathbf{P}_{i,i+1}, i = 1, \dots, n-1$ so that $\mathbf{P}_{n-1,n} \cdots \mathbf{P}_{1,2} \mathbf{H}_1 = \mathbf{R}_1$
lyche-numerical-linear-algebra- F04575	353	■ 1.000	$\mathbf{H}_1 = \mathbf{Q}_1 \mathbf{R}_1$	$\mathbf{H}_1 = \mathbf{Q}_1 \mathbf{R}_1$	is upper triangular. The details were described in Sect. 5.6. Thus $\mathbf{H}_1 = \mathbf{Q}_1 \mathbf{R}_1$,
lyche-numerical-linear-algebra- F04576	353	■ 0.999	$\mathbf{Q}_1 = \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$	$\mathbf{Q}_1 = \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$	where $\mathbf{Q}_1 = \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$ is a QR factorization of $\mathbf{H}_1$. To finish the QR step we
lyche-numerical-linear-algebra- F04577	353	■ 0.996	$\mathbf{R}_1 \mathbf{Q}_1 = \mathbf{R}_1 \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$	$\mathbf{R}_1 \mathbf{Q}_1 = \mathbf{R}_1 \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$	compute $\mathbf{R}_1 \mathbf{Q}_1 = \mathbf{R}_1 \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$. This postmultiplication step is illustrated by
lyche-numerical-linear-algebra- F04578	353	■ 0.999	$(i+1, i)$	$(i+1, i)$	The postmultiplication by $\mathbf{P}_{i,i+1}$ introduces a nonzero in position $(i+1, i)$
lyche-numerical-linear-algebra- F04579	354	■ 0.993	$\mathbf{R} \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$	$\mathbf{R} \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$	matrix $\mathbf{R} \mathbf{P}_{1,2}^* \cdots \mathbf{P}_{n-1,n}^*$ is upper Hessenberg and a QR step leaves the Hessenberg
lyche-numerical-linear-algebra- F04580	354	■ 1.000	$a_{i+1, i}$	$a_{i+1, i}$	If a subdiagonal element $a_{i+1, i}$ of an upper Hessenberg matrix $\mathbf{A}$ is equal to zero,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
lyche-numerical-linear-algebra- F04581	354	1.000	$A(1:i, 1:i)$	$A(1:i, 1:i)$	matrices $A(1:i, 1:i)$ and $A(i+1:n, i+1:n)$. Thus, if during the iteration
lyche-numerical-linear-algebra- F04582	354	1.000	$A(i+1:n, i+1:n)$	$A(i+1:n, i+1:n)$	matrices $A(1:i, 1:i)$ and $A(i+1:n, i+1:n)$. Thus, if during the iteration
lyche-numerical-linear-algebra- F04583	354	1.000	$ \hat{a}_{i+1,i}^{(k)} \leq \epsilon$	$ \hat{a}_{i+1,i}^{(k)} \leq \epsilon$	To see what effect this can have on the eigenvalues of A suppose $ \hat{a}_{i+1,i}^{(k)} \leq \epsilon$. Let
lyche-numerical-linear-algebra- F04584	354	0.992	$\hat{A}_k := A_k - a_{i+1,i}^{(k)} e_{i+1} e_i^T$	$\hat{A}_k := A_k - a_{i+1,i}^{(k)} e_{i+1} e_i^T$	$\hat{A}_k := A_k - a_{i+1,i}^{(k)} e_{i+1} e_i^T$ be the matrix obtained from A_k by setting the $(i+1, i)$
lyche-numerical-linear-algebra- F04585	354	1.000	$A_k = \tilde{Q}_{k-1}^* A \tilde{Q}_{k-1}$	$A_k = \tilde{Q}_{k-1}^* A \tilde{Q}_{k-1}$	element equal to zero. Since $A_k = \tilde{Q}_{k-1}^* A \tilde{Q}_{k-1}$ we have
lyche-numerical-linear-algebra- F04586	354	1.000	$\tilde{Q}_{k-1}$	$\tilde{Q}_{k-1}$	Since $\tilde{Q}_{k-1}$ is unitary, $\ E\ _F = \ a_{i+1,i}^{(k)} e_{i+1} e_i^T\ _F = a_{i+1,i}^{(k)} \leq \epsilon$ and setting
lyche-numerical-linear-algebra- F04587	354	1.000	$\ E\ _F = \ a_{i+1,i}^{(k)} e_{i+1} e_i^T\ _F = a_{i+1,i}^{(k)} \leq \epsilon$	$\ E\ _F = \ a_{i+1,i}^{(k)} e_{i+1} e_i^T\ _F = a_{i+1,i}^{(k)} \leq \epsilon$	Since $\tilde{Q}_{k-1}$ is unitary, $\ E\ _F = \ a_{i+1,i}^{(k)} e_{i+1} e_i^T\ _F = a_{i+1,i}^{(k)} \leq \epsilon$ and setting
lyche-numerical-linear-algebra- F04588	354	1.000	$a_{i+1,i}^{(k)} = 0$	$a_{i+1,i}^{(k)} = 0$	$a_{i+1,i}^{(k)} = 0$ amounts to a perturbation in the original A of at most ϵ . For how to
lyche-numerical-linear-algebra- F04589	355	1.000	$R_k = Q_k^* (A_k - s_k I)$	$R_k = Q_k^* (A_k - s_k I)$	Since $R_k = Q_k^* (A_k - s_k I)$ we find
lyche-numerical-linear-algebra- F04590	355	1.000	Q_k	Q_k	Theorem 15.3 and also the inverse power method. In fact the last column of Q_k
lyche-numerical-linear-algebra- F04591	355	0.998	$A - s_k I = Q_k R_k$	$A - s_k I = Q_k R_k$	since $A - s_k I = Q_k R_k$ we have $(A - s_k I)^* = R_k^* Q_k^*$ and $(A - s_k I)^* Q_k = R_k^*$.
lyche-numerical-linear-algebra- F04592	355	0.998	$(A - s_k I)^* = R_k^* Q_k^*$	$(A - s_k I)^* = R_k^* Q_k^*$	since $A - s_k I = Q_k R_k$ we have $(A - s_k I)^* = R_k^* Q_k^*$ and $(A - s_k I)^* Q_k = R_k^*$.
lyche-numerical-linear-algebra- F04593	355	0.998	$(A - s_k I)^* Q_k = R_k^*$	$(A - s_k I)^* Q_k = R_k^*$	since $A - s_k I = Q_k R_k$ we have $(A - s_k I)^* = R_k^* Q_k^*$ and $(A - s_k I)^* Q_k = R_k^*$.
lyche-numerical-linear-algebra- F04594	355	1.000	R_k^*	R_k^*	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A - s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04595	355	1.000	n, n	n, n	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A - s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04596	355	1.000	$\bar{r}_{nn}^{(k)}$	$\bar{r}_{nn}^{(k)}$	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A - s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04597	355	1.000	$(A - s_k I)^* Q_k e_n =$	$(A - s_k I)^* Q_k e_n =$	Thus, since R_k^* is lower triangular with n, n element $\bar{r}_{nn}^{(k)}$ we find $(A - s_k I)^* Q_k e_n =$
lyche-numerical-linear-algebra- F04598	355	1.000	$R_k^* e_n = \bar{r}_{nn}^{(k)} e_n$	$R_k^* e_n = \bar{r}_{nn}^{(k)} e_n$	$R_k^* e_n = \bar{r}_{nn}^{(k)} e_n$ from which the conclusion follows.
lyche-numerical-linear-algebra- F04599	355	1.000	$s_k := e_n^T A_k e_n$	$s_k := e_n^T A_k e_n$	The shift $s_k := e_n^T A_k e_n$ is called the Rayleigh quotient shift , while the
lyche-numerical-linear-algebra- F04600	355	0.999	$A := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$A := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	is given by $A := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. We obtain $A_k = A$ for all $k \in \mathbb{N}$. For convergence of the
lyche-numerical-linear-algebra- F04601	355	0.999	$A_k = A$	$A_k = A$	is given by $A := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. We obtain $A_k = A$ for all $k \in \mathbb{N}$. For convergence of the
lyche-numerical-linear-algebra- F04602	355	1.000	$Au - \lambda u$	$Au - \lambda u$	Exercise 15.1 (Orthogonal Vectors) Show that u and $Au - \lambda u$ are orthogonal
lyche-numerical-linear-algebra- F04603	355	1.000	$\lambda = \frac{u^* A u}{u^* u}$	$\lambda = \frac{u^* A u}{u^* u}$	when $\lambda = \frac{u^* A u}{u^* u}$.
lyche-numerical-linear-algebra- F04604	358	1.000	$f: \mathbb{R}^n \rightarrow \mathbb{R}$	$f: \mathbb{R}^n \rightarrow \mathbb{R}$	For any sufficiently differentiable $f: \mathbb{R}^n \rightarrow \mathbb{R}$ we recall that the partial derivative
lyche-numerical-linear-algebra- F04605	358	1.000	$\nabla f(x) \in \mathbb{R}^n$	$\nabla f(x) \in \mathbb{R}^n$	$\nabla f(x) \in \mathbb{R}^n$, and the hessian $Hf = \nabla \nabla^T f(x) \in \mathbb{R}^{n \times n}$ of f by

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lyche-numerical-linear-algebra-F04686	358	1.000	$\backslash\boldsymbol{\mathrm{H}} f=\backslash\nabla \backslash\nabla^{\mathrm{T}} f(\backslash\boldsymbol{\mathrm{x}}) \backslash\mathrm{in} \backslash\mathbb{R}^{\mathrm{n}} \backslash\times \mathrm{n}$	$\boldsymbol{H} f = \nabla \nabla^T f(\boldsymbol{x}) \in \mathbb{R}^{n \times n}$	$\nabla f(\boldsymbol{x}) \in \mathbb{R}^n$, and the hessian $\boldsymbol{H} f = \nabla \nabla^T f(\boldsymbol{x}) \in \mathbb{R}^{n \times n}$ of f by
lyche-numerical-linear-algebra-F04687	358	1.000	$\backslash\nabla^{\mathrm{T}} f:=\left(\nabla f\right)^{\mathrm{T}}$	$\nabla^T f := (\nabla f)^T$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla \nabla^T$ and $\nabla^T \nabla$ are
lyche-numerical-linear-algebra-F04688	358	1.000	$\backslash\nabla \backslash\nabla^{\mathrm{T}}$	$\nabla \nabla^T$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla \nabla^T$ and $\nabla^T \nabla$ are
lyche-numerical-linear-algebra-F04689	358	1.000	$\backslash\nabla^{\mathrm{T}} \backslash\nabla$	$\nabla^T \nabla$	where $\nabla^T f := (\nabla f)^T$ is the row vector gradient. The operators $\nabla \nabla^T$ and $\nabla^T \nabla$ are
lyche-numerical-linear-algebra-F04610	358	1.000	$\backslash\nabla^{\mathrm{T}} \backslash\nabla f=\mathrm{D}_{\mathrm{1}}^2 f+\cdots+\mathrm{D}_{\mathrm{n}}^2 f=:\backslash\nabla^2$	$\nabla^T \nabla f = D_1^2 f + \cdots + D_n^2 f =: \nabla^2$	quite different. Indeed, $\nabla^T \nabla f = D_1^2 f + \cdots + D_n^2 f =: \nabla^2$ the Laplacian of f ,
lyche-numerical-linear-algebra-F04611	358	1.000	$f, g: \backslash\mathbb{R}^{\mathrm{n}} \rightarrow \backslash\mathbb{R}$	$f, g: \mathbb{R}^n \rightarrow \mathbb{R}$	Lemma 16.1 (Product Rules) For $f, g: \mathbb{R}^n \rightarrow \mathbb{R}$ we have the product rules
lyche-numerical-linear-algebra-F04612	358	0.982	$\backslash\nabla(f g)=f \backslash\nabla g+g \backslash\nabla f, \quad \backslash\nabla^{\mathrm{T}}(f g)=f \backslash\nabla^{\mathrm{T}} g+g \backslash\nabla^{\mathrm{T}} f$	$\nabla(fg) = f\nabla g + g\nabla f, \quad \nabla^T(fg) = f\nabla^T g + g\nabla^T f$	1. $\nabla(fg) = f\nabla g + g\nabla f, \quad \nabla^T(fg) = f\nabla^T g + g\nabla^T f$.
lyche-numerical-linear-algebra-F04613	358	1.000	$\backslash\nabla \backslash\nabla^{\mathrm{T}}(f g)=\backslash\nabla f \backslash\nabla^{\mathrm{T}} g+\nabla g \nabla^{\mathrm{T}} f+f \nabla \nabla^{\mathrm{T}} g+g \nabla \nabla^{\mathrm{T}} f$	$\nabla \nabla^T(fg) = \nabla f \nabla^T g + \nabla g \nabla^T f + f \nabla \nabla^T g + g \nabla \nabla^T f$	2. $\nabla \nabla^T(fg) = \nabla f \nabla^T g + \nabla g \nabla^T f + f \nabla \nabla^T g + g \nabla \nabla^T f$.
lyche-numerical-linear-algebra-F04614	358	1.000	$\backslash\nabla^2(f g)=2 \backslash\nabla^{\mathrm{T}} f \nabla g+f \nabla^2 g+g \nabla^2 f$	$\nabla^2(fg) = 2\nabla^T f \nabla g + f \nabla^2 g + g \nabla^2 f$	3. $\nabla^2(fg) = 2\nabla^T f \nabla g + f \nabla^2 g + g \nabla^2 f$.
lyche-numerical-linear-algebra-F04615	359	1.000	$\backslash\boldsymbol{\mathrm{f}}=\left[\mathrm{f}_{\mathrm{1}}, \ldots \mathrm{f}_{\mathrm{m}}\right]^{\mathrm{T}}: \backslash\mathbb{R}^{\mathrm{n}} \rightarrow \backslash\mathbb{R}^{\mathrm{m}}$	$\boldsymbol{f} = [f_1, \dots f_m]^T: \mathbb{R}^n \rightarrow \mathbb{R}^m$	We define the Jacobian of a vector function $\boldsymbol{f} = [f_1, \dots f_m]^T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ as
lyche-numerical-linear-algebra-F04616	359	1.000	$f(\backslash\boldsymbol{\mathrm{x}})=f(x, y)=x^2-x y+y^2$	$f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$	As an example, if $f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$ and $\boldsymbol{g}(x, y) := [f(x, y), x - y]^T$
lyche-numerical-linear-algebra-F04617	359	1.000	$\backslash\boldsymbol{\mathrm{g}}(x, y):=\left[f(x, y), x-y\right]^{\mathrm{T}}$	$\boldsymbol{g}(x, y) := [f(x, y), x - y]^T$	As an example, if $f(\boldsymbol{x}) = f(x, y) = x^2 - xy + y^2$ and $\boldsymbol{g}(x, y) := [f(x, y), x - y]^T$
lyche-numerical-linear-algebra-F04618	359	1.000	$f \in C^2(\Omega)$	$f \in C^2(\Omega)$	Lemma 16.2 (Second Order Taylor Expansion) Suppose $f \in C^2(\Omega)$, where
lyche-numerical-linear-algebra-F04619	359	0.633	$\Omega \in \backslash\mathbb{R}^{\mathrm{n}}$	$\Omega \in \mathbb{R}^n$	$\Omega \in \mathbb{R}^n$ contains two points $\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$, such that the line segment $L := \{\boldsymbol{x} + t\boldsymbol{h} :$
lyche-numerical-linear-algebra-F04620	359	0.633	$\backslash\boldsymbol{\mathrm{x}}, \backslash\boldsymbol{\mathrm{x}}+\backslash\boldsymbol{\mathrm{h}} \in \Omega$	$\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$	$\Omega \in \mathbb{R}^n$ contains two points $\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$, such that the line segment $L := \{\boldsymbol{x} + t\boldsymbol{h} :$
lyche-numerical-linear-algebra-F04621	359	0.633	$L:=\left\{\backslash\boldsymbol{\mathrm{x}}+t \backslash\boldsymbol{\mathrm{h}}:\right.$	$L := \{\boldsymbol{x} + t\boldsymbol{h} :$	$\Omega \in \mathbb{R}^n$ contains two points $\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{h} \in \Omega$, such that the line segment $L := \{\boldsymbol{x} + t\boldsymbol{h} :$
lyche-numerical-linear-algebra-F04622	359	1.000	$t \in(0,1) \backslash\} \backslash\subset \Omega$	$t \in (0, 1)\} \subset \Omega$	$t \in (0, 1)\} \subset \Omega$. Then
lyche-numerical-linear-algebra-F04623	359	1.000	$g:[0,1] \rightarrow \backslash\mathbb{R}$	$g: [0, 1] \rightarrow \mathbb{R}$	Proof Let $g: [0, 1] \rightarrow \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra-F04624	359	1.000	$g(t):=f(\backslash\boldsymbol{\mathrm{x}}+t \backslash\boldsymbol{\mathrm{h}})$	$g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$	Proof Let $g: [0, 1] \rightarrow \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra-F04625	359	1.000	$g \in C^2[0,1]$	$g \in C^2[0, 1]$	Proof Let $g: [0, 1] \rightarrow \mathbb{R}$ be defined by $g(t) := f(\boldsymbol{x} + t\boldsymbol{h})$. Then $g \in C^2[0, 1]$ and
lyche-numerical-linear-algebra-F04626	359	1.000	$\backslash\boldsymbol{\mathrm{c}}=\backslash\boldsymbol{\mathrm{x}}+u \backslash\boldsymbol{\mathrm{h}}$	$\boldsymbol{c} = \boldsymbol{x} + u\boldsymbol{h}$	we obtain (16.2) with $\boldsymbol{c} = \boldsymbol{x} + u\boldsymbol{h}$.
lyche-numerical-linear-algebra-F04627	360	1.000	$m, n \in \backslash\mathbb{N}, \backslash\boldsymbol{\mathrm{B}} \in \backslash\mathbb{R}^{\mathrm{n} \times \mathrm{n}}, \backslash\boldsymbol{\mathrm{C}} \in \backslash\mathbb{R}^{\mathrm{m} \times \mathrm{n}}, \backslash\boldsymbol{\mathrm{C}} \in$	$m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in$	Lemma 16.3 (Functions Involving Matrices) For any $m, n \in \mathbb{N}, \boldsymbol{B} \in \mathbb{R}^{n \times n}, \boldsymbol{C} \in$
lyche-numerical-linear-algebra-F04628	360	1.000	$\backslash\mathbb{R}^{\mathrm{m} \times \mathrm{n}}, \backslash\boldsymbol{\mathrm{x}} \in \backslash\mathbb{R}^{\mathrm{n}}, \backslash\boldsymbol{\mathrm{y}} \in \backslash\mathbb{R}^{\mathrm{m}}$	$\mathbb{R}^{m \times n}, \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$	$\mathbb{R}^{m \times n}$, and $\boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^m$ we have
lyche-numerical-linear-algebra-F04629	360	1.000	$\backslash\nabla\left(\backslash\boldsymbol{\mathrm{y}}^{\mathrm{T}} \backslash\boldsymbol{\mathrm{C}}\right)=\backslash\nabla^{\mathrm{T}}(\backslash\boldsymbol{\mathrm{C}} \backslash\boldsymbol{\mathrm{x}})=\backslash\boldsymbol{\mathrm{C}}$	$\nabla(\boldsymbol{y}^T \boldsymbol{C}) = \nabla^T(\boldsymbol{C}\boldsymbol{x}) = \boldsymbol{C}$	1. $\nabla(\boldsymbol{y}^T \boldsymbol{C}) = \nabla^T(\boldsymbol{C}\boldsymbol{x}) = \boldsymbol{C}$,
lyche-numerical-linear-algebra-F04630	360	1.000	$\backslash\nabla\left(\backslash\boldsymbol{\mathrm{x}}^{\mathrm{T}} \backslash\boldsymbol{\mathrm{B}}\right)=\left(\backslash\boldsymbol{\mathrm{B}}+\backslash\boldsymbol{\mathrm{B}}^{\mathrm{T}}\right) \backslash\boldsymbol{\mathrm{x}}, \quad \backslash\nabla^{\mathrm{T}}\left(\backslash\boldsymbol{\mathrm{x}}^{\mathrm{T}} \backslash\boldsymbol{\mathrm{B}}\right)=\backslash\boldsymbol{\mathrm{x}}^{\mathrm{T}}\left(\backslash\boldsymbol{\mathrm{B}}+\backslash\boldsymbol{\mathrm{B}}^{\mathrm{T}}\right)$	$\nabla(\boldsymbol{x}^T \boldsymbol{B}\boldsymbol{x}) = (\boldsymbol{B} + \boldsymbol{B}^T) \boldsymbol{x}, \quad \nabla^T(\boldsymbol{x}^T \boldsymbol{B}\boldsymbol{x}) = \boldsymbol{x}^T (\boldsymbol{B} + \boldsymbol{B}^T)$	2. $\nabla(\boldsymbol{x}^T \boldsymbol{B}\boldsymbol{x}) = (\boldsymbol{B} + \boldsymbol{B}^T) \boldsymbol{x}, \quad \nabla^T(\boldsymbol{x}^T \boldsymbol{B}\boldsymbol{x}) = \boldsymbol{x}^T (\boldsymbol{B} + \boldsymbol{B}^T)$,

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lyche-numerical-linear-algebra- F04631	360	■ 1.000	$\nabla \nabla^T (\mathbf{x}^T \mathbf{B} \mathbf{x}) = \mathbf{B} + \mathbf{B}^T$	$\nabla \nabla^T (\mathbf{x}^T \mathbf{B} \mathbf{x}) = \mathbf{B} + \mathbf{B}^T$	3. $\nabla \nabla^T (\mathbf{x}^T \mathbf{B} \mathbf{x}) = \mathbf{B} + \mathbf{B}^T$.
lyche-numerical-linear-algebra- F04632	360	■ 1.000	$D_i (\mathbf{y}^T \mathbf{C}) = \lim_{h \rightarrow 0} \frac{1}{h} ((\mathbf{y} + h \mathbf{e}_i)^T \mathbf{C} - \mathbf{y}^T \mathbf{C}) = \mathbf{e}_i^T \mathbf{C}$	$D_i (\mathbf{y}^T \mathbf{C}) = \lim_{h \rightarrow 0} \frac{1}{h} ((\mathbf{y} + h \mathbf{e}_i)^T \mathbf{C} - \mathbf{y}^T \mathbf{C}) = \mathbf{e}_i^T \mathbf{C}$	1. We find $D_i (\mathbf{y}^T \mathbf{C}) = \lim_{h \rightarrow 0} \frac{1}{h} ((\mathbf{y} + h \mathbf{e}_i)^T \mathbf{C} - \mathbf{y}^T \mathbf{C}) = \mathbf{e}_i^T \mathbf{C}$ and $D_i (\mathbf{C} \mathbf{x}) =$
lyche-numerical-linear-algebra- F04633	360	■ 1.000	$D_i (\mathbf{C} \mathbf{x}) =$	$D_i (\mathbf{C} \mathbf{x}) =$	1. We find $D_i (\mathbf{y}^T \mathbf{C}) = \lim_{h \rightarrow 0} \frac{1}{h} ((\mathbf{y} + h \mathbf{e}_i)^T \mathbf{C} - \mathbf{y}^T \mathbf{C}) = \mathbf{e}_i^T \mathbf{C}$ and $D_i (\mathbf{C} \mathbf{x}) =$
lyche-numerical-linear-algebra- F04634	360	■ 1.000	$\lim_{h \rightarrow 0} \frac{1}{h} (\mathbf{C} (\mathbf{x} + h \mathbf{e}_i) - \mathbf{C} \mathbf{x}) = \mathbf{C} \mathbf{e}_i$	$\lim_{h \rightarrow 0} \frac{1}{h} (\mathbf{C} (\mathbf{x} + h \mathbf{e}_i) - \mathbf{C} \mathbf{x}) = \mathbf{C} \mathbf{e}_i$	$\lim_{h \rightarrow 0} \frac{1}{h} (\mathbf{C} (\mathbf{x} + h \mathbf{e}_i) - \mathbf{C} \mathbf{x}) = \mathbf{C} \mathbf{e}_i$ and 1. follows.
lyche-numerical-linear-algebra- F04635	363	■ 0.783	$\mathbf{x}^T \mathbf{A} \mathbf{y}$	$\mathbf{x}^T \mathbf{A} \mathbf{y}$	1979-3, $\mathbf{x}^T \mathbf{A} \mathbf{y}$ inequality, 119
lyche-numerical-linear-algebra- F04636	363	■ 1.000	$\mathbf{T} \mathbf{H} \mathbf{x} = \mathbf{b}$, 79	$\mathbf{T} \mathbf{H} \mathbf{x} = \mathbf{b}$, 79	1981-3, solving $\mathbf{T} \mathbf{H} \mathbf{x} = \mathbf{b}$, 79
lyche-numerical-linear-algebra- F04637	365	■ 0.995	$\mathbf{A}^* \mathbf{A}$, 86	$\mathbf{A}^* \mathbf{A}$, 86	$\mathbf{A}^* \mathbf{A}$, 86
lyche-numerical-linear-algebra- F04638	374	■ 1.000	$\mathbf{X} \mathbf{X}$	$\mathbf{X} \mathbf{X}$	Engineering $\mathbf{X} \mathbf{X}$.